



**TSL 0.18 micron, 1.8 volt
Fast Silicon SC Library
TSL18FS120**

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Tower Semiconductor/Synopsys, Inc

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Preface

This guide is intended for use with Production Release of Fast Silicon SC Library TSL18FS120, for the TSL 0.18 micron process technology (DR2 0018SL) and characterized with Tower 0.18 micron Hspice models Rev 1.2 from Tower's spec "DRS2_0018B_DIG", Rev 1.0. These are the most updated spice models intended for digital library characterization only.

Associated Guides and Documentation

Other publications you can consult for related information are:

TSL18FS120, 0.18 micron, 1.8 Volt, Fast Silicon SC Library-TSL visa(this datasheet)

TSL18CIO150, 0.18 micron, 1.8 Volt, Super Core Limited I/O Pad Library

TSL18CIO250, 0.18 micron, 3.3 Volt, Super Core Limited I/O Pad Library

TSL18CIO210, 0.18 micron, 3.3 Volt, Core Limited NAND I/O Pad Library

TSL18CIO310, 0.18 micron, 5.0 Volt, Core Limited HOST I/O Pad Library

TSL18CPC250, 0.18 micron, 3.3/5 Volt, Core Limited PCI I/O Pad Library

TSL18RD130, 0.18 micron, 1.8 Volt, Dual-Port Synchronous RAM Compiler

TSL18RS160, 0.18 micron, 1.8 Volt, High Density Synchronous RAM Compiler

TSL18RO160, 0.18 micron, 1.8 Volt, Synchronous Via ROM Compiler

TSL18CIO220, 0.18 micron, 3.3 Volt, Core Limited I/O Pad Library

TSL18SIO220, 0.18 micron, 3.3 Volt, Staggered I/O Pad Library

TSL18ASIO220, 0.18 micron, 3.3 Volt, Analog Staggered I/O Pad Library

TSL18SPC220, 0.18 micron, 3.3 Volt, PCI Staggered I/O Pad Library

TSL18PC220, 0.18 micron, 3.3 Volt, PCI Inline I/O Pad Library

TSL18IO231, 0.18 micron, 1.8/3.3 Volt, Inline Custom I/O Pad Library

TSL18IO210, 0.18 micron, 3.3 Volt, Inline Custom I/O Pad Library

TSL18IO310, 0.18 micron, 3.3/5 Volt, Inline Custom I/O Pad Library

Introduction

This manual addresses the design engineer who is doing a preliminary feasibility evaluation and wishes to make comparisons among the available technologies. Additionally, you can use this library manual while designing a chip, to see which cells are available, and to check the power consumption, critical timing values, propagation delay equations, and functions of a cell.

The datasheets only show individual pin-to-pin timings for the storage elements. For other cells, the delays in the datasheets are combined as typical-case delays for the purpose of readability.

Product Description

The Fast Silicon Standard Cell Library have the cell set functionality and drive strengths are optimized for industry standard synthesis design entry using Verilog or VHDL driving Synopsys. The cell layout is optimized for industry-leading, area-based routers.

The tsl18fs120 Library is a high-performance, standard cell library in 0.18 μm CMOS process (TS18SL).

Contents of This Manual

This introduction contains the following sections:

- The *General Information* section of this book gives basic information on the conditions under which this library was characterized and offers assistance in using derating factors and estimating propagation delay.
- The *Cells* section describes the contents of the datasheets and how to interpret them. It also explains how to decode the cell names.
- The tables in the *Cell Matrices* section give a quick reference to the features of storage elements in the library.

Following this introduction, there are three sections:

- Simple Logic Gates - AND, AND-OR-Invert, NAND, NOR, OR, OR-AND-Invert, delays, exclusive-OR and exclusive-NOR gates; buffers, balanced buffers, clock buffers and 3-state buffers with active-low enables.
- Storage Elements - D flip-flops, scan D flip-flops, JK flip-flops, scan JK flip-flops, multiplexed flip-flops, latches, tristate latches, scan latches and scan tristate latches.
- Special Functions - Adders, carry generators, multiplexers, clock gating cells and symbolic cells.

Within these divisions, the library cells are listed in alphabetical order where possible. Cells of a similar type have been combined. For example, the information for all the 2-input NAND gates - ND02D0, ND02D1, ND02D2, ND02D4, ND02D7 and ND02DA - has been combined into one datasheet.

For storage elements, there is a cover page listing the common information for all cells of that type, then the following pages give information specific to individual cells in the grouping. For example, there is a cover page for D flip-flops with set and clear, then a page each on DFBRB1, DFBRB2 and DFBRB4.

Buffers have been grouped together by type with different drive capabilities. For example, INV0D0, INV0D1, INV0D2, INV0D4, INV0D7 and INV0DA have been combined on a single datasheet.

Additional Information

For information on:

- RAMs, ROMs - Please see the individual Visa 0.18 μm cell compiler manuals.
- Synthesized Blocks - Please see the relevant Synopsys documentation.

General Information

Recommended Operating Conditions

This table gives the recommended operating conditions for integrated circuits developed with the tsl18fs120 Library

Parameters	Minimum	Maximum	Conditions
Power Supply	1.98V	1.62V	
VIL, low level input voltage CMOS Input	-0.33V	0.2 x VDD	Guaranteed Input Low Voltage
VIH, high level input voltage CMOS Input	0.7 x VDD	VDD + 0.5V	Guaranteed Input High Voltage
Junction Temperature	-40° C	125° C	

AC Characteristics

Timing Measurement Conditions

Unless otherwise specified:

VDD = 1.8 volts

Junction Temperature = 25 degrees C

Process = typical case

AC Timing Definitions

This section explains the timing definitions that are given in the critical timing values portion of the datasheets for storage elements.

tS - Setup Time

The time a signal must be maintained at a specified input before a transition occurs at another specified input. The value given is the necessary minimum.

tH - Hold Time

The time a signal must be retained at a specified input after a transition occurs at another specified input. The value given is the necessary minimum.

Width - Minimum Signal Widths

Cells

Reading the Datasheet

The first sheet of a standard datasheet contains the following elements:

Clock Buffer with Positive Clock Input
CLK2D2

CLK2D2 clock buffer with 2x drive capability. This buffer provides both a true and complementary clock output.

Function Table

INPUT	OUTPUTS	
CLK	C	CN
L	L	H
H	H	L

Cell Description

Macro Name:	CLK2D2
Drive Capability:	2x
Gate Equivalents:	5.25
Leakage Power (pW):	104430

Pin Description

Name	Capacitance (pF)	Description
	CLK2D2	
CLK	0.006	Input
	Maximum capacitance	
C	0.18	Output
CN	0.202	Inverted Output

Pin Powers for
Standard Load = 0.064 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)
	CLK2D2
C	0.18
CN	0.202

Header and Description

The cell header in the large font describes the cell type, such as Clock Buffer with Positive Clock Input. Under the header is a list of the cells included in the category, in a smaller font. The text block following the headers gives a brief description of the cells included in this datasheet.

Icon

The icon pictured on the datasheet is the one you will see in the SYNOPSYS Tools

when you place a schematic element.

Function Table

The function table gives all the possible combinations of input and output signals for this cell type. The following symbols are used in the function tables on the datasheets:

L =	Low level	Q =	Current Q
H =	High level	Qn =	Current QN, also complement of Q
↑ =	Low to High transition	Q ₀ =	Previous level of Q
↓ =	High to Low transition	QN ₀ =	Previous level of QN, also complement of Q ₀
X =	Any level (Don't Care)	HiZ =	High impedance state
U =	Unknown	Zrl =	3-state output with resistive pulldown
Rh =	Resistive High	Zrh =	3-state output with resistive pullup
RI =	Resistive Low	Z =	3-state output

Cell Information

This information is listed under the icon and function table for the cell; not all categories will be included for all cell types and libraries:

- Gate Equivalents - One gate is the equivalent in terms of area of one 2-input NAND. The Gate Equivalent is the ratio between the area of a cell and the one of the 2-input NAND gate. This is an indication of the area required by a cell.

Pin Description Table

The Pin Description table gives:

- The name of the pin.
- The total capacitance that a signal driving in to that pin will have to drive; this includes gate capacitance as well as interconnect capacitance within the cell. For outputs, the pin capacitance is not specified, only the maximum output load capacitance on that pin is given
- A description of the pin's usage.

Pin Power Table

The pin power table gives for each pin of the table a dissipated power parameter from the Synopsys look-up table models.

This power parameter is given for a standard load and a standard input transition. The power data provided are the internal power for input pin when outputs doesn't switch, and the internal power for output pins.

The power data for output pins is defined as defined in the synopsys power models

$$\text{internal power} = \text{total switching power} - C \cdot V_{dd}^2 / 2 - \text{input power}$$

In this equation, the input power is the internal power of the relative input that create the switching of the output.

Note that due to the fact that C includes both the output pin load and the external load, the output pin internal power may be negative for some cells; this is a modelisation effect.

The complete switching power when pin I makes the pin OUT switching is:

$$\text{total switching power} = \text{internal power(OUT)} + C(\text{OUT}) \cdot V_{dd}^2 / 2 + \text{input power(I)}$$

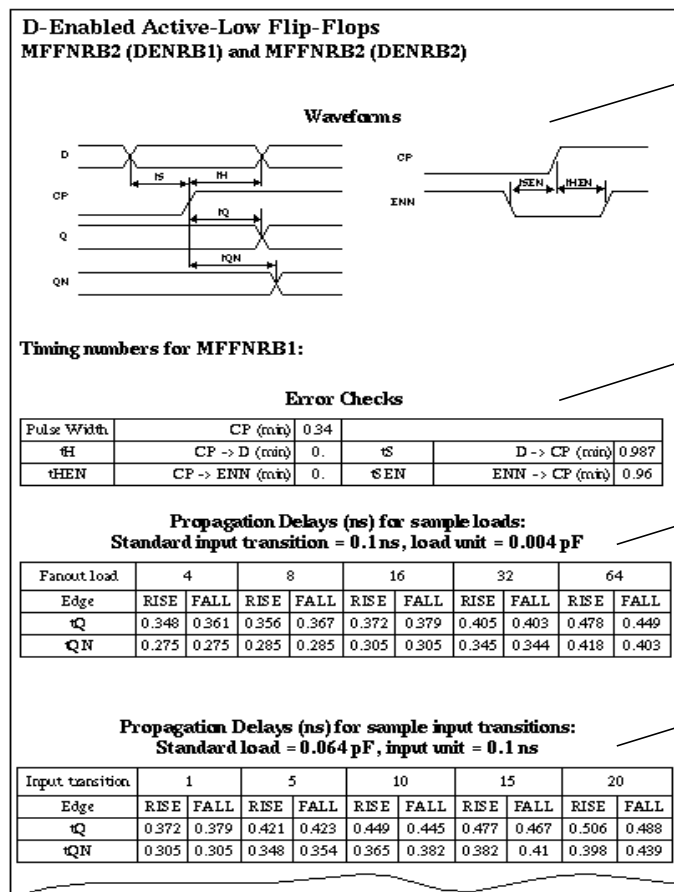
The internal power has been modelised for all output. The input power of the cells for which the input switching always create an output switching (i.e. buffer) is not modelised. Therefore only the internal power of output pin for this type of cells appears in the datasheet and includes the input power of the input pin.

In this case, the complete switching power when the input pin makes the output pin switching is:

$$\text{total switching power} = \text{internal power} + C(\text{OUT}) \cdot V_{dd}^2 / 2$$

Pin Powers for output pins gives for RISE transition (rise_power) in this datasheets.

The second page of a standard datasheet contains the following information:

Timing
WaveformError
ChecksPropagation
Delays for
Sample LoadsPropagation
Delays for
Sample Input
Transitions

Timing Waveforms

The waveforms graphically illustrate the critical timings in the Propagation Delays table.

Error Checks

For storage elements, a table of critical timing values is also included. This table gives the minimum conditions that control signals or data signals must meet

Propagation Delays for Sample Loads

The Propagation Delays for Sample Loads table are extrapolated from the characterized look-up table values using a number of standard loads multiplied by the value of the standard load for the term Cld as output load and a fixed input slew rate as specified in the table header. One standard load is the input load capacitance of a 1X inverter, which varies from library to library.

Propagation Delays for Sample Input Transitions

The Propagation Delays for Sample Input Transitions table are extrapolated from the characterized look-up table by fixing a Standard load value and using a number of input transition multiplied by the input unit. The values of Standard load and input unit are specified in the table header.

Note:

The RISE and FALL times represent the total delay time from the change of the input pin to the corresponding response on the output pin. Actual interconnect length and load cannot be determined until a design has completed placement and routing. When using these tables, you must estimate the interconnect load in units of standard loads and add that to the fanout. A rough rule of thumb is that, for every input load, there is a corresponding interconnect load approximately equal to it. For example, to estimate the delay of a NAND gate driving a fanout of two, use the column in the datasheet specifying four standard loads: two for fanout and two for the interconnect loading.

Decoding the Cell Name

This section describes the naming conventions for the cells in the tsl18fs120 library. Each cell name begins with either a two-, three-, or four-letter code that defines the type of cell. These codes are listed in the following table; the sections that follow give the detailed naming conventions for each cell type.

Code	Description
AD	Adder
AH	Half Adder
AN	AND Gate
AOI	AND-OR-Invert Gate
AOR	AND-OR Gate
BH	Bus Holder
BUFF	Non-Inverting Buffer
BUFB	Balanced Buffer
BUFT	Non-Inverting 3-State Buffer
CG	Carry Generator
CLK2	Clock Buffer
DE	D-Enabled Flip-Flop
DF	D Flip-Flop
DL	Delay cell
GC	Clock Gating
INVB	Balanced Inverter
INV0	Inverter
INVT	Inverting 3-State Buffer
JK	JK Flip-Flop
LA	D Latch
MFF	Buffered Enabled D Flip-Flop
MI	Inverting Multiplexer
MX	Multiplexer
ND	NAND Gate
NR	NOR Gate
OAI	OR-AND-Invert Gate
OR	OR Gate
ORA	OR-AND Gate
SD	Multiplexed Scan D Flip-Flop

TSL

Code	Description
SE	Multiplexed Scan Enable D Flip-Flop
SK	Scan JK Flip-Flop
SL	Multiplexed Scan Latch
SRLA	Set/Reset Latch
SU	Subtractor
XN	Exclusive NOR Gate
XR	Exclusive OR Gate

Buffers and Gates

Name Decoding Scheme: *aaaaDn*

aaaa = Name of the cell:

AN	=	AND Gate
AOI	=	AND-OR-Invert Gate
AOR	=	AND-OR Gate
BH	=	Bus Holder
BUFB	=	Balanced Buffer
BUFF	=	Non-Inverting Buffer
BUFT	=	Non-Inverting 3-State Buffer
CLK2	=	2-Phase Non-Overlapping Clocks
DL	=	Delay
INV0	=	Inverter
INVB	=	Balanced Inverter
INVT	=	Inverting 3-State Buffer
ND	=	NAND Gate
NR	=	NOR Gate
OAI	=	OR-AND-Invert Gate
OR	=	OR Gate
ORA	=	OR-AND Gate
XN	=	Exclusive NOR Gate
XR	=	Exclusive OR Gate

n = Drive Strength

0	=	Minimum drive
1	=	Basic drive speed
2	=	2 times basic drive speed
4	=	4 times basic drive speed
7	=	7 times basic drive speed
A	=	10 times basic drive speed
F	=	15 times basic drive speed
K	=	20 times basic drive speed

Multiplexers

Name Decoding Scheme: *aabcDn*

aa = Name of the Cell:

MX	=	Multiplexer
MI	=	Inverting Multiplexer

b = Number of Inversions in the Input

TSL

c = Number of Inputs

n = Drive Strength

Flip-Flops

Name Decoding Scheme: *aabcdn*

aa = Name of the Cell

DF = D Flip-Flop

DE = D Flip-Flop with D Enable

MFF = D Flip-Flop with D Enable

JK = JK Flip-Flop

b = Preset and Clear Notation

B = Both Preset and Clear

C = Clear

P = Preset

N = None

c = Clock Edge

R = Positive Rising Edge

F = Negative Falling Edge

d = Number of Output Pins:

B = Both Q and QN

Q = Q Only

N = QN Only

n = Drive Strength

Scan Flip-Flops and Scan Latches

Name Decoding Scheme: *aabcdn*

aa = Name of the Cell:

SD = Multiplexed Scan D Flip-Flop

SE = Multiplexed Scan D Flip-Flop with D Enable

SK = Scan JK Flip-Flop

SL = Scan Latch

b = Preset and Clear Notation:

B = Both Preset and Clear

C = Clear

P = Preset

N = None

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c = Enable:

H = Active High Enable
L = Active Low Enable

d = Number of Output Pins:

B = Both Q and QN
Q = Q Only
N = QN Only

n = Drive Strength

Clock Gating Cells

Name Decoding Scheme: *aabcdn*

aa = Name of the Cell:

GC = Clock Gating

b = Latch-Based or Latch-Free

L = Latch-Based

N = Latch-Free

c = For Negative or Positive-edge-triggered registers

F = For Negative-edge-triggered registers

R = For Positive-edge-triggered registers

d = Postcontrol or Precontrol Logic

S = Postcontrol Logic

R = Precontrol Logic

e = With or Without Observability Logic

N = Without Observability Logic

O = With Observability Logic

n = Drive Strength

Latches

Name Decoding Scheme: *aabcdn*

aa = Name of the Cell:

TSL

SRLA = SR Latch (borrowed *b* and *c*)
LA = D Latch

b = Preset and Clear Notation:

B = Both Preset and Clear
C = Clear
P = Preset
N = None

c = Enable:

H = Active High Enable
L = Active Low Enable

d = Number of Output Pins:

B = Both Q and QN
Q = Q Only
N = QN Only
T = Z Only

n = Drive Strength

Adders/Subtractors

Name Decoding Scheme: *aabcDn*

aa = Name of the Cell

AD = Adder
AH = Half Adder
SU = Subtractor

b = Number of Inversions in the Input

c = Number of Bits

n = Drive Strength

Carry Generators

Name Decoding Scheme: *aabcDn*

aa = Name of the Cell

CG = Carry Generator

b = Number of Inversions in the Input

c = Number of Bits

n = Drive Strength

Cell Matrices

This section gives a quick-reference guide to the storage elements in the tsl18fs120 library. Note that all storage elements feature buffered clock inputs and buffered outputs

Table 1: D Flip-Flops

Macro Name	Set	Clear	Enabled D Input	1x Drive	2x Drive	4x Drive	Single Output
DECFQ1		•	•	•			•
DECFQ2		•	•		•		•
DECFQ4		•	•			•	•
DECRQ1		•	•	•			•
DECRQ2		•	•		•		•
DECRQ4		•	•			•	•
DENRQ1			•	•			•
DENRQ2			•		•		•
DENRQ4			•			•	•
DEPFQ1	•		•	•			•
DEPFQ2	•		•		•		•
DEPRQ4	•		•			•	•
DEPRQ1	•		•	•			•
DEPRQ2	•		•		•		•
DEPRQ4	•		•			•	•
DFBFB1	•	•		•			
DFBFB2	•	•			•		
DFBFB4	•	•				•	
DFBRB1	•	•		•			
DFBRB2	•	•			•		
DFBRB4	•	•				•	
DFCRB1		•		•			
DFCRB2		•			•		
DFCRB4		•				•	
DFCFB1		•		•			
DFCFB2		•			•		
DFCFB4		•				•	
DFCRQ1		•		•			•

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Table 1: D Flip-Flops

Macro Name	Set	Clear	Enabled D Input	1x Drive	2x Drive	4x Drive	Single Output
DFCRQ2		•			•		•
DFCRQ4		•				•	•
DFCFQ1		•		•			•
DFCFQ2		•			•		•
DFCFQ4		•				•	•
DFCRN1		•		•			•
DFCRN2		•			•		•
DFCRN4		•				•	•
DFNRB1				•			
DFNRB2					•		
DFNRB4						•	
DFNFB1				•			
DFNFB2					•		
DFNFB4						•	
DFNRQ1				•			•
DFNRQ2					•		•
DFNRQ4						•	•
DFNRN1				•			•
DFNRN2					•		•
DFNRN4						•	•
DFPRB1	•			•			
DFPRB2	•				•		
DFPRB4	•					•	
DFPFB1	•			•			
DFPFB2	•				•		
DFPFB4	•					•	
MFFNRB1			•	•			
MFFNRB2			•		•		
MFFNRB4			•			•	

Table 2: JK Flip-Flops

Macro Name	Set	Clear	1x Drive	2x Drive	4x Drive
JKBRB1	•	•	•		
JKBRB2	•	•		•	
JKBRB4	•	•			•

Table 3: Multiplexed and Scan Flip-Flops

Macro Name	Set	Clear	1x Drive	2x Drive	4x Drive	Single Output
SDBFB1	•	•	•			
SDBFB2	•	•		•		
SDBFB4	•	•			•	
SDBRB1	•	•	•			
SDBRB2	•	•		•		
SDBRB4	•	•			•	
SDCFB1		•	•			
SDCFB2		•		•		
SDCFB4		•			•	
SDCRB1		•	•			
SDCRB2		•		•		
SDCRB4		•			•	
SDCRN1		•	•			•
SDCRN2		•		•		•
SDCRN4		•			•	•
SDCRQ1		•	•			•
SDCRQ2		•		•		•
SDCRQ4		•			•	•
SDCFQ1		•	•			•
SDCFQ2		•		•		•
SDCFQ4		•			•	•
SDNFB1			•			

Table 3: Multiplexed and Scan Flip-Flops

Macro Name	Set	Clear	1x Drive	2x Drive	4x Drive	Single Output
SDNFB2 SDNFB4				•	•	
SDNRB1 SDNRB2 SDNRB4			•	•	•	
SDNRN1 SDNRN2 SDNRN4			•	•	•	• • •
SDNRQ1 SDNRQ2 SDNRQ4			•	•	•	• • •
SDPFB1 SDPFB2 SDPFB4	• • •		•	•	•	
SDPRB1 SDPRB2 SDPRB4	• • •		•	•	•	
SECFQ1 SECFQ2 SECFQ4		• • •	•	•	•	• • •
SECRQ1 SECRQ2 SDPRB4		• •	•	•	•	• •
SENRB1 SENRB2 SENRB4			•	•	•	
SENRQ1 SENRQ2 SENRQ4			•	•	•	• • •
SEPFQ1 SEPFQ2	• •		•	•		• •

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Table 3: Multiplexed and Scan Flip-Flops

Macro Name	Set	Clear	1x Drive	2x Drive	4x Drive	Single Output
SEPFQ4	•				•	•
SEPRQ1	•		•			•
SEPRQ2	•			•		•
SEPRQ4	•				•	•

Table 4: Latches

Macro Name	Set	Clear	Enabled D Input	1x Drive	2x Drive	4x Drive	3-State Output	Single Output
LABHB1	•	•	•	•				
LABHB2	•	•	•		•			
LABHB4	•	•	•			•		
LACHQ1		•	•	•				•
LACHQ2		•	•		•			•
LACHQ4		•	•			•		•
LACLQ1		•	•	•				•
LACLQ2		•	•		•			•
LACLQ4		•	•			•		•
LANHB1			•	•				
LANHB2			•		•			
LANHB4			•			•		
LANHN1			•	•				•
LANHN2			•		•			•
LANHN4			•			•		•
LANHQ1			•	•				•
LANHQ2			•		•			•
LANHQ4			•			•		•
LANHT1			•	•			•	
LANHT2			•		•		•	
LANHT4			•			•	•	

Table 4: Latches

Macro Name	Set	Clear	Enabled D Input	1x Drive	2x Drive	4x Drive	3-State Output	Single Output
LANLB1			•	•				
LANLB2			•		•			
LANLB4			•			•		
LANLN1			•	•				•
LANLN2			•		•			•
LANLN4			•			•		•
LANLQ1			•	•				•
LANLQ2			•		•			•
LANLQ4			•			•		•

Table 5: Multiplexed and Scan Latches

Macro Name	Set	Clear	1x Drive	2x Drive	4x Drive	3-State Latch Output	Single latch Output
SLBHB1	•	•	•				
SLBHB2	•	•		•			
SLBHB4	•	•			•		
SLCHQ1		•	•				•
SLCHQ2		•		•			•
SLCHQ4		•			•		•
SLCLQ1		•	•				•
SLCLQ2		•		•			•
SLCLQ4		•			•		•
SLNHB1			•				
SLNHB2				•			
SLNHB4					•		
SLNHN1			•				•
SLNHN2				•			•
SLNHN4					•		•
SLNHQ1			•				•
SLNHQ2				•			•

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Table 5: Multiplexed and Scan Latches

Macro Name	Set	Clear	1x Drive	2x Drive	4x Drive	3-State Latch Output	Single latch Output
SLNHQ4					•		•
SLNHT1			•			•	
SLNHT2				•		•	
SLNHT4					•	•	
SLNLB1			•				
SLNLB2				•			
SLNLB4					•		
SLNLN1			•				•
SLNLN2				•			•
SLNLN4					•		•
SLNLQ1			•				•
SLNLQ2				•			•
SLNLQ4					•		•

TSL

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Characterization Information

Corner conditions

The tsl18fs120 library has been characterized using the following corner conditions:

	voltage	temp.	process
max	1.62	125	ss (slow-slow)
typ	1.8	25	tt (typical)
min	1.98	-40	ff (fast-fast)

max_slope = maximum_transition = 3ns

thresholds for output transitions measured from 30% to 70% and extrapolated to 0%-100%

thresholds for delays: 50% -> 50%

Note that this databook only reports typical values.

Table LookUp Information

Following is the description of the characteristics of all the table LookUp used to model the library in typical corner.

	Delay LookUp Table single stage 7x7						
input slope variations:	20 ps	100 ps	200 ps	400 ps	800 ps	1500ps	3000ps
output load variations:	0 ff	5.4 ff	15 ff	39 ff	75 ff	150 ff	300 ff

	Delay LookUp Table multi stage 7x7						
input slope variations:	20 ps	100ps	200ps	400ps	800ps	1500ps	3000ps
output load variations:	0 ff	5.4 ff	15 ff	39 ff	75 ff	150 ff	300 ff

	Setup/Hold LookUp Table				
data slope variations:	20 ps	500 ps	1000 ps	2000 ps	3000 ps
clock slope variations:	20 ps	500 ps	1000 ps	2000 ps	3000 ps

	Recovery/Removal LookUp Table				
clear/set slope variations:	20 ps	500 ps	1000 ps	2000 ps	3000 ps
clock slope variations:	20 ps	500 ps	1000 ps	2000 ps	3000 ps

output load variations are given for a 1X drive.

To have output load values for an other drive, multiply the previous values by the drive number. Input slope values are the same for all drives.

List of multi stage cells: All sequential cells + adders and subtractors

List of single stage cells: All other cells

Power Modeling

This chapter provides an overview of modelling static and dynamic power for tsl18fs120 library.

Terminology

The power a circuit dissipates falls into two broad categories:

- Static power
- Dynamic power

Static Power

Static power is the power dissipated by a gate when it is not switching, i.e., when it is inactive or static.

Static power is dissipated in a number of ways. The largest percentage of static power results from source-to-drain sub threshold leakage. This leakage is caused by reduced threshold voltage that prevent the gate from turning off completely. Static power also results when current leaks between the diffusion layers and substrate. For this reason, static power is often called leakage power.

Dynamic Power

Dynamic power is the power dissipated when a circuit is active. A circuit is active any time the voltage on a net changes due to some stimulus applied to the circuit. Because voltage on a net can change without necessarily resulting in a logic transition, dynamic power can result even when a net does not change its logic state.

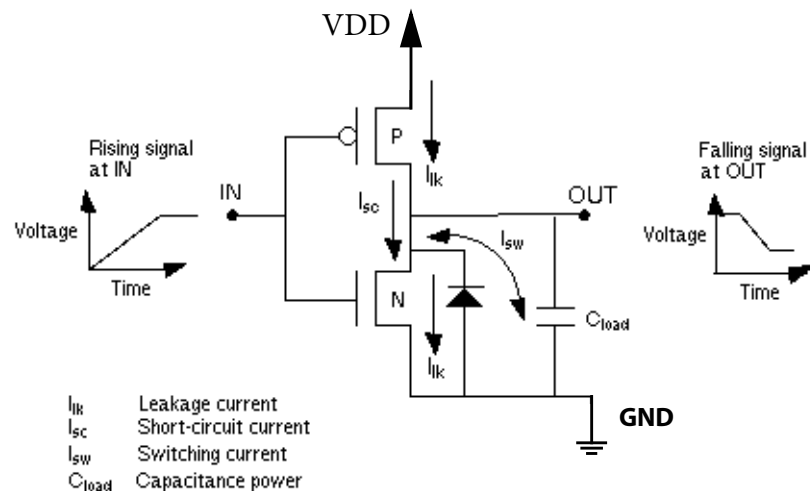
The dynamic power of a circuit is composed of

- Internal power
- Switching power

Internal Power

During switching, a circuit dissipates internal power by the charging or discharging of any existing capacitance internal to the cell. The definition of internal power includes power dissipated by a momentary short-circuit between the P and N transistors of a gate, called short-circuit power.

Figure 1: “Components of power dissipation” shows the cause of short-circuit power. In this figure, there is a slow rising signal at the gate input IN. As the signal makes a transition from low to high, the N type transistor turns on and the P type transistor turns off. However, during signal transition, both the P and N type transistors can be on simultaneously for a short time. During this time, current flows from VDD to GND, resulting in short-circuit power.

Figure 1: Components of power dissipation

Short-circuit power varies according to the circuit. For circuits with fast transition times, short-circuit power can be small. For circuits with slow transition times, short-circuit power can account for up to 30 percent of the total power dissipated. Short-circuit power is also affected by the dimensions of the transistors and the load capacitance at the output of the gate.

In most simple library cells, internal power is due primarily to short-circuit power. For this reason, the terms internal power and short-circuit power are often considered synonymous.

Switching Power

The switching power or capacitance power of a driving cell is the power dissipated by the charging and discharging of the load capacitance at the output of the cell. The total load capacitance at the output of a driving cell is the sum of the net and gate capacitance on the driver. Because such charging and discharging is the result of the logic transitions at the output of the cell, as logic transitions increase, switching power increases. The switching power of a cell is the function of both the total load capacitance at the cell output and the rate of logic transitions.

Figure 1: “Components of power dissipation” shows how the capacitance (C_{load}) is charged and discharged as the N or the P transistor turns on.

Switching Activity

Switching activity is a metric used to measure the number of transitions (0-to-1 and 1-to-0) for every net in a circuit when input stimuli are applied. Switching activity is the average activity of the circuit with a set of input stimuli.

A circuit with higher switching activity is likely to dissipate more power than a circuit with lower switching activity. Switching activity also has a correlation to random-pattern testability of the circuit. A circuit with higher switching activity implies that a randomly selected input pattern might have better coverage.

Note that switching activity is the only stimuli input needed by Synopsys Power compiler to evaluate dynamic power. One energy is associated to one net transition; so for example if pin A of cell U1 is toggling and creates the output Z of cell U1 to change, the resulting dissipated energy will be the energy associated to the input A pin toggling + the energy associated with the output Z pin toggling.

Modeling Cells for Power

These are the three components of power dissipation:

- Leakage power
- Internal (short-circuit) power
- Switching power

As this equation shows, leakage power is summed over all cells in the design to yield the design's total leakage power (total static power dissipation):

$$P_{leakage} = \sum_{\forall cells(i)} P_{cellleakage}$$

$P_{cellleakage(i)}$: Leakage power dissipation of each cell

$P_{leakage}$: Total leakage power dissipation of the design

As the following equation shows, the internal power of the cells and the switching power of the nets are used to compute the design's total dynamic power dissipation:

$$P_{dynamic} = P_{internal} + P_{switching}$$

where

$$P_{internal} = \sum_{\forall cell(i)} (E_i \times TR_i)$$

$$P_{switching} = \frac{VDD^2}{2} \sum_{\forall nets(i)} (C_{load(i)} \times TR_i)$$

$P_{dynamic}$ Total dynamic power dissipation of the design

E_i Internal energy dissipation of each cell

$C_{load(i)}$ Capacitive load of each net

VDD Supply voltage

TR_i Toggle rate or number of toggles per unit time for a net or a cell input or output

Modeling for Leakage Power

Normally, leakage power is only a very small component of the total power (typically, much less than 1 percent), but it is important to model for designs that are in an idle state most of the time.

Leakage power information is represented in Synopsys Library Compiler syntax with the cell-level `cell_leakage_power` attribute and associated library-level attributes that specify scaling factors, units, and a default value.

State-dependent leakage power can be modelled in the technology library, however that capability is not used as the precision gained in power dissipation is negligible. Therefore, for each cell, there is only one value of leakage power as shown in the following example:

Example

```
cell(AN02D1) {
...
cell_leakage_power: 31.7;
...
}
```

Modeling for Internal and Switching Power

These are two compatible definitions of internal or short-circuit power:

- Short-circuit power is the power dissipated by the instantaneous short-circuit connection between VDD and GND while the gate is in transition.
- Internal power is all the power dissipated within the boundary of the gate. This definition does not distinguish between the cell's short-circuit power and the component of switching power that is being dissipated internally to the cell as a result of the drain-to-substrate capacitance that is being charged and discharged. In this definition, the interconnect switching power is the power dissipated because of lumped wire capacitance and input pin capacitance, but not the output pin capacitance.

The effect of the output capacitance is included in the `internal_power` attribute, which gives the output pins zero capacitance; this is similar to the timing model approach.

Together, internal power and switching power contribute to the total dynamic

power dissipation. Like switching power, internal power is dissipated whenever an output pin makes a transition.

This description is not entirely accurate because some power is dissipated as a result of internal transitions that do not cause output transitions.

Figure 2: “Complex Gate Example” shows two examples of an input transition that does not cause a corresponding output transition.

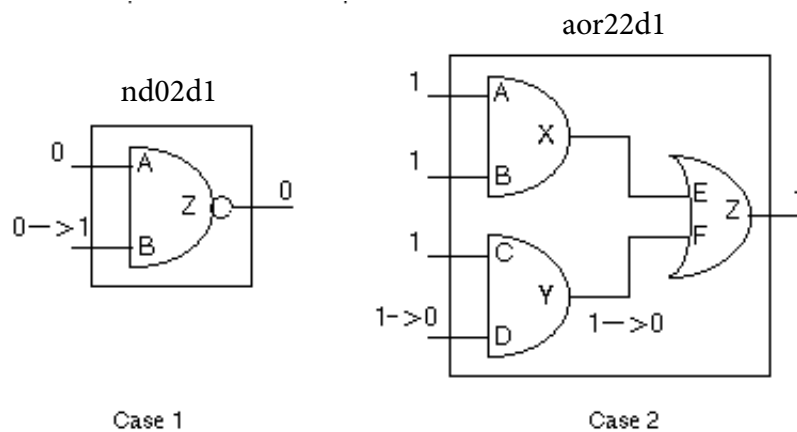


Figure 2:Complex Gate Example

In Case 1, input B of the nd02d1 gate undergoes a 0-to-1 transition but the output remains stable at 0. This might consume a small amount of power as one of the N-transistors opens, but the current flow will be very small.

In Case 2, input D of the multi level gate aor22d1 undergoes a 1-to-0 transition causing a 1-to-0 transition at internal pin Y. However, output Z remains stable at 1. The significance of the power dissipation in this case depends on the load of the internal wire connected to Y. In Case 1, power dissipation is negligible, but in Case 2, power dissipation may result in some inaccuracy.

In Synopsys Library Compiler syntax, the internal power information is represented with the internal_power group in a pin group within a cell.

The library group supports a one-, two-, or three-dimensional internal_power lookup table indexed by the total output load capacitance (best model), the input transition time, or both. The internal power lookup table uses the same syntax as the non-linear lookup table for delay.

You can set `internal_power` attributes for input pins, which are indexed by input transition time. In this way, you can model the gate `aor22d1` in Figure 2: or, more importantly, the power consumed by flip-flop clock or reset pins.

Note: The input pin power is added to the output pin power. Therefore, the input pin power is extracted from the output pin power when creating the internal power group table for the output pin.

Figure 3: “Internal Power for single-stage cells” shows how to calculate the input pin power information for the one-dimensional lookup table describing internal cell U1.

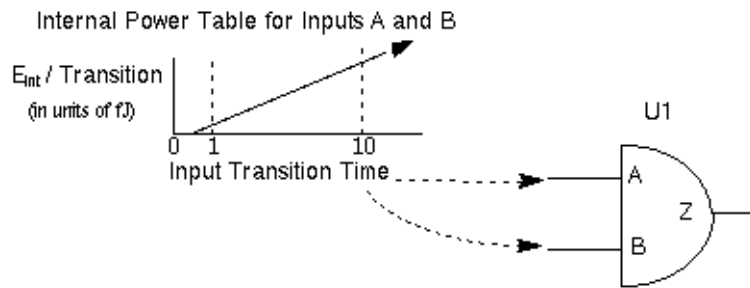


Figure 3: Internal Power for single-stage cells

Calculating the internal power for cell U1,

P_{int}	Total internal power for the cell
E	Internal energy for the pin
AF	Activity factor

Figure 4: “Internal Power for multi-stage cells” shows the same thing for internal

cell U2.

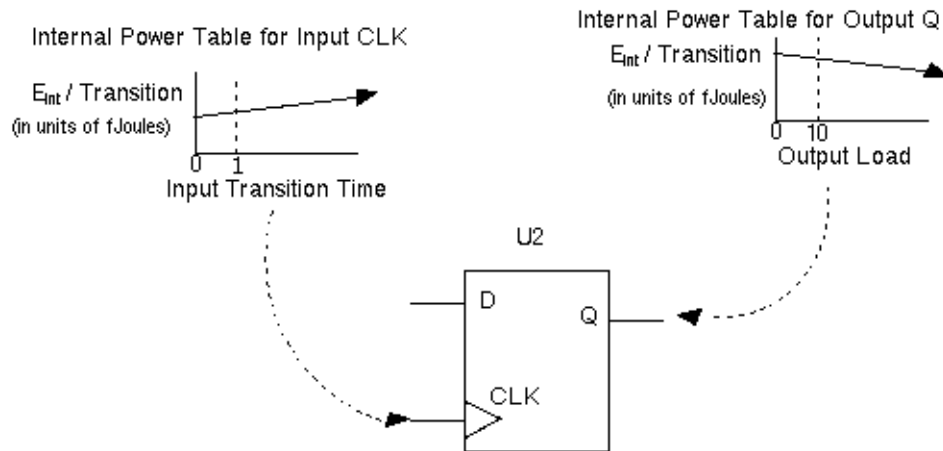


Figure 4: Internal Power for multi-stage cells

Calculating the internal power for cell U2,

Pint Total internal power for the cell

E Internal energy for the pin

AF Activity factor

Figure 5: “Internal Power Table for Cell Output” is an example of the two-dimensional lookup table for modelling output pin power in a cell.

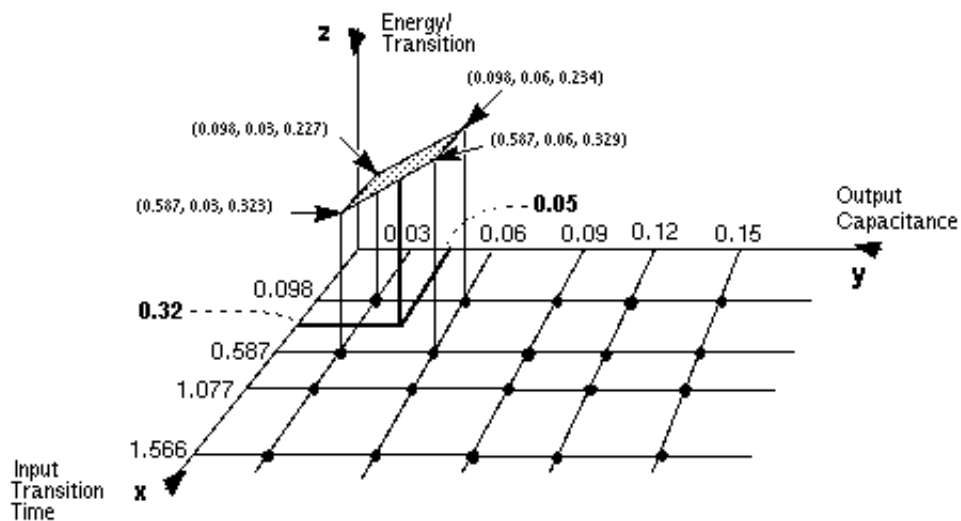


Figure 5: Internal Power Table for Cell Output

TSL18FS120 supply model state-dependent and path-dependent internal power as well as differentiating between rise and fall power for output pin internal power.

Calculating Switching Power

Switching (or interconnect) power is the power dissipated by the capacitive load on a net whenever the net makes a logical transition. Power is dissipated when the capacitive load is charged or discharged. With internal power, switching power is used to compute the design's total dynamic power dissipation.

Switching power information is a function of a net's toggle rate, capacitive loading, associated clock frequency, and the supply voltage level of the design. Except for toggle rate, these parameters are already supported by the Synopsys library model.

Because all toggle rates are internally adjusted to a period defined by the library-level `time_unit` attribute, the resulting value of switching power is defined in terms of joules per sec (or watts) multiplied by the appropriate power of 10 as determined by the units for time, capacitance, and voltage. For example, in a library with time units of 1 ns, capacitive units of 1 pF, and voltage units of 1 V, this is the calculation for the derived units for the library's switching power:

$$PowerUnits = \frac{(1V^2) \times (1pf)}{1nS} = 1pW$$

For a single net with a total load of 100ff, a toggle rate of 2 transitions every 100 ns, and a supply voltage of 1.8 V, this is the calculation of the net's power dissipation:

$$NetPower = \frac{VDD^2}{2} \sum_{\forall nets(i)} (C_{load_i} \times TR_i) = 0.14pW$$

TR Toggle rate (number of toggles per unit time)

C_{load} Capacitive load of each net

Characterization Methodology

This section presents the methodology used for internal and leakage power characterization. Figure 6 shows a circuit configuration which conceptualize how energy dissipation measurements for a single input, single output cell is done.

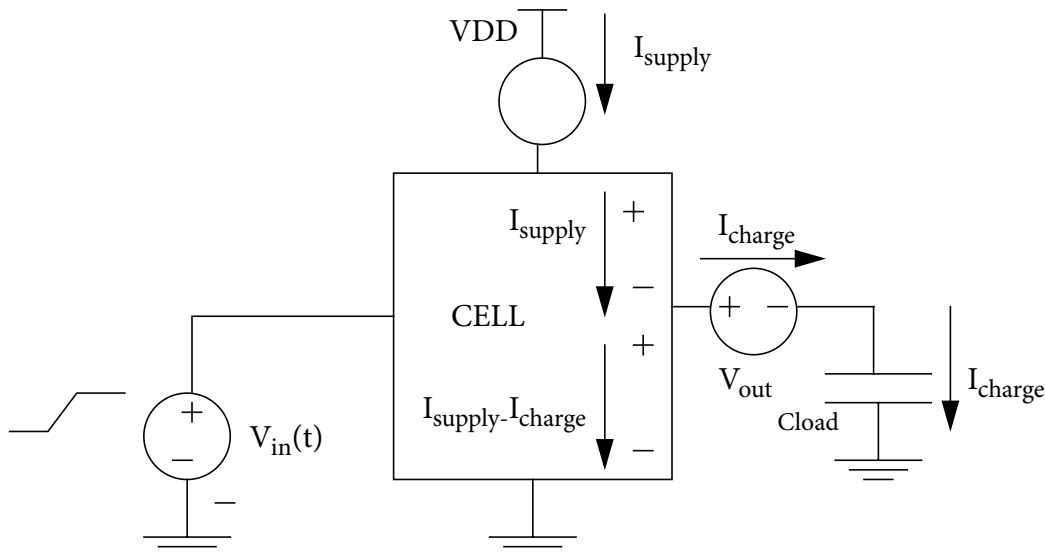


Figure 6:Characterization circuit example

Internal Power Characterization Methodology

Energy dissipated is measured by varying either input voltage transition or output load, holding the other constant. The same table indexes as those used in delay lookup tables are used 5 transitions X 7 output loads. [See “Table LookUp Information” on page 35.](#)

Input pin switching internal power

For each transition and for each input, energy dissipated during rising and falling input transitions that do not cause any output switching is measured.

The following formula is used to compute the energy values for a single cycle (T) of both rising and falling input transitions

$$E_{internal} = E_{total} = \int_0^T (VDD \times I_{supply}) dt$$

Note that I_{charge} being zero (no output switching), the total energy dissipated is the input pin internal energy.

Internal energy dissipation is calculated for each input states that do not cause any output switching. Although state dependent power modeling is supported in Synopsys, for performance (run time, memory) considerations, we use a single 1-dimensional internal power table at the input of each cell. In this case, the worst value of all the energy of the various states that do not cause any output switching is identified for each input transition and put in the 1-dimensional table.

Note that the value specified in the power model is for a single rising or falling edge and is therefore half the value measured above.

Output pin switching internal power

For each combination (input transition, output load), energy dissipated during rising and falling output transitions are measured.

The following formula is used to compute the energy values for a single cycle (T) of both rising and falling output transitions

$$E_{\text{total}} = \int_0^T (VDD \times I_{\text{supply}}) dt$$

E_{total} consists of energy dissipated in the load capacitance (E_{charge}) and within the cell (E_{internal}). E_{charge} can be computed using the following formula:

$$E_{\text{charge}} = \int_0^T (V_{\text{out}} \times I_{\text{charge}}) dt = CV^2$$

Thus, the equation for computing a cell's internal energy dissipation can be derived as follows:

$$E_{\text{internal}} = E_{\text{total}} - E_{\text{charge}} = \int_0^T (VDD \times I_{\text{supply}} - V_{\text{out}} \times I_{\text{charge}}) dt$$

$$E_{\text{internal}} = E_{\text{total}} - CV^2$$

In other words, internal energy dissipation is obtained by subtracting energy dissipated at the load capacitance from the total energy dissipation measured.

Internal energy dissipation is calculated for each input to output path of a cell; however, although path dependent power modeling is supported in Synopsys, for performance (run time, memory) considerations, we use a single 2-dimensional internal power table at the output of each cell. In this case, the worst value of all the energy of the various path from input to the specified output are identified for each input transition and output load combination. The same considerations apply for state-dependent power modeling, so that the internal power specified for an output pin is the worst internal power of all path dependent and state dependent internal powers.

Note that the value specified in the power model is for a single rising or falling edge and is therefore half the value measured above. Additionally, the measured value contains the energy dissipated by the input pin switching, in order to not double count this energy in Synopsys power calculation, the relevant input switching power is subtracted from the measured value.

It is therefore possible to see negative values for output power being modeled. This occurs because on some cells the power required to move the inputs under certain static conditions can be more than the power required to move the inputs + outputs under different conditions. We are unable to model a figure of power for each timing arc and therefore must subtract the worst input figure from the worst output figure. However, the total power that will be reported when the cell output switches will be correct (input + output).

Characterizing Leakage Power

Leakage power characterization requires the measurement of I supply at quiescent state of the cell. This requires a DC analysis of the circuit with steady state voltages at the inputs to the cell.

We characterize at multiple combinations of input states and used the median power value as the cell leakage value.

Characterizing For PVT Variations

The modeling of internal and leakage power variations due to process, voltage and temperature at different operating conditions is not supported. Only worst case power numbers are provided (usually best case delay conditions).

Derating Information

Derating Factors and Propagation Delays

The propagation delay values apply only to the specified operating conditions of VDD = 1.8 volts, junction temperature = 25 degrees C, and typical-case process. You can estimate the delay under different conditions by using the following timing constraint equations:

$$timing_rise (new) = timing_rise (datasheet) * Ev_rise * Et_rise * Ep_rise$$

$$timing_fall (new) = timing_fall (datasheet) * Ev_fall * Et_fall * Ep_fall$$

where:

$$Ev_rise = 1 + (VDD_{new} - VDD) * k_voltage_timing_rise$$

$$Et_rise = 1 + (T_{new} - T) * k_temperature_timing_rise$$

$$Ep_rise = 1 + (P_{new} - P) * k_process_timing_rise$$

$$Ev_fall = 1 + (VDD_{new} - VDD) * k_voltage_timing_fall$$

$$Et_fall = 1 + (T_{new} - T) * k_temperature_timing_fall$$

$$Ep_fall = 1 + (P_{new} - P) * k_process_timing_fall$$

timing: can be Delay propagation, Setup, Hold

k_voltage

Transition	Delay propagation	Setup	Hold
Rise	-0.6930	-0.6930	-0.6930
Fall	-0.6483	-0.6483	-0.6483

k_temperature

Transition	Delay propagation	Setup	Hold
Rise	0.0011	0.0011	0.0011
Fall	0.0012	0.0012	0.0012

k_process

Transition	Delay propagation	Setup	Hold
Rise	0.6597	0.6597	0.6597
Fall	0.5759	0.5759	0.5759

The process bias index are shown in the following table:

Process	Index (P)
Slow	1.2
Typical	1.0
Fast	0.8

Derating Example

The following example shows a delay propagation assessment in the below conditions:

Junction Temperature = 80 degrees C

VDD = 1.3 Volts

Process = slow

tDelay_Rise (datasheet, BUFFD1) = 0.121 ns

$$t\text{Delay_Rise (new)} = t\text{Delay_Rise (datasheet)} * E_v * E_t * E_p$$

where:

$$E_v = 1 + (1.3 - 1.8) * (-0.6930) = 1.347$$

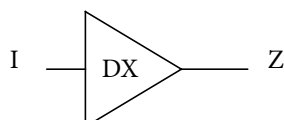
$$E_t = 1 + (80 - 25) * 0.0011 = 1.061$$

$$E_p = 1 + (1.2 - 1.0) * 0.6597 = 1.132$$

$$t\text{Delay_Rise (new)} = 0.121 * 1.347 * 1.061 * 1.132 = 0.196 \text{ ns}$$

BUFBD1, BUFBD2, BUFBD3, BUFBD4, BUFBD7, BUFBDA, BUFBDF and BUFBDK

Function Table

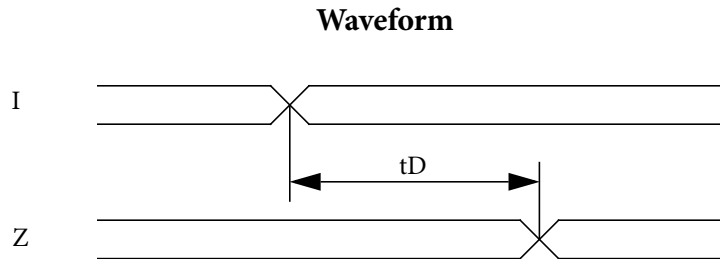


INPUT	OUTPUT
I	Z
L	L
H	H

Macro Name:	BUFBD1	BUFBD2	BUFBD3	BUFBD4	BUFBD7	BUFBDA	BUFBDF	BUFBDK
Drive Capability:	1x	2x	3x	4x	7x	10x	15x	20x
Gate Equivalents:	1.25	1.5	1.75	1.75	3.5	4.25	4.75	6.
Leakage Power (pW):	31.4	51.5	67.4	68.8	128.2	152.8	190.0	232.3

Name	Capacitance (pF)								Description
	BUFBD1	BUFBD2	BUFBD3	BUFBD4	BUFBD7	BUFBDA	BUFBDF	BUFBDK	
I	0.003	0.005	0.006	0.005	0.01	0.01	0.011	0.011	Input
Maximum capacitance									
Z	0.3	0.6	0.9	1.2	2.1	3	4.5	6	Output

Name	Pin Power (pW/Hz)							
	BUFBD1	BUFBD2	BUFBD3	BUFBD4	BUFBD7	BUFBDA	BUFBDF	BUFBDK
Z	0.025	0.039	0.054	0.065	0.112	0.173	0.275	0.392



Timing Numbers for BUFBD1, BUFBD2, BUFBD3, BUFBD4, BUFBD7, BUF-BDA, BUFBD F and BUFBDK:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
BUFBD1	0.113	0.127	0.146	0.160	0.212	0.223	0.344	0.346	0.605	0.593
BUFBD2	0.105	0.107	0.123	0.127	0.157	0.165	0.223	0.237	0.354	0.381
BUFBD3	0.093	0.096	0.106	0.108	0.132	0.130	0.181	0.172	0.280	0.250
BUFBD4	0.115	0.116	0.127	0.128	0.148	0.150	0.186	0.188	0.258	0.261
BUFBD7	0.094	0.094	0.103	0.102	0.116	0.115	0.140	0.137	0.186	0.178
BUFBDA	0.113	0.112	0.119	0.118	0.130	0.129	0.147	0.145	0.178	0.174
BUFBDF	0.148	0.136	0.152	0.140	0.161	0.149	0.175	0.164	0.200	0.189
BUFBDK	0.175	0.159	0.178	0.162	0.185	0.170	0.198	0.183	0.219	0.205

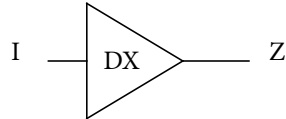
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input Transition	1		5		10		15		20	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
BUFBD1	0.146	0.160	0.192	0.227	0.220	0.284	0.241	0.332	0.249	0.370
BUFBD2	0.123	0.127	0.179	0.191	0.219	0.240	0.249	0.280	0.268	0.309
BUFBD3	0.106	0.108	0.156	0.173	0.187	0.224	0.210	0.266	0.222	0.298
BUFBD4	0.127	0.128	0.190	0.202	0.236	0.261	0.271	0.308	0.292	0.342
BUFBD7	0.103	0.102	0.160	0.167	0.198	0.216	0.228	0.256	0.246	0.285
BUFBDA	0.119	0.118	0.184	0.192	0.232	0.252	0.269	0.300	0.292	0.335
BUFBDF	0.152	0.140	0.227	0.219	0.292	0.285	0.343	0.337	0.378	0.372
BUFBDK	0.178	0.162	0.255	0.245	0.330	0.321	0.388	0.380	0.429	0.420

Non-Inverting Buffers

BUFFD1, BUFFD2, BUFFD3, BUFFD4, BUFFD7 and BUFFDA

BUFFD1, BUFFD2, BUFFD3, BUFFD4, BUFFD7 and BUFFDA are non-inverting buffers with 1x, 2x, 3x, 4x, 7x and 10x drive capabilities.



Function Table

INPUT	OUTPUT
I	Z
L	L
H	H

Cell Description

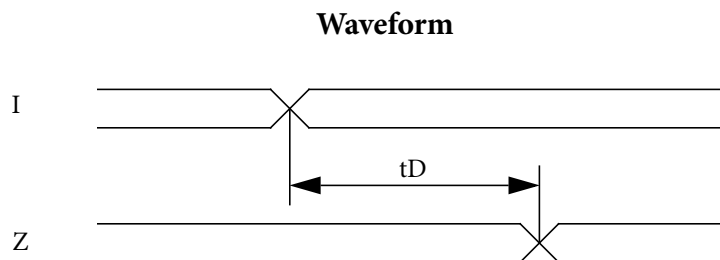
Macro Name:	BUFFD1	BUFFD2	BUFFD3	BUFFD4	BUFFD7	BUFFDA
Drive Capability:	1x	2x	3x	4x	7x	10x
Gate Equivalents:	1.	1.5	1.25	2.	2.5	3.25
Leakage Power (pW):	21.2	48.6	42.0	68.9	102.3	155.2

Pin Description

Name	Capacitance (pF)						Description
	BUFFD1	BUFFD2	BUFFD3	BUFFD4	BUFFD7	BUFFDA	
I	0.002	0.004	0.003	0.005	0.007	0.01	Input
Maximum capacitance							
Z	0.3	0.6	0.9	1.2	2.1	3	Output

Pin Powers for Standard Load = 0.032 pF Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)					
	BUFFD1	BUFFD2	BUFFD3	BUFFD4	BUFFD7	BUFFDA
Z	0.02	0.039	0.044	0.073	0.1	0.148



Timing Numbers for BUFFD1, BUFFD2, BUFFD3, BUFFD4, BUFFD7 and BUFFDA:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
BUFFD1	0.121	0.143	0.170	0.170	0.268	0.220	0.461	0.314	0.848	0.500
BUFFD2	0.103	0.110	0.122	0.130	0.155	0.168	0.221	0.240	0.352	0.384
BUFFD3	0.097	0.140	0.115	0.155	0.149	0.180	0.215	0.222	0.348	0.296
BUFFD4	0.101	0.126	0.111	0.138	0.129	0.160	0.162	0.199	0.225	0.272
BUFFD7	0.084	0.117	0.093	0.126	0.107	0.139	0.135	0.161	0.188	0.198
BUFFDA	0.082	0.112	0.088	0.117	0.099	0.128	0.119	0.144	0.158	0.170

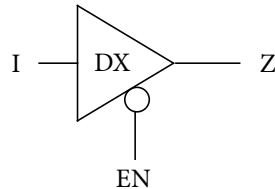
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
BUFFD1	0.170	0.170	0.200	0.258	0.207	0.342	0.204	0.416	0.189	0.480
BUFFD2	0.122	0.130	0.177	0.194	0.215	0.243	0.244	0.283	0.262	0.313
BUFFD3	0.115	0.155	0.157	0.243	0.174	0.330	0.180	0.404	0.172	0.466
BUFFD4	0.111	0.138	0.167	0.216	0.201	0.284	0.225	0.340	0.236	0.383
BUFFD7	0.093	0.126	0.137	0.209	0.156	0.285	0.164	0.350	0.160	0.404
BUFFDA	0.088	0.117	0.131	0.198	0.149	0.272	0.158	0.335	0.153	0.388

Non-inverting 3-State Buffers with Output Enable Bar

BUFTD1, BUFTD2, BUFTD4, BUFTD7 and BUFTDA

BUFTD1, BUFTD2, BUFTD4, BUFTD7 and BUFTDA are non-inverting 3-state buffers with output enable bar and with 1x, 2x, 4x, 7x and 10x drive capabilities.



Function Table

INPUT		OUTPUT
I	EN	Z
X	H	HiZ
L	L	L
H	L	H

Cell Description

Macro Name:	BUFTD1	BUFTD2	BUFTD4	BUFTD7	BUFTDA
Drive Capability:	1x	2x	4x	7x	10x
Gate Equivalents:	2.25	2.5	2.75	3.5	4.
Leakage Power (pW):	68.7	79.7	96.1	122.3	158.1

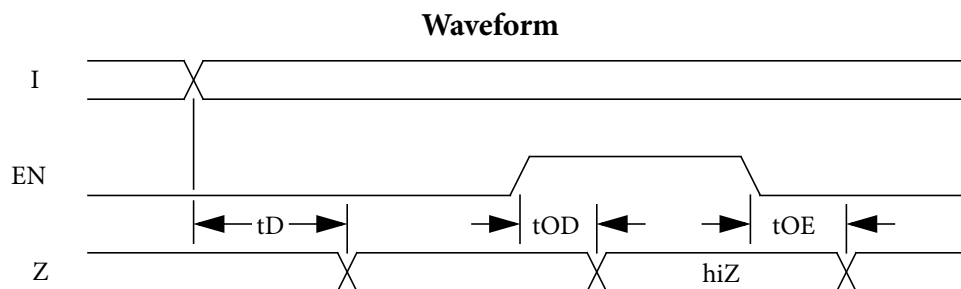
Pin Description

Name	Capacitance (pF)					Description
	BUFTD1	BUFTD2	BUFTD4	BUFTD7	BUFTDA	
I	0.005	0.005	0.005	0.005	0.005	Input
EN	0.005	0.006	0.006	0.006	0.006	Enable Input
Z	0.004	0.005	0.01	0.016	0.025	3-State Output

Pin Powers for Standard Load = 0.032 pF Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)				
	BUFTD1	BUFTD2	BUFTD4	BUFTD7	BUFTDA
Z	0.04	0.053	0.077	0.112	0.16

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Timing numbers for BUFTD1, BUFTD2, BUFTD4, BUFTD7 and BUFTDA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
BUFTD1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.144	0.153	0.190	0.184	0.280	0.240	0.460	0.344	0.820	0.546
tOE	0.186	0.126	0.231	0.155	0.321	0.210	0.500	0.312	0.860	0.514
tOD	0.034	0.117	0.034	0.117	0.033	0.117	0.034	0.117	0.034	0.117

BUFTD2	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.128	0.145	0.153	0.168	0.198	0.202	0.287	0.259	0.464	0.359
tOE	0.167	0.117	0.192	0.137	0.236	0.168	0.325	0.222	0.501	0.322
tOD	0.038	0.134	0.038	0.134	0.037	0.134	0.038	0.134	0.038	0.134

BUFTD4	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.134	0.158	0.150	0.174	0.178	0.201	0.226	0.240	0.319	0.305
tOE	0.167	0.117	0.182	0.131	0.209	0.154	0.257	0.190	0.350	0.252
tOD	0.045	0.156	0.045	0.156	0.045	0.156	0.045	0.156	0.045	0.156

BUFTD7	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.145	0.175	0.158	0.189	0.177	0.209	0.211	0.242	0.269	0.293
tOE	0.172	0.128	0.184	0.141	0.203	0.159	0.236	0.188	0.294	0.235
tOD	0.053	0.191	0.053	0.191	0.053	0.191	0.053	0.191	0.053	0.191

BUFTDA	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.171	0.206	0.182	0.217	0.200	0.235	0.227	0.263	0.272	0.304
tOE	0.184	0.147	0.194	0.157	0.211	0.174	0.239	0.199	0.283	0.237
tOD	0.065	0.228	0.065	0.228	0.065	0.228	0.065	0.228	0.065	0.228

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
BUFTD1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.190	0.184	0.226	0.260	0.244	0.328	0.254	0.387	0.253	0.437
tOE	0.231	0.155	0.315	0.237	0.392	0.316	0.459	0.389	0.516	0.456
tOD	0.034	0.117	0.034	0.149	0.011	0.158	-0.018	0.155	-0.054	0.142

BUFTD2	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.153	0.168	0.193	0.247	0.215	0.318	0.227	0.379	0.227	0.431
tOE	0.192	0.137	0.277	0.221	0.354	0.303	0.422	0.379	0.479	0.447
tOD	0.038	0.134	0.043	0.165	0.024	0.172	-0.003	0.171	-0.038	0.159

BUFTD4	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.150	0.174	0.198	0.256	0.226	0.332	0.243	0.397	0.246	0.450
tOE	0.182	0.131	0.268	0.214	0.346	0.295	0.415	0.367	0.474	0.432
tOD	0.045	0.156	0.061	0.188	0.050	0.195	0.031	0.195	0.002	0.183

BUFTD7	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.158	0.189	0.209	0.273	0.239	0.356	0.259	0.425	0.264	0.482
tOE	0.184	0.141	0.271	0.227	0.350	0.312	0.419	0.386	0.478	0.451
tOD	0.053	0.191	0.080	0.223	0.078	0.230	0.066	0.229	0.042	0.217

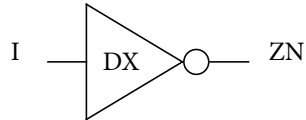
BUFTDA	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.182	0.217	0.239	0.303	0.279	0.395	0.307	0.471	0.318	0.532
tOE	0.194	0.157	0.281	0.245	0.362	0.332	0.432	0.407	0.491	0.472
tOD	0.065	0.228	0.105	0.260	0.113	0.268	0.109	0.267	0.091	0.255

Inverting Buffers

INV0D0, INV0D1, INV0D2, INV0D4, INV0D7 and INV0DA

INV0D0, INV0D1, INV0D2, INV0D4, INV0D7 and INV0DA are inverting buffers with 0.5x, 1x, 2x, 4x, 7x, and 10x drive capabilities.

Function Table



INPUT	OUTPUT
I	ZN
L	H
H	L

Cell Description

Macro Name:	INV0D0	INV0D1	INV0D2	INV0D4	INV0D7	INV0DA
Drive Capability:	0.5x	1x	2x	4x	7x	10x
Gate Equivalents:	0.5	0.75	1.	1.5	2.25	2.5
Leakage Power (pW):	8.8	21.5	47.2	94.3	160.0	231.2

Pin Description

Name	Capacitance (pF)						Description
	INV0D0	INV0D1	INV0D2	INV0D4	INV0D7	INV0DA	
I	0.001	0.004	0.008	0.017	0.027	0.038	Input
Maximum capacitance							
ZN	0.15	0.3	0.6	1.2	2.1	3	Output

Pin Powers for Standard Load = 0.032 pF Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)					
	INV0D0	INV0D1	INV0D2	INV0D4	INV0D7	INV0DA
ZN	0.003	0.007	0.012	0.023	0.037	0.054

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Waveform

Timing Numbers for INV0D0, INV0D1, INV0D2, INV0D4, INV0D7 and INV0DA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
INV0D0	0.234	0.125	0.422	0.220	0.797	0.408	1.546	0.785	3.045	1.540
INV0D1	0.086	0.055	0.134	0.081	0.231	0.132	0.423	0.236	0.807	0.441
INV0D2	0.059	0.036	0.083	0.051	0.130	0.075	0.225	0.121	0.412	0.215
INV0D4	0.046	0.027	0.059	0.036	0.083	0.051	0.130	0.075	0.225	0.121
INV0D7	0.037	0.024	0.046	0.031	0.060	0.041	0.087	0.059	0.140	0.090
INV0DA	0.036	0.020	0.042	0.025	0.054	0.034	0.073	0.047	0.113	0.069

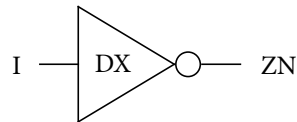
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
INV0D0	0.422	0.220	0.514	0.290	0.634	0.360	0.753	0.412	0.858	0.434
INV0D1	0.134	0.081	0.219	0.133	0.305	0.160	0.377	0.175	0.434	0.175
INV0D2	0.083	0.051	0.158	0.080	0.225	0.087	0.283	0.085	0.333	0.073
INV0D4	0.059	0.036	0.120	0.049	0.176	0.045	0.226	0.033	0.270	0.015
INV0D7	0.046	0.031	0.095	0.039	0.141	0.033	0.183	0.022	0.218	0.005
INV0DA	0.042	0.025	0.089	0.027	0.135	0.015	0.177	-0.001	0.214	-0.022

Inverting Buffers with Balanced Rise/Fall Time

INVBD2, INVBD4, INVBD7, INVBDA, INVBDF and INVBDK

INVBD2, INVBD4, INVBD7, INVBDA, INVBDF and INVBDK are inverting buffers with balanced rise/fall time and with 2x, 4x, 7x, 10x, 15x and 20x drive capabilities.



Function Table

INPUT	OUTPUT
I	ZN
L	H
H	L

Cell Description

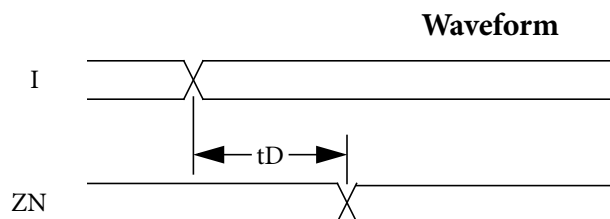
Macro Name:	INVBD2	INVBD4	INVBD7	INVBDA	INVBDF	INVBDK
Drive Capability:	2x	4x	7x	10x	15x	20x
Gate Equivalents:	1.25	1.75	2.5	3.	4.	5.25
Leakage Power (pW):	85.0	148.6	239.3	312.6	475.5	649.9

Pin Description

Name	Capacitance (pF)						Description
	INVBD2	INVBD4	INVBD7	INVBDA	INVBDF	INVBDK	
I	0.011	0.02	0.033	0.043	0.065	0.087	Input
Maximum capacitance							
ZN	0.6	1.2	2.1	3	4.5	6	Output

Pin Powers for Standard Load = 0.032 pF Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)					
	INVBD2	INVBD4	INVBD7	INVBDA	INVBDF	INVBDK
ZN	0.016	0.03	0.048	0.062	0.092	0.123



Timing Numbers for INVBD2, INVBD4, INVBD7, INVBDA, INVBDF and INVBDK:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
INVBD2	0.048	0.049	0.064	0.066	0.095	0.099	0.155	0.165	0.276	0.294
INVBD4	0.038	0.037	0.047	0.047	0.063	0.064	0.094	0.095	0.154	0.158
INVBD7	0.031	0.033	0.038	0.040	0.049	0.052	0.067	0.072	0.103	0.112
INVBDA	0.029	0.031	0.035	0.036	0.044	0.046	0.058	0.062	0.085	0.092
INVBDF	0.028	0.029	0.031	0.033	0.038	0.040	0.049	0.052	0.067	0.073
INVBDK	0.026	0.028	0.028	0.031	0.034	0.036	0.043	0.046	0.057	0.062

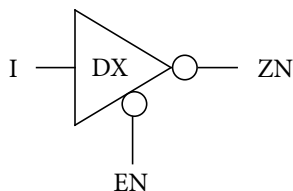
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
INVBD2	0.064	0.066	0.120	0.115	0.161	0.147	0.194	0.172	0.217	0.186
INVBD4	0.047	0.047	0.090	0.078	0.122	0.096	0.149	0.109	0.168	0.114
INVBD7	0.038	0.040	0.071	0.068	0.093	0.086	0.112	0.099	0.124	0.106
INVBDA	0.035	0.036	0.063	0.060	0.083	0.075	0.100	0.086	0.111	0.091
INVBDF	0.031	0.033	0.056	0.053	0.073	0.067	0.087	0.077	0.097	0.081
INVBKD	0.028	0.031	0.051	0.050	0.066	0.063	0.079	0.072	0.086	0.077

Inverting 3-State Buffers with Output Enable

INVTD1, INVTD2, INVTD4, INVTD7 and INVTD8

INVTD1, INVTD2, INVTD4, INVTD7 and INVTD8 are inverting 3-state buffers with output enable and with 1x, 2x, 4x, 7x and 10x drive capabilities.



Function Table

INPUTS		OUTPUT
EN	I	ZN
H	X	HiZ
L	L	H
L	H	L

Cell Description

Macro Name:	INVTD1	INVTD2	INVTD4	INVTD7	INVTD8
Drive Capability:	1x	2x	4x	7x	10x
Gate Equivalents:	1.25	1.75	3.5	4.	4.5
Leakage Power (pW):	20.8	27.1	79.8	108.7	128.3

Pin Description

Name	Capacitance (pF)					Description
	INVTD1	INVTD2	INVTD4	INVTD7	INVTD8	
I	0.004	0.008	0.002	0.002	0.002	Input
EN	0.006	0.008	0.006	0.006	0.006	Enable Input
ZN	0.007	0.008	0.008	0.016	0.022	3-State Output

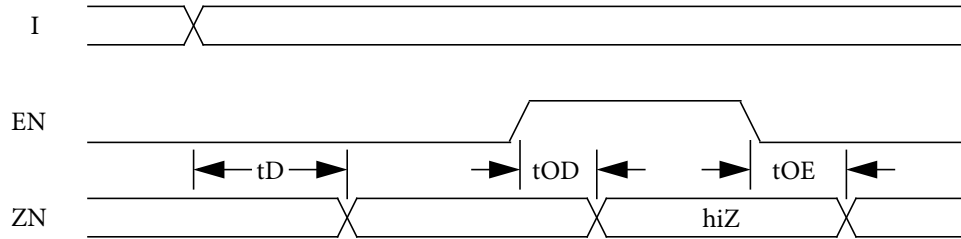
Pin Powers for Standard Load = 0.032 pF Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)				
	INVTD1	INVTD2	INVTD4	INVTD7	INVTD8
ZN	0.01	0.019	0.072	0.116	0.158

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Waveform**Timing Numbers for INVTD1, INVTD2, INVTD4, INVTD7 and INVTD7A:**

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
INVTD1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.120	0.063	0.192	0.094	0.334	0.155	0.618	0.276	1.183	0.519
tOE	0.125	0.110	0.195	0.141	0.336	0.201	0.619	0.323	1.184	0.566
tOD	0.049	0.049	0.049	0.049	0.048	0.049	0.049	0.049	0.049	0.049

INVTD2	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.098	0.046	0.146	0.063	0.239	0.094	0.425	0.157	0.794	0.280
tOE	0.093	0.102	0.139	0.121	0.230	0.154	0.414	0.217	0.781	0.340
tOD	0.059	0.040	0.059	0.040	0.059	0.040	0.059	0.040	0.059	0.041

INVTD4	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.237	0.200	0.258	0.217	0.294	0.245	0.362	0.285	0.495	0.351
tOE	0.171	0.121	0.191	0.136	0.227	0.160	0.294	0.196	0.426	0.259
tOD	0.043	0.144	0.043	0.144	0.043	0.144	0.043	0.144	0.043	0.144

INVTD7	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.265	0.238	0.280	0.252	0.303	0.272	0.342	0.305	0.413	0.356
tOE	0.176	0.139	0.191	0.152	0.213	0.171	0.252	0.200	0.322	0.246
tOD	0.053	0.178	0.053	0.178	0.053	0.178	0.053	0.178	0.053	0.179

INVTD7A	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.282	0.292	0.293	0.304	0.314	0.324	0.346	0.354	0.401	0.401
tOE	0.182	0.157	0.194	0.168	0.214	0.187	0.245	0.215	0.300	0.257
tOD	0.061	0.206	0.061	0.206	0.061	0.206	0.061	0.206	0.061	0.206

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**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
INVTD1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.192	0.094	0.270	0.144	0.360	0.169	0.440	0.182	0.506	0.179
tOE	0.195	0.141	0.238	0.206	0.270	0.266	0.287	0.319	0.280	0.366
tOD	0.049	0.049	0.064	0.083	0.062	0.137	0.053	0.191	0.036	0.245

INVTD2	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.146	0.063	0.211	0.097	0.283	0.104	0.348	0.101	0.407	0.086
tOE	0.139	0.121	0.189	0.194	0.215	0.259	0.223	0.317	0.205	0.366
tOD	0.059	0.040	0.086	0.072	0.092	0.122	0.090	0.174	0.077	0.225

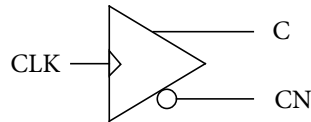
INVTD4	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.258	0.217	0.344	0.255	0.426	0.269	0.499	0.272	0.561	0.264
tOE	0.191	0.136	0.279	0.223	0.360	0.309	0.431	0.387	0.492	0.456
tOD	0.043	0.144	0.054	0.172	0.039	0.177	0.015	0.172	-0.019	0.157

INVTD7	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.280	0.252	0.367	0.290	0.452	0.305	0.525	0.310	0.588	0.302
tOE	0.191	0.152	0.279	0.243	0.362	0.333	0.434	0.413	0.496	0.482
tOD	0.053	0.178	0.080	0.207	0.076	0.213	0.060	0.210	0.032	0.195

INVTDA	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tD	0.293	0.304	0.380	0.343	0.466	0.359	0.541	0.365	0.604	0.358
tOE	0.194	0.168	0.282	0.259	0.365	0.350	0.438	0.430	0.500	0.499
tOD	0.061	0.206	0.096	0.234	0.098	0.239	0.087	0.235	0.062	0.220

Non-overlapping dual phase Clock Generator CLK2D2

CLK2D2 non-overlapping dual phase clock generator with 2x drive capability. This buffer provides both a true and complementary clock output.



Function Table

INPUT	OUTPUTS	
CLK	C	CN
L	L	H
H	H	L

Cell Description

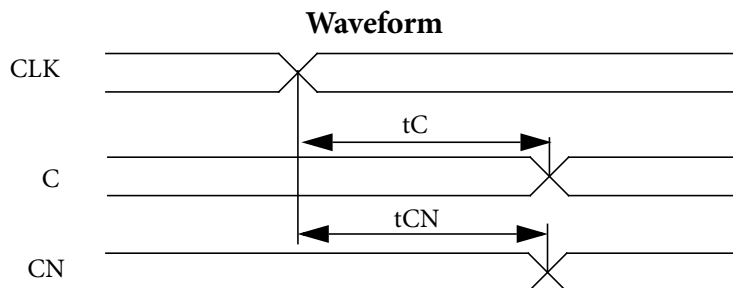
Macro Name:	CLK2D2
Drive Capability:	2x
Gate Equivalents:	5.25
Leakage Power (pW):	165.7

Pin Description

Name	Capacitance (pF)	Description
	CLK2D2	
CLK	0.004	Input
	Maximum capacitance	
C	0.6	Output
CN	0.6	Inverted Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)
	CLK2D2
C	0.075
CN	0.075



Timing Numbers for CLK2D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

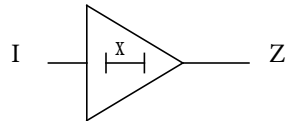
Fanout load	4		8		16		32		64	
CLK2D2	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tC	0.306	0.239	0.351	0.258	0.432	0.292	0.587	0.358	0.885	0.487
tCN	0.400	0.152	0.447	0.169	0.535	0.195	0.701	0.244	1.022	0.338

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
CLK2D2	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tC	0.351	0.258	0.391	0.341	0.406	0.418	0.412	0.485	0.405	0.542
tCN	0.447	0.169	0.530	0.208	0.607	0.224	0.674	0.230	0.731	0.223

DELAY BUFFERS**Non Inverting Delay Buffers with 1X Delay
DL01D1, DL01D2 and DL01D4**

DL01D1, DL01D2 and DL01D4 provides 1X logical delay of the single input with 1x, 2x and 4x drive capability.

**Function Table**

INPUT	OUTPUT
I	Z
L	L
H	H

Cell Description

Macro Name:	DL01D1	DL01D2	DL01D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.	2.25	2.75
Leakage Power (pW):	55.9	70.6	90.7

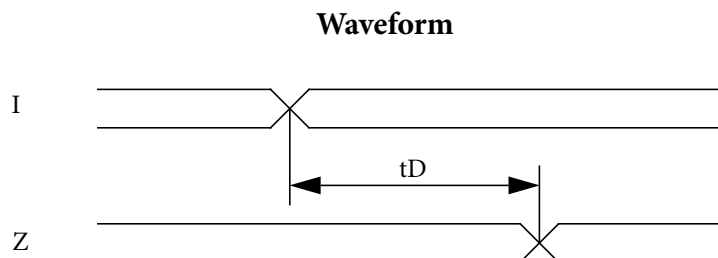
Pin Description

Name	Capacitance (pF)			Description
	DL01D1	DL01D2	DL01D4	
I	0.003	0.004	0.004	Input
Maximum capacitance				
Z	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	DL01D1	DL01D2	DL01D4
Z	0.042	0.059	0.114

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Timing Numbers for DL01D1, DL01D2 and DL01D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

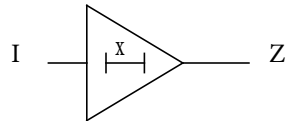
Fanout load	4		8		16		32		64	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
DL01D1	0.438	0.435	0.490	0.500	0.588	0.619	0.781	0.843	1.165	1.287
DL01D2	0.397	0.406	0.425	0.439	0.467	0.493	0.542	0.586	0.676	0.746
DL01D4	0.390	0.422	0.406	0.441	0.435	0.474	0.479	0.528	0.555	0.618

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
DL01D1	0.490	0.500	0.533	0.542	0.568	0.572	0.598	0.594	0.620	0.609
DL01D2	0.425	0.439	0.470	0.471	0.509	0.490	0.542	0.502	0.568	0.508
DL01D4	0.406	0.441	0.447	0.471	0.480	0.490	0.508	0.504	0.530	0.511

Non Inverting Delay Buffers with 2X Delay DL02D1, DL02D2 and DL02D4

DL02D1, DL02D2 and DL02D4 provides 2X logical delay of the single input with 1x, 2x and 4x drive capability.



Function Table

INPUT	OUTPUT
I	Z
L	L
H	H

Cell Description

Macro Name:	DL02D1	DL02D2	DL02D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.25	2.5	2.75
Leakage Power (pW):	54.1	61.9	82.2

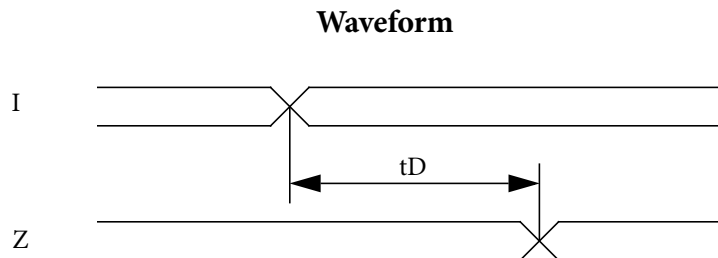
Pin Description

Name	Capacitance (pF)			Description
	DL02D1	DL02D2	DL02D4	
I	0.003	0.004	0.004	Input
Maximum capacitance				
Z	0.3	0.6	1.2	Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	DL02D1	DL02D2	DL02D4
Z	0.056	0.078	0.162

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Timing Numbers for DL02D1, DL02D2 and DL02D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

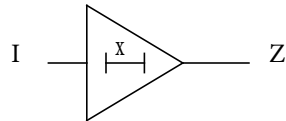
Fanout load	4		8		16		32		64	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
DL02D1	0.916	0.871	0.971	0.952	1.073	1.092	1.269	1.334	1.654	1.786
DL02D2	0.848	0.796	0.877	0.836	0.923	0.901	1.001	1.010	1.139	1.189
DL02D4	0.782	0.789	0.803	0.815	0.838	0.858	0.893	0.928	0.985	1.046

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
DL02D1	0.971	0.952	1.008	1.006	1.043	1.046	1.074	1.078	1.100	1.101
DL02D2	0.877	0.836	0.915	0.877	0.951	0.903	0.983	0.924	1.012	0.936
DL02D4	0.803	0.815	0.845	0.851	0.882	0.875	0.915	0.893	0.943	0.902

Non Inverting Delay Buffers with 3X Delay DL03D1, DL03D2 and DL03D4

DL03D1, DL03D2 and DL03D4 provides 3X logical delay of the single input with 1x, 2x and 4x drive capability.



Function Table

INPUT	OUTPUT
I	Z
L	L
H	H

Cell Description

Macro Name:	DL03D1	DL03D2	DL03D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.5	3.5	4.
Leakage Power (pW):	53.5	81.1	99.1

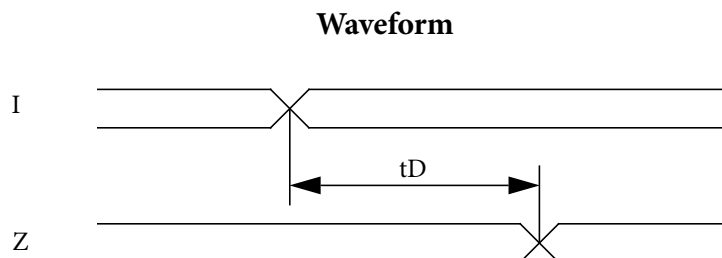
Pin Description

Name	Capacitance (pF)			Description
	DL03D1	DL03D2	DL03D4	
I	0.004	0.004	0.004	Input
Maximum capacitance				
Z	0.3	0.6	1.2	Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	DL03D1	DL03D2	DL03D4
Z	0.081	0.099	0.157

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Timing Numbers for DL03D1, DL03D2 and DL03D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

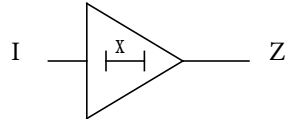
Fanout load	4		8		16		32		64	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
DL03D1	1.871	1.770	1.931	1.876	2.037	2.061	2.223	2.369	2.577	2.904
DL03D2	1.722	1.767	1.742	1.795	1.776	1.837	1.843	1.910	1.975	2.041
DL03D4	1.674	1.801	1.688	1.820	1.711	1.853	1.749	1.905	1.817	1.992

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
DL03D1	1.931	1.876	1.963	1.937	1.993	1.984	2.021	2.021	2.046	2.048
DL03D2	1.742	1.795	1.787	1.836	1.828	1.864	1.864	1.884	1.895	1.896
DL03D4	1.688	1.820	1.733	1.863	1.775	1.890	1.811	1.909	1.843	1.919

Non Inverting Delay Buffers with 4X Delay DL04D1, DL04D2 and DL04D4

DL04D1, DL04D2 and DL04D4 provides 4X logical delay of the single input with 1x, 2x and 4x drive capability.



Function Table

INPUT	OUTPUT
I	Z
L	L
H	H

Cell Description

Macro Name:	DL04D1	DL04D2	DL04D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	3.25	4.25	4.5
Leakage Power (pW):	52.2	81.2	96.3

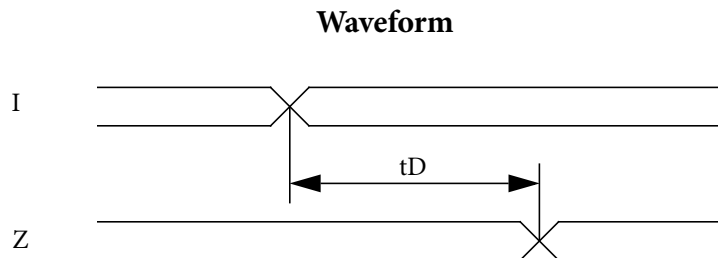
Pin Description

Name	Capacitance (pF)			Description
	DL04D1	DL04D2	DL04D4	
I	0.004	0.004	0.004	Input
Maximum capacitance				
Z	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	DL04D1	DL04D2	DL04D4
Z	0.122	0.126	0.178

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Timing Numbers for DL04D1, DL04D2 and DL04D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

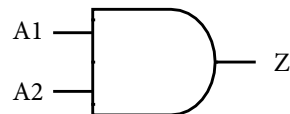
Fanout load	4		8		16		32		64	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
DL04D1	3.769	3.731	3.839	3.865	3.955	4.094	4.152	4.475	4.515	5.118
DL04D2	3.482	3.639	3.508	3.668	3.548	3.711	3.621	3.785	3.762	3.916
DL04D4	3.514	3.704	3.530	3.725	3.558	3.761	3.600	3.817	3.672	3.912

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
tD	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
DL04D1	3.839	3.865	3.873	3.932	3.900	3.987	3.923	4.030	3.943	4.062
DL04D2	3.508	3.668	3.553	3.714	3.593	3.745	3.629	3.768	3.659	3.781
DL04D4	3.530	3.725	3.576	3.769	3.615	3.797	3.651	3.819	3.682	3.831

GATES**2-Input AND****AN02D0, AN02D1, AN02D2, AN02D4, AN02D7 and AN02DA**

AN02D0, AN02D1, AN02D2, AN02D4, AN02D7 and AN02DA are 2-input AND gates with 0.5x, 1x, 2x, 4x, 7x and 10x drive capabilities.

Function Table

INPUTS		OUTPUT
A1	A2	Z
L	X	L
X	L	L
H	H	H

Cell Description

Macro Name:	AN02D0	AN02D1	AN02D2	AN02D4	AN02D7	AN02DA
Drive Capability:	0.5x	1x	2x	4x	7x	10x
Gate Equivalents:	1.25	1.25	1.5	1.75	2.75	3.25
Leakage Power (pW):	22.3	31.7	58.6	106.8	256.0	330.1

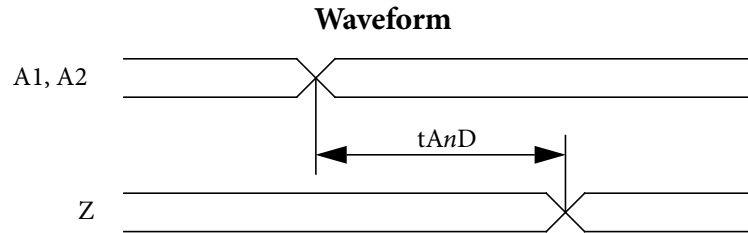
Pin Description

Name	Capacitance (pF)						Description
	AN02D0	AN02D1	AN02D2	AN02D4	AN02D7	AN02DA	
A1	0.003	0.003	0.003	0.004	0.005	0.005	Input
A2	0.003	0.003	0.003	0.004	0.005	0.005	Input
Maximum capacitance							
Z	0.15	0.3	0.6	1.2	2.1	3	Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)					
	AN02D0	AN02D1	AN02D2	AN02D4	AN02D7	AN02DA
A1	0.001	0.001	0.001	0.001	0.002	0.001
A2	-0	-0	-0	-0	-0	-0
Z	0.017	0.024	0.039	0.071	0.135	0.221

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Timing Numbers for AN02D0, AN02D1, AN02D2, AN02D4, AN02D7 and AN02DA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
AN02D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.177	0.146	0.279	0.199	0.483	0.303	0.890	0.509	1.704	0.922
tA2D	0.180	0.163	0.282	0.216	0.485	0.320	0.892	0.526	1.706	0.939
AN02D1										
tA1D	0.127	0.134	0.176	0.162	0.272	0.216	0.465	0.320	0.848	0.528
tA2D	0.129	0.150	0.178	0.179	0.275	0.233	0.467	0.337	0.851	0.545
AN02D2										
tA1D	0.134	0.163	0.160	0.182	0.207	0.213	0.300	0.264	0.483	0.356
tA2D	0.137	0.178	0.163	0.199	0.210	0.230	0.303	0.282	0.486	0.374
AN02D4										
tA1D	0.164	0.129	0.180	0.141	0.209	0.160	0.257	0.189	0.347	0.242
tA2D	0.167	0.141	0.182	0.153	0.211	0.172	0.259	0.202	0.350	0.255
AN02D7										
tA1D	0.164	0.169	0.174	0.179	0.190	0.196	0.217	0.225	0.264	0.274
tA2D	0.167	0.177	0.177	0.188	0.193	0.205	0.221	0.234	0.268	0.283
AN02DA										
tA1D	0.209	0.223	0.217	0.231	0.232	0.246	0.254	0.271	0.292	0.310
tA2D	0.213	0.229	0.221	0.238	0.236	0.253	0.258	0.278	0.296	0.317

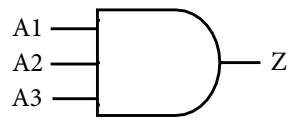
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
AN02D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.279	0.199	0.299	0.276	0.298	0.347	0.291	0.412	0.275	0.469
tA2D	0.282	0.216	0.296	0.300	0.291	0.380	0.280	0.452	0.260	0.515
AN02D1										
tA1D	0.176	0.162	0.202	0.244	0.207	0.321	0.204	0.389	0.191	0.448
tA2D	0.178	0.179	0.197	0.266	0.196	0.351	0.188	0.426	0.170	0.491
AN02D2										
tA1D	0.160	0.182	0.208	0.273	0.233	0.367	0.245	0.447	0.242	0.514
tA2D	0.163	0.199	0.195	0.291	0.211	0.390	0.215	0.476	0.205	0.547
AN02D4										
tA1D	0.180	0.141	0.246	0.219	0.302	0.286	0.344	0.340	0.372	0.379
tA2D	0.182	0.153	0.226	0.235	0.264	0.311	0.292	0.374	0.308	0.422
AN02D7										
tA1D	0.174	0.179	0.241	0.263	0.301	0.344	0.345	0.410	0.372	0.457
tA2D	0.177	0.188	0.221	0.273	0.261	0.359	0.291	0.429	0.308	0.481
AN02DA										
tA1D	0.217	0.231	0.287	0.318	0.358	0.415	0.412	0.496	0.445	0.555
tA2D	0.221	0.238	0.267	0.326	0.315	0.427	0.353	0.512	0.375	0.575

3-Input AND

AN03D0, AN03D1, AN03D2, AN03D4, AN03D7 and AN03DA

AN03D0, AN03D1, AN03D2, AN03D4, AN03D7 and AN03DA are 3-input AND gates with 0.5x, 1x, 2x, 4x, 7x and 10x drive capabilities.



Function Table

INPUTS			OUTPUT
A1	A2	A3	Z
L	X	X	L
X	L	X	L
X	X	L	L
H	H	H	H

Cell Description

Macro Name:	AN03D0	AN03D1	AN03D2	AN03D4	AN03D7	AN03DA
Drive Capability:	0.5x	1x	2x	4x	7x	10x
Gate Equivalents:	1.5	1.5	1.75	2.25	3.	3.5
Leakage Power (pW):	17.7	30.8	50.7	103.2	236.6	323.3

Pin Description

Name	Capacitance (pF)						Description
	AN03D0	AN03D1	AN03D2	AN03D4	AN03D7	AN03DA	
A1	0.002	0.002	0.002	0.004	0.004	0.004	Input
A2	0.003	0.003	0.003	0.004	0.004	0.004	Input
A3	0.003	0.003	0.003	0.004	0.005	0.005	Input
Maximum capacitance							
Z	0.15	0.3	0.6	1.2	2.1	3	Output

Pin Powers for

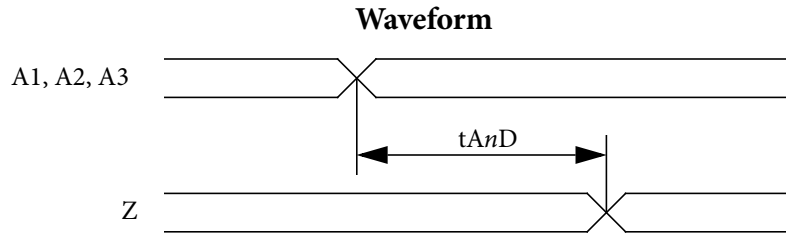
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)					
	AN03D0	AN03D1	AN03D2	AN03D4	AN03D7	AN03DA
A1	0.001	0.001	0.001	0.002	0.001	0.001
A2	-0	0	0	0	-0	-0
A3	0	0	0	0	-0	-0
Z	0.021	0.028	0.042	0.081	0.151	0.257

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Timing Numbers for AN03D0, AN03D1, AN03D2, AN03D4, AN03D7 and AN03DA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
AN03D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.213	0.153	0.321	0.197	0.536	0.281	0.965	0.449	1.821	0.783
tA2D	0.220	0.169	0.328	0.214	0.543	0.298	0.972	0.466	1.828	0.800
tA3D	0.224	0.184	0.332	0.229	0.547	0.314	0.976	0.482	1.832	0.816
AN03D1										
tA1D	0.161	0.145	0.213	0.171	0.315	0.217	0.516	0.301	0.917	0.468
tA2D	0.168	0.161	0.220	0.187	0.322	0.233	0.523	0.318	0.923	0.485
tA3D	0.172	0.174	0.224	0.201	0.326	0.248	0.527	0.334	0.928	0.501
AN03D2										
tA1D	0.161	0.160	0.192	0.179	0.247	0.209	0.352	0.260	0.559	0.349
tA2D	0.168	0.174	0.199	0.195	0.254	0.225	0.359	0.276	0.566	0.366
tA3D	0.173	0.188	0.204	0.209	0.259	0.240	0.364	0.292	0.571	0.382
AN03D4										
tA1D	0.160	0.136	0.177	0.148	0.206	0.167	0.255	0.195	0.346	0.246
tA2D	0.167	0.147	0.184	0.159	0.213	0.178	0.262	0.207	0.353	0.258
tA3D	0.171	0.155	0.188	0.167	0.217	0.187	0.266	0.216	0.357	0.267

AN03D7										
tA1D	0.220	0.187	0.232	0.199	0.250	0.216	0.281	0.246	0.333	0.298
tA2D	0.226	0.192	0.238	0.203	0.256	0.221	0.287	0.251	0.339	0.302
tA3D	0.232	0.202	0.244	0.214	0.262	0.232	0.293	0.262	0.345	0.314
AN03DA										
tA1D	0.297	0.226	0.306	0.235	0.323	0.250	0.351	0.275	0.394	0.314
tA2D	0.303	0.232	0.312	0.241	0.329	0.256	0.357	0.280	0.400	0.319
tA3D	0.309	0.239	0.318	0.248	0.335	0.263	0.363	0.288	0.406	0.327

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

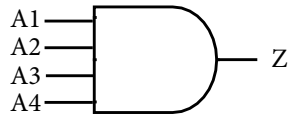
Input transition	1		5		10		15		20	
AN03D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.321	0.197	0.354	0.278	0.366	0.353	0.370	0.419	0.365	0.476
tA2D	0.328	0.214	0.354	0.300	0.362	0.384	0.362	0.456	0.353	0.519
tA3D	0.332	0.229	0.348	0.318	0.348	0.409	0.342	0.488	0.326	0.556
AN03D1										
tA1D	0.213	0.171	0.254	0.255	0.273	0.336	0.283	0.406	0.281	0.464
tA2D	0.220	0.187	0.251	0.275	0.265	0.363	0.270	0.439	0.263	0.503
tA3D	0.224	0.201	0.243	0.292	0.248	0.387	0.244	0.468	0.230	0.538
AN03D2										
tA1D	0.192	0.179	0.243	0.268	0.275	0.358	0.293	0.434	0.298	0.497
tA2D	0.199	0.195	0.237	0.285	0.260	0.382	0.273	0.463	0.271	0.531
tA3D	0.204	0.209	0.227	0.301	0.238	0.403	0.241	0.489	0.230	0.561
AN03D4										
tA1D	0.177	0.148	0.234	0.229	0.278	0.304	0.310	0.366	0.328	0.414
tA2D	0.184	0.159	0.226	0.242	0.260	0.323	0.283	0.390	0.294	0.443
tA3D	0.188	0.167	0.213	0.253	0.232	0.339	0.244	0.410	0.245	0.468

AN03D7					
tA1D	0.232 0.199	0.297 0.283	0.366 0.369	0.420 0.438	0.457 0.487
tA2D	0.238 0.203	0.284 0.289	0.335 0.379	0.377 0.451	0.406 0.505
tA3D	0.244 0.214	0.273 0.300	0.304 0.392	0.330 0.467	0.347 0.524
AN03DA					
tA1D	0.306 0.235	0.373 0.321	0.453 0.416	0.524 0.495	0.571 0.549
tA2D	0.312 0.241	0.361 0.327	0.421 0.424	0.474 0.506	0.512 0.564
tA3D	0.318 0.248	0.349 0.335	0.387 0.434	0.420 0.518	0.444 0.579

4-Input AND

AN04D0, AN04D1, AN04D2, AN04D4, AN04D7 and AN04DA

AN04D0, AN04D1, AN04D2, AN04D4, AN04D7 and AN04DA are 4-input AND gates with 0.5x, 1x, 2x, 4x, 7x and 10x drive capabilities.



Function Table

INPUTS				OUTPUT
A1	A2	A3	A4	Z
L	X	X	X	L
X	L	X	X	L
X	X	L	X	L
X	X	X	L	L
H	H	H	H	H

Cell Description

Macro Name:	AN04D0	AN04D1	AN04D2	AN04D4	AN04D7	AN04DA
Drive Capability:	0.5x	1x	2x	4x	7x	10x
Gate Equivalents:	2.	2.	2.25	2.5	3.25	3.75
Leakage Power (pW):	16.7	24.2	42.5	89.3	235.9	297.2

Pin Description

Name	Capacitance (pF)						Description
	AN04D0	AN04D1	AN04D2	AN04D4	AN04D7	AN04DA	
A1	0.003	0.003	0.003	0.004	0.004	0.004	Input
A2	0.003	0.003	0.003	0.004	0.004	0.004	Input
A3	0.003	0.003	0.003	0.004	0.004	0.004	Input
A4	0.003	0.003	0.003	0.004	0.005	0.005	Input
Maximum capacitance							
Z	0.15	0.3	0.6	1.2	2.1	3	Output

Pin Powers for

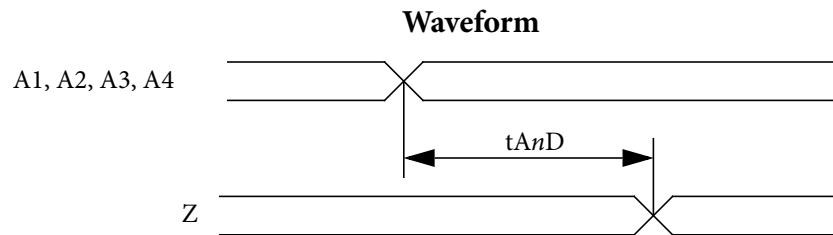
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)					
	AN04D0	AN04D1	AN04D2	AN04D4	AN04D7	AN04DA
A1	0.001	0.001	0.001	0.001	0.001	0.001
A2	-0	0	0	-0	-0	-0
A3	-0	-0	-0	-0	-0	-0
A4	-0	-0	-0	-0	-0	-0
Z	0.023	0.029	0.041	0.076	0.172	0.26

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Timing Numbers for AN04D0, AN04D1, AN04D2, AN04D4, AN04D7 and AN04DA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
AN04D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.239	0.165	0.358	0.219	0.595	0.323	1.067	0.529	2.011	0.941
tA2D	0.251	0.183	0.371	0.238	0.607	0.342	1.079	0.548	2.023	0.961
tA3D	0.259	0.199	0.379	0.254	0.616	0.358	1.088	0.565	2.032	0.978
tA4D	0.264	0.212	0.384	0.267	0.621	0.372	1.093	0.580	2.037	0.994
AN04D1										
tA1D	0.185	0.148	0.247	0.178	0.367	0.233	0.604	0.337	1.076	0.545
tA2D	0.197	0.164	0.258	0.195	0.378	0.250	0.615	0.355	1.087	0.563
tA3D	0.205	0.178	0.267	0.209	0.386	0.265	0.623	0.371	1.095	0.579
tA4D	0.210	0.191	0.272	0.223	0.392	0.280	0.629	0.386	1.101	0.595
AN04D2										
tA1D	0.175	0.152	0.212	0.174	0.275	0.207	0.396	0.265	0.633	0.371
tA2D	0.186	0.168	0.223	0.190	0.286	0.223	0.407	0.282	0.644	0.389
tA3D	0.194	0.182	0.232	0.205	0.295	0.239	0.416	0.298	0.653	0.405
tA4D	0.200	0.194	0.237	0.218	0.300	0.253	0.421	0.312	0.658	0.420
AN04D4										
tA1D	0.181	0.140	0.201	0.152	0.238	0.174	0.297	0.207	0.409	0.263
tA2D	0.193	0.151	0.213	0.165	0.249	0.187	0.309	0.220	0.421	0.277
tA3D	0.201	0.161	0.221	0.175	0.258	0.197	0.318	0.230	0.429	0.289
tA4D	0.207	0.172	0.228	0.186	0.264	0.208	0.324	0.242	0.435	0.301

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AN04D7									
tA1D	0.278	0.192	0.290	0.204	0.311	0.221	0.345	0.251	0.402 0.303
tA2D	0.289	0.199	0.301	0.211	0.322	0.229	0.356	0.259	0.413 0.311
tA3D	0.297	0.205	0.310	0.217	0.330	0.235	0.364	0.265	0.422 0.317
tA4D	0.303	0.213	0.315	0.225	0.336	0.243	0.370	0.274	0.427 0.326
AN04DA									
tA1D	0.360	0.215	0.370	0.224	0.390	0.239	0.420	0.264	0.471 0.303
tA2D	0.371	0.222	0.381	0.230	0.401	0.246	0.431	0.271	0.482 0.310
tA3D	0.380	0.227	0.390	0.235	0.410	0.251	0.440	0.276	0.491 0.315
tA4D	0.386	0.235	0.396	0.243	0.416	0.259	0.446	0.284	0.497 0.324

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

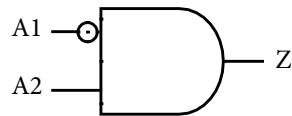
Input transition	1		5		10		15		20	
AN04D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.358	0.219	0.395	0.299	0.414	0.373	0.424	0.436	0.425	0.490
tA2D	0.371	0.238	0.400	0.323	0.416	0.405	0.423	0.476	0.421	0.535
tA3D	0.379	0.254	0.400	0.341	0.409	0.431	0.411	0.508	0.403	0.573
tA4D	0.384	0.267	0.397	0.357	0.397	0.453	0.391	0.536	0.376	0.605
AN04D1										
tA1D	0.247	0.178	0.288	0.259	0.311	0.336	0.325	0.403	0.327	0.457
tA2D	0.258	0.195	0.291	0.280	0.310	0.365	0.320	0.438	0.319	0.498
tA3D	0.267	0.209	0.289	0.298	0.300	0.390	0.304	0.468	0.297	0.534
tA4D	0.272	0.223	0.285	0.313	0.287	0.411	0.282	0.495	0.267	0.565

AN04D2									
tA1D	0.212	0.174	0.261	0.261	0.294	0.345	0.315	0.416	0.323 0.474
tA2D	0.223	0.190	0.260	0.278	0.287	0.370	0.303	0.446	0.306 0.509
tA3D	0.232	0.205	0.257	0.295	0.273	0.392	0.282	0.474	0.278 0.542
tA4D	0.237	0.218	0.252	0.309	0.256	0.411	0.255	0.498	0.243 0.571
AN04D4									
tA1D	0.201	0.152	0.258	0.235	0.304	0.311	0.339	0.373	0.360 0.421
tA2D	0.213	0.165	0.256	0.249	0.293	0.331	0.321	0.398	0.336 0.451
tA3D	0.221	0.175	0.250	0.261	0.275	0.348	0.293	0.420	0.300 0.477
tA4D	0.228	0.186	0.246	0.273	0.256	0.364	0.262	0.441	0.259 0.503
AN04D7									
tA1D	0.290	0.204	0.354	0.288	0.429	0.374	0.494	0.441	0.540 0.488
tA2D	0.301	0.211	0.348	0.296	0.407	0.385	0.459	0.455	0.498 0.505
tA3D	0.310	0.217	0.342	0.304	0.382	0.396	0.418	0.471	0.445 0.525
tA4D	0.315	0.225	0.335	0.311	0.359	0.404	0.380	0.481	0.396 0.538
AN04DA									
tA1D	0.370	0.224	0.433	0.309	0.514	0.401	0.591	0.474	0.647 0.523
tA2D	0.381	0.230	0.428	0.316	0.492	0.410	0.552	0.486	0.599 0.537
tA3D	0.390	0.235	0.421	0.322	0.465	0.419	0.508	0.499	0.541 0.555
tA4D	0.396	0.243	0.416	0.329	0.443	0.426	0.468	0.508	0.489 0.566

2-Input AND with 1 Inverted Input

AN12D1, AN12D2 and AN12D4

AN12D1, AN12D2 and AN12D4 are 2-input AND gates with one inverted input and with 1x, 2x and 4x drive capabilities.



Function Table

INPUTS		OUTPUT
A1	A2	Z
H	X	L
X	L	L
L	H	H

Cell Description

Macro Name:	AN12D1	AN12D2	AN12D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.75	2.	2.25
Leakage Power (pW):	47.2	58.3	90.0

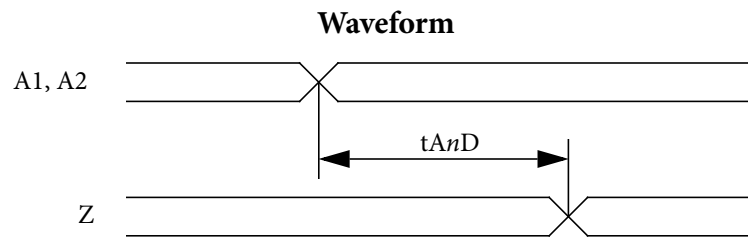
Pin Description

Name	Capacitance (pF)			Description
	AN12D1	AN12D2	AN12D4	
A1	0.002	0.002	0.002	Input
A2	0.003	0.003	0.004	Input
Maximum capacitance				
Z	0.3	0.6	1.2	Output

Pin Powers for Standard Load = 0.032 pF

Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	AN12D1	AN12D2	AN12D4
A1	0.009	0.009	0.011
A2	-0	-0	-0
Z	0.027	0.04	0.072



Timing Numbers for AN12D1, AN12D2 and AN12D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
AN12D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.196	0.164	0.242	0.190	0.332	0.237	0.511	0.326	0.869	0.504
tA2D	0.130	0.151	0.176	0.177	0.266	0.225	0.445	0.314	0.804	0.492
AN12D2										
tA1D	0.192	0.179	0.218	0.198	0.265	0.227	0.358	0.278	0.541	0.371
tA2D	0.125	0.163	0.151	0.183	0.198	0.213	0.290	0.265	0.474	0.358
AN12D4										
tA1D	0.193	0.173	0.208	0.185	0.234	0.205	0.280	0.236	0.369	0.289
tA2D	0.125	0.147	0.140	0.159	0.166	0.180	0.212	0.211	0.301	0.265

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

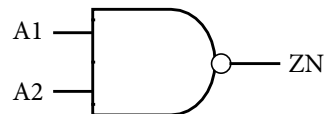
Input transition	1		5		10		15		20	
AN12D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.242	0.190	0.320	0.216	0.392	0.221	0.456	0.217	0.512	0.203
tA2D	0.176	0.177	0.195	0.266	0.196	0.354	0.188	0.431	0.169	0.498
AN12D2										
tA1D	0.218	0.198	0.298	0.224	0.371	0.229	0.437	0.226	0.494	0.213
tA2D	0.151	0.183	0.177	0.275	0.186	0.370	0.185	0.452	0.171	0.522
AN12D4										
tA1D	0.208	0.185	0.286	0.221	0.355	0.235	0.417	0.240	0.470	0.234
tA2D	0.140	0.159	0.173	0.246	0.192	0.332	0.201	0.404	0.197	0.465

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2-Input NAND

ND02D0, ND02D1, ND02D2, ND02D4, ND02D7 and ND02DA

ND02D0, ND02D1, ND02D2, ND02D4, ND02D7 and ND02DA are 2-input NAND gates with 0.5x, 1x, 2x, 4x 7x and 10x drive capabilities.



Function Table

INPUTS		OUTPUT
A1	A2	ZN
L	X	H
X	L	H
H	H	L

Cell Description

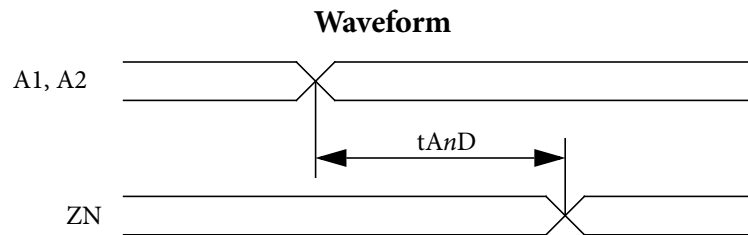
Macro Name:	ND02D0	ND02D1	ND02D2	ND02D4	ND02D7	ND02DA
Drive Capability:	0.5x	1x	2x	4x	7x	10x
Gate Equivalents:	1.	1.	1.5	2.5	3.25	4.
Leakage Power (pW):	9.0	10.0	20.6	67.8	100.3	118.0

Pin Description

Name	Capacitance (pF)						Description
	ND02D0	ND02D1	ND02D2	ND02D4	ND02D7	ND02DA	
A1	0.002	0.003	0.007	0.003	0.004	0.004	Input
A2	0.003	0.004	0.008	0.003	0.004	0.004	Input
Maximum capacitance							
ZN	0.15	0.3	0.6	1.2	2.1	3	Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)					
	ND02D0	ND02D1	ND02D2	ND02D4	ND02D7	ND02DA
A1	0	0.001	0.002	0	0.001	0.001
A2	-0	-0	-0	-0	-0	-0
ZN	0.006	0.009	0.017	0.078	0.147	0.214



Timing Numbers for ND02D0, ND02D1, ND02D2, ND02D4, ND02D7 and ND02DA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
ND02D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.127	0.127	0.209	0.212	0.372	0.382	0.697	0.721	1.347	1.398
tA2D	0.131	0.126	0.213	0.212	0.376	0.381	0.701	0.720	1.351	1.397
ND02D1										
tA1D	0.097	0.088	0.153	0.138	0.262	0.238	0.480	0.440	0.915	0.841
tA2D	0.103	0.088	0.158	0.139	0.267	0.239	0.485	0.440	0.920	0.842
ND02D2										
tA1D	0.068	0.058	0.095	0.081	0.149	0.127	0.257	0.219	0.471	0.402
tA2D	0.075	0.059	0.102	0.082	0.156	0.128	0.263	0.220	0.477	0.403
ND02D4										
tA1D	0.166	0.217	0.179	0.229	0.203	0.251	0.248	0.284	0.338	0.343
tA2D	0.168	0.217	0.181	0.229	0.205	0.251	0.250	0.284	0.340	0.343
ND02D7										
tA1D	0.189	0.211	0.198	0.221	0.212	0.237	0.235	0.265	0.277	0.312
tA2D	0.196	0.212	0.205	0.222	0.219	0.238	0.243	0.265	0.285	0.313
ND02DA										
tA1D	0.216	0.248	0.223	0.255	0.236	0.270	0.256	0.292	0.288	0.330
tA2D	0.224	0.248	0.231	0.255	0.243	0.270	0.263	0.293	0.295	0.331

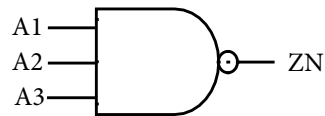
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**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
ND02D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.209	0.212	0.298	0.280	0.400	0.351	0.488	0.405	0.552	0.436
tA2D	0.213	0.212	0.304	0.258	0.407	0.305	0.498	0.342	0.566	0.362
ND02D1										
tA1D	0.153	0.138	0.238	0.200	0.325	0.251	0.397	0.285	0.452	0.302
tA2D	0.158	0.139	0.246	0.180	0.337	0.213	0.413	0.236	0.472	0.245
ND02D2										
tA1D	0.095	0.081	0.172	0.128	0.238	0.154	0.295	0.170	0.340	0.173
tA2D	0.102	0.082	0.183	0.113	0.255	0.129	0.317	0.135	0.368	0.131
ND02D4										
tA1D	0.179	0.229	0.252	0.279	0.314	0.310	0.366	0.333	0.407	0.345
tA2D	0.181	0.229	0.255	0.265	0.317	0.290	0.371	0.308	0.414	0.315
ND02D7										
tA1D	0.198	0.221	0.260	0.266	0.310	0.296	0.353	0.319	0.386	0.333
tA2D	0.205	0.222	0.274	0.255	0.331	0.276	0.381	0.290	0.422	0.295
ND02DA										
tA1D	0.223	0.255	0.285	0.300	0.336	0.331	0.379	0.354	0.412	0.368
tA2D	0.231	0.255	0.299	0.289	0.357	0.310	0.407	0.325	0.448	0.331

3-Input NAND**ND03D0, ND03D1, ND03D2, ND03D4, ND03D7 and ND03DA**

ND03D0, ND03D1, ND03D2, ND03D4, ND03D7 and ND03DA are 3-input NAND gates with 0.5x, 1x, 2x, 4x, 7x and 10x drive capabilities.

Function Table

INPUTS			OUTPUT
A1	A2	A3	ZN
L	X	X	H
X	L	X	H
X	X	L	H
H	H	H	L

Cell Description

Macro Name:	ND03D0	ND03D1	ND03D2	ND03D4	ND03D7	ND03DA
Drive Capability:	0.5x	1x	2x	4x	7x	10x
Gate Equivalents:	1.25	1.25	2.25	2.75	3.5	4.25
Leakage Power (pW):	11.2	6.2	22.8	70.7	92.4	114.1

Pin Description

Name	Capacitance (pF)						Description
	ND03D0	ND03D1	ND03D2	ND03D4	ND03D7	ND03DA	
A1	0.002	0.003	0.009	0.002	0.003	0.003	Input
A2	0.002	0.004	0.009	0.002	0.004	0.003	Input
A3	0.002	0.004	0.007	0.002	0.004	0.004	Input
Maximum capacitance							
ZN	0.15	0.3	0.6	1.2	2.1	3	Output

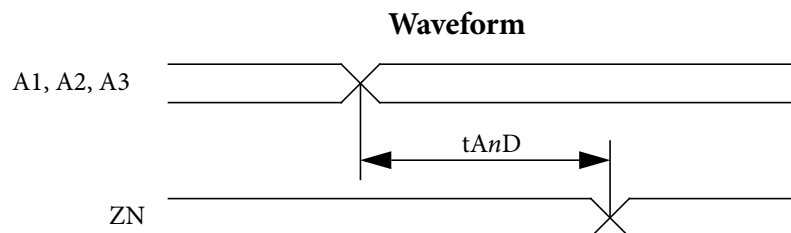
Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)					
	ND03D0	ND03D1	ND03D2	ND03D4	ND03D7	ND03DA
A1	-0	0.001	-0	-0	0.001	0.001
A2	-0	0	0	-0	0	0
A3	0.002	-0	0.008	0.002	0	0
ZN	0.006	0.014	0.028	0.079	0.15	0.218

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Timing Numbers for ND03D0, ND03D1, ND03D2, ND03D4, ND03D7 and ND03DA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
ND03D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.219	0.170	0.360	0.286	0.641	0.519	1.203	0.985	2.326	1.917
tA2D	0.209	0.166	0.350	0.283	0.630	0.516	1.192	0.982	2.315	1.914
tA3D	0.200	0.162	0.341	0.279	0.622	0.512	1.184	0.978	2.307	1.910
ND03D1										
tA1D	0.103	0.094	0.158	0.146	0.266	0.249	0.480	0.456	0.909	0.868
tA2D	0.113	0.099	0.168	0.151	0.277	0.255	0.493	0.461	0.926	0.874
tA3D	0.130	0.103	0.188	0.155	0.304	0.259	0.535	0.465	0.996	0.877
ND03D2										
tA1D	0.082	0.079	0.105	0.106	0.150	0.159	0.239	0.264	0.417	0.473
tA2D	0.074	0.076	0.097	0.102	0.140	0.155	0.227	0.260	0.400	0.469
tA3D	0.066	0.069	0.089	0.095	0.132	0.148	0.219	0.253	0.392	0.462
ND03D4										
tA1D	0.233	0.243	0.246	0.255	0.270	0.276	0.315	0.309	0.405	0.367
tA2D	0.220	0.239	0.233	0.251	0.258	0.272	0.303	0.305	0.393	0.363
tA3D	0.209	0.235	0.222	0.247	0.246	0.268	0.291	0.301	0.382	0.359

ND03D7										
tA1D	0.215	0.243	0.224	0.253	0.238	0.269	0.262	0.297	0.304	0.345
tA2D	0.226	0.250	0.235	0.260	0.249	0.276	0.273	0.304	0.315	0.352
tA3D	0.234	0.254	0.244	0.264	0.258	0.280	0.282	0.308	0.323	0.356
ND03DA										
tA1D	0.244	0.279	0.251	0.286	0.263	0.301	0.284	0.323	0.317	0.361
tA2D	0.255	0.286	0.262	0.293	0.274	0.308	0.295	0.331	0.328	0.368
tA3D	0.264	0.290	0.271	0.297	0.283	0.312	0.304	0.334	0.337	0.372

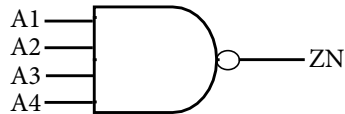
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
ND03D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.360	0.286	0.452	0.315	0.570	0.344	0.688	0.366	0.784	0.374
tA2D	0.350	0.283	0.441	0.328	0.559	0.379	0.674	0.420	0.767	0.442
tA3D	0.341	0.279	0.432	0.342	0.548	0.412	0.662	0.471	0.752	0.500
ND03D1										
tA1D	0.158	0.146	0.242	0.204	0.330	0.253	0.403	0.286	0.459	0.302
tA2D	0.168	0.151	0.255	0.194	0.348	0.229	0.426	0.252	0.487	0.261
tA3D	0.188	0.155	0.277	0.179	0.377	0.195	0.464	0.203	0.532	0.199
ND03D2										
tA1D	0.105	0.106	0.184	0.128	0.255	0.140	0.314	0.146	0.362	0.143
tA2D	0.097	0.102	0.172	0.139	0.238	0.164	0.292	0.180	0.335	0.186
tA3D	0.089	0.095	0.160	0.147	0.220	0.181	0.268	0.206	0.306	0.218
ND03D4										
tA1D	0.246	0.255	0.338	0.276	0.436	0.286	0.520	0.288	0.590	0.279
tA2D	0.233	0.251	0.323	0.286	0.415	0.307	0.493	0.318	0.558	0.317
tA3D	0.222	0.247	0.309	0.294	0.394	0.322	0.466	0.339	0.526	0.343

ND03D7										
tA1D	0.224	0.253	0.296	0.303	0.359	0.338	0.411	0.364	0.452	0.380
tA2D	0.235	0.260	0.312	0.298	0.381	0.326	0.439	0.346	0.486	0.356
tA3D	0.244	0.264	0.324	0.287	0.399	0.304	0.463	0.314	0.515	0.315
ND03DA										
tA1D	0.251	0.286	0.324	0.336	0.386	0.371	0.439	0.398	0.481	0.415
tA2D	0.262	0.293	0.339	0.331	0.408	0.359	0.467	0.380	0.514	0.390
tA3D	0.271	0.297	0.352	0.320	0.429	0.336	0.494	0.346	0.549	0.347

4-Input NAND**ND04D0, ND04D1, ND04D2, ND04D4, ND04D7 and ND04DA**

ND04D0, ND04D1, ND04D2, ND04D4, ND04D7 and ND04DA are 4-input NAND gates with 0.5x, 1x, 2x, 4x, 7x and 10x drive capabilities.

Function Table

INPUTS				OUTPUT
A1	A2	A3	A4	ZN
L	X	X	X	H
X	L	X	X	H
X	X	L	X	H
X	X	X	L	H
H	H	H	H	L

Cell Description

Macro Name:	ND04D0	ND04D1	ND04D2	ND04D4	ND04D7	ND04DA
Drive Capability:	0.5x	1x	2x	4x	7x	10x
Gate Equivalents:	1.5	1.5	2.75	3.	4.	4.5
Leakage Power (pW):	5.7	5.8	49.3	65.4	92.1	113.6

Pin Description

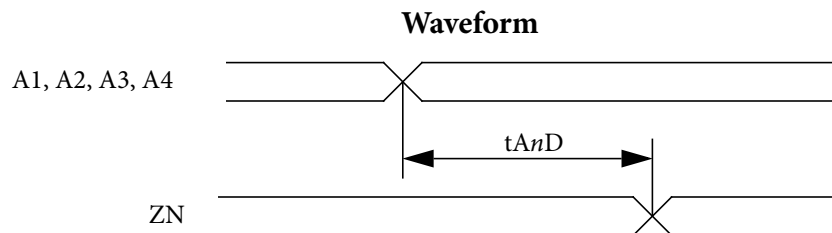
Name	Capacitance (pF)						Description
	ND04D0	ND04D1	ND04D2	ND04D4	ND04D7	ND04DA	
A1	0.002	0.004	0.004	0.004	0.003	0.003	Input
A2	0.002	0.003	0.003	0.003	0.003	0.003	Input
A3	0.002	0.004	0.004	0.004	0.003	0.003	Input
A4	0.002	0.004	0.004	0.004	0.004	0.003	Input
Maximum capacitance							
ZN	0.15	0.3	0.6	1.2	2.1	3	Output

Pin Powers for

Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)					
	ND04D0	ND04D1	ND04D2	ND04D4	ND04D7	ND04DA
A1	-0	-0	-0	-0	-0	-0
A2	0.001	0.001	0.001	0.001	0.001	0.001
A3	-0	-0	-0	-0	-0	-0
A4	-0	-0	-0	-0	-0	-0
ZN	0.01	0.017	0.062	0.089	0.155	0.226

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Timing Numbers for ND04D0, ND04D1, ND04D2, ND04D4, ND04D7 and ND04DA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
ND04D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.204	0.189	0.322	0.310	0.559	0.550	1.031	1.029	1.975	1.988
tA2D	0.181	0.170	0.299	0.291	0.536	0.531	1.007	1.010	1.949	1.969
tA3D	0.193	0.182	0.311	0.303	0.548	0.543	1.020	1.022	1.964	1.981
tA4D	0.212	0.193	0.332	0.313	0.569	0.553	1.041	1.033	1.985	1.994
ND04D1										
tA1D	0.142	0.131	0.206	0.196	0.333	0.327	0.584	0.588	1.086	1.110
tA2D	0.120	0.112	0.184	0.177	0.311	0.308	0.562	0.569	1.066	1.091
tA3D	0.132	0.123	0.195	0.189	0.322	0.320	0.573	0.581	1.076	1.103
tA4D	0.152	0.135	0.216	0.200	0.343	0.331	0.596	0.592	1.100	1.114
ND04D2										
tA1D	0.195	0.212	0.220	0.231	0.269	0.260	0.364	0.315	0.555	0.421
tA2D	0.169	0.193	0.194	0.212	0.243	0.242	0.338	0.297	0.529	0.403
tA3D	0.183	0.204	0.208	0.222	0.256	0.252	0.352	0.307	0.543	0.413
tA4D	0.205	0.216	0.230	0.234	0.279	0.264	0.374	0.319	0.565	0.425
ND04D4										
tA1D	0.203	0.239	0.216	0.251	0.240	0.273	0.285	0.306	0.375	0.363
tA2D	0.176	0.221	0.189	0.233	0.213	0.254	0.258	0.287	0.348	0.345
tA3D	0.190	0.231	0.203	0.243	0.227	0.265	0.272	0.298	0.362	0.355
tA4D	0.213	0.243	0.226	0.255	0.250	0.277	0.296	0.310	0.386	0.367

ND04D7										
tA1D	0.265	0.286	0.274	0.296	0.288	0.312	0.312	0.340	0.353	0.388
tA2D	0.239	0.267	0.248	0.277	0.262	0.293	0.286	0.321	0.328	0.369
tA3D	0.253	0.278	0.262	0.288	0.276	0.304	0.300	0.332	0.342	0.380
tA4D	0.277	0.292	0.286	0.302	0.300	0.318	0.324	0.346	0.366	0.394
ND04DA										
tA1D	0.295	0.322	0.302	0.329	0.314	0.344	0.335	0.367	0.368	0.405
tA2D	0.269	0.305	0.276	0.312	0.289	0.327	0.309	0.350	0.342	0.388
tA3D	0.282	0.314	0.289	0.321	0.302	0.336	0.322	0.359	0.355	0.397
tA4D	0.305	0.325	0.312	0.332	0.324	0.347	0.345	0.370	0.378	0.408

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

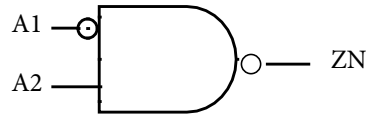
Input transition	1		5		10		15		20	
ND04D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.322	0.310	0.414	0.340	0.530	0.374	0.642	0.401	0.731	0.415
tA2D	0.299	0.291	0.390	0.350	0.503	0.418	0.611	0.477	0.693	0.508
tA3D	0.311	0.303	0.403	0.347	0.517	0.398	0.628	0.443	0.713	0.468
tA4D	0.332	0.313	0.424	0.332	0.540	0.348	0.655	0.360	0.746	0.361
ND04D1										
tA1D	0.206	0.196	0.295	0.225	0.397	0.251	0.486	0.268	0.555	0.273
tA2D	0.184	0.177	0.272	0.234	0.369	0.287	0.452	0.324	0.515	0.342
tA3D	0.195	0.189	0.283	0.232	0.382	0.272	0.467	0.302	0.533	0.316
tA4D	0.216	0.200	0.307	0.218	0.414	0.228	0.509	0.232	0.584	0.225
ND04D2										
tA1D	0.220	0.231	0.302	0.253	0.383	0.266	0.452	0.272	0.508	0.268
tA2D	0.194	0.212	0.270	0.253	0.336	0.277	0.393	0.294	0.440	0.299
tA3D	0.208	0.222	0.287	0.255	0.360	0.276	0.422	0.289	0.473	0.292
tA4D	0.230	0.234	0.318	0.248	0.406	0.250	0.482	0.246	0.545	0.232
ND04D4										
tA1D	0.216	0.251	0.299	0.275	0.381	0.288	0.451	0.295	0.508	0.292
tA2D	0.189	0.233	0.266	0.274	0.333	0.299	0.392	0.315	0.440	0.320
tA3D	0.203	0.243	0.283	0.277	0.357	0.298	0.421	0.312	0.473	0.315
tA4D	0.226	0.255	0.314	0.270	0.404	0.272	0.481	0.268	0.545	0.256

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ND04D7										
tA1D	0.274	0.296	0.360	0.321	0.449	0.340	0.525	0.351	0.586	0.352
tA2D	0.248	0.277	0.330	0.325	0.406	0.359	0.471	0.383	0.522	0.396
tA3D	0.262	0.288	0.346	0.326	0.430	0.353	0.500	0.372	0.556	0.380
tA4D	0.286	0.302	0.374	0.317	0.468	0.324	0.549	0.325	0.615	0.318
ND04DA										
tA1D	0.302	0.329	0.389	0.355	0.478	0.373	0.554	0.384	0.617	0.386
tA2D	0.276	0.312	0.357	0.360	0.433	0.393	0.498	0.416	0.549	0.429
tA3D	0.289	0.321	0.374	0.359	0.457	0.387	0.527	0.406	0.584	0.414
tA4D	0.312	0.332	0.401	0.349	0.495	0.356	0.576	0.357	0.643	0.350

2-Input NAND with 1 Inverted Input ND12D0, ND12D1, ND12D2 and ND12D4

ND12D0, ND12D1, ND12D2 and ND12D4 are 2-input NAND gates with one inverted input and with 0.5x, 1x, 2x and 4x drive capabilities.



Function Table

INPUTS		OUTPUT
A1	A2	ZN
H	X	H
X	L	H
L	H	L

Cell Description

Macro Name:	ND12D0	ND12D1	ND12D2	ND12D4
Drive Capability:	0.5x	1x	2x	4x
Gate Equivalents:	1.25	1.25	2.	2.75
Leakage Power (pW):	30.5	60.7	105.4	167.2

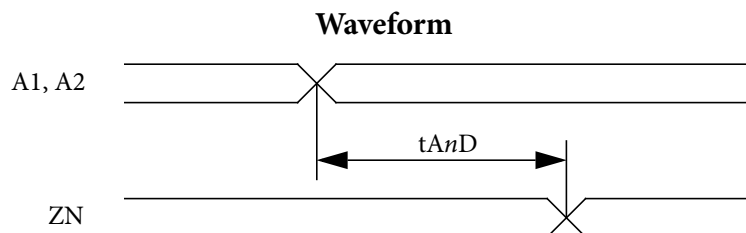
Pin Description

Name	Capacitance (pF)				Description
	ND12D0	ND12D1	ND12D2	ND12D4	
A1	0.002	0.003	0.004	0.003	Input
A2	0.002	0.004	0.008	0.004	Input
Maximum capacitance					
ZN	0.15	0.3	0.6	1.2	Output

Pin Powers for Standard Load = 0.032 pF Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)			
	ND12D0	ND12D1	ND12D2	ND12D4
A1	0.007	0.013	0.023	0.013
A2	0	0.001	0.002	0.001
ZN	0.005	0.011	0.02	0.082

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Timing Numbers for ND12D0, ND12D1, ND12D2 and ND12D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
ND12D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.201	0.183	0.321	0.269	0.560	0.438	1.038	0.777	1.996	1.454
tA2D	0.170	0.122	0.291	0.207	0.531	0.377	1.009	0.715	1.964	1.391
ND12D1										
tA1D	0.133	0.137	0.181	0.177	0.276	0.258	0.467	0.420	0.847	0.741
tA2D	0.089	0.076	0.138	0.116	0.233	0.197	0.424	0.359	0.804	0.680
ND12D2										
tA1D	0.104	0.119	0.127	0.141	0.172	0.181	0.261	0.261	0.440	0.419
tA2D	0.063	0.054	0.086	0.074	0.133	0.114	0.225	0.193	0.407	0.351
ND12D4										
tA1D	0.192	0.247	0.205	0.259	0.229	0.281	0.274	0.314	0.364	0.372
tA2D	0.146	0.186	0.159	0.198	0.184	0.220	0.229	0.253	0.319	0.312

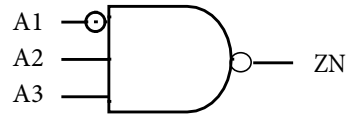
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
ND12D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.321	0.269	0.342	0.342	0.341	0.410	0.334	0.472	0.318	0.526
tA2D	0.291	0.207	0.381	0.275	0.495	0.341	0.604	0.390	0.687	0.413
ND12D1										
tA1D	0.181	0.177	0.218	0.244	0.235	0.301	0.245	0.351	0.245	0.393
tA2D	0.138	0.116	0.220	0.176	0.301	0.218	0.368	0.246	0.420	0.259
ND12D2										
tA1D	0.127	0.141	0.163	0.210	0.179	0.270	0.187	0.323	0.185	0.367
tA2D	0.086	0.074	0.160	0.118	0.222	0.141	0.275	0.154	0.318	0.157
ND12D4										
tA1D	0.205	0.259	0.240	0.327	0.256	0.385	0.264	0.436	0.262	0.479
tA2D	0.159	0.198	0.223	0.232	0.277	0.249	0.324	0.259	0.363	0.261

3-Input NAND with 1 Inverted Input

ND13D1, ND13D2 and ND13D4

ND13D1, ND13D2 and ND13D4 are 3-input NAND gates with one inverted input and with 1x, 2x and 4x drive capabilities.



Function Table

INPUTS			OUTPUT
A1	A2	A3	ZN
H	X	X	H
X	L	X	H
X	X	L	H
L	H	H	L

Cell Description

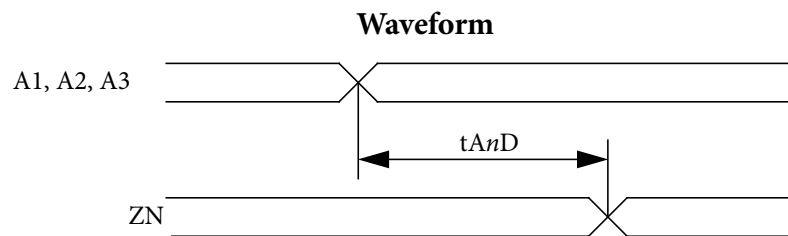
Macro Name:	ND13D1	ND13D2	ND13D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.75	2.5	3.25
Leakage Power (pW):	21.5	40.3	80.3

Pin Description

Name	Capacitance (pF)			Description
	ND13D1	ND13D2	ND13D4	
A1	0.004	0.003	0.004	Input
A2	0.003	0.009	0.003	Input
A3	0.004	0.008	0.004	Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	ND13D1	ND13D2	ND13D4
A1	0.014	0.02	0.014
A2	0.001	0	0.001
A3	0.001	0.008	0.001
ZN	0.014	0.028	0.086



Timing Numbers for ND13D1, ND13D2 and ND13D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
ND13D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.136	0.162	0.190	0.218	0.296	0.331	0.507	0.555	0.928	1.004
tA2D	0.101	0.101	0.155	0.157	0.261	0.270	0.472	0.495	0.893	0.944
tA3D	0.110	0.107	0.163	0.163	0.269	0.276	0.480	0.501	0.901	0.950
ND13D2										
tA1D	0.125	0.201	0.148	0.231	0.192	0.287	0.281	0.396	0.458	0.612
tA2D	0.074	0.076	0.097	0.103	0.141	0.158	0.230	0.266	0.406	0.482
tA3D	0.066	0.070	0.089	0.097	0.133	0.151	0.222	0.260	0.398	0.475
ND13D4										
tA1D	0.198	0.272	0.211	0.285	0.235	0.307	0.280	0.340	0.370	0.398
tA2D	0.159	0.212	0.172	0.224	0.196	0.246	0.241	0.279	0.331	0.338
tA3D	0.169	0.217	0.182	0.230	0.206	0.251	0.251	0.284	0.342	0.343

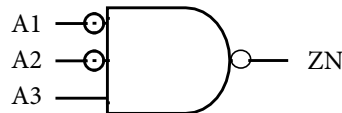
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
ND13D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.190	0.218	0.207	0.280	0.206	0.338	0.199	0.390	0.185	0.437
tA2D	0.155	0.157	0.239	0.216	0.325	0.269	0.396	0.306	0.449	0.326
tA3D	0.163	0.163	0.249	0.206	0.339	0.244	0.414	0.272	0.472	0.285
ND13D2										
tA1D	0.148	0.231	0.187	0.322	0.199	0.415	0.199	0.498	0.185	0.568
tA2D	0.097	0.103	0.173	0.141	0.238	0.167	0.292	0.185	0.335	0.193
tA3D	0.089	0.097	0.161	0.149	0.220	0.183	0.268	0.208	0.306	0.221
ND13D4										
tA1D	0.211	0.285	0.227	0.347	0.226	0.405	0.219	0.457	0.204	0.504
tA2D	0.172	0.224	0.238	0.266	0.295	0.292	0.344	0.310	0.383	0.319
tA3D	0.182	0.230	0.256	0.262	0.320	0.282	0.375	0.295	0.420	0.299

3-Input NAND with 2 Inverted Inputs

ND23D1, ND23D2 and ND23D4

ND23D1, ND23D2 and ND23D4 are 3-input NAND gates with two inverted inputs and with 1x, 2x and 4x drive capabilities.



Function Table

INPUTS			OUTPUT
A1	A2	A3	ZN
H	X	X	H
X	H	X	H
X	X	L	H
L	L	H	L

Cell Description

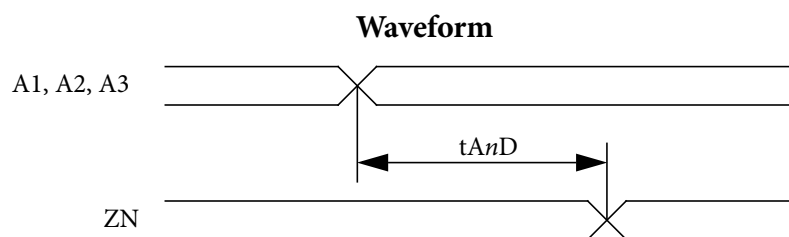
Macro Name:	ND23D1	ND23D2	ND23D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.25	2.5	2.75
Leakage Power (pW):	94.0	80.0	132.5

Pin Description

Name	Capacitance (pF)			Description
	ND23D1	ND23D2	ND23D4	
A1	0.002	0.003	0.003	Input
A2	0.002	0.003	0.003	Input
A3	0.004	0.002	0.002	Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	ND23D1	ND23D2	ND23D4
A1	0.011	0	0
A2	0.012	0	0
A3	-0	0.008	0.008
ZN	0.017	0.043	0.09



Timing Numbers for ND23D1, ND23D2 and ND23D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
ND23D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.128	0.182	0.174	0.234	0.265	0.338	0.446	0.543	0.808	0.952
tA2D	0.134	0.199	0.181	0.252	0.276	0.356	0.465	0.561	0.843	0.970
tA3D	0.106	0.110	0.152	0.162	0.243	0.266	0.424	0.471	0.784	0.880
ND23D2										
tA1D	0.150	0.304	0.177	0.333	0.226	0.377	0.321	0.449	0.511	0.570
tA2D	0.143	0.285	0.169	0.314	0.217	0.358	0.312	0.430	0.501	0.551
tA3D	0.212	0.271	0.238	0.300	0.286	0.343	0.381	0.416	0.570	0.536
ND23D4										
tA1D	0.170	0.399	0.186	0.419	0.215	0.452	0.262	0.503	0.354	0.586
tA2D	0.163	0.380	0.178	0.400	0.206	0.433	0.253	0.484	0.344	0.567
tA3D	0.233	0.366	0.248	0.386	0.275	0.419	0.322	0.470	0.414	0.553

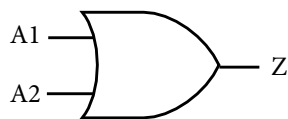
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
ND23D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.174	0.234	0.206	0.319	0.214	0.400	0.213	0.472	0.200	0.534
tA2D	0.181	0.252	0.206	0.340	0.204	0.427	0.194	0.504	0.173	0.572
tA3D	0.152	0.162	0.237	0.185	0.325	0.204	0.398	0.217	0.455	0.219
ND23D2										
tA1D	0.177	0.333	0.244	0.375	0.292	0.417	0.323	0.459	0.337	0.504
tA2D	0.169	0.314	0.232	0.365	0.272	0.432	0.297	0.498	0.304	0.559
tA3D	0.238	0.300	0.323	0.314	0.404	0.308	0.478	0.295	0.542	0.273
ND23D4										
tA1D	0.186	0.419	0.259	0.461	0.322	0.507	0.365	0.553	0.386	0.601
tA2D	0.178	0.400	0.249	0.452	0.306	0.523	0.343	0.595	0.358	0.662
tA3D	0.248	0.386	0.333	0.401	0.416	0.394	0.489	0.380	0.554	0.358

2-Input OR

OR02D0, OR02D1, OR02D2, OR02D4, OR02D7 and OR02DA

OR02D0, OR02D1, OR02D2, OR02D4, OR02D7 and OR02DA are 2-input OR gates with 0.5x, 1x, 2x, 4x, 7x and 10x drive capabilities.



Function Table

INPUTS		OUTPUT
A1	A2	Z
H	X	H
X	H	H
L	L	L

Cell Description

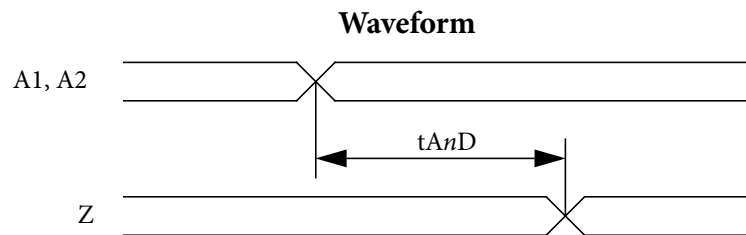
Macro Name:	OR02D0	OR02D1	OR02D2	OR02D4	OR02D7	OR02DA
Drive Capability:	0.5x	1x	2x	4x	7x	10x
Gate Equivalents:	1.25	1.25	1.5	2.	2.75	3.25
Leakage Power (pW):	25.9	27.7	38.4	54.2	90.9	111.4

Pin Description

Name	Capacitance (pF)						Description
	OR02D0	OR02D1	OR02D2	OR02D4	OR02D7	OR02DA	
A1	0.002	0.002	0.002	0.002	0.003	0.003	Input
A2	0.003	0.003	0.003	0.002	0.005	0.005	Input
Maximum capacitance							
Z	0.15	0.3	0.6	1.2	2.1	3	Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)					
	OR02D0	OR02D1	OR02D2	OR02D4	OR02D7	OR02DA
A1	0.001	0.001	0.001	0.001	0.002	0.002
A2	0	0	0	0	0	0
Z	0.015	0.023	0.037	0.078	0.153	0.257



Timing Numbers for OR02D0, OR02D1, OR02D2, OR02D4, OR02D7 and OR02DA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
OR02D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.175	0.184	0.276	0.240	0.478	0.345	0.879	0.552	1.681	0.965
tA2D	0.184	0.203	0.285	0.259	0.486	0.364	0.888	0.571	1.692	0.984
OR02D1										
tA1D	0.127	0.180	0.175	0.213	0.270	0.273	0.459	0.381	0.836	0.589
tA2D	0.136	0.199	0.183	0.233	0.278	0.292	0.467	0.400	0.844	0.608
OR02D2										
tA1D	0.121	0.213	0.147	0.238	0.194	0.278	0.287	0.343	0.473	0.454
tA2D	0.130	0.233	0.156	0.258	0.203	0.297	0.297	0.363	0.483	0.474
OR02D4										
tA1D	0.141	0.295	0.156	0.313	0.183	0.343	0.230	0.389	0.320	0.465
tA2D	0.150	0.313	0.165	0.331	0.192	0.362	0.239	0.408	0.329	0.484
OR02D7										
tA1D	0.207	0.244	0.217	0.256	0.234	0.276	0.262	0.310	0.310	0.368
tA2D	0.229	0.259	0.240	0.271	0.257	0.291	0.286	0.325	0.334	0.382
OR02DA										
tA1D	0.258	0.309	0.266	0.319	0.282	0.336	0.306	0.364	0.344	0.411
tA2D	0.280	0.323	0.288	0.333	0.304	0.350	0.329	0.378	0.367	0.425

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**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

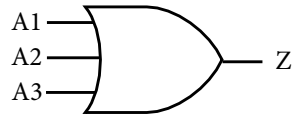
Input transition	1		5		10		15		20	
OR02D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.276	0.240	0.303	0.318	0.307	0.395	0.303	0.463	0.289	0.525
tA2D	0.285	0.259	0.323	0.311	0.337	0.371	0.343	0.428	0.338	0.479
OR02D1										
tA1D	0.175	0.213	0.213	0.295	0.224	0.378	0.227	0.452	0.217	0.516
tA2D	0.183	0.233	0.231	0.286	0.251	0.348	0.262	0.407	0.260	0.459
OR02D2										
tA1D	0.147	0.238	0.200	0.322	0.226	0.418	0.240	0.502	0.237	0.571
tA2D	0.156	0.258	0.215	0.313	0.249	0.380	0.270	0.443	0.274	0.499
OR02D4										
tA1D	0.156	0.313	0.223	0.395	0.267	0.502	0.294	0.601	0.300	0.679
tA2D	0.165	0.331	0.234	0.386	0.285	0.458	0.317	0.528	0.330	0.590
OR02D7										
tA1D	0.217	0.256	0.295	0.331	0.377	0.413	0.442	0.481	0.486	0.528
tA2D	0.240	0.271	0.319	0.319	0.408	0.369	0.481	0.412	0.534	0.443
OR02DA										
tA1D	0.266	0.319	0.344	0.393	0.436	0.483	0.513	0.565	0.565	0.620
tA2D	0.288	0.333	0.368	0.381	0.463	0.438	0.547	0.488	0.606	0.525

3-Input OR

OR03D0, OR03D1, OR03D2, OR03D4, OR03D7 and OR03DA

OR03D0, OR03D1, OR03D2, OR03D4, OR03D7 and OR03DA are 3-input OR gates with 0.5x, 1x, 2x, 4x, 7x and 10x drive capabilities.

Function Table



INPUTS			OUTPUT
A1	A2	A3	Z
H	X	X	H
X	H	X	H
X	X	H	H
L	L	L	L

Cell Description

Macro Name:	OR03D0	OR03D1	OR03D2	OR03D4	OR03D7	OR03DA
Drive Capability:	0.5x	1x	2x	4x	7x	10x
Gate Equivalents:	1.75	1.75	2.	2.25	3.	3.75
Leakage Power (pW):	32.4	33.6	44.0	59.9	91.3	113.1

Pin Description

Name	Capacitance (pF)						Description
	OR03D0	OR03D1	OR03D2	OR03D4	OR03D7	OR03DA	
A1	0.002	0.002	0.002	0.002	0.003	0.003	Input
A2	0.003	0.003	0.003	0.002	0.004	0.004	Input
A3	0.003	0.003	0.003	0.003	0.005	0.005	Input
Maximum capacitance							
Z	0.15	0.3	0.6	1.2	2.1	3	Output

Pin Powers for

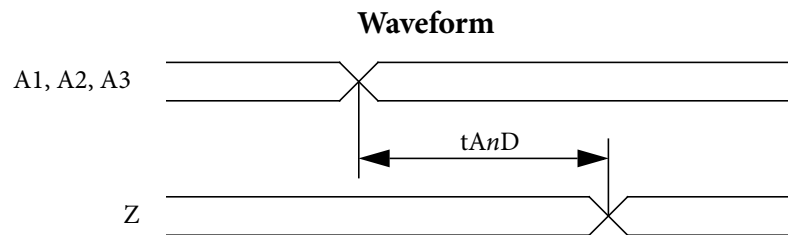
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)					
	OR03D0	OR03D1	OR03D2	OR03D4	OR03D7	OR03DA
A1	0.001	0.001	0.001	0.001	0.002	0.002
A2	0	0	0	0	0	0
A3	0	0	0	0	0	0
Z	0.021	0.026	0.041	0.093	0.176	0.307

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Timing Numbers for OR03D0, OR03D1, OR03D2, OR03D4, OR03D7 and OR03DA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
OR03D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.159	0.223	0.236	0.275	0.391	0.363	0.698	0.523	1.313	0.841
tA2D	0.168	0.260	0.245	0.312	0.400	0.400	0.707	0.560	1.322	0.878
tA3D	0.172	0.278	0.250	0.330	0.405	0.418	0.712	0.578	1.327	0.896
OR03D1										
tA1D	0.132	0.226	0.180	0.266	0.275	0.333	0.464	0.448	0.841	0.661
tA2D	0.140	0.263	0.188	0.303	0.283	0.370	0.473	0.485	0.850	0.698
tA3D	0.145	0.281	0.194	0.321	0.289	0.388	0.479	0.503	0.857	0.716
OR03D2										
tA1D	0.124	0.270	0.150	0.300	0.199	0.345	0.294	0.421	0.485	0.545
tA2D	0.133	0.306	0.159	0.337	0.208	0.382	0.303	0.458	0.494	0.582
tA3D	0.138	0.325	0.165	0.355	0.214	0.401	0.310	0.477	0.501	0.600
OR03D4										
tA1D	0.145	0.388	0.160	0.409	0.187	0.443	0.234	0.497	0.325	0.584
tA2D	0.154	0.424	0.169	0.445	0.196	0.479	0.243	0.533	0.334	0.620
tA3D	0.160	0.443	0.175	0.464	0.203	0.498	0.250	0.551	0.342	0.639

OR03D7										
tA1D	0.218	0.314	0.229	0.328	0.246	0.350	0.275	0.388	0.324	0.452
tA2D	0.243	0.345	0.255	0.359	0.272	0.381	0.302	0.419	0.351	0.483
tA3D	0.263	0.358	0.275	0.372	0.292	0.394	0.322	0.432	0.371	0.496
OR03DA										
tA1D	0.275	0.409	0.283	0.419	0.299	0.439	0.324	0.470	0.363	0.521
tA2D	0.301	0.439	0.309	0.450	0.325	0.469	0.350	0.500	0.389	0.551
tA3D	0.321	0.453	0.330	0.464	0.346	0.483	0.371	0.514	0.411	0.565

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
OR03D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.236	0.275	0.271	0.356	0.278	0.444	0.275	0.522	0.260	0.592
tA2D	0.245	0.312	0.289	0.364	0.306	0.431	0.311	0.499	0.304	0.561
tA3D	0.250	0.330	0.302	0.371	0.326	0.416	0.338	0.463	0.338	0.512
OR03D1										
tA1D	0.180	0.266	0.220	0.346	0.232	0.438	0.233	0.520	0.219	0.592
tA2D	0.188	0.303	0.237	0.355	0.258	0.423	0.267	0.492	0.261	0.555
tA3D	0.194	0.321	0.248	0.362	0.276	0.407	0.292	0.453	0.293	0.503
OR03D2										
tA1D	0.150	0.300	0.204	0.380	0.231	0.480	0.242	0.572	0.236	0.649
tA2D	0.159	0.337	0.219	0.389	0.252	0.462	0.271	0.534	0.271	0.601
tA3D	0.165	0.355	0.229	0.397	0.269	0.445	0.293	0.492	0.300	0.544
OR03D4										
tA1D	0.160	0.409	0.227	0.485	0.271	0.591	0.297	0.698	0.301	0.785
tA2D	0.169	0.445	0.239	0.497	0.289	0.573	0.321	0.651	0.330	0.723
tA3D	0.175	0.464	0.247	0.506	0.303	0.555	0.340	0.607	0.355	0.662
OR03D7										

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**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

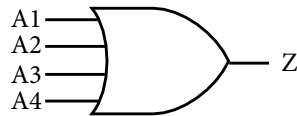
Input transition	1	5	10	15	20
tA1D	0.229 0.328	0.306 0.398	0.388 0.484	0.452 0.563	0.495 0.620
tA2D	0.255 0.359	0.333 0.406	0.421 0.464	0.493 0.516	0.543 0.558
tA3D	0.275 0.372	0.352 0.407	0.445 0.441	0.526 0.470	0.583 0.493
OR03DA					
tA1D	0.283 0.419	0.361 0.488	0.452 0.579	0.531 0.669	0.582 0.734
tA2D	0.309 0.450	0.386 0.498	0.481 0.558	0.567 0.617	0.624 0.665
tA3D	0.330 0.464	0.408 0.499	0.505 0.535	0.597 0.569	0.660 0.596

4-Input OR

OR04D0, OR04D1, OR04D2, OR04D4, OR04D7 and OR04DA

OR04D0, OR04D1, OR04D2, OR04D4, OR04D7 and OR04DA are 4-input OR gates with 0.5x, 1x, 2x, 4x, 7x and 10x drive capabilities.

Function Table



INPUTS				OUTPUT
A1	A2	A3	A4	Z
H	X	X	X	H
X	H	X	X	H
X	X	H	X	H
X	X	X	H	H
L	L	L	L	L

Cell Description

Macro Name:	OR04D0	OR04D1	OR04D2	OR04D4	OR04D7	OR04DA
Drive Capability:	0.5x	1x	2x	4x	7x	10x
Gate Equivalents:	2.	2.	2.25	2.5	3.5	4.
Leakage Power (pW):	45.4	47.2	57.6	73.7	101.5	123.3

Pin Description

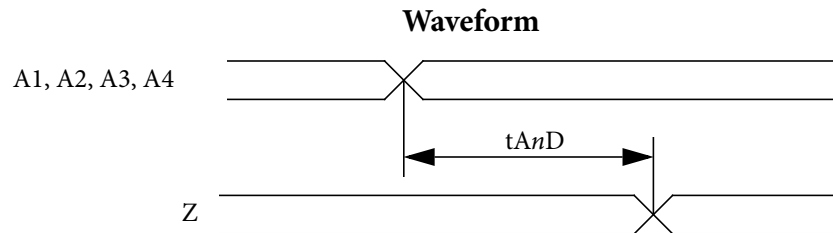
Name	Capacitance (pF)						Description
	OR04D0	OR04D1	OR04D2	OR04D4	OR04D7	OR04DA	
A1	0.003	0.003	0.003	0.003	0.003	0.003	Input
A2	0.003	0.003	0.003	0.003	0.004	0.004	Input
A3	0.003	0.003	0.003	0.003	0.004	0.004	Input
A4	0.004	0.004	0.004	0.004	0.005	0.005	Input
Maximum capacitance							
Z	0.15	0.3	0.6	1.2	2.1	3	Output

Pin Powers for

Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)					
	OR04D0	OR04D1	OR04D2	OR04D4	OR04D7	OR04DA
A1	0.003	0.003	0.003	0.003	0.005	0.005
A2	0	0	0	0	0	0
A3	0	0	0	0	0	0
A4	0	0	0	0	0	0
Z	0.024	0.032	0.047	0.095	0.203	0.362

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Timing Numbers for OR04D0, OR04D1, OR04D2, OR04D4, OR04D7 and OR04DA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
OR04D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.190	0.224	0.290	0.292	0.492	0.407	0.894	0.619	1.698	1.039
tA2D	0.204	0.278	0.305	0.345	0.507	0.460	0.909	0.672	1.713	1.090
tA3D	0.196	0.309	0.297	0.376	0.499	0.491	0.901	0.703	1.705	1.121
tA4D	0.201	0.326	0.304	0.394	0.506	0.509	0.910	0.721	1.715	1.141
OR04D1										
tA1D	0.141	0.215	0.188	0.257	0.283	0.327	0.472	0.446	0.849	0.662
tA2D	0.156	0.268	0.204	0.310	0.299	0.381	0.489	0.499	0.866	0.716
tA3D	0.146	0.299	0.195	0.341	0.290	0.412	0.480	0.530	0.858	0.747
tA4D	0.149	0.316	0.198	0.358	0.295	0.429	0.486	0.548	0.864	0.764
OR04D2										
tA1D	0.135	0.248	0.161	0.279	0.210	0.326	0.305	0.404	0.496	0.531
tA2D	0.149	0.301	0.175	0.332	0.224	0.379	0.320	0.458	0.511	0.585
tA3D	0.135	0.333	0.161	0.364	0.210	0.411	0.307	0.489	0.498	0.616
tA4D	0.138	0.350	0.165	0.381	0.215	0.428	0.312	0.506	0.504	0.633
OR04D4										
tA1D	0.154	0.344	0.169	0.365	0.197	0.399	0.244	0.452	0.335	0.540
tA2D	0.168	0.397	0.184	0.418	0.211	0.452	0.258	0.506	0.350	0.593
tA3D	0.149	0.429	0.164	0.450	0.190	0.484	0.237	0.537	0.328	0.624
tA4D	0.153	0.447	0.168	0.468	0.195	0.502	0.242	0.555	0.334	0.642

OR04D7									
tA1D	0.218	0.404	0.230	0.419	0.247	0.444	0.276	0.486	0.324 0.557
tA2D	0.248	0.459	0.259	0.474	0.276	0.499	0.306	0.542	0.355 0.613
tA3D	0.267	0.489	0.279	0.504	0.297	0.529	0.327	0.571	0.376 0.642
tA4D	0.282	0.503	0.294	0.518	0.312	0.543	0.342	0.586	0.392 0.657
OR04DA									
tA1D	0.275	0.535	0.284	0.547	0.299	0.567	0.324	0.602	0.363 0.659
tA2D	0.305	0.591	0.314	0.602	0.330	0.623	0.355	0.658	0.394 0.714
tA3D	0.326	0.620	0.335	0.631	0.351	0.652	0.376	0.687	0.416 0.743
tA4D	0.342	0.635	0.351	0.646	0.367	0.667	0.392	0.702	0.433 0.758

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
OR04D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.290	0.292	0.332	0.368	0.348	0.447	0.353	0.518	0.346	0.581
tA2D	0.305	0.345	0.357	0.392	0.384	0.454	0.398	0.516	0.399	0.573
tA3D	0.297	0.376	0.347	0.415	0.371	0.459	0.382	0.507	0.380	0.561
tA4D	0.304	0.394	0.358	0.424	0.388	0.452	0.406	0.482	0.411	0.521
OR04D1										
tA1D	0.188	0.257	0.238	0.334	0.261	0.419	0.271	0.494	0.267	0.559
tA2D	0.204	0.310	0.261	0.358	0.294	0.421	0.313	0.484	0.317	0.542
tA3D	0.195	0.341	0.249	0.381	0.277	0.424	0.293	0.471	0.293	0.525
tA4D	0.198	0.358	0.257	0.389	0.292	0.416	0.314	0.445	0.321	0.483
OR04D2										
tA1D	0.161	0.279	0.221	0.355	0.255	0.449	0.275	0.533	0.277	0.603
tA2D	0.175	0.332	0.241	0.381	0.284	0.448	0.312	0.514	0.321	0.576
tA3D	0.161	0.364	0.224	0.403	0.262	0.448	0.284	0.497	0.288	0.552
tA4D	0.165	0.381	0.232	0.411	0.276	0.440	0.304	0.469	0.316	0.508

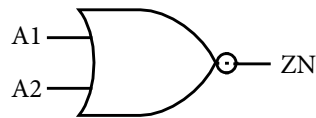
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
OR04D4										
tA1D	0.169	0.365	0.239	0.438	0.291	0.537	0.323	0.634	0.334	0.712
tA2D	0.184	0.418	0.256	0.467	0.315	0.537	0.355	0.609	0.371	0.676
tA3D	0.164	0.450	0.234	0.490	0.285	0.537	0.317	0.589	0.329	0.645
tA4D	0.168	0.468	0.241	0.497	0.298	0.527	0.336	0.558	0.353	0.598
OR04D7										
tA1D	0.230	0.419	0.306	0.489	0.386	0.583	0.446	0.676	0.483	0.749
tA2D	0.259	0.474	0.336	0.519	0.424	0.576	0.495	0.633	0.541	0.684
tA3D	0.279	0.504	0.356	0.542	0.447	0.579	0.526	0.615	0.578	0.649
tA4D	0.294	0.518	0.371	0.551	0.466	0.575	0.550	0.596	0.608	0.614
OR04DA										
tA1D	0.284	0.547	0.361	0.615	0.451	0.709	0.527	0.805	0.573	0.888
tA2D	0.314	0.602	0.391	0.646	0.485	0.704	0.569	0.765	0.624	0.821
tA3D	0.335	0.631	0.411	0.669	0.507	0.707	0.597	0.747	0.657	0.784
tA4D	0.351	0.646	0.427	0.678	0.526	0.703	0.620	0.727	0.685	0.747

2-Input NOR

NR02D0, NR02D1, NR02D2, NR02D4, NR02D7 and NR02DA

NR02D0, NR02D1, NR02D2, NR02D4, NR02D7 and NR02DA are 2-input NOR gates with 0.5x, 1x, 2x, 4x, 7x and 10x drive capabilities.



Function Table

INPUTS		OUTPUT
A1	A2	ZN
H	X	L
X	H	L
L	L	H

Cell Description

Macro Name:	NR02D0	NR02D1	NR02D2	NR02D4	NR02D7	NR02DA
Drive Capability:	0.5x	1x	2x	4x	7x	10x
Gate Equivalents:	1.	1.	1.5	2.5	3.25	4.
Leakage Power (pW):	17.1	29.9	61.5	115.4	266.5	370.5

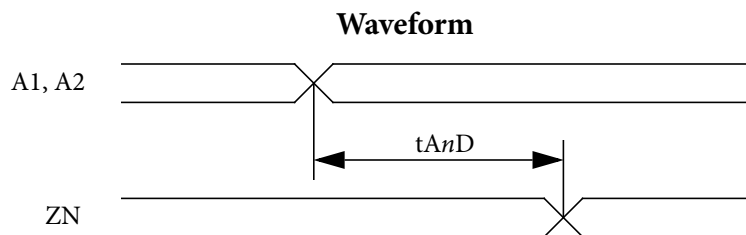
Pin Description

Name	Capacitance (pF)						Description
	NR02D0	NR02D1	NR02D2	NR02D4	NR02D7	NR02DA	
A1	0.002	0.004	0.007	0.002	0.005	0.005	Input
A2	0.002	0.004	0.009	0.002	0.004	0.004	Input
Maximum capacitance							
ZN	0.15	0.3	0.6	1.2	2.1	3	Output

Pin Powers for Standard Load = 0.032 pF Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)					
	NR02D0	NR02D1	NR02D2	NR02D4	NR02D7	NR02DA
A1	0.001	0	0.003	0	0	0
A2	0	0.002	0.001	0.001	0.002	0.002
ZN	0.006	0.011	0.02	0.082	0.146	0.209

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Timing Numbers for NR02D0, NR02D1, NR02D2, NR02D4, NR02D7 and NR02DA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

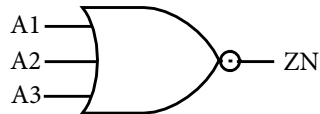
Fanout load	4		8		16		32		64	
NR02D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.230	0.088	0.403	0.142	0.745	0.248	1.428	0.462	2.794	0.891
tA2D	0.249	0.095	0.421	0.149	0.763	0.256	1.447	0.470	2.814	0.899
NR02D1										
tA1D	0.153	0.070	0.239	0.100	0.409	0.159	0.750	0.276	1.430	0.509
tA2D	0.136	0.063	0.222	0.092	0.392	0.151	0.733	0.268	1.413	0.501
NR02D2										
tA1D	0.093	0.040	0.138	0.053	0.226	0.077	0.403	0.123	0.753	0.215
tA2D	0.106	0.044	0.151	0.058	0.239	0.082	0.415	0.129	0.765	0.222
NR02D4										
tA1D	0.269	0.176	0.283	0.187	0.308	0.206	0.352	0.235	0.441	0.286
tA2D	0.250	0.169	0.264	0.180	0.289	0.199	0.333	0.228	0.422	0.279
NR02D7										
tA1D	0.215	0.237	0.224	0.246	0.238	0.262	0.261	0.290	0.301	0.338
tA2D	0.200	0.211	0.209	0.220	0.222	0.236	0.245	0.263	0.286	0.311
NR02DA										
tA1D	0.241	0.265	0.247	0.273	0.259	0.287	0.278	0.309	0.310	0.345
tA2D	0.225	0.239	0.231	0.247	0.243	0.260	0.263	0.282	0.295	0.319

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
NR02D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.403	0.142	0.482	0.213	0.589	0.269	0.696	0.305	0.786	0.317
tA2D	0.421	0.149	0.475	0.222	0.548	0.282	0.623	0.323	0.689	0.340
NR02D1										
tA1D	0.239	0.100	0.289	0.162	0.350	0.202	0.408	0.227	0.459	0.236
tA2D	0.222	0.092	0.300	0.150	0.393	0.183	0.477	0.201	0.544	0.202
NR02D2										
tA1D	0.138	0.053	0.219	0.083	0.303	0.085	0.376	0.078	0.441	0.058
tA2D	0.151	0.058	0.200	0.096	0.258	0.108	0.314	0.109	0.365	0.099
NR02D4										
tA1D	0.283	0.187	0.338	0.228	0.405	0.242	0.470	0.247	0.529	0.239
tA2D	0.264	0.180	0.349	0.213	0.439	0.219	0.518	0.215	0.589	0.199
NR02D7										
tA1D	0.224	0.246	0.261	0.317	0.290	0.380	0.313	0.430	0.326	0.470
tA2D	0.209	0.220	0.263	0.281	0.305	0.329	0.340	0.368	0.364	0.397
NR02DA										
tA1D	0.247	0.273	0.285	0.343	0.314	0.406	0.338	0.457	0.352	0.496
tA2D	0.231	0.247	0.287	0.307	0.329	0.355	0.364	0.394	0.388	0.423

3-Input NOR**NR03D0, NR03D1, NR03D2, NR03D4, NR03D7 and NR03DA**

NR03D0, NR03D1, NR03D2, NR03D4, NR03D7 and NR03DA are 3-input NOR gates with 0.5x, 1x, 2x, 4x, 7x and 10x drive capabilities.

Function Table

INPUTS			OUTPUT
A1	A2	A3	ZN
H	X	X	L
X	H	X	L
X	X	H	L
L	L	L	H

Cell Description

Macro Name:	NR03D0	NR03D1	NR03D2	NR03D4	NR03D7	NR03DA
Drive Capability:	0.5x	1x	2x	4x	7x	10x
Gate Equivalents:	1.25	1.25	2.5	2.75	3.5	4.25
Leakage Power (pW):	12.8	19.2	98.0	142.4	274.5	370.7

Pin Description

Name	Capacitance (pF)						Description
	NR03D0	NR03D1	NR03D2	NR03D4	NR03D7	NR03DA	
A1	0.002	0.004	0.004	0.004	0.003	0.003	Input
A2	0.002	0.004	0.004	0.004	0.004	0.004	Input
A3	0.002	0.004	0.005	0.005	0.005	0.005	Input
Maximum capacitance							
ZN	0.15	0.3	0.6	1.2	2.1	3	Output

Pin Powers for

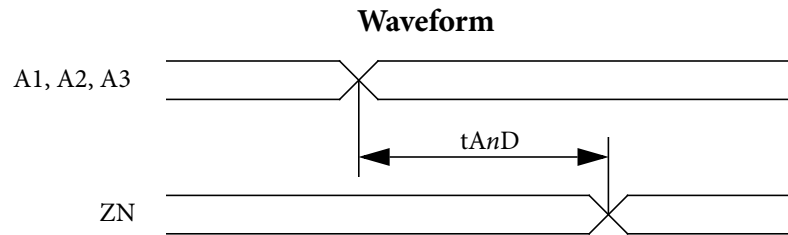
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)					
	NR03D0	NR03D1	NR03D2	NR03D4	NR03D7	NR03DA
A1	0	0	0.001	0.001	0.002	0.002
A2	0	0	0	0	0	0
A3	0.001	0.001	0	0	0	0
ZN	0.007	0.013	0.06	0.088	0.15	0.22

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Timing Numbers for NR03D0, NR03D1, NR03D2, NR03D4, NR03D7 and NR03DA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
NR03D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.432	0.095	0.718	0.147	1.286	0.247	2.419	0.447	4.686	0.844
tA2D	0.414	0.092	0.699	0.143	1.267	0.242	2.400	0.441	4.667	0.838
tA3D	0.377	0.086	0.663	0.136	1.231	0.236	2.365	0.434	4.633	0.830
NR03D1										
tA1D	0.290	0.073	0.454	0.106	0.780	0.169	1.430	0.292	2.728	0.537
tA2D	0.279	0.058	0.443	0.082	0.770	0.127	1.420	0.219	2.719	0.401
tA3D	0.248	0.053	0.413	0.075	0.740	0.118	1.391	0.204	2.690	0.376
NR03D2										
tA1D	0.214	0.137	0.239	0.155	0.288	0.184	0.383	0.239	0.574	0.345
tA2D	0.246	0.145	0.272	0.163	0.320	0.192	0.415	0.247	0.606	0.352
tA3D	0.260	0.147	0.285	0.165	0.333	0.193	0.429	0.248	0.619	0.354
NR03D4										
tA1D	0.227	0.159	0.241	0.171	0.266	0.192	0.311	0.223	0.402	0.280
tA2D	0.260	0.167	0.273	0.179	0.298	0.199	0.343	0.231	0.434	0.288
tA3D	0.273	0.169	0.287	0.181	0.312	0.201	0.357	0.233	0.448	0.290

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NR03D7										
tA1D	0.236	0.217	0.245	0.226	0.259	0.242	0.282	0.270	0.323	0.318
tA2D	0.269	0.243	0.278	0.253	0.292	0.269	0.315	0.296	0.356	0.344
tA3D	0.282	0.260	0.292	0.270	0.306	0.286	0.329	0.314	0.370	0.362
NR03DA										
tA1D	0.269	0.251	0.276	0.258	0.288	0.273	0.307	0.295	0.340	0.332
tA2D	0.302	0.277	0.309	0.285	0.321	0.299	0.340	0.321	0.372	0.359
tA3D	0.315	0.295	0.322	0.302	0.334	0.317	0.354	0.339	0.386	0.376

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

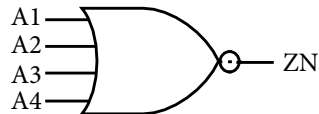
Input transition	1		5		10		15		20	
NR03D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.718	0.147	0.757	0.219	0.807	0.278	0.862	0.316	0.925	0.328
tA2D	0.699	0.143	0.750	0.214	0.824	0.270	0.905	0.304	0.989	0.313
tA3D	0.663	0.136	0.735	0.206	0.839	0.258	0.947	0.289	1.054	0.295
NR03D1										
tA1D	0.454	0.106	0.488	0.170	0.532	0.211	0.581	0.235	0.635	0.240
tA2D	0.443	0.082	0.495	0.134	0.574	0.155	0.658	0.161	0.738	0.147
tA3D	0.413	0.075	0.490	0.122	0.598	0.136	0.708	0.135	0.805	0.116
NR03D2										
tA1D	0.239	0.155	0.320	0.176	0.405	0.170	0.483	0.156	0.554	0.130
tA2D	0.272	0.163	0.322	0.194	0.389	0.200	0.455	0.195	0.517	0.178
tA3D	0.285	0.165	0.320	0.202	0.362	0.213	0.411	0.215	0.459	0.205
NR03D4										
tA1D	0.241	0.171	0.322	0.191	0.408	0.186	0.486	0.172	0.557	0.146
tA2D	0.273	0.179	0.324	0.210	0.391	0.215	0.459	0.210	0.521	0.193
tA3D	0.287	0.181	0.322	0.218	0.366	0.229	0.415	0.231	0.465	0.221

NR03D7										
tA1D	0.245	0.226	0.305	0.286	0.358	0.329	0.403	0.362	0.439	0.383
tA2D	0.278	0.253	0.319	0.322	0.359	0.379	0.394	0.424	0.422	0.456
tA3D	0.292	0.270	0.323	0.344	0.345	0.413	0.363	0.468	0.376	0.509
NR03DA										
tA1D	0.276	0.258	0.336	0.317	0.389	0.361	0.434	0.393	0.470	0.415
tA2D	0.309	0.285	0.350	0.353	0.390	0.410	0.426	0.455	0.456	0.487
tA3D	0.322	0.302	0.354	0.376	0.376	0.444	0.396	0.499	0.410	0.541

4-Input NOR

NR04D0, NR04D1, NR04D2, NR04D4, NR04D7 and NR04DA

NR04D0, NR04D1, NR04D2, NR04D4, NR04D7 and NR04DA are 4-input NOR gates with 0.5x, 1x, 2x, 4x, 7x and 10x drive capabilities



Function Table

INPUTS				OUTPUT
A1	A2	A3	A4	ZN
H	X	X	X	L
X	H	X	X	L
X	X	H	X	L
X	X	X	H	L
L	L	L	L	H

Cell Description

Macro Name:	NR04D0	NR04D1	NR04D2	NR04D4	NR04D7	NR04DA
Drive Capability:	0.5x	1x	2x	4x	7x	10x
Gate Equivalents:	1.5	1.5	2.75	3.	3.75	4.25
Leakage Power (pW):	13.6	24.4	96.6	141.1	282.7	382.2

Pin Description

Name	Capacitance (pF)						Description
	NR04D0	NR04D1	NR04D2	NR04D4	NR04D7	NR04DA	
A1	0.003	0.004	0.003	0.003	0.003	0.003	Input
A2	0.002	0.004	0.004	0.004	0.004	0.004	Input
A3	0.002	0.003	0.004	0.004	0.004	0.004	Input
A4	0.002	0.004	0.004	0.004	0.004	0.004	Input
Maximum capacitance							
ZN	0.15	0.3	0.6	1.2	2.1	3	Output

Pin Powers for

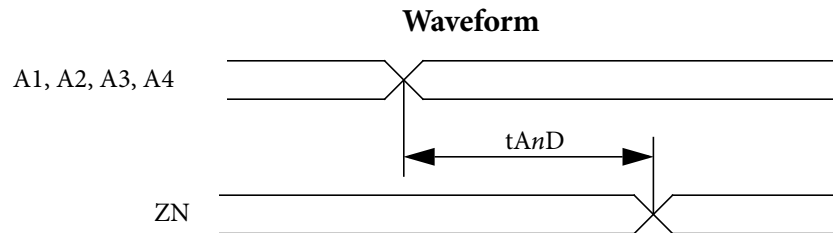
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)					
	NR04D0	NR04D1	NR04D2	NR04D4	NR04D7	NR04DA
A1	0	0.001	0.004	0.004	0.005	0.005
A2	0	0	0.001	0.001	0	0.001
A3	0	0.001	0	0	0	0
A4	0.001	0.001	0	0	0	0
ZN	0.009	0.017	0.062	0.092	0.158	0.225

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Timing Numbers for NR04D0, NR04D1, NR04D2, NR04D4, NR04D7 and NR04DA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004**

Fanout load	4		8		16		32		64	
NR04D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.566	0.105	0.919	0.162	1.621	0.272	3.022	0.489	5.822	0.921
tA2D	0.547	0.104	0.899	0.159	1.600	0.268	3.001	0.482	5.801	0.911
tA3D	0.510	0.100	0.862	0.154	1.563	0.261	2.964	0.475	5.764	0.901
tA4D	0.451	0.092	0.804	0.146	1.505	0.253	2.906	0.467	5.709	0.893
NR04D1										
tA1D	0.383	0.070	0.577	0.100	0.963	0.157	1.732	0.269	3.267	0.487
tA2D	0.368	0.071	0.562	0.099	0.948	0.155	1.716	0.264	3.251	0.482
tA3D	0.333	0.072	0.527	0.102	0.913	0.161	1.681	0.278	3.216	0.511
tA4D	0.282	0.066	0.476	0.095	0.862	0.154	1.631	0.270	3.166	0.504
NR04D2										
tA1D	0.263	0.147	0.288	0.165	0.336	0.193	0.431	0.248	0.622	0.354
tA2D	0.313	0.154	0.339	0.172	0.387	0.201	0.482	0.256	0.673	0.361
tA3D	0.348	0.155	0.374	0.173	0.422	0.201	0.517	0.256	0.708	0.362
tA4D	0.363	0.155	0.388	0.173	0.436	0.202	0.532	0.257	0.723	0.363
NR04D4										
tA1D	0.283	0.169	0.297	0.181	0.322	0.201	0.368	0.233	0.458	0.290
tA2D	0.334	0.177	0.348	0.189	0.373	0.209	0.419	0.241	0.509	0.298
tA3D	0.369	0.177	0.383	0.189	0.408	0.210	0.453	0.242	0.544	0.299
tA4D	0.383	0.179	0.397	0.191	0.423	0.211	0.468	0.243	0.559	0.300

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NR04D7										
tA1D	0.260	0.235	0.269	0.245	0.283	0.261	0.307	0.290	0.348	0.338
tA2D	0.314	0.263	0.324	0.273	0.338	0.289	0.361	0.318	0.403	0.366
tA3D	0.343	0.281	0.352	0.291	0.366	0.307	0.390	0.336	0.431	0.384
tA4D	0.357	0.294	0.367	0.304	0.381	0.320	0.404	0.349	0.445	0.397
NR04DA										
tA1D	0.307	0.238	0.314	0.246	0.327	0.259	0.348	0.281	0.381	0.317
tA2D	0.359	0.264	0.366	0.272	0.378	0.285	0.399	0.307	0.432	0.343
tA3D	0.392	0.284	0.399	0.292	0.411	0.305	0.432	0.327	0.465	0.363
tA4D	0.406	0.296	0.413	0.304	0.426	0.317	0.447	0.339	0.479	0.375

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
NR04D0	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.919	0.162	0.955	0.237	0.990	0.303	1.028	0.347	1.078	0.363
tA2D	0.899	0.159	0.940	0.232	0.988	0.296	1.043	0.338	1.110	0.351
tA3D	0.862	0.154	0.909	0.227	0.981	0.287	1.061	0.325	1.149	0.336
tA4D	0.804	0.146	0.872	0.218	0.972	0.274	1.076	0.309	1.184	0.316
NR04D1										
tA1D	0.577	0.100	0.609	0.164	0.642	0.202	0.679	0.223	0.729	0.223
tA2D	0.562	0.099	0.599	0.162	0.645	0.197	0.700	0.215	0.763	0.212
tA3D	0.527	0.102	0.574	0.163	0.648	0.198	0.729	0.216	0.809	0.212
tA4D	0.476	0.095	0.548	0.154	0.651	0.184	0.756	0.197	0.852	0.188
NR04D2										
tA1D	0.288	0.165	0.372	0.195	0.468	0.197	0.554	0.187	0.631	0.163
tA2D	0.339	0.172	0.389	0.211	0.461	0.221	0.535	0.218	0.605	0.200
tA3D	0.374	0.173	0.412	0.214	0.457	0.227	0.510	0.228	0.570	0.214
tA4D	0.388	0.173	0.422	0.219	0.454	0.236	0.492	0.241	0.542	0.231

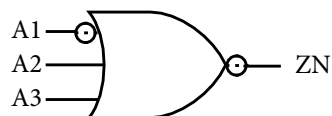
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1	5	10	15	20
NR04D4					
tA1D	0.297 0.181	0.380 0.212	0.476 0.213	0.564 0.204	0.642 0.181
tA2D	0.348 0.189	0.398 0.227	0.471 0.237	0.546 0.235	0.617 0.218
tA3D	0.383 0.189	0.421 0.230	0.467 0.244	0.521 0.245	0.582 0.232
tA4D	0.397 0.191	0.431 0.235	0.464 0.253	0.502 0.258	0.553 0.249
NR04D7					
tA1D	0.269 0.245	0.337 0.307	0.401 0.350	0.457 0.380	0.503 0.398
tA2D	0.324 0.273	0.368 0.343	0.416 0.399	0.461 0.441	0.501 0.468
tA3D	0.352 0.291	0.388 0.364	0.418 0.431	0.447 0.483	0.474 0.518
tA4D	0.367 0.304	0.397 0.380	0.415 0.455	0.430 0.514	0.442 0.557
NR04DA					
tA1D	0.314 0.246	0.381 0.307	0.446 0.348	0.502 0.379	0.550 0.395
tA2D	0.366 0.272	0.409 0.341	0.458 0.396	0.504 0.438	0.545 0.464
tA3D	0.399 0.292	0.434 0.364	0.465 0.431	0.493 0.483	0.522 0.517
tA4D	0.413 0.304	0.443 0.380	0.462 0.455	0.477 0.514	0.490 0.556

3-Input NOR with 1 Inverted Input

NR13D1, NR13D2 and NR13D4

NR13D1, NR13D2 and NR13D4 are 3-input NOR gates with one inverted input and with 1x, 2x, and 4x drive capabilities.



Function Table

INPUTS			OUTPUT
A1	A2	A3	ZN
L	X	X	L
X	H	X	L
X	X	H	L
H	L	L	H

Cell Description

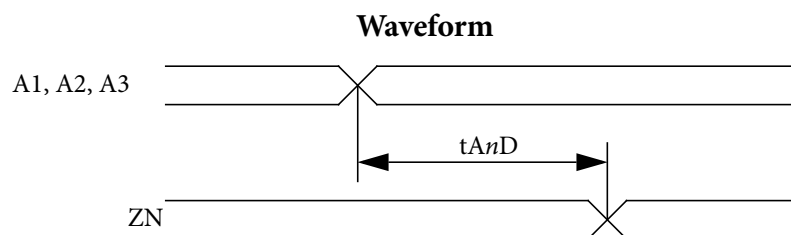
Macro Name:	NR13D1	NR13D2	NR13D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.75	2.75	3.
Leakage Power (pW):	20.3	72.8	120.6

Pin Description

Name	Capacitance (pF)			Description
	NR13D1	NR13D2	NR13D4	
A1	0.002	0.004	0.004	Input
A2	0.004	0.003	0.003	Input
A3	0.005	0.002	0.002	Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	NR13D1	NR13D2	NR13D4
A1	0.014	-0	-0
A2	0.001	0.013	0.013
A3	0.001	0.011	0.011
ZN	0.019	0.047	0.078



Timing Numbers for NR13D1, NR13D2 and NR13D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
NR13D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.265	0.139	0.371	0.162	0.583	0.203	1.006	0.281	1.850	0.435
tA2D	0.207	0.056	0.313	0.076	0.526	0.115	0.949	0.191	1.793	0.343
tA3D	0.176	0.052	0.282	0.071	0.496	0.109	0.920	0.185	1.764	0.337
NR13D2										
tA1D	0.128	0.147	0.155	0.168	0.204	0.199	0.301	0.255	0.492	0.361
tA2D	0.199	0.193	0.227	0.215	0.276	0.246	0.373	0.303	0.564	0.408
tA3D	0.220	0.171	0.247	0.190	0.296	0.221	0.393	0.276	0.584	0.381
NR13D4										
tA1D	0.138	0.172	0.154	0.186	0.182	0.209	0.230	0.244	0.323	0.304
tA2D	0.210	0.215	0.225	0.229	0.254	0.252	0.302	0.288	0.394	0.348
tA3D	0.231	0.188	0.247	0.202	0.275	0.224	0.323	0.258	0.415	0.317

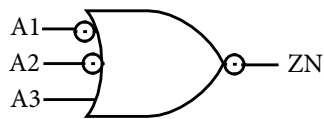
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
NR13D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.371	0.162	0.407	0.243	0.419	0.316	0.422	0.380	0.415	0.433
tA2D	0.313	0.076	0.365	0.125	0.439	0.144	0.515	0.149	0.586	0.137
tA3D	0.282	0.071	0.360	0.115	0.463	0.128	0.561	0.127	0.644	0.110
NR13D2										
tA1D	0.155	0.168	0.178	0.255	0.185	0.341	0.183	0.417	0.170	0.482
tA2D	0.227	0.215	0.298	0.247	0.362	0.258	0.418	0.262	0.466	0.256
tA3D	0.247	0.190	0.336	0.223	0.424	0.231	0.502	0.230	0.569	0.216
NR13D4										
tA1D	0.154	0.186	0.183	0.276	0.198	0.372	0.204	0.455	0.196	0.524
tA2D	0.225	0.229	0.297	0.262	0.360	0.273	0.416	0.276	0.465	0.270
tA3D	0.247	0.202	0.338	0.227	0.430	0.227	0.510	0.218	0.582	0.198

3-Input NOR with 2 Inverted Inputs

NR23D1, NR23D2 and NR23D4

NR23D1, NR23D2 and NR23D4 are 3-input NOR gates with two inverted inputs and with 1x, 2x and 4x drive capabilities.



Function Table

INPUTS			OUTPUT
A1	A2	A3	ZN
L	X	X	L
X	L	X	L
X	X	H	L
H	H	L	H

Cell Description

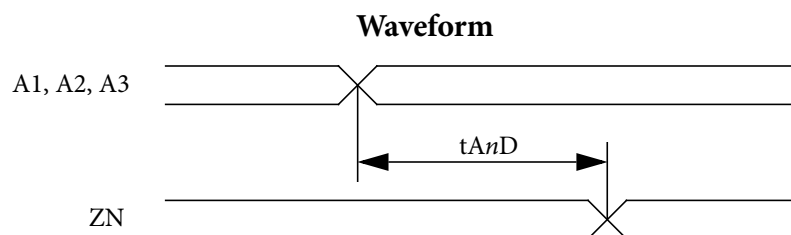
Macro Name:	NR23D1	NR23D2	NR23D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.	2.5	2.75
Leakage Power (pW):	26.0	59.4	112.0

Pin Description

Name	Capacitance (pF)			Description
	NR23D1	NR23D2	NR23D4	
A1	0.002	0.005	0.005	Input
A2	0.002	0.004	0.004	Input
A3	0.005	0.002	0.002	Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	NR23D1	NR23D2	NR23D4
A1	0.012	-0	-0
A2	0.015	0.001	0.001
A3	0.001	0.016	0.016
ZN	0.017	0.045	0.076



Timing Numbers for NR23D1, NR23D2 and NR23D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
NR23D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.257	0.181	0.365	0.209	0.582	0.262	1.012	0.362	1.872	0.557
tA2D	0.211	0.168	0.319	0.196	0.536	0.247	0.967	0.343	1.827	0.533
tA3D	0.204	0.067	0.312	0.093	0.529	0.145	0.960	0.247	1.820	0.451
NR23D2										
tA1D	0.132	0.160	0.159	0.181	0.208	0.213	0.304	0.270	0.493	0.375
tA2D	0.128	0.147	0.155	0.168	0.204	0.199	0.300	0.254	0.489	0.360
tA3D	0.213	0.170	0.240	0.190	0.289	0.221	0.385	0.276	0.574	0.381
NR23D4										
tA1D	0.142	0.183	0.157	0.197	0.186	0.221	0.234	0.257	0.327	0.317
tA2D	0.138	0.171	0.153	0.185	0.181	0.207	0.230	0.243	0.323	0.302
tA3D	0.223	0.194	0.238	0.208	0.267	0.230	0.315	0.264	0.408	0.323

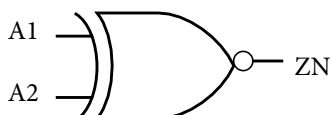
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
NR23D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.365	0.209	0.395	0.301	0.397	0.393	0.389	0.473	0.369	0.542
tA2D	0.319	0.196	0.346	0.287	0.348	0.379	0.342	0.460	0.325	0.529
tA3D	0.312	0.093	0.362	0.153	0.431	0.186	0.501	0.204	0.565	0.204
NR23D2										
tA1D	0.159	0.181	0.171	0.271	0.169	0.364	0.159	0.445	0.139	0.515
tA2D	0.155	0.168	0.178	0.254	0.185	0.341	0.183	0.417	0.170	0.482
tA3D	0.240	0.190	0.327	0.222	0.412	0.232	0.486	0.231	0.551	0.218
NR23D4										
tA1D	0.157	0.197	0.173	0.288	0.176	0.389	0.173	0.476	0.157	0.549
tA2D	0.153	0.185	0.183	0.274	0.198	0.370	0.204	0.452	0.196	0.521
tA3D	0.238	0.208	0.326	0.240	0.411	0.249	0.486	0.248	0.551	0.235

2-Input Exclusive NOR

XN02D1, XN02D2, XN02D4, XN02D7 and XN02DA

XN02D1, XN02D2, XN02D4, XN02D7 and XN02DA are 2-input XNOR gates with 1x, 2x, 4x, 7x and 10x drive capabilities.



Function Table

INPUTS		OUTPUT
A1	A2	ZN
H	L	L
L	H	L
L	L	H
H	H	H

Cell Description

Macro Name:	XN02D1	XN02D2	XN02D4	XN02D7	XN02DA
Drive Capability:	1x	2x	4x	7x	10x
Gate Equivalents:	2.75	3.	3.25	4.5	5.25
Leakage Power (pW):	79.4	101.1	147.8	334.3	436.0

Pin Description

Name	Capacitance (pF)					Description
	XN02D1	XN02D2	XN02D4	XN02D7	XN02DA	
A1	0.008	0.008	0.008	0.007	0.007	Input
A2	0.007	0.007	0.007	0.008	0.008	Input
Maximum capacitance						
ZN	0.3	0.6	1.2	2.1	3	Output

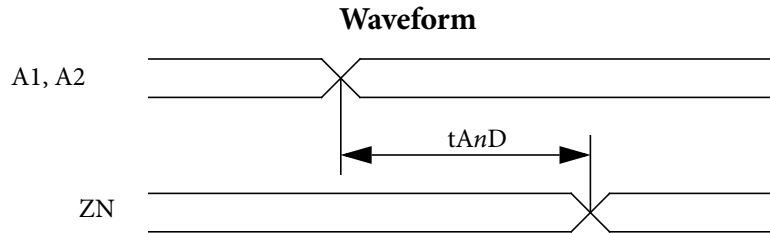
Pin Powers for Standard Load = 0.032 pF Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)				
	XN02D1	XN02D2	XN02D4	XN02D7	XN02DA
ZN	0.046	0.059	0.092	0.175	0.229

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Timing Numbers for XN02D1, XN02D2, XN02D4, XN02D7 and XN02DA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
XN02D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.237	0.182	0.286	0.215	0.382	0.273	0.573	0.379	0.954	0.587
tA2D	0.244	0.255	0.293	0.289	0.389	0.348	0.580	0.456	0.962	0.664
XN02D2										
tA1D	0.225	0.189	0.253	0.212	0.302	0.247	0.398	0.307	0.589	0.415
tA2D	0.233	0.259	0.260	0.284	0.309	0.321	0.406	0.384	0.597	0.493
XN02D4										
tA1D	0.237	0.227	0.252	0.243	0.280	0.269	0.329	0.309	0.422	0.376
tA2D	0.244	0.295	0.260	0.312	0.288	0.341	0.337	0.383	0.430	0.453
XN02D7										
tA1D	0.213	0.200	0.220	0.209	0.232	0.223	0.254	0.247	0.292	0.291
tA2D	0.238	0.224	0.246	0.233	0.258	0.247	0.279	0.271	0.317	0.315
XN02DA										
tA1D	0.235	0.220	0.241	0.226	0.252	0.238	0.269	0.257	0.298	0.290
tA2D	0.260	0.243	0.266	0.249	0.277	0.261	0.294	0.280	0.323	0.313

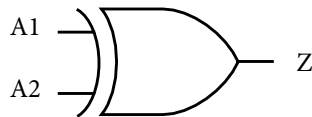
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
XN02D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.286	0.215	0.373	0.230	0.460	0.224	0.537	0.209	0.606	0.184
tA2D	0.293	0.289	0.379	0.315	0.462	0.316	0.536	0.307	0.600	0.291
XN02D2										
tA1D	0.253	0.212	0.342	0.226	0.428	0.220	0.506	0.205	0.575	0.180
tA2D	0.260	0.284	0.347	0.311	0.431	0.312	0.505	0.304	0.569	0.288
XN02D4										
tA1D	0.252	0.243	0.341	0.257	0.430	0.251	0.508	0.236	0.578	0.210
tA2D	0.260	0.312	0.347	0.341	0.431	0.344	0.504	0.336	0.569	0.320
XN02D7										
tA1D	0.220	0.209	0.290	0.270	0.355	0.318	0.409	0.355	0.451	0.378
tA2D	0.246	0.233	0.294	0.275	0.342	0.316	0.385	0.349	0.420	0.371
XN02DA										
tA1D	0.241	0.226	0.311	0.286	0.376	0.335	0.430	0.372	0.473	0.396
tA2D	0.266	0.249	0.314	0.292	0.363	0.333	0.406	0.367	0.442	0.389

2-Input Exclusive OR

XR02D1, XR02D2, XR02D4, XR02D7 and XR02DA

XR02D1, XR02D2, XR02D4, XR02D7 and XR02DA are 2-input XOR gates with 1x, 2x, 4x, 7x and 10x drive capabilities.



Function Table

INPUTS		OUTPUT
A1	A2	Z
H	L	H
L	H	H
L	L	L
H	H	L

Cell Description

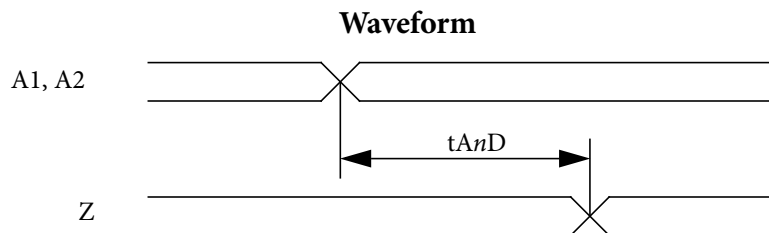
Macro Name:	XR02D1	XR02D2	XR02D4	XR02D7	XR02DA
Drive Capability:	1x	2x	4x	7x	10x
Gate Equivalents:	2.5	2.75	3.	4.5	5.25
Leakage Power (pW):	60.2	70.7	87.1	157.0	184.2

Pin Description

Name	Capacitance (pF)					Description
	XR02D1	XR02D2	XR02D4	XR02D7	XR02DA	
A1	0.005	0.005	0.005	0.007	0.007	Input
A2	0.004	0.004	0.004	0.007	0.007	Input
Maximum capacitance						
Z	0.3	0.6	1.2	2.1	3	Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)				
	XR02D1	XR02D2	XR02D4	XR02D7	XR02DA
Z	0.036	0.05	0.093	0.166	0.221



Timing Numbers for XR02D1, XR02D2, XR02D4, XR02D7 and XR02DA:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
XR02D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.144	0.193	0.193	0.229	0.290	0.292	0.481	0.402	0.862	0.617
tA2D	0.170	0.220	0.219	0.256	0.316	0.319	0.507	0.430	0.889	0.644
XR02D2										
tA1D	0.140	0.218	0.168	0.245	0.218	0.285	0.315	0.354	0.506	0.469
tA2D	0.169	0.244	0.198	0.272	0.249	0.312	0.347	0.380	0.538	0.496
XR02D4										
tA1D	0.164	0.293	0.181	0.312	0.212	0.343	0.263	0.391	0.359	0.470
tA2D	0.201	0.319	0.220	0.338	0.251	0.369	0.304	0.417	0.402	0.496
XR02D7										
tA1D	0.302	0.301	0.311	0.311	0.324	0.325	0.347	0.351	0.387	0.396
tA2D	0.321	0.311	0.330	0.321	0.344	0.335	0.367	0.360	0.407	0.405
XR02DA										
tA1D	0.315	0.316	0.321	0.323	0.333	0.335	0.350	0.354	0.379	0.388
tA2D	0.334	0.326	0.340	0.333	0.352	0.345	0.370	0.365	0.399	0.399

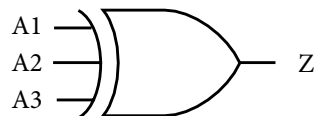
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
XR02D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.193	0.229	0.233	0.311	0.249	0.398	0.254	0.475	0.246	0.542
tA2D	0.219	0.256	0.256	0.312	0.280	0.376	0.294	0.436	0.296	0.490
XR02D2										
tA1D	0.168	0.245	0.218	0.328	0.246	0.424	0.260	0.508	0.259	0.579
tA2D	0.198	0.272	0.240	0.327	0.273	0.394	0.293	0.457	0.300	0.513
XR02D4										
tA1D	0.181	0.312	0.242	0.393	0.286	0.499	0.313	0.597	0.320	0.676
tA2D	0.220	0.338	0.265	0.394	0.309	0.465	0.340	0.533	0.354	0.594
XR02D7										
tA1D	0.311	0.311	0.349	0.375	0.372	0.428	0.388	0.474	0.394	0.511
tA2D	0.330	0.321	0.371	0.379	0.395	0.427	0.411	0.469	0.419	0.502
XR02DA										
tA1D	0.321	0.323	0.360	0.387	0.383	0.440	0.398	0.487	0.405	0.524
tA2D	0.340	0.333	0.380	0.393	0.402	0.443	0.417	0.486	0.423	0.521

3-Input Exclusive OR

XR03D1, XR03D2, XR03D4, XR03D7 and XR03DA

XR03D1, XR03D2, XR03D4, XR03D7 and XR03DA are 3-input XOR gates with 1x, 2x, 4x, 7x and 10x drive capabilities.



Function Table

INPUTS			OUTPUT
A1	A2	A3	Z
H	L	L	H
L	H	L	H
L	L	H	H
L	L	L	L
H	H	L	L
H	L	H	L
L	H	H	L
H	H	H	H

Cell Description

Macro Name:	XR03D1	XR03D2	XR03D4	XR03D7	XR03DA
Drive Capability:	1x	2x	4x	7x	10x
Gate Equivalents:	4.25	4.5	5.	6.	6.75
Leakage Power (pW):	166.2	176.8	192.9	270.3	296.0

Pin Description

Name	Capacitance (pF)					Description
	XR03D1	XR03D2	XR03D4	XR03D7	XR03DA	
A1	0.004	0.004	0.004	0.004	0.004	Input
A2	0.006	0.006	0.006	0.007	0.007	Input
A3	0.003	0.003	0.003	0.004	0.004	Input
Maximum capacitance						
Z	0.3	0.6	1.2	2.1	3	Output

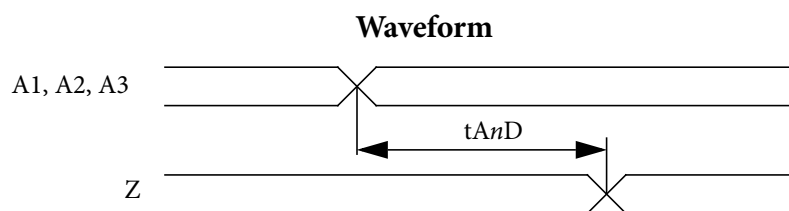
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Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)				
	XR03D1	XR03D2	XR03D4	XR03D7	XR03DA
Z	0.075	0.088	0.123	0.195	0.253



Timing Numbers for XR03D1, XR03D2, XR03D4, XR03D7 and XR03DA:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
XR03D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.186	0.181	0.235	0.216	0.331	0.275	0.522	0.383	0.902	0.591
tA2D	0.256	0.289	0.304	0.322	0.400	0.381	0.591	0.488	0.971	0.696
tA3D	0.288	0.329	0.336	0.362	0.432	0.420	0.623	0.527	1.003	0.735
XR03D2										
tA1D	0.172	0.175	0.201	0.200	0.251	0.238	0.348	0.301	0.539	0.411
tA2D	0.241	0.286	0.267	0.309	0.316	0.345	0.411	0.408	0.602	0.517
tA3D	0.273	0.326	0.299	0.349	0.347	0.385	0.443	0.447	0.634	0.557
XR03D4										
tA1D	0.185	0.195	0.202	0.211	0.232	0.240	0.282	0.281	0.376	0.350
tA2D	0.247	0.306	0.262	0.322	0.289	0.349	0.336	0.389	0.428	0.457
tA3D	0.279	0.347	0.294	0.363	0.321	0.390	0.368	0.430	0.460	0.498
XR03D7										
tA1D	0.291	0.286	0.300	0.296	0.313	0.310	0.336	0.335	0.376	0.380
tA2D	0.314	0.326	0.323	0.335	0.336	0.349	0.359	0.375	0.399	0.419
tA3D	0.360	0.361	0.369	0.371	0.382	0.385	0.405	0.410	0.445	0.455
XR03DA										
tA1D	0.309	0.302	0.315	0.308	0.327	0.320	0.344	0.340	0.373	0.374
tA2D	0.328	0.344	0.334	0.350	0.345	0.362	0.363	0.382	0.392	0.416
tA3D	0.378	0.381	0.384	0.388	0.396	0.400	0.413	0.420	0.442	0.454

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

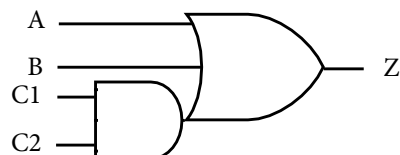
Input transition	1		5		10		15		20	
XR03D1	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.235	0.216	0.300	0.295	0.350	0.366	0.387	0.425	0.409	0.470
tA2D	0.304	0.322	0.343	0.402	0.357	0.485	0.360	0.558	0.349	0.621
tA3D	0.336	0.362	0.388	0.450	0.421	0.539	0.440	0.615	0.443	0.677
XR03D2										
tA1D	0.201	0.200	0.269	0.281	0.323	0.355	0.363	0.416	0.387	0.461
tA2D	0.267	0.309	0.306	0.390	0.320	0.473	0.323	0.546	0.312	0.610
tA3D	0.299	0.349	0.351	0.438	0.383	0.527	0.402	0.603	0.406	0.666
XR03D4										
tA1D	0.202	0.211	0.273	0.295	0.335	0.376	0.381	0.442	0.409	0.491
tA2D	0.262	0.322	0.301	0.403	0.315	0.488	0.319	0.562	0.309	0.626
tA3D	0.294	0.363	0.346	0.451	0.378	0.542	0.398	0.620	0.402	0.684
XR03D7										
tA1D	0.300	0.296	0.347	0.353	0.380	0.398	0.405	0.435	0.420	0.463
tA2D	0.323	0.335	0.374	0.406	0.410	0.470	0.436	0.526	0.450	0.571
tA3D	0.369	0.371	0.436	0.448	0.490	0.518	0.530	0.576	0.558	0.620
XR03DA										
tA1D	0.315	0.308	0.364	0.367	0.397	0.412	0.422	0.450	0.438	0.479
tA2D	0.334	0.350	0.386	0.420	0.421	0.485	0.447	0.541	0.461	0.586
tA3D	0.384	0.388	0.452	0.464	0.509	0.535	0.552	0.594	0.581	0.638

1-1-2 AND-OR Gates

AOR211D1, AOR211D2 and AOR211D4

The AOR211D1 (1x drive), AOR211D2 (2x drive) and AOR211D4 (4x drive) cells provide the complex function:

$$Z = [A + B + (C1 \cdot C2)]$$



Function Table

INPUTS				OUTPUT
A	B	C1	C2	Z
H	X	X	X	H
X	H	X	X	H
X	X	H	H	H
L	L	L	X	L
L	L	X	L	L

Cell Description

Macro Name:	AOR211D1	AOR211D2	AOR211D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.	2.25	2.75
Leakage Power (pW):	39.0	48.9	61.2

Pin Description

Name	Capacitance (pF)			Description
	AOR211D1	AOR211D2	AOR211D4	
A	0.004	0.005	0.004	Data Input
B	0.004	0.005	0.004	Data Input
C1	0.003	0.003	0.003	Data Input
C2	0.004	0.004	0.004	Data Input
Maximum capacitance				
Z	0.3	0.6	1.2	Data Output

Pin Powers for Standard Load = 0.032 pF
Standard input transition = 0.1 ns

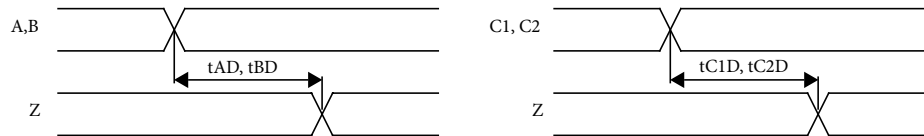
Name	Pin Power (pW/Hz)		
	AOR211D1	AOR211D2	AOR211D4
A	0.001	0.001	0.001
B	0.002	0.002	0.002
C1	-0	-0	-0
C2	0	0	0
Z	0.032	0.048	0.089

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Waveforms



Timing Numbers for AOR211D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.120	0.218	0.171	0.258	0.271	0.327	0.472	0.446	0.872	0.670
tBD	0.128	0.255	0.179	0.295	0.279	0.363	0.479	0.482	0.880	0.706
tC1D	0.154	0.254	0.207	0.294	0.309	0.362	0.510	0.480	0.911	0.704
tC2D	0.156	0.282	0.209	0.322	0.311	0.390	0.512	0.509	0.913	0.733

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.171	0.258	0.200	0.338	0.202	0.428	0.195	0.508	0.174	0.580
tBD	0.179	0.295	0.217	0.345	0.228	0.414	0.228	0.483	0.215	0.548
tC1D	0.207	0.294	0.254	0.342	0.277	0.391	0.289	0.439	0.286	0.490
tC2D	0.209	0.322	0.242	0.369	0.257	0.417	0.262	0.468	0.254	0.522

Timing Numbers for AOR211D2:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.097	0.246	0.121	0.275	0.166	0.318	0.257	0.390	0.437	0.510
tBD	0.105	0.284	0.129	0.313	0.175	0.356	0.266	0.428	0.446	0.548
tC1D	0.134	0.293	0.160	0.321	0.207	0.364	0.300	0.436	0.481	0.557
tC2D	0.136	0.312	0.162	0.341	0.210	0.384	0.302	0.456	0.483	0.576

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.121	0.275	0.158	0.356	0.166	0.455	0.161	0.545	0.141	0.623
tBD	0.129	0.313	0.174	0.363	0.189	0.437	0.192	0.511	0.179	0.580
tC1D	0.160	0.321	0.212	0.373	0.239	0.429	0.251	0.484	0.247	0.543
tC2D	0.162	0.341	0.198	0.388	0.217	0.440	0.223	0.493	0.214	0.551

Timing Numbers for AOR211D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.103	0.313	0.116	0.333	0.141	0.364	0.185	0.413	0.274	0.492
tBD	0.112	0.351	0.125	0.371	0.150	0.402	0.195	0.451	0.284	0.530
tC1D	0.152	0.364	0.167	0.383	0.195	0.414	0.243	0.462	0.334	0.541
tC2D	0.154	0.378	0.169	0.398	0.198	0.429	0.245	0.478	0.336	0.557

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

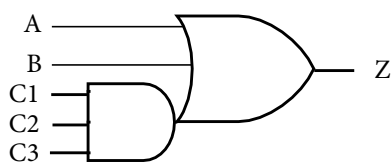
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.116	0.333	0.166	0.411	0.186	0.516	0.191	0.617	0.178	0.701
tBD	0.125	0.371	0.182	0.421	0.209	0.498	0.221	0.575	0.215	0.648
tC1D	0.167	0.383	0.229	0.433	0.270	0.489	0.294	0.545	0.299	0.604
tC2D	0.169	0.398	0.211	0.447	0.239	0.502	0.254	0.557	0.252	0.617

1-1-3 AND-OR Gates

AOR311D1, AOR311D2 and AOR311D4

The AOR311D1 (1x drive), AOR311D2 (2x drive) and AOR311D4 (4x drive) cells provide the complex function:

$$Z = [A + B + (C1 \cdot C2 \cdot C3)]$$



Function Table

INPUTS					OUTPUT
A	B	C1	C2	C3	Z
H	X	X	X	X	H
X	H	X	X	X	H
X	X	H	H	H	H
L	L	X	X	L	L
L	L	X	L	X	L
L	L	L	X	X	L

Cell Description

Macro Name:	AOR311D1	AOR311D2	AOR311D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.	2.25	2.75
Leakage Power (pW):	35.9	47.2	49.3

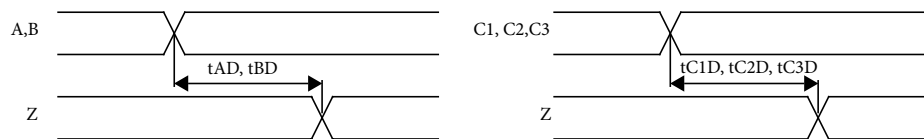
Pin Description

Name	Capacitance (pF)			Description
	AOR311D1	AOR311D2	AOR311D4	
A	0.004	0.004	0.003	Data Input
B	0.005	0.005	0.004	Data Input
C1	0.003	0.003	0.004	Data Input
C2	0.004	0.004	0.005	Data Input
C3	0.004	0.004	0.005	Data Input
Maximum capacitance				
Z	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	AOR311D1	AOR311D2	AOR311D4
A	0.001	0.001	0.001
B	0.002	0.002	0.003
C1	0.001	0.001	0.001
C2	0	0	0.001
C3	0	0	-0
Z	0.032	0.047	0.084

Waveforms



Timing Numbers for AOR311D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.124	0.238	0.178	0.275	0.284	0.336	0.496	0.438	0.920	0.625
tBD	0.129	0.272	0.183	0.309	0.290	0.370	0.502	0.472	0.926	0.659
tC1D	0.220	0.260	0.277	0.296	0.387	0.355	0.601	0.456	1.026	0.642
tC2D	0.227	0.283	0.283	0.319	0.393	0.379	0.608	0.481	1.032	0.668
tC3D	0.232	0.301	0.289	0.338	0.399	0.399	0.613	0.501	1.038	0.688

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.178	0.275	0.203	0.358	0.202	0.457	0.191	0.545	0.167	0.624
tBD	0.183	0.309	0.217	0.361	0.223	0.437	0.218	0.512	0.200	0.583
tC1D	0.277	0.296	0.334	0.341	0.382	0.381	0.415	0.417	0.434	0.452
tC2D	0.283	0.319	0.326	0.367	0.361	0.409	0.387	0.448	0.399	0.488
tC3D	0.289	0.338	0.314	0.382	0.336	0.422	0.351	0.460	0.354	0.503

Timing Numbers for AOR311D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.102	0.257	0.129	0.286	0.182	0.326	0.286	0.391	0.491	0.497
tBD	0.108	0.292	0.135	0.320	0.187	0.360	0.291	0.426	0.497	0.531
tC1D	0.213	0.283	0.248	0.308	0.304	0.347	0.412	0.411	0.620	0.516
tC2D	0.220	0.304	0.254	0.331	0.311	0.371	0.419	0.436	0.627	0.541
tC3D	0.225	0.320	0.260	0.348	0.316	0.388	0.424	0.453	0.632	0.559

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.129	0.286	0.165	0.367	0.173	0.471	0.168	0.566	0.148	0.647
tBD	0.135	0.320	0.178	0.371	0.192	0.448	0.192	0.525	0.178	0.597
tC1D	0.248	0.308	0.309	0.355	0.364	0.396	0.405	0.434	0.428	0.471
tC2D	0.254	0.331	0.298	0.379	0.340	0.423	0.371	0.463	0.386	0.504
tC3D	0.260	0.348	0.287	0.392	0.311	0.433	0.329	0.471	0.335	0.514

Timing Numbers for AOR311D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.124	0.427	0.140	0.450	0.169	0.488	0.218	0.544	0.314	0.636
tBD	0.105	0.455	0.120	0.478	0.147	0.516	0.195	0.572	0.290	0.664
tC1D	0.155	0.401	0.173	0.422	0.204	0.456	0.257	0.510	0.356	0.598
tC2D	0.161	0.437	0.180	0.459	0.211	0.496	0.264	0.551	0.363	0.640
tC3D	0.165	0.479	0.184	0.502	0.215	0.541	0.267	0.597	0.366	0.688

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

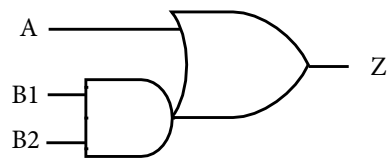
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.140	0.450	0.201	0.528	0.234	0.638	0.250	0.750	0.246	0.844
tBD	0.120	0.478	0.172	0.531	0.193	0.615	0.199	0.704	0.185	0.787
tC1D	0.173	0.422	0.229	0.451	0.268	0.486	0.293	0.523	0.302	0.569
tC2D	0.180	0.459	0.221	0.489	0.250	0.522	0.267	0.560	0.269	0.610
tC3D	0.184	0.502	0.208	0.535	0.222	0.571	0.226	0.612	0.217	0.667

1-2 AND-OR Gates

AOR21D1, AOR21D2 and AOR21D4

The AOR21D1 (1x drive), AOR21D2 (2x drive) and AOR21D4 (4x drive) cells provide the complex function:

$$Z = [A + (B1 \cdot B2)]$$



Function Table

INPUTS			OUTPUT
A	B1	B2	Z
H	X	X	H
X	H	H	H
L	L	X	L
L	X	L	L

Cell Description

Macro Name:	AOR21D1	AOR21D2	AOR21D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.75	2.	2.5
Leakage Power (pW):	32.7	43.2	55.6

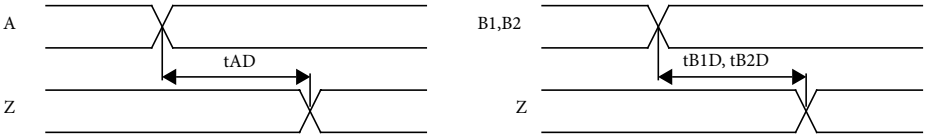
Pin Description

Name	Capacitance (pF)			Description
	AOR21D1	AOR21D2	AOR21D4	
A	0.004	0.004	0.004	Data Input
B1	0.003	0.003	0.003	Data Input
B2	0.004	0.004	0.004	Data Input
Maximum capacitance				
Z	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	AOR21D1	AOR21D2	AOR21D4
A	0.002	0.002	0.002
B1	-0	-0	-0
B2	0	0	0
Z	0.027	0.04	0.078

Waveforms



Timing Numbers for AOR21D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.113	0.192	0.161	0.227	0.258	0.287	0.450	0.397	0.834	0.612
tB1D	0.136	0.197	0.185	0.231	0.282	0.292	0.475	0.402	0.859	0.616
tB2D	0.139	0.218	0.188	0.253	0.284	0.314	0.476	0.424	0.861	0.639

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.161	0.227	0.183	0.311	0.181	0.401	0.169	0.482	0.146	0.554
tB1D	0.185	0.231	0.220	0.288	0.231	0.351	0.232	0.413	0.221	0.471
tB2D	0.188	0.253	0.214	0.307	0.220	0.370	0.217	0.433	0.204	0.492

Timing Numbers for AOR21D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.097	0.205	0.123	0.230	0.172	0.269	0.272	0.334	0.469	0.447
tB1D	0.125	0.213	0.152	0.238	0.203	0.276	0.303	0.341	0.500	0.454
tB2D	0.128	0.232	0.155	0.257	0.206	0.295	0.306	0.361	0.503	0.474

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.123	0.230	0.156	0.316	0.162	0.414	0.157	0.500	0.138	0.575
tB1D	0.152	0.238	0.198	0.296	0.217	0.362	0.225	0.427	0.219	0.487
tB2D	0.155	0.257	0.187	0.312	0.200	0.377	0.203	0.442	0.193	0.502

Timing Numbers for AOR21D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.101	0.262	0.114	0.279	0.139	0.308	0.183	0.350	0.272	0.420
tB1D	0.140	0.266	0.155	0.282	0.183	0.309	0.230	0.351	0.319	0.420
tB2D	0.142	0.287	0.157	0.304	0.184	0.333	0.231	0.375	0.321	0.445

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

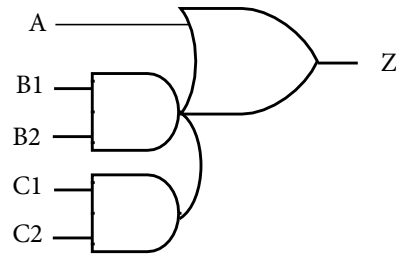
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.114	0.279	0.166	0.364	0.185	0.472	0.191	0.570	0.179	0.651
tB1D	0.155	0.282	0.213	0.332	0.251	0.396	0.274	0.457	0.280	0.514
tB2D	0.157	0.304	0.196	0.357	0.220	0.424	0.233	0.489	0.231	0.551

1-2-2 AND-OR Gates

AOR221D1, AOR221D2 and AOR221D4

The AOR221D1 (1x drive), AOR221D2 (2x drive) and AOR221D4 (4x drive) cells provide the complex function:

$$Z = [A + (B1 \cdot B2) + (C1 \cdot C2)]$$



Function Table

INPUTS					OUTPUT
A	B1	B2	C1	C2	Z
H	X	X	X	X	H
X	H	H	X	X	H
X	X	X	H	H	H
Any other combination					L

Cell Description

Macro Name:	AOR221D1	AOR221D2	AOR221D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.5	2.75	3.
Leakage Power (pW):	36.5	47.2	64.2

Pin Description

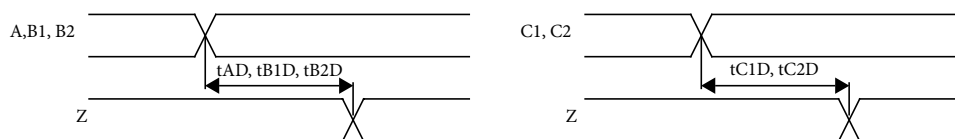
Name	Capacitance (pF)			Description
	AOR221D1	AOR221D2	AOR221D4	
A	0.004	0.004	0.004	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.004	0.004	0.004	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.004	0.004	0.004	Data Input
Maximum capacitance				
Z	0.3	0.6	1.2	Data Output

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**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	AOR221D1	AOR221D2	AOR221D4
A	0.008	0.008	0.008
B1	0.001	0.001	0.001
B2	-0	-0	-0
C1	0.001	0.001	0.001
C2	-0	-0	0
Z	0.035	0.051	0.09

Waveforms



Timing Numbers for AOR221D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.125	0.235	0.171	0.273	0.261	0.337	0.440	0.446	0.798	0.647
tB1D	0.164	0.267	0.211	0.305	0.302	0.368	0.481	0.476	0.840	0.677
tB2D	0.165	0.286	0.212	0.324	0.303	0.388	0.482	0.497	0.841	0.698
tC1D	0.164	0.286	0.212	0.323	0.305	0.386	0.485	0.494	0.845	0.694
tC2D	0.166	0.318	0.214	0.356	0.307	0.420	0.488	0.529	0.847	0.730

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.171	0.273	0.210	0.355	0.223	0.450	0.224	0.533	0.211	0.606
tB1D	0.211	0.305	0.263	0.354	0.293	0.414	0.310	0.474	0.313	0.532
tB2D	0.212	0.324	0.247	0.373	0.268	0.434	0.279	0.496	0.275	0.557
tC1D	0.212	0.323	0.267	0.366	0.303	0.404	0.326	0.440	0.335	0.480
tC2D	0.214	0.356	0.252	0.403	0.275	0.447	0.287	0.488	0.285	0.536

Timing Numbers for AOR221D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.111	0.256	0.136	0.285	0.180	0.327	0.267	0.396	0.442	0.509
tB1D	0.155	0.290	0.182	0.318	0.229	0.359	0.317	0.428	0.492	0.541
tB2D	0.156	0.307	0.183	0.336	0.229	0.378	0.318	0.447	0.493	0.560
tC1D	0.153	0.309	0.181	0.336	0.228	0.376	0.318	0.444	0.495	0.556
tC2D	0.156	0.339	0.183	0.368	0.230	0.410	0.321	0.479	0.497	0.593

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.136	0.285	0.185	0.365	0.205	0.466	0.212	0.556	0.203	0.631
tB1D	0.182	0.318	0.239	0.367	0.278	0.430	0.302	0.492	0.309	0.551
tB2D	0.183	0.336	0.223	0.385	0.249	0.447	0.264	0.511	0.263	0.573
tC1D	0.181	0.336	0.242	0.379	0.285	0.418	0.313	0.455	0.325	0.496
tC2D	0.183	0.368	0.224	0.416	0.253	0.460	0.269	0.502	0.270	0.550

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Timing Numbers for AOR221D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.116	0.312	0.130	0.331	0.156	0.363	0.201	0.410	0.290	0.489
tB1D	0.169	0.348	0.185	0.366	0.214	0.397	0.263	0.444	0.353	0.522
tB2D	0.170	0.362	0.187	0.381	0.216	0.413	0.264	0.460	0.354	0.539
tC1D	0.167	0.367	0.184	0.385	0.213	0.416	0.262	0.463	0.353	0.539
tC2D	0.169	0.395	0.185	0.414	0.214	0.446	0.263	0.493	0.355	0.571

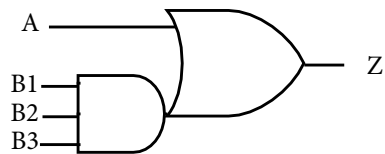
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.130	0.331	0.190	0.409	0.220	0.513	0.235	0.613	0.231	0.693
tB1D	0.185	0.366	0.249	0.416	0.298	0.482	0.330	0.548	0.342	0.610
tB2D	0.187	0.381	0.230	0.431	0.263	0.495	0.285	0.561	0.289	0.625
tC1D	0.184	0.385	0.249	0.428	0.302	0.469	0.338	0.508	0.354	0.551
tC2D	0.185	0.414	0.229	0.461	0.264	0.507	0.286	0.551	0.291	0.600

1-3 AND-OR Gates AOR31D1, AOR31D2 and AOR31D4

The AOR31D1 (1x drive), AOR31D2 (2x drive) and AOR31D4 (4x drive) cells provide the complex function:

$$Z = [A + (B1 \cdot B2 \cdot B3)]$$



Function Table

INPUTS				OUTPUT
A	B1	B2	B3	Z
H	X	X	X	H
X	H	H	H	H
L	X	X	L	L
L	X	L	X	L
L	L	X	X	L

Cell Description

Macro Name:	AOR31D1	AOR31D2	AOR31D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.75	2.	2.5
Leakage Power (pW):	31.8	43.6	68.2

Pin Description

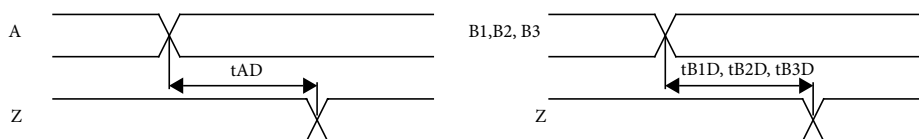
Name	Capacitance (pF)			Description
	AOR31D1	AOR31D2	AOR31D4	
A	0.004	0.004	0.004	Data Input
B1	0.003	0.003	0.003	Data Input
B2	0.004	0.003	0.003	Data Input
B3	0.004	0.004	0.004	Data Input
Maximum capacitance				
Z	0.3	0.6	1.2	Data Output

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**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	AOR31D1	AOR31D2	AOR31D4
A	0.003	0.003	0.002
B1	0.001	0.001	0.001
B2	0	-0	0
B3	-0	-0	0
Z	0.028	0.047	0.09

Waveforms



Timing Numbers for AOR31D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.113	0.233	0.157	0.271	0.245	0.334	0.420	0.448	0.770	0.666
tB1D	0.154	0.214	0.200	0.250	0.289	0.312	0.465	0.424	0.815	0.641
tB2D	0.161	0.235	0.207	0.271	0.296	0.334	0.472	0.445	0.822	0.663
tB3D	0.164	0.269	0.210	0.307	0.299	0.370	0.476	0.484	0.826	0.702

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.157	0.271	0.185	0.356	0.187	0.455	0.180	0.543	0.159	0.621
tB1D	0.200	0.250	0.240	0.306	0.258	0.369	0.267	0.429	0.262	0.486
tB2D	0.207	0.271	0.240	0.325	0.257	0.387	0.264	0.446	0.259	0.502
tB3D	0.210	0.307	0.229	0.367	0.232	0.438	0.227	0.507	0.210	0.574

Timing Numbers for AOR31D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.097	0.261	0.120	0.286	0.160	0.324	0.241	0.388	0.399	0.491
tB1D	0.150	0.236	0.176	0.259	0.219	0.295	0.301	0.355	0.461	0.457
tB2D	0.158	0.265	0.184	0.289	0.227	0.326	0.309	0.388	0.469	0.490
tB3D	0.162	0.299	0.188	0.324	0.231	0.362	0.313	0.426	0.473	0.529

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.120	0.286	0.161	0.372	0.172	0.477	0.172	0.574	0.155	0.655
tB1D	0.176	0.259	0.226	0.317	0.257	0.385	0.274	0.448	0.276	0.507
tB2D	0.184	0.289	0.222	0.347	0.247	0.416	0.260	0.481	0.258	0.544
tB3D	0.188	0.324	0.209	0.387	0.218	0.462	0.218	0.537	0.204	0.607

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Timing Numbers for AOR31D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.107	0.316	0.121	0.333	0.146	0.362	0.191	0.404	0.280	0.471
tB1D	0.181	0.294	0.198	0.309	0.227	0.334	0.276	0.374	0.368	0.438
tB2D	0.188	0.320	0.205	0.336	0.234	0.364	0.283	0.404	0.375	0.470
tB3D	0.192	0.354	0.209	0.371	0.239	0.400	0.288	0.442	0.380	0.509

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

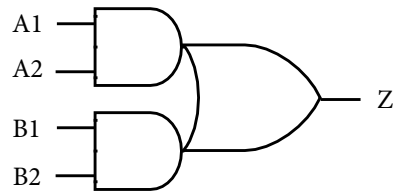
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.121	0.333	0.174	0.416	0.199	0.526	0.208	0.631	0.198	0.717
tB1D	0.198	0.309	0.256	0.367	0.301	0.439	0.330	0.507	0.341	0.570
tB2D	0.205	0.336	0.249	0.394	0.284	0.466	0.307	0.536	0.312	0.601
tB3D	0.209	0.371	0.234	0.433	0.250	0.510	0.258	0.588	0.250	0.661

2-2 AND-OR Gates

AOR22D1, AOR22D2 and AOR22D4

The AOR22D1 (1x drive), AOR22D2 (2x drive) and AOR22D4 (4x drive) cells provide the complex function:

$$Z = [(A1 \cdot A2) + (B1 \cdot B2)]$$



Function Table

INPUTS				OUTPUT
A1	A2	B1	B2	Z
H	H	X	X	H
X	X	H	H	H
L	X	L	X	L
X	L	X	L	L
L	X	X	L	L
X	L	L	X	L

Cell Description

Macro Name:	AOR22D1	AOR22D2	AOR22D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.	2.25	2.75
Leakage Power (pW):	44.2	61.5	112.9

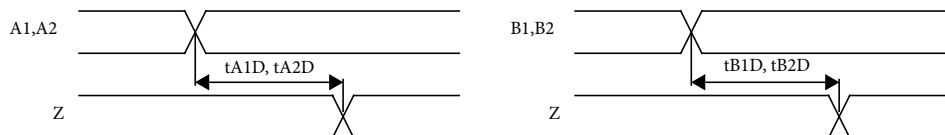
Pin Description

Name	Capacitance (pF)			Description
	AOR22D1	AOR22D2	AOR22D4	
A1	0.004	0.004	0.004	Data Input
A2	0.004	0.004	0.004	Data Input
B1	0.003	0.003	0.003	Data Input
B2	0.004	0.004	0.004	Data Input
Maximum capacitance				
Z	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	AOR22D1	AOR22D2	AOR22D4
A1	0.001	0.001	0.001
A2	-0	-0	-0
B1	0.002	0.001	0.001
B2	-0	-0	-0
Z	0.032	0.044	0.091

Waveforms



Timing Numbers for AOR22D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
t_{A1D}	0.153	0.180	0.211	0.213	0.328	0.273	0.561	0.381	1.025	0.590
t_{A2D}	0.155	0.206	0.213	0.241	0.330	0.300	0.562	0.409	1.026	0.619
t_{B1D}	0.166	0.219	0.225	0.253	0.343	0.313	0.575	0.421	1.039	0.631
t_{B2D}	0.168	0.237	0.227	0.272	0.344	0.332	0.577	0.440	1.041	0.650

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.211	0.213	0.249	0.291	0.265	0.372	0.271	0.442	0.265	0.503
tA2D	0.213	0.241	0.239	0.324	0.246	0.415	0.244	0.495	0.230	0.565
tB1D	0.225	0.253	0.274	0.309	0.299	0.367	0.314	0.423	0.316	0.476
tB2D	0.227	0.272	0.259	0.327	0.275	0.386	0.282	0.444	0.277	0.500

Timing Numbers for AOR22D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.136	0.183	0.170	0.208	0.231	0.245	0.354	0.308	0.597	0.416
tA2D	0.138	0.210	0.171	0.235	0.233	0.273	0.355	0.336	0.598	0.446
tB1D	0.150	0.224	0.183	0.249	0.245	0.287	0.367	0.350	0.611	0.459
tB2D	0.152	0.241	0.185	0.266	0.247	0.304	0.369	0.367	0.612	0.477

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.170	0.208	0.215	0.288	0.239	0.373	0.251	0.448	0.249	0.512
tA2D	0.171	0.235	0.203	0.318	0.216	0.414	0.219	0.499	0.208	0.570
tB1D	0.183	0.249	0.237	0.306	0.271	0.366	0.291	0.423	0.297	0.476
tB2D	0.185	0.266	0.221	0.322	0.242	0.382	0.254	0.441	0.252	0.497

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Timing Numbers for AOR22D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.153	0.237	0.168	0.251	0.197	0.276	0.244	0.314	0.335	0.375
tA2D	0.155	0.260	0.170	0.276	0.198	0.302	0.246	0.341	0.337	0.404
tB1D	0.161	0.280	0.176	0.295	0.203	0.320	0.250	0.358	0.340	0.421
tB2D	0.163	0.292	0.178	0.308	0.205	0.334	0.252	0.373	0.342	0.436

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

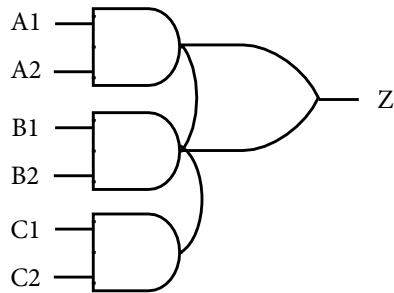
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.168	0.251	0.229	0.331	0.270	0.427	0.296	0.511	0.305	0.579
tA2D	0.170	0.276	0.210	0.358	0.237	0.461	0.252	0.555	0.251	0.630
tB1D	0.176	0.295	0.240	0.352	0.287	0.418	0.318	0.479	0.331	0.536
tB2D	0.178	0.308	0.219	0.364	0.249	0.428	0.269	0.490	0.273	0.549

2-2-2 AND-OR Gates

AOR222D1, AOR222D2 and AOR222D4

The AOR222D1 (1x drive), AOR222D2 (2x drive) and AOR222D4 (4x drive) cells provide the complex function:

$$Z = [(A1 \cdot A2) + (B1 \cdot B2) + (C1 \cdot C2)]$$



Function Table

INPUTS						OUTPUT
A1	A2	B1	B2	C1	C2	Z
H	H	X	X	X	X	H
X	X	H	H	X	X	H
X	X	X	X	H	H	H
Any other combination						L

Cell Description

Macro Name:	AOR222D1	AOR222D2	AOR222D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	3.	3.25	3.5
Leakage Power (pW):	53.3	74.5	124.8

Pin Description

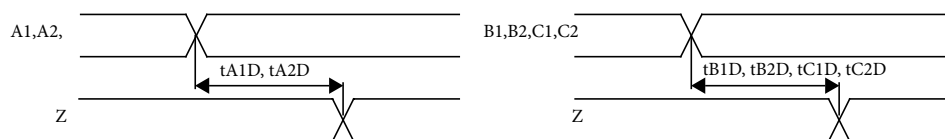
Name	Capacitance (pF)			Description
	AOR222D1	AOR222D2	AOR222D4	
A1	0.003	0.003	0.003	Data Input
A2	0.004	0.004	0.004	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.004	0.004	0.004	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.004	0.004	0.005	Data Input
Maximum capacitance				
Z	0.3	0.6	1.2	Data Output

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Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	AOR222D1	AOR222D2	AOR222D4
A1	0.001	0.001	0.001
A2	-0	-0	-0
B1	0.001	0.001	0.001
B2	-0	-0	-0
C1	0.001	0.001	0.003
C2	-0	-0	0
Z	0.042	0.055	0.112

Waveforms



Timing Numbers for AOR222D1:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.162	0.258	0.211	0.300	0.305	0.371	0.491	0.491	0.863	0.713
tA2D	0.163	0.274	0.212	0.316	0.306	0.388	0.492	0.510	0.864	0.732
tB1D	0.187	0.316	0.236	0.358	0.331	0.429	0.518	0.549	0.890	0.771
tB2D	0.188	0.342	0.237	0.384	0.332	0.456	0.519	0.578	0.891	0.800
tC1D	0.189	0.343	0.238	0.384	0.334	0.455	0.522	0.575	0.895	0.797
tC2D	0.191	0.375	0.241	0.418	0.337	0.490	0.524	0.611	0.898	0.834

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.211	0.300	0.259	0.379	0.285	0.475	0.298	0.559	0.295	0.631
tA2D	0.212	0.316	0.246	0.396	0.263	0.493	0.270	0.581	0.261	0.656
tB1D	0.236	0.358	0.295	0.406	0.335	0.464	0.362	0.521	0.373	0.578
tB2D	0.237	0.384	0.277	0.438	0.304	0.503	0.320	0.566	0.321	0.630
tC1D	0.238	0.384	0.300	0.428	0.345	0.467	0.375	0.502	0.390	0.542
tC2D	0.241	0.418	0.282	0.467	0.312	0.511	0.331	0.552	0.335	0.599

Timing Numbers for AOR222D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.149	0.268	0.179	0.299	0.230	0.345	0.327	0.421	0.518	0.546
tA2D	0.151	0.282	0.180	0.313	0.231	0.360	0.328	0.437	0.519	0.563
tB1D	0.176	0.325	0.205	0.356	0.256	0.401	0.353	0.477	0.544	0.602
tB2D	0.177	0.350	0.206	0.381	0.257	0.428	0.354	0.505	0.545	0.631
tC1D	0.176	0.352	0.205	0.383	0.257	0.428	0.355	0.504	0.547	0.628
tC2D	0.179	0.383	0.208	0.415	0.260	0.462	0.358	0.538	0.550	0.664

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.179	0.299	0.234	0.378	0.266	0.477	0.284	0.568	0.285	0.643
tA2D	0.180	0.313	0.218	0.392	0.240	0.493	0.250	0.586	0.245	0.663
tB1D	0.205	0.356	0.268	0.405	0.315	0.464	0.346	0.523	0.360	0.580
tB2D	0.206	0.381	0.248	0.436	0.280	0.501	0.299	0.565	0.303	0.629
tC1D	0.205	0.383	0.270	0.426	0.321	0.466	0.356	0.502	0.374	0.542
tC2D	0.208	0.415	0.251	0.464	0.285	0.509	0.307	0.550	0.314	0.596

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Timing Numbers for AOR222D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.181	0.371	0.198	0.390	0.228	0.422	0.278	0.469	0.371	0.546
tA2D	0.182	0.379	0.199	0.399	0.229	0.431	0.279	0.480	0.372	0.557
tB1D	0.208	0.425	0.225	0.444	0.255	0.476	0.304	0.523	0.396	0.599
tB2D	0.209	0.449	0.226	0.469	0.256	0.502	0.305	0.551	0.398	0.628
tC1D	0.134	0.438	0.149	0.456	0.176	0.487	0.222	0.533	0.312	0.607
tC2D	0.137	0.486	0.152	0.506	0.178	0.539	0.225	0.587	0.314	0.665

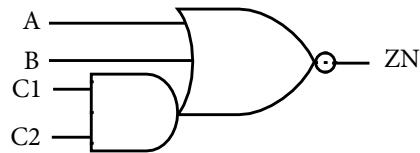
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.198	0.390	0.263	0.465	0.313	0.568	0.346	0.672	0.357	0.756
tA2D	0.199	0.399	0.243	0.474	0.278	0.577	0.299	0.681	0.303	0.766
tB1D	0.225	0.444	0.293	0.494	0.355	0.558	0.399	0.624	0.420	0.687
tB2D	0.226	0.469	0.271	0.524	0.313	0.593	0.342	0.665	0.352	0.734
tC1D	0.149	0.456	0.205	0.499	0.235	0.547	0.249	0.598	0.244	0.658
tC2D	0.152	0.506	0.188	0.555	0.206	0.607	0.210	0.661	0.197	0.727

1-1-2 AND-OR-Invert Gates AOI211D1, AOI211D2 and AOI211D4

The AOI211D1 (1x drive), AOI211D2 (2x drive) and AOI211D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A + B + (C1 \cdot C2)]$$



Function Table

INPUTS				OUTPUT
A	B	C1	C2	ZN
H	X	X	X	L
X	H	X	X	L
X	X	H	H	L
L	L	L	X	H
L	L	X	L	H

Cell Description

Macro Name:	AOI211D1	AOI211D2	AOI211D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.5	2.75	3.
Leakage Power (pW):	26.4	83.2	129.6

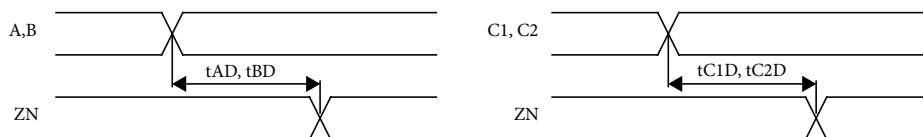
Pin Description

Name	Capacitance (pF)			Description
	AOI211D1	AOI211D2	AOI211D4	
A	0.004	0.004	0.004	Data Input
B	0.004	0.004	0.004	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	AOI211D1	AOI211D2	AOI211D4
A	0.001	0.001	0.001
B	0.001	0.001	0.001
C1	-0	-0	-0
C2	0	0	0
ZN	0.015	0.063	0.098

Waveforms



Timing Numbers for AOI211D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
t_{AD}	0.267	0.062	0.429	0.088	0.751	0.141	1.393	0.247	2.675	0.457
t_{BD}	0.312	0.062	0.474	0.086	0.796	0.134	1.438	0.228	2.719	0.415
t_{C1D}	0.313	0.091	0.476	0.131	0.799	0.212	1.442	0.368	2.726	0.677
t_{C2D}	0.342	0.092	0.503	0.133	0.824	0.213	1.465	0.369	2.746	0.678

Propagation Delays (ns) for sample input transitions:**Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.429	0.088	0.501	0.144	0.604	0.171	0.708	0.182	0.801	0.172
tBD	0.474	0.086	0.522	0.143	0.597	0.170	0.679	0.181	0.758	0.173
tC1D	0.476	0.131	0.527	0.194	0.586	0.238	0.648	0.264	0.711	0.267
tC2D	0.503	0.133	0.557	0.172	0.616	0.198	0.679	0.211	0.747	0.205

Timing Numbers for AOI211D2:**Propagation Delays (ns) for sample loads:****Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.284	0.158	0.307	0.175	0.349	0.201	0.433	0.247	0.601	0.333
tBD	0.327	0.160	0.350	0.177	0.392	0.203	0.476	0.249	0.644	0.336
tC1D	0.318	0.184	0.341	0.201	0.383	0.227	0.467	0.273	0.635	0.360
tC2D	0.354	0.185	0.377	0.202	0.419	0.228	0.503	0.275	0.671	0.361

Propagation Delays (ns) for sample input transitions:**Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.307	0.175	0.389	0.202	0.486	0.204	0.573	0.194	0.650	0.171
tBD	0.350	0.177	0.402	0.209	0.477	0.213	0.554	0.207	0.626	0.186
tC1D	0.341	0.201	0.391	0.244	0.443	0.260	0.497	0.265	0.554	0.255
tC2D	0.377	0.202	0.429	0.230	0.484	0.238	0.543	0.234	0.606	0.216

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Timing Numbers for AOI211D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.307	0.147	0.320	0.157	0.344	0.174	0.388	0.200	0.477	0.246
tBD	0.347	0.155	0.360	0.165	0.384	0.182	0.428	0.208	0.517	0.255
tC1D	0.340	0.182	0.353	0.193	0.378	0.210	0.422	0.236	0.510	0.282
tC2D	0.376	0.184	0.389	0.194	0.414	0.211	0.458	0.237	0.546	0.284

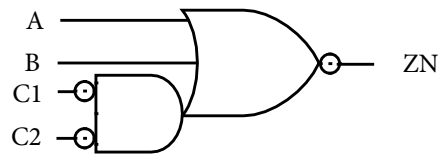
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.320	0.157	0.405	0.180	0.507	0.176	0.600	0.161	0.683	0.132
tBD	0.360	0.165	0.412	0.205	0.488	0.214	0.565	0.211	0.637	0.193
tC1D	0.353	0.193	0.404	0.238	0.457	0.258	0.513	0.265	0.572	0.257
tC2D	0.389	0.194	0.442	0.224	0.498	0.234	0.557	0.233	0.621	0.217

1-1-2 AND-OR-Invert Gates with Inverted C Inputs AOIM211D1, AOIM211D2 and AOIM211D4

The AOIM211D1 (1x drive), AOIM211D2 (2x drive) and AOIM211D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A + B + \text{NOT}(C1) \cdot \text{NOT}(C2)]$$



Function Table

INPUTS				OUTPUT
A	B	C1	C2	ZN
H	X	X	X	L
X	H	X	X	L
X	X	L	L	L
L	L	X	H	H
L	L	H	X	H

Cell Description

Macro Name:	AOIM211D1	AOIM211D2	AOIM211D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.75	3.	3.25
Leakage Power (pW):	33.5	94.1	136.2

Pin Description

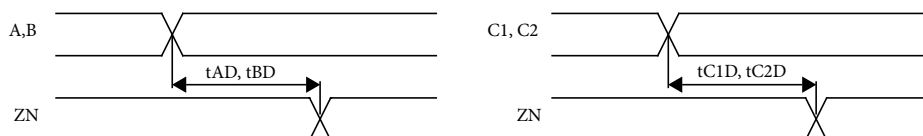
Name	Capacitance (pF)			Description
	AOIM211D1	AOIM211D2	AOIM211D4	
A	0.004	0.004	0.004	Data Input
B	0.004	0.004	0.004	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

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**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	AOIM211D1	AOIM211D2	AOIM211D4
A	0.001	0.001	0.001
B	0	0	0
C1	0.017	0.017	0.017
C2	0	0	0
ZN	0.014	0.06	0.089

Waveforms



Timing Numbers for AOIM211D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.213	0.058	0.354	0.082	0.635	0.130	1.196	0.227	2.314	0.420
tBD	0.247	0.064	0.388	0.090	0.669	0.141	1.229	0.242	2.346	0.444
tC1D	0.286	0.174	0.426	0.203	0.707	0.257	1.267	0.356	2.384	0.547
tC2D	0.292	0.192	0.432	0.222	0.713	0.275	1.273	0.374	2.390	0.565

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.354	0.082	0.429	0.135	0.531	0.156	0.633	0.163	0.719	0.151
tBD	0.388	0.090	0.438	0.147	0.512	0.176	0.591	0.189	0.663	0.183
tC1D	0.426	0.203	0.436	0.285	0.422	0.369	0.399	0.445	0.367	0.514
tC2D	0.432	0.222	0.452	0.278	0.447	0.346	0.432	0.412	0.409	0.473

Timing Numbers for AOIM211D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.219	0.152	0.242	0.169	0.287	0.197	0.376	0.246	0.554	0.338
tBD	0.253	0.159	0.277	0.177	0.321	0.204	0.411	0.254	0.589	0.346
tC1D	0.292	0.267	0.315	0.285	0.360	0.312	0.449	0.362	0.627	0.454
tC2D	0.298	0.285	0.322	0.303	0.366	0.330	0.455	0.380	0.633	0.472

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.242	0.169	0.322	0.188	0.405	0.182	0.482	0.167	0.552	0.140
tBD	0.277	0.177	0.327	0.207	0.393	0.210	0.460	0.204	0.522	0.185
tC1D	0.315	0.285	0.325	0.363	0.311	0.447	0.288	0.523	0.256	0.591
tC2D	0.322	0.303	0.341	0.356	0.336	0.423	0.322	0.488	0.298	0.547

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Timing Numbers for AOIM211D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.236	0.149	0.250	0.161	0.274	0.180	0.318	0.211	0.407	0.267
tBD	0.271	0.157	0.284	0.169	0.308	0.188	0.352	0.219	0.441	0.275
tC1D	0.309	0.266	0.323	0.278	0.347	0.297	0.391	0.328	0.480	0.384
tC2D	0.315	0.284	0.329	0.296	0.353	0.315	0.398	0.346	0.487	0.402

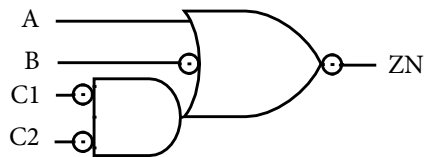
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.250	0.161	0.330	0.183	0.416	0.179	0.495	0.166	0.567	0.141
tBD	0.284	0.169	0.335	0.201	0.403	0.207	0.472	0.203	0.536	0.186
tC1D	0.323	0.278	0.332	0.358	0.318	0.440	0.295	0.516	0.263	0.584
tC2D	0.329	0.296	0.349	0.350	0.343	0.417	0.329	0.482	0.305	0.541

1-1-2 AND-OR-Invert Gates with Inverted B and C Inputs AOIM2M11D1, AOIM2M11D2 and AOIM2M11D4

The AOIM2M11D1 (1x drive), AOIM2M11D2 (2x drive) and AOIM2M11D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A + \text{NOT}(B) + \text{NOT}(C1) \cdot \text{NOT}(C2)]$$



Function Table

INPUTS				OUTPUT
A	B	C1	C2	ZN
H	X	X	X	L
X	L	X	X	L
X	X	L	L	L
L	H	H	X	H
L	H	X	H	H

Cell Description

Macro Name:	AOIM2M11D1	AOIM2M11D2	AOIM2M11D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.25	3.5	3.75
Leakage Power (pW):	38.9	99.2	140.7

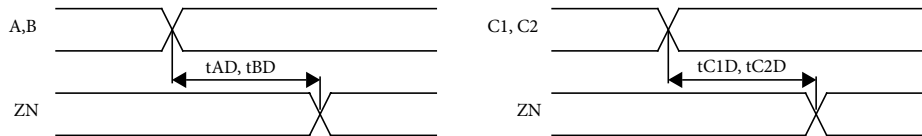
Pin Description

Name	Capacitance (pF)			Description
	AOIM2M11D1	AOIM2M11D2	AOIM2M11D4	
A	0.004	0.004	0.004	Data Input
B	0.004	0.004	0.004	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	AOIM2M11D1	AOIM2M11D2	AOIM2M11D4
A	0.004	0.004	0.004
B	0.016	0.016	0.016
C1	0.016	0.016	0.016
C2	0	0	0
ZN	0.016	0.061	0.089

Waveforms



Timing Numbers for AOIM2M11D1:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.212	0.058	0.352	0.082	0.633	0.131	1.190	0.229	2.303	0.423
tBD	0.273	0.116	0.413	0.142	0.691	0.194	1.248	0.297	2.360	0.501
tC1D	0.288	0.170	0.428	0.200	0.707	0.254	1.264	0.355	2.376	0.551
tC2D	0.295	0.187	0.435	0.217	0.714	0.271	1.271	0.372	2.382	0.568

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.352	0.082	0.429	0.135	0.533	0.156	0.637	0.163	0.725	0.151
tBD	0.413	0.142	0.431	0.211	0.432	0.271	0.426	0.324	0.414	0.371
tC1D	0.428	0.200	0.444	0.278	0.436	0.357	0.420	0.429	0.394	0.494
tC2D	0.435	0.217	0.460	0.268	0.461	0.330	0.454	0.390	0.437	0.445

Timing Numbers for AOIM2M11D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.222	0.151	0.246	0.169	0.290	0.196	0.380	0.246	0.558	0.338
tBD	0.283	0.211	0.306	0.229	0.351	0.256	0.440	0.306	0.618	0.398
tC1D	0.299	0.262	0.322	0.280	0.367	0.307	0.456	0.357	0.634	0.449
tC2D	0.305	0.279	0.328	0.297	0.373	0.325	0.462	0.374	0.640	0.466

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.246	0.169	0.327	0.186	0.412	0.179	0.490	0.164	0.561	0.137
tBD	0.306	0.229	0.324	0.294	0.325	0.352	0.320	0.404	0.307	0.448
tC1D	0.322	0.280	0.338	0.358	0.330	0.435	0.313	0.506	0.287	0.570
tC2D	0.328	0.297	0.354	0.346	0.356	0.407	0.348	0.465	0.331	0.517

Timing Numbers for AOIM2M11D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.241	0.168	0.255	0.180	0.280	0.201	0.325	0.234	0.414	0.291
tBD	0.302	0.229	0.316	0.241	0.341	0.261	0.385	0.294	0.474	0.351
tC1D	0.318	0.280	0.332	0.292	0.357	0.313	0.401	0.346	0.490	0.402
tC2D	0.324	0.297	0.338	0.309	0.363	0.330	0.407	0.363	0.496	0.420

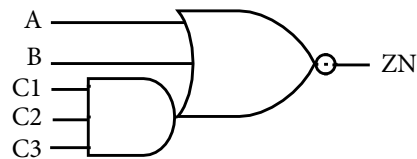
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.255	0.180	0.337	0.200	0.423	0.193	0.501	0.178	0.573	0.152
tBD	0.316	0.241	0.333	0.307	0.335	0.365	0.330	0.416	0.317	0.461
tC1D	0.332	0.292	0.348	0.370	0.340	0.447	0.323	0.518	0.297	0.582
tC2D	0.338	0.309	0.364	0.359	0.365	0.419	0.358	0.478	0.341	0.530

1-1-3 AND-OR-Invert Gates AOI311D1, AOI311D2 and AOI311D4

The AOI311D1 (1x drive), AOI311D2 (2x drive) and AOI311D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT} [A + B + (C1 \cdot C2 \cdot C3)]$$



Function Table

INPUTS					OUTPUT
A	B	C1	C2	C3	ZN
H	X	X	X	X	L
X	H	X	X	X	L
X	X	H	H	H	L
L	L	X	X	L	H
L	L	X	L	X	H
L	L	L	X	X	H

Cell Description

Macro Name:	AOI311D1	AOI311D2	AOI311D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.75	3.	3.25
Leakage Power (pW):	25.6	93.5	128.9

Pin Description

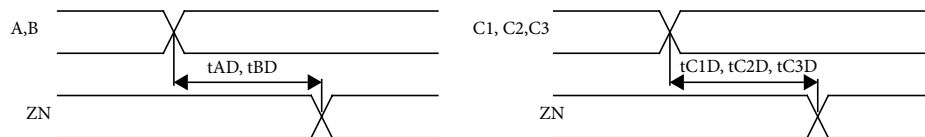
Name	Capacitance (pF)			Description
	AOI311D1	AOI311D2	AOI311D4	
A	0.004	0.004	0.004	Data Input
B	0.005	0.004	0.004	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.005	0.005	0.004	Data Input
C3	0.005	0.005	0.005	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

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**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	AOI311D1	AOI311D2	AOI311D4
A	0.001	0.001	0.001
B	0.003	0.003	0.003
C1	0.002	0.001	0.001
C2	0.001	-0	0
C3	-0	-0	-0
ZN	0.017	0.068	0.093

Waveforms



Timing Numbers for AOI311D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.276	0.059	0.413	0.083	0.686	0.132	1.229	0.230	2.313	0.425
tBD	0.307	0.055	0.444	0.075	0.717	0.115	1.260	0.195	2.345	0.353
tC1D	0.277	0.101	0.416	0.149	0.692	0.241	1.240	0.420	2.335	0.774
tC2D	0.308	0.107	0.446	0.155	0.721	0.247	1.269	0.426	2.363	0.780
tC3D	0.333	0.111	0.469	0.159	0.742	0.251	1.284	0.430	2.367	0.784

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.413	0.083	0.487	0.137	0.592	0.162	0.698	0.172	0.790	0.164
tBD	0.444	0.075	0.494	0.125	0.572	0.143	0.656	0.147	0.735	0.132
tC1D	0.416	0.149	0.455	0.208	0.497	0.257	0.540	0.289	0.586	0.300
tC2D	0.446	0.155	0.487	0.196	0.530	0.228	0.576	0.248	0.627	0.251
tC3D	0.469	0.159	0.510	0.183	0.552	0.198	0.598	0.205	0.651	0.198

Timing Numbers for AOI311D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.277	0.169	0.296	0.185	0.335	0.209	0.411	0.250	0.561	0.326
tBD	0.308	0.170	0.328	0.187	0.366	0.211	0.442	0.252	0.593	0.328
tC1D	0.282	0.228	0.302	0.245	0.340	0.269	0.416	0.311	0.567	0.387
tC2D	0.308	0.235	0.328	0.252	0.366	0.276	0.442	0.318	0.593	0.393
tC3D	0.337	0.238	0.356	0.255	0.395	0.279	0.471	0.321	0.621	0.397

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.296	0.185	0.377	0.221	0.472	0.232	0.556	0.232	0.630	0.216
tBD	0.328	0.187	0.383	0.226	0.458	0.239	0.531	0.241	0.600	0.227
tC1D	0.302	0.245	0.342	0.298	0.377	0.335	0.409	0.360	0.443	0.372
tC2D	0.328	0.252	0.366	0.291	0.400	0.320	0.435	0.340	0.473	0.347
tC3D	0.356	0.255	0.399	0.279	0.439	0.293	0.479	0.301	0.526	0.297

Timing Numbers for AOI311D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.288	0.198	0.301	0.210	0.325	0.232	0.369	0.265	0.458	0.321
tBD	0.319	0.200	0.332	0.212	0.356	0.234	0.400	0.267	0.489	0.323
tC1D	0.292	0.259	0.305	0.271	0.329	0.293	0.373	0.326	0.462	0.383
tC2D	0.319	0.266	0.332	0.278	0.356	0.300	0.400	0.333	0.489	0.390
tC3D	0.348	0.269	0.361	0.282	0.385	0.303	0.429	0.336	0.518	0.393

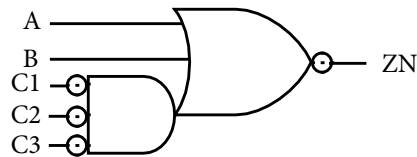
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.301	0.210	0.381	0.246	0.477	0.257	0.561	0.256	0.635	0.242
tBD	0.332	0.212	0.387	0.251	0.463	0.264	0.537	0.265	0.606	0.252
tC1D	0.305	0.271	0.346	0.324	0.381	0.361	0.415	0.386	0.450	0.399
tC2D	0.332	0.278	0.371	0.318	0.406	0.347	0.442	0.366	0.481	0.374
tC3D	0.361	0.282	0.404	0.306	0.444	0.321	0.485	0.328	0.532	0.325

1-1-3 AND-OR-Invert Gates with Inverted C Inputs AOIM311D1, AOIM311D2 and AOIM311D4

The AOIM311D1 (1x drive), AOIM311D2 (2x drive) and AOIM311D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A + B + \text{NOT}(C1) \cdot \text{NOT}(C2) \cdot \text{NOT}(C3)]$$



Function Table

INPUTS					OUTPUT
A	B	C1	C2	C3	ZN
H	X	X	X	X	L
X	H	X	X	X	L
X	X	L	L	L	L
L	L	X	X	H	H
L	L	X	H	X	H
L	L	H	X	X	H

Cell Description

Macro Name:	AOIM311D1	AOIM311D2	AOIM311D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.5	3.25	3.75
Leakage Power (pW):	43.9	104.8	158.8

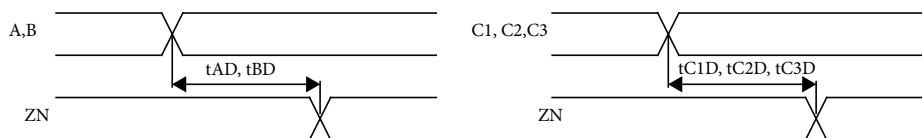
Pin Description

Name	Capacitance (pF)			Description
	AOIM311D1	AOIM311D2	AOIM311D4	
A	0.005	0.004	0.005	Data Input
B	0.004	0.004	0.004	Data Input
C1	0.004	0.003	0.003	Data Input
C2	0.004	0.004	0.004	Data Input
C3	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	AOIM311D1	AOIM311D2	AOIM311D4
A	0.002	0.001	0.002
B	0.001	0	0
C1	0.001	0.001	0.001
C2	0	0	0
C3	0	0	0
ZN	0.021	0.071	0.102

Waveforms



Timing Numbers for AOIM311D1:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.163	0.057	0.267	0.080	0.472	0.125	0.880	0.216	1.694	0.397
tBD	0.197	0.066	0.300	0.090	0.505	0.138	0.912	0.232	1.726	0.421
tC1D	0.240	0.237	0.343	0.270	0.547	0.328	0.954	0.429	1.768	0.616
tC2D	0.243	0.272	0.347	0.305	0.551	0.363	0.958	0.464	1.772	0.651
tC3D	0.260	0.286	0.363	0.319	0.568	0.377	0.976	0.477	1.789	0.664

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.267	0.080	0.343	0.131	0.441	0.154	0.534	0.162	0.610	0.153
tBD	0.300	0.090	0.350	0.147	0.419	0.178	0.488	0.194	0.552	0.192
tC1D	0.343	0.270	0.359	0.355	0.346	0.452	0.324	0.539	0.290	0.618
tC2D	0.347	0.305	0.369	0.360	0.362	0.437	0.345	0.513	0.316	0.585
tC3D	0.363	0.319	0.407	0.359	0.423	0.405	0.427	0.455	0.418	0.506

Timing Numbers for AOIM311D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.277	0.123	0.300	0.136	0.343	0.160	0.429	0.202	0.599	0.284
tBD	0.314	0.126	0.337	0.140	0.380	0.163	0.466	0.206	0.637	0.287
tC1D	0.354	0.291	0.377	0.305	0.420	0.328	0.506	0.371	0.677	0.452
tC2D	0.360	0.326	0.383	0.340	0.426	0.363	0.512	0.406	0.683	0.487
tC3D	0.364	0.341	0.387	0.354	0.430	0.378	0.516	0.420	0.687	0.502

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.300	0.136	0.387	0.155	0.492	0.144	0.587	0.121	0.672	0.086
tBD	0.337	0.140	0.393	0.162	0.475	0.155	0.557	0.137	0.635	0.106
tC1D	0.377	0.305	0.387	0.391	0.370	0.488	0.344	0.575	0.306	0.655
tC2D	0.383	0.340	0.402	0.394	0.394	0.469	0.375	0.544	0.344	0.616
tC3D	0.387	0.354	0.416	0.396	0.417	0.446	0.408	0.501	0.386	0.559

Timing Numbers for AOIM311D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.231	0.135	0.244	0.145	0.267	0.163	0.311	0.191	0.398	0.245
tBD	0.265	0.144	0.278	0.154	0.301	0.172	0.345	0.200	0.432	0.254
tC1D	0.309	0.318	0.322	0.329	0.345	0.347	0.389	0.375	0.476	0.429
tC2D	0.312	0.357	0.325	0.367	0.349	0.385	0.393	0.413	0.479	0.467
tC3D	0.315	0.373	0.328	0.383	0.352	0.401	0.395	0.429	0.482	0.483

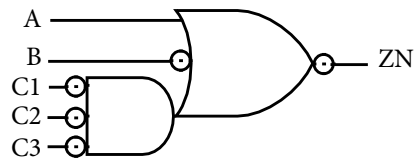
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.244	0.145	0.323	0.172	0.409	0.172	0.487	0.163	0.557	0.141
tBD	0.278	0.154	0.329	0.191	0.394	0.202	0.461	0.202	0.522	0.189
tC1D	0.322	0.329	0.339	0.412	0.328	0.507	0.307	0.592	0.274	0.670
tC2D	0.325	0.367	0.349	0.422	0.344	0.500	0.328	0.577	0.299	0.650
tC3D	0.328	0.383	0.359	0.424	0.361	0.476	0.352	0.533	0.330	0.593

1-1-3 AND-OR-Invert Gates with Inverted B and C Inputs AOIM3M11D1, AOIM3M11D2 and AOIM3M11D4

The AOIM3M11D1 (1x drive), AOIM3M11D2 (2x drive) and AOIM3M11D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A + \text{NOT}(B) + \text{NOT}(C1) \cdot \text{NOT}(C2) \cdot \text{NOT}(C3)]$$



Function Table

INPUTS					OUTPUT
A	B	C1	C2	C3	ZN
H	X	X	X	X	L
X	L	X	X	X	L
X	X	L	L	L	L
L	H	X	X	H	H
L	H	X	H	X	H
L	H	H	X	X	H

Cell Description

Macro Name:	AOIM3M11D1	AOIM3M11D2	AOIM3M11D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	3.	4.	4.5
Leakage Power (pW):	52.4	113.3	154.9

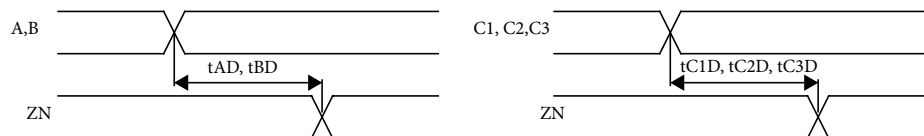
Pin Description

Name	Capacitance (pF)			Description
	AOIM3M11D1	AOIM3M11D2	AOIM3M11D4	
A	0.004	0.004	0.004	Data Input
B	0.004	0.004	0.004	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.005	0.005	0.005	Data Input
C3	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	AOIM3M11D1	AOIM3M11D2	AOIM3M11D4
A	0.002	0.002	0.002
B	0.016	0.016	0.015
C1	0.004	0.004	0.004
C2	0.026	0.026	0.026
C3	0	0	0
ZN	0.019	0.066	0.093

Waveforms



Timing Numbers for AOIM3M11D1:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.173	0.057	0.284	0.081	0.503	0.128	0.940	0.222	1.812	0.409
tBD	0.230	0.142	0.340	0.167	0.558	0.216	0.994	0.310	1.866	0.498
tC1D	0.251	0.210	0.361	0.243	0.580	0.303	1.016	0.409	1.888	0.608
tC2D	0.257	0.244	0.367	0.278	0.586	0.337	1.022	0.443	1.894	0.643
tC3D	0.275	0.258	0.385	0.292	0.604	0.351	1.040	0.457	1.912	0.656

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.284	0.081	0.360	0.133	0.459	0.156	0.555	0.164	0.634	0.155
tBD	0.340	0.167	0.352	0.252	0.342	0.332	0.324	0.403	0.298	0.467
tC1D	0.361	0.243	0.381	0.325	0.374	0.411	0.358	0.488	0.331	0.558
tC2D	0.367	0.278	0.394	0.330	0.395	0.399	0.385	0.468	0.364	0.532
tC3D	0.385	0.292	0.434	0.329	0.457	0.369	0.470	0.411	0.470	0.453

Timing Numbers for AOIM3M11D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.208	0.148	0.231	0.165	0.275	0.192	0.364	0.241	0.542	0.333
tBD	0.266	0.234	0.289	0.251	0.334	0.277	0.423	0.327	0.601	0.419
tC1D	0.287	0.301	0.310	0.318	0.355	0.345	0.444	0.394	0.622	0.486
tC2D	0.292	0.335	0.316	0.352	0.360	0.379	0.449	0.428	0.627	0.520
tC3D	0.310	0.349	0.333	0.366	0.378	0.392	0.467	0.441	0.645	0.533

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.231	0.165	0.307	0.189	0.383	0.189	0.454	0.181	0.517	0.161
tBD	0.289	0.251	0.301	0.333	0.292	0.412	0.274	0.483	0.248	0.545
tC1D	0.310	0.318	0.330	0.398	0.323	0.484	0.307	0.561	0.279	0.630
tC2D	0.316	0.352	0.343	0.403	0.344	0.471	0.334	0.539	0.312	0.602
tC3D	0.333	0.366	0.383	0.402	0.406	0.440	0.418	0.481	0.417	0.522

Timing Numbers for AOIM3M11D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.215	0.173	0.228	0.185	0.252	0.207	0.296	0.240	0.385	0.298
tBD	0.273	0.259	0.286	0.271	0.311	0.293	0.355	0.326	0.444	0.384
tC1D	0.295	0.327	0.308	0.339	0.332	0.361	0.377	0.394	0.466	0.452
tC2D	0.300	0.361	0.313	0.374	0.338	0.395	0.382	0.428	0.471	0.487
tC3D	0.318	0.375	0.331	0.387	0.356	0.409	0.400	0.442	0.489	0.500

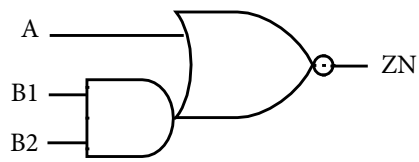
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.228	0.185	0.304	0.210	0.382	0.210	0.453	0.201	0.517	0.182
tBD	0.286	0.271	0.298	0.354	0.289	0.433	0.271	0.503	0.245	0.566
tC1D	0.308	0.339	0.328	0.420	0.321	0.505	0.305	0.582	0.278	0.652
tC2D	0.313	0.374	0.341	0.425	0.342	0.493	0.332	0.561	0.311	0.624
tC3D	0.331	0.387	0.381	0.424	0.405	0.463	0.416	0.503	0.416	0.544

1-2 AND-OR-Invert Gates AOI21D1 , AOI21D2 and AOI21D4

The AOI21D1 (1x drive), AOI21D2 (2x drive) and AOI21D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A + (B1 \cdot B2)]$$



Function Table

INPUTS			OUTPUT
A	B1	B2	ZN
H	X	X	L
X	H	H	L
L	L	X	H
L	X	L	H

Cell Description

Macro Name:	AOI21D1	AOI21D2	AOI21D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.25	2.5	2.5
Leakage Power (pW):	17.1	86.5	119.5

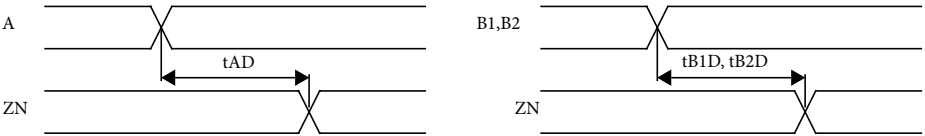
Pin Description

Name	Capacitance (pF)			Description
	AOI21D1	AOI21D2	AOI21D4	
A	0.004	0.004	0.003	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	AOI21D1	AOI21D2	AOI21D4
A	0.002	0.002	0.002
B1	-0	-0	-0
B2	0	0	0
ZN	0.011	0.06	0.09

Waveforms



Timing Numbers for AOI21D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.214	0.054	0.341	0.078	0.595	0.125	1.100	0.218	2.106	0.405
tB1D	0.215	0.080	0.342	0.120	0.596	0.197	1.101	0.352	2.110	0.660
tB2D	0.237	0.081	0.364	0.121	0.616	0.199	1.119	0.354	2.125	0.662

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.341	0.078	0.424	0.127	0.536	0.146	0.647	0.150	0.740	0.136
tB1D	0.342	0.120	0.394	0.179	0.460	0.218	0.528	0.240	0.591	0.243
tB2D	0.364	0.121	0.417	0.158	0.485	0.180	0.557	0.191	0.624	0.185

Timing Numbers for AOI21D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.234	0.145	0.254	0.161	0.294	0.186	0.371	0.230	0.527	0.314
tB1D	0.230	0.166	0.250	0.183	0.290	0.207	0.368	0.252	0.523	0.336
tB2D	0.258	0.166	0.278	0.183	0.318	0.207	0.396	0.250	0.551	0.334

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.254	0.161	0.341	0.180	0.435	0.175	0.519	0.160	0.594	0.133
tB1D	0.250	0.183	0.301	0.216	0.357	0.227	0.413	0.230	0.466	0.220
tB2D	0.278	0.183	0.331	0.206	0.394	0.209	0.456	0.203	0.516	0.186

Timing Numbers for AOI21D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.316	0.231	0.331	0.243	0.358	0.265	0.404	0.296	0.493	0.350
tB1D	0.309	0.212	0.324	0.224	0.351	0.245	0.397	0.277	0.486	0.331
tB2D	0.341	0.213	0.356	0.225	0.384	0.246	0.430	0.278	0.519	0.332

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

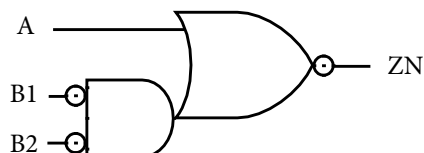
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.331	0.243	0.415	0.290	0.504	0.312	0.582	0.323	0.648	0.320
tB1D	0.324	0.224	0.376	0.243	0.437	0.239	0.501	0.226	0.562	0.205
tB2D	0.356	0.225	0.406	0.239	0.470	0.231	0.537	0.215	0.602	0.190

1-2 AND-OR-Invert Gates with Inverted B Inputs

AOIM21D1, AOIM21D2 and AOIM21D4

The AOIM21D1 (1x drive), AOIM21D2 (2x drive) and AOIM21D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A + \text{NOT}(B1) \cdot \text{NOT}(B2)]$$



Function Table

INPUTS			OUTPUT
A	B1	B2	ZN
H	X	X	L
X	L	L	L
L	H	X	H
L	X	H	H

Cell Description

Macro Name:	AOIM21D1	AOIM21D2	AOIM21D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.5	2.75	3.
Leakage Power (pW):	31.6	92.2	134.3

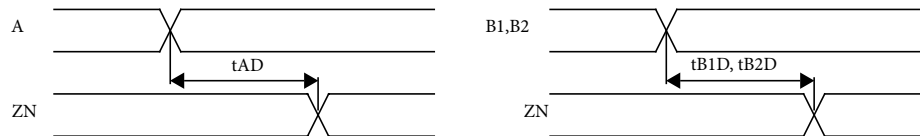
Pin Description

Name	Capacitance (pF)			Description
	AOIM21D1	AOIM21D2	AOIM21D4	
A	0.004	0.004	0.004	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	AOIM21D1	AOIM21D2	AOIM21D4
A	0.001	0.001	0.001
B1	0.016	0.016	0.016
B2	0	0	0
ZN	0.012	0.056	0.085

Waveforms



Timing Numbers for AOIM21D1:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
t_{AD}	0.145	0.058	0.237	0.084	0.419	0.135	0.784	0.238	1.511	0.443
t_{B1D}	0.182	0.165	0.274	0.194	0.456	0.248	0.820	0.349	1.546	0.545
t_{B2D}	0.189	0.182	0.280	0.212	0.462	0.265	0.826	0.366	1.553	0.562

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
t_{AD}	0.237	0.084	0.317	0.136	0.416	0.161	0.507	0.172	0.580	0.166
t_{B1D}	0.274	0.194	0.286	0.275	0.274	0.355	0.254	0.429	0.225	0.497
t_{B2D}	0.280	0.212	0.303	0.266	0.300	0.332	0.289	0.396	0.268	0.455

Timing Numbers for AOIM21D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.176	0.149	0.200	0.167	0.244	0.194	0.334	0.244	0.512	0.336
tB1D	0.214	0.254	0.237	0.272	0.281	0.299	0.371	0.349	0.549	0.441
tB2D	0.220	0.271	0.243	0.289	0.288	0.317	0.377	0.366	0.555	0.458

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.200	0.167	0.272	0.185	0.343	0.183	0.408	0.173	0.466	0.153
tB1D	0.237	0.272	0.248	0.351	0.236	0.430	0.216	0.504	0.187	0.570
tB2D	0.243	0.289	0.265	0.342	0.262	0.406	0.251	0.468	0.230	0.524

Timing Numbers for AOIM21D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.187	0.147	0.200	0.159	0.224	0.179	0.268	0.210	0.357	0.265
tB1D	0.224	0.253	0.237	0.265	0.261	0.284	0.305	0.315	0.395	0.371
tB2D	0.231	0.270	0.244	0.282	0.268	0.301	0.312	0.332	0.401	0.388

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

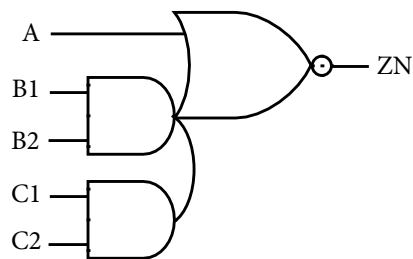
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.200	0.159	0.275	0.181	0.349	0.180	0.416	0.173	0.476	0.154
tB1D	0.237	0.265	0.249	0.342	0.237	0.423	0.217	0.497	0.188	0.563
tB2D	0.244	0.282	0.265	0.335	0.263	0.398	0.252	0.461	0.232	0.517

1-2-2 AND-OR-Invert Gates

AOI221D1, AOI221D2 and AOI221D4

The AOI221D1 (1x drive), AOI221D2 (2x drive) and AOI221D4(4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A + (B1 \cdot B2) + (C1 \cdot C2)]$$



Function Table

INPUTS					OUTPUT
A	B1	B2	C1	C2	ZN
H	X	X	X	X	L
X	H	H	X	X	L
X	X	X	H	H	L
Any other combination					H

Cell Description

Macro Name:	AOI221D1	AOI221D2	AOI221D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.	3.25	3.75
Leakage Power (pW):	26.3	98.4	142.1

Pin Description

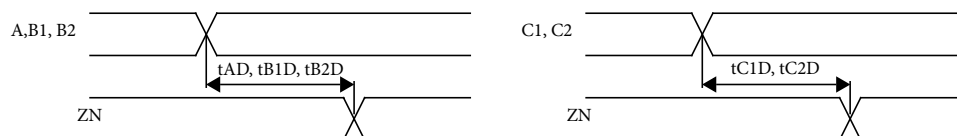
Name	Capacitance (pF)			Description
	AOI221D1	AOI221D2	AOI221D4	
A	0.004	0.004	0.004	Data Input
B1	0.003	0.004	0.004	Data Input
B2	0.004	0.004	0.004	Data Input
C1	0.004	0.005	0.005	Data Input
C2	0.004	0.005	0.005	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

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Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	AOI221D1	AOI221D2	AOI221D4
A	0.008	0.008	0.008
B1	0.001	0.001	0.001
B2	-0	-0	-0
C1	-0	0.001	0.001
C2	0.001	-0	-0
ZN	0.018	0.08	0.112

Waveforms



Timing Numbers for AOI221D1:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.261	0.062	0.402	0.088	0.683	0.142	1.244	0.246	2.361	0.454
tB1D	0.296	0.096	0.443	0.139	0.735	0.224	1.316	0.392	2.475	0.725
tB2D	0.311	0.100	0.451	0.143	0.731	0.228	1.290	0.396	2.407	0.729
tC1D	0.342	0.106	0.482	0.153	0.762	0.242	1.321	0.415	2.438	0.754
tC2D	0.316	0.105	0.456	0.152	0.733	0.241	1.286	0.413	2.391	0.753

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.402	0.088	0.477	0.146	0.582	0.175	0.686	0.188	0.777	0.181
tB1D	0.443	0.139	0.496	0.202	0.568	0.249	0.644	0.278	0.717	0.284
tB2D	0.451	0.143	0.502	0.186	0.570	0.218	0.643	0.237	0.714	0.238
tC1D	0.482	0.153	0.531	0.197	0.576	0.234	0.621	0.258	0.670	0.265
tC2D	0.456	0.152	0.498	0.218	0.538	0.275	0.579	0.314	0.622	0.331

Timing Numbers for AOI221D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.278	0.137	0.297	0.149	0.337	0.169	0.414	0.203	0.568	0.269
tB1D	0.297	0.174	0.316	0.186	0.356	0.206	0.433	0.241	0.587	0.307
tB2D	0.326	0.175	0.345	0.188	0.385	0.208	0.462	0.243	0.616	0.309
tC1D	0.332	0.192	0.352	0.205	0.391	0.225	0.468	0.261	0.622	0.327
tC2D	0.356	0.194	0.375	0.207	0.415	0.227	0.492	0.262	0.646	0.328

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.297	0.149	0.380	0.187	0.479	0.196	0.567	0.193	0.644	0.176
tB1D	0.316	0.186	0.365	0.238	0.426	0.266	0.487	0.281	0.546	0.281
tB2D	0.345	0.188	0.399	0.223	0.466	0.241	0.533	0.246	0.599	0.237
tC1D	0.352	0.205	0.395	0.267	0.431	0.310	0.463	0.339	0.500	0.353
tC2D	0.375	0.207	0.420	0.249	0.458	0.278	0.494	0.296	0.537	0.300

Timing Numbers for AOI221D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.300	0.155	0.313	0.164	0.336	0.179	0.380	0.203	0.466	0.243
tB1D	0.320	0.194	0.333	0.203	0.356	0.218	0.399	0.242	0.486	0.282
tB2D	0.349	0.195	0.362	0.204	0.385	0.219	0.429	0.243	0.516	0.284
tC1D	0.352	0.213	0.365	0.222	0.388	0.238	0.431	0.262	0.518	0.303
tC2D	0.379	0.214	0.392	0.223	0.415	0.239	0.459	0.263	0.546	0.304

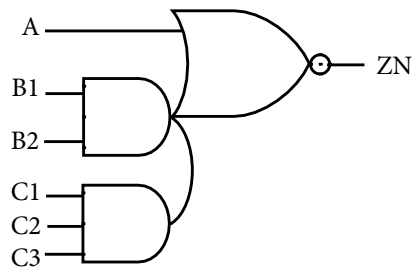
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.313	0.164	0.395	0.202	0.495	0.211	0.585	0.208	0.663	0.191
tB1D	0.333	0.203	0.381	0.254	0.442	0.282	0.505	0.298	0.564	0.299
tB2D	0.362	0.204	0.415	0.239	0.482	0.256	0.551	0.262	0.618	0.254
tC1D	0.365	0.222	0.406	0.283	0.442	0.326	0.476	0.356	0.513	0.370
tC2D	0.392	0.223	0.439	0.264	0.480	0.293	0.519	0.310	0.564	0.313

1-2-3 AND-OR-Invert Gates AOI321D1, AOI321D2 and AOI321D4

The AOI321D1 (1x drive), AOI321D2 (2x drive) and AOI321D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A + (B1 \cdot B2) + (C1 \cdot C2 \cdot C3)]$$



Function Table

INPUTS						OUTPUT
A	B1	B2	C1	C2	C3	ZN
H	X	X	X	X	X	L
X	H	H	X	X	X	L
X	X	X	H	H	H	L
L	L	X	X	X	L	H
L	L	X	X	L	X	H
L	L	X	L	X	X	H
L	X	L	X	X	L	H
L	X	L	X	L	X	H
L	X	L	L	X	X	H

Cell Description

Macro Name:	AOI321D1	AOI321D2	AOI321D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.5	3.5	3.75
Leakage Power (pW):	23.4	94.0	134.3

Pin Description

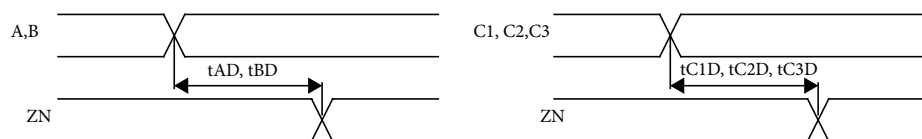
Name	Capacitance (pF)			Description
	AOI321D1	AOI321D2	AOI321D4	
A	0.003	0.003	0.003	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.004	0.004	0.004	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.004	0.004	0.004	Data Input
C3	0.005	0.005	0.005	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

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**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	AOI321D1	AOI321D2	AOI321D4
A	0.008	0.008	0.008
B1	0.001	0.001	0.001
B2	-0	-0	-0
C1	0.001	0.001	0.001
C2	-0	-0	-0
C3	-0	-0	-0
ZN	0.016	0.069	0.1

Waveforms



Timing Numbers for AOI321D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.301	0.089	0.453	0.134	0.755	0.223	1.359	0.400	2.563	0.750
tB1D	0.321	0.095	0.471	0.138	0.770	0.224	1.363	0.394	2.547	0.731
tB2D	0.350	0.096	0.501	0.140	0.803	0.226	1.407	0.395	2.611	0.732
tC1D	0.333	0.135	0.485	0.199	0.787	0.321	1.390	0.558	2.595	1.024
tC2D	0.362	0.141	0.516	0.205	0.821	0.327	1.431	0.564	2.648	1.031
tC3D	0.382	0.146	0.534	0.210	0.835	0.332	1.438	0.568	2.642	1.035

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.453	0.134	0.526	0.205	0.625	0.261	0.727	0.298	0.815	0.311
tB1D	0.471	0.138	0.519	0.201	0.584	0.249	0.655	0.277	0.723	0.285
tB2D	0.501	0.140	0.554	0.181	0.625	0.211	0.700	0.228	0.775	0.227
tC1D	0.485	0.199	0.526	0.262	0.563	0.327	0.599	0.376	0.636	0.402
tC2D	0.516	0.205	0.562	0.251	0.604	0.297	0.644	0.332	0.686	0.348
tC3D	0.534	0.210	0.579	0.237	0.617	0.264	0.655	0.282	0.699	0.287

Timing Numbers for AOI321D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.303	0.199	0.322	0.216	0.358	0.241	0.430	0.285	0.573	0.363
tB1D	0.322	0.208	0.341	0.225	0.378	0.249	0.450	0.293	0.592	0.371
tB2D	0.353	0.209	0.372	0.226	0.408	0.250	0.480	0.294	0.623	0.372
tC1D	0.331	0.248	0.350	0.265	0.386	0.290	0.458	0.334	0.601	0.412
tC2D	0.363	0.254	0.382	0.271	0.418	0.296	0.490	0.340	0.633	0.419
tC3D	0.385	0.258	0.404	0.275	0.440	0.300	0.512	0.344	0.655	0.423

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.322	0.216	0.401	0.275	0.496	0.310	0.579	0.333	0.648	0.337
tB1D	0.341	0.225	0.390	0.275	0.451	0.304	0.513	0.320	0.574	0.320
tB2D	0.372	0.226	0.426	0.260	0.493	0.277	0.562	0.284	0.630	0.275
tC1D	0.350	0.265	0.391	0.323	0.423	0.368	0.453	0.401	0.484	0.419
tC2D	0.382	0.271	0.428	0.313	0.466	0.347	0.501	0.370	0.540	0.379
tC3D	0.404	0.275	0.449	0.302	0.486	0.321	0.520	0.333	0.563	0.333

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Timing Numbers for AOI321D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.312	0.229	0.325	0.241	0.348	0.261	0.392	0.290	0.480	0.341
tB1D	0.331	0.237	0.344	0.249	0.367	0.269	0.411	0.299	0.499	0.350
tB2D	0.361	0.239	0.374	0.251	0.398	0.270	0.442	0.300	0.530	0.351
tC1D	0.339	0.281	0.352	0.293	0.375	0.313	0.419	0.343	0.508	0.394
tC2D	0.372	0.287	0.385	0.299	0.408	0.319	0.452	0.349	0.540	0.400
tC3D	0.394	0.291	0.407	0.303	0.430	0.323	0.474	0.353	0.562	0.404

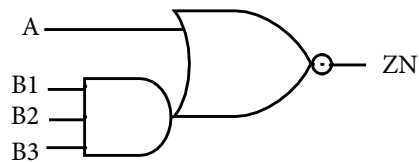
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.325	0.241	0.402	0.301	0.497	0.338	0.582	0.362	0.651	0.368
tB1D	0.344	0.249	0.391	0.301	0.454	0.331	0.517	0.348	0.577	0.350
tB2D	0.374	0.251	0.427	0.286	0.496	0.304	0.566	0.312	0.633	0.304
tC1D	0.352	0.293	0.393	0.351	0.426	0.397	0.456	0.431	0.487	0.450
tC2D	0.385	0.299	0.430	0.341	0.469	0.376	0.504	0.400	0.543	0.410
tC3D	0.407	0.303	0.451	0.330	0.488	0.350	0.523	0.363	0.566	0.364

1-3 AND-OR-Invert Gates AOI31D1, AOI31D2 and AOI31D4

The AOI31D1 (1x drive), AOI31D2 (2x drive) and AOI31D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A + (B1 \cdot B2 \cdot B3)]$$



Function Table

INPUTS				OUTPUT
A	B1	B2	B3	ZN
H	X	X	X	L
X	H	H	H	L
L	X	X	L	H
L	X	L	X	H
L	L	X	X	H

Cell Description

Macro Name:	AOI31D1	AOI31D2	AOI31D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.5	2.5	3.
Leakage Power (pW):	16.7	77.1	129.1

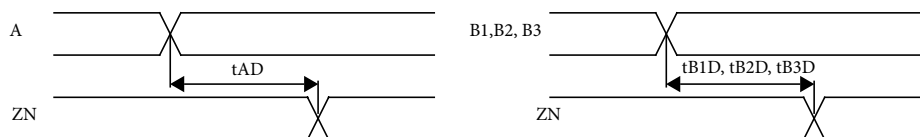
Pin Description

Name	Capacitance (pF)			Description
	AOI31D1	AOI31D2	AOI31D4	
A	0.004	0.004	0.004	Data Input
B1	0.003	0.003	0.003	Data Input
B2	0.004	0.004	0.004	Data Input
B3	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	AOI31D1	AOI31D2	AOI31D4
A	0.003	0.003	0.003
B1	0.002	0.001	0.001
B2	0	0	0
B3	0	-0	-0
ZN	0.011	0.057	0.091

Waveforms



Timing Numbers for AOI31D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.251	0.054	0.387	0.078	0.658	0.125	1.196	0.219	2.271	0.406
tB1D	0.224	0.104	0.356	0.160	0.617	0.269	1.137	0.486	2.174	0.919
tB2D	0.249	0.111	0.380	0.167	0.640	0.276	1.159	0.493	2.197	0.926
tB3D	0.280	0.114	0.416	0.170	0.685	0.279	1.222	0.496	2.296	0.929

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.387	0.078	0.469	0.127	0.581	0.146	0.693	0.150	0.791	0.136
tB1D	0.356	0.160	0.407	0.219	0.472	0.269	0.537	0.301	0.596	0.313
tB2D	0.380	0.167	0.433	0.209	0.499	0.245	0.566	0.269	0.629	0.274
tB3D	0.416	0.170	0.475	0.196	0.547	0.213	0.623	0.221	0.694	0.215

Timing Numbers for AOI31D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.274	0.157	0.298	0.175	0.343	0.203	0.432	0.255	0.611	0.352
tB1D	0.241	0.196	0.265	0.215	0.309	0.243	0.399	0.295	0.578	0.392
tB2D	0.272	0.203	0.296	0.222	0.340	0.250	0.430	0.302	0.609	0.399
tB3D	0.306	0.207	0.330	0.225	0.374	0.253	0.464	0.305	0.643	0.402

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.298	0.175	0.386	0.200	0.490	0.199	0.582	0.188	0.662	0.163
tB1D	0.265	0.215	0.315	0.254	0.369	0.272	0.423	0.281	0.473	0.278
tB2D	0.296	0.222	0.348	0.253	0.408	0.266	0.467	0.270	0.523	0.262
tB3D	0.330	0.225	0.389	0.244	0.457	0.246	0.525	0.240	0.590	0.222

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Timing Numbers for AOI31D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.271	0.188	0.284	0.200	0.308	0.220	0.352	0.250	0.441	0.301
tB1D	0.251	0.226	0.264	0.238	0.288	0.258	0.332	0.288	0.421	0.339
tB2D	0.280	0.232	0.294	0.244	0.318	0.264	0.362	0.294	0.451	0.345
tB3D	0.300	0.236	0.313	0.248	0.337	0.268	0.381	0.299	0.470	0.350

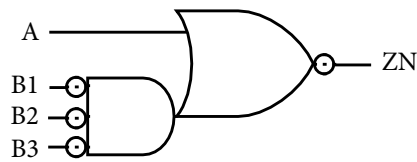
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.284	0.200	0.370	0.223	0.466	0.222	0.553	0.212	0.629	0.189
tB1D	0.264	0.238	0.321	0.275	0.382	0.289	0.441	0.295	0.496	0.288
tB2D	0.294	0.244	0.352	0.273	0.419	0.283	0.483	0.284	0.544	0.273
tB3D	0.313	0.248	0.368	0.267	0.434	0.270	0.499	0.266	0.561	0.251

1-3 AND-OR-Invert Gates with Inverted B Inputs AOIM31D1, AOIM31D2 and AOIM31D4

The AOIM31D1 (1x drive), AOIM31D2 (2x drive) and AOIM31D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A + \text{NOT}(B1) \cdot \text{NOT}(B2) \cdot \text{NOT}(B3)]$$



Function Table

INPUTS				OUTPUT
A	B1	B2	B3	ZN
H	X	X	X	L
X	L	L	L	L
L	X	X	H	H
L	X	H	X	H
L	H	X	X	H

Cell Description

Macro Name:	AOIM31D1	AOIM31D2	AOIM31D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.75	3.	3.25
Leakage Power (pW):	42.9	103.5	144.6

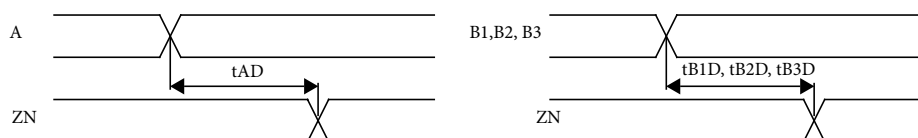
Pin Description

Name	Capacitance (pF)			Description
	AOIM31D1	AOIM31D2	AOIM31D4	
A	0.004	0.004	0.004	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.004	0.004	0.004	Data Input
B3	0.005	0.005	0.005	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	AOIM31D1	AOIM31D2	AOIM31D4
A	0.002	0.002	0.002
B1	0.001	0.001	0.001
B2	0	0	0
B3	0	0	0
ZN	0.017	0.061	0.088

Waveforms



Timing Numbers for AOIM31D1:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.150	0.053	0.245	0.076	0.435	0.121	0.812	0.209	1.563	0.388
tB1D	0.193	0.204	0.288	0.237	0.476	0.296	0.852	0.397	1.603	0.586
tB2D	0.199	0.236	0.294	0.270	0.482	0.328	0.859	0.429	1.610	0.618
tB3D	0.200	0.251	0.295	0.284	0.483	0.343	0.859	0.444	1.610	0.633

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.245	0.076	0.326	0.124	0.426	0.143	0.520	0.149	0.596	0.137
tB1D	0.288	0.237	0.304	0.320	0.293	0.409	0.273	0.490	0.242	0.564
tB2D	0.294	0.270	0.320	0.324	0.318	0.396	0.306	0.468	0.282	0.535
tB3D	0.295	0.284	0.326	0.321	0.329	0.370	0.323	0.425	0.305	0.480

Timing Numbers for AOIM31D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.181	0.147	0.204	0.164	0.249	0.192	0.338	0.242	0.516	0.334
tB1D	0.223	0.292	0.246	0.309	0.291	0.337	0.380	0.387	0.558	0.479
tB2D	0.230	0.324	0.253	0.342	0.297	0.369	0.387	0.419	0.565	0.511
tB3D	0.230	0.338	0.253	0.356	0.298	0.384	0.387	0.433	0.565	0.525

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.204	0.164	0.278	0.178	0.351	0.171	0.420	0.156	0.481	0.132
tB1D	0.246	0.309	0.262	0.391	0.252	0.479	0.231	0.560	0.200	0.633
tB2D	0.253	0.342	0.279	0.394	0.277	0.464	0.265	0.536	0.241	0.601
tB3D	0.253	0.356	0.283	0.392	0.287	0.439	0.281	0.492	0.263	0.545

Timing Numbers for AOIM31D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.196	0.170	0.209	0.183	0.234	0.205	0.278	0.238	0.367	0.297
tB1D	0.238	0.315	0.252	0.328	0.276	0.350	0.321	0.383	0.410	0.442
tB2D	0.245	0.348	0.258	0.360	0.283	0.382	0.327	0.416	0.416	0.475
tB3D	0.245	0.362	0.259	0.375	0.284	0.397	0.328	0.430	0.417	0.489

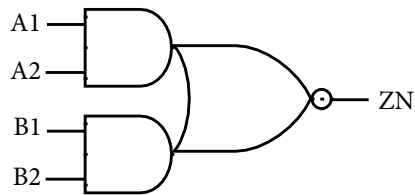
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.209	0.183	0.285	0.197	0.359	0.189	0.428	0.174	0.491	0.151
tB1D	0.252	0.328	0.268	0.410	0.257	0.498	0.236	0.578	0.205	0.652
tB2D	0.258	0.360	0.285	0.413	0.282	0.484	0.270	0.554	0.247	0.620
tB3D	0.259	0.375	0.290	0.412	0.293	0.458	0.286	0.511	0.269	0.564

2-2 AND-OR-Invert Gates AOI22D1, AOI22D2 and AOI22D4

The AOI22D1 (1x drive), AOI22D2 (2x drive) and AOI22D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[(A1 \cdot A2) + (B1 \cdot B2)]$$



Function Table

INPUTS				OUTPUT
A1	A2	B1	B2	ZN
H	H	X	X	L
X	X	H	H	L
Any other combination				H

Cell Description

Macro Name:	AOI22D1	AOI22D2	AOI22D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.5	2.75	3.
Leakage Power (pW):	16.9	58.1	84.7

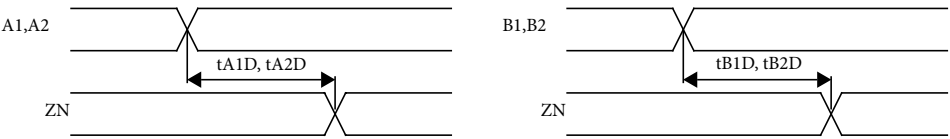
Pin Description

Name	Capacitance (pF)			Description
	AOI22D1	AOI22D2	AOI22D4	
A1	0.004	0.004	0.004	Data Input
A2	0.004	0.004	0.004	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.005	0.005	0.005	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	AOI22D1	AOI22D2	AOI22D4
A1	0.001	0.001	0.001
A2	-0	-0	-0
B1	0.002	0.001	0.001
B2	-0	-0	-0
ZN	0.018	0.067	0.099

Waveforms



Timing Numbers for AOI22D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.178	0.081	0.275	0.121	0.469	0.202	0.854	0.361	1.621	0.679
tA2D	0.187	0.083	0.280	0.123	0.466	0.204	0.836	0.363	1.575	0.681
tB1D	0.204	0.099	0.298	0.139	0.484	0.218	0.855	0.374	1.597	0.684
tB2D	0.220	0.100	0.312	0.140	0.498	0.219	0.868	0.375	1.607	0.686

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.275	0.121	0.355	0.181	0.456	0.220	0.552	0.243	0.628	0.247
tA2D	0.280	0.123	0.359	0.163	0.459	0.189	0.555	0.203	0.631	0.202
tB1D	0.298	0.139	0.352	0.203	0.416	0.254	0.476	0.289	0.531	0.304
tB2D	0.312	0.140	0.369	0.181	0.435	0.212	0.499	0.232	0.559	0.237

Timing Numbers for AOI22D2:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.218	0.183	0.238	0.200	0.278	0.225	0.355	0.269	0.511	0.348
tA2D	0.233	0.185	0.253	0.202	0.293	0.226	0.370	0.270	0.526	0.350
tB1D	0.246	0.200	0.266	0.217	0.306	0.242	0.383	0.286	0.539	0.365
tB2D	0.265	0.201	0.285	0.218	0.325	0.243	0.403	0.287	0.558	0.367

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.238	0.200	0.317	0.236	0.396	0.252	0.466	0.257	0.527	0.249
tA2D	0.253	0.202	0.332	0.228	0.415	0.238	0.488	0.238	0.552	0.226
tB1D	0.266	0.217	0.318	0.267	0.370	0.296	0.420	0.315	0.466	0.320
tB2D	0.285	0.218	0.340	0.250	0.398	0.268	0.453	0.276	0.506	0.272

Timing Numbers for AOI22D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.227	0.186	0.240	0.197	0.263	0.215	0.307	0.242	0.395	0.289
tA2D	0.241	0.187	0.254	0.198	0.278	0.217	0.322	0.244	0.410	0.291
tB1D	0.254	0.202	0.267	0.213	0.290	0.232	0.334	0.259	0.423	0.306
tB2D	0.274	0.204	0.287	0.215	0.310	0.233	0.354	0.260	0.443	0.307

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

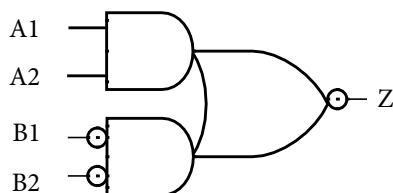
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.240	0.197	0.319	0.236	0.401	0.253	0.474	0.259	0.536	0.253
tA2D	0.254	0.198	0.334	0.226	0.419	0.238	0.494	0.239	0.559	0.228
tB1D	0.267	0.213	0.320	0.265	0.374	0.296	0.425	0.316	0.472	0.323
tB2D	0.287	0.215	0.342	0.248	0.400	0.267	0.457	0.275	0.511	0.272

2-2 AND-OR-Invert Gates with Inverted B Inputs

AOIM22D1 , AOIM22D2 and AOIM22D4

The AOIM22D1 (1x drive), AOIM22D2 (2x drive) and AOIM22D4 (4x drive) cells provide the complex function:

$$Z = \text{NOT}\{(A1 \cdot A2) + [\text{NOT}(B1) \cdot \text{NOT}(B2)]\}$$



Function Table

INPUTS				OUTPUT
A1	A2	B1	B2	Z
H	H	X	X	L
X	X	L	L	L
Any other combination				H

Cell Description

Macro Name:	AOIM22D1	AOIM22D2	AOIM22D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.75	2.5	2.75
Leakage Power (pW):	41.2	77.4	97.6

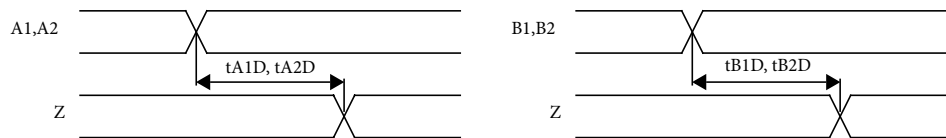
Pin Description

Name	Capacitance (pF)			Description
	AOIM22D1	AOIM22D2	AOIM22D4	
A1	0.005	0.003	0.003	Data Input
A2	0.005	0.004	0.004	Data Input
B1	0.003	0.005	0.005	Data Input
B2	0.004	0.005	0.005	Data Input
Maximum capacitance				
Z	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	AOIM22D1	AOIM22D2	AOIM22D4
A1	0.002	0.001	0.001
A2	-0	-0	-0
B1	0.001	0	0
B2	0	0.001	0.001
Z	0.029	0.058	0.089

Waveforms



Timing Numbers for AOIM22D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.126	0.062	0.197	0.088	0.337	0.140	0.615	0.245	1.169	0.455
tA2D	0.144	0.063	0.214	0.089	0.352	0.142	0.629	0.247	1.182	0.457
tB1D	0.220	0.170	0.290	0.196	0.429	0.246	0.706	0.340	1.258	0.525
tB2D	0.234	0.187	0.304	0.213	0.443	0.263	0.720	0.357	1.273	0.542

Propagation Delays (ns) for sample input transitions:**Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.197	0.088	0.275	0.134	0.369	0.154	0.453	0.161	0.523	0.152
tA2D	0.214	0.089	0.295	0.118	0.393	0.129	0.482	0.128	0.556	0.112
tB1D	0.290	0.196	0.335	0.266	0.360	0.329	0.375	0.383	0.380	0.429
tB2D	0.304	0.213	0.360	0.259	0.397	0.304	0.423	0.346	0.438	0.379

Timing Numbers for AOIM22D2:**Propagation Delays (ns) for sample loads:****Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.206	0.197	0.232	0.216	0.277	0.243	0.367	0.293	0.545	0.385
tA2D	0.220	0.199	0.246	0.218	0.291	0.245	0.381	0.295	0.559	0.387
tB1D	0.120	0.193	0.145	0.215	0.191	0.248	0.280	0.305	0.459	0.404
tB2D	0.112	0.178	0.137	0.200	0.182	0.233	0.271	0.290	0.450	0.388

Propagation Delays (ns) for sample input transitions:**Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.232	0.216	0.312	0.253	0.387	0.269	0.453	0.276	0.509	0.271
tA2D	0.246	0.218	0.331	0.244	0.413	0.253	0.486	0.254	0.548	0.242
tB1D	0.145	0.215	0.189	0.266	0.210	0.330	0.220	0.391	0.217	0.446
tB2D	0.137	0.200	0.174	0.281	0.186	0.368	0.188	0.446	0.176	0.514

Timing Numbers for AOIM22D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.218	0.210	0.233	0.222	0.260	0.243	0.306	0.275	0.396	0.331
tA2D	0.232	0.212	0.247	0.224	0.274	0.245	0.320	0.277	0.409	0.333
tB1D	0.129	0.224	0.144	0.240	0.171	0.265	0.217	0.304	0.306	0.370
tB2D	0.120	0.209	0.135	0.224	0.161	0.249	0.208	0.289	0.297	0.355

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

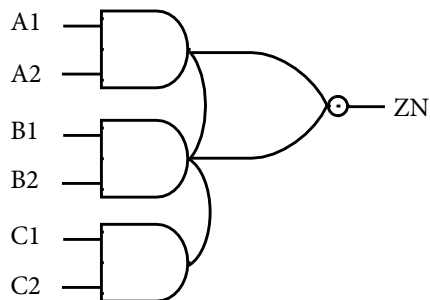
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.233	0.222	0.314	0.261	0.390	0.278	0.457	0.285	0.514	0.281
tA2D	0.247	0.224	0.333	0.252	0.416	0.261	0.489	0.262	0.551	0.251
tB1D	0.144	0.240	0.199	0.291	0.231	0.356	0.251	0.417	0.255	0.472
tB2D	0.135	0.224	0.184	0.305	0.210	0.399	0.222	0.480	0.218	0.549

2-2-2 AND-OR-Invert Gates

AOI222D1, AOI222D2 and AOI222D4

The AOI222D1 (1x drive), AOI222D2 (2x drive) and AOI222D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[(A1 \cdot A2) + (B1 \cdot B2) + (C1 \cdot C2)]$$



Function Table

INPUTS						OUTPUT
A1	A2	B1	B2	C1	C2	ZN
H	H	X	X	X	X	L
X	X	H	H	X	X	L
X	X	X	X	H	H	L
Any other combination						H

Cell Description

Macro Name:	AOI222D1	AOI222D2	AOI222D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.5	3.5	4.
Leakage Power (pW):	27.9	81.3	97.8

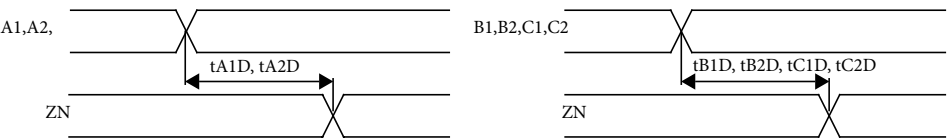
Pin Description

Name	Capacitance (pF)			Description
	AOI222D1	AOI222D2	AOI222D4	
A1	0.004	0.004	0.004	Data Input
A2	0.004	0.004	0.004	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.005	0.004	0.005	Data Input
C1	0.005	0.004	0.004	Data Input
C2	0.004	0.005	0.005	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	AOI222D1	AOI222D2	AOI222D4
A1	-0	0.001	0.001
A2	0.001	-0	-0
B1	0.001	0.001	0.001
B2	-0	-0	-0
C1	-0	0.001	0.001
C2	0.001	-0	-0
ZN	0.027	0.086	0.111

Waveforms



Timing Numbers for AOI222D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.289	0.083	0.421	0.121	0.685	0.196	1.210	0.344	2.258	0.636
tA2D	0.273	0.082	0.409	0.120	0.679	0.194	1.217	0.342	2.291	0.634
tB1D	0.331	0.100	0.466	0.139	0.734	0.214	1.267	0.361	2.331	0.654
tB2D	0.351	0.103	0.483	0.141	0.746	0.216	1.271	0.364	2.319	0.656
tC1D	0.378	0.113	0.510	0.153	0.774	0.230	1.298	0.379	2.346	0.674
tC2D	0.356	0.110	0.490	0.150	0.756	0.227	1.284	0.377	2.340	0.671

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.421	0.121	0.497	0.161	0.601	0.186	0.708	0.198	0.802	0.190
tA2D	0.409	0.120	0.484	0.179	0.589	0.217	0.695	0.237	0.788	0.236
tB1D	0.466	0.139	0.518	0.202	0.586	0.250	0.658	0.280	0.728	0.288
tB2D	0.483	0.141	0.535	0.183	0.600	0.215	0.671	0.235	0.741	0.235
tC1D	0.510	0.153	0.552	0.197	0.592	0.235	0.633	0.261	0.680	0.270
tC2D	0.490	0.150	0.532	0.216	0.572	0.273	0.614	0.312	0.659	0.329

Timing Numbers for AOI222D2:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.294	0.179	0.316	0.192	0.356	0.213	0.438	0.249	0.600	0.317
tA2D	0.311	0.181	0.332	0.194	0.373	0.214	0.454	0.251	0.617	0.318
tB1D	0.352	0.201	0.374	0.214	0.414	0.234	0.496	0.271	0.658	0.338
tB2D	0.372	0.203	0.393	0.216	0.434	0.236	0.515	0.273	0.678	0.341
tC1D	0.375	0.201	0.397	0.213	0.437	0.234	0.519	0.271	0.681	0.338
tC2D	0.398	0.203	0.419	0.216	0.460	0.237	0.541	0.273	0.704	0.341

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.316	0.192	0.395	0.244	0.493	0.271	0.582	0.284	0.654	0.281
tA2D	0.332	0.194	0.412	0.230	0.512	0.248	0.602	0.253	0.677	0.244
tB1D	0.374	0.214	0.426	0.273	0.489	0.313	0.551	0.337	0.611	0.345
tB2D	0.393	0.216	0.445	0.257	0.507	0.285	0.570	0.300	0.632	0.301
tC1D	0.397	0.213	0.442	0.276	0.483	0.320	0.519	0.349	0.559	0.362
tC2D	0.419	0.216	0.464	0.258	0.502	0.289	0.539	0.309	0.583	0.314

Timing Numbers for AOI222D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.302	0.199	0.315	0.209	0.338	0.226	0.382	0.252	0.471	0.299
tA2D	0.319	0.200	0.332	0.210	0.356	0.227	0.400	0.253	0.488	0.300
tB1D	0.361	0.221	0.374	0.231	0.398	0.248	0.442	0.274	0.530	0.321
tB2D	0.383	0.223	0.396	0.233	0.420	0.251	0.464	0.277	0.552	0.324
tC1D	0.387	0.222	0.400	0.232	0.424	0.250	0.468	0.276	0.556	0.323
tC2D	0.409	0.224	0.422	0.234	0.445	0.251	0.489	0.277	0.578	0.324

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

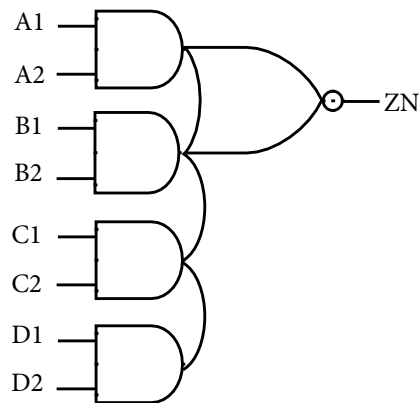
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.315	0.209	0.394	0.259	0.492	0.285	0.580	0.298	0.653	0.295
tA2D	0.332	0.210	0.412	0.245	0.512	0.262	0.602	0.268	0.677	0.260
tB1D	0.374	0.231	0.427	0.289	0.491	0.328	0.553	0.353	0.613	0.360
tB2D	0.396	0.233	0.447	0.273	0.510	0.301	0.573	0.317	0.636	0.317
tC1D	0.400	0.232	0.444	0.293	0.484	0.337	0.521	0.366	0.562	0.379
tC2D	0.422	0.234	0.465	0.275	0.503	0.306	0.541	0.326	0.585	0.331

2-2-2-2 AND-OR-Invert Gates

AOI2222D1, AOI2222D2 and AOI2222D4

The AOI2222D1 (1x drive), AOI2222D2 (2x drive) and AOI2222D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[(A1 \cdot A2) + (B1 \cdot B2) + (C1 \cdot C2) + (D1 \cdot D2)]$$



Function Table

INPUTS								OUTPUT
A1	A2	B1	B2	C1	C2	D1	D2	ZN
H	H	X	X	X	X	X	X	L
X	X	H	H	X	X	X	X	L
X	X	X	X	H	H	X	X	L
X	X	X	X	X	X	H	H	L
Any other combination								H

Cell Description

Macro Name:	AOI2222D1	AOI2222D2	AOI2222D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	4.25	4.5	5.
Leakage Power (pW):	76.1	87.4	110.9

Pin Description

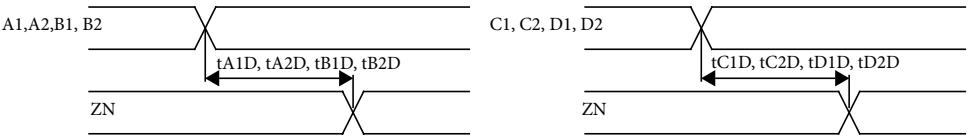
Name	Capacitance (pF)			Description
	AOI2222D1	AOI2222D2	AOI2222D4	
A1	0.004	0.004	0.004	Data Input
A2	0.004	0.004	0.004	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.005	0.005	0.005	Data Input
C1	0.003	0.003	0.003	Data Input
C2	0.004	0.004	0.004	Data Input
D1	0.004	0.004	0.004	Data Input
D2	0.005	0.005	0.005	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

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Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	AOI2222D1	AOI2222D2	AOI2222D4
A1	0.001	0.001	0.001
A2	-0	-0	-0
B1	0.001	0.001	0.001
B2	-0	-0	-0
C1	0.001	0.001	0.001
C2	-0	-0	-0
D1	0.001	0.001	0.001
D2	-0	-0	-0
ZN	0.055	0.071	0.114

Waveforms



Timing Numbers for AOI2222D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.218	0.225	0.253	0.250	0.321	0.296	0.458	0.385	0.731	0.560
tA2D	0.225	0.226	0.261	0.251	0.329	0.297	0.466	0.386	0.738	0.561
tB1D	0.238	0.222	0.273	0.247	0.341	0.293	0.478	0.382	0.751	0.557
tB2D	0.258	0.224	0.293	0.250	0.361	0.296	0.498	0.385	0.771	0.560
tC1D	0.213	0.197	0.249	0.221	0.317	0.267	0.454	0.355	0.726	0.530
tC2D	0.221	0.198	0.257	0.222	0.325	0.268	0.462	0.356	0.735	0.531
tD1D	0.235	0.209	0.271	0.233	0.339	0.279	0.476	0.368	0.749	0.543
tD2D	0.253	0.210	0.289	0.235	0.357	0.281	0.494	0.369	0.767	0.544

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.253	0.250	0.328	0.301	0.398	0.334	0.458	0.357	0.507	0.366
tA2D	0.261	0.251	0.336	0.288	0.409	0.312	0.472	0.327	0.524	0.330
tB1D	0.273	0.247	0.317	0.300	0.357	0.335	0.396	0.360	0.431	0.373
tB2D	0.293	0.250	0.342	0.287	0.389	0.312	0.435	0.328	0.478	0.332
tC1D	0.249	0.221	0.323	0.269	0.395	0.297	0.457	0.315	0.508	0.320
tC2D	0.257	0.222	0.333	0.256	0.407	0.276	0.472	0.288	0.526	0.287
tD1D	0.271	0.233	0.315	0.290	0.352	0.331	0.387	0.361	0.417	0.379
tD2D	0.289	0.235	0.337	0.273	0.382	0.301	0.425	0.320	0.465	0.328

Timing Numbers for AOI2222D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.217	0.237	0.237	0.256	0.274	0.284	0.344	0.333	0.483	0.422
tA2D	0.225	0.238	0.245	0.257	0.281	0.285	0.352	0.335	0.490	0.424
tB1D	0.237	0.234	0.257	0.253	0.293	0.281	0.364	0.331	0.502	0.420
tB2D	0.257	0.236	0.277	0.255	0.314	0.283	0.384	0.333	0.523	0.422
tC1D	0.214	0.209	0.234	0.226	0.270	0.254	0.341	0.302	0.479	0.391
tC2D	0.223	0.210	0.243	0.227	0.279	0.255	0.349	0.304	0.488	0.393
tD1D	0.236	0.221	0.256	0.239	0.293	0.266	0.363	0.315	0.502	0.404
tD2D	0.255	0.222	0.274	0.240	0.311	0.268	0.381	0.316	0.520	0.405

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.237	0.256	0.313	0.307	0.383	0.340	0.443	0.363	0.493	0.373
tA2D	0.245	0.257	0.321	0.293	0.395	0.318	0.458	0.334	0.511	0.337
tB1D	0.257	0.253	0.302	0.305	0.343	0.340	0.382	0.365	0.418	0.378
tB2D	0.277	0.255	0.327	0.292	0.375	0.317	0.421	0.334	0.465	0.338
tC1D	0.234	0.226	0.310	0.274	0.382	0.302	0.445	0.321	0.497	0.327
tC2D	0.243	0.227	0.320	0.261	0.395	0.282	0.460	0.294	0.515	0.294
tD1D	0.256	0.239	0.301	0.295	0.340	0.336	0.375	0.366	0.407	0.384
tD2D	0.274	0.240	0.323	0.278	0.369	0.306	0.412	0.326	0.454	0.334

Timing Numbers for AOI2222D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.244	0.285	0.257	0.297	0.279	0.319	0.317	0.352	0.389	0.407
tA2D	0.252	0.286	0.265	0.299	0.287	0.321	0.325	0.354	0.397	0.408
tB1D	0.263	0.280	0.275	0.293	0.297	0.315	0.335	0.348	0.407	0.402
tB2D	0.284	0.283	0.296	0.295	0.318	0.317	0.356	0.350	0.428	0.405
tC1D	0.239	0.252	0.252	0.264	0.274	0.285	0.312	0.316	0.384	0.370
tC2D	0.248	0.253	0.261	0.265	0.283	0.286	0.321	0.318	0.393	0.372
tD1D	0.261	0.264	0.274	0.276	0.296	0.297	0.334	0.329	0.406	0.383
tD2D	0.280	0.266	0.292	0.278	0.314	0.299	0.352	0.330	0.424	0.384

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

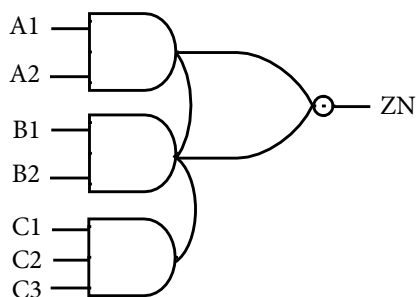
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.257	0.297	0.332	0.350	0.404	0.384	0.467	0.407	0.518	0.418
tA2D	0.265	0.299	0.341	0.336	0.416	0.361	0.481	0.378	0.535	0.383
tB1D	0.275	0.293	0.320	0.346	0.362	0.381	0.403	0.406	0.440	0.420
tB2D	0.296	0.295	0.346	0.334	0.394	0.359	0.442	0.375	0.487	0.380
tC1D	0.252	0.264	0.328	0.312	0.402	0.340	0.465	0.359	0.518	0.366
tC2D	0.261	0.265	0.338	0.299	0.414	0.320	0.480	0.333	0.535	0.334
tD1D	0.274	0.276	0.319	0.332	0.358	0.373	0.396	0.403	0.429	0.422
tD2D	0.292	0.278	0.341	0.316	0.389	0.344	0.434	0.363	0.476	0.372

2-2-3 AND-OR-Invert Gates

AOI322D1, AOI322D2 and AOI322D4

The AOI322D1 (1x drive), AOI322D2 (2x drive) and AOI322D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[(A1 \cdot A2) + (B1 \cdot B2) + (C1 \cdot C2 \cdot C3)]$$



Function Table

INPUTS							OUTPUT
A1	A2	B1	B2	C1	C2	C3	ZN
H	H	X	X	X	X	X	L
X	X	H	H	X	X	X	L
X	X	X	X	H	H	H	L
Any other combination							H

Cell Description

Macro Name:	AOI322D1	AOI322D2	AOI322D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.75	3.75	4.
Leakage Power (pW):	26.7	72.8	94.5

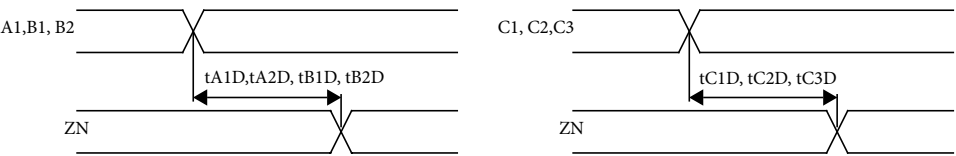
Pin Description

Name	Capacitance (pF)			Description
	AOI322D1	AOI322D2	AOI322D4	
A1	0.003	0.003	0.003	Data Input
A2	0.004	0.004	0.004	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.004	0.004	0.005	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.005	0.005	0.005	Data Input
C3	0.005	0.005	0.005	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	AOI322D1	AOI322D2	AOI322D4
A1	0.001	0.001	0.001
A2	-0	-0	-0
B1	0.001	0.002	0.002
B2	-0	-0	-0
C1	0.001	0.001	0.001
C2	0	0	0
C3	0	0	0
ZN	0.025	0.074	0.108

Waveforms



Timing Numbers for AOI322D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.291	0.104	0.428	0.156	0.702	0.259	1.248	0.461	2.338	0.862
tA2D	0.299	0.105	0.434	0.158	0.703	0.261	1.237	0.464	2.306	0.868
tB1D	0.348	0.106	0.486	0.146	0.760	0.226	1.306	0.383	2.396	0.696
tB2D	0.366	0.108	0.501	0.148	0.770	0.229	1.304	0.386	2.373	0.699
tC1D	0.352	0.157	0.488	0.215	0.757	0.327	1.292	0.544	2.361	0.976
tC2D	0.372	0.164	0.506	0.222	0.771	0.334	1.300	0.552	2.358	0.983
tC3D	0.398	0.167	0.533	0.225	0.801	0.337	1.336	0.555	2.405	0.986

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.428	0.156	0.500	0.222	0.600	0.277	0.701	0.315	0.788	0.330
tA2D	0.434	0.158	0.507	0.202	0.606	0.239	0.707	0.265	0.795	0.271
tB1D	0.486	0.146	0.541	0.210	0.613	0.260	0.689	0.292	0.763	0.302
tB2D	0.501	0.148	0.555	0.191	0.625	0.225	0.700	0.247	0.773	0.249
tC1D	0.488	0.215	0.534	0.278	0.576	0.346	0.616	0.399	0.656	0.429
tC2D	0.506	0.222	0.550	0.268	0.589	0.318	0.627	0.358	0.667	0.381
tC3D	0.533	0.225	0.583	0.253	0.627	0.281	0.669	0.303	0.714	0.312

Timing Numbers for AOI322D2:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.317	0.216	0.341	0.233	0.387	0.258	0.479	0.302	0.661	0.381
tA2D	0.331	0.217	0.355	0.234	0.401	0.259	0.493	0.303	0.675	0.382
tB1D	0.374	0.214	0.398	0.231	0.444	0.255	0.536	0.299	0.719	0.378
tB2D	0.398	0.217	0.422	0.234	0.468	0.258	0.560	0.302	0.743	0.381
tC1D	0.381	0.267	0.405	0.284	0.451	0.309	0.543	0.353	0.726	0.432
tC2D	0.404	0.274	0.427	0.291	0.474	0.315	0.566	0.359	0.748	0.439
tC3D	0.431	0.277	0.455	0.294	0.501	0.318	0.593	0.362	0.775	0.442

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.341	0.233	0.418	0.289	0.513	0.324	0.599	0.347	0.669	0.352
tA2D	0.355	0.234	0.432	0.273	0.529	0.298	0.617	0.312	0.690	0.310
tB1D	0.398	0.231	0.454	0.286	0.522	0.320	0.589	0.340	0.655	0.342
tB2D	0.422	0.234	0.477	0.272	0.544	0.296	0.612	0.309	0.680	0.305
tC1D	0.405	0.284	0.450	0.344	0.489	0.397	0.522	0.436	0.556	0.457
tC2D	0.427	0.291	0.471	0.334	0.508	0.376	0.541	0.407	0.578	0.423
tC3D	0.455	0.294	0.504	0.320	0.546	0.344	0.583	0.360	0.627	0.363

Timing Numbers for AOI322D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.336	0.254	0.350	0.266	0.374	0.285	0.418	0.315	0.506	0.366
tA2D	0.350	0.255	0.364	0.267	0.388	0.286	0.432	0.316	0.521	0.367
tB1D	0.393	0.250	0.406	0.262	0.431	0.282	0.475	0.312	0.563	0.363
tB2D	0.418	0.253	0.431	0.265	0.456	0.285	0.500	0.314	0.588	0.365
tC1D	0.399	0.305	0.412	0.317	0.437	0.337	0.481	0.367	0.569	0.418
tC2D	0.422	0.312	0.435	0.324	0.460	0.343	0.504	0.373	0.592	0.424
tC3D	0.450	0.315	0.464	0.327	0.488	0.346	0.532	0.376	0.620	0.427

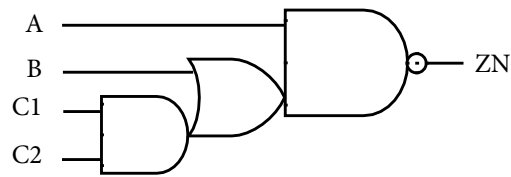
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.350	0.266	0.427	0.321	0.522	0.357	0.609	0.381	0.680	0.387
tA2D	0.364	0.267	0.441	0.305	0.538	0.331	0.627	0.346	0.700	0.345
tB1D	0.406	0.262	0.462	0.317	0.531	0.352	0.599	0.372	0.666	0.374
tB2D	0.431	0.265	0.486	0.303	0.554	0.327	0.623	0.340	0.691	0.336
tC1D	0.412	0.317	0.457	0.377	0.497	0.430	0.531	0.468	0.567	0.490
tC2D	0.435	0.324	0.479	0.367	0.516	0.409	0.550	0.440	0.588	0.456
tC3D	0.464	0.327	0.513	0.354	0.555	0.377	0.594	0.394	0.639	0.398

1-1-2 AND-OR-AND-Invert Gates AON211D1, AON211D2 and AON211D4

The AON211D1 (1x drive), AON211D2 (2x drive) and AON211D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A \cdot (B + (C1 \cdot C2))]$$



Function Table

INPUTS				OUTPUT
A	B	C1	C2	ZN
H	H	X	X	L
H	X	H	H	L
X	L	L	X	H
X	L	X	L	H
L	X	X	X	H

Cell Description

Macro Name:	AON211D1	AON211D2	AON211D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.5	2.75	3.
Leakage Power (pW):	20.6	64.9	79.1

Pin Description

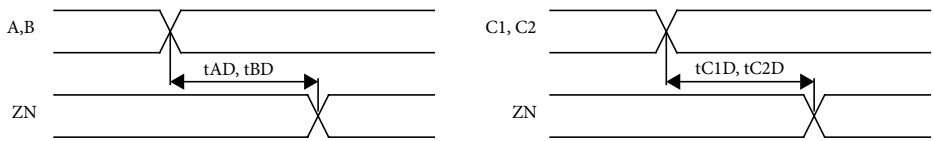
Name	Capacitance (pF)			Description
	AON211D1	AON211D2	AON211D4	
A	0.004	0.004	0.004	Data Input
B	0.004	0.004	0.004	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.004	0.005	0.005	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

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**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	AON211D1	AON211D2	AON211D4
A	0.003	0.003	0.003
B	-0	-0	-0
C1	0	0	0
C2	-0	-0	-0
ZN	0.014	0.061	0.086

Waveforms



Timing Numbers for AON211D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.088	0.114	0.134	0.172	0.225	0.285	0.406	0.511	0.767	0.959
tBD	0.189	0.085	0.282	0.124	0.467	0.201	0.836	0.356	1.571	0.664
tC1D	0.199	0.128	0.291	0.185	0.476	0.298	0.845	0.523	1.582	0.971
tC2D	0.213	0.132	0.305	0.189	0.490	0.302	0.859	0.527	1.593	0.976

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.134	0.172	0.214	0.233	0.291	0.292	0.354	0.336	0.400	0.363
tBD	0.282	0.124	0.362	0.164	0.465	0.187	0.564	0.198	0.643	0.192
tC1D	0.291	0.185	0.343	0.230	0.403	0.273	0.461	0.305	0.512	0.321
tC2D	0.305	0.189	0.358	0.219	0.420	0.245	0.480	0.264	0.533	0.270

Timing Numbers for AON211D2:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.153	0.200	0.173	0.217	0.213	0.242	0.291	0.288	0.446	0.372
tBD	0.229	0.177	0.249	0.194	0.289	0.219	0.367	0.264	0.522	0.348
tC1D	0.237	0.213	0.257	0.230	0.297	0.255	0.374	0.301	0.530	0.385
tC2D	0.254	0.217	0.274	0.234	0.314	0.259	0.392	0.304	0.547	0.388

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.173	0.217	0.234	0.265	0.284	0.299	0.326	0.325	0.359	0.340
tBD	0.249	0.194	0.331	0.220	0.417	0.225	0.492	0.220	0.558	0.204
tC1D	0.257	0.230	0.305	0.263	0.356	0.285	0.405	0.299	0.450	0.302
tC2D	0.274	0.234	0.325	0.257	0.380	0.270	0.433	0.275	0.482	0.271

Timing Numbers for AON211D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.164	0.229	0.177	0.241	0.201	0.263	0.245	0.296	0.334	0.354
tBD	0.241	0.205	0.255	0.217	0.279	0.239	0.323	0.272	0.412	0.330
tC1D	0.248	0.242	0.261	0.254	0.286	0.276	0.330	0.309	0.418	0.367
tC2D	0.267	0.246	0.280	0.258	0.304	0.280	0.348	0.313	0.437	0.371

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

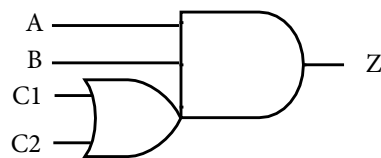
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.177	0.241	0.238	0.290	0.291	0.324	0.335	0.350	0.370	0.366
tBD	0.255	0.217	0.336	0.244	0.422	0.249	0.499	0.245	0.565	0.230
tC1D	0.261	0.254	0.310	0.289	0.362	0.310	0.412	0.324	0.457	0.329
tC2D	0.280	0.258	0.330	0.282	0.387	0.295	0.441	0.300	0.491	0.297

1-1-2 OR-AND Gates

ORA211D1, ORA211D2 and ORA211D4

The ORA211D1 (1x drive), ORA211D2 (2x drive) and ORA211D4 (4x drive) cells provide the complex function:

$$Z = [A \cdot B \cdot (C1 + C2)]$$



Function Table

INPUTS				OUTPUT
A	B	C1	C2	Z
L	X	X	X	L
X	L	X	X	L
X	X	L	L	L
H	H	H	X	H
H	H	X	H	H

Cell Description

Macro Name:	ORA211D1	ORA211D2	ORA211D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.	2.25	2.75
Leakage Power (pW):	37.8	55.5	108.6

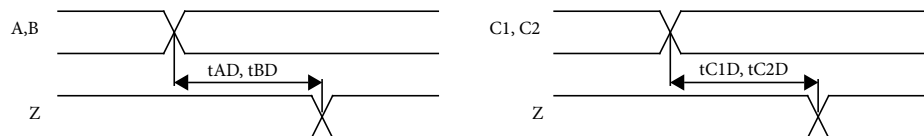
Pin Description

Name	Capacitance (pF)			Description
	ORA211D1	ORA211D2	ORA211D4	
A	0.003	0.003	0.003	Data Input
B	0.003	0.003	0.003	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.004	0.004	0.004	Data Input
Maximum capacitance				
Z	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	ORA211D1	ORA211D2	ORA211D4
A	0.001	0.001	0.001
B	-0	-0	-0
C1	-0	-0	-0
C2	0	0	0
Z	0.031	0.045	0.081

Waveforms



Timing Numbers for ORA211D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.172	0.168	0.222	0.204	0.319	0.264	0.510	0.372	0.891	0.581
tBD	0.179	0.185	0.229	0.221	0.326	0.282	0.517	0.390	0.899	0.599
tC1D	0.163	0.204	0.212	0.238	0.308	0.297	0.499	0.405	0.880	0.614
tC2D	0.194	0.220	0.244	0.253	0.341	0.312	0.532	0.420	0.914	0.629

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.222	0.204	0.265	0.291	0.287	0.380	0.298	0.457	0.295	0.521
tBD	0.229	0.221	0.263	0.311	0.280	0.407	0.287	0.490	0.280	0.559
tC1D	0.212	0.238	0.230	0.318	0.233	0.407	0.229	0.485	0.214	0.552
tC2D	0.244	0.253	0.277	0.304	0.301	0.362	0.317	0.417	0.322	0.464

Timing Numbers for ORA211D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.167	0.173	0.196	0.198	0.248	0.237	0.344	0.301	0.534	0.412
tBD	0.174	0.190	0.204	0.216	0.255	0.255	0.352	0.320	0.542	0.431
tC1D	0.156	0.213	0.185	0.238	0.235	0.276	0.331	0.339	0.521	0.448
tC2D	0.190	0.229	0.219	0.254	0.270	0.291	0.367	0.354	0.557	0.464

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.196	0.198	0.247	0.288	0.279	0.383	0.297	0.465	0.300	0.532
tBD	0.204	0.216	0.242	0.307	0.266	0.408	0.279	0.496	0.277	0.568
tC1D	0.185	0.238	0.207	0.320	0.214	0.413	0.214	0.495	0.202	0.565
tC2D	0.219	0.254	0.255	0.305	0.283	0.365	0.302	0.421	0.311	0.469

Timing Numbers for ORA211D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.190	0.214	0.208	0.232	0.239	0.261	0.291	0.304	0.385	0.375
tBD	0.198	0.229	0.216	0.247	0.247	0.277	0.299	0.321	0.394	0.392
tC1D	0.174	0.257	0.191	0.274	0.221	0.302	0.271	0.344	0.365	0.414
tC2D	0.213	0.273	0.231	0.290	0.262	0.319	0.314	0.361	0.408	0.430

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

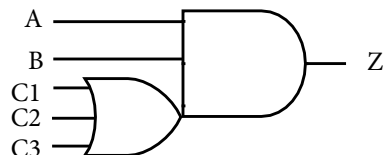
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.208	0.232	0.265	0.323	0.310	0.430	0.340	0.522	0.350	0.596
tBD	0.216	0.247	0.258	0.339	0.292	0.449	0.314	0.546	0.318	0.624
tC1D	0.191	0.274	0.215	0.355	0.227	0.456	0.233	0.547	0.225	0.622
tC2D	0.231	0.290	0.269	0.344	0.304	0.407	0.329	0.466	0.342	0.518

1-1-3 OR-AND Gates

ORA311D1, ORA311D2 and ORA311D4

The ORA311D1 (1x drive), ORA311D2 (2x drive) and ORA311D4 (4x drive) cells provide the complex function:

$$Z = [A \cdot B \cdot (C1 + C2 + C3)]$$



Function Table

INPUTS					OUTPUT
A	B	C1	C2	C3	Z
L	X	X	X	X	L
X	L	X	X	X	L
X	X	L	L	L	L
H	H	H	X	X	H
H	H	X	H	X	H
H	H	X	X	H	H

Cell Description

Macro Name:	ORA311D1	ORA311D2	ORA311D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.25	2.5	2.75
Leakage Power (pW):	34.4	64.2	108.7

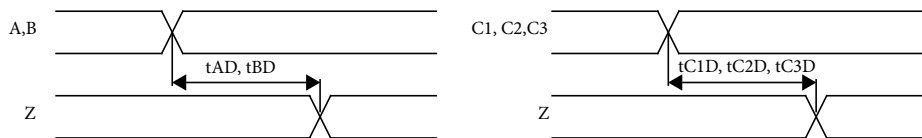
Pin Description

Name	Capacitance (pF)			Description
	ORA311D1	ORA311D2	ORA311D4	
A	0.003	0.003	0.003	Data Input
B	0.003	0.003	0.003	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.005	0.005	0.005	Data Input
C3	0.004	0.004	0.004	Data Input
Maximum capacitance				
Z	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	ORA311D1	ORA311D2	ORA311D4
A	0.001	0.001	0.001
B	-0	-0	-0
C1	0	0	0
C2	0	0	-0
C3	0	0	0
Z	0.031	0.045	0.082

Waveforms



Timing Numbers for ORA311D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.194	0.148	0.245	0.186	0.342	0.251	0.532	0.362	0.910	0.572
tBD	0.198	0.162	0.249	0.200	0.346	0.266	0.536	0.378	0.914	0.589
tC1D	0.178	0.257	0.227	0.297	0.322	0.363	0.512	0.478	0.889	0.690
tC2D	0.195	0.291	0.244	0.331	0.340	0.398	0.530	0.512	0.907	0.725
tC3D	0.209	0.304	0.259	0.344	0.356	0.410	0.547	0.525	0.925	0.738

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.245	0.186	0.303	0.265	0.345	0.338	0.375	0.400	0.390	0.450
tBD	0.249	0.200	0.291	0.284	0.324	0.365	0.346	0.435	0.353	0.492
tC1D	0.227	0.297	0.243	0.374	0.244	0.469	0.237	0.555	0.218	0.630
tC2D	0.244	0.331	0.259	0.382	0.261	0.454	0.257	0.525	0.241	0.592
tC3D	0.259	0.344	0.277	0.382	0.285	0.428	0.287	0.478	0.280	0.531

Timing Numbers for ORA311D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.191	0.149	0.222	0.174	0.274	0.214	0.372	0.282	0.563	0.397
tBD	0.195	0.165	0.225	0.191	0.277	0.231	0.376	0.300	0.567	0.416
tC1D	0.175	0.278	0.204	0.308	0.256	0.352	0.353	0.425	0.544	0.545
tC2D	0.191	0.312	0.221	0.343	0.273	0.387	0.370	0.459	0.561	0.580
tC3D	0.205	0.325	0.236	0.356	0.288	0.400	0.387	0.472	0.578	0.593

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.222	0.174	0.284	0.257	0.335	0.336	0.371	0.402	0.391	0.454
tBD	0.225	0.191	0.270	0.277	0.308	0.364	0.334	0.438	0.346	0.497
tC1D	0.204	0.308	0.222	0.385	0.227	0.484	0.224	0.576	0.208	0.654
tC2D	0.221	0.343	0.237	0.393	0.243	0.466	0.241	0.539	0.228	0.608
tC3D	0.236	0.356	0.255	0.394	0.266	0.441	0.270	0.490	0.264	0.543

Timing Numbers for ORA311D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.217	0.178	0.236	0.194	0.267	0.222	0.320	0.264	0.416	0.335
tBD	0.220	0.203	0.239	0.220	0.270	0.249	0.323	0.293	0.419	0.367
tC1D	0.200	0.340	0.219	0.360	0.250	0.394	0.302	0.443	0.397	0.524
tC2D	0.216	0.375	0.234	0.395	0.265	0.428	0.317	0.477	0.412	0.558
tC3D	0.231	0.389	0.250	0.409	0.282	0.442	0.334	0.491	0.431	0.572

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

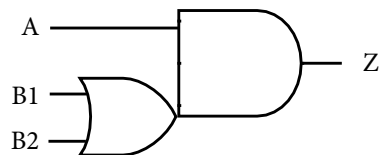
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.236	0.194	0.300	0.281	0.363	0.372	0.409	0.447	0.435	0.505
tBD	0.239	0.220	0.285	0.311	0.329	0.414	0.361	0.501	0.375	0.570
tC1D	0.219	0.360	0.239	0.436	0.249	0.538	0.252	0.638	0.242	0.722
tC2D	0.234	0.395	0.252	0.447	0.262	0.520	0.265	0.596	0.257	0.667
tC3D	0.250	0.409	0.270	0.447	0.284	0.495	0.292	0.545	0.291	0.600

1-2 OR-AND Gates

ORA21D1, ORA21D2 and ORA21D4

The ORA21D1 (1x drive), ORA21D2 (2x drive) and ORA21D4 (4x drive) cells provide the complex function:

$$Z = [A \cdot (B1 + B2)]$$



Function Table

INPUTS			OUTPUT
A	B1	B2	Z
L	X	X	L
X	L	L	L
H	H	X	H
H	X	H	H

Cell Description

Macro Name:	ORA21D1	ORA21D2	ORA21D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.	2.25	2.5
Leakage Power (pW):	47.6	72.7	122.3

Pin Description

Name	Capacitance (pF)			Description
	ORA21D1	ORA21D2	ORA21D4	
A	0.004	0.004	0.004	Data Input
B1	0.004	0.005	0.005	Data Input
B2	0.005	0.005	0.005	Data Input
Maximum capacitance				
Z	0.3	0.6	1.2	Data Output

Pin Powers for

Standard Load = 0.032 pF, Standard input transition = 0.1 ns

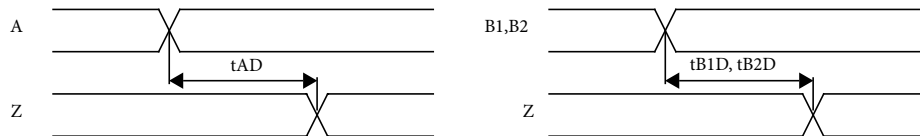
Name	Pin Power (pW/Hz)		
	ORA21D1	ORA21D2	ORA21D4
A	0.003	0.003	0.004
B1	-0	-0	-0
B2	0	0	0
Z	0.028	0.041	0.074

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Waveforms



Timing Numbers for ORA21D1:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.130	0.156	0.178	0.190	0.273	0.250	0.462	0.359	0.840	0.572
tB1D	0.129	0.185	0.177	0.218	0.272	0.276	0.461	0.385	0.838	0.598
tB2D	0.135	0.200	0.183	0.233	0.278	0.291	0.468	0.399	0.845	0.612

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.178	0.190	0.209	0.279	0.214	0.367	0.210	0.446	0.193	0.515
tB1D	0.177	0.218	0.197	0.299	0.196	0.385	0.187	0.461	0.168	0.528
tB2D	0.183	0.233	0.206	0.286	0.210	0.352	0.205	0.416	0.191	0.474

Timing Numbers for ORA21D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.117	0.160	0.143	0.185	0.192	0.223	0.287	0.286	0.477	0.397
tB1D	0.116	0.193	0.142	0.217	0.190	0.253	0.285	0.315	0.475	0.425
tB2D	0.123	0.207	0.149	0.231	0.197	0.267	0.292	0.330	0.482	0.440

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.143	0.185	0.185	0.276	0.199	0.372	0.201	0.455	0.189	0.525
tB1D	0.142	0.217	0.170	0.299	0.177	0.390	0.173	0.470	0.157	0.540
tB2D	0.149	0.231	0.177	0.285	0.187	0.352	0.187	0.417	0.176	0.476

Timing Numbers for ORA21D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.125	0.195	0.140	0.211	0.166	0.240	0.213	0.281	0.304	0.350
tB1D	0.126	0.233	0.141	0.250	0.167	0.277	0.214	0.317	0.304	0.385
tB2D	0.131	0.248	0.146	0.264	0.172	0.291	0.219	0.331	0.310	0.399

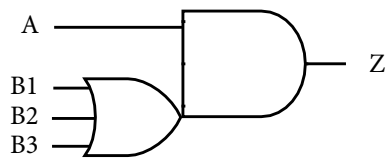
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.140	0.211	0.193	0.305	0.219	0.412	0.231	0.504	0.225	0.580
tB1D	0.141	0.250	0.175	0.332	0.190	0.432	0.194	0.520	0.184	0.594
tB2D	0.146	0.264	0.180	0.318	0.197	0.389	0.204	0.457	0.197	0.519

1-3 OR-AND Gates ORA31D1, ORA31D2 and ORA31D4

The ORA31D1 (1x drive), ORA31D2 (2x drive) and ORA31D4 (4x drive) cells provide the complex function:

$$Z = [A \cdot (B1 + B2 + B3)]$$



Function Table

INPUTS				OUTPUT
A	B1	B2	B3	Z
L	X	X	X	L
X	L	L	L	L
H	H	X	X	H
H	X	H	X	H
H	X	X	H	H

Cell Description

Macro Name:	ORA31D1	ORA31D2	ORA31D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.	2.25	2.5
Leakage Power (pW):	55.7	85.2	120.8

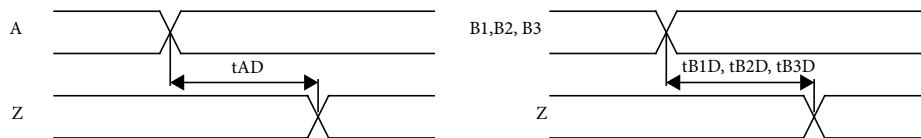
Pin Description

Name	Capacitance (pF)			Description
	ORA31D1	ORA31D2	ORA31D4	
A	0.003	0.003	0.003	Data Input
B1	0.004	0.005	0.005	Data Input
B2	0.004	0.004	0.004	Data Input
B3	0.004	0.005	0.005	Data Input
Maximum capacitance				
Z	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	ORA31D1	ORA31D2	ORA31D4
A	0.004	0.004	0.004
B1	0.001	0.001	0.001
B2	-0	-0	-0
B3	0	0	0
Z	0.029	0.044	0.068

Waveforms



Timing Numbers for ORA31D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.151	0.140	0.200	0.173	0.296	0.231	0.486	0.338	0.864	0.550
tB1D	0.137	0.220	0.185	0.258	0.280	0.322	0.469	0.435	0.847	0.649
tB2D	0.151	0.252	0.199	0.290	0.294	0.354	0.484	0.467	0.861	0.681
tB3D	0.159	0.267	0.208	0.305	0.304	0.370	0.494	0.482	0.872	0.696

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.200	0.173	0.251	0.252	0.279	0.326	0.296	0.390	0.298	0.444
tB1D	0.185	0.258	0.206	0.338	0.205	0.429	0.194	0.510	0.172	0.583
tB2D	0.199	0.290	0.225	0.342	0.233	0.411	0.231	0.480	0.217	0.543
tB3D	0.208	0.305	0.238	0.343	0.252	0.389	0.256	0.438	0.249	0.489

Timing Numbers for ORA31D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.141	0.142	0.167	0.165	0.213	0.199	0.303	0.258	0.481	0.362
tB1D	0.127	0.240	0.152	0.267	0.198	0.307	0.287	0.374	0.464	0.486
tB2D	0.140	0.272	0.166	0.299	0.212	0.339	0.301	0.406	0.479	0.518
tB3D	0.149	0.287	0.175	0.314	0.222	0.354	0.311	0.421	0.489	0.533

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.167	0.165	0.225	0.249	0.262	0.330	0.286	0.399	0.293	0.455
tB1D	0.152	0.267	0.178	0.346	0.185	0.444	0.181	0.532	0.163	0.608
tB2D	0.166	0.299	0.197	0.351	0.211	0.422	0.215	0.492	0.205	0.557
tB3D	0.175	0.314	0.208	0.352	0.228	0.398	0.237	0.448	0.233	0.499

Timing Numbers for ORA31D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.151	0.160	0.169	0.176	0.199	0.205	0.254	0.246	0.359	0.315
tB1D	0.137	0.279	0.154	0.299	0.184	0.332	0.238	0.381	0.342	0.463
tB2D	0.151	0.311	0.168	0.331	0.198	0.364	0.252	0.413	0.357	0.495
tB3D	0.160	0.326	0.177	0.346	0.208	0.380	0.262	0.429	0.367	0.510

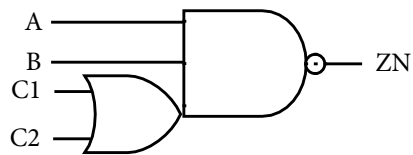
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.169	0.176	0.233	0.264	0.279	0.353	0.309	0.427	0.321	0.487
tB1D	0.154	0.299	0.184	0.377	0.198	0.479	0.200	0.574	0.187	0.653
tB2D	0.168	0.331	0.202	0.383	0.222	0.456	0.232	0.529	0.226	0.596
tB3D	0.177	0.346	0.213	0.385	0.238	0.433	0.252	0.482	0.252	0.534

1-1-2 OR-AND-Invert Gates OAI211D1, OAI211D2 and OAI211D4

The OAI211D1 (1x drive), OAI211D2 (2x drive) and OAI211D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A \cdot B \cdot (C1 + C2)]$$



Function Table

INPUTS				OUTPUT
A	B	C1	C2	ZN
L	X	X	X	H
X	L	X	X	H
X	X	L	L	H
H	H	H	X	L
H	H	X	H	L

Cell Description

Macro Name:	OAI211D1	OAI211D2	OAI211D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.5	2.75	3.
Leakage Power (pW):	7.1	49.9	66.0

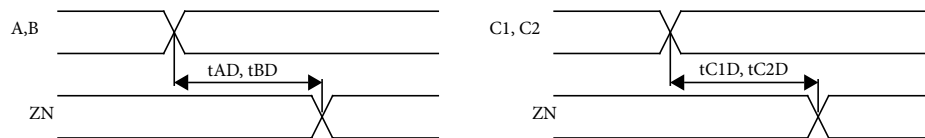
Pin Description

Name	Capacitance (pF)			Description
	OAI211D1	OAI211D2	OAI211D4	
A	0.003	0.003	0.003	Data Input
B	0.004	0.004	0.004	Data Input
C1	0.005	0.005	0.005	Data Input
C2	0.005	0.005	0.005	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	OAI211D1	OAI211D2	OAI211D4
A	0.001	0.001	0.001
B	-0	-0	-0
C1	-0	-0	-0
C2	0	0	0
ZN	0.019	0.062	0.09

Waveforms



Timing Numbers for OAI211D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.115	0.114	0.173	0.166	0.290	0.269	0.522	0.473	0.986	0.881
tBD	0.116	0.120	0.170	0.172	0.276	0.275	0.490	0.479	0.915	0.887
tC1D	0.151	0.110	0.217	0.161	0.348	0.260	0.609	0.458	1.129	0.853
tC2D	0.163	0.130	0.230	0.182	0.361	0.285	0.622	0.490	1.142	0.898

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.173	0.166	0.259	0.226	0.353	0.279	0.432	0.316	0.494	0.333
tBD	0.170	0.172	0.258	0.216	0.353	0.255	0.434	0.283	0.497	0.295
tC1D	0.217	0.161	0.294	0.185	0.385	0.203	0.464	0.214	0.528	0.213
tC2D	0.230	0.182	0.279	0.213	0.337	0.241	0.390	0.262	0.435	0.273

Timing Numbers for OAI211D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.171	0.203	0.196	0.222	0.244	0.252	0.340	0.307	0.531	0.413
tBD	0.179	0.210	0.204	0.229	0.252	0.258	0.348	0.313	0.539	0.419
tC1D	0.202	0.195	0.227	0.214	0.276	0.244	0.371	0.299	0.562	0.405
tC2D	0.216	0.220	0.240	0.239	0.289	0.268	0.384	0.323	0.575	0.430

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.196	0.222	0.271	0.270	0.338	0.300	0.397	0.319	0.444	0.326
tBD	0.204	0.229	0.282	0.266	0.352	0.290	0.413	0.307	0.463	0.311
tC1D	0.227	0.214	0.299	0.234	0.372	0.243	0.436	0.245	0.490	0.237
tC2D	0.240	0.239	0.286	0.266	0.334	0.287	0.379	0.302	0.417	0.309

Timing Numbers for OAI211D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.181	0.233	0.194	0.245	0.218	0.267	0.263	0.300	0.354	0.359
tBD	0.189	0.239	0.202	0.251	0.226	0.273	0.271	0.306	0.361	0.365
tC1D	0.211	0.224	0.224	0.237	0.249	0.258	0.294	0.291	0.384	0.350
tC2D	0.225	0.249	0.238	0.261	0.262	0.283	0.307	0.316	0.397	0.375

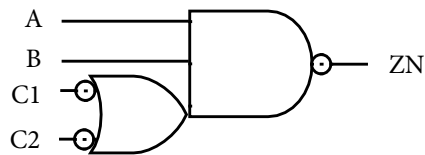
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.194	0.245	0.269	0.294	0.338	0.325	0.397	0.345	0.445	0.354
tBD	0.202	0.251	0.282	0.288	0.356	0.313	0.420	0.328	0.473	0.332
tC1D	0.224	0.237	0.298	0.257	0.371	0.266	0.435	0.268	0.490	0.262
tC2D	0.238	0.261	0.284	0.289	0.333	0.311	0.379	0.326	0.418	0.334

1-1-2 OR-AND-Invert Gates with Inverted C Inputs OAIM211D1, OAIM211D2 and OAIM211D4

The OAIM211D1 (1x drive), OAIM211D2 (2x drive) and OAIM211D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A \cdot B \cdot (\text{NOT}(C1) + \text{NOT}(C2))]$$



Function Table

INPUTS				OUTPUT
A	B	C1	C2	ZN
L	X	X	X	H
X	L	X	X	H
X	X	H	H	H
H	H	L	X	L
H	H	X	L	L

Cell Description

Macro Name:	OAIM211D1	OAIM211D2	OAIM211D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.	3.	3.25
Leakage Power (pW):	22.5	65.5	72.8

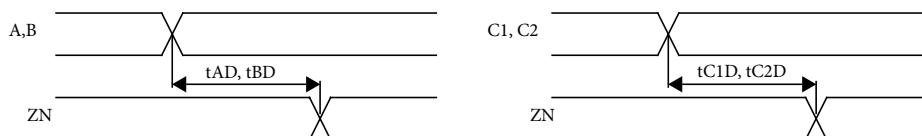
Pin Description

Name	Capacitance (pF)			Description
	OAIM211D1	OAIM211D2	OAIM211D4	
A	0.003	0.003	0.003	Data Input
B	0.004	0.004	0.004	Data Input
C1	0.003	0.003	0.003	Data Input
C2	0.003	0.003	0.003	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	OAIM211D1	OAIM211D2	OAIM211D4
A	0.002	0.002	0.002
B	-0	-0	-0
C1	0.001	0.001	0.001
C2	0.017	0.017	0.017
ZN	0.014	0.058	0.082

Waveforms



Timing Numbers for OAIM211D1:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.118	0.083	0.182	0.127	0.309	0.216	0.560	0.394	1.064	0.750
tBD	0.134	0.089	0.199	0.133	0.330	0.222	0.590	0.401	1.109	0.757
tC1D	0.188	0.152	0.254	0.196	0.384	0.286	0.645	0.464	1.164	0.820
tC2D	0.190	0.169	0.256	0.214	0.386	0.303	0.647	0.482	1.166	0.838

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.182	0.127	0.270	0.181	0.371	0.218	0.457	0.239	0.526	0.242
tBD	0.199	0.133	0.290	0.172	0.394	0.198	0.486	0.212	0.559	0.210
tC1D	0.254	0.196	0.287	0.263	0.301	0.321	0.309	0.372	0.307	0.415
tC2D	0.256	0.214	0.278	0.292	0.283	0.364	0.281	0.428	0.269	0.483

Timing Numbers for OAIM211D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.163	0.170	0.188	0.189	0.234	0.218	0.326	0.272	0.510	0.375
tBD	0.182	0.176	0.206	0.195	0.252	0.224	0.344	0.278	0.528	0.381
tC1D	0.238	0.240	0.262	0.259	0.308	0.288	0.400	0.342	0.584	0.445
tC2D	0.240	0.257	0.264	0.276	0.311	0.305	0.403	0.359	0.586	0.462

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.188	0.189	0.264	0.218	0.335	0.228	0.397	0.230	0.451	0.221
tBD	0.206	0.195	0.288	0.218	0.368	0.224	0.438	0.222	0.498	0.210
tC1D	0.262	0.259	0.294	0.324	0.309	0.382	0.317	0.432	0.314	0.475
tC2D	0.264	0.276	0.286	0.354	0.291	0.425	0.288	0.489	0.275	0.544

Timing Numbers for OAIM211D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.169	0.194	0.183	0.208	0.209	0.230	0.260	0.266	0.360	0.329
tBD	0.187	0.200	0.202	0.214	0.228	0.236	0.279	0.272	0.379	0.335
tC1D	0.244	0.264	0.259	0.278	0.285	0.300	0.336	0.336	0.436	0.399
tC2D	0.246	0.282	0.261	0.296	0.287	0.318	0.338	0.353	0.438	0.416

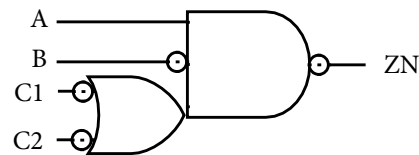
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.183	0.208	0.261	0.237	0.332	0.246	0.396	0.249	0.450	0.241
tBD	0.202	0.214	0.286	0.237	0.366	0.244	0.437	0.243	0.497	0.231
tC1D	0.259	0.278	0.291	0.343	0.306	0.401	0.313	0.452	0.311	0.495
tC2D	0.261	0.296	0.283	0.373	0.288	0.444	0.285	0.509	0.272	0.564

1-1-2 OR-AND-Invert Gates with Inverted B and C Inputs OAIM2M11D1, OAIM2M11D2 and OAIM2M11D4

The OAIM2M11D1 (1x drive), OAIM2M11D2 (2x drive) and OAIM2M11D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A \cdot \text{NOT}(B) \cdot (\text{NOT}(C1) + \text{NOT}(C2))]$$



Function Table

INPUTS				OUTPUT
A	B	C1	C2	ZN
L	X	X	X	H
X	H	X	X	H
X	X	H	H	H
H	L	L	X	L
H	L	X	L	L

Cell Description

Macro Name:	OAIM2M11D1	OAIM2M11D2	OAIM2M11D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.5	3.75	4.
Leakage Power (pW):	73.7	136.0	180.9

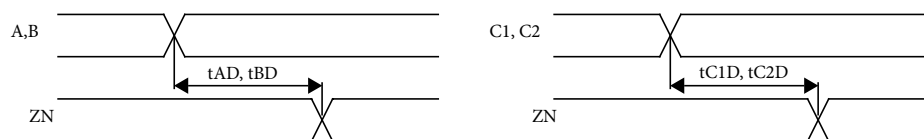
Pin Description

Name	Capacitance (pF)			Description
	OAIM2M11D1	OAIM2M11D2	OAIM2M11D4	
A	0.003	0.004	0.004	Data Input
B	0.002	0.002	0.002	Data Input
C1	0.003	0.003	0.003	Data Input
C2	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	OAIM2M11D1	OAIM2M11D2	OAIM2M11D4
A	0.002	0.002	0.002
B	0.011	0.012	0.012
C1	0.001	0.001	0.001
C2	0.017	0.017	0.017
ZN	0.014	0.056	0.085

Waveforms



Timing Numbers for OAIM2M11D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.119	0.102	0.186	0.161	0.318	0.279	0.582	0.516	1.109	0.988
tBD	0.163	0.193	0.229	0.253	0.359	0.371	0.620	0.608	1.140	1.080
tC1D	0.181	0.175	0.252	0.235	0.392	0.353	0.671	0.590	1.228	1.062
tC2D	0.183	0.188	0.254	0.247	0.394	0.366	0.673	0.603	1.230	1.075

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.186	0.161	0.273	0.220	0.369	0.271	0.452	0.307	0.515	0.323
tBD	0.229	0.253	0.260	0.337	0.267	0.416	0.265	0.486	0.253	0.548
tC1D	0.252	0.235	0.279	0.303	0.287	0.365	0.288	0.421	0.280	0.469
tC2D	0.254	0.247	0.274	0.323	0.276	0.392	0.273	0.454	0.259	0.509

Timing Numbers for OAIM2M11D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.161	0.181	0.187	0.200	0.235	0.230	0.330	0.285	0.521	0.391
tBD	0.210	0.276	0.235	0.295	0.284	0.324	0.379	0.379	0.570	0.486
tC1D	0.225	0.254	0.250	0.273	0.299	0.302	0.394	0.357	0.585	0.464
tC2D	0.228	0.266	0.253	0.285	0.301	0.315	0.396	0.370	0.587	0.476

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.187	0.200	0.259	0.238	0.323	0.260	0.378	0.273	0.424	0.275
tBD	0.235	0.295	0.269	0.379	0.278	0.459	0.278	0.531	0.265	0.593
tC1D	0.250	0.273	0.277	0.340	0.284	0.402	0.285	0.458	0.276	0.506
tC2D	0.253	0.285	0.272	0.360	0.274	0.430	0.269	0.492	0.255	0.547

Timing Numbers for OAIM2M11D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.167	0.210	0.180	0.222	0.205	0.244	0.250	0.277	0.340	0.336
tBD	0.218	0.307	0.231	0.320	0.255	0.341	0.300	0.374	0.391	0.433
tC1D	0.233	0.283	0.246	0.295	0.270	0.317	0.315	0.350	0.405	0.408
tC2D	0.235	0.295	0.248	0.308	0.272	0.329	0.317	0.362	0.407	0.421

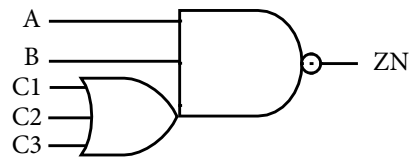
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.180	0.222	0.254	0.262	0.319	0.283	0.376	0.296	0.422	0.299
tBD	0.231	0.320	0.265	0.405	0.276	0.486	0.276	0.558	0.264	0.620
tC1D	0.246	0.295	0.272	0.363	0.280	0.425	0.280	0.481	0.271	0.529
tC2D	0.248	0.308	0.267	0.383	0.269	0.452	0.264	0.515	0.250	0.569

1-1-3 OR-AND-Invert Gates OAI311D1, OAI311D2 and OAI311D4

The OAI311D1 (1x drive), OAI311D2 (2x drive) and OAI311D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A \cdot B \cdot (C1 + C2 + C3)]$$



Function Table

INPUTS					OUTPUT
A	B	C1	C2	C3	ZN
L	X	X	X	X	H
X	L	X	X	X	H
X	X	L	L	L	H
H	H	H	X	X	L
H	H	X	H	X	L
H	H	X	X	H	L

Cell Description

Macro Name:	OAI311D1	OAI311D2	OAI311D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.	3.25	3.5
Leakage Power (pW):	12.8	55.6	71.7

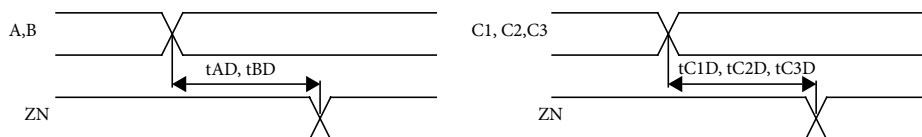
Pin Description

Name	Capacitance (pF)			Description
	OAI311D1	OAI311D2	OAI311D4	
A	0.003	0.003	0.003	Data Input
B	0.003	0.003	0.003	Data Input
C1	0.005	0.005	0.005	Data Input
C2	0.005	0.005	0.005	Data Input
C3	0.005	0.005	0.005	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	OAI311D1	OAI311D2	OAI311D4
A	0.001	0.001	0.001
B	0	-0	-0
C1	0.001	0.001	0.001
C2	0	0	0
C3	0	0	0
ZN	0.018	0.062	0.091

Waveforms



Timing Numbers for OAI311D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout loads	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.134	0.133	0.207	0.191	0.353	0.306	0.644	0.531	1.225	0.978
tBD	0.145	0.138	0.218	0.196	0.363	0.311	0.654	0.536	1.235	0.983
tC1D	0.215	0.121	0.319	0.177	0.524	0.290	0.932	0.513	1.746	0.960
tC2D	0.245	0.139	0.349	0.195	0.554	0.308	0.962	0.531	1.775	0.978
tC3D	0.257	0.150	0.361	0.208	0.566	0.322	0.974	0.547	1.787	0.994

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.207	0.191	0.295	0.255	0.397	0.316	0.488	0.360	0.558	0.381
tBD	0.218	0.196	0.307	0.242	0.412	0.286	0.507	0.319	0.580	0.332
tC1D	0.319	0.177	0.391	0.200	0.488	0.213	0.583	0.218	0.661	0.212
tC2D	0.349	0.195	0.399	0.219	0.467	0.236	0.536	0.245	0.600	0.243
tC3D	0.361	0.208	0.398	0.232	0.441	0.251	0.485	0.265	0.531	0.268

Timing Numbers for OAI311D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout loads	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.189	0.226	0.214	0.245	0.262	0.275	0.358	0.330	0.549	0.437
tBD	0.202	0.231	0.227	0.250	0.275	0.280	0.371	0.335	0.562	0.441
tC1D	0.253	0.205	0.278	0.224	0.326	0.254	0.421	0.309	0.612	0.415
tC2D	0.286	0.229	0.311	0.248	0.360	0.278	0.455	0.333	0.646	0.439
tC3D	0.298	0.243	0.323	0.262	0.371	0.292	0.467	0.347	0.658	0.454

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.214	0.245	0.294	0.302	0.369	0.341	0.434	0.367	0.487	0.377
tBD	0.227	0.250	0.311	0.292	0.393	0.322	0.464	0.341	0.523	0.345
tC1D	0.278	0.224	0.354	0.242	0.440	0.245	0.516	0.242	0.583	0.226
tC2D	0.311	0.248	0.361	0.268	0.423	0.277	0.484	0.280	0.541	0.272
tC3D	0.323	0.262	0.359	0.285	0.398	0.299	0.439	0.308	0.481	0.307

Timing Numbers for OAI311D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout loads	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.203	0.257	0.217	0.269	0.242	0.291	0.287	0.324	0.377	0.383
tBD	0.217	0.261	0.231	0.274	0.256	0.296	0.301	0.329	0.391	0.388
tC1D	0.266	0.235	0.280	0.248	0.305	0.269	0.350	0.302	0.440	0.361
tC2D	0.300	0.259	0.314	0.271	0.338	0.293	0.383	0.326	0.473	0.385
tC3D	0.312	0.274	0.325	0.286	0.350	0.308	0.395	0.341	0.485	0.400

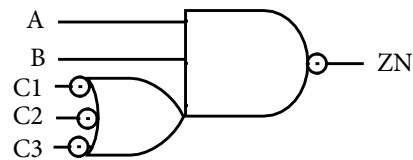
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.217	0.269	0.299	0.326	0.378	0.366	0.445	0.391	0.501	0.402
tBD	0.231	0.274	0.316	0.316	0.402	0.347	0.475	0.366	0.536	0.371
tC1D	0.280	0.248	0.356	0.266	0.442	0.270	0.519	0.266	0.587	0.252
tC2D	0.314	0.271	0.363	0.292	0.426	0.302	0.489	0.305	0.547	0.298
tC3D	0.325	0.286	0.362	0.309	0.402	0.324	0.444	0.333	0.487	0.334

1-1-3 OR-AND-Invert Gates with Inverted C Inputs OAIM311D1, OAIM311D2 and OAIM311D4

The OAIM311D1 (1x drive), OAIM311D2 (2x drive) and OAIM311D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A \cdot B \cdot (\text{NOT}(C1) + \text{NOT}(C2) + \text{NOT}(C3))]$$



Function Table

INPUTS					OUTPUT
A	B	C1	C2	C3	ZN
L	X	X	X	X	H
X	L	X	X	X	H
X	X	H	H	H	H
H	H	L	X	X	L
H	H	X	L	X	L
H	H	X	X	L	L

Cell Description

Macro Name:	OAIM311D1	OAIM311D2	OAIM311D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.25	3.25	3.75
Leakage Power (pW):	20.9	65.1	81.5

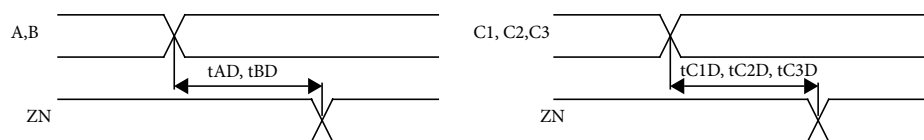
Pin Description

Name	Capacitance (pF)			Description
	OAIM311D1	OAIM311D2	OAIM311D4	
A	0.003	0.004	0.003	Data Input
B	0.004	0.004	0.004	Data Input
C1	0.003	0.003	0.003	Data Input
C2	0.004	0.004	0.004	Data Input
C3	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	OAIM311D1	OAIM311D2	OAIM311D4
A	0.001	0.001	0.001
B	-0	-0	-0
C1	0.001	0.001	0.001
C2	-0	-0	-0
C3	0	0	0
ZN	0.016	0.06	0.088

Waveforms



Timing Numbers for OAIM311D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.107	0.104	0.163	0.163	0.276	0.281	0.502	0.518	0.953	0.990
tBD	0.115	0.109	0.171	0.168	0.284	0.287	0.510	0.524	0.961	0.996
tC1D	0.195	0.175	0.256	0.234	0.377	0.352	0.618	0.589	1.100	1.061
tC2D	0.200	0.188	0.261	0.248	0.382	0.366	0.624	0.603	1.106	1.075
tC3D	0.205	0.193	0.266	0.253	0.387	0.371	0.628	0.608	1.110	1.080

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.163	0.163	0.248	0.223	0.336	0.277	0.410	0.315	0.465	0.336
tBD	0.171	0.168	0.257	0.212	0.350	0.251	0.427	0.279	0.487	0.293
tC1D	0.256	0.234	0.298	0.297	0.324	0.349	0.343	0.394	0.353	0.430
tC2D	0.261	0.248	0.293	0.321	0.311	0.385	0.322	0.441	0.323	0.488
tC3D	0.266	0.253	0.286	0.328	0.297	0.395	0.303	0.453	0.299	0.502

Timing Numbers for OAIM311D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.156	0.178	0.181	0.196	0.229	0.225	0.325	0.280	0.516	0.385
tBD	0.166	0.183	0.191	0.201	0.239	0.230	0.335	0.285	0.526	0.390
tC1D	0.246	0.249	0.271	0.267	0.319	0.296	0.414	0.351	0.605	0.456
tC2D	0.251	0.262	0.276	0.280	0.325	0.309	0.420	0.364	0.611	0.470
tC3D	0.256	0.267	0.281	0.285	0.329	0.314	0.425	0.369	0.616	0.475

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.181	0.196	0.250	0.237	0.307	0.264	0.356	0.283	0.395	0.291
tBD	0.191	0.201	0.267	0.233	0.333	0.252	0.390	0.264	0.437	0.265
tC1D	0.271	0.267	0.312	0.329	0.338	0.382	0.356	0.426	0.365	0.462
tC2D	0.276	0.280	0.307	0.353	0.324	0.417	0.333	0.473	0.333	0.520
tC3D	0.281	0.285	0.301	0.360	0.310	0.427	0.314	0.486	0.309	0.535

Timing Numbers for OAIM311D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.163	0.203	0.176	0.215	0.200	0.235	0.246	0.268	0.337	0.325
tBD	0.174	0.208	0.187	0.220	0.211	0.241	0.257	0.273	0.348	0.330
tC1D	0.254	0.274	0.267	0.286	0.291	0.307	0.337	0.339	0.428	0.396
tC2D	0.259	0.287	0.272	0.299	0.297	0.320	0.342	0.352	0.433	0.409
tC3D	0.264	0.293	0.277	0.305	0.301	0.325	0.347	0.358	0.438	0.415

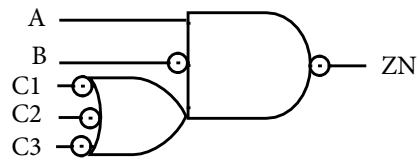
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.176	0.215	0.246	0.257	0.305	0.284	0.356	0.304	0.397	0.313
tBD	0.187	0.220	0.263	0.253	0.331	0.272	0.390	0.284	0.438	0.286
tC1D	0.267	0.286	0.308	0.348	0.334	0.401	0.353	0.445	0.362	0.481
tC2D	0.272	0.299	0.303	0.372	0.320	0.437	0.331	0.493	0.331	0.540
tC3D	0.277	0.305	0.297	0.379	0.307	0.446	0.311	0.505	0.306	0.554

1-1-3 OR-AND-Invert Gates with Inverted B and C Inputs OAIM3M11D1, OAIM3M11D2 and OAIM3M11D4

The OAIM3M11D1 (1x drive), OAIM3M11D2 (2x drive) and OAIM3M11D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A \cdot \text{NOT}(B) \cdot (\text{NOT}(C1) + \text{NOT}(C2) + \text{NOT}(C3))]$$



Function Table

INPUTS					OUTPUT
A	B	C1	C2	C3	ZN
L	X	X	X	X	H
X	H	X	X	X	H
X	X	H	H	H	H
H	L	L	X	X	L
H	L	X	L	X	L
H	L	X	X	L	L

Cell Description

Macro Name:	OAIM3M11D1	OAIM3M11D2	OAIM3M11D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.75	4.	4.25
Leakage Power (pW):	70.0	132.4	176.9

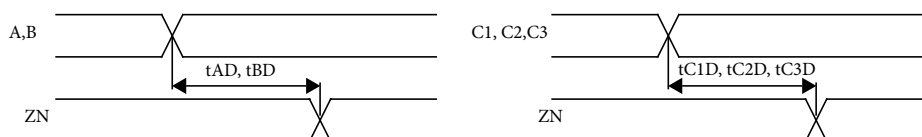
Pin Description

Name	Capacitance (pF)			Description
	OAIM3M11D1	OAIM3M11D2	OAIM3M11D4	
A	0.003	0.004	0.004	Data Input
B	0.002	0.002	0.002	Data Input
C1	0.003	0.003	0.003	Data Input
C2	0.003	0.003	0.003	Data Input
C3	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	OAIM3M11D1	OAIM3M11D2	OAIM3M11D4
A	0.002	0.003	0.003
B	0.011	0.011	0.011
C1	0.001	0.001	0.001
C2	-0	-0	-0
C3	0	0	0
ZN	0.017	0.061	0.089

Waveforms



Timing Numbers for OAIM3M11D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.122	0.098	0.193	0.156	0.334	0.272	0.614	0.504	1.172	0.967
tBD	0.168	0.197	0.239	0.255	0.380	0.372	0.660	0.604	1.222	1.066
tC1D	0.211	0.178	0.287	0.236	0.437	0.353	0.736	0.585	1.334	1.047
tC2D	0.217	0.195	0.293	0.253	0.443	0.369	0.742	0.601	1.340	1.064
tC3D	0.222	0.202	0.298	0.260	0.448	0.377	0.747	0.609	1.345	1.072

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.193	0.156	0.281	0.215	0.380	0.265	0.466	0.297	0.532	0.311
tBD	0.239	0.255	0.268	0.343	0.272	0.427	0.267	0.502	0.251	0.568
tC1D	0.287	0.236	0.323	0.308	0.340	0.370	0.350	0.425	0.350	0.472
tC2D	0.293	0.253	0.321	0.332	0.332	0.406	0.336	0.470	0.331	0.526
tC3D	0.298	0.260	0.316	0.343	0.320	0.421	0.318	0.490	0.307	0.549

Timing Numbers for OAIM3M11D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.166	0.173	0.191	0.191	0.239	0.220	0.335	0.275	0.525	0.381
tBD	0.215	0.273	0.240	0.291	0.288	0.320	0.383	0.375	0.574	0.480
tC1D	0.255	0.252	0.280	0.271	0.328	0.299	0.424	0.354	0.614	0.460
tC2D	0.261	0.269	0.286	0.287	0.335	0.316	0.430	0.371	0.621	0.476
tC3D	0.266	0.276	0.291	0.294	0.339	0.323	0.435	0.378	0.625	0.484

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.191	0.191	0.267	0.229	0.333	0.248	0.392	0.259	0.441	0.259
tBD	0.240	0.291	0.270	0.378	0.275	0.462	0.271	0.538	0.254	0.604
tC1D	0.280	0.271	0.316	0.341	0.333	0.403	0.343	0.458	0.342	0.504
tC2D	0.286	0.287	0.314	0.367	0.325	0.439	0.329	0.503	0.322	0.558
tC3D	0.291	0.294	0.309	0.377	0.313	0.454	0.309	0.523	0.296	0.582

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Timing Numbers for OAIM3M11D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.174	0.196	0.187	0.208	0.211	0.228	0.257	0.260	0.348	0.317
tBD	0.224	0.296	0.237	0.308	0.261	0.328	0.306	0.360	0.397	0.417
tC1D	0.264	0.276	0.277	0.288	0.302	0.308	0.347	0.340	0.438	0.397
tC2D	0.271	0.292	0.284	0.304	0.308	0.324	0.353	0.356	0.444	0.413
tC3D	0.276	0.299	0.289	0.311	0.313	0.332	0.358	0.364	0.449	0.421

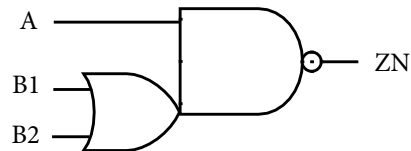
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.187	0.208	0.263	0.246	0.333	0.265	0.393	0.276	0.443	0.276
tBD	0.237	0.308	0.266	0.395	0.271	0.479	0.267	0.555	0.251	0.620
tC1D	0.277	0.288	0.313	0.358	0.331	0.420	0.340	0.475	0.339	0.521
tC2D	0.284	0.304	0.311	0.383	0.322	0.456	0.326	0.521	0.319	0.576
tC3D	0.289	0.311	0.306	0.394	0.310	0.471	0.307	0.540	0.294	0.599

1-2 OR-AND-Invert Gates OAI21D1, OAI21D2 and OAI21D4

The OAI21D1 (1x drive), OAI21D2 (2x drive) and OAI21D4 (4x drive) cells provide the complex function:

$$Z_N = \text{NOT}[A \cdot (B_1 + B_2)]$$



Function Table

INPUTS			OUTPUT
A	B1	B2	ZN
L	X	X	H
X	L	L	H
H	H	X	L
H	X	H	L

Cell Description

Macro Name:	OAI21D1	OAI21D2	OAI21D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.25	2.5	2.75
Leakage Power (pW):	24.5	68.7	85.4

Pin Description

Name	Capacitance (pF)			Description
	OAI21D1	OAI21D2	OAI21D4	
A	0.004	0.004	0.004	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

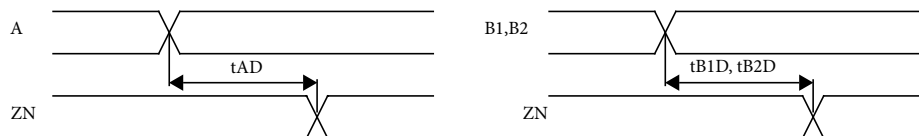
Pin Powers for

Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	OAI21D1	OAI21D2	OAI21D4
A	0.003	0.003	0.003
B1	-0	-0	-0
B2	0	0	0
ZN	0.012	0.055	0.084

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Waveforms



Timing Numbers for OAI21D1:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.103	0.081	0.160	0.121	0.273	0.198	0.499	0.354	0.949	0.664
tB1D	0.206	0.080	0.319	0.119	0.543	0.196	0.990	0.352	1.882	0.661
tB2D	0.221	0.086	0.333	0.126	0.557	0.203	1.004	0.359	1.896	0.669

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.160	0.121	0.246	0.180	0.338	0.221	0.416	0.246	0.477	0.254
tB1D	0.319	0.119	0.399	0.158	0.506	0.179	0.610	0.189	0.695	0.180
tB2D	0.333	0.126	0.387	0.166	0.460	0.193	0.534	0.207	0.601	0.204

Timing Numbers for OAI21D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.157	0.160	0.182	0.178	0.230	0.207	0.325	0.262	0.516	0.367
tB1D	0.229	0.157	0.254	0.175	0.302	0.203	0.398	0.258	0.589	0.364
tB2D	0.245	0.165	0.270	0.184	0.318	0.212	0.413	0.267	0.604	0.373

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.182	0.178	0.253	0.214	0.317	0.231	0.374	0.240	0.421	0.237
tB1D	0.254	0.175	0.337	0.196	0.427	0.196	0.507	0.186	0.577	0.165
tB2D	0.270	0.184	0.324	0.211	0.390	0.218	0.454	0.216	0.512	0.203

Timing Numbers for OAI21D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.167	0.183	0.180	0.195	0.204	0.216	0.250	0.248	0.340	0.304
tB1D	0.243	0.180	0.256	0.192	0.281	0.212	0.326	0.244	0.417	0.301
tB2D	0.258	0.189	0.271	0.201	0.296	0.221	0.341	0.253	0.432	0.310

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

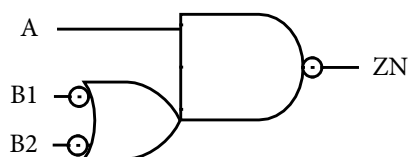
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.180	0.195	0.253	0.232	0.320	0.249	0.379	0.258	0.428	0.256
tB1D	0.256	0.192	0.339	0.213	0.430	0.214	0.511	0.205	0.582	0.185
tB2D	0.271	0.201	0.326	0.228	0.393	0.236	0.458	0.235	0.517	0.223

1-2 OR-AND-Invert Gates with Inverted B Inputs

OAIM21D1, OAIM21D2 and OAIM21D4

The OAIM21D1 (1x drive), OAIM21D2 (2x drive) and OAIM21D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A \cdot (\text{NOT}(B1) + \text{NOT}(B2))]$$



Function Table

INPUTS			OUTPUT
A	B1	B2	ZN
L	X	X	H
X	H	H	H
H	L	X	L
H	X	L	L

Cell Description

Macro Name:	OAIM21D1	OAIM21D2	OAIM21D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.5	2.75	3.
Leakage Power (pW):	46.7	109.0	153.9

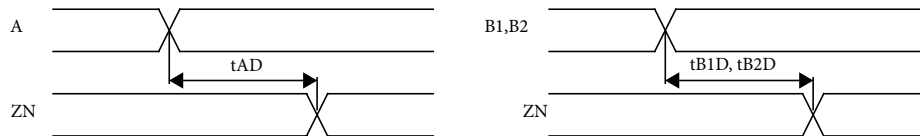
Pin Description

Name	Capacitance (pF)			Description
	OAIM21D1	OAIM21D2	OAIM21D4	
A	0.003	0.004	0.004	Data Input
B1	0.003	0.003	0.003	Data Input
B2	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	OAIM21D1	OAIM21D2	OAIM21D4
A	0.001	0.001	0.001
B1	0.001	0.001	0.001
B2	0.017	0.017	0.017
ZN	0.009	0.052	0.08

Waveforms



Timing Numbers for OAIM21D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.114	0.076	0.182	0.118	0.318	0.202	0.589	0.370	1.129	0.706
tB1D	0.160	0.141	0.228	0.183	0.364	0.268	0.634	0.437	1.174	0.773
tB2D	0.163	0.152	0.231	0.194	0.367	0.279	0.636	0.448	1.176	0.784

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.182	0.118	0.271	0.175	0.369	0.213	0.455	0.236	0.522	0.241
tB1D	0.228	0.183	0.253	0.258	0.258	0.326	0.256	0.387	0.244	0.440
tB2D	0.231	0.194	0.251	0.272	0.252	0.344	0.248	0.408	0.234	0.464

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Timing Numbers for OAIM21D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.154	0.155	0.179	0.174	0.227	0.204	0.322	0.259	0.513	0.365
tB1D	0.202	0.221	0.227	0.239	0.275	0.269	0.370	0.324	0.561	0.430
tB2D	0.205	0.232	0.229	0.250	0.278	0.280	0.373	0.335	0.564	0.441

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.179	0.174	0.251	0.202	0.317	0.210	0.376	0.211	0.426	0.202
tB1D	0.227	0.239	0.251	0.312	0.256	0.380	0.253	0.440	0.241	0.492
tB2D	0.229	0.250	0.248	0.327	0.250	0.397	0.244	0.461	0.229	0.516

Timing Numbers for OAIM21D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.160	0.185	0.173	0.197	0.197	0.219	0.243	0.252	0.333	0.310
tB1D	0.209	0.250	0.222	0.262	0.246	0.284	0.291	0.317	0.381	0.376
tB2D	0.212	0.261	0.225	0.273	0.249	0.295	0.294	0.328	0.384	0.387

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

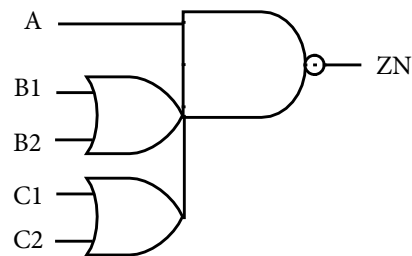
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.173	0.197	0.247	0.225	0.315	0.233	0.374	0.234	0.425	0.226
tB1D	0.222	0.262	0.246	0.336	0.251	0.403	0.249	0.463	0.236	0.515
tB2D	0.225	0.273	0.243	0.350	0.246	0.421	0.240	0.484	0.225	0.539

1-2-2 OR-AND-Invert Gates

OAI221D1, OAI221D2 and OAI221D4

The OAI221D1 (1x drive), OAI221D2 (2x drive) and OAI221D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A \cdot (B1 + B2) \cdot (C1 + C2)]$$



Function Table

INPUTS					OUTPUT
A	B1	B2	C1	C2	ZN
L	X	X	X	X	H
X	L	L	X	X	H
X	X	X	L	L	H
H	X	H	X	H	L
H	H	X	X	H	L
H	X	H	H	X	L
H	H	X	H	X	L

Cell Description

Macro Name:	OAI221D1	OAI221D2	OAI221D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.25	3.25	3.5
Leakage Power (pW):	16.7	58.6	70.6

Pin Description

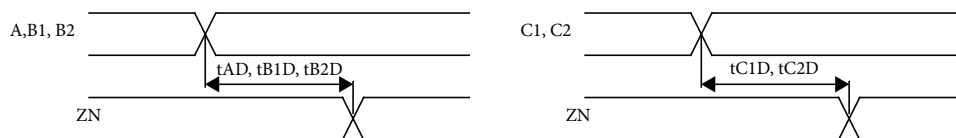
Name	Capacitance (pF)			Description
	OAI221D1	OAI221D2	OAI221D4	
A	0.003	0.003	0.003	Data Input
B1	0.005	0.004	0.004	Data Input
B2	0.004	0.004	0.004	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

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**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	OAI221D1	OAI221D2	OAI221D4
A	0.004	0.004	0.004
B1	-0	0	0
B2	0	0	0
C1	0	-0	-0
C2	0	0	0
ZN	0.02	0.065	0.088

Waveforms



Timing Numbers for OAI221D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.118	0.113	0.180	0.164	0.303	0.264	0.549	0.462	1.039	0.858
tB1D	0.213	0.111	0.319	0.160	0.530	0.257	0.950	0.448	1.788	0.831
tB2D	0.227	0.125	0.333	0.176	0.544	0.276	0.964	0.474	1.802	0.870
tC1D	0.259	0.127	0.365	0.178	0.577	0.278	0.996	0.476	1.831	0.871
tC2D	0.274	0.137	0.380	0.188	0.592	0.288	1.010	0.486	1.846	0.882

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.180	0.164	0.268	0.224	0.366	0.278	0.451	0.313	0.517	0.328
tB1D	0.319	0.160	0.398	0.197	0.503	0.224	0.607	0.239	0.692	0.236
tB2D	0.333	0.176	0.385	0.218	0.456	0.255	0.527	0.280	0.591	0.287
tC1D	0.365	0.178	0.444	0.208	0.552	0.226	0.660	0.235	0.749	0.228
tC2D	0.380	0.188	0.434	0.219	0.507	0.240	0.582	0.253	0.650	0.251

Timing Numbers for OAI221D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.182	0.193	0.208	0.211	0.258	0.241	0.358	0.297	0.557	0.407
tB1D	0.257	0.190	0.283	0.208	0.333	0.237	0.433	0.294	0.632	0.404
tB2D	0.271	0.205	0.297	0.224	0.347	0.253	0.447	0.310	0.646	0.420
tC1D	0.299	0.205	0.325	0.223	0.375	0.253	0.475	0.309	0.674	0.419
tC2D	0.314	0.217	0.340	0.236	0.390	0.265	0.490	0.322	0.689	0.432

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.208	0.211	0.287	0.260	0.361	0.289	0.424	0.308	0.476	0.313
tB1D	0.283	0.208	0.365	0.235	0.458	0.245	0.541	0.245	0.613	0.231
tB2D	0.297	0.224	0.350	0.260	0.415	0.281	0.477	0.292	0.533	0.290
tC1D	0.325	0.223	0.408	0.249	0.509	0.256	0.600	0.255	0.676	0.239
tC2D	0.340	0.236	0.395	0.263	0.466	0.276	0.534	0.279	0.596	0.270

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Timing Numbers for OAI221D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.186	0.225	0.201	0.239	0.228	0.262	0.281	0.298	0.386	0.362
tB1D	0.262	0.222	0.277	0.236	0.304	0.258	0.357	0.294	0.462	0.358
tB2D	0.276	0.238	0.291	0.252	0.319	0.274	0.372	0.310	0.477	0.374
tC1D	0.303	0.237	0.318	0.251	0.345	0.274	0.398	0.310	0.503	0.374
tC2D	0.318	0.250	0.333	0.264	0.360	0.286	0.413	0.322	0.518	0.386

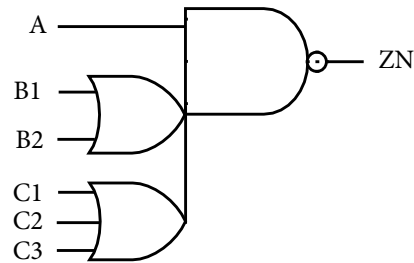
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.201	0.239	0.280	0.288	0.355	0.316	0.419	0.335	0.472	0.340
tB1D	0.277	0.236	0.359	0.263	0.452	0.273	0.535	0.273	0.608	0.259
tB2D	0.291	0.252	0.344	0.288	0.409	0.309	0.472	0.321	0.528	0.320
tC1D	0.318	0.251	0.400	0.277	0.500	0.285	0.591	0.284	0.667	0.270
tC2D	0.333	0.264	0.387	0.291	0.457	0.305	0.525	0.309	0.587	0.301

1-2-3 OR-AND-Invert Gates OAI321D1, OAI321D2 and OAI321D4

The OAI321D1 (1x drive), OAI321D2 (2x drive) and OAI321D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A \cdot (B1 + B2) \cdot (C1 + C2 + C3)]$$



Function Table

INPUTS						OUTPUT
A	B1	B2	C1	C2	C3	ZN
L	X	X	X	X	X	H
X	L	L	X	X	X	H
X	X	X	L	L	L	H
H	H	X	H	X	X	L
H	H	X	X	H	X	L
H	H	X	X	X	H	L
H	X	H	H	X	X	L
H	X	H	X	H	X	L
H	X	H	X	X	H	L

Cell Description

Macro Name:	OAI321D1	OAI321D2	OAI321D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.5	3.75	4.
Leakage Power (pW):	17.6	60.4	76.4

Pin Description

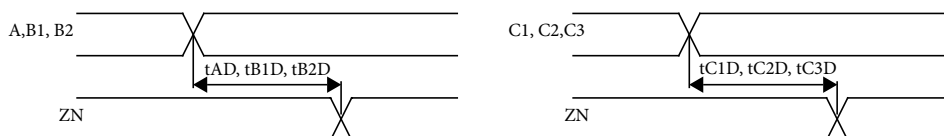
Name	Capacitance (pF)			Description
	OAI321D1	OAI321D2	OAI321D4	
A	0.004	0.004	0.004	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.004	0.004	0.004	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.004	0.004	0.004	Data Input
C3	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.3	1.2	Data Output

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**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	OAI321D1	OAI321D2	OAI321D4
A	0.004	0.004	0.004
B1	0	-0	-0
B2	-0	0	-0
C1	0	0	0
C2	0	0	0
C3	0	0	0
ZN	0.019	0.064	0.093

Waveforms



Timing Numbers for OAI321D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.108	0.134	0.164	0.192	0.276	0.308	0.499	0.535	0.946	0.988
tB1D	0.241	0.116	0.363	0.168	0.606	0.267	1.088	0.464	2.051	0.854
tB2D	0.257	0.149	0.378	0.207	0.620	0.323	1.102	0.550	2.065	1.002
tC1D	0.429	0.142	0.613	0.199	0.976	0.312	1.700	0.538	3.143	0.989
tC2D	0.461	0.154	0.645	0.212	1.008	0.327	1.732	0.555	3.175	1.011
tC3D	0.479	0.160	0.663	0.218	1.026	0.334	1.750	0.561	3.193	1.013

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.164	0.192	0.249	0.256	0.338	0.319	0.412	0.367	0.469	0.393
tB1D	0.363	0.168	0.442	0.199	0.551	0.221	0.661	0.233	0.755	0.225
tB2D	0.378	0.207	0.432	0.257	0.502	0.306	0.574	0.342	0.637	0.357
tC1D	0.613	0.199	0.684	0.225	0.789	0.239	0.899	0.243	1.006	0.231
tC2D	0.645	0.212	0.696	0.241	0.771	0.258	0.853	0.267	0.938	0.260
tC3D	0.663	0.218	0.704	0.246	0.754	0.263	0.811	0.271	0.876	0.265

Timing Numbers for OAI321D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.168	0.225	0.193	0.244	0.242	0.274	0.337	0.329	0.528	0.435
tB1D	0.267	0.206	0.293	0.225	0.341	0.255	0.436	0.310	0.627	0.417
tB2D	0.284	0.240	0.310	0.259	0.358	0.289	0.453	0.344	0.644	0.451
tC1D	0.417	0.228	0.443	0.247	0.491	0.277	0.586	0.332	0.777	0.438
tC2D	0.451	0.243	0.477	0.262	0.525	0.292	0.620	0.347	0.811	0.453
tC3D	0.470	0.251	0.496	0.270	0.544	0.300	0.639	0.355	0.830	0.462

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.193	0.244	0.264	0.301	0.326	0.342	0.379	0.372	0.421	0.389
tB1D	0.293	0.225	0.377	0.250	0.477	0.257	0.566	0.254	0.645	0.236
tB2D	0.310	0.259	0.363	0.304	0.426	0.338	0.487	0.360	0.542	0.366
tC1D	0.443	0.247	0.521	0.269	0.631	0.273	0.739	0.267	0.831	0.245
tC2D	0.477	0.262	0.529	0.287	0.608	0.296	0.691	0.295	0.769	0.279
tC3D	0.496	0.270	0.538	0.295	0.590	0.305	0.649	0.307	0.713	0.294

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Timing Numbers for OAI321D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.180	0.255	0.194	0.268	0.218	0.290	0.263	0.323	0.354	0.382
tB1D	0.285	0.236	0.299	0.249	0.324	0.270	0.369	0.303	0.459	0.362
tB2D	0.302	0.270	0.316	0.283	0.341	0.305	0.386	0.338	0.476	0.397
tC1D	0.437	0.257	0.451	0.270	0.477	0.292	0.522	0.325	0.612	0.384
tC2D	0.471	0.272	0.485	0.285	0.511	0.307	0.556	0.340	0.645	0.399
tC3D	0.490	0.281	0.504	0.294	0.529	0.316	0.574	0.349	0.664	0.408

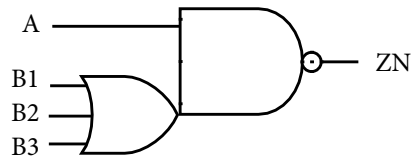
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.194	0.268	0.266	0.324	0.331	0.366	0.387	0.395	0.431	0.412
tB1D	0.299	0.249	0.383	0.274	0.484	0.281	0.575	0.278	0.655	0.261
tB2D	0.316	0.283	0.369	0.329	0.434	0.363	0.496	0.384	0.552	0.391
tC1D	0.451	0.270	0.530	0.293	0.640	0.297	0.748	0.292	0.840	0.271
tC2D	0.485	0.285	0.538	0.311	0.617	0.320	0.701	0.320	0.780	0.305
tC3D	0.504	0.294	0.546	0.320	0.599	0.330	0.659	0.332	0.724	0.320

1-3 OR-AND-Invert Gates OAI31D1, OAI31D2 and OAI31D4

The OAI31D1 (1x drive), OAI31D2 (2x drive) and OAI31D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A \cdot (B1 + B2 + B3)]$$



Function Table

INPUTS				OUTPUT
A	B1	B2	B3	ZN
L	X	X	X	H
X	L	L	L	H
H	H	X	X	L
H	X	H	X	L
H	X	X	H	L

Cell Description

Macro Name:	OAI31D1	OAI31D2	OAI31D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.75	2.75	3.
Leakage Power (pW):	32.9	77.6	94.2

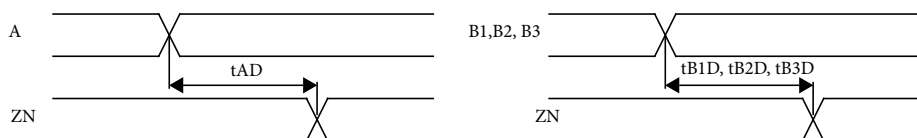
Pin Description

Name	Capacitance (pF)			Description
	OAI31D1	OAI31D2	OAI31D4	
A	0.003	0.003	0.003	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.005	0.005	0.005	Data Input
B3	0.005	0.005	0.005	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	OAI31D1	OAI31D2	OAI31D4
A	0.004	0.004	0.004
B1	0.001	0.001	0.001
B2	0	-0	-0
B3	0	0	0
ZN	0.013	0.058	0.087

Waveforms



Timing Numbers for OAI31D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
t_{AD}	0.133	0.100	0.214	0.147	0.373	0.238	0.692	0.417	1.327	0.772
t_{B1D}	0.216	0.095	0.337	0.141	0.579	0.233	1.061	0.416	2.024	0.782
t_{B2D}	0.247	0.101	0.369	0.145	0.610	0.234	1.092	0.408	2.055	0.757
t_{B3D}	0.260	0.108	0.381	0.155	0.622	0.246	1.104	0.425	2.067	0.781

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.214	0.147	0.304	0.212	0.411	0.264	0.508	0.298	0.584	0.309
tB1D	0.337	0.141	0.410	0.182	0.508	0.210	0.605	0.227	0.684	0.225
tB2D	0.369	0.145	0.419	0.183	0.490	0.211	0.562	0.227	0.630	0.226
tB3D	0.381	0.155	0.419	0.196	0.464	0.229	0.511	0.251	0.560	0.257

Timing Numbers for OAI31D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.161	0.176	0.186	0.194	0.235	0.223	0.330	0.278	0.521	0.384
tB1D	0.246	0.165	0.271	0.184	0.319	0.212	0.415	0.267	0.605	0.373
tB2D	0.280	0.178	0.305	0.196	0.353	0.225	0.448	0.280	0.639	0.386
tB3D	0.292	0.184	0.317	0.202	0.365	0.231	0.461	0.286	0.651	0.392

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.186	0.194	0.259	0.244	0.324	0.271	0.382	0.288	0.429	0.292
tB1D	0.271	0.184	0.350	0.210	0.436	0.215	0.514	0.210	0.582	0.193
tB2D	0.305	0.196	0.355	0.227	0.420	0.238	0.485	0.240	0.545	0.229
tB3D	0.317	0.202	0.354	0.237	0.397	0.255	0.442	0.263	0.489	0.259

Timing Numbers for OAI31D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.200	0.199	0.214	0.211	0.239	0.232	0.284	0.264	0.375	0.321
tB1D	0.261	0.189	0.275	0.201	0.299	0.220	0.345	0.252	0.435	0.309
tB2D	0.293	0.201	0.307	0.213	0.331	0.233	0.377	0.265	0.467	0.322
tB3D	0.305	0.208	0.319	0.220	0.344	0.240	0.389	0.272	0.480	0.329

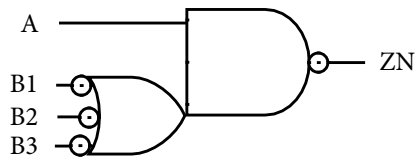
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.214	0.211	0.300	0.258	0.385	0.280	0.460	0.290	0.524	0.285
tB1D	0.275	0.201	0.353	0.228	0.440	0.234	0.518	0.231	0.587	0.215
tB2D	0.307	0.213	0.357	0.244	0.424	0.256	0.490	0.258	0.551	0.248
tB3D	0.319	0.220	0.357	0.254	0.401	0.272	0.447	0.281	0.495	0.278

1-3 OR-AND-Invert Gates with Inverted B Inputs OAIM31D1, OAIM31D2 and OAIM31D4

The OAIM31D1 (1x drive), OAIM31D2 (2x drive) and OAIM31D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A \cdot (\text{NOT}(B1) + \text{NOT}(B2) + \text{NOT}(B3))]$$



Function Table

INPUTS				OUTPUT
A	B1	B2	B3	ZN
L	X	X	X	H
X	H	H	H	H
H	L	X	X	L
H	X	L	X	L
H	X	X	L	L

Cell Description

Macro Name:	OAIM31D1	OAIM31D2	OAIM31D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.	3.25	3.5
Leakage Power (pW):	53.0	115.4	159.9

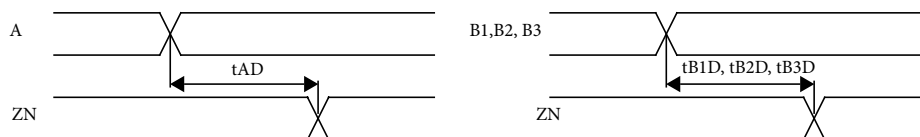
Pin Description

Name	Capacitance (pF)			Description
	OAIM31D1	OAIM31D2	OAIM31D4	
A	0.004	0.004	0.004	Data Input
B1	0.003	0.003	0.003	Data Input
B2	0.004	0.004	0.004	Data Input
B3	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	OAIM31D1	OAIM31D2	OAIM31D4
A	0.001	0.001	0.001
B1	0.001	0.001	0.001
B2	-0	-0	-0
B3	0	0	0
ZN	0.015	0.059	0.087

Waveforms



Timing Numbers for OAIM31D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.096	0.085	0.150	0.134	0.256	0.229	0.470	0.421	0.895	0.804
tB1D	0.182	0.152	0.236	0.201	0.343	0.297	0.556	0.488	0.981	0.871
tB2D	0.188	0.158	0.243	0.207	0.350	0.303	0.563	0.494	0.988	0.877
tB3D	0.192	0.169	0.247	0.218	0.354	0.314	0.567	0.506	0.992	0.889

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.150	0.134	0.234	0.195	0.319	0.244	0.389	0.278	0.442	0.294
tB1D	0.236	0.201	0.284	0.265	0.317	0.318	0.343	0.363	0.358	0.398
tB2D	0.243	0.207	0.280	0.275	0.307	0.332	0.329	0.380	0.340	0.419
tB3D	0.247	0.218	0.269	0.294	0.283	0.362	0.292	0.419	0.293	0.468

Timing Numbers for OAIM31D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.146	0.159	0.171	0.176	0.219	0.205	0.314	0.260	0.505	0.366
tB1D	0.232	0.225	0.257	0.243	0.305	0.272	0.400	0.327	0.591	0.433
tB2D	0.238	0.231	0.263	0.249	0.312	0.277	0.407	0.332	0.598	0.438
tB3D	0.242	0.241	0.267	0.260	0.316	0.288	0.411	0.343	0.601	0.449

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.171	0.176	0.235	0.214	0.288	0.236	0.333	0.250	0.371	0.254
tB1D	0.257	0.243	0.304	0.307	0.337	0.359	0.361	0.403	0.375	0.437
tB2D	0.263	0.249	0.298	0.317	0.326	0.373	0.345	0.421	0.355	0.459
tB3D	0.267	0.260	0.288	0.336	0.301	0.403	0.308	0.461	0.306	0.508

Timing Numbers for OAIM31D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.153	0.181	0.166	0.193	0.190	0.213	0.235	0.245	0.326	0.302
tB1D	0.239	0.247	0.252	0.259	0.276	0.280	0.321	0.312	0.412	0.369
tB2D	0.245	0.253	0.258	0.265	0.282	0.286	0.328	0.318	0.419	0.374
tB3D	0.249	0.264	0.262	0.276	0.286	0.296	0.332	0.328	0.423	0.385

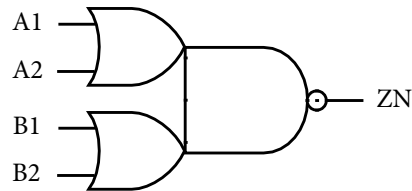
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.166	0.193	0.231	0.230	0.286	0.252	0.334	0.266	0.373	0.271
tB1D	0.252	0.259	0.299	0.324	0.332	0.376	0.356	0.420	0.370	0.454
tB2D	0.258	0.265	0.294	0.333	0.321	0.390	0.341	0.437	0.351	0.475
tB3D	0.262	0.276	0.283	0.352	0.296	0.419	0.303	0.477	0.301	0.525

2-2 OR-AND-Invert Gates OAI22D1, OAI22D2 and OAI22D4

The OAI22D1 (1x drive), OAI22D2 (2x drive) and OAI22D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[(A1 + A2) \cdot (B1 + B2)]$$



Function Table

INPUTS				OUTPUT
A1	A2	B1	B2	ZN
L	L	X	X	H
X	X	L	L	H
H	X	H	X	L
H	X	X	H	L
X	H	H	X	L
X	H	X	H	L

Cell Description

Macro Name:	OAI22D1	OAI22D2	OAI22D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.5	2.75	3.
Leakage Power (pW):	22.3	66.4	83.1

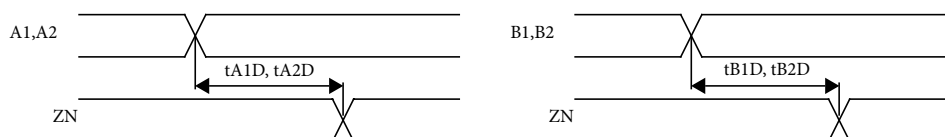
Pin Description

Name	Capacitance (pF)			Description
	OAI22D1	OAI22D2	OAI22D4	
A1	0.003	0.003	0.003	Data Input
A2	0.004	0.004	0.004	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	OAI22D1	OAI22D2	OAI22D4
A1	0.001	0.001	0.001
A2	0	0	0
B1	0	-0	-0
B2	0	0	0
ZN	0.017	0.062	0.091

Waveforms



Timing Numbers for OAI22D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout loads	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.174	0.079	0.281	0.115	0.493	0.187	0.913	0.329	1.751	0.613
tA2D	0.190	0.086	0.296	0.122	0.508	0.194	0.928	0.337	1.767	0.621
tB1D	0.225	0.090	0.330	0.127	0.540	0.200	0.958	0.345	1.792	0.635
tB2D	0.239	0.095	0.344	0.132	0.554	0.204	0.972	0.346	1.806	0.630

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.281	0.115	0.360	0.173	0.463	0.209	0.562	0.227	0.642	0.227
tA2D	0.296	0.122	0.348	0.183	0.419	0.224	0.489	0.248	0.552	0.254
tB1D	0.330	0.127	0.410	0.168	0.517	0.193	0.623	0.205	0.709	0.198
tB2D	0.344	0.132	0.398	0.172	0.473	0.199	0.548	0.214	0.617	0.210

Timing Numbers for OAI22D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout loads	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.209	0.162	0.234	0.180	0.282	0.209	0.377	0.264	0.568	0.369
tA2D	0.225	0.171	0.250	0.188	0.298	0.217	0.393	0.272	0.584	0.378
tB1D	0.258	0.173	0.283	0.191	0.331	0.219	0.426	0.274	0.617	0.380
tB2D	0.273	0.179	0.298	0.197	0.346	0.226	0.441	0.281	0.632	0.387

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.234	0.180	0.314	0.214	0.395	0.225	0.469	0.225	0.534	0.212
tA2D	0.250	0.188	0.301	0.231	0.362	0.249	0.421	0.258	0.473	0.253
tB1D	0.283	0.191	0.367	0.220	0.462	0.228	0.546	0.225	0.618	0.208
tB2D	0.298	0.197	0.352	0.229	0.422	0.242	0.489	0.244	0.550	0.233

Timing Numbers for OAI22D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout loads	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.224	0.185	0.238	0.197	0.263	0.217	0.308	0.249	0.399	0.306
tA2D	0.240	0.194	0.254	0.206	0.279	0.226	0.324	0.258	0.415	0.315
tB1D	0.271	0.195	0.285	0.207	0.309	0.228	0.355	0.260	0.445	0.317
tB2D	0.286	0.202	0.300	0.214	0.325	0.235	0.370	0.266	0.461	0.323

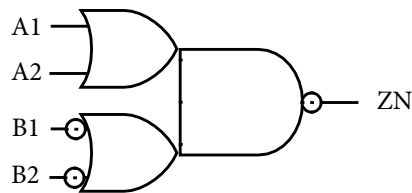
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.238	0.197	0.318	0.231	0.402	0.242	0.477	0.242	0.544	0.229
tA2D	0.254	0.206	0.306	0.248	0.368	0.266	0.429	0.275	0.483	0.271
tB1D	0.285	0.207	0.369	0.237	0.465	0.246	0.550	0.243	0.624	0.227
tB2D	0.300	0.214	0.355	0.247	0.425	0.260	0.493	0.262	0.555	0.252

2-2 OR-AND-Invert Gates with Inverted B Inputs OAIM22D1, OAIM22D2 and OAIM22D4

The OAIM22D1 (1x drive), OAIM22D2 (2x drive) and OAIM22D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[(A1 + A2) \cdot (\text{NOT}(B1) + \text{NOT}(B2))]$$



Function Table

INPUTS				OUTPUT
A1	A2	B1	B2	ZN
L	L	X	X	H
X	X	H	H	H
H	X	L	X	L
H	X	X	L	L
X	H	L	X	L
X	H	X	L	L

Cell Description

Macro Name:	OAIM22D1	OAIM22D2	OAIM22D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.	3.	3.25
Leakage Power (pW):	45.1	108.7	153.2

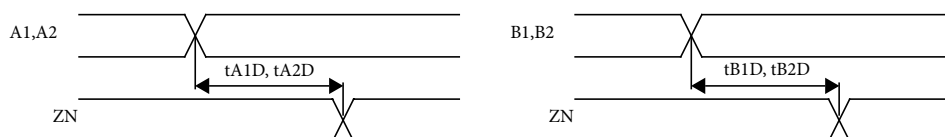
Pin Description

Name	Capacitance (pF)			Description
	OAIM22D1	OAIM22D2	OAIM22D4	
A1	0.004	0.004	0.004	Data Input
A2	0.004	0.004	0.004	Data Input
B1	0.003	0.003	0.003	Data Input
B2	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	OAIM22D1	OAIM22D2	OAIM22D4
A1	0.001	0.001	0.001
A2	0.002	0.002	0.002
B1	0.001	0.001	0.001
B2	0.018	0.018	0.018
ZN	0.018	0.062	0.09

Waveforms



Timing Numbers for OAIM22D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.156	0.074	0.251	0.111	0.439	0.184	0.812	0.331	1.557	0.624
tA2D	0.170	0.083	0.265	0.122	0.453	0.198	0.826	0.349	1.571	0.652
tB1D	0.185	0.145	0.252	0.184	0.384	0.261	0.647	0.412	1.172	0.715
tB2D	0.188	0.157	0.254	0.196	0.386	0.273	0.650	0.425	1.175	0.727

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.251	0.111	0.329	0.167	0.428	0.201	0.520	0.220	0.593	0.221
tA2D	0.265	0.122	0.316	0.183	0.383	0.226	0.446	0.253	0.502	0.262
tB1D	0.252	0.184	0.285	0.251	0.299	0.308	0.308	0.360	0.307	0.402
tB2D	0.254	0.196	0.278	0.270	0.287	0.335	0.289	0.394	0.282	0.444

Timing Numbers for OAIM22D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.188	0.156	0.213	0.174	0.261	0.204	0.356	0.259	0.547	0.365
tA2D	0.201	0.167	0.226	0.185	0.275	0.214	0.370	0.269	0.561	0.375
tB1D	0.228	0.228	0.253	0.246	0.301	0.275	0.397	0.330	0.588	0.437
tB2D	0.230	0.239	0.255	0.257	0.303	0.286	0.399	0.341	0.590	0.447

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.213	0.174	0.288	0.202	0.361	0.209	0.427	0.207	0.485	0.194
tA2D	0.226	0.185	0.274	0.225	0.329	0.243	0.380	0.253	0.425	0.252
tB1D	0.253	0.246	0.288	0.310	0.305	0.365	0.316	0.413	0.317	0.453
tB2D	0.255	0.257	0.279	0.327	0.290	0.390	0.294	0.445	0.288	0.492

Timing Numbers for OAIM22D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.188	0.184	0.201	0.196	0.226	0.218	0.275	0.252	0.372	0.313
tA2D	0.202	0.194	0.215	0.207	0.240	0.229	0.289	0.263	0.386	0.323
tB1D	0.227	0.256	0.240	0.268	0.266	0.290	0.315	0.324	0.411	0.385
tB2D	0.229	0.266	0.242	0.279	0.268	0.301	0.317	0.335	0.413	0.395

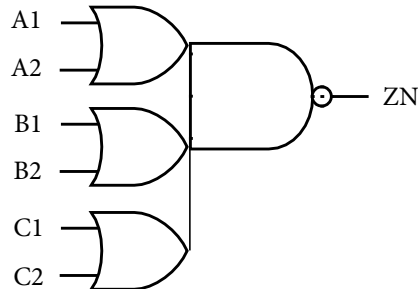
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.201	0.196	0.276	0.227	0.351	0.235	0.418	0.234	0.477	0.223
tA2D	0.215	0.207	0.263	0.248	0.318	0.267	0.371	0.277	0.417	0.277
tB1D	0.240	0.268	0.275	0.332	0.293	0.387	0.303	0.435	0.304	0.475
tB2D	0.242	0.279	0.268	0.350	0.278	0.412	0.281	0.467	0.276	0.514

2-2-2 OR-AND-Invert Gates OAI222D1, OAI222D2 and OAI222D4

The OAI222D1 (1x drive), OAI222D2 (2x drive) and OAI222D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[(A1 + A2) \cdot (B1 + B2) \cdot (C1 + C2)]$$



Function Table

INPUTS						OUTPUT
A1	A2	B1	B2	C1	C2	ZN
L	L	X	X	X	X	H
X	X	L	L	X	X	H
X	X	X	X	L	L	H
Any other combination						L

Cell Description

Macro Name:	OAI222D1	OAI222D2	OAI222D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.25	3.5	3.75
Leakage Power (pW):	12.4	54.9	65.7

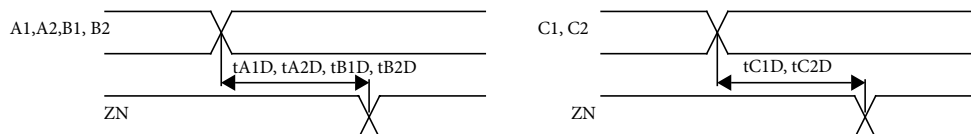
Pin Description

Name	Capacitance (pF)			Description
	OAI222D1	OAI222D2	OAI222D4	
A1	0.003	0.004	0.004	Data Input
A2	0.004	0.004	0.004	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.004	0.004	0.004	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.005	0.005	0.005	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	OAI222D1	OAI222D2	OAI222D4
A1	0.003	0.003	0.003
A2	0.005	0.005	0.005
B1	0	-0	-0
B2	0	0	0
C1	0	-0	-0
C2	0	0	0
ZN	0.025	0.068	0.093

Waveforms



Timing Numbers for OAI222D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.189	0.117	0.295	0.167	0.505	0.265	0.923	0.459	1.753	0.847
tA2D	0.203	0.129	0.308	0.179	0.517	0.279	0.935	0.476	1.765	0.869
tB1D	0.234	0.138	0.338	0.189	0.543	0.289	0.952	0.489	1.768	0.887
tB2D	0.250	0.146	0.353	0.197	0.558	0.297	0.967	0.494	1.783	0.888
tC1D	0.262	0.146	0.360	0.197	0.554	0.297	0.942	0.497	1.715	0.895
tC2D	0.276	0.155	0.374	0.205	0.569	0.305	0.956	0.502	1.729	0.896

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.295	0.167	0.370	0.228	0.470	0.281	0.566	0.316	0.643	0.328
tA2D	0.308	0.179	0.357	0.242	0.424	0.301	0.488	0.343	0.546	0.362
tB1D	0.338	0.189	0.417	0.233	0.521	0.272	0.625	0.299	0.709	0.305
tB2D	0.353	0.197	0.405	0.241	0.477	0.282	0.549	0.311	0.614	0.321
tC1D	0.360	0.197	0.438	0.228	0.543	0.250	0.648	0.263	0.734	0.261
tC2D	0.374	0.205	0.426	0.236	0.498	0.259	0.571	0.273	0.637	0.274

Timing Numbers for OAI222D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.229	0.211	0.254	0.229	0.302	0.259	0.398	0.314	0.589	0.421
tA2D	0.242	0.224	0.268	0.243	0.316	0.272	0.411	0.327	0.602	0.434
tB1D	0.274	0.230	0.299	0.249	0.348	0.279	0.443	0.334	0.634	0.440
tB2D	0.290	0.242	0.315	0.260	0.363	0.290	0.458	0.345	0.649	0.452
tC1D	0.300	0.239	0.325	0.257	0.373	0.287	0.469	0.342	0.660	0.449
tC2D	0.315	0.250	0.340	0.269	0.388	0.299	0.483	0.354	0.674	0.460

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.254	0.229	0.331	0.279	0.412	0.308	0.483	0.325	0.545	0.326
tA2D	0.268	0.243	0.317	0.298	0.373	0.336	0.427	0.362	0.474	0.373
tB1D	0.299	0.249	0.380	0.288	0.474	0.313	0.557	0.327	0.626	0.324
tB2D	0.315	0.260	0.367	0.301	0.433	0.330	0.497	0.348	0.554	0.351
tC1D	0.325	0.257	0.406	0.286	0.505	0.300	0.593	0.306	0.666	0.297
tC2D	0.340	0.269	0.392	0.297	0.461	0.314	0.527	0.323	0.587	0.319

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Timing Numbers for OAI222D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.253	0.221	0.268	0.233	0.296	0.255	0.348	0.289	0.452	0.351
tA2D	0.267	0.233	0.282	0.246	0.309	0.268	0.361	0.302	0.465	0.364
tB1D	0.297	0.240	0.312	0.253	0.339	0.275	0.391	0.309	0.495	0.371
tB2D	0.312	0.251	0.327	0.264	0.354	0.286	0.406	0.320	0.510	0.382
tC1D	0.322	0.248	0.337	0.261	0.365	0.283	0.417	0.317	0.520	0.379
tC2D	0.337	0.260	0.352	0.272	0.379	0.294	0.431	0.328	0.535	0.390

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

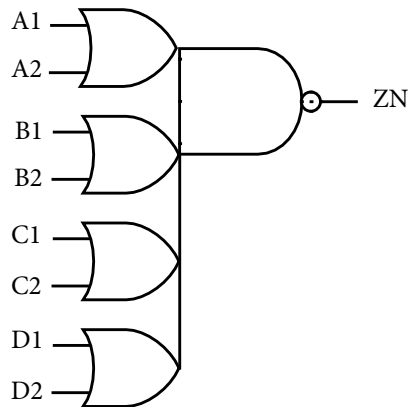
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.268	0.233	0.346	0.284	0.431	0.313	0.505	0.330	0.570	0.332
tA2D	0.282	0.246	0.331	0.302	0.391	0.341	0.447	0.367	0.497	0.379
tB1D	0.312	0.253	0.393	0.293	0.489	0.318	0.574	0.332	0.646	0.330
tB2D	0.327	0.264	0.380	0.305	0.448	0.334	0.513	0.352	0.571	0.356
tC1D	0.337	0.261	0.418	0.290	0.518	0.305	0.607	0.311	0.682	0.303
tC2D	0.352	0.272	0.405	0.301	0.474	0.319	0.541	0.328	0.602	0.325

2-2-2-2 OR-AND-Invert Gates

OAI2222D1, OAI2222D2 and OAI2222D4

The OAI2222D1 (1x drive), OAI2222D2 (2x drive) and OAI2222D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[(A1 + A2) \cdot (B1 + B2) \cdot (C1 + C2) \cdot (D1 + D2)]$$



Function Table

INPUTS								OUTPUT
A1	A2	B1	B2	C1	C2	D1	D2	ZN
L	L	X	X	X	X	X	X	H
X	X	L	L	X	X	X	X	H
X	X	X	X	L	L	X	X	H
X	X	X	X	X	X	L	L	H
Any other combination								L

Cell Description

Macro Name:	OAI2222D1	OAI2222D2	OAI2222D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	4.75	4.75	5.
Leakage Power (pW):	53.9	64.4	80.1

Pin Description

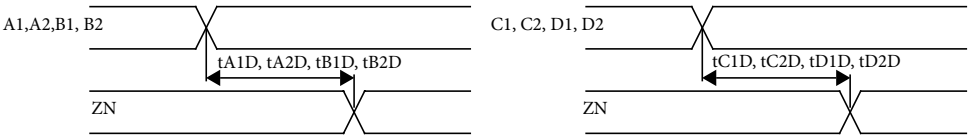
Name	Capacitance (pF)			Description
	OAI2222D1	OAI2222D2	OAI2222D4	
A1	0.003	0.003	0.003	Data Input
A2	0.004	0.004	0.003	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.005	0.005	0.004	Data Input
C1	0.003	0.003	0.003	Data Input
C2	0.004	0.004	0.003	Data Input
D1	0.004	0.004	0.004	Data Input
D2	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

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**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	OAI2222D1	OAI2222D2	OAI2222D4
A1	0.004	0.004	0.004
A2	0.004	0.004	0.004
B1	0	0	0
B2	0	0	0
C1	0.004	0.004	0.004
C2	0.03	0.03	0.028
D1	0	0	0
D2	0	0	0
ZN	0.05	0.063	0.1

Waveforms



Timing Numbers for OAI2222D1:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.248	0.203	0.296	0.235	0.391	0.292	0.580	0.399	0.957	0.611
tA2D	0.259	0.211	0.307	0.243	0.402	0.301	0.591	0.408	0.968	0.620
tB1D	0.261	0.206	0.309	0.239	0.404	0.297	0.593	0.404	0.971	0.616
tB2D	0.277	0.220	0.324	0.252	0.419	0.309	0.608	0.416	0.986	0.628
tC1D	0.239	0.228	0.287	0.260	0.382	0.317	0.571	0.424	0.948	0.636
tC2D	0.253	0.236	0.301	0.268	0.396	0.325	0.585	0.432	0.962	0.644
tD1D	0.276	0.235	0.323	0.267	0.418	0.325	0.608	0.432	0.985	0.643
tD2D	0.291	0.244	0.339	0.276	0.434	0.334	0.623	0.441	1.000	0.653

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.296	0.235	0.382	0.269	0.476	0.276	0.559	0.272	0.632	0.254
tA2D	0.307	0.243	0.350	0.288	0.403	0.308	0.457	0.317	0.505	0.315
tB1D	0.309	0.239	0.392	0.269	0.483	0.276	0.563	0.272	0.632	0.256
tB2D	0.324	0.252	0.378	0.283	0.443	0.296	0.506	0.300	0.563	0.292
tC1D	0.287	0.260	0.365	0.298	0.445	0.312	0.516	0.314	0.578	0.305
tC2D	0.301	0.268	0.351	0.314	0.409	0.336	0.465	0.347	0.515	0.346
tD1D	0.323	0.267	0.406	0.294	0.498	0.299	0.579	0.295	0.649	0.278
tD2D	0.339	0.276	0.393	0.308	0.458	0.320	0.521	0.323	0.578	0.314

Timing Numbers for OAI2222D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.240	0.214	0.265	0.236	0.313	0.272	0.408	0.333	0.597	0.442
tA2D	0.251	0.222	0.276	0.245	0.324	0.280	0.419	0.341	0.608	0.450
tB1D	0.250	0.221	0.275	0.243	0.323	0.279	0.418	0.339	0.607	0.449
tB2D	0.265	0.231	0.290	0.253	0.338	0.289	0.433	0.349	0.622	0.459
tC1D	0.229	0.239	0.254	0.261	0.301	0.297	0.396	0.358	0.586	0.467
tC2D	0.242	0.247	0.267	0.269	0.315	0.305	0.410	0.366	0.599	0.475
tD1D	0.263	0.246	0.288	0.268	0.336	0.304	0.431	0.365	0.620	0.474
tD2D	0.279	0.256	0.304	0.278	0.352	0.314	0.447	0.374	0.636	0.484

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.265	0.236	0.352	0.271	0.447	0.277	0.532	0.273	0.606	0.255
tA2D	0.276	0.245	0.319	0.288	0.374	0.308	0.429	0.318	0.478	0.316
tB1D	0.275	0.243	0.358	0.270	0.450	0.277	0.531	0.273	0.601	0.258
tB2D	0.290	0.253	0.344	0.284	0.410	0.298	0.474	0.301	0.532	0.293
tC1D	0.254	0.261	0.332	0.300	0.414	0.314	0.486	0.316	0.549	0.307
tC2D	0.267	0.269	0.318	0.316	0.378	0.338	0.435	0.349	0.485	0.349
tD1D	0.288	0.268	0.371	0.296	0.465	0.301	0.546	0.296	0.616	0.280
tD2D	0.304	0.278	0.358	0.309	0.424	0.322	0.488	0.325	0.546	0.316

Timing Numbers for OAI2222D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.243	0.307	0.257	0.323	0.282	0.352	0.328	0.394	0.419	0.465
tA2D	0.259	0.307	0.273	0.323	0.298	0.352	0.343	0.394	0.435	0.465
tB1D	0.291	0.307	0.305	0.323	0.330	0.352	0.376	0.394	0.467	0.465
tB2D	0.308	0.315	0.322	0.331	0.347	0.360	0.392	0.402	0.484	0.473
tC1D	0.259	0.324	0.273	0.340	0.298	0.368	0.344	0.410	0.435	0.482
tC2D	0.274	0.329	0.288	0.345	0.313	0.374	0.359	0.416	0.450	0.487
tD1D	0.307	0.331	0.321	0.347	0.346	0.375	0.391	0.417	0.482	0.488
tD2D	0.323	0.338	0.337	0.354	0.362	0.382	0.407	0.424	0.498	0.496

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

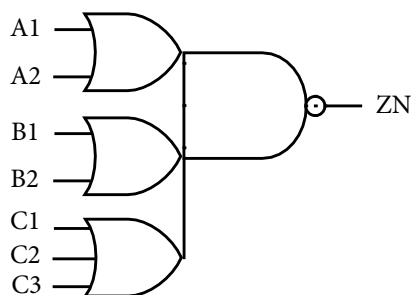
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.257	0.323	0.339	0.362	0.424	0.377	0.500	0.380	0.567	0.371
tA2D	0.273	0.323	0.327	0.362	0.394	0.376	0.458	0.379	0.517	0.371
tB1D	0.305	0.323	0.391	0.349	0.491	0.353	0.580	0.346	0.658	0.326
tB2D	0.322	0.331	0.377	0.361	0.450	0.370	0.521	0.369	0.586	0.356
tC1D	0.273	0.340	0.357	0.375	0.447	0.382	0.528	0.378	0.600	0.362
tC2D	0.288	0.345	0.343	0.386	0.411	0.399	0.477	0.402	0.537	0.393
tD1D	0.321	0.347	0.406	0.372	0.509	0.373	0.601	0.364	0.680	0.342
tD2D	0.337	0.354	0.393	0.384	0.467	0.392	0.538	0.390	0.604	0.376

2-2-3 OR-AND-Invert Gates

OAI322D1, OAI322D2 and OAI322D4

The OAI322D1 (1x drive), OAI322D2 (2x drive) and OAI322D4 (4x drive) cells provide the complex function:

$$Z_N = \text{NOT}[(A1 + A2) \cdot (B1 + B2) \cdot (C1 + C2 + C3)]$$



Function Table

INPUTS							OUTPUT
A1	A2	B1	B2	C1	C2	C3	ZN
L	L	X	X	X	X	X	H
X	X	L	L	X	X	X	H
X	X	X	X	L	L	L	H
Any other combination							L

Cell Description

Macro Name:	OAI322D1	OAI322D2	OAI322D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.75	3.75	4.
Leakage Power (pW):	23.8	59.8	75.3

Pin Description

Name	Capacitance (pF)			Description
	OAI322D1	OAI322D2	OAI322D4	
A1	0.003	0.003	0.003	Data Input
A2	0.003	0.003	0.003	Data Input
B1	0.004	0.004	0.004	Data Input
B2	0.004	0.004	0.004	Data Input
C1	0.005	0.004	0.004	Data Input
C2	0.004	0.004	0.004	Data Input
C3	0.004	0.005	0.005	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

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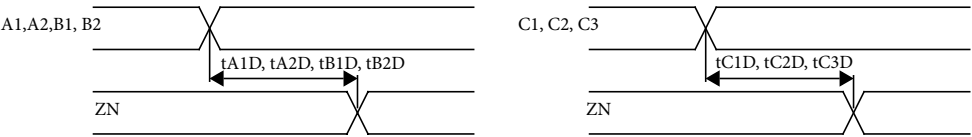
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**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	OAI322D1	OAI322D2	OAI322D4
A1	0.005	0.005	0.005
A2	0.004	0.004	0.004
B1	-0	-0	-0
B2	0	0	0
C1	0	0	0
C2	0	0	0
C3	0	0	0
ZN	0.024	0.071	0.093

Waveforms



Timing Numbers for OAI322D1:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.243	0.115	0.386	0.164	0.669	0.259	1.232	0.446	2.349	0.818
tA2D	0.259	0.125	0.401	0.174	0.684	0.271	1.247	0.462	2.364	0.840
tB1D	0.287	0.133	0.414	0.183	0.668	0.281	1.173	0.472	2.180	0.853
tB2D	0.304	0.141	0.431	0.190	0.685	0.287	1.190	0.478	2.196	0.856
tC1D	0.446	0.139	0.610	0.187	0.935	0.282	1.583	0.469	2.875	0.844
tC2D	0.480	0.153	0.644	0.202	0.968	0.299	1.616	0.490	2.908	0.873
tC3D	0.493	0.156	0.657	0.205	0.982	0.302	1.630	0.493	2.921	0.871

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.386	0.164	0.463	0.227	0.572	0.279	0.680	0.312	0.774	0.320
tA2D	0.401	0.174	0.454	0.239	0.528	0.296	0.605	0.334	0.676	0.348
tB1D	0.414	0.183	0.493	0.219	0.603	0.248	0.716	0.264	0.814	0.261
tB2D	0.431	0.190	0.483	0.226	0.561	0.255	0.643	0.274	0.720	0.273
tC1D	0.610	0.187	0.682	0.220	0.787	0.239	0.897	0.247	1.004	0.237
tC2D	0.644	0.202	0.694	0.238	0.768	0.263	0.850	0.277	0.934	0.274
tC3D	0.657	0.205	0.696	0.241	0.746	0.265	0.802	0.280	0.868	0.277

Timing Numbers for OAI322D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.261	0.208	0.288	0.227	0.341	0.258	0.447	0.318	0.657	0.433
tA2D	0.277	0.220	0.304	0.239	0.357	0.270	0.463	0.329	0.673	0.444
tB1D	0.317	0.226	0.344	0.245	0.397	0.276	0.502	0.335	0.712	0.450
tB2D	0.333	0.236	0.360	0.255	0.414	0.286	0.519	0.345	0.729	0.460
tC1D	0.454	0.230	0.481	0.249	0.534	0.280	0.639	0.339	0.849	0.454
tC2D	0.487	0.245	0.514	0.264	0.568	0.295	0.673	0.355	0.883	0.470
tC3D	0.506	0.253	0.533	0.272	0.586	0.303	0.691	0.362	0.901	0.477

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.288	0.227	0.372	0.281	0.466	0.310	0.550	0.326	0.623	0.323
tA2D	0.304	0.239	0.358	0.296	0.426	0.333	0.491	0.355	0.550	0.360
tB1D	0.344	0.245	0.428	0.278	0.533	0.295	0.630	0.300	0.712	0.288
tB2D	0.360	0.255	0.415	0.287	0.491	0.307	0.566	0.316	0.635	0.308
tC1D	0.481	0.249	0.558	0.279	0.667	0.290	0.776	0.289	0.869	0.272
tC2D	0.514	0.264	0.566	0.298	0.644	0.317	0.726	0.323	0.804	0.313
tC3D	0.533	0.272	0.572	0.305	0.623	0.324	0.681	0.331	0.744	0.324

Timing Numbers for OAI322D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.272	0.227	0.288	0.241	0.317	0.263	0.371	0.299	0.480	0.362
tA2D	0.288	0.238	0.304	0.252	0.333	0.274	0.387	0.310	0.496	0.374
tB1D	0.327	0.245	0.342	0.259	0.371	0.281	0.425	0.316	0.534	0.380
tB2D	0.343	0.254	0.359	0.268	0.388	0.290	0.442	0.326	0.550	0.390
tC1D	0.465	0.248	0.481	0.261	0.510	0.283	0.564	0.319	0.672	0.383
tC2D	0.499	0.264	0.515	0.277	0.543	0.299	0.597	0.335	0.706	0.399
tC3D	0.517	0.271	0.533	0.285	0.562	0.307	0.616	0.343	0.724	0.407

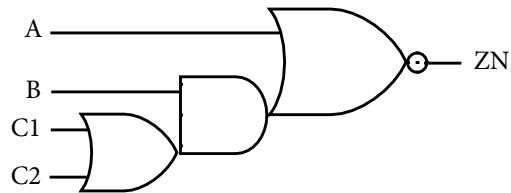
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tA1D	0.288	0.241	0.372	0.293	0.468	0.323	0.554	0.338	0.628	0.335
tA2D	0.304	0.252	0.359	0.309	0.428	0.345	0.494	0.368	0.554	0.372
tB1D	0.342	0.259	0.427	0.291	0.533	0.308	0.631	0.314	0.715	0.302
tB2D	0.359	0.268	0.414	0.301	0.492	0.321	0.567	0.330	0.637	0.322
tC1D	0.481	0.261	0.558	0.291	0.667	0.303	0.777	0.304	0.871	0.287
tC2D	0.515	0.277	0.566	0.312	0.644	0.330	0.727	0.338	0.805	0.328
tC3D	0.533	0.285	0.572	0.318	0.624	0.337	0.682	0.346	0.746	0.339

1-1-2 OR-AND-OR-Invert Gates OAN211D1, OAN211D2 and OAN211D4

The OAN211D1 (1x drive), OAN211D2 (2x drive) and OAN211D4 (4x drive) cells provide the complex function:

$$ZN = \text{NOT}[A + (B \cdot (C1 + C2))]$$



Function Table

INPUTS				OUTPUT
A	B	C1	C2	ZN
H	X	X	X	L
L	H	H	X	L
L	H	X	H	L
L	L	X	X	H
L	H	L	L	H

Cell Description

Macro Name:	OAN211D1	OAN211D2	OAN211D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	1.5	2.75	3.
Leakage Power (pW):	19.4	81.7	123.4

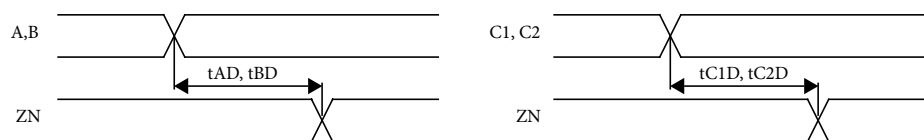
Pin Description

Name	Capacitance (pF)			Description
	OAN211D1	OAN211D2	OAN211D4	
A	0.004	0.004	0.004	Data Input
B	0.004	0.004	0.004	Data Input
C1	0.004	0.004	0.004	Data Input
C2	0.004	0.004	0.004	Data Input
Maximum capacitance				
ZN	0.3	0.6	1.2	Data Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	OAN211D1	OAN211D2	OAN211D4
A	0.002	0.002	0.002
B	0	0	0
C1	-0	-0	-0
C2	0	0	0
ZN	0.013	0.057	0.085

Waveforms



Timing Numbers for OAN211D1:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.272	0.056	0.440	0.081	0.771	0.130	1.431	0.228	2.749	0.425
tBD	0.204	0.085	0.312	0.127	0.528	0.209	0.957	0.372	1.813	0.697
tC1D	0.328	0.092	0.494	0.134	0.824	0.216	1.483	0.379	2.799	0.704
tC2D	0.342	0.094	0.508	0.136	0.838	0.220	1.497	0.382	2.813	0.705

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.440	0.081	0.516	0.133	0.624	0.155	0.732	0.161	0.831	0.148
tBD	0.312	0.127	0.369	0.188	0.432	0.232	0.492	0.260	0.547	0.269
tC1D	0.494	0.134	0.548	0.173	0.619	0.199	0.697	0.211	0.774	0.204
tC2D	0.508	0.136	0.549	0.176	0.597	0.203	0.650	0.218	0.710	0.214

Timing Numbers for OAN211D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.269	0.141	0.294	0.159	0.342	0.188	0.438	0.243	0.629	0.349
tBD	0.226	0.168	0.251	0.187	0.299	0.214	0.395	0.268	0.586	0.374
tC1D	0.325	0.170	0.350	0.188	0.399	0.216	0.494	0.270	0.685	0.376
tC2D	0.339	0.174	0.364	0.193	0.412	0.222	0.508	0.277	0.699	0.383

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.294	0.159	0.380	0.185	0.480	0.184	0.570	0.171	0.650	0.145
tBD	0.251	0.187	0.305	0.233	0.356	0.256	0.402	0.268	0.446	0.268
tC1D	0.350	0.188	0.404	0.216	0.472	0.222	0.542	0.216	0.612	0.197
tC2D	0.364	0.193	0.405	0.222	0.452	0.232	0.502	0.231	0.559	0.217

Timing Numbers for OAN211D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.282	0.164	0.296	0.176	0.320	0.197	0.365	0.228	0.455	0.285
tBD	0.235	0.192	0.248	0.204	0.272	0.224	0.317	0.256	0.407	0.312
tC1D	0.338	0.194	0.352	0.206	0.377	0.226	0.421	0.257	0.511	0.313
tC2D	0.352	0.198	0.366	0.210	0.391	0.231	0.435	0.263	0.525	0.319

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

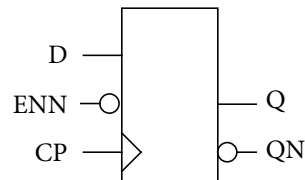
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAD	0.296	0.176	0.381	0.202	0.481	0.200	0.572	0.188	0.652	0.163
tBD	0.248	0.204	0.302	0.250	0.354	0.272	0.402	0.284	0.447	0.284
tC1D	0.352	0.206	0.406	0.234	0.475	0.239	0.546	0.234	0.616	0.215
tC2D	0.366	0.210	0.407	0.240	0.454	0.250	0.505	0.249	0.563	0.235

FLIP-FLOPS

D-Enabled Active-Low Flip-Flops

MFFNRB1 (DENRB1), MFFNRB2 (DENRB2) and MFFNRB4 (DENRB4)

The MFFNRB1 (1x drive), MFFNRB2 (2x drive) and MFFNRB4 (4x drive) cells are positive edge triggered D flip-flops with an active-Low enable (ENN). When ENN is Low, data at the D input is transferred to the Q and QN outputs on the rising transition of the clock (CP). When ENN is High, data is retained.



Function Table

INPUTS			OUTPUTS	
ENN	CP	D	Q	QN
L	↑	L	L	H
L	↑	H	H	L
L	L	X	Q	QN
L	H	X	Q	QN
H	X	X	Q _o	QN _o

Cell Description

Macro Name:	MFFNRB1	MFFNRB2	MFFNRB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.25	7.5	9.75
Leakage Power (pW):	181.3	221.8	342.9

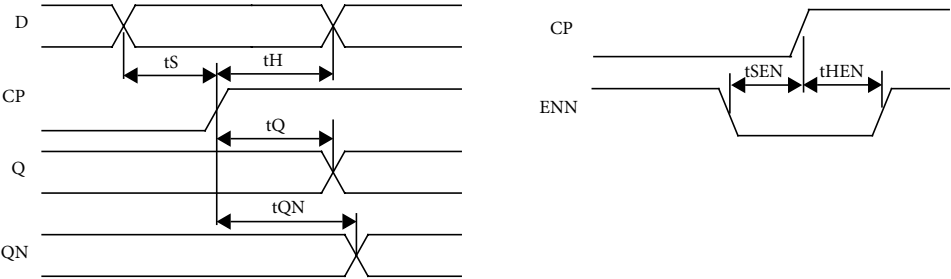
Pin Description

Name	Capacitance (pF)			Description
	MFFNRB1	MFFNRB2	MFFNRB4	
D	0.003	0.003	0.004	Data Input
CP	0.002	0.002	0.004	Clock Input
ENN	0.002	0.002	0.004	Enable Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	MFFNRB1	MFFNRB2	MFFNRB4
D	0.021	0.021	0.028
CP	0.039	0.039	0.051
ENN	0.023	0.023	0.036
Q	0.041	0.055	0.109
QN	0.041	0.055	0.109

Waveforms



Timing numbers for MFFNRB1:

Error Checks

Pulse Width	CP (min)	0.258			
tH	CP -> D (min)	-0.081	tS	D -> CP (min)	0.244
tHEN	CP -> ENN (min)	-0.196	tSEN	ENN -> CP (min)	0.291

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.303	0.300	0.350	0.329	0.445	0.383	0.633	0.487	1.010	0.695
tQN	0.257	0.263	0.305	0.303	0.400	0.370	0.590	0.481	0.967	0.692

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.350	0.329	0.392	0.370	0.409	0.385	0.414	0.388	0.406	0.378
tQN	0.305	0.303	0.346	0.345	0.361	0.363	0.365	0.369	0.355	0.360

Timing numbers for MFFNRB2:

Error Checks

Pulse Width	CP (min)	0.26			
tH	CP -> D (min)	-0.081	tS	D -> CP (min)	0.246
tHEN	CP -> ENN (min)	-0.196	tSEN	ENN -> CP (min)	0.292

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.326	0.321	0.349	0.340	0.396	0.371	0.490	0.427	0.680	0.534
tQN	0.242	0.266	0.269	0.297	0.318	0.341	0.414	0.414	0.605	0.532

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.349	0.340	0.392	0.380	0.409	0.396	0.416	0.400	0.408	0.390
tQN	0.269	0.297	0.310	0.340	0.325	0.356	0.329	0.363	0.319	0.355

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Timing numbers for MFFNRB4:**Error Checks**

Pulse Width	CP (min)	0.178			
tH	CP -> D (min)	-0.054	tS	D -> CP (min)	0.150
tHEN	CP -> ENN (min)	-0.158	tSEN	ENN -> CP (min)	0.202

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

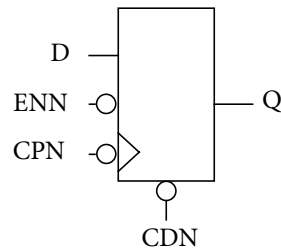
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.507	0.482	0.516	0.494	0.533	0.513	0.565	0.551	0.629	0.624
tQN	0.304	0.348	0.322	0.369	0.352	0.407	0.400	0.465	0.481	0.563

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.516	0.494	0.575	0.554	0.615	0.595	0.644	0.626	0.661	0.643
tQN	0.322	0.369	0.382	0.428	0.424	0.468	0.454	0.497	0.471	0.514

D-Enabled, Active-Low Flip-Flops with Clear, Negative Clock DECFQ1, DECFQ2 and DECFQ4

The DECFQ1 (1x drive), DECFQ2 (2x drive) and DECFQ4 (4x drive) cells are negative edge triggered D flip-flops with active-low enable (ENN) and clear (CDN). When ENN is Low, data present at the D input is transferred to the Q output on the negative edge of the clock, CPN. When ENN is High, the Q output is retained.



Function Table

INPUTS				OUTPUT
CDN	CPN	ENN	D	Q
L	X	X	X	L
H	↓	L	L	L
H	↓	L	H	H
H	L	L	X	Q
H	H	L	X	Q
H	X	H	X	Q _o

Cell Description

Macro Name:	DECFQ1	DECFQ2	DECFQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.25	7.5	9.25
Leakage Power (pW):	149.1	180.0	306.3

Pin Description

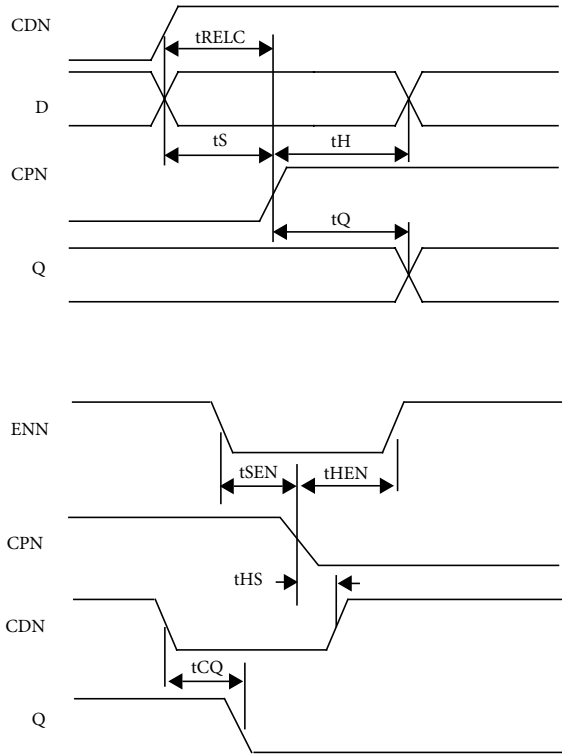
Name	Capacitance (pF)			Description
	DECFQ1	DECFQ2	DECFQ4	
D	0.002	0.002	0.004	Data Input
CPN	0.002	0.002	0.005	Clock Input
ENN	0.002	0.002	0.004	Enable Input
CDN	0.006	0.006	0.01	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

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**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	DECFQ1	DECFQ2	DECFQ4
D	0.015	0.015	0.026
CPN	0.032	0.032	0.049
ENN	0.021	0.021	0.036
CDN	0.001	0.001	-0
Q	0.078	0.093	0.166

Waveforms



Timing numbers for DECFQ1:**Error Checks**

Pulse Width	CPN (min)	0.167	Pulse Width	CDN (min)	0.161
tH	CPN -> D (min)	0.028	tS	D -> CPN (min)	0.205
tHS	CPN -> CDN (min)	0.220	tRelC	CDN -> CPN (min)	-0.187
tHEN	CPN -> ENN (min)	-0.048	tSEN	ENN -> CPN (min)	0.233

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.381	0.319	0.429	0.350	0.524	0.405	0.712	0.512	1.087	0.722
tCQ		0.163		0.195		0.253		0.359		0.569

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.429	0.350	0.519	0.439	0.609	0.527	0.688	0.605	0.753	0.669
tCQ		0.195		0.284		0.374		0.453		0.520

Timing numbers for DECFQ2:**Error Checks**

Pulse Width	CPN (min)	0.178	Pulse Width	CDN (min)	0.164
tH	CPN -> D (min)	0.027	tS	D -> CPN (min)	0.203
tHS	CPN -> CDN (min)	0.220	tRelC	CDN -> CPN (min)	-0.186
tHEN	CPN -> ENN (min)	-0.049	tSEN	ENN -> CPN (min)	0.232

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.377	0.328	0.403	0.349	0.450	0.382	0.543	0.438	0.724	0.541
tCQ		0.170		0.192		0.225		0.284		0.386

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.403	0.349	0.493	0.438	0.583	0.527	0.662	0.604	0.727	0.668
tCQ		0.192		0.284		0.382		0.465		0.536

Timing numbers for DECFQ4:**Error Checks**

Pulse Width	CPN (min)	0.139	Pulse Width	CDN (min)	0.136
tH	CPN -> D (min)	-0.042	tS	D -> CPN (min)	0.167
tHS	CPN -> CDN (min)	0.170	tRelC	CDN -> CPN (min)	-0.138
tHEN	CPN -> ENN (min)	-0.094	tSEN	ENN -> CPN (min)	0.223

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

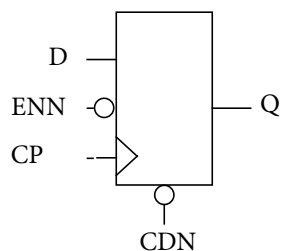
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.347	0.285	0.361	0.300	0.385	0.326	0.425	0.368	0.495	0.444
tCQ		0.163		0.178		0.205		0.247		0.322

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.361	0.300	0.427	0.367	0.480	0.420	0.525	0.465	0.559	0.499
tCQ		0.178		0.262		0.343		0.410		0.461

D-Enabled, Active-Low Flip-Flops with Clear, Positive Clock DECRQ1, DECRQ2 and DECRQ4

The DECRQ1 (1x drive), DECRQ2 (2x drive) and DECRQ4 (4x drive) cells are positive edge triggered D flip-flops with active-low enable (ENN) and clear (CDN). When ENN is Low, data present at the D input is transferred to the Q output on the positive edge of the clock, CP. When ENN is High, the Q output is retained.



Function Table

INPUTS				OUTPUT
CDN	CP	ENN	D	Q
L	X	X	X	L
H	↑	L	L	L
H	↑	L	H	H
H	L	L	X	Q
H	H	L	X	Q
H	X	H	X	Q ₀

Cell Description

Macro Name:	DECRQ1	DECRQ2	DECRQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.	7.25	9.25
Leakage Power (pW):	153.1	182.8	310.9

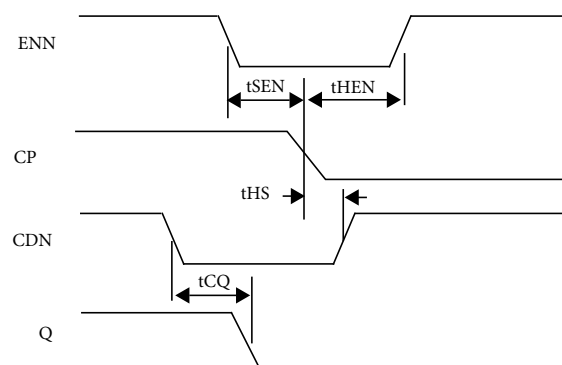
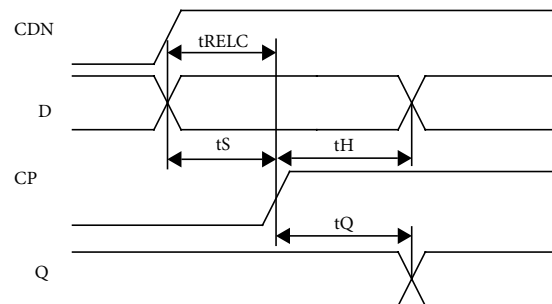
Pin Description

Name	Capacitance (pF)			Description
	DECRQ1	DECRQ2	DECRQ4	
D	0.002	0.002	0.004	Data Input
CP	0.002	0.002	0.005	Clock Input
ENN	0.003	0.003	0.004	Enable Input
CDN	0.006	0.006	0.01	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	DECRQ1	DECRQ2	DECRQ4
D	0.032	0.033	0.051
CP	0.032	0.032	0.05
ENN	0.023	0.023	0.041
CDN	0.001	0.001	0.002
Q	0.065	0.079	0.142

Waveforms



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Timing numbers for DECRQ1:**Error Checks**

Pulse Width	CP (min)	0.226	Pulse Width	CDN (min)	0.141
tH	CP -> D (min)	-0.119	tS	D -> CP (min)	0.215
tHS	CP -> CDN (min)	0.159	tRelC	CDN -> CP (min)	-0.122
tHEN	CPN -> ENN (min)	-0.184	tSEN	ENN -> CPN (min)	0.239

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.308	0.314	0.357	0.344	0.452	0.399	0.642	0.504	1.019	0.712
tCQ		0.158		0.189		0.246		0.352		0.560

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.357	0.344	0.400	0.387	0.419	0.404	0.426	0.411	0.420	0.404
tCQ		0.189		0.277		0.366		0.443		0.509

Timing numbers for DECRQ2:**Error Checks**

Pulse Width	CP (min)	0.226	Pulse Width	CDN (min)	0.171
tH	CP -> D (min)	-0.120	tS	D -> CP (min)	0.216
tHS	CP -> CDN (min)	0.161	tRelC	CDN -> CP (min)	-0.123
tHEN	CPN -> ENN (min)	-0.185	tSEN	ENN -> CPN (min)	0.240

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.302	0.321	0.329	0.342	0.378	0.375	0.475	0.433	0.666	0.539
tCQ		0.162		0.184		0.218		0.277		0.382

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.329	0.342	0.372	0.385	0.391	0.403	0.399	0.410	0.393	0.403
tCQ		0.184		0.276		0.371		0.453		0.523

Timing numbers for DECRQ4:**Error Checks**

Pulse Width	CP (min)	0.166	Pulse Width	CDN (min)	0.162
tH	CP -> D (min)	-0.073	tS	D -> CP (min)	0.144
tHS	CP -> CDN (min)	0.153	tRelC	CDN -> CP (min)	-0.118
tHEN	CPN -> ENN (min)	-0.171	tSEN	ENN -> CPN (min)	0.190

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

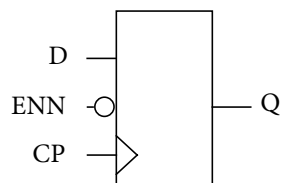
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.312	0.329	0.326	0.344	0.351	0.370	0.391	0.411	0.461	0.487
tCQ		0.164		0.179		0.205		0.247		0.322

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.326	0.344	0.382	0.400	0.421	0.438	0.449	0.466	0.465	0.480
tCQ		0.179		0.263		0.345		0.413		0.465

D-Enabled, Active-Low Flip-Flops with Q Output Only DENRQ1, DENRQ2 and DENRQ4

The DENRQ1 (1x drive), DENRQ2 (2x drive) and DENRQ4 (4x drive) cells are positive edge triggered D flip-flops with active-low enable (ENN). When ENN is Low, data present at the D input is transferred to the Q output on the positive edge of the clock, CP. When ENN is High, the Q output is retained.



Function Table

INPUTS			OUTPUT
CP	ENN	D	Q
↑	L	L	L
↑	L	H	H
L	L	X	Q
H	L	X	Q
X	H	X	Q _o

Cell Description

Macro Name:	DENRQ1	DENRQ2	DENRQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.	7.25	8.
Leakage Power (pW):	157.9	183.7	330.3

Pin Description

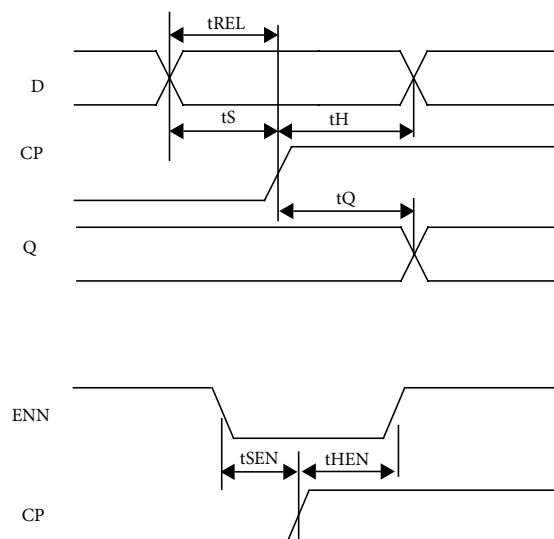
Name	Capacitance (pF)			Description
	DENRQ1	DENRQ2	DENRQ4	
D	0.002	0.002	0.004	Data Input
CP	0.002	0.002	0.004	Clock Input
ENN	0.002	0.002	0.004	Enable Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

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Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	DENRQ1	DENRQ2	DENRQ4
D	0.018	0.018	0.028
CP	0.031	0.031	0.046
ENN	0.022	0.022	0.035
Q	0.055	0.067	0.123

Waveforms



Timing numbers for DENRQ1:**Error Checks**

Pulse Width	CP (min)	0.251			
tH	CP -> D (min)	-0.078	tS	D -> CP (min)	0.224
tHEN	CP -> ENN (min)	-0.175	tSEN	ENN -> CP (min)	0.263

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.334	0.312	0.382	0.342	0.478	0.397	0.669	0.503	1.049	0.714

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.382	0.342	0.428	0.385	0.447	0.402	0.455	0.406	0.447	0.396

Timing numbers for DENRQ2:**Error Checks**

Pulse Width	CP (min)	0.251			
tH	CP -> D (min)	-0.078	tS	D -> CP (min)	0.224
tHEN	CP -> ENN (min)	-0.175	tSEN	ENN -> CP (min)	0.263

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.325	0.317	0.351	0.338	0.399	0.370	0.495	0.427	0.686	0.534

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.351	0.338	0.397	0.381	0.416	0.397	0.423	0.402	0.416	0.392

Timing numbers for DENRQ4:**Error Checks**

Pulse Width	CP (min)	0.152			
tH	CP -> D (min)	-0.070	tS	D -> CP (min)	0.149
tHEN	CP -> ENN (min)	-0.168	tSEN	ENN -> CP (min)	0.181

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

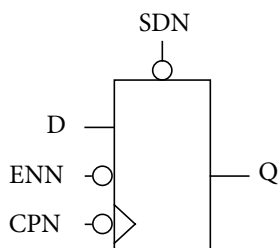
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.265	0.316	0.277	0.330	0.297	0.356	0.331	0.398	0.396	0.474

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.277	0.330	0.327	0.381	0.360	0.414	0.385	0.437	0.397	0.448

D-Enabled, Active-Low Flip-Flops with Set, Negative Clock DEPFQ1, DEPFQ2 and DEPFQ4

The DEPFQ1 (1x drive), DEPFQ2 (2x drive) and DEPFQ4 (4x drive) cells are negative edge triggered D flip-flops with active-low enable (ENN) and set (SDN). When ENN is Low, data present at the D input is transferred to the Q output on the negative edge of the clock, CPN. When ENN is High, the Q output is retained.



Function Table

INPUTS				OUTPUT
SDN	CPN	ENN	D	Q
L	X	X	X	H
H	↓	L	L	L
H	↓	L	H	H
H	L	L	X	Q
H	H	L	X	Q
H	X	H	X	Q ₀

Cell Description

Macro Name:	DEPFQ1	DEPFQ2	DEPFQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.5	7.5	8.5
Leakage Power (pW):	168.6	178.9	264.8

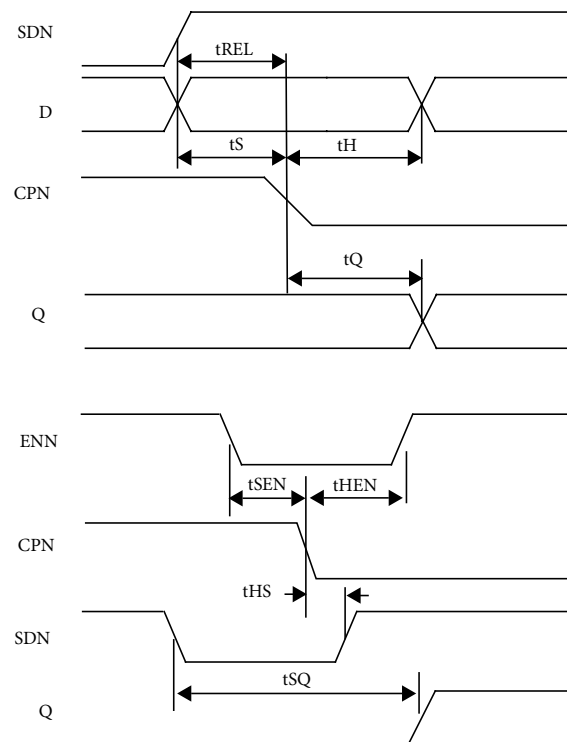
Pin Description

Name	Capacitance (pF)			Description
	DEPFQ1	DEPFQ2	DEPFQ4	
D	0.002	0.002	0.004	Data Input
CPN	0.002	0.002	0.004	Clock Input
ENN	0.003	0.003	0.004	Enable Input
SDN	0.006	0.006	0.009	Preset Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	DEPFQ1	DEPFQ2	DEPFQ4
D	0.018	0.018	0.032
CPN	0.032	0.032	0.047
SDN	-0	-0	-0
ENN	0.018	0.018	0.035
Q	0.075	0.085	0.159

Waveforms



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Timing numbers for DEPFQ1:**Error Checks**

Pulse Width	CPN (min)	0.196	Pulse Width	SDN (min)	0.156
tH	CPN -> D (min)	0.028	tS	D -> CPN (min)	0.218
tHS	CPN -> SDN (min)	0.092	tRelS	SDN -> CPN (min)	-0.062
tHEN	CPN -> ENN (min)	-0.035	tSEN	ENN -> CPN (min)	0.242

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.437	0.343	0.507	0.381	0.645	0.452	0.919	0.590	1.467	0.867
tSQ	0.364		0.434		0.572		0.846		1.394	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.507	0.381	0.600	0.473	0.696	0.570	0.781	0.654	0.851	0.725
tSQ	0.434		0.528		0.635		0.730		0.806	

Timing numbers for DEPFQ2:**Error Checks**

Pulse Width	CPN (min)	0.205	Pulse Width	SDN (min)	0.158
tH	CPN -> D (min)	0.029	tS	D -> CPN (min)	0.217
tHS	CPN -> SDN (min)	0.092	tRelS	SDN -> CPN (min)	-0.064
tHEN	CPN -> ENN (min)	-0.036	tSEN	ENN -> CPN (min)	0.241

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.411	0.336	0.446	0.359	0.512	0.395	0.644	0.462	0.907	0.590
tSQ	0.339		0.373		0.440		0.572		0.834	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.446	0.359	0.538	0.451	0.635	0.548	0.718	0.632	0.788	0.702
tSQ	0.373		0.467		0.575		0.670		0.747	

Timing numbers for DEPFQ4:**Error Checks**

Pulse Width	CPN (min)	0.161	Pulse Width	SDN (min)	0.149
tH	CPN -> D (min)	-0.050	tS	D -> CPN (min)	0.194
tHS	CPN -> SDN (min)	0.038	tRelS	SDN -> CPN (min)	-0.001
tHEN	CPN -> ENN (min)	-0.100	tSEN	ENN -> CPN (min)	0.246

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

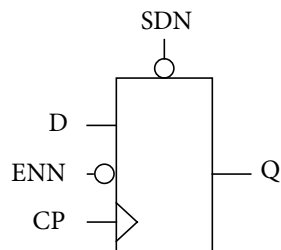
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.317	0.299	0.330	0.313	0.352	0.339	0.388	0.380	0.456	0.456
tSQ	0.306		0.319		0.341		0.377		0.443	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.330	0.313	0.394	0.379	0.445	0.432	0.488	0.477	0.520	0.511
tSQ	0.319		0.405		0.492		0.566		0.626	

D-Enabled, Active-Low Flip-Flops with Set, Positive Clock DEPRQ1, DEPRQ2 and DEPRQ4

The DEPRQ1 (1x drive), DEPRQ2 (2x drive) and DEPRQ4 (4x drive) cells are positive edge triggered D flip-flops with active-low enable (ENN) and set (SDN). When ENN is Low, data present at the D input is transferred to the Q output on the positive edge of the clock, CP. When ENN is High, the Q output is retained.



Function Table

INPUTS				OUTPUT
SDN	CP	ENN	D	Q
L	X	X	X	H
H	↑	L	L	L
H	↑	L	H	H
H	L	L	X	Q
H	H	L	X	Q
H	X	H	X	Q _o

Cell Description

Macro Name:	DEPRQ1	DEPRQ2	DEPRQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.5	7.75	9.25
Leakage Power (pW):	175.9	186.5	286.6

Pin Description

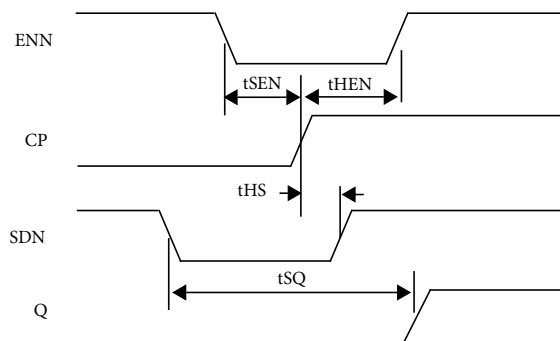
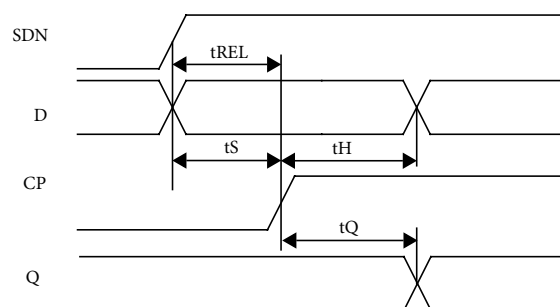
Name	Capacitance (pF)			Description
	DEPRQ1	DEPRQ2	DEPRQ4	
D	0.002	0.002	0.004	Data Input
CP	0.002	0.002	0.004	Clock Input
ENN	0.002	0.002	0.004	Enable Input
SDN	0.006	0.006	0.01	Preset Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

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**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	DEPRQ1	DEPRQ2	DEPRQ4
D	0.017	0.017	0.033
CP	0.032	0.032	0.043
ENN	0.021	0.021	0.035
SDN	-0	-0	0
Q	0.074	0.087	0.149

Waveforms



Timing numbers for DEPRQ1:**Error Checks**

Pulse Width	CP (min)	0.246	Pulse Width	SDN (min)	0.182
tH	CP -> D (min)	-0.103	tS	D -> CP (min)	0.216
tHS	CP -> SDN (min)	0.075	tRelS	SDN -> CP (min)	-0.053
tHEN	CP -> ENN (min)	-0.180	tSEN	ENN -> CP (min)	0.250

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.336	0.337	0.384	0.368	0.480	0.423	0.671	0.528	1.052	0.736
tSQ	0.329		0.378		0.473		0.664		1.045	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.384	0.368	0.429	0.410	0.450	0.429	0.460	0.437	0.456	0.430
tSQ	0.378		0.471		0.576		0.668		0.743	

Timing numbers for DEPRQ2:**Error Checks**

Pulse Width	CP (min)	0.245	Pulse Width	SDN (min)	0.194
tH	CP -> D (min)	-0.103	tS	D -> CP (min)	0.216
tHS	CP -> SDN (min)	0.075	tRelS	SDN -> CP (min)	-0.053
tHEN	CP -> ENN (min)	-0.180	tSEN	ENN -> CP (min)	0.249

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.327	0.342	0.353	0.363	0.401	0.395	0.497	0.453	0.688	0.558
tSQ	0.325		0.351		0.399		0.494		0.685	

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**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.353	0.363	0.398	0.405	0.419	0.424	0.429	0.432	0.425	0.426
tSQ	0.351		0.443		0.550		0.644		0.720	

Timing numbers for DEPRQ4:

Error Checks

Pulse Width	CP (min)	0.187	Pulse Width	SDN (min)	0.157
tH	CP -> D (min)	-0.085	tS	D -> CP (min)	0.186
tHS	CP -> SDN (min)	0.036	tRelS	SDN -> CP (min)	-0.015
tHEN	CPN -> ENN (min)	-0.181	tSEN	ENN -> CPN (min)	0.236

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.295	0.336	0.307	0.351	0.329	0.377	0.366	0.420	0.434	0.499
tSQ	0.328		0.341		0.363		0.399		0.466	

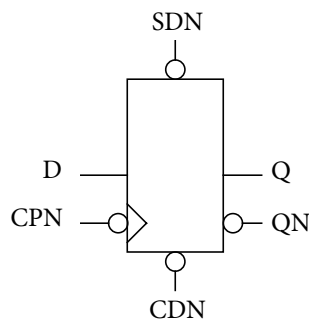
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.307	0.351	0.359	0.403	0.393	0.437	0.419	0.463	0.435	0.477
tSQ	0.341		0.430		0.526		0.609		0.675	

D Flip-Flops with Set, Clear and Negative Clock

DFBFB1, DFBFB2 and DFBFB4

The DFBFB1 (1x drive), DFBFB2 (2x drive) and DFBFB4 (4x drive) cells are negative edge triggered D flip-flops with active-low clear, CDN, and set, SDN. Data present at the D input is transferred to the Q and QN outputs on the negative edge of the clock, CPN.



Function Table

INPUTS				OUTPUTS	
CDN	SDN	CPN	D	Q	QN
L	H	X	X	L	H
H	L	X	X	H	L
H	H	↓	L	L	H
H	H	↓	H	H	L
H	H	L	X	Q	QN
H	H	H	X	Q	QN
L	L	X	X	L	L

Cell Description

Macro Name:	DFBFB1	DFBFB2	DFBFB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	6.75	7.25	8.5
Leakage Power (pW):	146.5	190.3	389.5

Pin Description

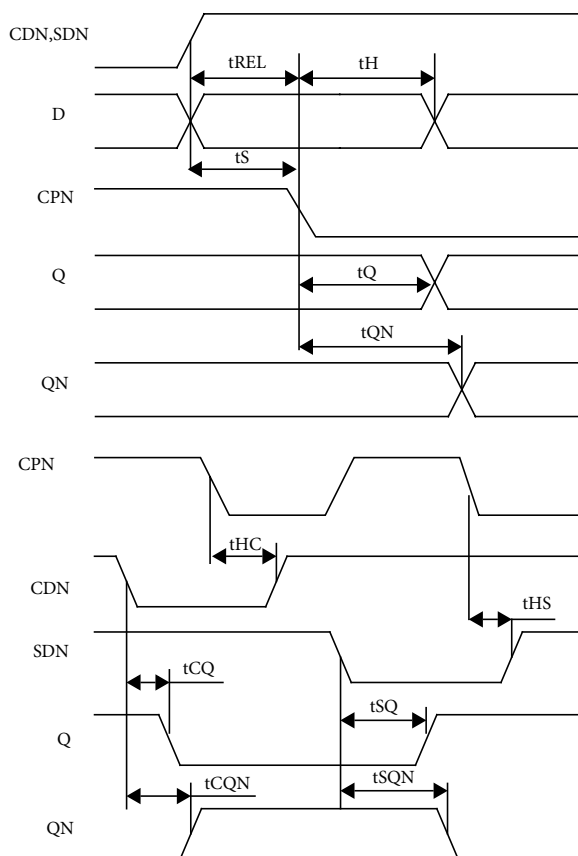
Name	Capacitance (pF)			Description
	DFBFB1	DFBFB2	DFBFB4	
D	0.002	0.002	0.004	Data Input
CPN	0.002	0.002	0.004	Clock Input
SDN	0.007	0.007	0.009	Preset Input
CDN	0.008	0.008	0.01	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

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Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	DFBFB1	DFBFB2	DFBFB4
D	0.015	0.015	0.019
CPN	0.043	0.044	0.055
SDN	0.006	0.006	0.004
CDN	0.001	0.001	-0
Q	0.038	0.054	0.099
QN	0.039	0.057	0.108

Waveforms



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Timing numbers for DFBFB1:**Error Checks**

Pulse Width	CPN (min)	0.201			
Pulse Width	CDN (min)	0.162	Pulse Width	SDN (min)	0.19
tH	CPN -> D (min)	0.074	tS	D -> CPN (min)	0.130
tHS	CPN -> SDN (min)	0.108	tRelS	SDN -> CPN (min)	-0.056
tHC	CPN -> CDN (min)	0.231	tRelC	CDN -> CPN (min)	-0.201

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.426	0.350	0.478	0.378	0.581	0.431	0.786	0.530	1.198	0.726
tQN	0.296	0.323	0.350	0.365	0.451	0.434	0.648	0.544	1.039	0.738
tSQ/tCQ	0.368	0.140	0.419	0.168	0.522	0.221	0.727	0.321	1.139	0.519
tCQN/tSQN	0.369	0.264	0.422	0.308	0.523	0.377	0.719	0.484	1.110	0.676

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.478	0.378	0.568	0.468	0.659	0.558	0.738	0.635	0.803	0.699
tQN	0.350	0.365	0.439	0.455	0.529	0.547	0.606	0.626	0.670	0.691
tSQ/tCQ	0.419	0.168	0.510	0.248	0.609	0.323	0.696	0.387	0.766	0.442
tCQN/tSQN	0.422	0.308	0.507	0.399	0.591	0.499	0.665	0.586	0.728	0.656

Timing numbers for DFBFB2:**Error Checks**

Pulse Width	CPN (min)	0.239			
Pulse Width	CDN (min)	0.172	Pulse Width	SDN (min)	0.221
tH	CPN -> D (min)	0.076	tS	D -> CPN (min)	0.128
tHS	CPN -> SDN (min)	0.106	tRelS	SDN -> CPN (min)	-0.061
tHC	CPN -> CDN (min)	0.219	tRelC	CDN -> CPN (min)	-0.196

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.467	0.378	0.490	0.396	0.534	0.426	0.623	0.481	0.805	0.586
tQN	0.297	0.331	0.329	0.365	0.383	0.417	0.480	0.499	0.666	0.627
tSQ/tCQ	0.416	0.141	0.439	0.161	0.482	0.193	0.571	0.251	0.753	0.357
tCQN/tSQN	0.367	0.280	0.398	0.315	0.451	0.365	0.548	0.445	0.732	0.570

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.490	0.396	0.580	0.486	0.673	0.577	0.753	0.656	0.819	0.721
tQN	0.329	0.365	0.418	0.455	0.510	0.548	0.589	0.628	0.655	0.694
tSQ/tCQ	0.439	0.161	0.530	0.244	0.636	0.323	0.731	0.391	0.805	0.447
tCQN/tSQN	0.398	0.315	0.484	0.407	0.568	0.511	0.642	0.603	0.706	0.677

Timing numbers for DFBFB4:**Error Checks**

Pulse Width	CPN (min)	0.315			
Pulse Width	CDN (min)	0.191	Pulse Width	SDN (min)	0.215
tH	CPN -> D (min)	0.020	tS	D -> CPN (min)	0.103
tHS	CPN -> SDN (min)	0.036	tRelS	SDN -> CPN (min)	-0.005
tHC	CPN -> CDN (min)	0.134	tRelC	CDN -> CPN (min)	-0.111

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.506	0.486	0.517	0.498	0.534	0.519	0.568	0.556	0.634	0.628
tQN	0.281	0.340	0.299	0.362	0.329	0.400	0.376	0.460	0.457	0.561
tSQ/tCQ	0.475	0.177	0.485	0.193	0.503	0.220	0.537	0.263	0.603	0.340
tCQN/tSQN	0.402	0.323	0.417	0.344	0.443	0.379	0.486	0.434	0.562	0.528

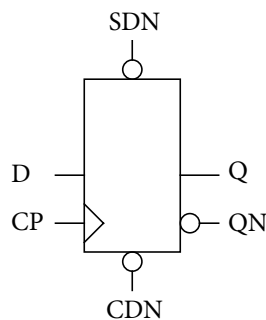
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.517	0.498	0.583	0.567	0.638	0.623	0.684	0.670	0.720	0.707
tQN	0.299	0.362	0.368	0.429	0.424	0.483	0.471	0.530	0.508	0.566
tSQ/tCQ	0.485	0.193	0.571	0.280	0.667	0.368	0.753	0.443	0.819	0.501
tCQN/tSQN	0.417	0.344	0.502	0.430	0.593	0.527	0.670	0.613	0.731	0.680

D Flip-Flops with Set, Clear and Positive Clock

DFBRB1, DFBRB2 and DFBRB4

The DFBRB1 (1x drive), DFBRB2 (2x drive) and DFBRB4 (4x drive) cells are positive edge triggered D flip-flops with active-low clear, CDN, and set, SDN. Data present at the D input is transferred to the Q and QN outputs on the positive edge of the clock, CP.



Function Table

INPUTS				OUTPUTS	
CDN	SDN	CP	D	Q	QN
L	H	X	X	L	H
H	L	X	X	H	L
H	H	↑	L	L	H
H	H	↑	H	H	L
H	H	L	X	Q	QN
H	H	H	X	Q	QN
L	L	X	X	L	L

Cell Description

Macro Name:	DFBRB1	DFBRB2	DFBRB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	6.75	7.25	8.25
Leakage Power (pW)	146.5	190.3	382.5

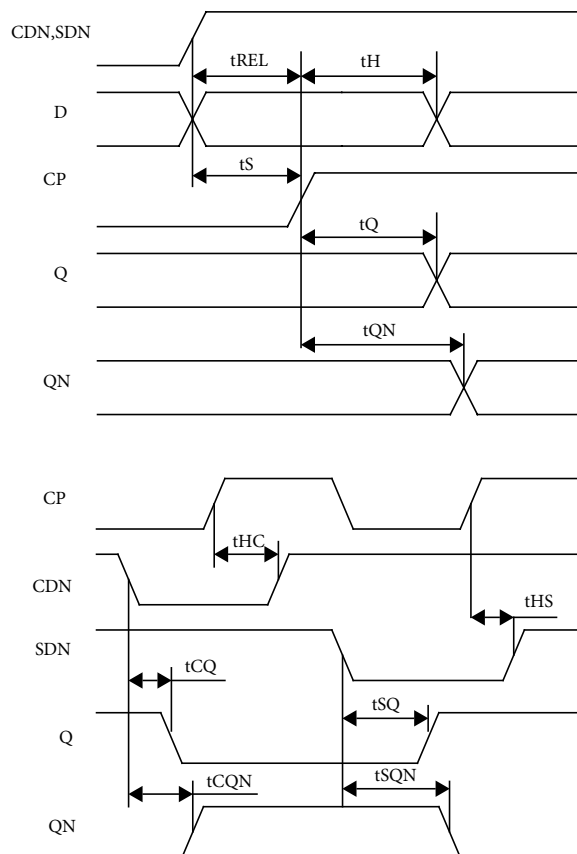
Pin Description

Name	Capacitance (pF)			Description
	DFBRB1	DFBRB2	DFBRB4	
D	0.002	0.002	0.004	Data Input
CP	0.002	0.002	0.004	Clock Input
SDN	0.006	0.006	0.008	Preset Input
CDN	0.008	0.008	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	DFBRB1	DFBRB2	DFBRB4
D	0.019	0.019	0.024
CP	0.042	0.042	0.052
SDN	0.012	0.012	0.013
CDN	0.002	0.002	0.003
Q	0.035	0.05	0.094
QN	0.035	0.054	0.104

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Timing numbers for DFBRB1:**Error Checks**

Pulse Width	CP (min)	0.223			
Pulse Width	CDN (min)	0.128	Pulse Width	SDN (min)	0.162
tH	CP -> D (min)	-0.053	tS	D -> CP (min)	0.118
tHS	CP -> SDN (min)	0.093	tRelS	SDN -> CP (min)	-0.065
tHC	CP -> CDN (min)	0.186	tRelC	CDN -> CP (min)	-0.157

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.399	0.364	0.451	0.392	0.554	0.445	0.759	0.543	1.171	0.740
tQN	0.309	0.297	0.363	0.338	0.464	0.406	0.661	0.516	1.052	0.711
tSQ/tCQ	0.368	0.140	0.419	0.168	0.522	0.221	0.727	0.321	1.139	0.519
tCQN/tSQN	0.370	0.264	0.423	0.308	0.524	0.377	0.720	0.484	1.111	0.676

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.451	0.392	0.502	0.444	0.528	0.468	0.543	0.481	0.543	0.479
tQN	0.363	0.338	0.415	0.389	0.440	0.416	0.453	0.431	0.450	0.430
tSQ/tCQ	0.419	0.168	0.510	0.248	0.609	0.323	0.696	0.387	0.766	0.442
tCQN/tSQN	0.423	0.308	0.508	0.399	0.592	0.499	0.666	0.586	0.729	0.656

Timing numbers for DFBRB2:**Error Checks**

Pulse Width	CP (min)	0.225			
Pulse Width	CDN (min)	0.161	Pulse Width	SDN (min)	0.225
tH	CP -> D (min)	-0.051	tS	D -> CP (min)	0.120
tHS	CP -> SDN (min)	0.091	tRelS	SDN -> CP (min)	-0.063
tHC	CP -> CDN (min)	0.179	tRelC	CDN -> CP (min)	-0.157

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.448	0.384	0.470	0.402	0.514	0.433	0.603	0.488	0.785	0.592
tQN	0.304	0.312	0.335	0.346	0.389	0.397	0.486	0.479	0.671	0.607
tSQ/tCQ	0.416	0.141	0.439	0.161	0.482	0.193	0.571	0.251	0.753	0.358
tCQN/tSQN	0.368	0.280	0.399	0.315	0.451	0.365	0.548	0.445	0.733	0.569

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.470	0.402	0.522	0.454	0.548	0.479	0.563	0.491	0.562	0.489
tQN	0.335	0.346	0.387	0.397	0.411	0.423	0.424	0.438	0.422	0.438
tSQ/tCQ	0.439	0.161	0.530	0.244	0.636	0.323	0.730	0.391	0.804	0.447
tCQN/tSQN	0.399	0.315	0.484	0.407	0.568	0.511	0.642	0.603	0.706	0.677

Timing numbers for DFBRB4:**Error Checks**

Pulse Width	CP (min)	0.158			
Pulse Width	CDN (min)	0.216	Pulse Width	SDN (min)	0.334
tH	CP -> D (min)	-0.008	tS	D -> CP (min)	0.089
tHS	CP -> SDN (min)	0.036	tRelS	SDN -> CP (min)	-0.019
tHC	CP -> CDN (min)	0.116	tRelC	CDN -> CP (min)	-0.097

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.502	0.500	0.513	0.512	0.530	0.533	0.563	0.570	0.629	0.642
tQN	0.297	0.331	0.315	0.352	0.346	0.390	0.394	0.449	0.475	0.548
tSQ/tCQ	0.481	0.174	0.490	0.190	0.507	0.217	0.540	0.261	0.604	0.339
tCQN/tSQN	0.397	0.322	0.415	0.342	0.445	0.376	0.492	0.430	0.573	0.523

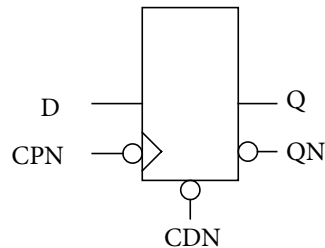
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.513	0.512	0.566	0.567	0.603	0.604	0.631	0.632	0.647	0.648
tQN	0.315	0.352	0.370	0.406	0.407	0.443	0.435	0.470	0.451	0.487
tSQ/tCQ	0.490	0.190	0.576	0.277	0.671	0.365	0.758	0.438	0.828	0.496
tCQN/tSQN	0.415	0.342	0.492	0.429	0.563	0.526	0.624	0.613	0.674	0.680

D Flip-Flops with Clear and Negative Clock

DFCFB1, DFCFB2 and DFCFB4

The DFCFB1 (1x drive), DFCFB2 (2x drive) and DFCFB4 (4x drive) cells are negative edge triggered D flip-flops with active-low clear, CDN. Data present at the D input is transferred to the Q and QN outputs on the negative edge of the clock, CPN.



Function Table

INPUTS			OUTPUTS	
CDN	CPN	D	Q	QN
L	X	X	L	H
H	↓	L	L	H
H	↓	H	H	L
H	L	X	Q	QN
H	H	X	Q	QN

Cell Description

Macro Name:	DFCFB1	DFCFB2	DFCFB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	5.75	6.25	7.25
Leakage Power (pW):	130.4	174.7	284.8

Pin Description

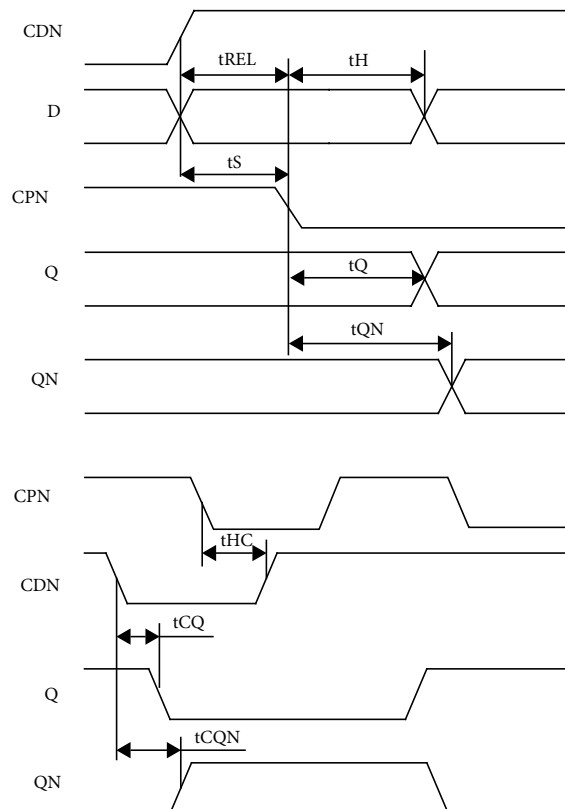
Name	Capacitance (pF)			Description
	DFCFB1	DFCFB2	DFCFB4	
D	0.002	0.002	0.004	Data Input
CPN	0.002	0.002	0.004	Clock Input
CDN	0.009	0.008	0.01	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

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Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	DFCFB1	DFCFB2	DFCFB4
D	0.009	0.009	0.013
CPN	0.038	0.038	0.05
CDN	0.001	0.001	-0
Q	0.048	0.065	0.112
QN	0.048	0.065	0.112

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Timing numbers for DFCFB1:**Error Checks**

Pulse Width	CPN (min)	0.212	Pulse Width	CDN (min)	0.153
tH	CPN -> D (min)	0.080	tS	D -> CPN (min)	0.094
tHC	CPN -> CDN (min)	0.199	tRelC	CDN -> CPN (min)	-0.176

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.398	0.370	0.450	0.398	0.554	0.451	0.759	0.549	1.171	0.746
tQN	0.306	0.333	0.358	0.375	0.459	0.442	0.656	0.549	1.046	0.744
tCQ		0.143		0.171		0.223		0.324		0.520
tCQN	0.342		0.394		0.494		0.690		1.081	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.450	0.398	0.540	0.488	0.631	0.580	0.710	0.659	0.775	0.725
tQN	0.358	0.375	0.448	0.464	0.539	0.555	0.618	0.634	0.684	0.698
tCQ		0.171		0.256		0.340		0.414		0.477
tCQN	0.394		0.468		0.533		0.589		0.632	

Timing numbers for DFCFB2:**Error Checks**

Pulse Width	CPN (min)	0.251	Pulse Width	CDN (min)	0.165
tH	CPN -> D (min)	0.080	tS	D -> CPN (min)	0.093
tHC	CPN -> CDN (min)	0.187	tRelC	CDN -> CPN (min)	-0.169

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.432	0.399	0.453	0.416	0.497	0.445	0.588	0.498	0.770	0.598
tQN	0.307	0.338	0.338	0.370	0.391	0.418	0.489	0.495	0.678	0.614
tCQ		0.143		0.162		0.193		0.248		0.349
tCQN	0.341		0.372		0.424		0.520		0.709	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.453	0.416	0.543	0.506	0.634	0.599	0.714	0.679	0.780	0.745
tQN	0.338	0.370	0.428	0.460	0.520	0.552	0.600	0.631	0.667	0.697
tCQ		0.162		0.249		0.336		0.412		0.477
tCQN	0.372		0.445		0.511		0.566		0.610	

Timing numbers for DFCFB4:**Error Checks**

Pulse Width	CPN (min)	0.253	Pulse Width	CDN (min)	0.158
tH	CPN -> D (min)	0.030	tS	D -> CPN (min)	0.095
tHC	CPN -> CDN (min)	0.124	tRelC	CDN -> CPN (min)	-0.104

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.494	0.418	0.505	0.430	0.522	0.450	0.556	0.488	0.622	0.560
tQN	0.251	0.313	0.267	0.334	0.295	0.371	0.338	0.429	0.414	0.526
tCQ		0.160		0.175		0.201		0.243		0.320
tCQN	0.378		0.394		0.420		0.462		0.538	

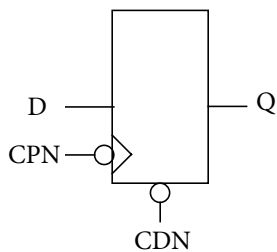
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.505	0.430	0.571	0.498	0.626	0.554	0.672	0.600	0.708	0.636
tQN	0.267	0.334	0.335	0.401	0.391	0.456	0.437	0.501	0.473	0.537
tCQ		0.175		0.259		0.339		0.405		0.456
tCQN	0.394		0.477		0.562		0.633		0.688	

D Flip-Flops with Clear, Negative Clock and Q Output Only

DFCFQ1, DFCFQ2 and DFCFQ4

The DFCFQ1 (1x drive), DFCFQ2 (2x drive) and DFCFQ4 (4x drive) cells are negative edge triggered D flip-flops with active-low clear, CDN. Data present at the D input is transferred to the Q output on the negative edge of the clock, CPN.



Function Table

INPUTS			OUTPUT
CDN	CPN	D	Q
L	X	X	L
H	↓	L	L
H	↓	H	H
H	L	X	Q
H	H	X	Q

Cell Description

Macro Name:	DFCFQ1	DFCFQ2	DFCFQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	5.25	5.5	6.
Leakage Power (pW):	126.7	149.0	244.2

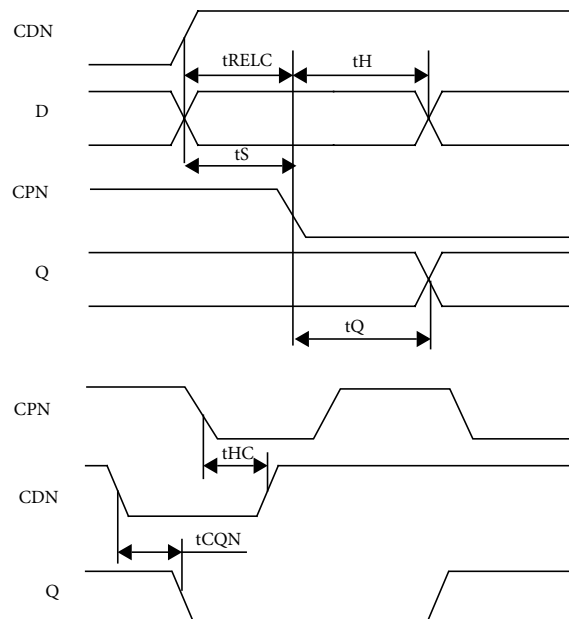
Pin Description

Name	Capacitance (pF)			Description
	DFCFQ1	DFCFQ2	DFCFQ4	
D	0.002	0.002	0.004	Data Input
CPN	0.002	0.002	0.004	Clock Input
CDN	0.009	0.008	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	DFCFQ1	DFCFQ2	DFCFQ4
D	0.009	0.009	0.013
CPN	0.038	0.038	0.05
CDN	0.001	0.001	-0
Q	0.082	0.095	0.14

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Timing numbers for DFCFQ1:**Error Checks**

Pulse Width	CPN (min)	0.173	Pulse Width	CDN (min)	0.138
tH	CPN -> D (min)	0.077	tS	D -> CPN (min)	0.093
tHC	CPN -> CDN (min)	0.208	tRelC	CDN -> CPN (min)	-0.184

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.362	0.318	0.407	0.345	0.496	0.395	0.671	0.489	1.021	0.676
tCQ		0.145		0.172		0.223		0.319		0.507

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.407	0.345	0.496	0.434	0.586	0.524	0.663	0.601	0.727	0.665
tCQ		0.172		0.258		0.343		0.419		0.479

Timing numbers for DFCFQ2:**Error Checks**

Pulse Width	CPN (min)	0.178	Pulse Width	CDN (min)	0.142
tH	CPN -> D (min)	0.077	tS	D -> CPN (min)	0.093
tHC	CPN -> CDN (min)	0.208	tRelC	CDN -> CPN (min)	-0.182

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.352	0.318	0.377	0.338	0.423	0.371	0.514	0.429	0.693	0.542
tCQ		0.151		0.173		0.207		0.268		0.380

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.377	0.338	0.467	0.428	0.557	0.517	0.634	0.594	0.698	0.658
tCQ		0.173		0.265		0.359		0.440		0.498

Timing numbers for DFCFQ4:**Error Checks**

Pulse Width	CPN (min)	0.135	Pulse Width	CDN (min)	0.132
tH	CPN -> D (min)	0.024	tS	D -> CPN (min)	0.090
tHC	CPN -> CDN (min)	0.157	tRelC	CDN -> CPN (min)	-0.123

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.341	0.293	0.355	0.308	0.379	0.334	0.419	0.376	0.488	0.452
tCQ		0.159		0.174		0.201		0.243		0.319

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

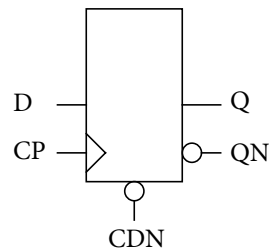
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.355	0.308	0.423	0.375	0.477	0.430	0.523	0.476	0.558	0.510
tCQ		0.174		0.258		0.340		0.407		0.458

D Flip-Flops with Clear and Positive Clock

DFCRB1, DFCRB2 and DFCRB4

The DFCRB1 (1x drive), DFCRB2 (2x drive) and DFCRB4 (4x drive) cells are positive edge triggered D flip-flops with active-low clear, CDN. Data present at the D input is transferred to the Q and QN outputs on the positive edge of the clock, CP.

Function Table



INPUTS			OUTPUTS	
CDN	CP	D	Q	QN
L	X	X	L	H
H	↑	L	L	H
H	↑	H	H	L
H	L	X	Q	QN
H	H	X	Q	QN

Cell Description

Macro Name:	DFCRB1	DFCRB2	DFCRB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	5.75	6.25	7.
Leakage Power (pW):	130.4	174.7	278.4

Pin Description

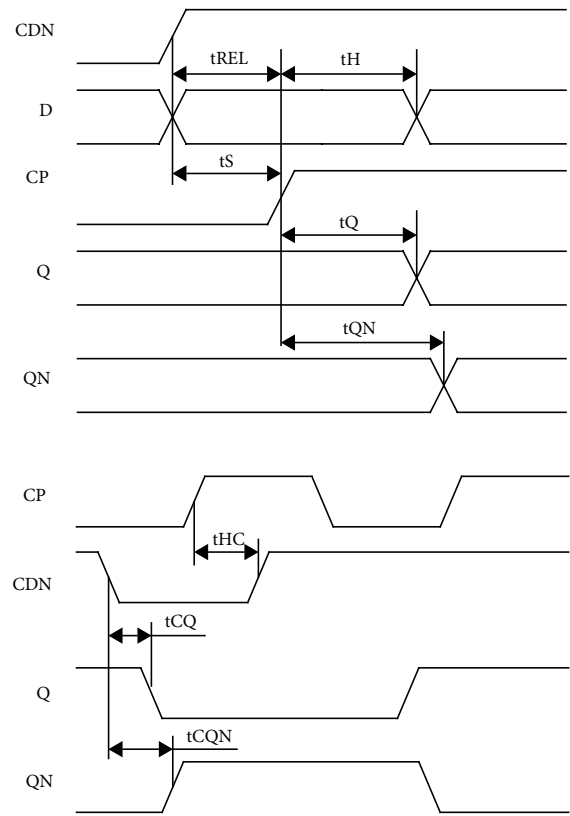
Name	Capacitance (pF)			Description
	DFCRB1	DFCRB2	DFCRB4	
D	0.002	0.002	0.003	Data Input
CP	0.002	0.002	0.004	Clock Input
CDN	0.009	0.009	0.01	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

footer

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	DFCRB1	DFCRB2	DFCRB4
D	0.029	0.029	0.038
CP	0.041	0.041	0.051
CDN	0.002	0.002	0.003
Q	0.04	0.057	0.101
QN	0.04	0.057	0.101

Waveforms



footer

Timing numbers for DFCRB1:**Error Checks**

Pulse Width	CP (min)	0.214	Pulse Width	CDN (min)	0.131
tH	CP -> D (min)	-0.018	tS	D -> CP (min)	0.090
tHC	CP -> CDN (min)	0.149	tRelC	CDN -> CP (min)	-0.122

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.361	0.377	0.413	0.405	0.517	0.458	0.722	0.556	1.134	0.753
tQN	0.313	0.296	0.365	0.337	0.465	0.404	0.662	0.512	1.052	0.706
tCQ		0.142		0.170		0.223		0.323		0.520
tCQN	0.342		0.394		0.494		0.690		1.081	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.413	0.405	0.463	0.456	0.489	0.481	0.503	0.494	0.501	0.492
tQN	0.365	0.337	0.416	0.388	0.441	0.413	0.454	0.427	0.452	0.426
tCQ		0.170		0.255		0.339		0.414		0.477
tCQN	0.394		0.468		0.533		0.589		0.632	

Timing numbers for DFCRB2:**Error Checks**

Pulse Width	CP (min)	0.217	Pulse Width	CDN (min)	0.168
tH	CP -> D (min)	-0.020	tS	D -> CP (min)	0.093
tHC	CP -> CDN (min)	0.142	tRelC	CDN -> CP (min)	-0.123

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.405	0.399	0.427	0.416	0.471	0.445	0.561	0.498	0.743	0.598
tQN	0.307	0.311	0.338	0.343	0.391	0.391	0.489	0.468	0.677	0.587
tCQ		0.143		0.163		0.194		0.248		0.350
tCQN	0.341		0.372		0.424		0.520		0.709	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.427	0.416	0.477	0.467	0.502	0.492	0.517	0.506	0.516	0.504
tQN	0.338	0.343	0.389	0.394	0.414	0.419	0.427	0.433	0.425	0.432
tCQ		0.163		0.251		0.339		0.416		0.481
tCQN	0.372		0.445		0.511		0.566		0.610	

Timing numbers for DFICRB4:**Error Checks**

Pulse Width	CP (min)	0.149	Pulse Width	CDN (min)	0.21
tH	CP -> D (min)	-0.005	tS	D -> CP (min)	0.076
tHC	CP -> CDN (min)	0.105	tRelC	CDN -> CP (min)	-0.092

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.494	0.430	0.505	0.442	0.523	0.462	0.556	0.500	0.622	0.572
tQN	0.263	0.312	0.279	0.333	0.307	0.371	0.350	0.428	0.426	0.525
tCQ		0.161		0.176		0.202		0.244		0.320
tCQN	0.380		0.395		0.421		0.463		0.538	

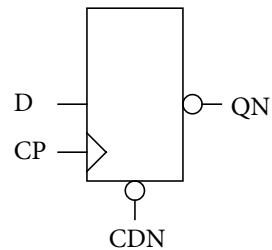
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.505	0.442	0.559	0.497	0.596	0.534	0.624	0.563	0.641	0.579
tQN	0.279	0.333	0.335	0.388	0.372	0.425	0.400	0.453	0.417	0.470
tCQ		0.176		0.259		0.339		0.405		0.455
tCQN	0.395		0.478		0.563		0.634		0.688	

D Flip-Flops with Clear and QN Output Only

DFCRN1, DFCRN2 and DFCRN4

The DFCRN1 (1x drive), DFCRN2 (2x drive) and DFCRN4 (4x drive) cells are positive edge triggered D flip-flops with active-low clear, CDN. Data present at the D input is transferred to the QN output on the positive edge of the clock, CP.



Function Table

INPUTS			OUTPUT
CDN	CP	D	QN
L	X	X	H
H	↑	L	H
H	↑	H	L
H	L	X	QN
H	H	X	QN

Cell Description

Macro Name:	DFCRN1	DFCRN2	DFCRN4
Drive Capability:	1x	2x	4x
Gate Equivalents:	5.25	5.5	6.
Leakage Power (pW):	109.5	115.2	142.3

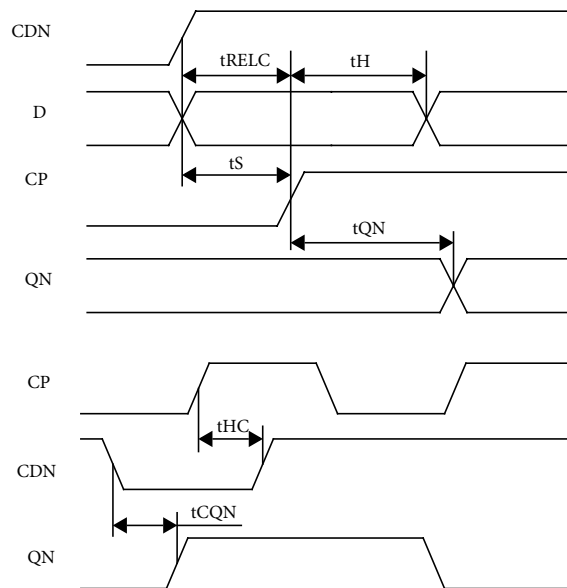
Pin Description

Name	Capacitance (pF)			Description
	DFCRN1	DFCRN2	DFCRN4	
D	0.002	0.002	0.004	Data Input
CP	0.002	0.002	0.004	Clock Input
CDN	0.009	0.009	0.009	Clear Input
Maximum capacitance				
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	DFCRN1	DFCRN2	DFCRN4
D	0.029	0.029	0.039
CP	0.041	0.04	0.051
CDN	0.002	0.002	0.003
QN	0.066	0.083	0.135

Waveforms



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Timing numbers for DFCRN1:**Error Checks**

Pulse Width	CP (min)	0.213	Pulse Width	CDN (min)	0.109
tH	CP -> D (min)	-0.020	tS	D -> CP (min)	0.090
tHC	CP -> CDN (min)	0.148	tRelC	CDN -> CP (min)	-0.124

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.307	0.307	0.354	0.347	0.444	0.414	0.620	0.522	0.970	0.716
tCQN	0.338		0.385		0.474		0.649		0.999	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.354	0.347	0.405	0.398	0.430	0.423	0.442	0.437	0.440	0.436
tCQN	0.385		0.457		0.523		0.579		0.622	

Timing numbers for DFCRN2:**Error Checks**

Pulse Width	CP (min)	0.215	Pulse Width	CDN (min)	0.124
tH	CP -> D (min)	-0.022	tS	D -> CP (min)	0.092
tHC	CP -> CDN (min)	0.142	tRelC	CDN -> CP (min)	-0.123

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.300	0.318	0.330	0.352	0.380	0.401	0.473	0.480	0.652	0.610
tCQN	0.333		0.362		0.411		0.504		0.683	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.330	0.352	0.381	0.402	0.406	0.428	0.419	0.441	0.417	0.440
tCQN	0.362		0.435		0.501		0.557		0.601	

Timing numbers for DFCRN4:**Error Checks**

Pulse Width	CP (min)	0.153	Pulse Width	CDN (min)	0.134
tH	CP -> D (min)	-0.007	tS	D -> CP (min)	0.080
tHC	CP -> CDN (min)	0.105	tRelC	CDN -> CP (min)	-0.091

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.263	0.309	0.279	0.330	0.307	0.368	0.350	0.424	0.425	0.519
tCQN	0.340		0.356		0.383		0.425		0.500	

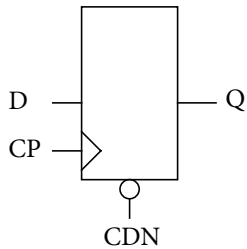
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.279	0.330	0.333	0.384	0.369	0.419	0.396	0.446	0.412	0.463
tCQN	0.356		0.429		0.497		0.555		0.602	

D Flip-Flops with Clear, Positive Clock and Q Output Only

DFCRQ1, DFCRQ2 and DFCRQ4

The DFCRQ1 (1x drive), DFCRQ2 (2x drive) and DFCRQ4 (4x drive) cells are positive edge triggered D flip-flops with active-low clear, CDN. Data present at the D input is transferred to the Q output on the positive edge of the clock, CP.



Function Table

INPUTS			OUTPUT
CDN	CP	D	Q
L	X	X	L
H	↑	L	L
H	↑	H	H
H	L	X	Q
H	H	X	Q

Cell Description

Macro Name:	DFCRQ1	DFCRQ2	DFCRQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	5.25	5.5	6.
Leakage Power (pW)	126.1	148.5	257.8

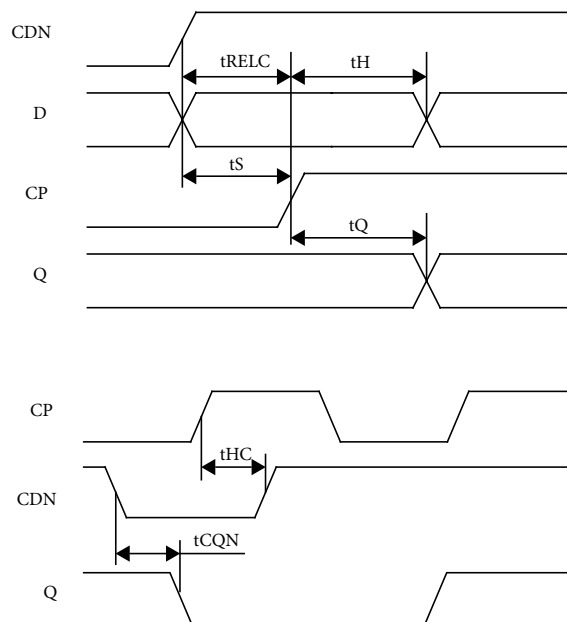
Pin Description

Name	Capacitance (pF)			Description
	DFCRQ1	DFCRQ2	DFCRQ4	
D	0.002	0.002	0.003	Data Input
CP	0.002	0.002	0.004	Clock Input
CDN	0.009	0.009	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	DFCRQ1	DFCRQ2	DFCRQ4
D	0.029	0.029	0.039
CP	0.04	0.04	0.052
CDN	0.002	0.002	0.003
Q	0.066	0.078	0.116

Waveforms



Timing numbers for DFCRQ1:**Error Checks**

Pulse Width	CP (min)	0.208	Pulse Width	CDN (min)	0.117
tH	CP -> D (min)	-0.018	tS	D -> CP (min)	0.085
tHC	CP -> CDN (min)	0.155	tRelC	CDN -> CP (min)	-0.124

Propagation Delays (ns) for sample loads:**Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.308	0.333	0.353	0.359	0.442	0.409	0.618	0.504	0.968	0.691
tCQ		0.145		0.172		0.223		0.319		0.507

Propagation Delays (ns) for sample input transitions:**Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.353	0.359	0.404	0.411	0.429	0.435	0.442	0.448	0.441	0.445
tCQ		0.172		0.258		0.343		0.419		0.479

Timing numbers for DFCRQ2:**Error Checks**

Pulse Width	CP (min)	0.208	Pulse Width	CDN (min)	0.14
tH	CP -> D (min)	-0.018	tS	D -> CP (min)	0.085
tHC	CP -> CDN (min)	0.155	tRelC	CDN -> CP (min)	-0.123

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.298	0.330	0.323	0.350	0.369	0.383	0.459	0.441	0.638	0.554
tCQ		0.151		0.173		0.206		0.267		0.380

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.323	0.350	0.373	0.402	0.398	0.426	0.412	0.439	0.410	0.436
tCQ		0.173		0.264		0.358		0.439		0.497

Timing numbers for DFCRQ4:**Error Checks**

Pulse Width	CP (min)	0.148	Pulse Width	CDN (min)	0.151
tH	CP -> D (min)	-0.008	tS	D -> CP (min)	0.083
tHC	CP -> CDN (min)	0.126	tRelC	CDN -> CP (min)	-0.094

Propagation Delays (ns) for sample loads:**Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.312	0.320	0.326	0.335	0.350	0.361	0.389	0.402	0.458	0.478
tCQ		0.160		0.175		0.202		0.243		0.319

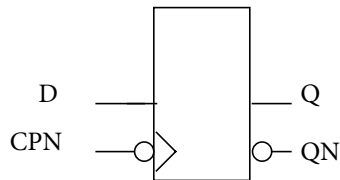
Propagation Delays (ns) for sample input transitions:**Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.326	0.335	0.380	0.389	0.417	0.426	0.445	0.453	0.462	0.468
tCQ		0.175		0.260		0.342		0.409		0.460

D Flip-Flops with Negative Clock

DFNFB1, DFNFB2 and DFNFB4

The DFNFB1 (1x drive), DFNFB2 (2x drive) and DFNFB4 (4x drive) cells are negative edge triggered D flip-flops. Data present at the D input is transferred to the Q and QN outputs on the negative edge of the clock, CPN.



Function Table

INPUTS		OUTPUTS	
CPN	D	Q	QN
↓	L	L	H
↓	H	H	L
L	X	Q	QN
H	X	Q	QN

Cell Description

Macro Name:	DFNFB1	DFNFB2	DFNFB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	5.	5.75	6.75
Leakage Power (pW):	133.6	174.6	315.5

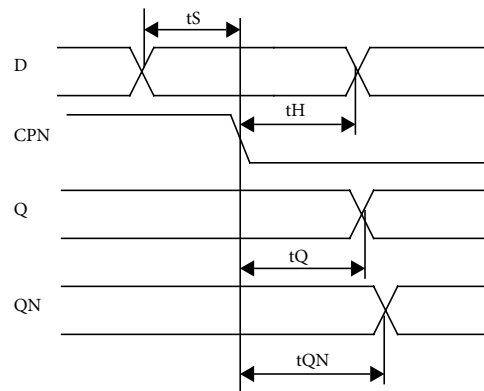
Pin Description

Name	Capacitance (pF)			Description
	DFNFB1	DFNFB2	DFNFB4	
D	0.002	0.002	0.004	Data Input
CPN	0.002	0.002	0.004	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	DFNFB1	DFNFB2	DFNFB4
D	0.012	0.012	0.019
CPN	0.039	0.039	0.055
Q	0.036	0.053	0.106
QN	0.036	0.053	0.106

Waveforms



Timing numbers for DFNFB1:

Error Checks

Pulse Width	CPN (min)	0.213			
t_H	CPN -> D (min)	0.083	t_S	D -> CPN (min)	0.109

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
t_Q	0.403	0.395	0.450	0.426	0.545	0.480	0.735	0.579	1.115	0.772
t_{QN}	0.290	0.315	0.340	0.353	0.436	0.416	0.626	0.522	1.004	0.722

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**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.450	0.426	0.540	0.516	0.631	0.608	0.709	0.687	0.773	0.753
tQN	0.340	0.353	0.430	0.443	0.522	0.534	0.600	0.611	0.666	0.676

Timing numbers for DFNFB2:

Error Checks

Pulse Width	CPN (min)	0.261			
tH	CPN -> D (min)	0.084	tS	D -> CPN (min)	0.107

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.457	0.443	0.480	0.464	0.522	0.497	0.610	0.555	0.792	0.658
tQN	0.290	0.316	0.321	0.345	0.372	0.387	0.468	0.456	0.657	0.568

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.480	0.464	0.570	0.555	0.661	0.648	0.740	0.728	0.805	0.795
tQN	0.321	0.345	0.411	0.435	0.504	0.527	0.585	0.606	0.652	0.671

Timing numbers for DFNFB4:**Error Checks**

Pulse Width	CPN (min)	0.242			
tH	CPN -> D (min)	0.024	tS	D -> CPN (min)	0.092

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.441	0.394	0.450	0.406	0.467	0.426	0.498	0.464	0.560	0.536
tQN	0.244	0.302	0.260	0.323	0.288	0.359	0.330	0.415	0.405	0.508

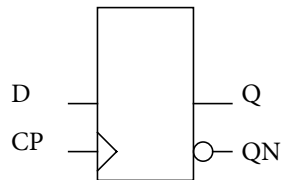
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transitiont	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.450	0.406	0.517	0.474	0.571	0.529	0.616	0.575	0.651	0.610
tQN	0.260	0.323	0.327	0.389	0.383	0.443	0.429	0.488	0.464	0.523

D Flip-Flops with Positive Clock

DFNRB1, DFNRB2 and DFNRB4

The DFNRB1 (1x drive), DFNRB2 (2x drive) and DFNRB4 (4x drive) cells are positive edge triggered D flip-flops. Data present at the D input is transferred to the Q and QN outputs on the positive edge of the clock, CP.



Function Table

INPUTS		OUTPUTS	
CP	D	Q	QN
↑	L	L	H
↑	H	H	L
L	X	Q	QN
X	H	Q	QN

Cell Description

Macro Name:	DFNRB1	DFNRB2	DFNRB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	5.	5.75	6.5
Leakage Power (pW):	133.9	174.8	301.9

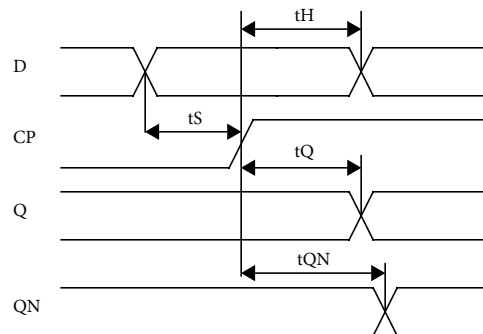
Pin Description

Name	Capacitance (pF)			Description
	DFNRB1	DFNRB2	DFNRB4	
D	0.002	0.002	0.003	Data Input
CP	0.002	0.002	0.004	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	DFNRB1	DFNRB2	DFNRB4
D	0.012	0.012	0.019
CP	0.041	0.041	0.053
Q	0.03	0.047	0.096
QN	0.03	0.047	0.096

Waveform



Timing numbers for DFNRB1:

Error Checks

Pulse Width	CP (min)	0.221			
tH	CP -> D (min)	-0.025	tS	D -> CP (min)	0.108

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.349	0.396	0.396	0.427	0.491	0.483	0.681	0.583	1.061	0.779
tQN	0.290	0.257	0.339	0.294	0.435	0.355	0.625	0.458	1.003	0.648

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**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.396	0.427	0.445	0.478	0.468	0.502	0.481	0.514	0.478	0.511
tQN	0.339	0.294	0.390	0.343	0.414	0.367	0.426	0.380	0.423	0.377

Timing numbers for DFNRB2:

Error Checks

Pulse Width	CP (min)	0.226			
tH	CP -> D (min)	-0.030	tS	D -> CP (min)	0.111

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.416	0.435	0.438	0.456	0.480	0.489	0.569	0.548	0.750	0.650
tQN	0.284	0.275	0.314	0.303	0.365	0.345	0.461	0.414	0.649	0.526

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.438	0.456	0.487	0.507	0.512	0.531	0.525	0.543	0.523	0.540
tQN	0.314	0.303	0.364	0.352	0.388	0.377	0.400	0.390	0.398	0.388

Timing numbers for DFNRB4:**Error Checks**

Pulse Width	CP (min)	0.145			
tH	CP -> D (min)	-0.012	tS	D -> CP (min)	0.070

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.438	0.408	0.447	0.420	0.463	0.440	0.494	0.476	0.557	0.548
tQN	0.260	0.299	0.276	0.320	0.304	0.357	0.347	0.413	0.422	0.506

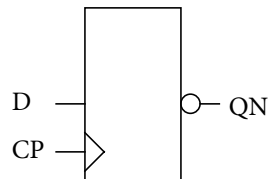
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transitiont	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.447	0.420	0.502	0.474	0.538	0.512	0.566	0.540	0.583	0.555
tQN	0.276	0.320	0.332	0.374	0.368	0.411	0.396	0.439	0.412	0.456

D Flip-Flops with QN Output Only

DFNRN1, DFNRN2 and DFNRN4

The DFNRN1 (1x drive), DFNRN2 (2x drive) and DFNRN4 (4x drive) cells are positive edge triggered D flip-flops. Data present at the D input is transferred to the QN output on the positive edge of the clock, CP.



Function Table

INPUTS		OUTPUT
CP	D	QN
↑	L	H
↑	H	L
L	X	QN
H	X	QN

Cell Description

Macro Name:	DFNRN1	DFNRN2	DFNRN4
Drive Capability:	1x	2x	4x
Gate Equivalents:	4.75	5.	5.5
Leakage Power (pW)	108.5	119.4	182.5

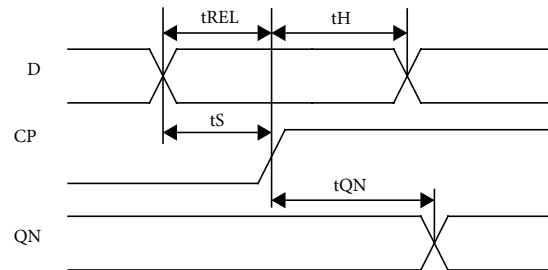
Pin Description

Name	Capacitance (pF)			Description
	DFNRN1	DFNRN2	DFNRN4	
D	0.002	0.002	0.004	Data Input
CP	0.002	0.002	0.004	Clock Input
Maximum capacitance				
QN	0.3	0.6	1.2	Inverted Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	DFNRN1	DFNRN2	DFNRN4
D	0.012	0.011	0.019
CP	0.041	0.04	0.056
QN	0.042	0.057	0.126

Waveforms



Timing numbers for DFNRN1:

Error Checks

Pulse Width	CP (min)	0.216			
tH	CP -> D (min)	-0.027	tS	D -> CP (min)	0.106

Propagation Delays (ns) for sample loads:

Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.284	0.259	0.333	0.297	0.428	0.360	0.618	0.470	0.996	0.681

Propagation Delays (ns) for sample input transitions:

Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.333	0.297	0.383	0.345	0.407	0.369	0.419	0.382	0.415	0.379

Timing numbers for DFNRN2:**Error Checks**

Pulse Width	CP (min)	0.219			
tH	CP -> D (min)	-0.031	tS	D -> CP (min)	0.109

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.278	0.271	0.308	0.300	0.360	0.341	0.459	0.408	0.653	0.520

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.308	0.300	0.358	0.349	0.382	0.373	0.394	0.386	0.391	0.384

Timing numbers for DFNRN4:**Error Checks**

Pulse Width	CP (min)	0.147			
tH	CP -> D (min)	-0.011	tS	D -> CP (min)	0.067

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.265	0.309	0.281	0.330	0.309	0.367	0.351	0.423	0.425	0.516

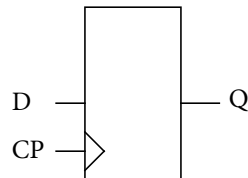
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.281	0.330	0.337	0.386	0.375	0.424	0.404	0.452	0.421	0.470

D Flip-Flops with Q Output Only

DFNRQ1, DFNRQ2 and DFNRQ4

The DFNRQ1 (1x drive), DFNRQ2 (2x drive) and DFNRQ4 (4x drive) cells are positive edge triggered D flip-flops. Data present at the D input is transferred to the Q output on the positive edge of the clock, CP.



Function Table

INPUTS		OUTPUT
CP	D	Q
↑	L	L
↑	H	H
L	X	Q
H	X	Q

Cell Description

Macro Name:	DFNRQ1	DFNRQ2	DFNRQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	4.75	5.	5.5
Leakage Power (pW)	123.8	144.4	277.4

Pin Description

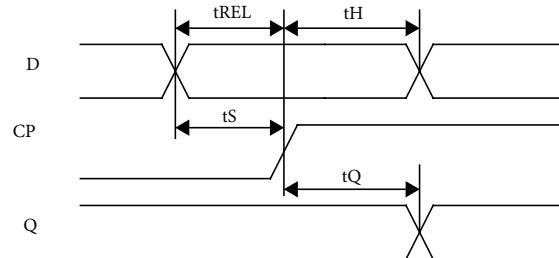
Name	Capacitance (pF)			Description
	DFNRQ1	DFNRQ2	DFNRQ4	
D	0.002	0.002	0.004	Data Input
CP	0.002	0.002	0.004	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

Pin Powers for

Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	DFNRQ1	DFNRQ2	DFNRQ4
D	0.011	0.011	0.019
CP	0.04	0.04	0.053
Q	0.043	0.057	0.107

Waveforms



Timing numbers for DFNRQ1:

Error Checks

Pulse Width	CP (min)	0.21			
tH	CP -> D (min)	-0.035	tS	D -> CP (min)	0.099

Propagation Delays (ns) for sample loads: Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.284	0.350	0.332	0.383	0.427	0.441	0.616	0.547	0.994	0.755

Propagation Delays (ns) for sample input transitions: Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.332	0.383	0.381	0.437	0.405	0.460	0.416	0.471	0.413	0.468

Timing numbers for DFNRQ2:**Error Checks**

Pulse Width	CP (min)	0.209			
tH	CP -> D (min)	-0.035	tS	D -> CP (min)	0.098

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.279	0.369	0.306	0.394	0.356	0.431	0.455	0.492	0.651	0.598

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.306	0.394	0.356	0.447	0.380	0.470	0.391	0.482	0.387	0.479

Timing numbers for DFNRQ4:**Error Checks**

Pulse Width	CP (min)	0.138			
tH	CP -> D (min)	-0.013	tS	D -> CP (min)	0.070

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.278	0.298	0.290	0.312	0.311	0.338	0.346	0.378	0.411	0.453

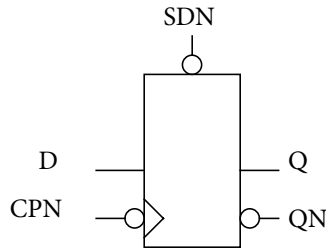
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transitiont	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.290	0.312	0.344	0.367	0.382	0.403	0.411	0.431	0.428	0.447

D Flip-Flops with Set and Negative Clock

DFPFB1, DFPFB2 and DFPFB4

The DFPFB1 (1x drive), DFPFB2 (2x drive) and DFPFB4 (4x drive) cells are negative edge triggered D flip-flops with active-low set, SDN. Data present at the D input is transferred to the Q and QN outputs on the negative edge of the clock, CPN.



Function Table

INPUTS			OUTPUTS	
SDN	CPN	D	Q	QN
L	X	X	H	L
H	↓	L	L	H
H	↓	H	H	L
H	L	X	Q	QN
H	H	X	Q	QN

Cell Description

Macro Name:	DFPFB1	DFPFB2	DFPFB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	6.	6.5	7.5
Leakage Power (pW):	145.9	179.4	322.9

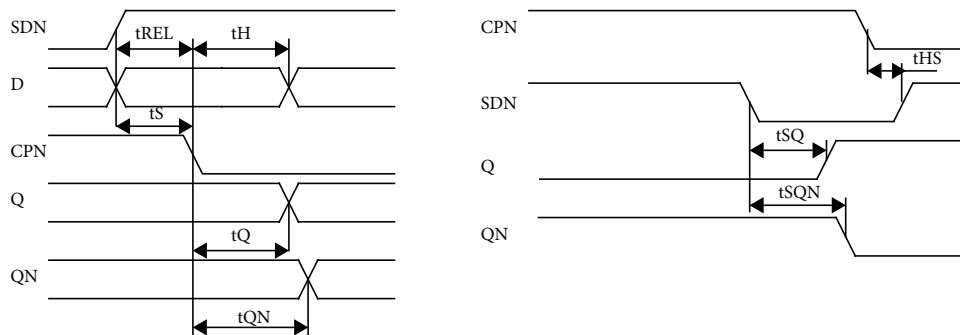
Pin Description

Name	Capacitance (pF)			Description
	DFPFB1	DFPFB2	DFPFB4	
D	0.002	0.002	0.003	Data Input
CPN	0.002	0.002	0.004	Clock Input
SDN	0.006	0.006	0.008	Preset Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	DFPFB1	DFPFB2	DFPFB4
D	0.018	0.018	0.021
CPN	0.043	0.043	0.052
SDN	-0	-0	-0
Q	0.043	0.059	0.116
QN	0.043	0.059	0.116

Waveforms



Timing numbers for DFPFB1:

Error Checks

Pulse Width	CPN (min)	0.195	Pulse Width	SDN (min)	0.204
tH	CPN -> D (min)	0.081	tS	D -> CPN (min)	0.120
tHS	CPN -> SDN (min)	0.116	tRelS	SDN -> CPN (min)	-0.079

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.423	0.335	0.475	0.362	0.577	0.414	0.782	0.512	1.194	0.708
tQN	0.296	0.331	0.349	0.372	0.450	0.438	0.647	0.545	1.037	0.738
tSQ	0.370		0.420		0.522		0.727		1.139	
t/SQN		0.274		0.318		0.387		0.492		0.683

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**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.475	0.362	0.565	0.452	0.659	0.543	0.740	0.621	0.806	0.686
tQN	0.349	0.372	0.439	0.463	0.530	0.557	0.608	0.637	0.673	0.703
tSQ	0.420		0.513		0.619		0.712		0.790	
t/SQN		0.318		0.412		0.517		0.611		0.687

Timing numbers for DFPFB2:

Error Checks

Pulse Width	CPN (min)	0.237	Pulse Width	SDN (min)	0.232
tH	CPN -> D (min)	0.082	tS	D -> CPN (min)	0.120
tHS	CPN -> SDN (min)	0.115	tRelS	SDN -> CPN (min)	-0.082

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.458	0.378	0.479	0.396	0.521	0.425	0.610	0.478	0.791	0.578
tQN	0.290	0.328	0.321	0.359	0.374	0.405	0.473	0.480	0.662	0.597
tSQ	0.417		0.438		0.480		0.567		0.749	
tSQN		0.280		0.315		0.362		0.437		0.552

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.479	0.396	0.570	0.485	0.665	0.578	0.747	0.658	0.814	0.723
tQN	0.321	0.359	0.411	0.450	0.503	0.544	0.583	0.626	0.649	0.693
tSQ	0.438		0.531		0.642		0.742		0.823	
tSQN		0.315		0.408		0.517		0.615		0.694

Timing numbers for DFPFB4:**Error Checks**

Pulse Width	CPN (min)	0.299	Pulse Width	SDN (min)	0.192
tH	CPN -> D (min)	0.020	tS	D -> CPN (min)	0.094
tHS	CPN -> SDN (min)	0.041	tRelS	SDN -> CPN (min)	-0.014

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.483	0.454	0.492	0.464	0.506	0.483	0.537	0.519	0.599	0.588
tQN	0.290	0.343	0.308	0.365	0.339	0.403	0.388	0.463	0.470	0.563
tSQ	0.450		0.458		0.472		0.502		0.564	
tSQN		0.318		0.338		0.372		0.427		0.520

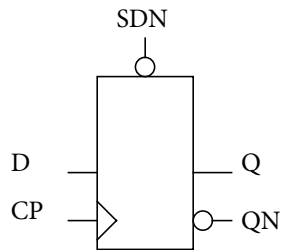
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.492	0.464	0.559	0.534	0.614	0.591	0.661	0.638	0.696	0.675
tQN	0.308	0.365	0.377	0.432	0.434	0.487	0.481	0.534	0.518	0.570
tSQ	0.458		0.545		0.641		0.729		0.800	
tSQN		0.338		0.426		0.522		0.606		0.672

D Flip-Flops with Set and Positive Clock

DFPRB1, DFPRB2 and DFPRB4

The DFPRB1 (1x drive), DFPRB2 (2x drive) and DFPRB4 (4x drive) cells are positive edge triggered D flip-flops with active-low set, SDN. Data present at the D input is transferred to the Q and QN outputs on the positive edge of the clock, CP.



Function Table

INPUTS			OUTPUTS	
SDN	CP	D	Q	QN
L	X	X	H	L
H	↑	L	L	H
H	↑	H	H	L
H	L	X	Q	QN
H	H	X	Q	QN

Cell Description

Macro Name:	DFPRB1	DFPRB2	DFPRB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	6.	6.5	7.5
Leakage Power (pW):	150.6	184.1	319.5

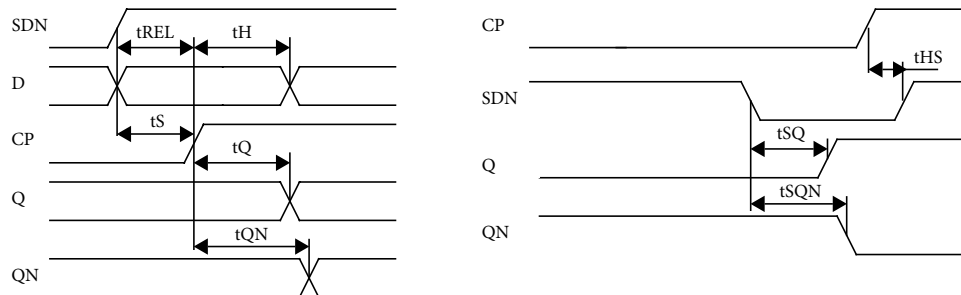
Pin Description

Name	Capacitance (pF)			Description
	DFPRB1	DFPRB2	DFPRB4	
D	0.002	0.002	0.003	Data Input
CP	0.002	0.002	0.004	Clock Input
SDN	0.006	0.006	0.008	Preset Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	DFPRB1	DFPRB2	DFPRB4
D	0.018	0.018	0.021
CP	0.044	0.043	0.052
SDN	-0	-0	-0
Q	0.04	0.056	0.112
QN	0.04	0.056	0.112

Waveforms



Timing numbers for DFPRB1:

Error Checks

Pulse Width	CP (min)	0.22	Pulse Width	SDN (min)	0.16
tH	CP -> D (min)	-0.045	tS	D -> CP (min)	0.122
tHS	CP -> SDN (min)	0.118	tRelS	SDN -> CP (min)	-0.091

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.392	0.347	0.444	0.375	0.546	0.426	0.751	0.524	1.163	0.721
tQN	0.308	0.300	0.361	0.341	0.461	0.407	0.658	0.514	1.048	0.707
tSQ	0.370		0.421		0.523		0.728		1.140	
t/SQN		0.275		0.319		0.387		0.493		0.684

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**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.444	0.375	0.496	0.427	0.524	0.451	0.539	0.464	0.540	0.462
tQN	0.361	0.341	0.413	0.393	0.437	0.421	0.450	0.437	0.448	0.437
tSQ	0.421		0.514		0.620		0.713		0.791	
t/SQN		0.319		0.412		0.518		0.611		0.688

Timing numbers for DFPRB2:

Error Checks

Pulse Width	CP (min)	0.222	Pulse Width	SDN (min)	0.215
tH	CP -> D (min)	-0.044	tS	D -> CP (min)	0.123
tHS	CP -> SDN (min)	0.116	tRelS	SDN -> CP (min)	-0.089

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.437	0.385	0.458	0.403	0.500	0.432	0.588	0.485	0.770	0.585
tQN	0.297	0.306	0.328	0.337	0.381	0.384	0.479	0.458	0.668	0.575
tSQ	0.418		0.439		0.480		0.568		0.749	
tSQN		0.281		0.316		0.363		0.438		0.553

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.458	0.403	0.510	0.454	0.538	0.479	0.554	0.492	0.554	0.490
tQN	0.328	0.337	0.380	0.390	0.405	0.418	0.418	0.433	0.416	0.434
tSQ	0.439		0.532		0.643		0.743		0.824	
tSQN		0.316		0.409		0.518		0.615		0.695

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Timing numbers for DFPRB4:**Error Checks**

Pulse Width	CP (min)	0.154	Pulse Width	SDN (min)	0.325
tH	CP -> D (min)	-0.008	tS	D -> CP (min)	0.082
tHS	CP -> SDN (min)	0.035	tRelS	SDN -> CP (min)	-0.022

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.479	0.467	0.487	0.477	0.501	0.495	0.532	0.530	0.593	0.599
tQN	0.302	0.338	0.320	0.361	0.351	0.399	0.400	0.459	0.483	0.558
tSQ	0.460		0.467		0.481		0.511		0.572	
tSQN		0.323		0.344		0.378		0.433		0.525

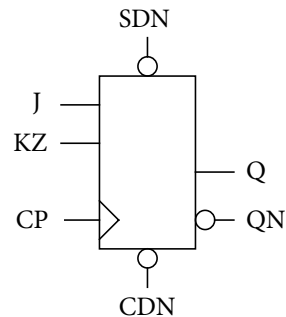
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.487	0.477	0.540	0.532	0.577	0.569	0.605	0.597	0.621	0.613
tQN	0.320	0.361	0.375	0.414	0.412	0.450	0.440	0.478	0.457	0.495
tSQ	0.467		0.555		0.652		0.741		0.813	
tSQN		0.344		0.431		0.527		0.612		0.678

J/K-Bar Flip-Flops with Set and Clear

JKBRB1, JKBRB2 and JKBRB4

The JKBRB1 (1x drive), JKBRB2 (2x drive) and JKBRB4 (4x drive) cells are positive edge triggered J/K-bar flip-flops with active-low clear, CDN, and set, SDN. Data present at the J/K-bar inputs is transferred to the Q and QN outputs on the positive edge of the clock (CP).



Function Table

INPUTS					OUTPUTS	
CDN	SDN	CP	J	KZ	Q	QN
L	H	X	X	X	L	H
H	L	X	X	X	H	L
H	H	↑	L	H	Q _o	QN _o
H	H	↑	L	L	L	H
H	H	↑	H	H	H	L
H	H	↑	H	L	QN _o	Q _o
H	H	L	X	X	Q	QN
H	H	H	X	X	Q	QN
L	L	X	X	X	L	L

Cell Description

Macro Name:	JKBRB1	JKBRB2	JKBRB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	8.75	9.	9.5
Leakage Power (pW):	166.5	218.3	408.2

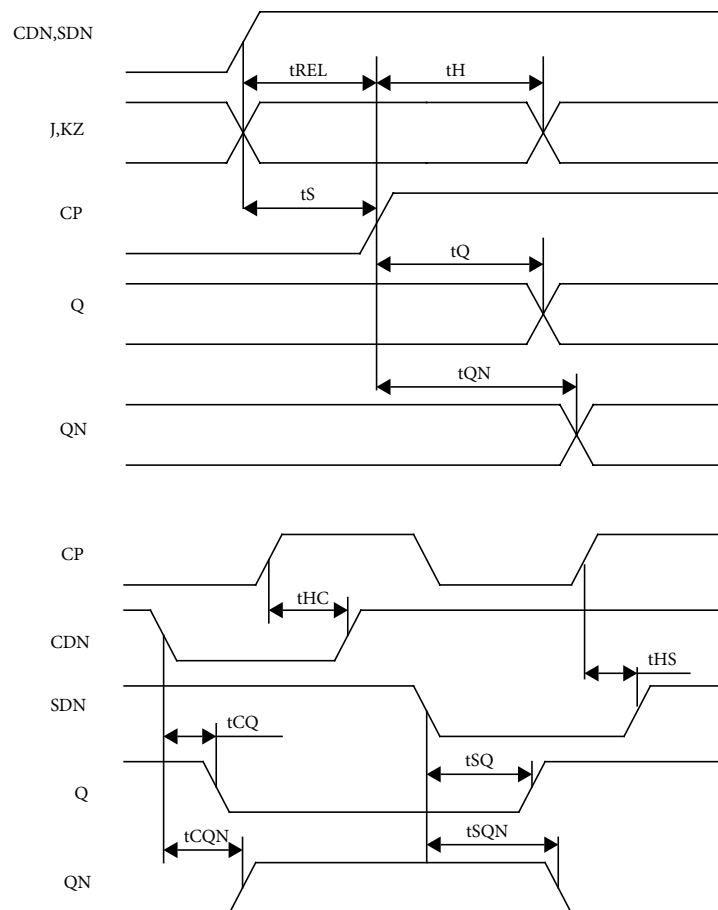
Pin Description

Name	Capacitance (pF)			Description
	JKBRB1	JKBRB2	JKBRB4	
KZ	0.003	0.003	0.004	Data Input
J	0.003	0.003	0.003	Data Select
CP	0.002	0.002	0.003	Clock Input
SDN	0.006	0.007	0.007	Preset Input
CDN	0.008	0.008	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	JKBRB1	JKBRB2	JKBRB4
KZ	0.02	0.02	0.024
J	0.009	0.009	0.011
CP	0.045	0.045	0.047
SDN	0.009	0.008	0.009
CDN	0.001	0.001	0.002
Q	0.047	0.066	0.113
QN	0.064	0.087	0.143

Waveforms



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Timing numbers for JKBRB1:**Error Checks**

Pulse Width	CP (min)	0.24			
Pulse Width	CDN (min)	0.165	Pulse Width	SDN (min)	0.335
tH	CP -> J/KZ (min)	-0.101	tS	J/KZ -> CP (min)	0.220
tHS	CP -> SDN (min)	0.112	tRelS	SDN -> CP (min)	-0.078
tHC	CP -> CDN (min)	0.167	tRelC	CDN -> CP (min)	-0.127

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.386	0.360	0.437	0.390	0.537	0.447	0.738	0.557	1.139	0.776
tQN	0.314	0.303	0.368	0.349	0.470	0.425	0.671	0.547	1.072	0.771
tSQ/tCQ	0.500	0.140	0.549	0.171	0.649	0.228	0.849	0.340	1.249	0.560
tCQN/tSQN	0.394	0.409	0.448	0.469	0.551	0.557	0.752	0.686	1.153	0.912

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.437	0.390	0.486	0.439	0.510	0.461	0.522	0.472	0.519	0.467
tQN	0.368	0.349	0.417	0.398	0.440	0.422	0.450	0.434	0.445	0.431
tSQ/tCQ	0.549	0.171	0.642	0.250	0.761	0.323	0.876	0.387	0.967	0.441
tCQN/tSQN	0.448	0.469	0.540	0.562	0.638	0.681	0.723	0.795	0.794	0.886

Timing numbers for JKBRB2:**Error Checks**

Pulse Width	CP (min)	0.244			
Pulse Width	CDN (min)	0.197	Pulse Width	SDN (min)	0.445
tH	CP -> J/KZ (min)	-0.104	tS	J/KZ -> CP (min)	0.216
tHS	CP -> SDN (min)	0.108	tRelS	SDN -> CP (min)	-0.076
tHC	CP -> CDN (min)	0.159	tRelC	CDN -> CP (min)	-0.131

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.459	0.374	0.482	0.391	0.527	0.416	0.622	0.462	0.815	0.550
tQN	0.310	0.309	0.343	0.340	0.399	0.386	0.501	0.461	0.697	0.574
tSQ/tCQ	0.608	0.127	0.631	0.145	0.673	0.171	0.764	0.219	0.955	0.308
tCQN/tSQN	0.400	0.409	0.435	0.450	0.493	0.510	0.598	0.603	0.796	0.730

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.482	0.391	0.532	0.440	0.556	0.463	0.568	0.474	0.565	0.469
tQN	0.343	0.340	0.393	0.390	0.416	0.415	0.427	0.427	0.422	0.424
tSQ/tCQ	0.631	0.145	0.723	0.226	0.844	0.302	0.965	0.366	1.066	0.419
tCQN/tSQN	0.435	0.450	0.516	0.542	0.593	0.661	0.661	0.779	0.716	0.873

Timing numbers for JKBRB4:**Error Checks**

Pulse Width	CP (min)	0.183			
Pulse Width	CDN (min)	0.302	Pulse Width	SDN (min)	0.425
tH	CP -> J/KZ (min)	-0.094	tS	J/KZ -> CP (min)	0.146
tHS	CP -> SDN (min)	0.076	tRelS	SDN -> CP (min)	-0.057
tHC	CP -> CDN (min)	0.135	tRelC	CDN -> CP (min)	-0.117

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.560	0.623	0.571	0.637	0.590	0.662	0.623	0.702	0.689	0.775
tQN	0.330	0.367	0.348	0.389	0.379	0.427	0.428	0.487	0.509	0.587
tSQ/tCQ	0.598	0.227	0.609	0.244	0.626	0.273	0.659	0.319	0.724	0.398
tCQN/tSQN	0.517	0.395	0.532	0.419	0.558	0.459	0.602	0.522	0.681	0.627

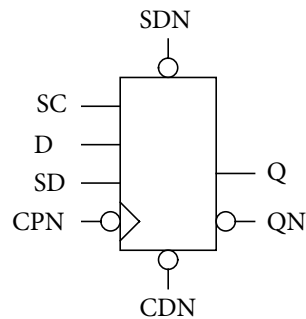
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.571	0.637	0.629	0.697	0.668	0.737	0.694	0.764	0.706	0.777
tQN	0.348	0.389	0.408	0.447	0.448	0.485	0.476	0.512	0.488	0.524
tSQ/tCQ	0.609	0.244	0.696	0.334	0.801	0.439	0.899	0.528	0.976	0.597
tCQN/tSQN	0.532	0.419	0.621	0.507	0.724	0.612	0.816	0.711	0.888	0.787

SCAN FLIP - FLOPS

Muxed Scan D Flip-Flops with Clear, Preset SDBFB1 , SDBFB2 and SDBFB4

The SDBFB1 (1x drive), SDBFB2 (2x drive) and SDBFB4 (4x drive) cells are negative edge triggered multiplexed scan D flip-flops with active-low clear, CDN, and set, SDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q and QN outputs on the falling edge of the clock, CPN.



Function Table

INPUTS						OUTPUTS	
CDN	SDN	CPN	SC	D	SD	Q	QN
L	H	X	X	X	X	L	H
H	L	X	X	X	X	H	L
H	H	↓	L	L	X	L	H
H	H	↓	L	H	X	H	L
H	H	↓	H	X	L	L	H
H	H	↓	H	X	H	H	L
H	H	L	X	X	X	Q	QN
H	H	H	X	X	X	Q	QN
L	L	X	X	X	X	L	L

Cell Description

Macro Name:	SDBFB1	SDBFB2	SDBFB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	8.5	8.75	10.
Leakage Power (pW):	242.3	312.5	448.3

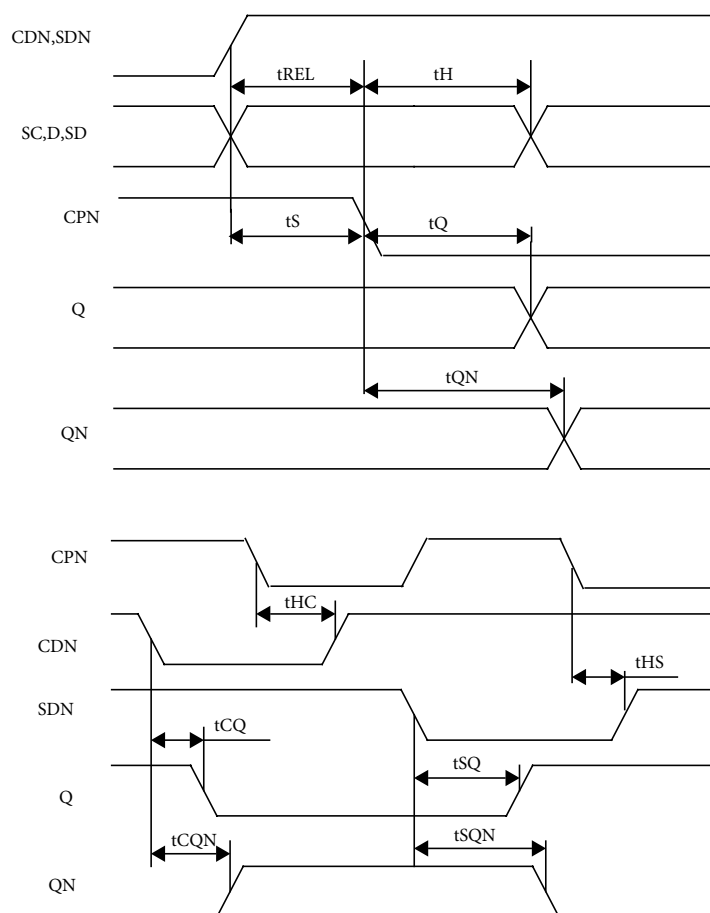
Pin Description

Name	Capacitance (pF)			Description
	SDBFB1	SDBFB2	SDBFB4	
D	0.004	0.004	0.004	Data Input
SD	0.004	0.004	0.004	Scan Data Input
SC	0.004	0.004	0.004	Scan Control
CPN	0.004	0.004	0.004	Clock Input
SDN	0.009	0.009	0.009	Preset Input
CDN	0.009	0.009	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SDBFB1	SDBFB2	SDBFB4
D	0.017	0.017	0.017
SD	0.014	0.014	0.014
SC	0.056	0.056	0.056
CPN	0.057	0.057	0.059
SDN	0.008	0.008	0.008
CDN	-0	-0	-0
Q	0.037	0.053	0.096
QN	0.037	0.056	0.107

Waveforms



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Timing Numbers for SDBFB1:**Error Checks**

Pulse Width	CPN (min)	0.178			
Pulse Width	CDN (min)	0.154	Pulse Width	SDN (min)	0.16
tH	CPN -> D/SD/SC (min)	-0.055	tS	D/SD/SC -> CPN (min)	0.204
tHS	CPN -> SDN (min)	0.041	tRelS	SDN -> CPN (min)	0.004
tHC	CPN -> CDN (min)	0.155	tRelC	CDN -> CPN (min)	-0.124

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.390	0.355	0.429	0.395	0.504	0.471	0.651	0.621	0.945	0.921
tQN	0.260	0.315	0.304	0.368	0.386	0.458	0.536	0.617	0.832	0.920
tSQ/tCQ	0.345	0.169	0.383	0.210	0.458	0.288	0.605	0.440	0.899	0.741
tCQN/tSQN	0.370	0.276	0.414	0.329	0.493	0.418	0.644	0.574	0.940	0.876

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.429	0.395	0.496	0.462	0.550	0.516	0.595	0.562	0.629	0.595
tQN	0.304	0.368	0.372	0.435	0.426	0.489	0.471	0.535	0.505	0.569
tSQ/tCQ	0.383	0.210	0.465	0.290	0.545	0.364	0.614	0.427	0.669	0.479
tCQN/tSQN	0.414	0.329	0.490	0.411	0.562	0.494	0.624	0.564	0.674	0.619

Timing Numbers for SDBFB2:**Error Checks**

Pulse Width	CPN (min)	0.236			
Pulse Width	CDN (min)	0.168	Pulse Width	SDN (min)	0.18
tH	CPN -> D/SD/SC (min)	-0.053	tS	D/SD/SC -> CPN (min)	0.203
tHS	CPN -> SDN (min)	0.038	tRelS	SDN -> CPN (min)	0.003
tHC	CPN -> CDN (min)	0.146	tRelC	CDN -> CPN (min)	-0.120

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.442	0.400	0.460	0.421	0.495	0.459	0.561	0.529	0.691	0.667
tQN	0.257	0.315	0.286	0.350	0.330	0.405	0.407	0.497	0.544	0.651
tSQ/tCQ	0.394	0.162	0.412	0.188	0.446	0.228	0.512	0.302	0.642	0.442
tCQN/tSQN	0.368	0.276	0.396	0.310	0.439	0.362	0.515	0.451	0.652	0.601

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.460	0.421	0.527	0.489	0.581	0.543	0.626	0.588	0.661	0.623
tQN	0.286	0.350	0.353	0.417	0.407	0.471	0.453	0.516	0.487	0.551
tSQ/tCQ	0.412	0.188	0.495	0.271	0.580	0.352	0.653	0.420	0.711	0.474
tCQN/tSQN	0.396	0.310	0.472	0.393	0.545	0.479	0.607	0.552	0.657	0.610

Timing Numbers for SDBFB4:**Error Checks**

Pulse Width	CPN (min)	0.346			
Pulse Width	CDN (min)	0.199	Pulse Width	SDN (min)	0.209
tH	CPN -> D/SD/SC (min)	-0.053	tS	D/SD/SC -> CPN (min)	0.198
tHS	CPN -> SDN (min)	0.038	tRelS	SDN -> CPN (min)	-0.006
tHC	CPN -> CDN (min)	0.136	tRelC	CDN -> CPN (min)	-0.120

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.584	0.530	0.595	0.543	0.612	0.565	0.645	0.603	0.711	0.675
tQN	0.300	0.371	0.319	0.395	0.350	0.436	0.399	0.499	0.484	0.604
tSQ/tCQ	0.521	0.191	0.532	0.208	0.550	0.237	0.583	0.281	0.649	0.361
tCQN/tSQN	0.431	0.327	0.446	0.349	0.472	0.387	0.517	0.445	0.597	0.545

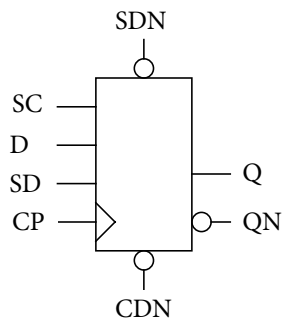
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.595	0.543	0.662	0.611	0.717	0.667	0.763	0.713	0.799	0.749
tQN	0.319	0.395	0.387	0.463	0.442	0.518	0.489	0.564	0.524	0.599
tSQ/tCQ	0.532	0.208	0.615	0.295	0.705	0.389	0.784	0.466	0.846	0.526
tCQN/tSQN	0.414	0.349	0.532	0.433	0.626	0.524	0.706	0.604	0.768	0.666

Multiplexed Scan D Flip-Flops with Set and Clear

SDBRB1, SDBRB2 and SDBRB4

The SDBRB1 (1x drive), SDBRB2 (2x drive) and SDBRB4 (4x drive) cells are positive edge triggered multiplexed scan D flip-flops with active-low clear, CDN, and set, SDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q and QN outputs on the rising edge of the clock, CP.



Function Table

INPUTS						OUTPUTS	
CDN	SDN	CP	SC	D	SD	Q	QN
L	H	X	X	X	X	L	H
H	L	X	X	X	X	H	L
H	H	↑	L	L	X	L	H
H	H	↑	L	H	X	H	L
H	H	↑	H	X	L	L	H
H	H	↑	H	X	H	H	L
H	H	L	X	X	X	Q	QN
H	H	H	X	X	X	Q	QN
L	L	X	X	X	X	L	L

Cell Description

Macro Name:	SDBRB1	SDBRB2	SDBRB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	9.25	9.5	10.
Leakage Power (pW):	199.8	256.2	448.3

Pin Description

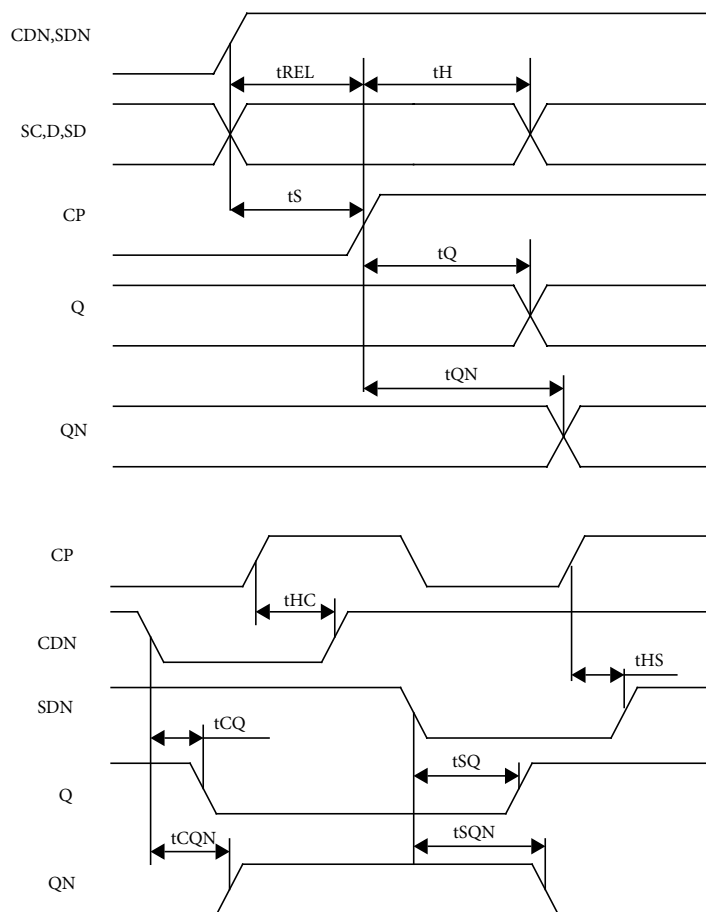
Name	Capacitance (pF)			Description
	SDBRB1	SDBRB2	SDBRB4	
D	0.002	0.002	0.004	Data Input
SD	0.002	0.002	0.004	Scan Data Input
SC	0.003	0.003	0.004	Scan Control
CP	0.003	0.003	0.004	Clock Input
SDN	0.006	0.006	0.009	Preset Input
CDN	0.008	0.008	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

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Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SDBRB1	SDBRB2	SDBRB4
D	0.016	0.016	0.02
SD	0.028	0.028	0.015
SC	0.03	0.03	0.058
CP	0.039	0.039	0.06
SDN	0.008	0.008	0.008
CDN	0.003	0.003	0.002
Q	0.029	0.044	0.094
QN	0.032	0.049	0.106

Waveforms



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Timing Numbers for SDBRB1:**Error Checks**

Pulse Width	CP (min)	0.236			
Pulse Width	CDN (min)	0.142	Pulse Width	SDN (min)	0.162
tH	CP -> D/SD/SC (min)	-0.103	tS	D/SD/SC -> CP (min)	0.236
tHS	CP -> SDN (min)	0.068	tRelS	SDN -> CP (min)	-0.047
tHC	CP -> CDN (min)	0.174	tRelC	CDN -> CP (min)	-0.148

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.375	0.371	0.423	0.399	0.518	0.451	0.706	0.549	1.083	0.741
tQN	0.306	0.296	0.359	0.340	0.458	0.412	0.649	0.526	1.027	0.728
tSQ/tCQ	0.326	0.139	0.373	0.167	0.468	0.219	0.656	0.318	1.033	0.512
tCQN/tSQN	0.398	0.248	0.450	0.290	0.549	0.356	0.738	0.463	1.116	0.659

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.423	0.399	0.474	0.452	0.503	0.480	0.522	0.497	0.527	0.501
tQN	0.359	0.340	0.412	0.392	0.440	0.421	0.457	0.439	0.461	0.445
tSQ/tCQ	0.373	0.167	0.459	0.249	0.545	0.326	0.619	0.394	0.677	0.452
tCQN/tSQN	0.450	0.290	0.540	0.375	0.633	0.463	0.715	0.538	0.786	0.598

Timing Numbers for SDBRB2:**Error Checks**

Pulse Width	CP (min)	0.24			
Pulse Width	CDN (min)	0.179	Pulse Width	SDN (min)	0.212
tH	CP -> D/SD/SC (min)	-0.102	tS	D/SD/SC -> CP (min)	0.240
tHS	CP -> SDN (min)	0.064	tRelS	SDN -> CP (min)	-0.044
tHC	CP -> CDN (min)	0.166	tRelC	CDN -> CP (min)	-0.143

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.410	0.412	0.433	0.430	0.477	0.461	0.565	0.515	0.744	0.616
tQN	0.297	0.304	0.329	0.338	0.383	0.389	0.480	0.470	0.661	0.596
tSQ/tCQ	0.355	0.141	0.378	0.162	0.422	0.193	0.510	0.249	0.688	0.352
tCQN/tSQN	0.393	0.254	0.425	0.286	0.477	0.331	0.572	0.405	0.753	0.522

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.433	0.430	0.484	0.482	0.513	0.511	0.532	0.528	0.537	0.532
tQN	0.329	0.338	0.382	0.389	0.410	0.419	0.427	0.437	0.432	0.443
tSQ/tCQ	0.378	0.162	0.465	0.247	0.556	0.330	0.636	0.401	0.698	0.461
tCQN/tSQN	0.425	0.286	0.515	0.373	0.608	0.465	0.690	0.544	0.761	0.606

Timing Numbers for SDBRB4:**Error Checks**

Pulse Width	CP (min)	0.171			
Pulse Width	CDN (min)	0.241	Pulse Width	SDN (min)	0.36
tH	CP -> D/SD/SC (min)	-0.083	tS	D/SD/SC -> CP (min)	0.173
tHS	CP -> SDN (min)	0.046	tRelS	SDN -> CP (min)	-0.027
tHC	CP -> CDN (min)	0.125	tRelC	CDN -> CP (min)	-0.113

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

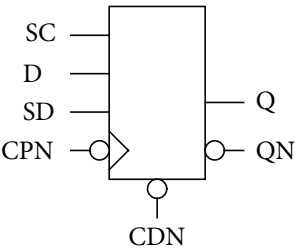
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.593	0.550	0.604	0.562	0.622	0.584	0.655	0.620	0.719	0.690
tQN	0.322	0.377	0.341	0.400	0.372	0.440	0.421	0.502	0.504	0.605
tSQ/tCQ	0.524	0.190	0.535	0.206	0.553	0.234	0.586	0.278	0.651	0.355
tCQN/tSQN	0.433	0.327	0.448	0.348	0.475	0.385	0.519	0.442	0.599	0.539

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.604	0.562	0.663	0.622	0.705	0.664	0.737	0.696	0.758	0.716
tQN	0.341	0.400	0.400	0.459	0.442	0.501	0.475	0.533	0.494	0.554
tSQ/tCQ	0.535	0.206	0.618	0.294	0.708	0.386	0.788	0.464	0.850	0.523
tCQN/tSQN	0.448	0.348	0.535	0.432	0.629	0.522	0.709	0.602	0.771	0.664

Buffered Muxed Scan D Flip-Flops with Clear SDCFB1 , SDCFB2 and SDCFB4

The SDCFB1 (1x drive), SDCFB2 (2x drive) and SDCFB4 (4x drive) cells are negative edge triggered multiplexed scan D flip-flops with active-low clear, CDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q and QN outputs on the falling edge of the clock, CPN.



Function Table

INPUTS					OUTPUTS	
CDN	CPN	SC	D	SD	Q	QN
L	X	X	X	X	L	H
H	↓	L	L	X	L	H
H	↓	L	H	X	H	L
H	↓	H	X	L	L	H
H	↓	H	X	H	H	L
H	L	X	X	X	Q	QN
H	H	X	X	X	Q	QN

Cell Description

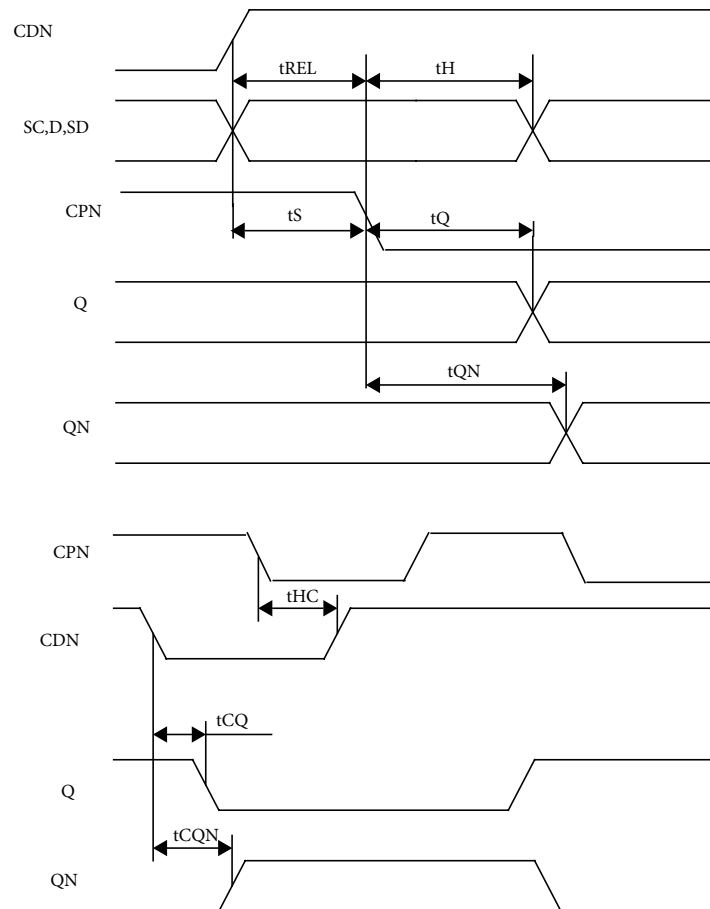
Macro Name:	SDCFB1	SDCFB2	SDCFB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.5	7.75	9.
Leakage Power (pW):	237.3	275.2	362.9

Pin Description

Name	Capacitance (pF)			Description
	SDCFB1	SDCFB2	SDCFB4	
D	0.004	0.004	0.004	Data Input
SD	0.004	0.004	0.004	Scan Data Input
SC	0.004	0.004	0.004	Scan Control
CPN	0.004	0.004	0.004	Clock Input
CDN	0.008	0.008	0.008	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	SDCFB1	SDCFB2	SDCFB4
D	0.022	0.022	0.022
SD	0.004	0.004	0.004
SC	0.044	0.044	0.044
CPN	0.054	0.055	0.055
CDN	-0	-0	-0
Q	0.055	0.071	0.115
QN	0.055	0.071	0.115

Waveforms

Timing Numbers for SDCFB1:**Error Checks**

Pulse Width	CPN (min)	0.165	Pulse Width	CDN (min)	0.135
tH	CPN -> D/SD/SC (min)	-0.045	tS	D/SD/SC -> CPN (min)	0.176
tHC	CPN -> CDN (min)	0.161	tRelC	CDN -> CPN (min)	-0.130

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.361	0.340	0.397	0.380	0.464	0.456	0.596	0.606	0.859	0.906
tQN	0.235	0.290	0.273	0.337	0.343	0.422	0.477	0.576	0.740	0.878
tCQ		0.172		0.213		0.292		0.443		0.744
tCQN	0.318		0.355		0.424		0.556		0.819	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.397	0.380	0.464	0.448	0.517	0.504	0.563	0.550	0.596	0.584
tQN	0.273	0.337	0.342	0.404	0.397	0.458	0.443	0.503	0.478	0.536
tCQ		0.213		0.295		0.370		0.435		0.487
tCQN	0.355		0.431		0.502		0.563		0.612	

Timing Numbers for SDCFB2:**Error Checks**

Pulse Width	CPN (min)	0.209	Pulse Width	CDN (min)	0.147
tH	CPN -> D/SD/SC (min)	-0.045	tS	D/SD/SC -> CPN (min)	0.174
tHC	CPN -> CDN (min)	0.157	tRelC	CDN -> CPN (min)	-0.129

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.396	0.375	0.417	0.397	0.453	0.435	0.520	0.509	0.651	0.653
tQN	0.238	0.284	0.264	0.316	0.304	0.367	0.375	0.455	0.509	0.612
tCQ		0.167		0.193		0.235		0.311		0.456
tCQN	0.317		0.342		0.382		0.452		0.585	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.417	0.397	0.484	0.466	0.539	0.523	0.584	0.569	0.619	0.605
tQN	0.264	0.316	0.332	0.383	0.388	0.438	0.436	0.483	0.471	0.517
tCQ		0.193		0.277		0.360		0.430		0.484
tCQN	0.342		0.419		0.489		0.549		0.598	

Timing Numbers for SDCFB4:**Error Checks**

Pulse Width	CPN (min)	0.302	Pulse Width	CDN (min)	0.176
tH	CPN -> D/SD/SC (min)	-0.045	tS	D/SD/SC -> CPN (min)	0.168
tHC	CPN -> CDN (min)	0.149	tRelC	CDN -> CPN (min)	-0.128

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

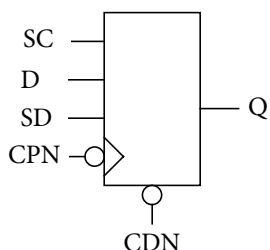
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.509	0.483	0.521	0.497	0.541	0.520	0.577	0.559	0.643	0.633
tQN	0.272	0.321	0.288	0.342	0.317	0.377	0.361	0.433	0.439	0.526
tCQ		0.195		0.212		0.241		0.285		0.364
tCQN	0.376		0.390		0.415		0.456		0.530	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.521	0.497	0.589	0.566	0.644	0.623	0.689	0.671	0.724	0.707
tQN	0.288	0.342	0.358	0.410	0.415	0.465	0.462	0.511	0.498	0.546
tCQ		0.212		0.300		0.394		0.473		0.531
tCQN	0.390		0.477		0.571		0.652		0.714	

Multiplexed Scan D Flip-Flops with Clear and Q Output Only SDCFQ1 , SDCFQ2 and SDCFQ4

The SDCFQ1 (1x drive), SDCFQ2 (2x drive) and SDCFQ4 (4x drive) cells are negative edge triggered multiplexed scan D flip-flops with active-low clear, CDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q output on the falling edge of the clock, CPN.



Function Table

INPUTS					OUTPUT
CDN	CPN	SC	D	SD	Q
L	X	X	X	X	L
H	↓	L	L	X	L
H	↓	L	H	X	H
H	↓	H	X	L	L
H	↓	H	X	H	H
H	L	X	X	X	Q
H	H	X	X	X	Q

Cell Description

Macro Name:	SDCFQ1	SDCFQ2	SDCFQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.75	8.	8.
Leakage Power (pW):	177.8	202.6	313.6

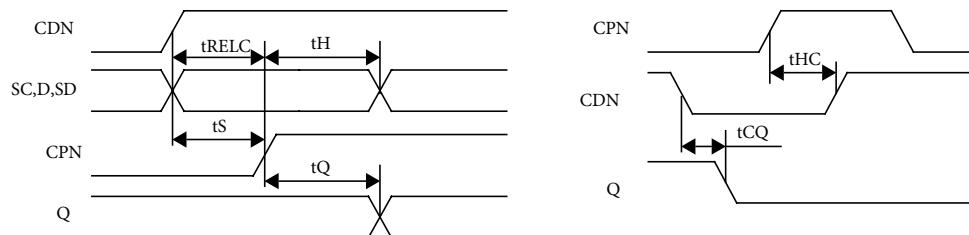
Pin Description

Name	Capacitance (pF)			Description
	SDCFQ1	SDCFQ2	SDCFQ4	
D	0.002	0.002	0.004	Data Input
SD	0.002	0.002	0.004	Scan Data Input
SC	0.002	0.002	0.004	Scan Control
CPN	0.002	0.002	0.004	Clock Input
CDN	0.007	0.007	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SDCFQ1	SDCFQ2	SDCFQ4
D	0.014	0.014	0.023
SD	0.02	0.02	0.004
SC	0.021	0.021	0.045
CPN	0.031	0.031	0.046
CDN	0.003	0.003	-0
Q	0.066	0.082	0.142

Waveforms



Timing Numbers for SDCFQ1:

Error Checks

Pulse Width	CPN (min)	0.161	Pulse Width	CDN (min)	0.185
tH	CPN -> D/SD/SC (min)	0.015	tS	D/SD/SC -> CPN (min)	0.224
tHC	CPN -> CDN (min)	0.208	tRelC	CDN -> CPN (min)	-0.174

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.364	0.309	0.413	0.337	0.508	0.390	0.698	0.490	1.075	0.688
tCQ		0.133		0.161		0.214		0.315		0.515

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.413	0.337	0.500	0.425	0.587	0.511	0.662	0.586	0.724	0.648
tCQ		0.161		0.240		0.314		0.379		0.433

Timing Numbers for SDCFQ2:

Error Checks

Pulse Width	CPN (min)	0.168	Pulse Width	CDN (min)	0.187
tH	CPN -> D/SD/SC (min)	0.015	tS	D/SD/SC -> CPN (min)	0.225
tHC	CPN -> CDN(min)	0.206	tRelC	CDN -> CPN (min)	-0.173

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.350	0.305	0.376	0.324	0.424	0.353	0.519	0.408	0.706	0.511
tCQ		0.122		0.141		0.171		0.226		0.330

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.376	0.324	0.464	0.411	0.550	0.498	0.625	0.573	0.688	0.635
tCQ		0.141		0.221		0.295		0.359		0.413

Timing Numbers for SDCFQ4:**Error Checks**

Pulse Width	CPN (min)	0.147	Pulse Width	CDN (min)	0.147
tH	CPN -> D/SD/SC (min)	-0.068	tS	D/SD/SC -> CPN (min)	0.175
tHC	CPN -> CDN(min)	0.157	tRelC	CDN -> CPN (min)	-0.128

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

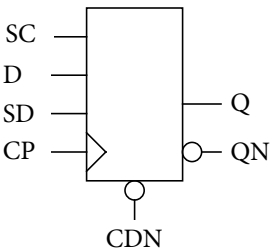
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.347	0.310	0.361	0.326	0.387	0.354	0.427	0.397	0.497	0.476
tCQ		0.185		0.201		0.230		0.272		0.348

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.361	0.326	0.428	0.393	0.480	0.445	0.524	0.488	0.556	0.520
tCQ		0.201		0.288		0.378		0.452		0.508

Multiplexed Scan D Flip-Flops with Clear **SDCRB1 , SDCRB2 and SDCRB4**

The SDCRB1 (1x drive), SDCRB2 (2x drive) and SDCRB4 (4x drive) cells are positive edge triggered multiplexed scan D flip-flops with active-low clear, CDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q output on the rising edge of the clock, CP.



Function Table

INPUTS					OUTPUTS	
CDN	CP	SC	D	SD	Q	QN
L	X	X	X	X	L	H
H	↑	L	L	X	L	H
H	↑	L	H	X	H	L
H	↑	H	X	L	L	H
H	↑	H	X	H	H	L
H	L	X	X	X	Q	QN
H	H	X	X	X	Q	QN

Cell Description

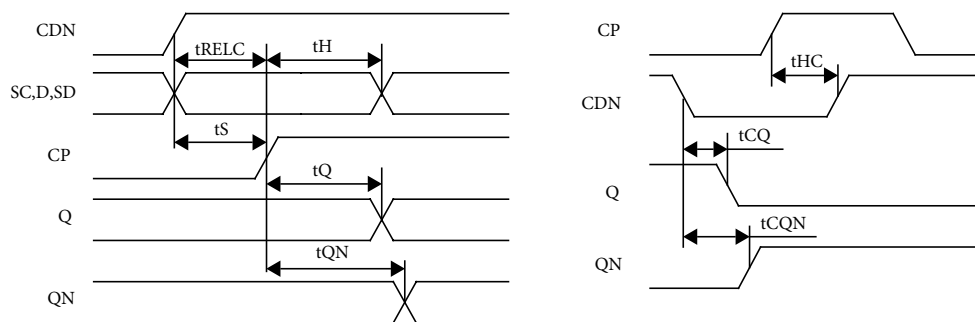
Macro Name:	SDCRB1	SDCRB2	SDCRB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	8.25	8.75	9.
Leakage Power (pW):	191.8	229.6	341.4

Pin Description

Name	Capacitance (pF)			Description
	SDCRB1	SDCRB2	SDCRB4	
D	0.002	0.002	0.004	Data Input
SD	0.002	0.002	0.004	Scan Data Input
SC	0.002	0.002	0.004	Scan Control
CP	0.003	0.003	0.004	Clock Input
CDN	0.007	0.007	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SDCRB1	SDCRB2	SDCRB4
D	0.035	0.035	0.05
SD	0.023	0.023	0.007
SC	0.026	0.026	0.051
CP	0.035	0.034	0.046
CDN	0.003	0.003	0.002
Q	0.036	0.052	0.105
QN	0.036	0.052	0.105

Waveforms

Timing Numbers for SDCRB1:**Error Checks**

Pulse Width	CP (min)	0.229	Pulse Width	CDN (min)	0.141
tH	CP -> D/SD/SC (min)	-0.098	tS	D/SD/SC -> CP (min)	0.233
tHC	CP -> CDN (min)	0.173	tRelC	CDN -> CP (min)	-0.147

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.365	0.344	0.414	0.368	0.513	0.413	0.710	0.497	1.104	0.661
tQN	0.285	0.278	0.337	0.318	0.438	0.381	0.638	0.480	1.035	0.654
tCQ		0.136		0.161		0.206		0.292		0.458
tCQN	0.378		0.430		0.530		0.729		1.126	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.414	0.368	0.466	0.420	0.495	0.447	0.512	0.464	0.518	0.467
tQN	0.337	0.318	0.388	0.369	0.416	0.398	0.432	0.416	0.436	0.421
tCQ		0.161		0.245		0.326		0.398		0.459
tCQN	0.430		0.522		0.623		0.712		0.788	

Timing Numbers for SDCRB2:**Error Checks**

Pulse Width	CP (min)	0.232	Pulse Width	CDN (min)	0.17
tH	CP -> D/SD/SC (min)	-0.097	tS	D/SD/SC -> CP (min)	0.236
tHC	CP -> CDN(min)	0.164	tRelC	CDN -> CP (min)	-0.143

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.393	0.372	0.416	0.391	0.459	0.421	0.546	0.475	0.721	0.576
tQN	0.269	0.295	0.298	0.328	0.346	0.376	0.437	0.454	0.614	0.577
tCQ		0.138		0.158		0.189		0.245		0.348
tCQN	0.365		0.394		0.442		0.532		0.709	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.416	0.391	0.467	0.443	0.495	0.470	0.514	0.487	0.519	0.491
tQN	0.298	0.328	0.349	0.379	0.377	0.408	0.394	0.426	0.398	0.431
tCQ		0.158		0.244		0.325		0.396		0.455
tCQN	0.394		0.487		0.588		0.677		0.753	

Timing Numbers for SDCRB4:**Error Checks**

Pulse Width	CP (min)	0.135	Pulse Width	CDN (min)	0.202
tH	CP -> D/SD/SC (min)	-0.087	tS	D/SD/SC -> CP (min)	0.157
tHC	CP -> CDN (min)	0.114	tRelC	CDN -> CP (min)	-0.104

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.539	0.463	0.550	0.476	0.568	0.498	0.601	0.535	0.666	0.606
tQN	0.271	0.336	0.287	0.358	0.315	0.397	0.359	0.457	0.434	0.558
tCQ		0.180		0.196		0.223		0.266		0.343
tCQN	0.362		0.375		0.400		0.441		0.515	

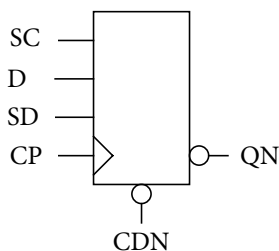
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.550	0.476	0.605	0.532	0.644	0.571	0.673	0.600	0.692	0.618
tQN	0.287	0.358	0.343	0.414	0.382	0.452	0.411	0.481	0.429	0.500
tCQ		0.196		0.283		0.372		0.446		0.503
tCQN	0.375		0.461		0.551		0.627		0.686	

Multiplexed Scan D Flip-Flops with Clear and QN Output Only

SDCRN1 , SDCRN2 and SDCRN4

The SDCRN1 (1x drive), SDCRN2 (2x drive) and SDCRN4 (4x drive) cells are positive edge triggered multiplexed scan D flip-flops with active-low clear, CDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the QN output on the rising edge of the clock, CP.



Function Table

INPUTS					OUTPUT
CDN	CP	SC	D	SD	QN
L	X	X	X	X	H
H	↑	L	L	X	H
H	↑	L	H	X	L
H	↑	H	X	L	H
H	↑	H	X	H	L
H	L	X	X	X	QN
H	H	X	X	X	QN

Cell Description

Macro Name:	SDCRN1	SDCRN2	SDCRN4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.75	8.	7.75
Leakage Power (pW):	164.3	170.2	220.7

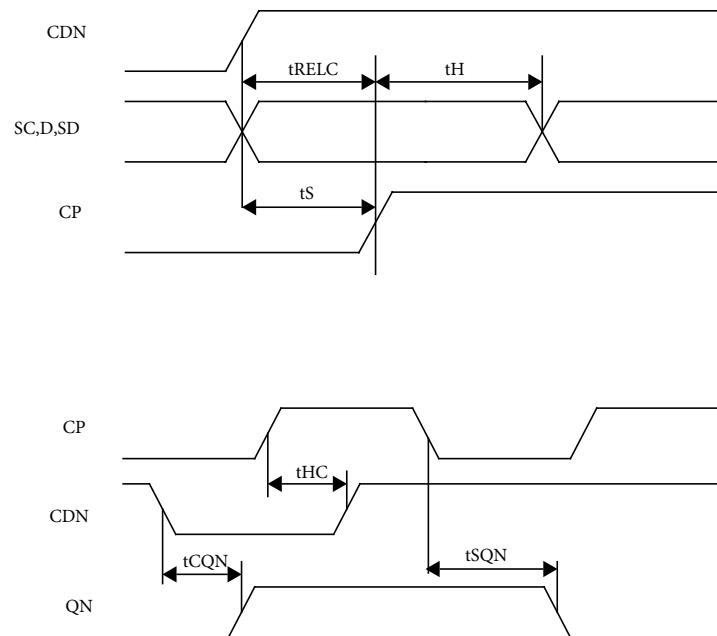
Pin Description

Name	Capacitance (pF)			Description
	SDCRN1	SDCRN2	SDCRN4	
D	0.002	0.002	0.004	Data Input
SD	0.002	0.002	0.004	Scan Data Input
SC	0.002	0.002	0.004	Scan Control
CP	0.003	0.003	0.004	Clock Input
CDN	0.006	0.006	0.009	Clear Input
Maximum capacitance				
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SDCRN1	SDCRN2	SDCRN4
D	0.034	0.034	0.049
SD	0.023	0.023	0.007
SC	0.026	0.026	0.051
CP	0.035	0.035	0.052
CDN	0.003	0.003	0.002
QN	0.054	0.07	0.138

Waveforms



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Timing Numbers for SDCRN1:**Error Checks**

Pulse Width	CP (min)	0.226	Pulse Width	CDN (min)	0.113
tH	CP -> D/SD/SC (min)	-0.097	tS	D/SD/SC -> CP (min)	0.227
tHC	CP -> CDN (min)	0.170	tRelC	CDN -> CP (min)	-0.146

Propagation Delays (ns) for sample loads:**Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.280	0.293	0.327	0.335	0.420	0.405	0.601	0.519	0.963	0.727
tCQN	0.365		0.413		0.505		0.686		1.048	

Propagation Delays (ns) for sample input transitions:**Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.327	0.335	0.379	0.387	0.407	0.416	0.425	0.434	0.429	0.439
tCQN	0.413		0.506		0.605		0.693		0.769	

Timing Numbers for SDCRN2:**Error Checks**

Pulse Width	CP (min)	0.229	Pulse Width	CDN (min)	0.126
tH	CP -> D/SD/SC (min)	-0.096	tS	D/SD/SC -> CP (min)	0.230
tHC	CP -> CDN (min)	0.162	tRelC	CDN -> CP (min)	-0.140

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.272	0.297	0.302	0.331	0.354	0.381	0.453	0.462	0.647	0.593
tCQN	0.361		0.391		0.443		0.541		0.734	

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.302	0.331	0.354	0.382	0.382	0.412	0.399	0.430	0.404	0.435
tCQN	0.391		0.484		0.584		0.672		0.748	

Timing Numbers for SDCRN4:**Error Checks**

Pulse Width	CP (min)	0.137	Pulse Width	CDN (min)	0.114
tH	CP -> D/SD/SC (min)	-0.086	tS	D/SD/SC -> CP (min)	0.147
tHC	CP -> CDN (min)	0.127	tRelC	CDN -> CP (min)	-0.117

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.288	0.355	0.304	0.377	0.333	0.416	0.375	0.475	0.450	0.575
tCQN	0.346		0.362		0.390		0.432		0.506	

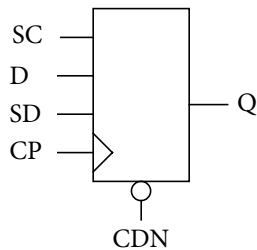
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.304	0.377	0.364	0.436	0.407	0.478	0.438	0.510	0.458	0.530
tCQN	0.362		0.439		0.511		0.572		0.622	

Multiplexed Scan D Flip-Flops with Clear and Positive Clock

SDCRQ1 , SDCRQ2 and SDCRQ4

The SDCRQ1 (1x drive), SDCRQ2 (2x drive) and SDCRQ4 (4x drive) cells are positive edge triggered multiplexed scan D flip-flops with active-low clear, CDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q output on the rising edge of the clock, CP.



Function Table

INPUTS					OUTPUT
CDN	CP	SC	D	SD	Q
L	X	X	X	X	L
H	↑	L	L	X	L
H	↑	L	H	X	H
H	↑	H	X	L	L
H	↑	H	X	H	H
H	L	X	X	X	Q
H	H	X	X	X	Q

Cell Description

Macro Name:	SDCRQ1	SDCRQ2	SDCRQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.75	8.	8.
Leakage Power (pW):	179.6	204.7	321.6

Pin Description

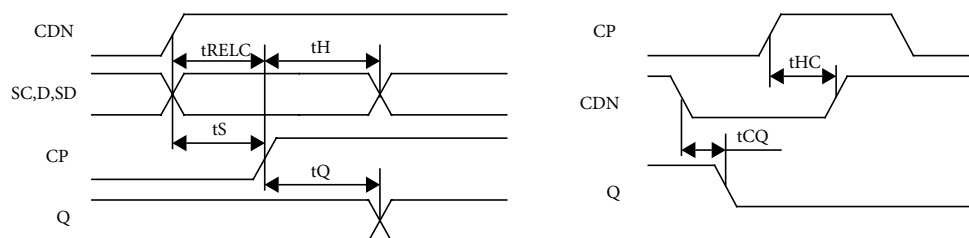
Name	Capacitance (pF)			Description
	SDCRQ1	SDCRQ2	SDCRQ4	
D	0.002	0.002	0.004	Data Input
SD	0.002	0.002	0.004	Scan Data Input
SC	0.002	0.002	0.004	Scan Control
CP	0.003	0.003	0.004	Clock Input
CDN	0.006	0.006	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

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Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SDCRQ1	SDCRQ2	SDCRQ4
D	0.034	0.034	0.052
SD	0.023	0.023	0.007
SC	0.026	0.026	0.052
CP	0.034	0.034	0.047
CDN	0.003	0.003	0.002
Q	0.053	0.066	0.119

Waveforms



Timing Numbers for SDCRQ1:

Error Checks

Pulse Width	CP (min)	0.219	Pulse Width	CDN (min)	0.123
t_H	CP -> D/SD/SC (min)	-0.099	t_S	D/SD/SC -> CP (min)	0.219
t_{HC}	CP -> CDN (min)	0.183	t_{Relc}	CDN -> CP (min)	-0.148

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.301	0.329	0.347	0.360	0.439	0.416	0.620	0.523	0.982	0.735
tCQ		0.148		0.179		0.236		0.345		0.557

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.347	0.360	0.398	0.415	0.427	0.442	0.446	0.460	0.450	0.463
tCQ		0.179		0.265		0.349		0.424		0.488

Timing Numbers for SDCRQ2:

Error Checks

Pulse Width	CP (min)	0.218	Pulse Width	CDN (min)	0.151
tH	CP -> D/SD/SC (min)	-0.099	tS	D/SD/SC -> CP (min)	0.219
tHC	CP -> CDN(min)	0.183	tRelC	CDN -> CP (min)	-0.144

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.291	0.332	0.319	0.354	0.368	0.389	0.466	0.450	0.659	0.564
tCQ		0.153		0.175		0.210		0.272		0.387

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.319	0.354	0.370	0.409	0.399	0.437	0.417	0.453	0.422	0.457
tCQ		0.175		0.264		0.355		0.434		0.500

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Timing Numbers for SDCRQ4:**Error Checks**

Pulse Width	CP (min)	0.126	Pulse Width	CDN (min)	0.158
tH	CP -> D/SD/SC (min)	-0.096	tS	D/SD/SC -> CP (min)	0.168
tHC	CP -> CDN (min)	0.130	tRelC	CDN -> CP (min)	-0.104

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.332	0.332	0.347	0.348	0.374	0.375	0.414	0.418	0.485	0.496
tCQ		0.184		0.200		0.229		0.270		0.345

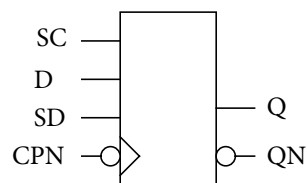
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.347	0.348	0.402	0.403	0.441	0.440	0.472	0.468	0.490	0.485
tCQ		0.200		0.288		0.378		0.452		0.508

Muxed Scan D Flip-Flops

SDNFB1 , SDNFB2 and SDNFB4

The SDNFB1 (1x drive), SDNFB2 (2x drive) and SDNFB4 (4x drive) cells are negative edge triggered multiplexed scan D flip-flops. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q and QN outputs on the falling edge of the clock, CPN.



Function Table

INPUTS				OUTPUTS	
CPN	SC	D	SD	Q	QN
↓	L	L	X	L	H
↓	L	H	X	H	L
↓	H	X	L	L	H
↓	H	X	H	H	L
L	X	X	X	Q	QN
H	X	X	X	Q	QN

Cell Description

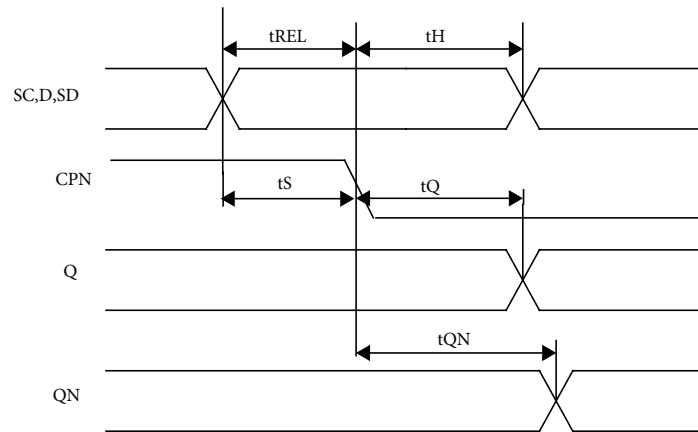
Macro Name:	SDNFB1	SDNFB2	SDNFB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.	7.5	8.75
Leakage Power (pW):	241.1	280.4	363.6

Pin Description

Name	Capacitance (pF)			Description
	SDNFB1	SDNFB2	SDNFB4	
D	0.004	0.004	0.004	Data Input
SD	0.004	0.004	0.004	Scan Data Input
SC	0.004	0.004	0.004	Scan Control
CPN	0.004	0.004	0.004	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	SDNFB1	SDNFB2	SDNFB4
D	0.026	0.026	0.026
SD	0.004	0.004	0.004
SC	0.046	0.046	0.046
CPN	0.047	0.047	0.047
Q	0.052	0.067	0.113
QN	0.052	0.067	0.113

Waveforms

Timing Numbers for SDNFB1:**Error Checks**

Pulse Width	CPN (min)	0.144			
tH	CPN -> D/SD/SC (min)	-0.065	tS	D/SD/SC -> CPN (min)	0.178

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.342	0.301	0.375	0.337	0.440	0.406	0.571	0.544	0.833	0.821
tQN	0.230	0.296	0.269	0.347	0.340	0.432	0.474	0.577	0.736	0.856

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.375	0.337	0.440	0.402	0.492	0.454	0.535	0.498	0.567	0.529
tQN	0.269	0.347	0.335	0.412	0.387	0.463	0.430	0.507	0.462	0.539

Timing Numbers for SDNFB2:**Error Checks**

Pulse Width	CPN (min)	0.178			
tH	CPN -> D/SD/SC (min)	-0.066	tS	D/SD/SC -> CPN (min)	0.177

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.372	0.323	0.389	0.343	0.421	0.381	0.485	0.453	0.614	0.598
tQN	0.227	0.292	0.253	0.327	0.293	0.383	0.365	0.475	0.499	0.631

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.389	0.343	0.455	0.410	0.506	0.463	0.550	0.506	0.582	0.538
tQN	0.253	0.327	0.318	0.393	0.371	0.445	0.415	0.488	0.447	0.521

Timing Numbers for SDNFB4:**Error Checks**

Pulse Width	CPN (min)	0.258			
tH	CPN -> D/SD/SC (min)	-0.068	tS	D/SD/SC -> CPN (min)	0.172

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.491	0.414	0.500	0.425	0.514	0.444	0.543	0.480	0.604	0.549
tQN	0.259	0.334	0.275	0.357	0.304	0.397	0.348	0.457	0.426	0.560

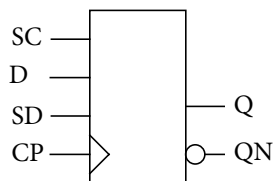
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.500	0.425	0.565	0.491	0.617	0.544	0.661	0.588	0.694	0.622
tQN	0.275	0.357	0.342	0.423	0.395	0.475	0.438	0.518	0.472	0.551

Multiplexed Scan D Flip-Flops

SDNRB1 , SDNRB2 and SDNRB4

The SDNRB1 (1x drive), SDNRB2 (2x drive) and SDNRB4 (4x drive) cells are positive edge triggered multiplexed scan D flip-flops. The Scan Control input, SC, selects either the data input, D, when the SC is low, or the Scan Data input, SD, when the SC is high.



Function Table

INPUTS				OUTPUTS	
CP	SC	D	SD	Q	QN
↑	L	L	X	L	H
↑	L	H	X	H	L
↑	H	X	L	L	H
↑	H	X	H	H	L
L	X	X	X	Q	QN
H	X	X	X	Q	QN

Cell Description

Macro Name:	SDNRB1	SDNRB2	SDNRB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.75	8.5	8.25
Leakage Power (pW):	203.7	245.7	355.5

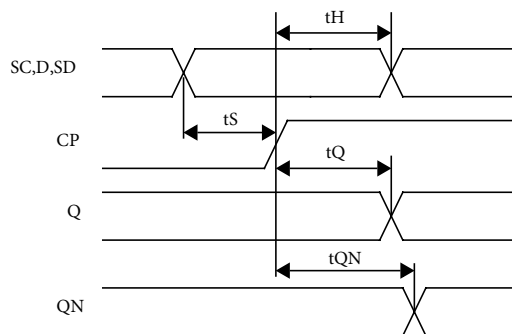
Pin Description

Name	Capacitance (pF)			Description
	SDNRB1	SDNRB2	SDNRB4	
D	0.002	0.002	0.004	Data Input
SD	0.002	0.002	0.004	Scan Data Input
SC	0.002	0.002	0.004	Scan Control
CP	0.003	0.003	0.004	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SDNRB1	SDNRB2	SDNRB4
D	0.016	0.016	0.023
SD	0.021	0.021	0.004
SC	0.022	0.022	0.046
CP	0.034	0.033	0.054
Q	0.036	0.056	0.104
QN	0.036	0.056	0.104

Waveform



Timing Numbers for SDNRB1:

Error Checks

Pulse Width	CP (min)	0.212			
tH	CP -> D/SD/SC (min)	-0.096	tS	D/SD/SC -> CP (min)	0.217

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.346	0.323	0.386	0.346	0.466	0.385	0.627	0.457	0.948	0.602
tQN	0.272	0.269	0.315	0.305	0.398	0.361	0.559	0.450	0.880	0.604

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.386	0.346	0.436	0.395	0.465	0.423	0.482	0.439	0.487	0.443
tQN	0.315	0.305	0.365	0.355	0.392	0.383	0.408	0.401	0.412	0.406

Timing Numbers for SDNRB2:

Error Checks

Pulse Width	CP (min)	0.214			
tH	CP -> D/SD/SC (min)	-0.094	tS	D/SD/SC -> CP (min)	0.219

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.406	0.368	0.423	0.384	0.459	0.408	0.536	0.451	0.695	0.526
tQN	0.269	0.279	0.296	0.306	0.340	0.346	0.425	0.410	0.587	0.508

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.423	0.384	0.474	0.435	0.502	0.462	0.520	0.480	0.525	0.483
tQN	0.296	0.306	0.347	0.357	0.374	0.385	0.391	0.403	0.394	0.408

Timing Numbers for SDNRB4:**Error Checks**

Pulse Width	CP (min)	0.138			
tH	CP -> D/SD/SC (min)	-0.081	tS	D/SD/SC -> CP (min)	0.148

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

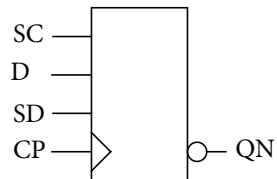
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.504	0.442	0.513	0.454	0.528	0.473	0.559	0.509	0.621	0.580
tQN	0.286	0.345	0.302	0.368	0.331	0.407	0.375	0.467	0.451	0.568

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.513	0.454	0.572	0.513	0.614	0.555	0.646	0.587	0.667	0.606
tQN	0.302	0.368	0.362	0.427	0.404	0.469	0.436	0.501	0.456	0.521

Multiplexed Scan D Flip-Flops with QN Output Only SDNRN1 , SDNRN2 and SDNRN4

The SDNRN1 (1x drive), SDNRN2 (2x drive) and SDNRN4 (4x drive) cells are positive edge triggered multiplexed scan D flip-flops. The scan control input, SC, selects either the data input, D, or scan data input, SD, which is clocked to the QN output on the rising edge of the clock, CP.



Function Table

INPUTS				OUTPUT
CP	SC	D	SD	QN
↑	L	L	X	H
↑	L	H	X	L
↑	H	X	L	H
↑	H	X	H	L
L	X	X	X	QN
H	X	X	X	QN

Cell Description

Macro Name:	SDNRN1	SDNRN2	SDNRN4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.5	7.75	7.
Leakage Power (pW):	171.7	188.5	228.7

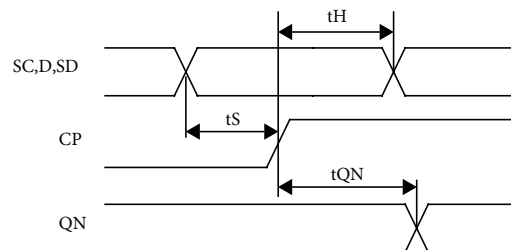
Pin Description

Name	Capacitance (pF)			Description
	SDNRN1	SDNRN2	SDNRN4	
D	0.002	0.002	0.004	Data Input
SD	0.002	0.002	0.004	Scan Data Input
SC	0.002	0.002	0.004	Scan Control
CP	0.003	0.003	0.004	Clock Input
Maximum capacitance				
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SDNRN1	SDNRN2	SDNRN4
D	0.016	0.016	0.024
SD	0.021	0.021	0.004
SC	0.022	0.022	0.046
CP	0.033	0.033	0.051
QN	0.045	0.075	0.135

Waveforms



Timing Numbers for SDNRN1:

Error Checks

Pulse Width	CP (min)	0.211			
tH	CP -> D/SD/SC (min)	-0.096	tS	D/SD/SC -> CP (min)	0.216

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.263	0.264	0.307	0.302	0.395	0.363	0.568	0.464	0.914	0.651

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.307	0.302	0.359	0.353	0.386	0.381	0.403	0.400	0.407	0.405

Timing Numbers for SDNRN2:

Error Checks

Pulse Width	CP (min)	0.215			
tH	CP -> D/SD/SC (min)	-0.094	tS	D/SD/SC -> CP (min)	0.219

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.269	0.280	0.296	0.308	0.340	0.347	0.424	0.409	0.586	0.505

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.296	0.308	0.347	0.359	0.374	0.388	0.391	0.405	0.394	0.410

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Timing Numbers for SDNRN4:**Error Checks**

Pulse Width	CP (min)	0.132			
tH	CP -> D/SD/SC (min)	-0.087	tS	D/SD/SC -> CP (min)	0.148

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.285	0.348	0.302	0.372	0.331	0.412	0.375	0.475	0.453	0.578

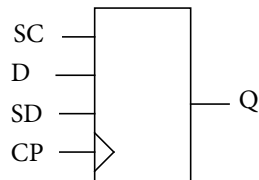
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.302	0.372	0.360	0.429	0.400	0.468	0.430	0.499	0.448	0.518

Multiplexed Scan D Flip-Flops with Q Output Only

SDNRQ1 , SDNRQ2 and SDNRQ4

The SDNRQ1 (1x drive), SDNRQ2 (2x drive) and SDNRQ4 (4x drive) cells are positive edge triggered multiplexed scan D flip-flops. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q output on the rising edge of the clock, CP.



Function Table

INPUTS				OUTPUT
CP	SC	D	SD	Q
↑	L	L	X	L
↑	L	H	X	H
↑	H	X	L	L
↑	H	X	H	H
L	X	X	X	Q
H	X	X	X	Q

Cell Description

Macro Name:	SDNRQ1	SDNRQ2	SDNRQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.5	7.75	7.
Leakage Power (pW):	187.4	209.7	313.7

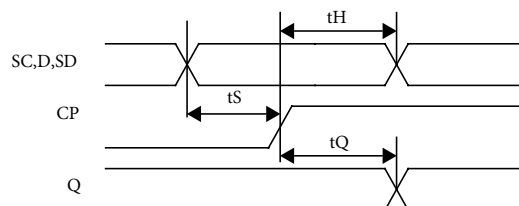
Pin Description

Name	Capacitance (pF)			Description
	SDNRQ1	SDNRQ2	SDNRQ4	
D	0.003	0.003	0.004	Data Input
SD	0.002	0.002	0.004	Scan Data Input
SC	0.002	0.002	0.004	Scan Control
CP	0.003	0.003	0.004	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SDNRQ1	SDNRQ2	SDNRQ4
D	0.02	0.02	0.024
SD	0.022	0.022	0.004
SC	0.023	0.023	0.047
CP	0.037	0.037	0.052
Q	0.046	0.061	0.11

Waveforms



Timing Numbers for SDNRQ1:**Error Checks**

Pulse Width	CP (min)	0.2			
tH	CP -> D/SD/SC (min)	-0.057	tS	D/SD/SC -> CP (min)	0.206

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.306	0.297	0.350	0.323	0.439	0.369	0.617	0.457	0.971	0.630

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.350	0.323	0.407	0.377	0.445	0.410	0.471	0.434	0.485	0.444

Timing Numbers for SDNRQ2:**Error Checks**

Pulse Width	CP (min)	0.2			
tH	CP -> D/SD/SC (min)	-0.057	tS	D/SD/SC -> CP (min)	0.205

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.306	0.301	0.331	0.319	0.376	0.346	0.467	0.394	0.647	0.481

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.331	0.319	0.388	0.373	0.425	0.407	0.452	0.430	0.465	0.440

Timing Numbers for SDNRQ4:**Error Checks**

Pulse Width	CP (min)	0.131			
tH	CP -> D/SD/SC (min)	-0.080	tS	D/SD/SC -> CP (min)	0.149

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.301	0.315	0.313	0.329	0.335	0.355	0.371	0.395	0.438	0.471

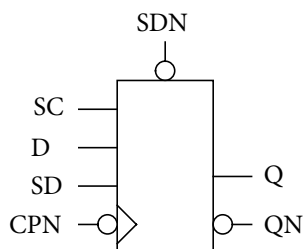
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.313	0.329	0.372	0.387	0.413	0.428	0.446	0.458	0.466	0.476

Muxed Scan D Flip-Flops with Preset

SDPFB1 , SDPFB2 and SDPFB4

The SDPFB1 (1x drive), SDPFB2 (2x drive) and SDPFB4 (4x drive) cells are negative edge triggered multiplexed scan D flip-flops with active-low set, SDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q and QN outputs on the falling edge of the clock, CPN.



Function Table

INPUTS					OUTPUTS	
SDN	CPN	SC	D	SD	Q	QN
L	X	X	X	X	H	L
H	↓	L	L	X	L	H
H	↓	L	H	X	H	L
H	↓	H	X	L	L	H
H	↓	H	X	H	H	L
H	L	X	X	X	Q	QN
H	H	X	X	X	Q	QN

Cell Description

Macro Name:	SDPFB1	SDPFB2	SDPFB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	8.	9.	10.25
Leakage Power (pW):	261.0	307.3	390.6

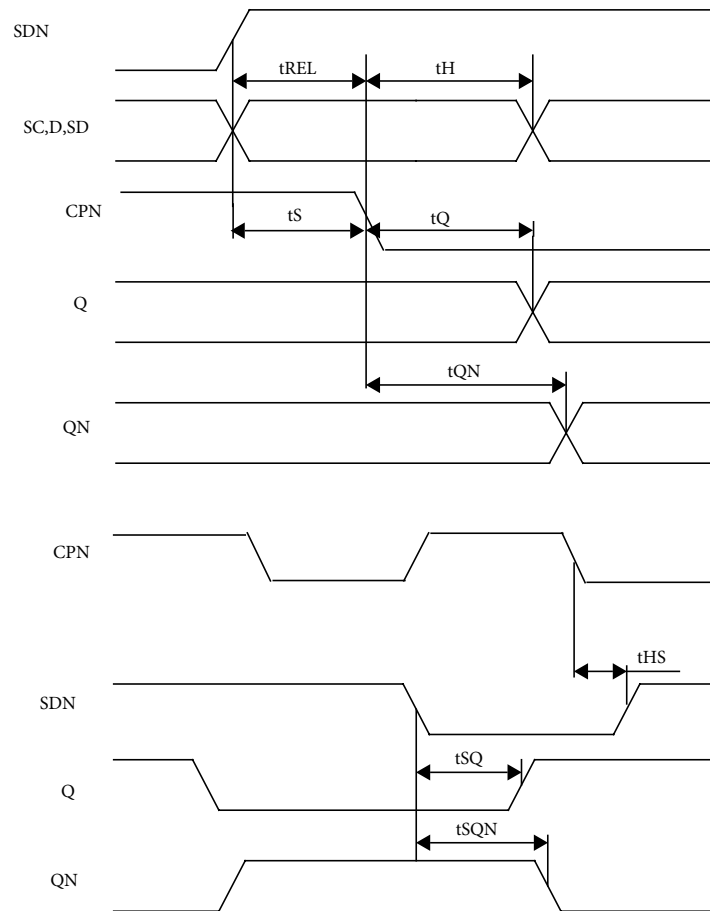
Pin Description

Name	Capacitance (pF)			Description
	SDPFB1	SDPFB2	SDPFB4	
D	0.004	0.004	0.004	Data Input
SD	0.004	0.005	0.005	Scan Data Input
SC	0.004	0.004	0.004	Scan Control
CPN	0.004	0.004	0.004	Clock Input
SDN	0.008	0.008	0.008	Preset Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SDPFB1	SDPFB2	SDPFB4
D	0.029	0.028	0.028
SD	0.002	0.002	0.002
SC	0.057	0.056	0.056
CPN	0.054	0.056	0.058
SDN	-0	-0	-0
Q	0.053	0.069	0.116
QN	0.053	0.069	0.116

Waveforms



Timing Numbers for SDPFB1:**Error Checks**

Pulse Width	CPN (min)	0.149	Pulse Width	SDN (min)	0.149
tH	CPN -> D/SD/SC (min)	-0.064	tS	D/SD/SC -> CPN (min)	0.216
tHS	CPN -> SDN (min)	0.034	tRelS	SDN -> CPN (min)	0.016

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.331	0.299	0.366	0.339	0.434	0.415	0.572	0.565	0.847	0.865
tQN	0.243	0.301	0.283	0.351	0.356	0.437	0.490	0.592	0.753	0.894
tSQ	0.309		0.343		0.411		0.549		0.824	
tSQN		0.283		0.331		0.415		0.569		0.870

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.366	0.339	0.432	0.406	0.485	0.460	0.529	0.505	0.562	0.539
tQN	0.283	0.351	0.350	0.417	0.404	0.470	0.449	0.514	0.483	0.547
tSQ	0.343		0.427		0.512		0.585		0.642	
tSQN		0.331		0.415		0.501		0.575		0.632

Timing Numbers for SDPFB2:**Error Checks**

Pulse Width	CPN (min)	0.197	Pulse Width	SDN (min)	0.164
tH	CPN -> D/SD/SC (min)	-0.056	tS	D/SD/SC -> CPN (min)	0.211
tHS	CPN -> SDN (min)	0.031	tRelS	SDN -> CPN (min)	0.016

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.371	0.337	0.388	0.356	0.421	0.393	0.486	0.465	0.616	0.609
tQN	0.247	0.300	0.275	0.335	0.319	0.390	0.395	0.481	0.531	0.640
tSQ	0.344		0.361		0.394		0.459		0.589	
tSQN		0.279		0.312		0.364		0.453		0.609

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.388	0.356	0.455	0.424	0.509	0.479	0.554	0.525	0.588	0.560
tQN	0.275	0.335	0.343	0.402	0.398	0.456	0.444	0.501	0.478	0.536
tSQ	0.361		0.446		0.534		0.609		0.668	
tSQN		0.312		0.397		0.485		0.560		0.619

Timing Numbers for SDPFB4:**Error Checks**

Pulse Width	CPN (min)	0.293	Pulse Width	SDN (min)	0.193
tH	CPN -> D/SD/SC (min)	-0.054	tS	D/SD/SC -> CPN (min)	0.206
tHS	CPN -> SDN (min)	0.029	tRelS	SDN -> CPN (min)	0.008

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

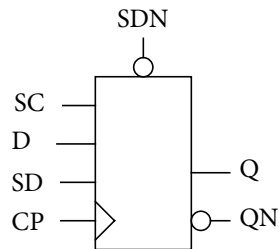
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.492	0.444	0.500	0.453	0.514	0.470	0.544	0.504	0.606	0.572
tQN	0.291	0.343	0.309	0.366	0.340	0.404	0.389	0.463	0.470	0.560
tSQ	0.453		0.461		0.475		0.504		0.566	
tSQN		0.318		0.339		0.373		0.429		0.521

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.500	0.453	0.569	0.521	0.624	0.577	0.669	0.624	0.704	0.660
tQN	0.309	0.366	0.377	0.433	0.434	0.488	0.480	0.534	0.516	0.569
tSQ	0.461		0.544		0.633		0.713		0.777	
tSQN		0.339		0.424		0.518		0.599		0.661

Multiplexed Scan D Flip-Flops with Set SDPRB1 , SDPRB2 and SDPRB4

The SDPRB1 (1x drive), SDPRB2 (2x drive) and SDPRB4 (4x drive) cells are positive edge triggered multiplexed D flip-flops with active-low set, SDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q and QN outputs on the rising edge of the clock, CP.



Function Table

INPUTS					OUTPUTS	
SDN	CP	SC	D	SD	Q	QN
L	X	X	X	X	H	L
H	↑	L	L	X	L	H
H	↑	L	H	X	H	L
H	↑	H	X	L	L	H
H	↑	H	X	H	H	L
H	L	X	X	X	Q	QN
H	H	X	X	X	Q	QN

Cell Description

Macro Name:	SDPRB1	SDPRB2	SDPRB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	9.	9.25	9.5
Leakage Power (pW):	211.5	244.2	371.8

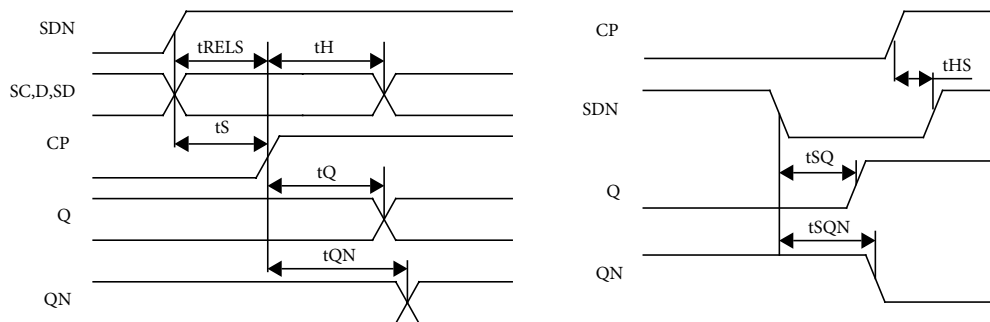
Pin Description

Name	Capacitance (pF)			Description
	SDPRB1	SDPRB2	SDPRB4	
D	0.003	0.003	0.004	Data Input
SD	0.002	0.002	0.004	Scan Data Input
SC	0.002	0.002	0.004	Scan Control
CP	0.003	0.003	0.004	Clock Input
SDN	0.006	0.006	0.009	Preset Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SDPRB1	SDPRB2	SDPRB4
D	0.023	0.023	0.028
SD	0.022	0.022	0.002
SC	0.028	0.028	0.056
CP	0.042	0.042	0.049
SDN	0	-0	-0
Q	0.04	0.056	0.111
QN	0.04	0.056	0.111

Waveforms



Timing Numbers for SDPRB1:

Error Checks

Pulse Width	CP (min)	0.221	Pulse Width	SDN (min)	0.154
tH	CP -> D/SD/SC (min)	-0.063	tS	D/SD/SC -> CP (min)	0.222
tHS	CP -> SDN (min)	0.102	tRelS	SDN -> CP (min)	-0.081

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.363	0.349	0.410	0.377	0.505	0.429	0.693	0.526	1.070	0.719
tQN	0.297	0.294	0.348	0.336	0.446	0.403	0.636	0.513	1.013	0.712
tSQ	0.311		0.358		0.452		0.640		1.018	
tSQN		0.242		0.282		0.347		0.452		0.647

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**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.410	0.377	0.468	0.433	0.507	0.469	0.535	0.494	0.549	0.506
tQN	0.348	0.336	0.405	0.394	0.441	0.432	0.466	0.460	0.477	0.474
tSQ	0.358		0.445		0.535		0.613		0.676	
tSQN		0.282		0.370		0.461		0.539		0.603

Timing Numbers for SDPRB2:

Error Checks

Pulse Width	CP (min)	0.223	Pulse Width	SDN (min)	0.202
tH	CP -> D/SD/SC (min)	-0.063	tS	D/SD/SC -> CP (min)	0.224
tHS	CP -> SDN (min)	0.100	tRelS	SDN -> CP (min)	-0.078

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.399	0.379	0.420	0.398	0.465	0.429	0.558	0.482	0.745	0.580
tQN	0.288	0.294	0.319	0.325	0.372	0.372	0.471	0.446	0.662	0.563
tSQ	0.343		0.365		0.409		0.502		0.690	
tSQN		0.241		0.272		0.315		0.385		0.495

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.420	0.398	0.478	0.454	0.517	0.490	0.544	0.516	0.558	0.527
tQN	0.319	0.325	0.375	0.383	0.411	0.421	0.436	0.449	0.448	0.463
tSQ	0.365		0.454		0.549		0.631		0.697	
tSQN		0.272		0.360		0.455		0.537		0.603

Timing Numbers for SDPRB4:**Error Checks**

Pulse Width	CP (min)	0.163	Pulse Width	SDN (min)	0.316
tH	CP -> D/SD/SC (min)	-0.093	tS	D/SD/SC -> CP (min)	0.180
tHS	CP -> SDN (min)	0.035	tRelS	SDN -> CP (min)	-0.015

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

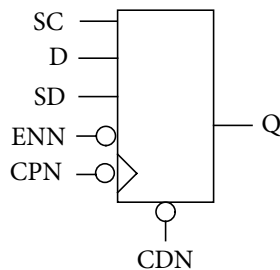
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.501	0.461	0.510	0.471	0.525	0.489	0.556	0.525	0.619	0.596
tQN	0.295	0.342	0.313	0.365	0.344	0.405	0.393	0.466	0.477	0.569
tSQ	0.467		0.476		0.491		0.523		0.586	
tSQN		0.322		0.343		0.380		0.437		0.534

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.510	0.471	0.566	0.527	0.603	0.564	0.632	0.593	0.650	0.610
tQN	0.313	0.365	0.369	0.420	0.407	0.458	0.436	0.487	0.453	0.505
tSQ	0.476		0.561		0.653		0.733		0.796	
tSQN		0.343		0.428		0.521		0.601		0.664

Muxed Scan Enable D Flip-Flops with Clear and Q only SECFQ1 , SECFQ2 and SECFQ4

The SECFQ1 (1x drive), SECFQ2 (2x drive) and SECFQ4 (4x drive) cells are negative edge triggered multiplexed scan D flip-flops with active-low enable, ENN, and clear, CDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q output on the falling edge of the clock, CPN.



Function Table

INPUTS						OUTPUT
CDN	CPN	ENN	SC	D	SD	Q
L	X	X	X	X	X	L
H	↓	L	L	L	X	L
H	↓	L	L	H	X	H
H	↓	X	H	X	L	L
H	↓	X	H	X	H	H
H	L	L	X	X	X	Q
H	H	L	X	X	X	Q
H	X	H	L	X	X	Q _o

Cell Description

Macro Name:	SECFQ1	SECFQ2	SECFQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	11.	11.5	12.25
Leakage Power (pW):	232.0	257.0	340.9

Pin Description

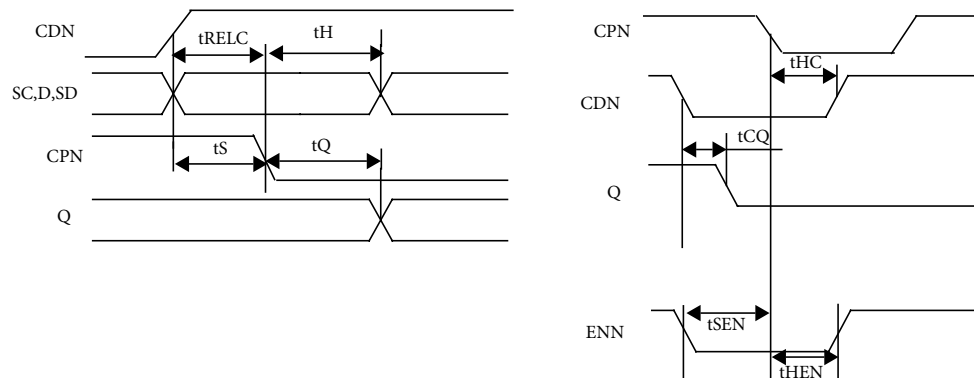
Name	Capacitance (pF)			Description
	SECFQ1	SECFQ2	SECFQ4	
D	0.004	0.004	0.004	Data Input
SD	0.003	0.003	0.003	Scan Data Input
SC	0.005	0.004	0.005	Scan Control Input
ENN	0.009	0.009	0.009	Enable Input
CPN	0.005	0.005	0.005	Clock Input
CDN	0.009	0.009	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

**Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	SECFQ1	SECFQ2	SECFQ4
D	0.029	0.029	0.029
SD	0.005	0.004	0.005
SC	0.056	0.052	0.056
ENN	0.018	0.018	0.018
CPN	0.057	0.06	0.059
CDN	0.001	0	0
Q	0.125	0.138	0.176

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Waveforms



Timing Numbers for SECFQ1:

Error Checks

Pulse Width	CPN (min)	0.129	Pulse Width	CDN (min)	0.136
t_H	CPN -> D (min)	-0.119	t_S	D -> CPN (min)	0.333
t_{HSC}	CPN -> SC (min)	-0.174	t_{SSC}	SC -> CPN (min)	0.360
t_{HSD}	CPN -> SD (min)	-0.086	t_{SSD}	SD -> CPN (min)	0.300
t_{HC}	CPN -> CDN (min)	0.137	t_{RelC}	CDN -> CPN (min)	-0.101
t_{HEN}	CPN -> ENN (min)	-0.113	t_{SEN}	ENN -> CPN (min)	0.353

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
t_Q	0.326	0.290	0.362	0.330	0.428	0.407	0.558	0.558	0.818	0.858
t_{CQ}		0.158		0.199		0.277		0.427		0.728

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.362	0.330	0.430	0.399	0.485	0.455	0.532	0.502	0.567	0.538
tCQ		0.199		0.275		0.343		0.400		0.447

Timing Numbers for SECFQ2:

Error Checks

Pulse Width	CPN (min)	0.131	Pulse Width	CDN (min)	0.135
tH	CPN -> D (min)	-0.116	tS	D -> CPN (min)	0.323
tHSC	CPN -> SC (min)	-0.184	tSSC	SC -> CPN (min)	0.362
tHSD	CPN -> SD (min)	-0.085	tSSD	SD -> CPN (min)	0.297
tHC	CPN -> CDN (min)	0.142	tRelC	CDN -> CPN (min)	-0.107
tHEN	CPN -> ENN (min)	-0.131	tSEN	ENN -> CPN (min)	0.343

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.318	0.275	0.340	0.300	0.377	0.340	0.446	0.416	0.578	0.564
tCQ		0.147		0.172		0.213		0.290		0.438

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.340	0.300	0.408	0.367	0.463	0.423	0.509	0.470	0.544	0.504
tCQ		0.172		0.253		0.329		0.394		0.446

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Timing Numbers for SECFQ4:**Error Checks**

Pulse Width	CPN (min)	0.152	Pulse Width	CDN (min)	0.144
tH	CPN -> D (min)	-0.119	tS	D -> CPN (min)	0.328
tHSC	CPN -> SC (min)	-0.175	tSSC	SC -> CPN (min)	0.357
tHSD	CPN -> SD (min)	-0.087	tSSD	SD -> CPN (min)	0.294
tHC	CPN -> CDN (min)	0.138	tRelC	CDN -> CPN (min)	-0.105
tHEN	CPN -> ENN (min)	-0.113	tSEN	ENN -> CPN (min)	0.348

Propagation Delays (ns) for sample loads:**Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.343	0.298	0.357	0.313	0.381	0.339	0.421	0.381	0.492	0.456
tCQ		0.163		0.178		0.204		0.246		0.321

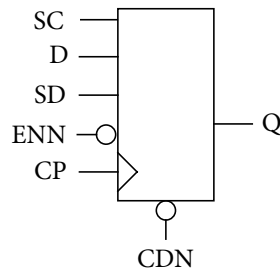
Propagation Delays (ns) for sample input transitions:**Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.357	0.313	0.425	0.381	0.480	0.437	0.526	0.484	0.561	0.518
tCQ		0.178		0.262		0.345		0.413		0.466

Multiplexed Scan, D-Enabled Flip-Flops with Clear and Q Output

SECRQ1 , SECRQ2 and SECRQ4

The SECRQ1 (1x drive), SECRQ2 (2x drive) and SECRQ4 (4x drive) cells are positive edge triggered multiplexed scan D flip-flops with active-low enable, ENN, and clear, CDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q output on the rising edge of the clock, CP.



Function Table

INPUTS						OUTPUT
CDN	CP	ENN	SC	D	SD	Q
L	X	X	X	X	X	L
H	↑	L	L	L	X	L
H	↑	L	L	H	X	H
H	↑	X	H	X	L	L
H	↑	X	H	X	H	H
H	L	L	X	X	X	Q
H	H	L	X	X	X	Q
H	X	H	L	X	X	Q ₀

Cell Description

Macro Name:	SECRQ1	SECRQ2	SECRQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	12.	12.25	9.5
Leakage Power (pW):	256.4	283.5	326.5

Pin Description

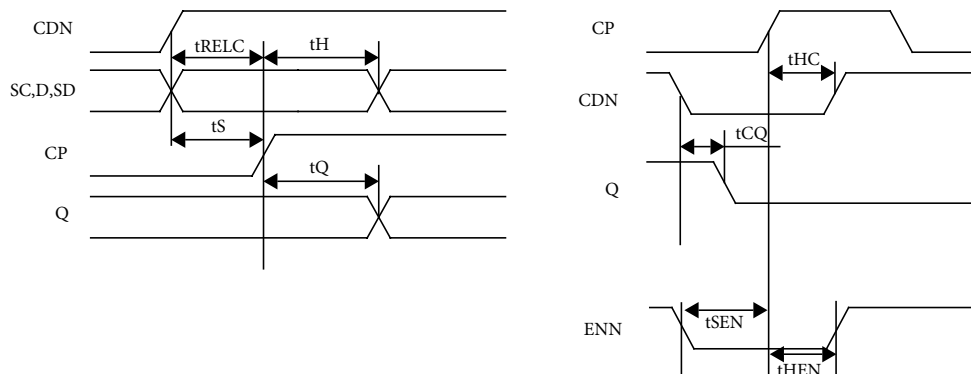
Name	Capacitance (pF)			Description
	SECRQ1	SECRQ2	SECRQ4	
D	0.002	0.002	0.004	Data Input
SD	0.003	0.003	0.003	Scan Data Input
SC	0.002	0.002	0.005	Scan Control Input
ENN	0.005	0.005	0.009	Enable Input
CP	0.004	0.004	0.004	Clock Input
CDN	0.007	0.007	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

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**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	SECRQ1	SECRQ2	SECRQ4
D	0.036	0.036	0.057
SD	0.028	0.028	0.007
SC	0.051	0.051	0.062
ENN	0.045	0.045	0.023
CP	0.037	0.037	0.049
CDN	0.002	0.002	0.001
Q	0.061	0.076	0.141

Waveforms



Timing Numbers for SECRQ1:

Error Checks

Pulse Width	CP (min)	0.211	Pulse Width	CDN (min)	0.109
tH	CP -> D (min)	-0.146	tS	D -> CP (min)	0.254
tHSC	CP -> SC (min)	-0.228	tSSC	SC -> CP (min)	0.429
tHSD	CP -> SD (min)	-0.219	tSSD	SD -> CP (min)	0.352
tHC	CP -> CDN (min)	0.110	tRelC	CDN -> CP (min)	-0.089
tHEN	CP -> ENN (min)	-0.240	tSEN	ENN -> CP (min)	0.368

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.304	0.287	0.353	0.316	0.448	0.371	0.638	0.475	1.015	0.683
tCQ		0.126		0.155		0.210		0.315		0.523

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.353	0.316	0.380	0.344	0.384	0.348	0.380	0.342	0.364	0.326
tCQ		0.155		0.230		0.295		0.353		0.401

Timing Numbers for SECRQ2:

Error Checks

Pulse Width	CP (min)	0.211	Pulse Width	CDN (min)	0.135
tH	CP -> D (min)	-0.146	tS	D -> CP (min)	0.254
tHSC	CP -> SC (min)	-0.229	tSSC	SC -> CP (min)	0.427
tHSD	CP -> SD (min)	-0.220	tSSD	SD -> CP (min)	0.352
tHC	CP -> CDN (min)	0.110	tRelC	CDN -> CP (min)	-0.088
tHEN	CP -> ENN (min)	-0.240	tSEN	ENN -> CP (min)	0.368

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.298	0.288	0.325	0.307	0.373	0.337	0.467	0.389	0.652	0.484
tCQ		0.126		0.145		0.174		0.226		0.320

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.325	0.307	0.352	0.335	0.356	0.339	0.352	0.334	0.337	0.317
tCQ		0.145		0.225		0.297		0.359		0.409

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Timing Numbers for SECRQ4:**Error Checks**

Pulse Width	CP (min)	0.183	Pulse Width	CDN (min)	0.208
tH	CP -> D (min)	-0.173	tS	D -> CP (min)	0.307
tHSC	CP -> SC (min)	-0.228	tSSC	SC -> CP (min)	0.335
tHSD	CP -> SD (min)	-0.151	tSSD	SD -> CP (min)	0.277
tHC	CP -> CDN (min)	0.146	tRelC	CDN -> CP (min)	-0.110
tHEN	CP -> ENN (min)	-0.189	tSEN	ENN -> CP (min)	0.333

Propagation Delays (ns) for sample loads:**Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.320	0.319	0.334	0.334	0.358	0.360	0.398	0.401	0.468	0.476
tCQ		0.207		0.223		0.252		0.293		0.367

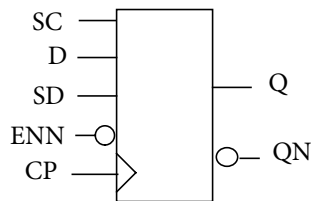
Propagation Delays (ns) for sample input transitions:**Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.334	0.334	0.389	0.388	0.429	0.426	0.459	0.453	0.478	0.469
tCQ		0.223		0.314		0.411		0.489		0.544

Muxed Scan Enable D Flip-Flops

SENRB1 , SENRB2 and SENRB4

The SENRB1 (1x drive), SENRB2 (2x drive) and SENRB4 (4x drive) cells are positive edge triggered multiplexed scan D flip-flops with active-low enable, ENN. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q and QN outputs on the rising edge of the clock, CP.



Function Table

INPUTS					OUTPUTS	
CP	ENN	SC	D	SD	Q	QN
↑	L	L	L	X	L	H
↑	L	L	H	X	H	L
↑	X	H	X	L	L	H
↑	X	H	X	H	H	L
L	L	X	X	X	Q	QN
H	L	X	X	X	Q	QN
X	H	L	X	X	Q _o	QN _o

Cell Description

Macro Name:	SENRB1	SENRB2	SENRB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	8.5	8.75	10.
Leakage Power (pW):	249.0	282.9	364.0

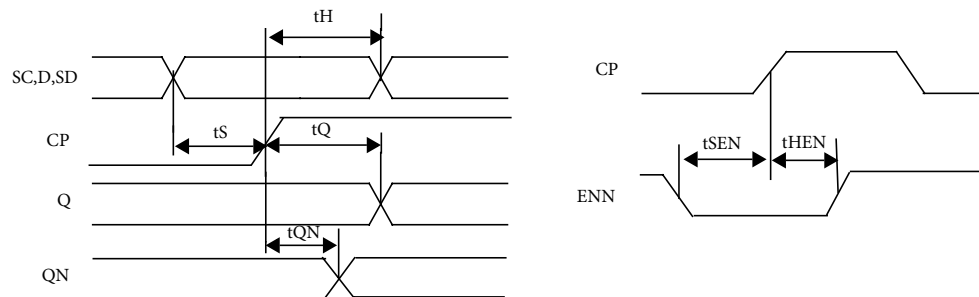
Pin Description

Name	Capacitance (pF)			Description
	SENRB1	SENRB2	SENRB4	
D	0.004	0.004	0.004	Data Input
SD	0.003	0.003	0.003	Scan Data Input
SC	0.004	0.004	0.004	Scan Control Input
ENN	0.009	0.009	0.009	Enable Input
CP	0.004	0.004	0.004	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SENRB1	SENRB2	SENRB4
D	0.029	0.029	0.029
SD	0.005	0.005	0.005
SC	0.054	0.054	0.054
ENN	0.018	0.018	0.018
CP	0.047	0.047	0.047
Q	0.052	0.068	0.11
QN	0.052	0.068	0.11

Waveforms



Timing Numbers for SENRB1:

Error Checks

Pulse Width	CP (min)	0.162			
t_H	CP -> D (min)	-0.185	t_S	D -> CP (min)	0.344
t_{HSC}	CP -> SC (min)	-0.218	t_{SSC}	SC -> CP (min)	0.308
t_{HSD}	CP -> SD (min)	-0.139	t_{SSD}	SD -> CP (min)	0.239
t_{HEN}	CP -> ENN (min)	-0.197	t_{SEN}	ENN -> CP (min)	0.376

Propagation Delays (ns) for sample loads: Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
t_Q	0.333	0.319	0.367	0.359	0.433	0.435	0.564	0.585	0.827	0.885
t_{QN}	0.256	0.291	0.294	0.341	0.363	0.428	0.496	0.582	0.758	0.884

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.367	0.359	0.423	0.414	0.462	0.451	0.492	0.480	0.511	0.497
tQN	0.294	0.341	0.349	0.397	0.387	0.436	0.415	0.467	0.433	0.486

Timing Numbers for SENRB2:

Error Checks

Pulse Width	CP (min)	0.165			
tH	CP -> D (min)	-0.184	tS	D -> CP (min)	0.347
tHSC	CP -> SC (min)	-0.223	tSSC	SC -> CP (min)	0.311
tHSD	CP -> SD (min)	-0.144	tSSD	SD -> CP (min)	0.242
tHEN	CP -> ENN (min)	-0.196	tSEN	ENN -> CP (min)	0.379

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.368	0.331	0.385	0.351	0.418	0.390	0.483	0.464	0.612	0.611
tQN	0.246	0.280	0.269	0.313	0.308	0.365	0.378	0.454	0.511	0.608

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.385	0.351	0.441	0.407	0.480	0.445	0.510	0.473	0.529	0.491
tQN	0.269	0.313	0.325	0.369	0.363	0.408	0.391	0.438	0.409	0.457

Timing Numbers for SENRB4:**Error Checks**

Pulse Width	CP (min)	0.167			
tH	CP -> D (min)	-0.183	tS	D -> CP (min)	0.348
tHSC	CP -> SC (min)	-0.228	tSSC	SC -> CP (min)	0.312
tHSD	CP -> SD (min)	-0.146	tSSD	SD -> CP (min)	0.243
tHEN	CP -> ENN (min)	-0.195	tSEN	ENN -> CP (min)	0.381

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.467	0.396	0.476	0.407	0.493	0.427	0.525	0.465	0.588	0.538
tQN	0.266	0.317	0.282	0.338	0.309	0.376	0.351	0.434	0.426	0.531

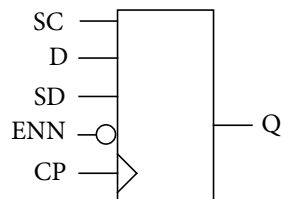
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.476	0.407	0.532	0.462	0.571	0.500	0.601	0.530	0.620	0.547
tQN	0.282	0.338	0.337	0.394	0.375	0.433	0.404	0.462	0.422	0.481

Multiplexed Scan D-Enabled Flip-Flops with Q Output Only

SENQR1 , SENQR2 and SENQR4

The SENQR1 (1x drive), SENQR2 (2x drive) and SENQR4 (4x drive) cells are positive edge triggered multiplexed scan D flip-flops with an active-low enable, ENN. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q output on the rising edge of the clock, CP.



Function Table

INPUTS					OUTPUT
CP	ENN	SC	D	SD	Q
↑	L	L	L	X	L
↑	L	L	H	X	H
↑	X	H	X	L	L
↑	X	H	X	H	H
L	L	X	X	X	Q
H	L	X	X	X	Q
X	H	L	X	X	Q ₀

Cell Description

Macro Name:	SENQR1	SENQR2	SENQR4
Drive Capability:	1x	2x	4x
Gate Equivalents:	11.5	11.75	8.75
Leakage Power (pW):	272.9	303.9	332.9

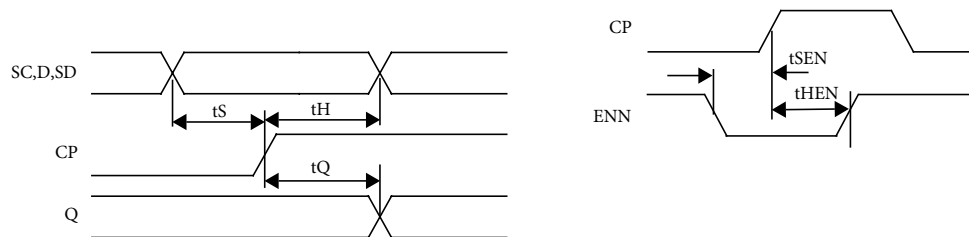
Pin Description

Name	Capacitance (pF)			Description
	SENQR1	SENQR2	SENQR4	
D	0.002	0.002	0.004	Data Input
SD	0.003	0.003	0.003	Scan Data Input
SC	0.002	0.002	0.004	Scan Control Input
ENN	0.005	0.005	0.009	Enable Input
CP	0.004	0.004	0.004	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SENQ1	SENQ2	SENQ4
D	0.019	0.019	0.029
SD	0.027	0.027	0.005
SC	0.048	0.048	0.054
CP	0.038	0.038	0.018
ENN	0.042	0.042	0.047
Q	0.06	0.075	0.133

Waveforms



Timing Numbers for SENQ1:

Error Checks

Pulse Width	CP (min)	0.201			
tH	CP -> D (min)	-0.123	tS	D -> CP (min)	0.261
tHSC	CP -> SC (min)	-0.237	tSSC	SC -> CP (min)	0.383
tHSD	CP -> SD (min)	-0.192	tSSD	SD -> CP (min)	0.361
tHEN	CP -> ENN (min)	-0.212	tSEN	ENN -> CP (min)	0.376

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.277	0.246	0.316	0.270	0.393	0.315	0.544	0.401	0.846	0.573

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.316	0.270	0.346	0.297	0.354	0.306	0.352	0.304	0.340	0.290

Timing Numbers for SENRQ2:

Error Checks

Pulse Width	CP (min)	0.201			
tH	CP -> D (min)	-0.123	tS	D -> CP (min)	0.261
tHSC	CP -> SC (min)	-0.239	tSSC	SC -> CP (min)	0.382
tHSD	CP -> SD (min)	-0.192	tSSD	SD -> CP (min)	0.361
tHEN	CP -> ENN (min)	-0.212	tSEN	ENN -> CP (min)	0.375

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.274	0.249	0.294	0.265	0.332	0.289	0.407	0.334	0.557	0.419

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.294	0.265	0.324	0.294	0.332	0.301	0.330	0.299	0.318	0.285

Timing Numbers for SENRQ4:**Error Checks**

Pulse Width	CP (min)	0.157			
tH	CP -> D (min)	-0.187	tS	D -> CP (min)	0.340
tHSC	CP -> SC (min)	-0.217	tSSC	SC -> CP (min)	0.303
tHSD	CP -> SD (min)	-0.137	tSSD	SD -> CP (min)	0.234
tHEN	CP -> ENN (min)	-0.199	tSEN	ENN -> CP (min)	0.371

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.300	0.294	0.312	0.308	0.334	0.331	0.370	0.371	0.437	0.445

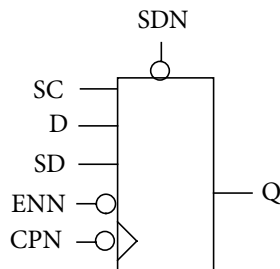
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.312	0.308	0.368	0.366	0.407	0.404	0.437	0.432	0.456	0.449

Muxed Scan Enable D Flip-Flops with Preset and Q only

SEPFQ1 , SEPFQ2 and SEPFQ4

The SEPFQ1 (1x drive), SEPFQ2 (2x drive) and SEPFQ4 (4x drive) cells are negative edge triggered multiplexed scan D flip-flops with active-low enable, ENN, and set, SDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q output on the falling edge of the clock, CPN.



Function Table

INPUTS						OUTPUT
SDN	CPN	ENN	SC	D	SD	Q
L	X	X	X	X	X	H
H	↓	L	L	L	X	L
H	↓	L	L	H	X	H
H	↓	X	H	X	L	L
H	↓	X	H	X	H	H
H	L	L	X	X	X	Q
H	H	L	X	X	X	Q
H	X	H	L	X	X	Q ₀

Cell Description

Macro Name:	SEPFQ1	SEPFQ2	SEPFQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	9.5	9.	9.75
Leakage Power (pW):	254.7	242.1	267.8

Pin Description

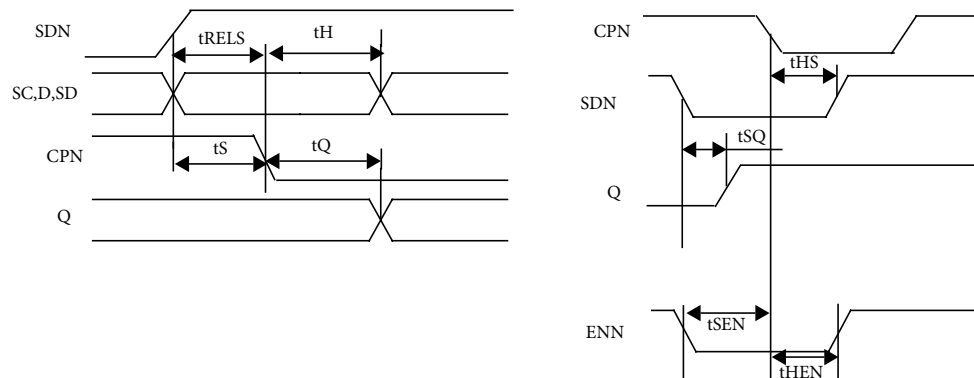
Name	Capacitance (pF)			Description
	SEPFQ1	SEPFQ2	SEPFQ4	
D	0.004	0.004	0.004	Data Input
SD	0.003	0.003	0.003	Scan Data Input
SC	0.005	0.005	0.005	Scan Control Input
ENN	0.008	0.008	0.008	Enable Input
CPN	0.005	0.005	0.005	Clock Input
SDN	0.009	0.009	0.009	Set Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SEPFQ1	SEPFQ2	SEPFQ4
D	0.029	0.029	0.029
SD	0.003	0.003	0.003
SC	0.07	0.07	0.07
ENN	0.018	0.019	0.019
CPN	0.057	0.054	0.054
SDN	-0	-0	-0
Q	0.106	0.118	0.155

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Waveforms



Timing Numbers for SEPFPQ1:

Error Checks

Pulse Width	CPN (min)	0.125	Pulse Width	SDN (min)	0.117
tH	CPN -> D (min)	-0.122	tS	D -> CPN (min)	0.348
tHSC	CPN -> SC (min)	-0.178	tSSC	SC -> CPN (min)	0.372
tHSD	CPN -> SD (min)	-0.092	tSSD	SD -> CPN (min)	0.318
tHS	CPN -> SDN (min)	0.028	tRelS	SDN -> CPN (min)	0.016
tHEN	CPN -> ENN (min)	-0.120	tSEN	ENN -> CPN (min)	0.373

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.306	0.279	0.341	0.319	0.407	0.396	0.539	0.546	0.802	0.847
tSQ	0.282		0.316		0.382		0.514		0.776	

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.341	0.319	0.406	0.385	0.459	0.439	0.504	0.484	0.537	0.518
tSQ	0.316		0.398		0.476		0.542		0.594	

Timing Numbers for SEPFQ2:

Error Checks

Pulse Width	CPN (min)	0.134	Pulse Width	SDN (min)	0.116
tH	CPN -> D (min)	-0.126	tS	D -> CPN (min)	0.338
tHSC	CPN -> SC (min)	-0.185	tSSC	SC -> CPN (min)	0.368
tHSD	CPN -> SD (min)	-0.096	tSSD	SD -> CPN (min)	0.308
tHS	CPN -> SDN (min)	0.030	tRelS	SDN -> CPN (min)	0.007
tHEN	CPN -> ENN (min)	-0.126	tSEN	ENN -> CPN (min)	0.362

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.301	0.275	0.321	0.300	0.357	0.340	0.424	0.415	0.556	0.562
tSQ	0.277		0.297		0.333		0.399		0.530	

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.321	0.300	0.386	0.364	0.438	0.417	0.481	0.462	0.514	0.495
tSQ	0.297		0.378		0.456		0.521		0.573	

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Timing Numbers for SEPFQ4:**Error Checks**

Pulse Width	CPN (min)	0.146	Pulse Width	SDN (min)	0.121
tH	CPN -> D (min)	-0.126	tS	D -> CPN (min)	0.336
tHSC	CPN -> SC (min)	-0.185	tSSC	SC -> CPN (min)	0.366
tHSD	CPN -> SD (min)	-0.096	tSSD	SD -> CPN (min)	0.305
tHS	CPN -> SDN (min)	0.029	tRelS	SDN -> CPN (min)	0.005
tHEN	CPN -> ENN (min)	-0.126	tSEN	ENN -> CPN (min)	0.358

Propagation Delays (ns) for sample loads:**Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.303	0.280	0.315	0.294	0.336	0.319	0.372	0.360	0.438	0.438
tSQ	0.279		0.291		0.312		0.347		0.413	

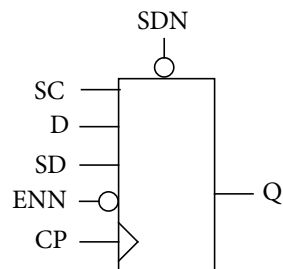
Propagation Delays (ns) for sample input transitions:**Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.315	0.294	0.380	0.359	0.432	0.413	0.476	0.457	0.508	0.491
tSQ	0.291		0.373		0.452		0.518		0.571	

Multiplexed Scan, D-Enabled Flip-Flops with Set and Q Output SEPRQ1 , SEPRQ2 and SEPRQ4

The SEPRQ1 (1x drive), SEPRQ2 (2x drive) and SEPRQ4 (4x drive) cells are positive edge triggered multiplexed scan D flip-flops with active-low enable, ENN, and set, SDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD, which is clocked to the Q output on the rising edge of the clock, CP.

Function Table



INPUTS						OUTPUT
SDN	CP	ENN	SC	D	SD	Q
L	X	X	X	X	X	H
H	↑	L	L	L	X	L
H	↑	L	L	H	X	H
H	↑	X	H	X	L	L
H	↑	X	H	X	H	H
H	L	L	X	X	X	Q
H	H	L	X	X	X	Q
H	X	H	L	X	X	Q _o

Cell Description

Macro Name:	SEPRQ1	SEPRQ2	SEPRQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	12.5	12.75	10.5
Leakage Power (pW):	283.5	294.5	284.0

Pin Description

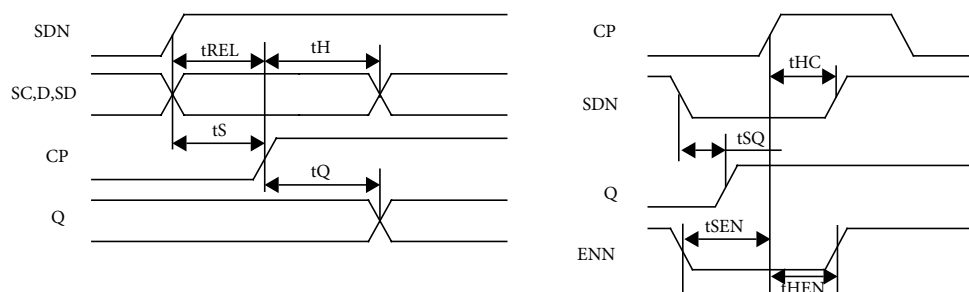
Name	Capacitance (pF)			Description
	SEPRQ1	SEPRQ2	SEPRQ4	
D	0.002	0.002	0.004	Data Input
SD	0.003	0.003	0.003	Scan Data Input
SC	0.002	0.002	0.005	Scan Control Input
ENN	0.005	0.005	0.009	Enable Input
CP	0.004	0.004	0.005	Clock Input
SDN	0.007	0.007	0.008	Set Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

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**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	SEPRQ1	SEPRQ2	SEPRQ4
D	0.02	0.02	0.029
SD	0.026	0.026	0.003
SC	0.054	0.054	0.07
ENN	0.042	0.042	0.017
CP	0.041	0.041	0.061
SDN	-0	-0	-0
Q	0.061	0.076	0.148

Waveforms



Timing Numbers for SEPRQ1:

Error Checks

Pulse Width	CP (min)	0.225	Pulse Width	SDN (min)	0.2
tH	CP -> D (min)	-0.143	tS	D -> CP (min)	0.279
tHSC	CP -> SC (min)	-0.251	tSSC	SC -> CP (min)	0.397
tHSD	CP -> SD (min)	-0.211	tSSD	SD -> CP (min)	0.381
tHS	CP -> SDN (min)	0.039	tRelC	SDN -> CP (min)	-0.019
tHEN	CP -> ENN (min)	-0.231	tSEN	ENN -> CP (min)	0.392

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.275	0.297	0.323	0.329	0.418	0.385	0.608	0.484	0.985	0.677
tSQ	0.354		0.402		0.497		0.686		1.063	

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.323	0.329	0.356	0.361	0.367	0.371	0.369	0.371	0.360	0.360
tSQ	0.402		0.496		0.604		0.700		0.780	

Timing Numbers for SEPRQ2:

Error Checks

Pulse Width	CP (min)	0.223	Pulse Width	SDN (min)	0.219
tH	CP -> D (min)	-0.144	tS	D -> CP (min)	0.278
tHSC	CP -> SC (min)	-0.253	tSSC	SC -> CP (min)	0.396
tHSD	CP -> SD (min)	-0.212	tSSD	SD -> CP (min)	0.379
tHS	CP -> SDN (min)	0.039	tRelC	SDN -> CP (min)	-0.021
tHEN	CP -> ENN (min)	-0.232	tSEN	ENN -> CP (min)	0.390

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.272	0.317	0.298	0.340	0.345	0.376	0.438	0.435	0.623	0.535
tSQ	0.357		0.383		0.430		0.522		0.707	

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.298	0.340	0.331	0.372	0.342	0.382	0.344	0.382	0.335	0.371
tSQ	0.383		0.477		0.587		0.685		0.766	

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Timing Numbers for SEPRQ4:**Error Checks**

Pulse Width	CP (min)	0.205	Pulse Width	SDN (min)	0.16
tH	CP -> D (min)	-0.166	tS	D -> CP (min)	0.333
tHSC	CP -> SC (min)	-0.217	tSSC	SC -> CP (min)	0.362
tHSD	CP -> SD (min)	-0.143	tSSD	SD -> CP (min)	0.302
tHS	CP -> SDN (min)	0.023	tRelS	SDN -> CP (min)	-0.001
tHEN	CP -> ENN (min)	-0.210	tSEN	ENN -> CP (min)	0.365

Propagation Delays (ns) for sample loads:**Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.288	0.307	0.300	0.321	0.320	0.346	0.356	0.387	0.422	0.465
tSQ	0.286		0.298		0.318		0.354		0.420	

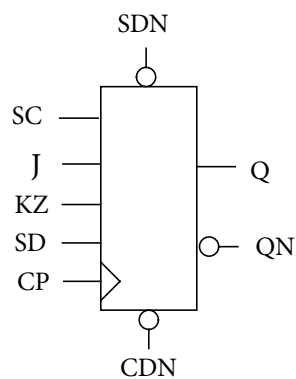
Propagation Delays (ns) for sample input transitions:**Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.300	0.321	0.355	0.376	0.394	0.413	0.424	0.441	0.442	0.456
tSQ	0.298		0.381		0.464		0.533		0.587	

Scan JK Flip-Flops with Clear, Preset SKBRB1 , SKBRB2 and SKBRB4

The SKBRB1 (1x drive), SKBRB2 (2x drive) and SKBRB4 (4x drive) cells are positive edge triggered multiplexed scan JK flip-flops with active-low clear, CDN, and set, SDN. The scan control input, SC, selects either the data input, JK, or the scan data input, SD, which is clocked to the Q and QN outputs on the rising edge of the clock, CP.

Function Table



INPUTS							OUTPUTS	
CDN	SDN	CP	SC	J	KZ	SD	Q	QN
L	H	X	X	X	X	X	L	H
H	L	X	X	X	X	X	H	L
H	H	↑	L	L	H	X	Q ₀	QN ₀
H	H	↑	L	L	L	X	L	H
H	H	↑	L	H	H	X	H	L
H	H	↑	L	H	L	X	QN ₀	Q ₀
H	H	↑	H	X	X	L	L	H
H	H	↑	H	X	X	H	H	L
H	H	L	X	X	X	X	Q	QN
H	H	H	X	X	X	X	Q	QN
L	L	X	X	X	X	X	L	L

Cell Description

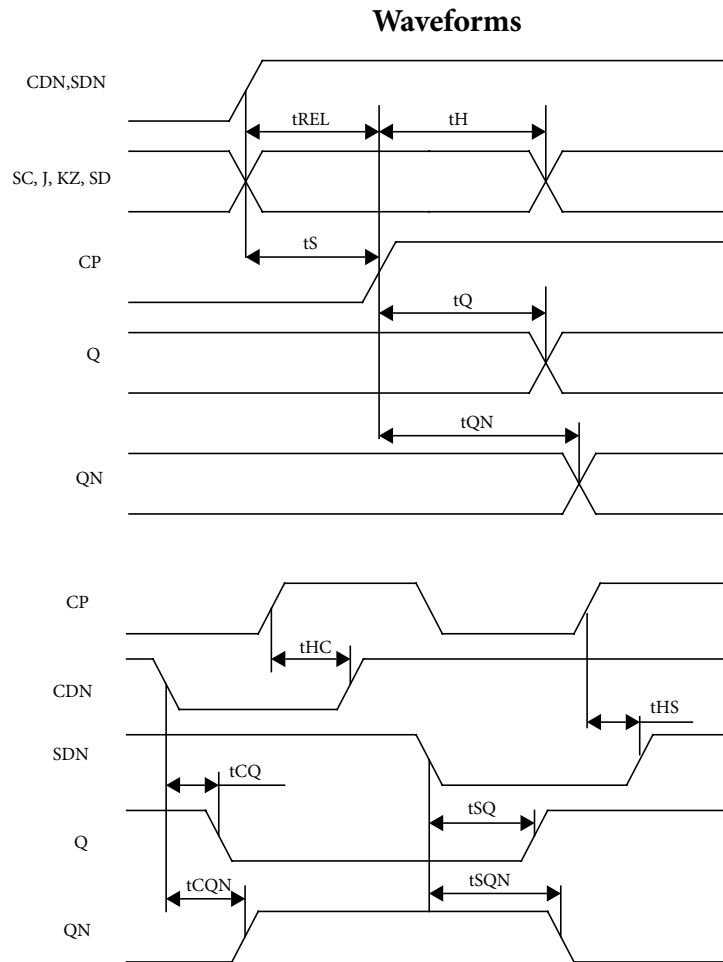
Macro Name:	SKBRB1	SKBRB2	SKBRB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	11.25	11.75	13.
Leakage Power (pW):	256.2	323.9	459.4

Pin Description

Name	Capacitance (pF)			Description
	SKBRB1	SKBRB2	SKBRB4	
J	0.003	0.003	0.003	Data Input
KZ	0.004	0.004	0.004	Data Input
SD	0.004	0.004	0.004	Scan Data Input
SC	0.002	0.002	0.002	Scan Control
CP	0.003	0.003	0.003	Clock Input
SDN	0.007	0.007	0.007	Preset Input
CDN	0.007	0.007	0.007	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF
Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SKBRB1	SKBRB2	SKBRB4
J	0.005	0.005	0.005
KZ	0.018	0.018	0.018
SD	0.025	0.025	0.025
SC	0.051	0.05	0.051
CP	0.051	0.051	0.051
SDN	0.008	0.008	0.008
CDN	0.001	0.001	0.001
Q	0.045	0.062	0.108
QN	0.066	0.084	0.141



Timing Numbers for SKBRB1:**Error Checks**

Pulse Width	CP (min)	0.248			
Pulse Width	CDN (min)	0.18	Pulse Width	SDN (min)	0.284
tH	CP -> J/KZ/SD/SC (min)	-0.180	tS	J/KZ/SD/SC -> CP (min)	0.286
tHS	CP -> SDN (min)	0.051	tRelS	SDN -> CP (min)	-0.029
tHC	CP -> CDN (min)	0.172	tRelC	CDN -> CP (min)	-0.136

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.412	0.477	0.447	0.517	0.514	0.595	0.647	0.746	0.910	1.046
tQN	0.341	0.345	0.384	0.399	0.461	0.490	0.598	0.649	0.863	0.951
tSQ/tCQ	0.457	0.178	0.493	0.220	0.559	0.299	0.691	0.452	0.954	0.753
tCQN/tSQN	0.421	0.397	0.464	0.455	0.540	0.549	0.677	0.707	0.941	1.009

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.447	0.517	0.511	0.583	0.560	0.632	0.596	0.668	0.616	0.687
tQN	0.384	0.399	0.449	0.463	0.498	0.512	0.534	0.548	0.553	0.568
tSQ/tCQ	0.493	0.220	0.585	0.304	0.697	0.385	0.801	0.453	0.882	0.510
tCQN/tSQN	0.464	0.455	0.544	0.548	0.622	0.661	0.690	0.768	0.745	0.852

Timing Numbers for SKBRB2:**Error Checks**

Pulse Width	CP (min)	0.251			
Pulse Width	CDN (min)	0.225	Pulse Width	SDN (min)	0.356
tH	CP -> J/KZ/SD/SC (min)	-0.182	tS	J/KZ/SD/SC -> CP (min)	0.278
tHS	CP -> SDN (min)	0.045	tRelS	SDN -> CP (min)	-0.026
tHC	CP -> CDN (min)	0.165	tRelC	CDN -> CP (min)	-0.136

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.476	0.535	0.495	0.557	0.530	0.597	0.597	0.671	0.730	0.813
tQN	0.341	0.352	0.371	0.386	0.418	0.442	0.498	0.535	0.639	0.691
tSQ/tCQ	0.529	0.177	0.547	0.204	0.582	0.246	0.648	0.322	0.778	0.467
tCQN/tSQN	0.426	0.400	0.455	0.440	0.502	0.500	0.582	0.598	0.722	0.756

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.495	0.557	0.560	0.623	0.608	0.673	0.644	0.708	0.665	0.728
tQN	0.371	0.386	0.436	0.451	0.485	0.500	0.521	0.535	0.541	0.556
tSQ/tCQ	0.547	0.204	0.640	0.290	0.754	0.379	0.862	0.454	0.948	0.514
tCQN/tSQN	0.455	0.440	0.536	0.533	0.614	0.648	0.681	0.758	0.736	0.845

Timing Numbers for SKBRB4:**Error Checks**

Pulse Width	CP (min)	0.257			
Pulse Width	CDN (min)	0.317	Pulse Width	SDN (min)	0.504
tH	CP -> J/KZ/SD/SC (min)	-0.188	tS	J/KZ/SD/SC -> CP (min)	0.272
tHS	CP -> SDN (min)	0.035	tRelS	SDN -> CP (min)	-0.022
tHC	CP -> CDN (min)	0.155	tRelC	CDN -> CP (min)	-0.136

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.635	0.700	0.647	0.714	0.666	0.738	0.699	0.777	0.764	0.850
tQN	0.387	0.413	0.406	0.435	0.439	0.473	0.491	0.532	0.580	0.631
tSQ/tCQ	0.704	0.217	0.715	0.234	0.733	0.263	0.764	0.310	0.828	0.391
tCQN/tSQN	0.534	0.469	0.550	0.493	0.578	0.534	0.622	0.599	0.702	0.704

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

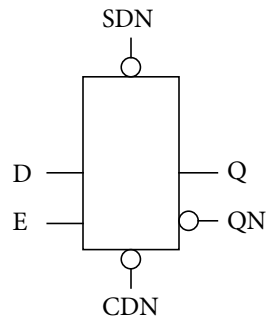
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.647	0.714	0.711	0.780	0.759	0.829	0.794	0.865	0.814	0.885
tQN	0.406	0.435	0.471	0.499	0.521	0.548	0.557	0.582	0.577	0.603
tSQ/tCQ	0.715	0.234	0.807	0.324	0.923	0.427	1.035	0.514	1.130	0.581
tCQN/tSQN	0.550	0.493	0.639	0.585	0.743	0.702	0.835	0.815	0.906	0.908

LATCHES

Latches with Active-High Enable, Set and Clear

LABHB1, LABHB2 and LABHB4

The LABHB1 (1x drive), LABHB2 (2x drive) and LABHB4 (4x drive) cells are active-high latches with active-low clear, CDN, and set, SDN. When E is High, the latch is transparent and data present at the D input is transferred to the Q and QN outputs. When E is Low, data is retained.



Function Table

INPUTS				OUTPUTS	
CDN	SDN	E	D	Q	QN
L	H	X	X	L	H
H	L	X	X	H	L
L	L	X	X	H	H
H	H	H	L	L	H
H	H	H	H	H	L
H	H	L	X	Q _o	QN _o

Cell Description

Macro Name:	LABHB1	LABHB2	LABHB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	6.25	6.75	7.25
Leakage Power (pW):	105.8	126.7	200.4

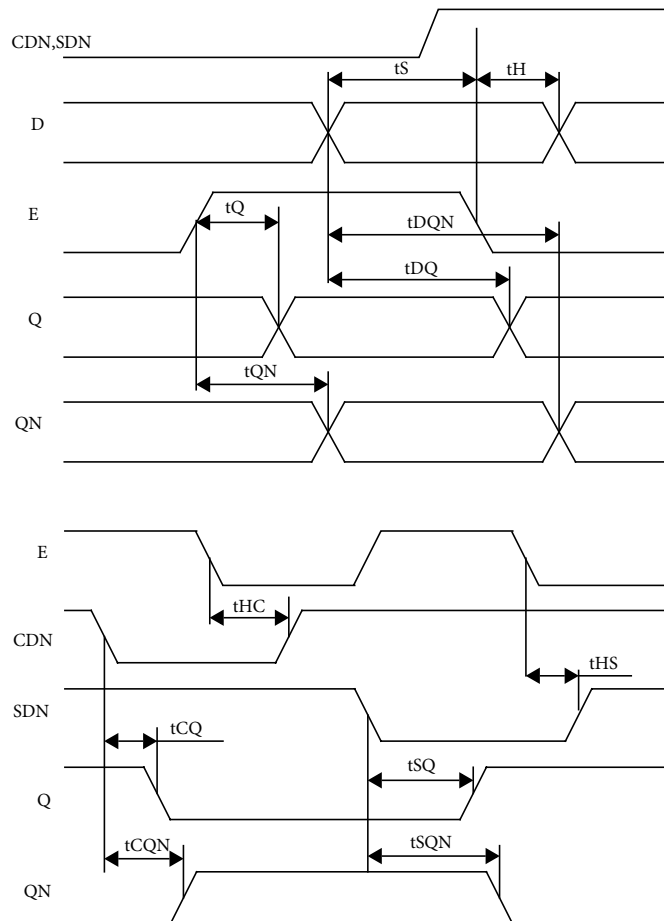
Pin Description

Name	Capacitance (pF)			Description
	LABHB1	LABHB2	LABHB4	
D	0.003	0.003	0.004	Data Input
E	0.002	0.002	0.004	Clock Input
SDN	0.004	0.004	0.005	Preset Input
CDN	0.007	0.007	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	LABHB1	LABHB2	LABHB4
D	0.007	0.007	0.01
E	0.022	0.022	0.04
SDN	0	0	-0
CDN	0.003	0.003	0.005
Q	0.039	0.052	0.097
QN	0.042	0.055	0.106

Waveforms



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Timing Numbers for LABHB1:**Error Checks**

Pulse Width	E (min)	0.145			
Pulse Width	CDN (min)	0.156	Pulse Width	SDN (min)	0.129
tH	E -> D (min)	-0.051	tS	D -> E (min)	0.139
tHC	E -> CDN (min)	-0.050	tRelC	CDN -> E (min)	0.062
tHS	E -> SDN (min)	0.046	tRelC	SDN -> E (min)	-0.036

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.344	0.376	0.393	0.407	0.489	0.462	0.681	0.567	1.064	0.775
tQN	0.432	0.436	0.480	0.466	0.575	0.521	0.764	0.627	1.141	0.835
tDQ	0.302	0.366	0.351	0.396	0.448	0.451	0.639	0.556	1.022	0.764
tDQN	0.421	0.394	0.469	0.424	0.564	0.479	0.753	0.585	1.130	0.794
tSQ/tCQ	0.204	0.369	0.253	0.399	0.349	0.454	0.541	0.558	0.924	0.766
tCQN/tSQN	0.210	0.335	0.258	0.366	0.353	0.421	0.543	0.527	0.920	0.735

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.393	0.407	0.427	0.444	0.438	0.458	0.439	0.462	0.427	0.454
tQN	0.480	0.466	0.517	0.501	0.531	0.511	0.535	0.513	0.527	0.501
tDQ	0.351	0.396	0.385	0.482	0.408	0.566	0.421	0.639	0.421	0.698
tDQN	0.469	0.424	0.555	0.459	0.639	0.481	0.711	0.495	0.770	0.495
tSQ/tCQ	0.253	0.399	0.338	0.484	0.421	0.568	0.494	0.641	0.556	0.700
tCQN/tSQN	0.258	0.366	0.342	0.452	0.421	0.538	0.489	0.615	0.544	0.680

Timing Numbers for LABHB2:**Error Checks**

Pulse Width	E (min)	0.147			
Pulse Width	CDN (min)	0.157	Pulse Width	SDN (min)	0.13
tH	E -> D (min)	-0.052	tS	D -> E (min)	0.141
tHC	E -> CDN (min)	-0.051	tRelC	CDN -> E (min)	0.068
tHS	E -> SDN (min)	0.045	tRelC	SDN -> E (min)	-0.034

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.335	0.402	0.360	0.424	0.408	0.459	0.503	0.519	0.692	0.628
tQN	0.422	0.459	0.447	0.481	0.494	0.516	0.589	0.576	0.779	0.685
tDQ	0.293	0.391	0.318	0.413	0.366	0.448	0.461	0.508	0.650	0.617
tDQN	0.411	0.417	0.436	0.439	0.484	0.475	0.579	0.534	0.768	0.643
tSQ/tCQ	0.195	0.391	0.219	0.413	0.267	0.448	0.362	0.508	0.551	0.617
tCQN/tSQN	0.201	0.359	0.226	0.382	0.273	0.417	0.368	0.477	0.558	0.585

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.360	0.424	0.395	0.461	0.405	0.475	0.406	0.479	0.395	0.471
tQN	0.447	0.481	0.484	0.516	0.498	0.527	0.502	0.528	0.494	0.517
tDQ	0.318	0.413	0.353	0.500	0.375	0.584	0.389	0.656	0.389	0.716
tDQN	0.436	0.439	0.522	0.475	0.606	0.497	0.678	0.510	0.737	0.510
tSQ/tCQ	0.219	0.413	0.307	0.499	0.392	0.583	0.468	0.655	0.532	0.714
tCQN/tSQN	0.226	0.382	0.311	0.469	0.393	0.554	0.463	0.630	0.520	0.696

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Timing Numbers for LABHB4:**Error Checks**

Pulse Width	E (min)	0.134			
Pulse Width	CDN (min)	0.155	Pulse Width	SDN (min)	0.117
tH	E -> D (min)	-0.113	tS	D -> E (min)	0.135
tHC	E -> CDN (min)	-0.119	tRelC	CDN -> E (min)	0.133
tHS	E -> SDN (min)	0.027	tRelC	SDN -> E (min)	-0.017

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.371	0.338	0.383	0.352	0.404	0.377	0.440	0.417	0.506	0.491
tQN	0.401	0.423	0.413	0.436	0.434	0.458	0.469	0.496	0.535	0.569
tDQ	0.351	0.319	0.363	0.333	0.384	0.357	0.420	0.397	0.486	0.471
tDQN	0.382	0.403	0.394	0.415	0.414	0.437	0.449	0.476	0.515	0.549
tSQ/tCQ	0.206	0.351	0.218	0.365	0.239	0.389	0.275	0.429	0.341	0.503
tCQN/tSQN	0.238	0.306	0.250	0.319	0.271	0.341	0.306	0.379	0.372	0.452

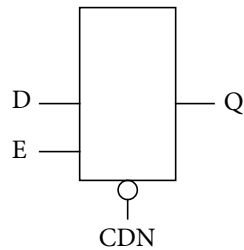
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.383	0.352	0.433	0.402	0.465	0.436	0.488	0.462	0.502	0.477
tQN	0.413	0.436	0.463	0.485	0.497	0.517	0.523	0.541	0.538	0.554
tDQ	0.363	0.333	0.409	0.403	0.452	0.462	0.488	0.511	0.513	0.546
tDQN	0.394	0.415	0.464	0.461	0.523	0.504	0.572	0.540	0.607	0.564
tSQ/tCQ	0.218	0.365	0.295	0.446	0.364	0.523	0.424	0.588	0.473	0.639
tCQN/tSQN	0.250	0.319	0.333	0.392	0.413	0.456	0.480	0.514	0.533	0.561

Latches with Active-High Enable, Clear and Q Output Only

LACHQ1, LACHQ2 and LACHQ4

The LACHQ1 (1x drive), LACHQ2 (2x drive) and LACHQ4 (4x drive) cells are active-high latches with active-low clear, CDN. When E is High, the latch is transparent and data present at the D input is transferred to the Q output. When E is Low, data is retained.



Function Table

INPUTS			OUTPUT
CDN	E	D	Q
L	X	X	L
H	H	L	L
H	H	H	H
H	L	X	Q ₀

Cell Description

Macro Name:	LACHQ1	LACHQ2	LACHQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	4.25	4.5	5.
Leakage Power (pW):	88.6	114.3	230.8

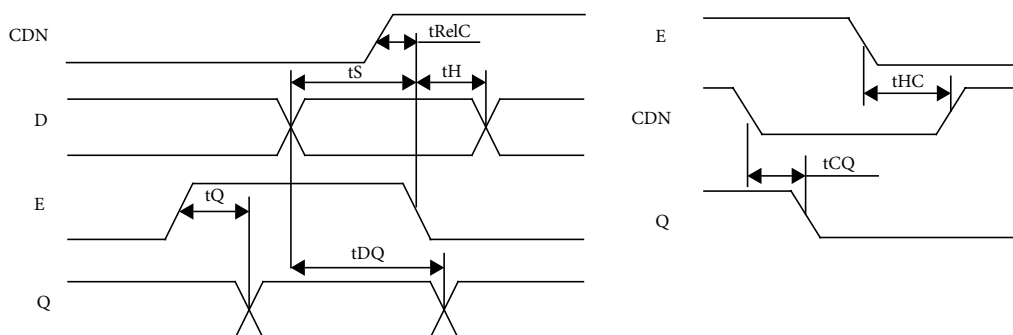
Pin Description

Name	Capacitance (pF)			Description
	LACHQ1	LACHQ2	LACHQ4	
D	0.005	0.005	0.004	Data Input
E	0.003	0.003	0.004	Clock Input
CDN	0.007	0.007	0.008	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	LACHQ1	LACHQ2	LACHQ4
D	0.002	0.002	0.003
E	0.029	0.029	0.041
CDN	0.004	0.004	0.004
Q	0.058	0.073	0.144

Waveforms



Timing Numbers for LACHQ1

Error Checks

Pulse Width	E (min)	0.099	Pulse Width	CDN (min)	0.144
tH	E -> D (min)	0.023	tS	D -> E (min)	0.110
tHC	E -> CDN (min)	0.022	tRelC	CDN -> E (min)	-0.019

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.227	0.210	0.276	0.248	0.373	0.311	0.564	0.422	0.945	0.636
tDQ	0.175	0.220	0.224	0.258	0.321	0.323	0.512	0.435	0.893	0.649
tCQ		0.215		0.255		0.319		0.430		0.643

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.276	0.248	0.293	0.264	0.284	0.256	0.265	0.238	0.236	0.209
tDQ	0.224	0.258	0.250	0.348	0.259	0.442	0.260	0.524	0.247	0.593
tCQ		0.255		0.340		0.425		0.498		0.558

Timing Numbers for LACHQ2:

Error Checks

Pulse Width	E (min)	0.128	Pulse Width	CDN (min)	0.175
tH	E -> D (min)	0.012	tS	D -> E (min)	0.136
tHC	E -> CDN (min)	0.011	tRelC	CDN -> E (min)	-0.003

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.215	0.211	0.244	0.240	0.294	0.281	0.391	0.349	0.582	0.464
tDQ	0.163	0.217	0.192	0.246	0.243	0.287	0.340	0.356	0.531	0.471
tCQ		0.214		0.245		0.288		0.358		0.472

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.244	0.240	0.261	0.257	0.252	0.248	0.233	0.231	0.203	0.202
tDQ	0.192	0.246	0.220	0.335	0.231	0.431	0.234	0.515	0.224	0.585
tCQ		0.245		0.332		0.422		0.499		0.561

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Timing Numbers for LACHQ4:**Error Checks**

Pulse Width	E (min)	0.212	Pulse Width	CDN (min)	0.201
tH	E -> D (min)	-0.152	tS	D -> E (min)	0.190
tHC	E -> CDN (min)	-0.152	tRelC	CDN -> E (min)	0.179

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

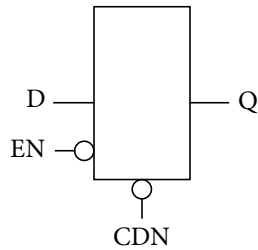
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.284	0.263	0.303	0.282	0.335	0.316	0.385	0.368	0.469	0.456
tDQ	0.267	0.236	0.286	0.255	0.317	0.289	0.367	0.341	0.452	0.429
tCQ		0.240		0.260		0.293		0.343		0.427

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.303	0.282	0.354	0.334	0.388	0.369	0.413	0.395	0.428	0.412
tDQ	0.286	0.255	0.334	0.335	0.386	0.412	0.430	0.476	0.461	0.524
tCQ		0.260		0.342		0.425		0.497		0.553

Latches with Active-Low Enable, Clear and Q Output Only LACLQ1, LACLQ2 and LACLQ4

The LACLQ1 (1x drive), LACLQ2 (2x drive) and LACLQ4 (4x drive) cells are active-low latches with active-low clear, CDN. When EN is Low, the latch is transparent and data present at the D input is transferred to the Q output. When EN is High, data is retained.



Function Table

INPUTS			OUTPUT
CDN	EN	D	Q
L	X	X	L
H	L	L	L
H	L	H	H
H	H	X	Q _o

Cell Description

Macro Name:	LACLQ1	LACLQ2	LACLQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	4.25	4.5	5.
Leakage Power (pW):	77.4	98.1	229.1

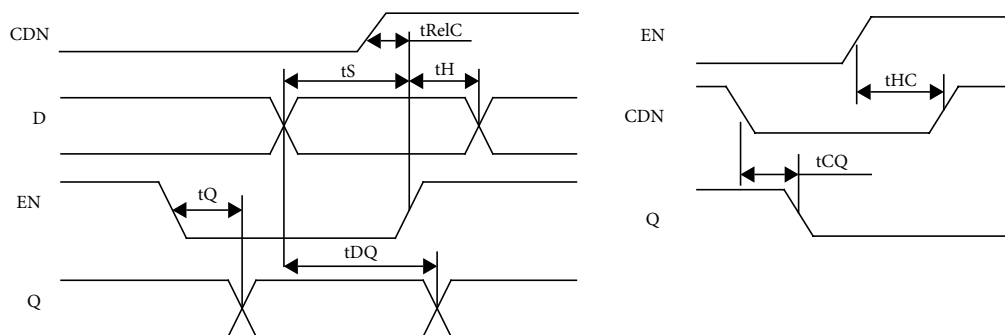
Pin Description

Name	Capacitance (pF)			Description
	LACLQ1	LACLQ2	LACLQ4	
D	0.004	0.003	0.004	Data Input
EN	0.002	0.002	0.004	Clock Input
CDN	0.006	0.006	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	LACLQ1	LACLQ2	LACLQ4
D	0.006	0.006	0.007
EN	0.021	0.022	0.037
CDN	0.005	0.005	0.007
Q	0.051	0.071	0.144

Waveforms



Timing Numbers for LACLQ1:

Error Checks

Pulse Width	EN (min)	0.145	Pulse Width	CDN (min)	0.123
tH	EN -> D (min)	-0.089	tS	D -> EN (min)	0.131
tHC	EN -> CDN (min)	-0.103	tRelC	CDN -> EN (min)	0.128

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.255	0.258	0.304	0.295	0.398	0.355	0.581	0.462	0.947	0.664
tDQ	0.204	0.203	0.253	0.240	0.347	0.301	0.530	0.408	0.895	0.610
tCQ		0.201		0.239		0.300		0.405		0.606

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.304	0.295	0.389	0.379	0.471	0.459	0.542	0.529	0.603	0.589
tDQ	0.253	0.240	0.290	0.322	0.318	0.398	0.337	0.464	0.344	0.518
tCQ		0.239		0.320		0.397		0.464		0.517

Timing Numbers for LACLQ2:

Error Checks

Pulse Width	EN (min)	0.164	Pulse Width	CDN (min)	0.138
tH	EN -> D (min)	-0.120	tS	D -> EN (min)	0.165
tHC	EN -> CDN (min)	-0.128	tRelC	CDN -> EN (min)	0.164

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.262	0.269	0.294	0.299	0.348	0.341	0.447	0.410	0.637	0.526
tDQ	0.210	0.222	0.242	0.253	0.295	0.295	0.394	0.365	0.584	0.482
tCQ		0.219		0.250		0.294		0.366		0.484

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.294	0.299	0.381	0.384	0.465	0.466	0.538	0.539	0.601	0.600
tDQ	0.242	0.253	0.283	0.341	0.317	0.432	0.339	0.510	0.349	0.572
tCQ		0.250		0.338		0.431		0.509		0.571

Timing Numbers for LACLQ4:**Error Checks**

Pulse Width	EN (min)	0.186	Pulse Width	CDN (min)	0.19
tH	EN -> D (min)	-0.128	tS	D -> EN (min)	0.227
tHC	EN -> CDN (min)	-0.195	tRelC	CDN -> EN (min)	0.227

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.269	0.268	0.288	0.287	0.320	0.321	0.369	0.372	0.453	0.459
tDQ	0.265	0.232	0.284	0.251	0.315	0.284	0.364	0.336	0.448	0.423
tCQ		0.232		0.251		0.284		0.333		0.415

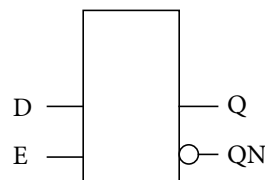
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.288	0.287	0.352	0.350	0.403	0.400	0.448	0.442	0.482	0.474
tDQ	0.284	0.251	0.332	0.329	0.385	0.404	0.429	0.467	0.460	0.512
tCQ		0.251		0.330		0.409		0.476		0.529

Latches with Active-High Enable

LANHB1, LANHB2 and LANHB4

The LANHB1 (1x drive), LANHB2 (2x drive) and LANHB4 (4x drive) cells are active-high latches. When E is High, the latch is transparent and data present at the D input is transferred to the Q and QN outputs. When E is Low, data is retained.



Function Table

INPUTS		OUTPUTS	
E	D	Q	QN
H	L	L	H
H	H	H	L
L	X	Qo	QNo

Cell Description

Macro Name:	LANHB1	LANHB2	LANHB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	4.	4.5	5.
Leakage Power (pW):	120.3	151.3	263.4

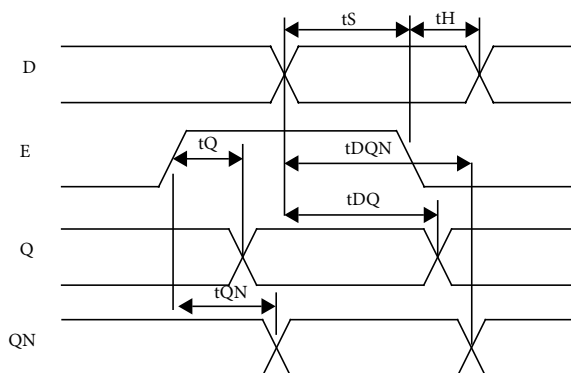
Pin Description

Name	Capacitance (pF)			Description
	LANHB1	LANHB2	LANHB4	
D	0.004	0.004	0.003	Data Input
E	0.003	0.003	0.004	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	.0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	LANHB1	LANHB2	LANHB4
D	0.01	0.01	0.008
E	0.031	0.03	0.033
Q	0.032	0.046	0.102
QN	0.032	0.046	0.102

Waveform



Timing Numbers for LANHB1:

Error Checks

Pulse Width	E (min)	0.142			
tH	E -> D (min)	0.036	tS	D -> E (min)	0.107

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.224	0.245	0.273	0.285	0.368	0.352	0.558	0.466	0.935	0.679
tQN	0.309	0.253	0.356	0.281	0.451	0.335	0.639	0.441	1.016	0.652
tDQ	0.150	0.206	0.199	0.247	0.295	0.314	0.484	0.428	0.861	0.641
tDQN	0.270	0.180	0.317	0.208	0.412	0.262	0.600	0.367	0.977	0.578

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.273	0.285	0.321	0.336	0.344	0.362	0.356	0.377	0.354	0.378
tQN	0.356	0.281	0.406	0.329	0.433	0.353	0.448	0.364	0.449	0.362
tDQ	0.199	0.247	0.238	0.333	0.255	0.419	0.263	0.494	0.257	0.558
tDQN	0.317	0.208	0.403	0.246	0.488	0.262	0.562	0.268	0.625	0.260

Timing Numbers for LANHB2:

Error Checks

Pulse Width	E (min)	0.182			
tH	E ->D (min)	0.027	tS	D ->E (min)	0.151

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.208	0.236	0.235	0.267	0.284	0.311	0.380	0.384	0.570	0.502
tQN	0.335	0.268	0.358	0.286	0.403	0.315	0.496	0.370	0.684	0.476
tDQ	0.135	0.199	0.162	0.230	0.211	0.274	0.307	0.347	0.497	0.466
tDQN	0.299	0.195	0.321	0.213	0.366	0.242	0.460	0.297	0.647	0.403

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.235	0.267	0.283	0.316	0.307	0.342	0.318	0.357	0.316	0.357
tQN	0.358	0.286	0.407	0.334	0.433	0.357	0.448	0.369	0.449	0.367
tDQ	0.162	0.230	0.204	0.318	0.226	0.408	0.236	0.486	0.232	0.552
tDQN	0.321	0.213	0.410	0.254	0.501	0.277	0.582	0.287	0.649	0.282

Timing Numbers for LANHB4:**Error Checks**

Pulse Width	E (min)	0.339			
tH	E -> D (min)	-0.130	tS	D -> E (min)	0.314

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.275	0.326	0.294	0.347	0.326	0.384	0.375	0.441	0.460	0.537
tQN	0.487	0.455	0.496	0.465	0.513	0.483	0.543	0.518	0.604	0.589
tDQ	0.248	0.298	0.267	0.319	0.298	0.356	0.348	0.413	0.432	0.510
tDQN	0.460	0.427	0.469	0.438	0.485	0.456	0.516	0.491	0.576	0.561

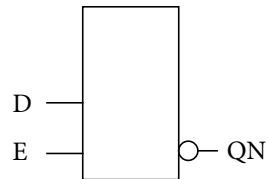
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.294	0.347	0.346	0.397	0.379	0.429	0.403	0.452	0.416	0.466
tQN	0.496	0.465	0.546	0.517	0.579	0.550	0.602	0.574	0.615	0.587
tDQ	0.267	0.319	0.339	0.407	0.408	0.505	0.461	0.593	0.493	0.660
tDQN	0.469	0.438	0.556	0.511	0.656	0.582	0.748	0.642	0.818	0.679

Latches with Active-High Enable and QN Output Only

LANHN1, LANHN2 and LANHN4

The LANHN1 (1x drive), LANHN2 (2x drive) and LANHN4 (4x drive) cells are active-high latches. When E is High, the latch is transparent and data present at the D input is transferred to the QN output. When E is Low, data is retained.



Function Table

INPUTS		OUTPUT
E	D	QN
H	H	L
H	L	H
L	X	QN _o

Cell Description

Macro Name:	LANHN1	LANHN2	LANHN4
Drive Capability:	1x	2x	4x
Gate Equivalents:	4.	4.25	4.5
Leakage Power (pW):	107.9	129.7	284.3

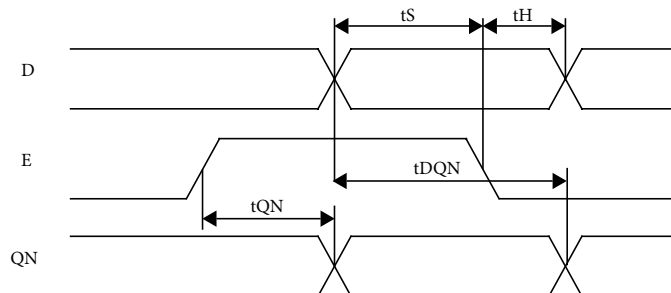
Pin Description

Name	Capacitance (pF)			Description
	LANHN1	LANHN2	LANHN4	
D	0.002	0.002	0.003	Data Input
E	0.002	0.002	0.003	Clock Input
Maximum capacitance				
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	LANHN1	LANHN2	LANHN4
D	0.014	0.014	0.018
E	0.021	0.021	0.033
QN	0.039	0.055	0.139

Waveforms



Timing Numbers for LANHN1:

Error Checks

Pulse Width	E (min)	0.09			
tH	E ->D (min)	-0.050	tS	D ->E (min)	0.113

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.258	0.272	0.307	0.313	0.403	0.382	0.592	0.497	0.970	0.710
tDQN	0.249	0.259	0.298	0.301	0.394	0.370	0.584	0.485	0.961	0.698

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.307	0.313	0.342	0.349	0.351	0.359	0.350	0.359	0.335	0.347
tDQN	0.298	0.301	0.381	0.320	0.459	0.318	0.529	0.308	0.591	0.288

Timing Numbers for LANHN2:

Error Checks

Pulse Width	E (min)	0.109			
tH	E ->D (min)	-0.067	tS	D ->E (min)	0.150

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.251	0.278	0.281	0.311	0.332	0.358	0.429	0.435	0.618	0.557
tDQN	0.244	0.266	0.274	0.299	0.325	0.346	0.422	0.423	0.611	0.545

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.281	0.311	0.316	0.347	0.326	0.357	0.324	0.357	0.310	0.344
tDQN	0.274	0.299	0.357	0.319	0.435	0.317	0.506	0.307	0.568	0.287

Timing Numbers for LANHN4:**Error Checks**

Pulse Width	E (min)	0.202			
tH	E -> D (min)	-0.174	tS	D -> E (min)	0.278

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.289	0.351	0.308	0.373	0.340	0.411	0.390	0.470	0.474	0.568
tDQN	0.303	0.350	0.322	0.371	0.354	0.409	0.405	0.468	0.489	0.566

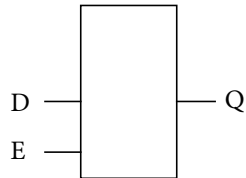
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.308	0.373	0.364	0.426	0.401	0.464	0.429	0.490	0.444	0.506
tDQN	0.322	0.371	0.385	0.408	0.437	0.429	0.483	0.442	0.520	0.447

Latches with Active-High Enable and Q Output Only

LANHQ1, LANHQ2 and LANHQ4

The LANHQ1 (1x drive), LANHQ2 (2x drive) and LANHQ4 (4x drive) cells are active-high latches. When E is High, the latch is transparent and data present at the D input is transferred to the Q output. When E is Low, data is retained.



Function Table

INPUTS		OUTPUT
E	D	Q
H	H	H
H	L	L
L	X	Q _o

Cell Description

Macro Name:	LANHQ1	LANHQ2	LANHQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	3.5	3.75	4.
Leakage Power (pW):	90.7	101.1	163.2

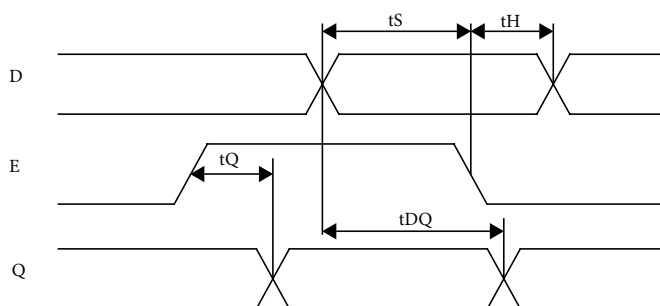
Pin Description

Name	Capacitance (pF)			Description
	LANHQ1	LANHQ2	LANHQ4	
D	0.003	0.003	0.003	Data Input
E	0.002	0.002	0.004	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	LANHQ1	LANHQ2	LANHQ4
D	0.007	0.007	0.008
E	0.022	0.022	0.034
Q	0.042	0.058	0.135

Waveforms



Timing Numbers for LANHQ1:

Error Checks

Pulse Width	E (min)	0.123			
tH	E ->D (min)	0.030	tS	D ->E (min)	0.095

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.230	0.259	0.277	0.299	0.372	0.366	0.559	0.481	0.934	0.693
tDQ	0.153	0.230	0.201	0.270	0.296	0.338	0.483	0.453	0.857	0.665

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.277	0.299	0.315	0.339	0.329	0.356	0.332	0.362	0.323	0.356
tDQ	0.201	0.270	0.237	0.363	0.251	0.461	0.254	0.548	0.243	0.621

Timing Numbers for LANHQ2:

Error Checks

Pulse Width	E (min)	0.146			
tH	E ->D (min)	0.017	tS	D ->E (min)	0.122

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.219	0.261	0.247	0.294	0.296	0.340	0.393	0.415	0.584	0.537
tDQ	0.143	0.235	0.171	0.268	0.220	0.314	0.317	0.389	0.508	0.511

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.247	0.294	0.285	0.333	0.299	0.349	0.303	0.356	0.294	0.350
tDQ	0.171	0.268	0.213	0.361	0.234	0.465	0.242	0.556	0.234	0.632

Timing Numbers for LANHQ4:**Error Checks**

Pulse Width	E (min)	0.268			
tH	E -> D (min)	-0.125	tS	D -> E (min)	0.243

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.275	0.333	0.294	0.354	0.325	0.391	0.373	0.448	0.454	0.545
tDQ	0.246	0.304	0.265	0.325	0.296	0.362	0.344	0.419	0.425	0.515

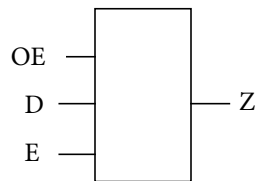
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.294	0.354	0.348	0.406	0.383	0.440	0.408	0.465	0.421	0.479
tDQ	0.265	0.325	0.337	0.413	0.406	0.512	0.459	0.601	0.491	0.668

Minimum Tristate Latches with Z-output only and Active Higher Enable

LANHT1, LANHT2 and LANHT4

The LANHT1 (1x drive), LANHT2 (2x drive) and LANHT4 (4x drive) cells are active-high latches. When E is High, the latch is transparent and data present at the D input is transferred to the Z output. When E is Low, data is retained.



Function Table

INPUTS			OUTPUT
E	OE	D	Z
H	H	L	L
H	H	H	H
L	H	X	Qo
X	L	X	HiZ

Cell Description

Macro Name:	LANHT1	LANHT2	LANHT4
Drive Capability:	1x	2x	4x
Gate Equivalents:	3.75	4.5	5.5
Leakage Power (pW):	78.2	84.8	167.4

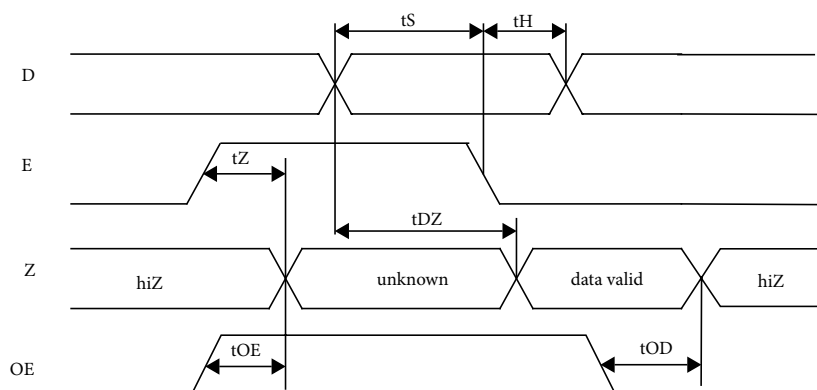
Pin Description

Name	Capacitance (pF)			Description
	LANHT1	LANHT2	LANHT4	
D	0.003	0.003	0.003	Data Input
E	0.002	0.002	0.003	Clock Input
OE	0.004	0.007	0.008	Enable Input
Z	0.005	0.007	0.011	3-State Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	LANHT1	LANHT2	LANHT4
D	0.016	0.02	0.025
E	0.025	0.028	0.041
Z	0.03	0.061	0.122

Waveforms



Timing Numbers for LANHT1:

Error Checks

Pulse Width	E (min)	0.13			
tH	E ->D (min)	0.027	tS	D ->E (min)	0.111

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tZ	0.298	0.263	0.393	0.316	0.583	0.408	0.960	0.578	1.711	0.909
tDZ	0.230	0.241	0.326	0.293	0.516	0.385	0.892	0.555	1.644	0.886
tOE	0.178	0.072	0.272	0.116	0.461	0.201	0.836	0.368	1.586	0.698
tOD	0.018	0.106	0.018	0.106	0.018	0.106	0.018	0.106	0.018	0.106

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tZ	0.393	0.316	0.431	0.353	0.443	0.367	0.446	0.371	0.437	0.363
tDZ	0.326	0.293	0.371	0.382	0.396	0.472	0.410	0.550	0.411	0.615
tOE	0.272	0.116	0.289	0.147	0.284	0.156	0.272	0.150	0.252	0.121
tOD	0.018	0.106	0.060	0.179	0.112	0.244	0.166	0.303	0.219	0.355

Timing Numbers for LANHT2:

Error Checks

Pulse Width	E (min)	0.168			
tH	E ->D (min)	0.008	tS	D ->E (min)	0.149

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tZ	0.304	0.285	0.353	0.315	0.448	0.366	0.637	0.457	1.013	0.627
tDZ	0.238	0.261	0.287	0.291	0.382	0.342	0.570	0.433	0.947	0.604
tOE	0.120	0.047	0.167	0.070	0.261	0.112	0.449	0.195	0.824	0.360
tOD	0.011	0.079	0.011	0.079	0.011	0.079	0.011	0.079	0.011	0.079

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tZ	0.353	0.315	0.391	0.352	0.403	0.366	0.407	0.370	0.398	0.362
tDZ	0.287	0.291	0.337	0.382	0.369	0.479	0.389	0.562	0.393	0.631
tOE	0.167	0.070	0.186	0.105	0.186	0.099	0.179	0.078	0.163	0.031
tOD	0.011	0.079	0.043	0.145	0.085	0.205	0.129	0.259	0.174	0.308

Timing Numbers for LANHT4:**Error Checks**

Pulse Width	E (min)	0.135			
tH	E -> D (min)	-0.006	tS	D -> E (min)	0.106

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tZ	0.396	0.392	0.413	0.409	0.441	0.438	0.481	0.482	0.552	0.560
tDZ	0.350	0.359	0.367	0.376	0.394	0.405	0.434	0.449	0.506	0.526
tOE	0.128	0.113	0.144	0.129	0.170	0.156	0.210	0.198	0.280	0.275
tOD	0.098	0.092	0.098	0.092	0.099	0.092	0.099	0.092	0.099	0.093

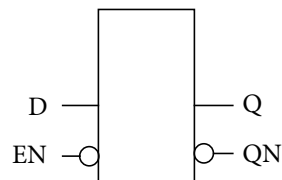
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tZ	0.413	0.409	0.469	0.466	0.505	0.503	0.530	0.531	0.543	0.547
tDZ	0.367	0.376	0.430	0.457	0.474	0.532	0.507	0.595	0.524	0.645
tOE	0.144	0.129	0.201	0.179	0.244	0.206	0.277	0.223	0.297	0.232
tOD	0.098	0.092	0.152	0.160	0.194	0.209	0.228	0.248	0.255	0.275

Latches with Active-Low Enable

LANLB1, LANLB2 and LANLB4

The LANLB1 (1x drive), LANLB2 (2x drive) and LANLB4 (4x drive) cells are active-low latches. When EN is Low, the latch is transparent and data present at the D input is transferred to the Q and QN outputs. When EN is High, data is retained.



Function Table

INPUTS		OUTPUTS	
EN	D	Q	QN
L	L	L	H
L	H	H	L
H	X	Q _o	QN _o

Cell Description

Macro Name:	LANLB1	LANLB2	LANLB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	3.75	4.5	5.5
Leakage Power (pW):	107.1	143.4	261.3

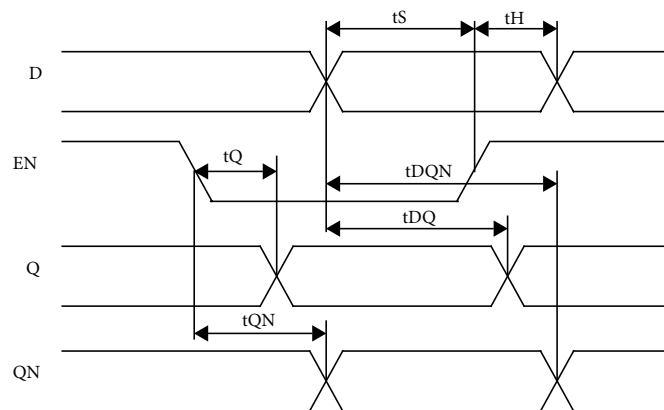
Pin Description

Name	Capacitance (pF)			Description
	LANLB1	LANLB2	LANLB4	
D	0.003	0.003	0.003	Data Input
EN	0.003	0.003	0.003	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	LANLB1	LANLB2	LANLB4
D	0.008	0.007	0.008
EN	0.025	0.025	0.036
Q	0.031	0.046	0.106
QN	0.031	0.046	0.106

Waveforms



Timing Numbers for LANLB1:

Error Checks

Pulse Width	EN (min)	0.224			
tH	EN -> D (min)	-0.066	tS	D -> EN (min)	0.141

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.277	0.317	0.326	0.359	0.423	0.429	0.614	0.544	0.994	0.760
tQN	0.387	0.360	0.434	0.392	0.529	0.450	0.720	0.557	1.100	0.768
tDQ	0.160	0.219	0.209	0.261	0.306	0.331	0.497	0.447	0.877	0.662
tDQN	0.289	0.243	0.336	0.275	0.431	0.332	0.622	0.439	1.001	0.650

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**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.326	0.359	0.418	0.455	0.521	0.555	0.610	0.643	0.688	0.720
tQN	0.434	0.392	0.530	0.485	0.630	0.588	0.718	0.677	0.795	0.754
tDQ	0.209	0.261	0.238	0.351	0.247	0.442	0.248	0.523	0.235	0.592
tDQN	0.336	0.275	0.425	0.304	0.514	0.314	0.593	0.315	0.661	0.303

Timing Numbers for LANLB2:

Error Checks

Pulse Width	EN (min)	0.279			
tH	EN ->D (min)	-0.082	tS	D ->EN (min)	0.198

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.272	0.318	0.301	0.350	0.352	0.397	0.449	0.474	0.640	0.596
tQN	0.431	0.406	0.453	0.428	0.497	0.463	0.590	0.522	0.779	0.631
tDQ	0.152	0.223	0.181	0.255	0.232	0.302	0.329	0.379	0.520	0.502
tDQN	0.336	0.286	0.358	0.308	0.402	0.342	0.496	0.402	0.685	0.511

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.301	0.350	0.394	0.445	0.497	0.546	0.588	0.635	0.666	0.713
tQN	0.453	0.428	0.548	0.522	0.649	0.624	0.738	0.715	0.815	0.793
tDQ	0.181	0.255	0.214	0.345	0.229	0.441	0.233	0.525	0.224	0.597
tDQN	0.358	0.308	0.448	0.341	0.544	0.358	0.629	0.365	0.701	0.358

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Timing Numbers for LANLB4:**Error Checks**

Pulse Width	EN (min)	0.328			
tH	EN -> D (min)	-0.171	tS	D -> EN (min)	0.285

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.287	0.347	0.306	0.368	0.337	0.404	0.386	0.460	0.471	0.555
tQN	0.510	0.464	0.520	0.475	0.537	0.494	0.568	0.529	0.631	0.598
tDQ	0.247	0.297	0.266	0.318	0.297	0.353	0.346	0.409	0.431	0.504
tDQN	0.459	0.424	0.469	0.435	0.486	0.454	0.518	0.489	0.581	0.559

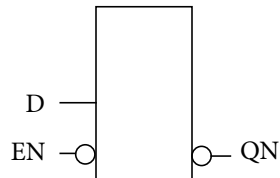
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.306	0.368	0.377	0.437	0.436	0.494	0.487	0.542	0.527	0.580
tQN	0.520	0.475	0.588	0.546	0.645	0.606	0.694	0.656	0.732	0.696
tDQ	0.266	0.318	0.338	0.405	0.407	0.503	0.459	0.591	0.491	0.657
tDQN	0.469	0.435	0.556	0.508	0.656	0.579	0.747	0.638	0.817	0.675

Latches with Active-Low Enable and QN Output Only

LANLN1, LANLN2 and LANLN4

The LANLN1 (1x drive), LANLN2 (2x drive) and LANLN4 (4x drive) cells are active-low latches. When EN is Low, the latch is transparent and data present at the D input is transferred to the QN output. When EN is High, data is retained.



Function Table

INPUTS		OUTPUT
EN	D	QN
L	H	L
L	L	H
H	X	QN _o

Cell Description

Macro Name:	LANLN1	LANLN2	LANLN4
Drive Capability:	1x	2x	4x
Gate Equivalents:	4.	4.25	4.5
Leakage Power (pW):	93.3	103.8	138.5

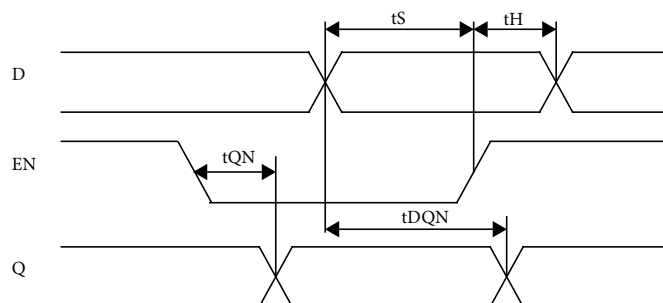
Pin Description

Name	Capacitance (pF)			Description
	LANLN1	LANLN2	LANLN4	
D	0.002	0.002	0.003	Data Input
EN	0.002	0.002	0.003	Clock Input
Maximum capacitance				
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	LANLN1	LANLN2	LANLN4
D	0.018	0.018	0.018
EN	0.023	0.023	0.034
QN	0.043	0.057	0.142

Waveforms



Timing Numbers for LANLN1:

Error Checks

Pulse Width	EN (min)	0.141			
tH	EN -> D (min)	-0.088	tS	D -> EN (min)	0.157

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.240	0.267	0.289	0.304	0.384	0.367	0.574	0.476	0.951	0.685
tDQN	0.248	0.221	0.296	0.258	0.391	0.321	0.581	0.430	0.958	0.639

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.289	0.304	0.375	0.390	0.457	0.473	0.529	0.544	0.589	0.604
tDQN	0.296	0.258	0.382	0.284	0.465	0.286	0.540	0.280	0.604	0.262

Timing Numbers for LANLN2:

Error Checks

Pulse Width	EN (min)	0.164			
tH	EN ->D (min)	-0.108	tS	D ->EN (min)	0.180

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.235	0.266	0.264	0.295	0.315	0.336	0.413	0.404	0.607	0.518
tDQN	0.240	0.222	0.269	0.251	0.320	0.292	0.418	0.360	0.611	0.475

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.264	0.295	0.351	0.382	0.435	0.465	0.508	0.538	0.570	0.599
tDQN	0.269	0.251	0.356	0.277	0.439	0.279	0.513	0.273	0.578	0.255

Timing Numbers for LANLN4:**Error Checks**

Pulse Width	EN (min)	0.243			
tH	EN -> D (min)	-0.215	tS	D -> EN (min)	0.259

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.294	0.357	0.313	0.378	0.345	0.415	0.394	0.471	0.477	0.566
tDQN	0.303	0.340	0.322	0.361	0.354	0.398	0.403	0.454	0.486	0.549

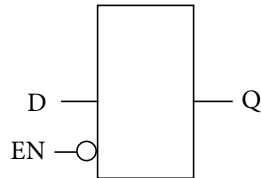
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQN	0.313	0.378	0.385	0.449	0.446	0.508	0.498	0.556	0.538	0.595
tDQN	0.322	0.361	0.386	0.398	0.439	0.419	0.487	0.432	0.525	0.437

Latches with Active-Low Enable and Q Output Only

LANLQ1, LANLQ2 and LANLQ4

The LANLQ1 (1x drive), LANLQ2 (2x drive) and LANLQ4 (4x drive) cells are active-low latches. When EN is Low, the latch is transparent and data present at the D input is transferred to the Q output. When EN is High, data is retained.



Function Table

INPUTS		OUTPUT
EN	D	Q
L	H	H
L	L	L
H	X	Q ₀

Cell Description

Macro Name:	LANLQ1	LANLQ2	LANLQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	3.5	3.75	4.25
Leakage Power (pW):	85.2	110.4	231.8

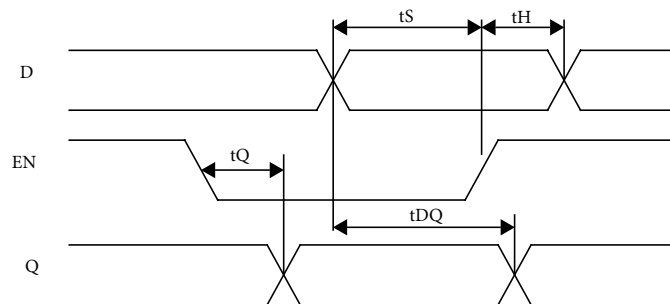
Pin Description

Name	Capacitance (pF)			Description
	LANLQ1	LANLQ2	LANLQ4	
D	0.004	0.004	0.003	Data Input
EN	0.002	0.002	0.004	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	LANLQ1	LANLQ2	LANLQ4
D	0.008	0.008	0.007
EN	0.024	0.024	0.033
Q	0.046	0.06	0.142

Waveforms



Timing Numbers for LANLQ1:

Error Checks

Pulse Width	EN (min)	0.151			
tH	EN -> D (min)	-0.049	tS	D -> EN (min)	0.088

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.261	0.286	0.310	0.325	0.406	0.389	0.595	0.500	0.972	0.711
tDQ	0.155	0.200	0.204	0.239	0.299	0.304	0.489	0.415	0.866	0.626

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.310	0.325	0.400	0.413	0.490	0.500	0.568	0.574	0.633	0.636
tDQ	0.204	0.239	0.234	0.322	0.246	0.406	0.249	0.479	0.240	0.542

Timing Numbers for LANLQ2:

Error Checks

Pulse Width	EN (min)	0.17			
tH	EN ->D (min)	-0.064	tS	D ->EN (min)	0.112

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.256	0.287	0.285	0.318	0.335	0.360	0.433	0.431	0.627	0.549
tDQ	0.147	0.202	0.176	0.233	0.227	0.276	0.325	0.347	0.519	0.464

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.285	0.318	0.375	0.406	0.466	0.494	0.544	0.571	0.610	0.634
tDQ	0.176	0.233	0.210	0.318	0.227	0.405	0.234	0.481	0.228	0.546

Timing Numbers for LANLQ4:**Error Checks**

Pulse Width	EN (min)	0.255			
tH	EN -> D (min)	-0.176	tS	D -> EN (min)	0.216

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

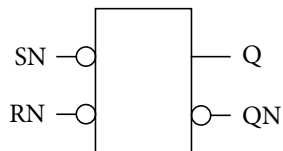
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.285	0.350	0.304	0.371	0.336	0.408	0.385	0.464	0.467	0.557
tDQ	0.251	0.304	0.270	0.325	0.301	0.361	0.350	0.417	0.433	0.511

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tQ	0.304	0.371	0.372	0.437	0.428	0.491	0.476	0.537	0.514	0.573
tDQ	0.270	0.325	0.343	0.412	0.412	0.512	0.465	0.601	0.498	0.668

Set/Reset Latches with NAND Input SRLAB1, SRLAB2 and SRLAB4

The SRLAB1 (1x drive), SRLAB2 (2x drive) and SRLAB4 (4x drive) cells are active-low SR latches. Latch mode (SN = RN = 1) can be entered only when complementary output states have been established. An input condition (SN = RN = 0) generates Low outputs for both outputs Q and QN.



Function Table

INPUTS		OUTPUTS	
SN	RN	Q	QN
L	L	L	L
L	H	H	L
H	L	L	H
H	H	Q _o	QN _o

Cell Description

Macro Name:	SRLAB1	SRLAB2	SRLAB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	2.5	2.75	3.75
Leakage Power (pW):	80.3	135.9	297.0

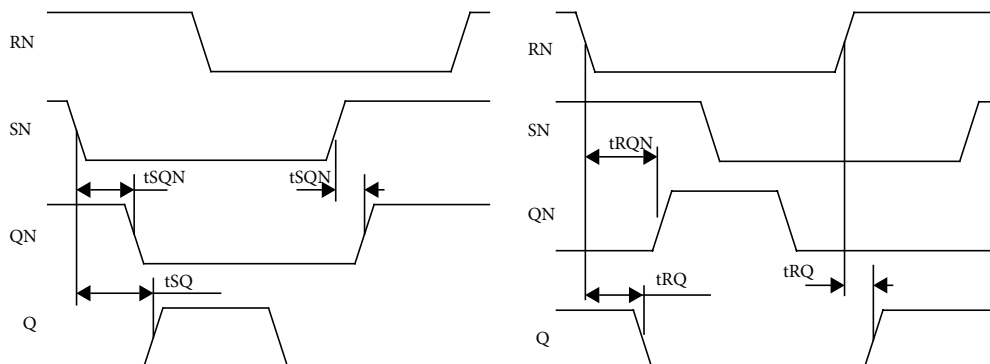
Pin Description

Name	Capacitance (pF)			Description
	SRLAB1	SRLAB2	SRLAB4	
SN	0.004	0.004	0.004	Set Input
RN	0.003	0.003	0.004	Reset Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SRLAB1	SRLAB2	SRLAB4
SN	-0	0	-0
RN	0.002	0.002	0.001
Q	0.034	0.048	0.085
QN	0.036	0.051	0.086

Waveforms



Timing Numbers for SRLAB1:

Error Checks

Pulse Width	RN (min)	0.115	Pulse Width	SN (min)	0.131
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Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSQ/tRQ	0.253	0.156	0.299	0.187	0.391	0.242	0.573	0.348	0.938	0.558
tRQN/tSQN	0.223	0.185	0.267	0.217	0.353	0.273	0.526	0.379	0.870	0.588

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSQ/tRQ	0.299	0.187	0.391	0.274	0.488	0.362	0.574	0.439	0.647	0.505
tRQN/tSQN	0.267	0.217	0.352	0.311	0.437	0.412	0.512	0.500	0.577	0.574

Timing Numbers for SRLAB2:

Error Checks

Pulse Width	RN (min)	0.153	Pulse Width	SN (min)	0.169
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**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSQ/tRQ	0.271	0.161	0.294	0.184	0.338	0.219	0.426	0.280	0.605	0.394
tRQN/tSQN	0.244	0.185	0.267	0.207	0.311	0.240	0.398	0.293	0.574	0.387

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSQ/tRQ	0.294	0.184	0.388	0.275	0.493	0.369	0.585	0.451	0.662	0.521
tRQN/tSQN	0.267	0.207	0.357	0.302	0.451	0.408	0.533	0.501	0.602	0.579

Timing Numbers for SRLAB4:**Error Checks**

Pulse Width	RN (min)	0.184	Pulse Width	SN (min)	0.165
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Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSQ/tRQ	0.288	0.158	0.300	0.173	0.322	0.199	0.360	0.241	0.429	0.317
tRQN/tSQN	0.295	0.162	0.307	0.177	0.328	0.204	0.365	0.246	0.433	0.322

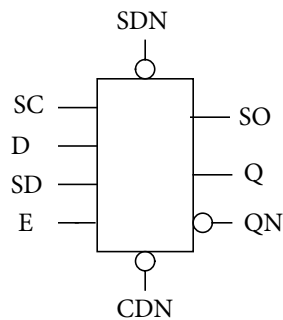
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSQ/tRQ	0.300	0.173	0.384	0.255	0.466	0.332	0.536	0.395	0.589	0.442
tRQN/tSQN	0.307	0.177	0.389	0.262	0.471	0.344	0.538	0.412	0.589	0.464

SCAN LATCHES

Muxed Scan Latches with Clear, Preset SLBHB1 , SLBHB2 and SLBHB4

The SLBHB1 (1x drive), SLBHB2 (2x drive) and SLBHB4 (4x drive) are active-high multiplexed latch flip-flop scan cells with active-low clear, CDN, and set, SDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD. These cells are level-sensitive in capture and edge-sensitive in scan shift. When E is High, the latch is transparent and data present at the D or SD inputs is transferred to the Q and QN outputs. When E is Low, data on Q and QN outputs is retained. Also data present at the D or SD inputs is clocked to the SO output on the falling edge of the clock, E.



Function Table for SO output

INPUTS						OUTPUT
CDN	SDN	E	SC	D	SD	SO
L	H	↓	X	X	X	L
H	L	↓	X	X	X	H
H	H	↓	L	L	X	L
H	H	↓	L	H	X	H
H	H	↓	H	X	L	L
H	H	↓	H	X	H	H
H	H	L	X	X	X	Q
H	H	H	X	X	X	Q

Function Table for Q and QN outputs

INPUTS						OUTPUTS	
CDN	SDN	E	SC	D	SD	Q	QN
L	H	X	X	X	X	L	H
H	L	X	X	X	X	H	L
H	H	H	L	L	X	L	H
H	H	H	L	H	X	H	L
H	H	H	H	X	L	L	H
H	H	H	H	X	H	H	L
H	H	L	X	X	X	Q	QN
L	L	X	X	X	X	H	H

Cell Description

Macro Name:	SLBHB1	SLBHB2	SLBHB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	10.	10.75	12.75
Leakage Power (pW):	337.6	411.9	546.7

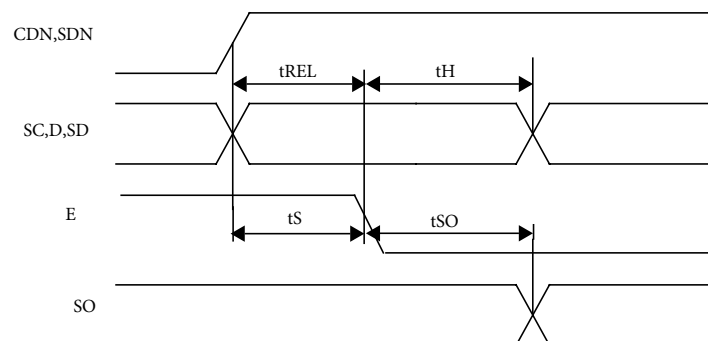
Pin Description

Name	Capacitance (pF)			Description
	SLBHB1	SLBHB2	SLBHB4	
D	0.004	0.004	0.004	Data Input
SD	0.003	0.003	0.003	Scan Data Input
SC	0.004	0.004	0.004	Scan Control
E	0.004	0.004	0.004	Clock Input
SDN	0.005	0.004	0.005	Preset Input
CDN	0.009	0.009	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output
SO	0.3	0.6	1.2	Scan Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

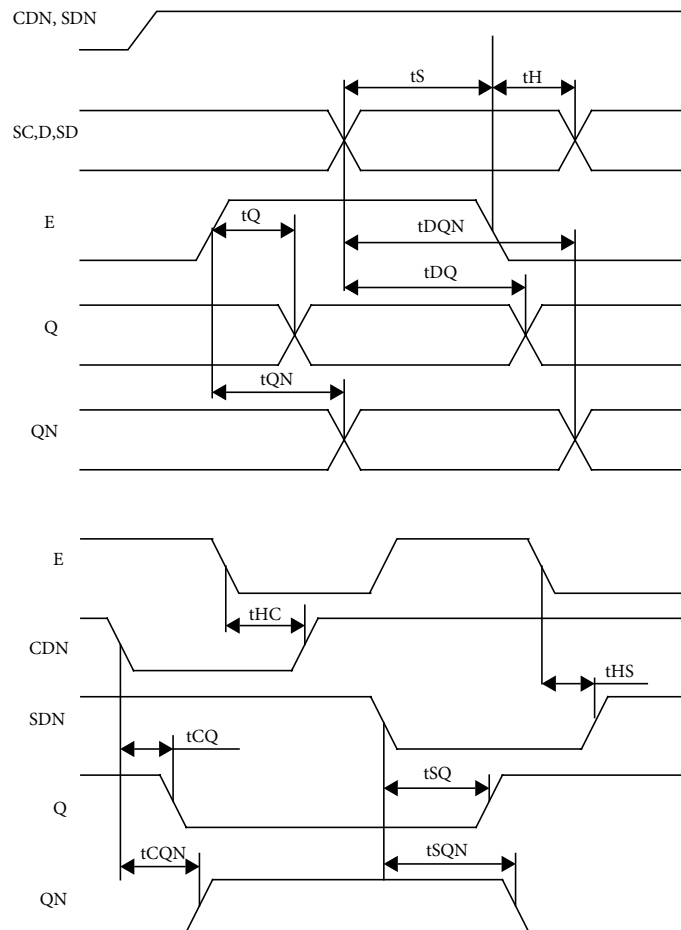
Name	Pin Power (pW/Hz)		
	SLBHB1	SLBHB2	SLBHB4
D	0.012	0.013	0.014
SD	0.007	0.007	0.007
SC	0.047	0.047	0.047
E	0.059	0.061	0.067
SDN	0	0	0
CDN	0.003	0.003	0.003
Q	0.066	0.085	0.125
QN	0.06	0.076	0.11
SO	0.105	0.122	0.159

Waveforms for SO output



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Waveforms for Q and QN outputs



Timing Numbers for SLBHB1:**Error Checks**

Pulse Width	E (min)	0.249			
Pulse Width	CDN (min)	0.163	Pulse Width	SDN (min)	0.155
tH	E -> D/SD/SC (min)	-0.096	tS	D/SD/SC -> E (min)	0.274
tHS	E -> SDN (min)	0.042	tRelS	SDN -> E (min)	-0.032
tHC	E -> CDN (min)	-0.201	tRelC	CDN -> E (min)	0.222

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.324	0.298	0.358	0.338	0.424	0.414	0.557	0.564	0.820	0.865
tQ	0.407	0.413	0.440	0.451	0.507	0.527	0.639	0.677	0.902	0.978
tQN	0.470	0.485	0.503	0.524	0.569	0.600	0.701	0.750	0.964	1.051
tDQ	0.415	0.431	0.448	0.470	0.514	0.546	0.646	0.696	0.909	0.997
tDQN	0.488	0.493	0.522	0.532	0.588	0.607	0.720	0.757	0.983	1.058
tSQ/tCQ	0.207	0.379	0.241	0.418	0.307	0.494	0.439	0.644	0.702	0.945
tCQN/tSQN	0.191	0.335	0.224	0.374	0.291	0.450	0.423	0.600	0.686	0.901

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.358	0.338	0.435	0.397	0.504	0.483	0.561	0.540	0.604	0.583
tQ	0.440	0.451	0.502	0.512	0.544	0.555	0.575	0.587	0.592	0.605
tQN	0.503	0.524	0.564	0.585	0.607	0.628	0.639	0.659	0.657	0.676
tDQ	0.448	0.470	0.511	0.545	0.575	0.624	0.626	0.691	0.660	0.743
tDQN	0.522	0.532	0.597	0.595	0.676	0.658	0.743	0.709	0.795	0.743
tSQ/tCQ	0.241	0.418	0.319	0.495	0.393	0.566	0.450	0.627	0.494	0.673
tCQN/tSQN	0.224	0.374	0.303	0.449	0.372	0.525	0.429	0.586	0.473	0.631

Timing Numbers for SLBHB2:**Error Checks**

Pulse Width	E (min)	0.255			
Pulse Width	CDN (min)	0.167	Pulse Width	SDN (min)	0.158
tH	E -> D/SD/SC (min)	-0.096	tS	D/SD/SC -> E (min)	0.280
tHS	E -> SDN (min)	0.047	tRelS	SDN -> E (min)	-0.031
tHC	E -> CDN (min)	-0.204	tRelC	CDN -> E (min)	0.230

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.322	0.295	0.341	0.319	0.376	0.359	0.442	0.437	0.571	0.589
tQ	0.415	0.418	0.434	0.439	0.468	0.478	0.534	0.554	0.666	0.704
tQN	0.474	0.491	0.493	0.513	0.527	0.552	0.594	0.628	0.727	0.778
tDQ	0.420	0.434	0.439	0.455	0.472	0.495	0.539	0.571	0.671	0.721
tDQN	0.491	0.496	0.510	0.517	0.544	0.557	0.611	0.633	0.744	0.783
tSQ/tCQ	0.207	0.382	0.226	0.403	0.259	0.443	0.326	0.520	0.458	0.670
tCQN/tSQN	0.194	0.337	0.213	0.359	0.247	0.398	0.314	0.475	0.447	0.625

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.341	0.319	0.420	0.397	0.489	0.465	0.546	0.523	0.590	0.566
tQ	0.434	0.439	0.496	0.501	0.540	0.545	0.571	0.577	0.588	0.596
tQN	0.493	0.513	0.554	0.575	0.598	0.618	0.631	0.649	0.650	0.667
tDQ	0.439	0.455	0.502	0.531	0.567	0.610	0.618	0.677	0.652	0.730
tDQN	0.510	0.517	0.585	0.580	0.664	0.645	0.732	0.696	0.784	0.730
tSQ/tCQ	0.226	0.403	0.307	0.480	0.380	0.553	0.440	0.613	0.485	0.660
tCQN/tSQN	0.213	0.359	0.292	0.438	0.363	0.512	0.421	0.572	0.466	0.618

Timing Numbers for SLBHB4:**Error Checks**

Pulse Width	E (min)	0.254			
Pulse Width	CDN (min)	0.166	Pulse Width	SDN (min)	0.158
tH	E -> D/SD/SC (min)	-0.091	tS	D/SD/SC -> E (min)	0.269
tHS	E -> SDN (min)	0.056	tRelS	SDN -> E (min)	-0.031
tHC	E -> CDN (min)	-0.154	tRelC	CDN -> E (min)	0.172

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.348	0.321	0.360	0.336	0.382	0.362	0.417	0.403	0.483	0.481
tQ	0.445	0.450	0.457	0.464	0.478	0.488	0.514	0.528	0.580	0.604
tQN	0.504	0.527	0.516	0.541	0.537	0.566	0.572	0.607	0.638	0.685
tDQ	0.442	0.456	0.454	0.470	0.475	0.494	0.510	0.534	0.577	0.611
tDQN	0.510	0.523	0.522	0.537	0.543	0.563	0.578	0.604	0.644	0.681
tSQ/tCQ	0.234	0.405	0.246	0.419	0.267	0.443	0.303	0.483	0.369	0.561
tCQN/tSQN	0.218	0.373	0.230	0.387	0.251	0.413	0.287	0.454	0.353	0.532

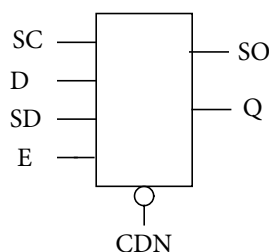
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.360	0.336	0.439	0.414	0.510	0.485	0.568	0.542	0.612	0.586
tQ	0.457	0.464	0.522	0.527	0.569	0.575	0.604	0.611	0.623	0.632
tQN	0.516	0.541	0.579	0.606	0.627	0.653	0.663	0.687	0.684	0.707
tDQ	0.454	0.470	0.517	0.544	0.582	0.624	0.634	0.692	0.668	0.744
tDQN	0.522	0.537	0.597	0.601	0.677	0.666	0.745	0.717	0.797	0.751
tSQ/tCQ	0.246	0.419	0.327	0.496	0.401	0.567	0.462	0.627	0.508	0.673
tCQN/tSQN	0.224	0.387	0.309	0.466	0.380	0.540	0.439	0.600	0.484	0.645

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Muxed Scan Latches with High Enable, Clear and Q only SLCHQ1 , SLCHQ2 and SLCHQ4

The SLCHQ1 (1x drive), SLCHQ2 (2x drive) and SLCHQ4 (4x drive) are active-high multiplexed latch flip-flop scan cells with active-low clear, CDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD. These cells are level-sensitive in capture and edge-sensitive in scan shift. When E is High, the latch is transparent and data present at the D or SD inputs is transferred to the Q output. When E is Low, data on Q output is retained. Also data present at the D or SD inputs is clocked to the SO output on the falling edge of the clock, E.



Function Table for SO output

INPUTS					OUTPUT
CDN	E	SC	D	SD	SO
L	↓	X	X	X	L
H	↓	L	L	X	L
H	↓	L	H	X	H
H	↓	H	X	L	L
H	↓	H	X	H	H
H	L	X	X	X	Q
H	H	X	X	X	Q

Function Table for Q output

INPUTS					OUTPUT
CDN	E	SC	D	SD	Q
L	X	X	X	X	L
H	H	L	L	X	L
H	H	L	H	X	H
H	H	H	X	L	L
H	H	H	X	H	H
H	L	X	X	X	Q

Cell Description

Macro Name:	SLCHQ1	SLCHQ2	SLCHQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	8.	8.5	9.75
Leakage Power (pW):	262.5	330.5	462.1

Pin Description

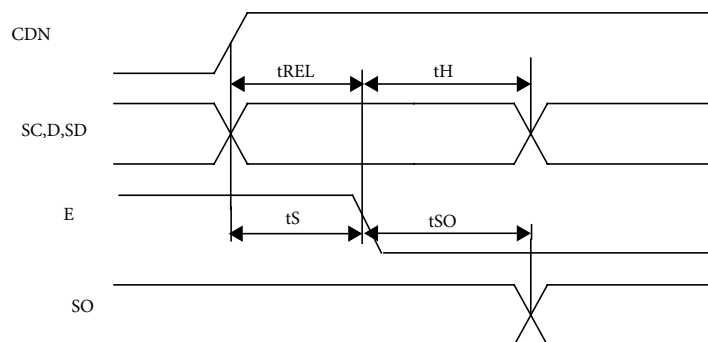
Name	Capacitance (pF)			Description
	SLCHQ1	SLCHQ2	SLCHQ4	
D	0.005	0.005	0.005	Data Input
SD	0.003	0.003	0.003	Scan Data Input
SC	0.004	0.004	0.004	Scan Control
E	0.004	0.004	0.004	Clock Input
CDN	0.009	0.009	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
SO	0.3	0.6	1.2	Scan Output

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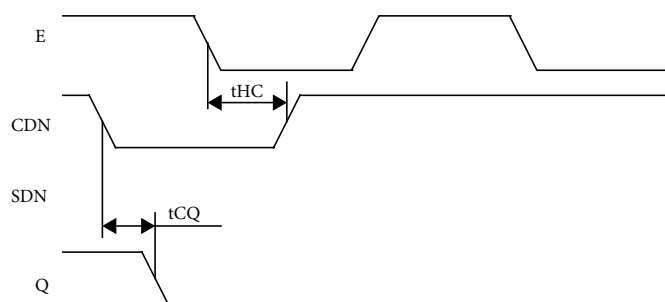
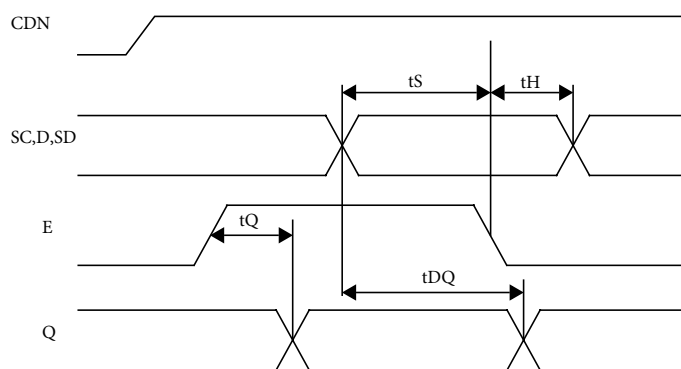
Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SLCHQ1	SLCHQ2	SLCHQ4
D	0.017	0.018	0.022
SD	0.003	0.003	0.003
SC	0.04	0.04	0.04
E	0.047	0.049	0.054
CDN	0.003	0.003	0.003
Q	0.093	0.116	0.179
SO	0.089	0.103	0.136

Waveforms for SO output



Waveforms for Q output



Timing Numbers for SLCHQ1:**Error Checks**

Pulse Width	E (min)	0.194	Pulse Width	CDN (min)	0.126
tH	E -> D/SD/SC (min)	-0.130	tS	D/SD/SC -> E (min)	0.263
tHC	E -> CDN (min)	-0.182	tRelC	CDN -> E (min)	0.237

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.297	0.271	0.331	0.310	0.397	0.386	0.529	0.537	0.792	0.837
tQ	0.294	0.293	0.341	0.349	0.422	0.442	0.563	0.604	0.829	0.910
tDQ	0.333	0.345	0.380	0.401	0.462	0.495	0.603	0.657	0.869	0.962
tCQ		0.238		0.286		0.370		0.523		0.825

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.331	0.310	0.402	0.359	0.460	0.441	0.510	0.491	0.548	0.529
tQ	0.341	0.349	0.396	0.402	0.430	0.437	0.455	0.463	0.469	0.478
tDQ	0.380	0.401	0.445	0.475	0.514	0.559	0.570	0.632	0.609	0.687
tCQ		0.286		0.359		0.426		0.480		0.518

Timing Numbers for SLCHQ2:**Error Checks**

Pulse Width	E (min)	0.25	Pulse Width	CDN (min)	0.155
tH	E -> D/SD/SC (min)	-0.151	tS	D/SD/SC -> E (min)	0.320
tHC	E -> CDN (min)	-0.206	tRelC	CDN -> E (min)	0.284

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.293	0.264	0.313	0.287	0.347	0.327	0.413	0.403	0.545	0.553
tQ	0.295	0.300	0.327	0.339	0.379	0.398	0.465	0.498	0.610	0.664
tDQ	0.336	0.354	0.369	0.394	0.421	0.454	0.507	0.555	0.652	0.725
tCQ		0.233		0.265		0.316		0.403		0.558

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.313	0.287	0.384	0.359	0.443	0.418	0.492	0.468	0.530	0.507
tQ	0.327	0.339	0.381	0.392	0.416	0.427	0.442	0.454	0.456	0.468
tDQ	0.369	0.394	0.434	0.469	0.505	0.556	0.565	0.633	0.606	0.691
tCQ		0.265		0.340		0.409		0.464		0.504

Timing Numbers for SLCHQ4:**Error Checks**

Pulse Width	E (min)	0.355	Pulse Width	CDN (min)	0.208
tH	E -> D/SD/SC (min)	-0.192	tS	D/SD/SC -> E (min)	0.422
tHC	E -> CDN (min)	-0.263	tRelC	CDN -> E (min)	0.368

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

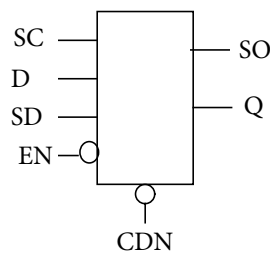
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.311	0.282	0.323	0.296	0.345	0.322	0.381	0.364	0.448	0.441
tQ	0.337	0.360	0.358	0.384	0.395	0.425	0.451	0.490	0.544	0.597
tDQ	0.383	0.411	0.404	0.436	0.440	0.478	0.496	0.543	0.589	0.653
tCQ		0.262		0.283		0.317		0.373		0.465

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.323	0.296	0.394	0.368	0.453	0.428	0.503	0.478	0.542	0.517
tQ	0.358	0.384	0.413	0.437	0.449	0.472	0.475	0.498	0.489	0.513
tDQ	0.404	0.436	0.469	0.512	0.544	0.603	0.609	0.687	0.655	0.750
tCQ		0.283		0.358		0.432		0.492		0.535

Muxed Scan Latches with Low Enable, Clear and Q only SLCLQ1 , SLCLQ2 and SLCLQ4

The SLCLQ1 (1x drive), SLCLQ2 (2x drive) and SLCLQ4 (4x drive) are active-low multiplexed latch flip-flop scan cells with active-low clear, CDN. The scan control input, SC, selects either the data input, D, or the scan data input, SD. These cells are level-sensitive in capture and edge-sensitive in scan shift. When EN is Low, the latch is transparent and data present at the D or SD inputs is transferred to the Q output. When EN is High, data on Q output is retained. Also data present at the D or SD inputs is clocked to the SO output on the rising edge of the clock, EN.



Function Table for SO output

INPUTS					OUTPUT
CDN	EN	SC	D	SD	SO
L	↑	X	X	X	L
H	↑	L	L	X	L
H	↑	L	H	X	H
H	↑	H	X	L	L
H	↑	H	X	H	H
H	L	X	X	X	Q
H	H	X	X	X	Q

Function Table for Q output

INPUTS					OUTPUT
CDN	EN	SC	D	SD	Q
L	X	X	X	X	L
H	L	L	L	X	L
H	L	L	H	X	H
H	L	H	X	L	L
H	L	H	X	H	H
H	H	X	X	X	Q

Cell Description

Macro Name:	SLCLQ1	SLCLQ2	SLCLQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	8.25	8.75	9.75
Leakage Power (pW):	262.0	330.0	461.8

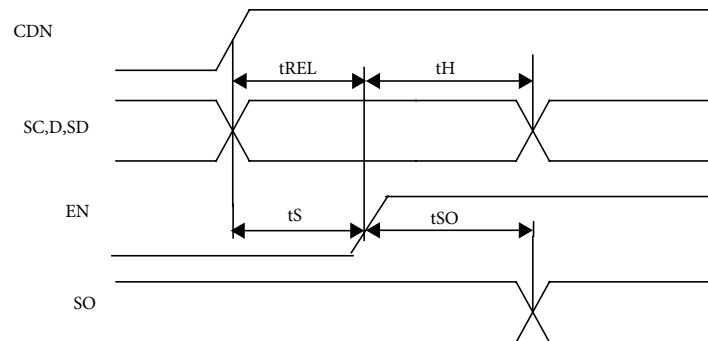
Pin Description

Name	Capacitance (pF)			Description
	SLCLQ1	SLCLQ2	SLCLQ4	
D	0.005	0.005	0.005	Data Input
SD	0.003	0.003	0.003	Scan Data Input
SC	0.004	0.004	0.004	Scan Control
EN	0.004	0.004	0.004	Clock Input
CDN	0.009	0.009	0.009	Clear Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
SO	0.3	0.6	1.2	Scan Output

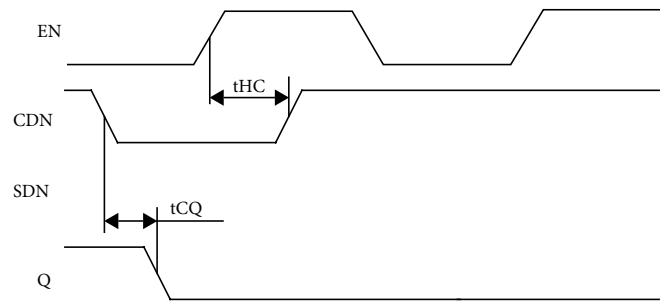
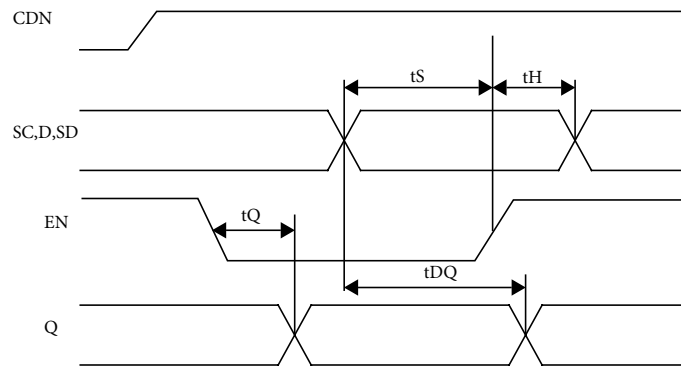
**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	SLCLQ1	SLCLQ2	SLCLQ4
D	0.017	0.019	0.024
SD	0.002	0.002	0.002
SC	0.04	0.04	0.04
EN	0.069	0.05	0.056
CDN	0.005	0.005	0.006
Q	0.092	0.118	0.183
SO	0.067	0.086	0.118

Waveforms for SO output



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Waveforms for Q output

Timing Numbers for SLCLQ1:**Error Checks**

Pulse Width	EN (min)	0.203	Pulse Width	CDN (min)	0.126
tH	EN -> D/SD/SC (min)	-0.147	tS	D/SD/SC -> EN (min)	0.279
tHC	EN -> CDN (min)	-0.233	tRelC	CDN -> EN (min)	0.267

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.280	0.293	0.313	0.333	0.379	0.409	0.511	0.559	0.774	0.860
tQ	0.283	0.331	0.330	0.387	0.411	0.483	0.552	0.646	0.817	0.952
tDQ	0.333	0.362	0.380	0.418	0.462	0.514	0.603	0.677	0.869	0.983
tCQ		0.240		0.288		0.372		0.526		0.828

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.313	0.333	0.368	0.363	0.404	0.422	0.431	0.447	0.447	0.460
tQ	0.330	0.387	0.401	0.459	0.462	0.518	0.513	0.568	0.553	0.606
tDQ	0.380	0.418	0.444	0.493	0.512	0.581	0.567	0.658	0.604	0.717
tCQ		0.288		0.361		0.428		0.481		0.519

Timing Numbers for SLCLQ2:**Error Checks**

Pulse Width	EN (min)	0.255	Pulse Width	CDN (min)	0.155
tH	EN -> D/SD/SC (min)	-0.171	tS	D/SD/SC -> EN (min)	0.329
tHC	EN -> CDN (min)	-0.256	tRelC	CDN -> EN (min)	0.311

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.275	0.286	0.294	0.309	0.329	0.349	0.395	0.425	0.527	0.575
tQ	0.287	0.336	0.320	0.376	0.372	0.436	0.457	0.536	0.602	0.705
tDQ	0.336	0.371	0.369	0.411	0.420	0.473	0.506	0.576	0.651	0.748
tCQ		0.235		0.267		0.319		0.406		0.561

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.294	0.309	0.349	0.363	0.386	0.399	0.413	0.424	0.428	0.437
tQ	0.320	0.376	0.391	0.447	0.452	0.506	0.503	0.556	0.544	0.594
tDQ	0.369	0.411	0.433	0.487	0.503	0.577	0.561	0.658	0.600	0.720
tCQ		0.267		0.341		0.410		0.465		0.504

Timing Numbers for SLCLQ4:**Error Checks**

Pulse Width	EN (min)	0.354	Pulse Width	CDN (min)	0.209
tH	EN -> D/SD/SC (min)	-0.222	tS	D/SD/SC -> EN (min)	0.426
tHC	EN -> CDN (min)	-0.316	tRelC	CDN -> EN (min)	0.395

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.294	0.306	0.306	0.321	0.328	0.347	0.364	0.388	0.431	0.465
tQ	0.336	0.393	0.357	0.418	0.394	0.461	0.450	0.526	0.543	0.635
tDQ	0.383	0.433	0.404	0.458	0.441	0.501	0.497	0.567	0.590	0.677
tCQ		0.265		0.286		0.320		0.376		0.469

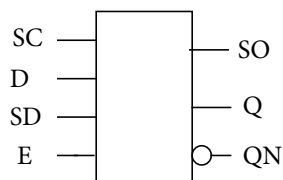
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.306	0.321	0.361	0.375	0.398	0.412	0.426	0.437	0.442	0.451
tQ	0.357	0.418	0.428	0.490	0.489	0.550	0.540	0.600	0.581	0.639
tDQ	0.404	0.458	0.469	0.535	0.543	0.628	0.607	0.716	0.651	0.783
tCQ		0.286		0.360		0.434		0.494		0.537

Muxed Scan Latches

SLNHB1 , SLNHB2 and SLNHB4

The SLNHB1 (1x drive), SLNHB2 (2x drive) and SLNHB4 (4x drive) are active-high multiplexed latch flip-flop scan cells. The scan control input, SC, selects either the data input, D, or the scan data input, SD. These cells are level-sensitive in capture and edge-sensitive in scan shift. When E is High, the latch is transparent and data present at the D or SD inputs is transferred to the Q and QN outputs. When E is Low, data on Q and QN outputs is retained. Also data present at the D or SD inputs is clocked to the SO output on the falling edge of the clock, E.



Function Table for SO output

INPUTS				OUTPUT
E	SC	D	SD	SO
↓	L	L	X	L
↓	L	H	X	H
↓	H	X	L	L
↓	H	X	H	H
L	X	X	X	Q
H	X	X	X	Q

Function Table for Q and QN outputs

INPUTS				OUTPUTS	
E	SC	D	SD	Q	QN
H	L	L	X	L	H
H	L	H	X	H	L
H	H	X	L	L	H
H	H	X	H	H	L
L	X	X	X	Q	QN

Cell Description

Macro Name:	SLNHB1	SLNHB2	SLNHB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.25	7.75	9.75
Leakage Power (pW):	273.3	356.7	529.6

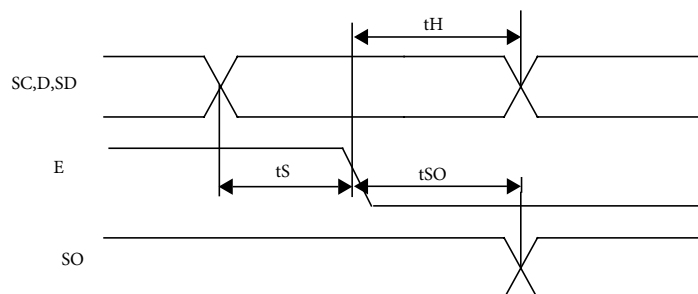
Pin Description

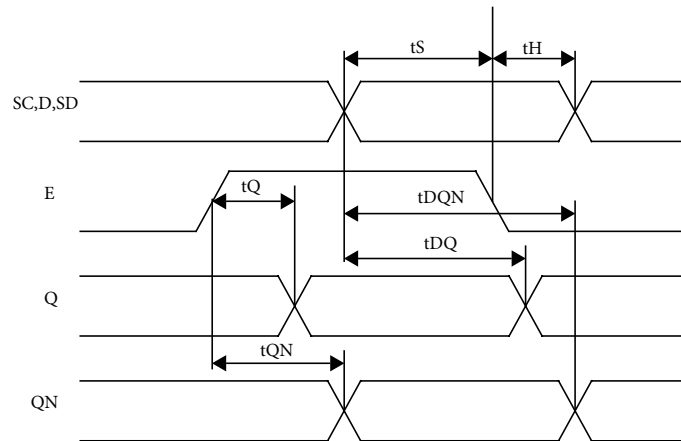
Name	Capacitance (pF)			Description
	SLNHB1	SLNHB2	SLNHB4	
D	0.004	0.004	0.004	Data Input
SD	0.004	0.004	0.004	Scan Data Input
SC	0.004	0.004	0.004	Scan Control
E	0.004	0.004	0.004	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output
SO	0.3	0.6	1.2	Scan Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SLNHB1	SLNHB2	SLNHB4
D	0.025	0.028	0.037
SD	0.002	0.002	0.002
SC	0.045	0.045	0.045
E	0.069	0.072	0.092
Q	0.041	0.06	0.107
QN	0.041	0.06	0.107
SO	0.086	0.097	0.131

Waveforms for SO output



Waveforms for Q and QN outputs

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Timing Numbers for SLNHB1:**Error Checks**

Pulse Width	E (min)	0.208			
tH	E -> D/SD/SC (min)	-0.077	tS	D/SD/SC -> E (min)	0.243

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.307	0.275	0.341	0.315	0.408	0.391	0.540	0.541	0.802	0.842
tQ	0.284	0.289	0.328	0.342	0.408	0.432	0.553	0.591	0.834	0.894
tQN	0.329	0.356	0.365	0.395	0.433	0.471	0.571	0.621	0.846	0.921
tDQ	0.268	0.309	0.312	0.364	0.392	0.458	0.536	0.620	0.818	0.924
tDQN	0.348	0.340	0.384	0.379	0.452	0.455	0.590	0.605	0.864	0.906

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.341	0.315	0.410	0.354	0.466	0.438	0.512	0.483	0.546	0.516
tQ	0.328	0.342	0.387	0.400	0.428	0.441	0.459	0.473	0.478	0.494
tQN	0.365	0.395	0.423	0.454	0.464	0.494	0.497	0.525	0.517	0.544
tDQ	0.312	0.364	0.378	0.438	0.440	0.516	0.488	0.584	0.518	0.637
tDQN	0.384	0.379	0.457	0.445	0.534	0.506	0.600	0.554	0.651	0.585

Timing Numbers for SLNHB2:**Error Checks**

Pulse Width	E (min)	0.307			
tH	E -> D/SD/SC (min)	-0.101	tS	D/SD/SC -> E (min)	0.340

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.301	0.262	0.321	0.286	0.357	0.326	0.425	0.402	0.560	0.554
tQ	0.282	0.302	0.312	0.340	0.359	0.398	0.440	0.497	0.581	0.663
tQN	0.387	0.402	0.406	0.423	0.440	0.462	0.507	0.538	0.640	0.689
tDQ	0.269	0.321	0.298	0.359	0.345	0.419	0.425	0.520	0.566	0.689
tDQN	0.406	0.388	0.425	0.409	0.459	0.448	0.526	0.524	0.659	0.675

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.321	0.286	0.390	0.354	0.446	0.409	0.492	0.455	0.526	0.488
tQ	0.312	0.340	0.371	0.397	0.413	0.439	0.444	0.471	0.463	0.491
tQN	0.406	0.423	0.464	0.482	0.505	0.523	0.538	0.555	0.558	0.574
tDQ	0.298	0.359	0.365	0.434	0.430	0.516	0.481	0.588	0.514	0.644
tDQN	0.425	0.409	0.500	0.476	0.582	0.543	0.654	0.595	0.710	0.631

Timing Numbers for SLNHB4:**Error Checks**

Pulse Width	E (min)	0.48			
tH	E -> D/SD/SC (min)	-0.140	tS	D/SD/SC -> E (min)	0.503

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

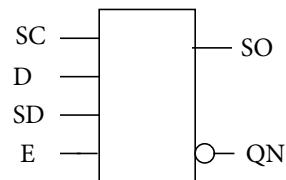
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.333	0.289	0.346	0.304	0.368	0.330	0.404	0.371	0.472	0.449
tQ	0.329	0.368	0.349	0.392	0.383	0.432	0.436	0.495	0.525	0.603
tQN	0.521	0.524	0.530	0.535	0.547	0.554	0.580	0.591	0.644	0.665
tDQ	0.309	0.379	0.329	0.403	0.362	0.444	0.415	0.509	0.504	0.618
tDQN	0.533	0.502	0.543	0.514	0.560	0.533	0.593	0.570	0.657	0.644

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.346	0.304	0.417	0.374	0.476	0.432	0.524	0.479	0.560	0.514
tQ	0.349	0.392	0.410	0.451	0.455	0.494	0.488	0.528	0.509	0.549
tQN	0.530	0.535	0.590	0.596	0.634	0.641	0.667	0.674	0.688	0.695
tDQ	0.329	0.403	0.397	0.478	0.467	0.566	0.525	0.645	0.563	0.706
tDQN	0.543	0.514	0.618	0.582	0.706	0.654	0.786	0.716	0.848	0.759

Muxed Scan Latches with QN only SLNHN1 , SLNHN2 and SLNHN4

The SLNHN1 (1x drive), SLNHN2 (2x drive) and SLNHN4 (4x drive) are active-high multiplexed latch flip-flop scan cells. The scan control input, SC, selects either the data input, D, or the scan data input, SD. These cells are level-sensitive in capture and edge-sensitive in scan shift. When E is High, the latch is transparent and data present at the D or SD inputs is transferred to the QN output. When E is Low, data on QN output is retained. Also data present at the D or SD inputs is clocked to the SO output on the falling edge of the clock, E.



Function Table for SO output

INPUTS				OUTPUT
E	SC	D	SD	SO
↓	L	L	X	L
↓	L	H	X	H
↓	H	X	L	L
↓	H	X	H	H
L	X	X	X	Q
H	X	X	X	Q

Function Table for QN output

INPUTS				OUTPUT
E	SC	D	SD	QN
H	L	L	X	H
H	L	H	X	L
H	H	X	L	H
H	H	X	H	L
L	X	X	X	QN

Cell Description

Macro Name:	SLNHN1	SLNHN2	SLNHN4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.25	7.5	8.75
Leakage Power (pW):	242.6	284.1	369.1

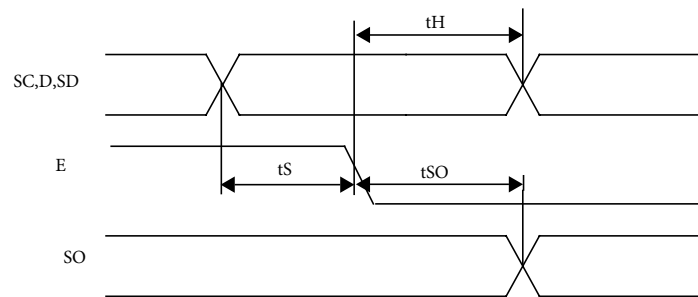
Pin Description

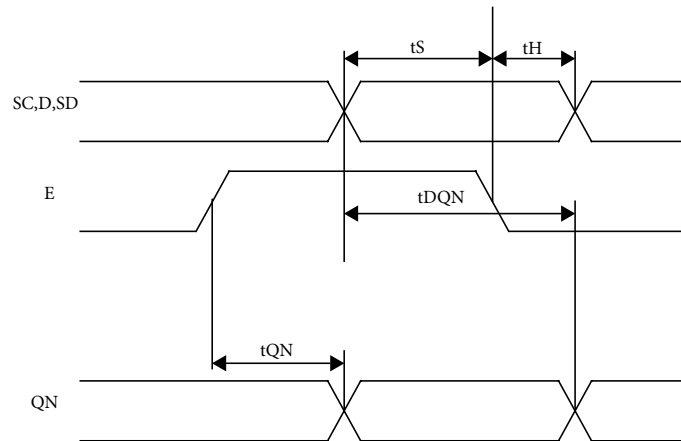
Name	Capacitance (pF)			Description
	SLNHN1	SLNHN2	SLNHN4	
D	0.005	0.005	0.005	Data Input
SD	0.004	0.004	0.004	Scan Data Input
SC	0.004	0.004	0.004	Scan Control
E	0.005	0.005	0.005	Clock Input
Maximum capacitance				
QN	0.3	0.6	1.2	Inverted Output
SO	0.3	0.6	1.2	Scan Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	SLNHN1	SLNHN2	SLNHN4
D	0.024	0.026	0.029
SD	0.002	0.002	0.002
SC	0.044	0.044	0.044
E	0.06	0.064	0.055
QN	0.067	0.08	0.117
SO	0.083	0.097	0.132

Waveforms for SO output



Waveforms for QN output

Timing Numbers for SLNHN1:**Error Checks**

Pulse Width	E (min)	0.155			
tH	E -> D/SD/SC (min)	-0.069	tS	D/SD/SC -> E (min)	0.200

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.297	0.268	0.331	0.308	0.397	0.383	0.530	0.534	0.793	0.835
tQN	0.285	0.321	0.319	0.361	0.385	0.438	0.516	0.588	0.777	0.889
tDQN	0.313	0.316	0.348	0.356	0.414	0.432	0.544	0.582	0.806	0.883

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.331	0.308	0.399	0.353	0.453	0.430	0.498	0.474	0.531	0.507
tQN	0.319	0.361	0.377	0.419	0.418	0.459	0.449	0.489	0.468	0.506
tDQN	0.348	0.356	0.420	0.421	0.494	0.481	0.557	0.526	0.606	0.554

Timing Numbers for SLNHN2:**Error Checks**

Pulse Width	E (min)	0.175			
tH	E -> D/SD/SC (min)	-0.069	tS	D/SD/SC -> E (min)	0.220

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.294	0.262	0.314	0.285	0.348	0.325	0.414	0.401	0.545	0.551
tQN	0.281	0.317	0.301	0.341	0.336	0.381	0.402	0.458	0.533	0.608
tDQN	0.310	0.310	0.331	0.334	0.366	0.374	0.432	0.451	0.563	0.601

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.314	0.285	0.382	0.353	0.436	0.408	0.481	0.453	0.514	0.486
tQN	0.301	0.341	0.359	0.399	0.400	0.439	0.431	0.469	0.450	0.486
tDQN	0.331	0.334	0.404	0.399	0.478	0.459	0.541	0.504	0.590	0.534

Timing Numbers for SLNHN4:**Error Checks**

Pulse Width	E (min)	0.223				
tH	E -> D/SD/SC (min)	-0.075	tS	D/SD/SC -> E (min)	0.265	

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.310	0.279	0.323	0.293	0.345	0.319	0.381	0.361	0.447	0.439
tQN	0.303	0.340	0.316	0.355	0.338	0.382	0.374	0.424	0.441	0.502
tDQN	0.332	0.332	0.346	0.347	0.368	0.374	0.405	0.416	0.472	0.494

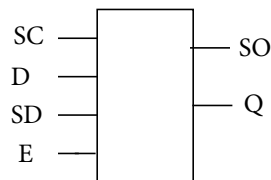
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.323	0.293	0.391	0.361	0.446	0.416	0.491	0.461	0.525	0.494
tQN	0.316	0.355	0.373	0.413	0.415	0.453	0.447	0.484	0.466	0.502
tDQN	0.346	0.347	0.419	0.412	0.494	0.473	0.558	0.519	0.608	0.549

Muxed Scan Latches with Q only

SLNHQ1 , SLNHQ2 and SLNHQ4

The SLNHQ1 (1x drive), SLNHQ2 (2x drive) and SLNHQ4 (4x drive) are active-high multiplexed latch flip-flop scan cells. The scan control input, SC, selects either the data input, D, or the scan data input, SD. These cells are level-sensitive in capture and edge-sensitive in scan shift. When E is High, the latch is transparent and data present at the D or SD inputs is transferred to the Q output. When E is Low, data on Q output is retained. Also data present at the D or SD inputs is clocked to the SO output on the falling edge of the clock, E.



Function Table for SO output

INPUTS				OUTPUT
E	SC	D	SD	SO
↓	L	L	X	L
↓	L	H	X	H
↓	H	X	L	L
↓	H	X	H	H
L	X	X	X	Q
H	X	X	X	Q

Function Table for Q output

INPUTS				OUTPUT
E	SC	D	SD	Q
H	L	L	X	L
H	L	H	X	H
H	H	X	L	L
H	H	X	H	H
L	X	X	X	Q

Cell Description

Macro Name:	SLNHQ1	SLNHQ2	SLNHQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.	7.25	8.5
Leakage Power (pW):	266.2	324.1	450.7

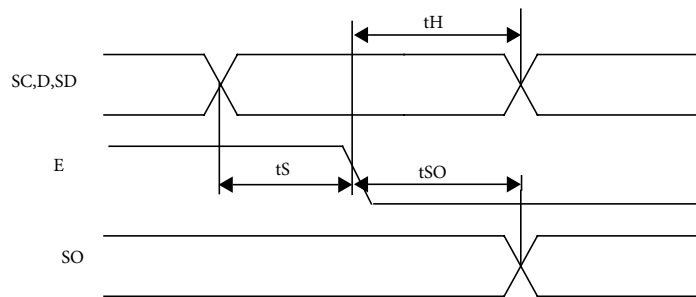
Pin Description

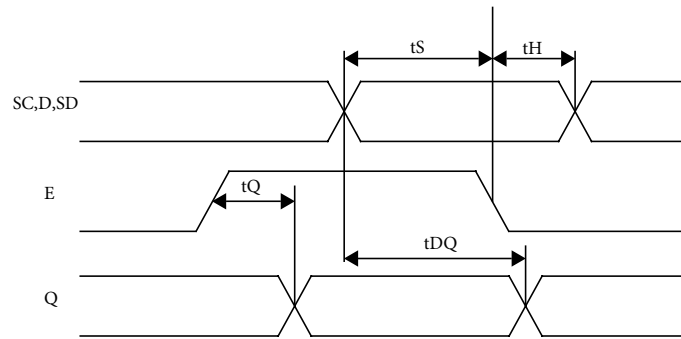
Name	Capacitance (pF)			Description
	SLNHQ1	SLNHQ2	SLNHQ4	
D	0.004	0.004	0.004	Data Input
SD	0.004	0.004	0.004	Scan Data Input
SC	0.004	0.004	0.004	Scan Control
E	0.004	0.004	0.004	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
SO	0.3	0.6	1.2	Scan Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SLNHQ1	SLNHQ2	SLNHQ4
D	0.023	0.025	0.03
SD	0.002	0.002	0.002
SC	0.045	0.045	0.045
E	0.063	0.055	0.063
Q	0.068	0.091	0.149
SO	0.086	0.102	0.134

Waveforms for SO output



Waveforms for Q output

Timing Numbers for SLNHQ1:**Error Checks**

Pulse Width	E (min)	0.193			
tH	E -> D/SD/SC (min)	-0.075	tS	D/SD/SC -> E (min)	0.230

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.305	0.270	0.339	0.310	0.405	0.386	0.537	0.536	0.799	0.837
tQ	0.285	0.296	0.329	0.349	0.406	0.440	0.543	0.599	0.812	0.902
tDQ	0.271	0.317	0.314	0.372	0.391	0.466	0.528	0.628	0.797	0.933

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.339	0.310	0.407	0.355	0.461	0.432	0.507	0.476	0.541	0.509
tQ	0.329	0.349	0.386	0.406	0.427	0.448	0.457	0.480	0.476	0.500
tDQ	0.314	0.372	0.379	0.446	0.441	0.525	0.490	0.593	0.521	0.646

Timing Numbers for SLNHQ2:**Error Checks**

Pulse Width	E (min)	0.253			
tH	E -> D/SD/SC (min)	-0.090	tS	D/SD/SC -> E (min)	0.285

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.305	0.262	0.325	0.286	0.361	0.326	0.430	0.402	0.568	0.554
tQ	0.285	0.302	0.316	0.340	0.365	0.398	0.448	0.496	0.592	0.660
tDQ	0.269	0.318	0.300	0.358	0.348	0.417	0.431	0.517	0.575	0.684

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.325	0.286	0.395	0.355	0.452	0.411	0.499	0.458	0.535	0.491
tQ	0.316	0.340	0.375	0.397	0.416	0.439	0.447	0.472	0.466	0.492
tDQ	0.300	0.358	0.367	0.432	0.431	0.514	0.482	0.586	0.515	0.642

Timing Numbers for SLNHQ4:**Error Checks**

Pulse Width	E (min)	0.361			
tH	E -> D/SD/SC (min)	-0.120	tS	D/SD/SC -> E (min)	0.386

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

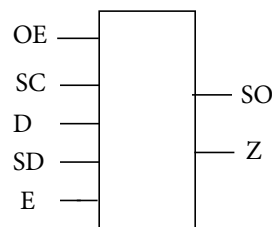
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.327	0.282	0.340	0.296	0.362	0.322	0.399	0.363	0.468	0.441
tQ	0.324	0.361	0.345	0.385	0.379	0.427	0.434	0.491	0.525	0.597
tDQ	0.306	0.372	0.327	0.397	0.361	0.440	0.415	0.505	0.505	0.612

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.340	0.296	0.410	0.367	0.468	0.424	0.517	0.471	0.553	0.506
tQ	0.345	0.385	0.405	0.444	0.448	0.486	0.480	0.519	0.500	0.539
tDQ	0.327	0.397	0.394	0.473	0.465	0.560	0.522	0.639	0.560	0.699

Muxed Scan Latches with Z-output only and Active Higher Enable SLNHT1 , SLNHT2 and SLNHT4

The SLNHT1 (1x drive), SLNHT2 (2x drive) and SLNHT4 (4x drive) are active-high multiplexed latch flip-flop scan cells with Z-output only. The scan control input, SC, selects either the data input, D, or the scan data input, SD. These cells are level-sensitive in capture and edge-sensitive in scan shift. When E is High, the latch is transparent and data present at the D or SD inputs is transferred to the Z output. When E is Low, data on Z output is retained. Also data present at the D or SD inputs is clocked to the SO output on the falling edge of the clock, E.



Function Table for SO output

INPUTS					OUTPUT
OE	E	SC	D	SD	SO
X	↓	L	L	X	L
X	↓	L	H	X	H
X	↓	H	X	L	L
X	↓	H	X	H	H
X	L	X	X	X	Q
X	H	X	X	X	Q

Function Table for Z output

INPUTS					OUTPUT
OE	E	SC	D	SD	Z
H	H	L	L	X	L
H	H	L	H	X	H
H	H	H	X	L	L
H	H	H	X	H	H
H	L	X	X	X	Qo
L	X	X	X	X	HiZ

Cell Description

Macro Name:	SLNHT1	SLNHT2	SLNHT4
Drive Capability:	1x	2x	4x
Gate Equivalents:	8.25	9.25	10.25
Leakage Power (pW):	258.0	332.3	413.6

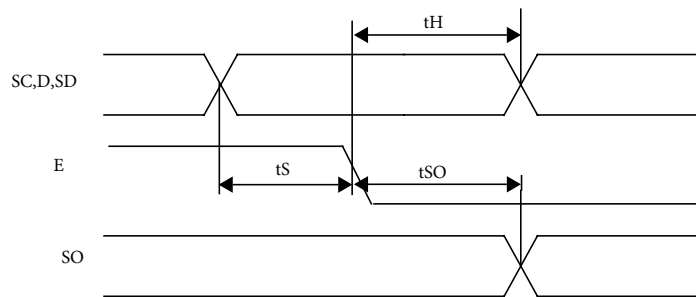
Pin Description

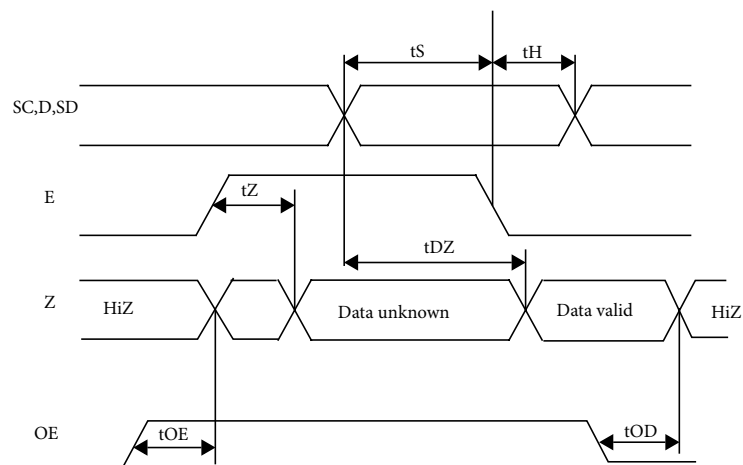
Name	Capacitance (pF)			Description
	SLNHT1	SLNHT2	SLNHT4	
D	0.004	0.004	0.004	Data Input
SD	0.004	0.004	0.004	Scan Data Input
SC	0.004	0.004	0.004	Scan Control
E	0.004	0.004	0.004	Clock Input
Maximum capacitance				
Z	0.006	0.006	0.011	Output
SO	0.3	0.6	1.2	Scan Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	SLNHT1	SLNHT2	SLNHT4
D	0.025	0.024	0.024
SD	0.028	0.027	0.027
SC	0.068	0.067	0.066
E	0.08	0.084	0.099
Z	0.089	0.107	0.131
SO	0.067	0.083	0.115

Waveforms for SO output



Waveforms for Z output

Timing Numbers for SLNHT1:**Error Checks**

Pulse Width	E (min)	0.238			
tH	E -> D/SD/SC (min)	-0.100	tS	D/SD/SC -> E (min)	0.270

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.303	0.266	0.337	0.306	0.403	0.382	0.535	0.532	0.798	0.833
tZ	0.325	0.351	0.364	0.391	0.437	0.459	0.571	0.586	0.832	0.828
tDZ	0.318	0.370	0.354	0.406	0.425	0.474	0.560	0.601	0.822	0.845
tOE	0.111	0.056	0.144	0.083	0.208	0.139	0.337	0.258	0.596	0.500
tOD	0.007	0.063	0.007	0.063	0.008	0.063	0.007	0.063	0.007	0.063

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.337	0.306	0.405	0.347	0.459	0.427	0.504	0.471	0.538	0.504
tQ	0.364	0.391	0.423	0.443	0.463	0.484	0.494	0.515	0.511	0.536
tDQ	0.354	0.406	0.424	0.483	0.487	0.566	0.541	0.636	0.573	0.693
tOE	0.144	0.083	0.189	0.121	0.221	0.130	0.245	0.121	0.261	0.085
tOD	0.007	0.063	0.034	0.109	0.074	0.138	0.113	0.162	0.157	0.180

Timing Numbers for SLNHT2:**Error Checks**

Pulse Width	E (min)	0.161			
tH	E -> D/SD/SC (min)	-0.061	tS	D/SD/SC -> E (min)	0.202

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.305	0.257	0.325	0.280	0.360	0.320	0.426	0.397	0.557	0.547
tQ	0.415	0.379	0.441	0.405	0.476	0.449	0.544	0.530	0.675	0.682
tDQ	0.408	0.408	0.433	0.437	0.466	0.479	0.534	0.558	0.665	0.710
tOE	0.116	0.129	0.138	0.154	0.175	0.196	0.242	0.274	0.373	0.425
tOD	0.097	0.072	0.097	0.072	0.097	0.072	0.096	0.072	0.094	0.072

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.325	0.280	0.392	0.347	0.446	0.401	0.491	0.445	0.524	0.477
tQ	0.441	0.405	0.496	0.462	0.537	0.503	0.564	0.534	0.582	0.553
tDQ	0.433	0.437	0.492	0.506	0.551	0.578	0.596	0.639	0.625	0.687
tOE	0.138	0.154	0.194	0.205	0.238	0.238	0.274	0.262	0.299	0.277
tOD	0.097	0.072	0.142	0.111	0.177	0.140	0.208	0.164	0.228	0.177

Timing Numbers for SLNHT4:**Error Checks**

Pulse Width	E (min)	0.162			
tH	E -> D/SD/SC (min)	-0.059	tS	D/SD/SC -> E (min)	0.200

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.328	0.277	0.340	0.291	0.361	0.317	0.397	0.359	0.464	0.437
tQ	0.435	0.394	0.452	0.412	0.475	0.443	0.515	0.485	0.585	0.566
tDQ	0.419	0.420	0.436	0.436	0.465	0.467	0.504	0.512	0.576	0.593
tOE	0.124	0.119	0.140	0.134	0.166	0.160	0.206	0.205	0.277	0.285
tOD	0.098	0.084	0.098	0.084	0.098	0.084	0.098	0.084	0.098	0.084

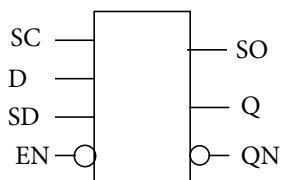
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.340	0.291	0.407	0.359	0.462	0.413	0.508	0.458	0.542	0.491
tQ	0.452	0.412	0.508	0.470	0.549	0.511	0.577	0.542	0.595	0.562
tDQ	0.436	0.436	0.504	0.509	0.559	0.581	0.604	0.644	0.633	0.692
tOE	0.140	0.134	0.201	0.188	0.247	0.222	0.285	0.244	0.311	0.260
tOD	0.098	0.084	0.148	0.139	0.182	0.174	0.211	0.207	0.231	0.225

Muxed Scan Latches with Low Enable

SLNLB1 , SLNLB2 and SLNLB4

The SLNLB1 (1x drive), SLNLB2 (2x drive) and SLNLB4 (4x drive) are active-Low multiplexed latch flip-flop scan cells. The scan control input, SC, selects either the data input, D, or the scan data input, SD. These cells are level-sensitive in capture and edge-sensitive in scan shift. When EN is Low, the latch is transparent and data present at the D or SD inputs is transferred to the Q and QN outputs. When EN is High, data on Q and QN outputs is retained. Also data present at the D or SD inputs is clocked to the SO output on the rising edge of the clock, EN.



Function Table for SO output

INPUTS				OUTPUT
EN	SC	D	SD	SO
↑	L	L	X	L
↑	L	H	X	H
↑	H	X	L	L
↑	H	X	H	H
L	X	X	X	Q
H	X	X	X	Q

Function Table for Q and QN outputs

INPUTS				OUTPUTS	
EN	SC	D	SD	Q	QN
L	L	L	X	L	H
L	L	H	X	H	L
L	H	X	L	L	H
L	H	X	H	H	L
H	X	X	X	Q	QN

Cell Description

Macro Name:	SLNLB1	SLNLB2	SLNLB4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.25	7.75	9.25
Leakage Power (pW):	281.2	347.4	486.5

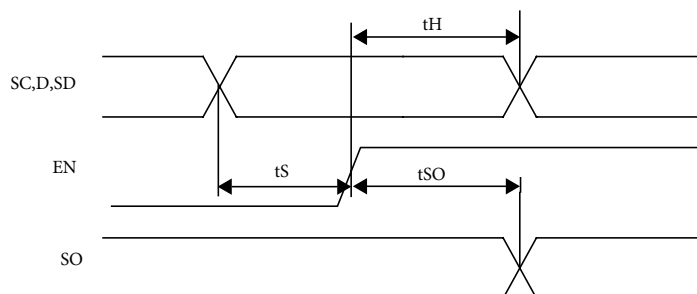
Pin Description

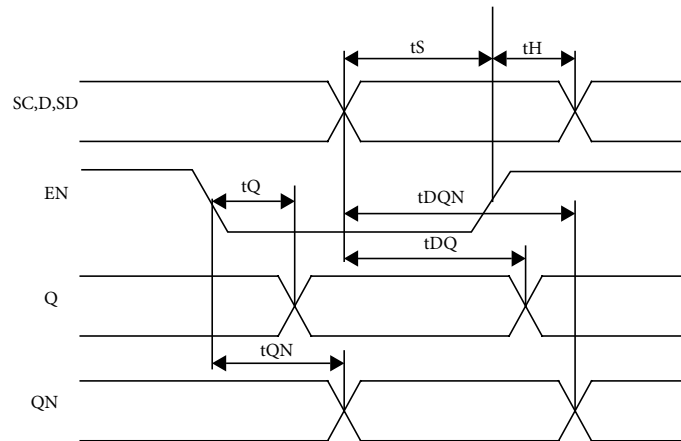
Name	Capacitance (pF)			Description
	SLNLB1	SLNLB2	SLNLB4	
D	0.004	0.004	0.004	Data Input
SD	0.004	0.004	0.004	Scan Data Input
SC	0.004	0.004	0.004	Scan Control
EN	0.004	0.004	0.004	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
QN	0.3	0.6	1.2	Inverted Output
SO	0.3	0.6	1.2	Scan Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	SLNLB1	SLNLB2	SLNLB4
D	0.026	0.03	0.039
SD	0.002	0.002	0.002
SC	0.044	0.044	0.044
EN	0.065	0.065	0.089
Q	0.05	0.067	0.113
QN	0.05	0.067	0.113
SO	0.057	0.071	0.096

Waveforms for SO output



Waveforms for Q and QN outputs

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Timing Numbers for SLNLB1:**Error Checks**

Pulse Width	EN (min)	0.223			
tH	EN -> D/SD/SC (min)	-0.129	tS	D/SD/SC -> EN (min)	0.247

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.283	0.289	0.317	0.328	0.383	0.404	0.515	0.555	0.778	0.856
tQ	0.273	0.324	0.318	0.380	0.396	0.474	0.534	0.636	0.799	0.940
tQN	0.370	0.360	0.403	0.400	0.469	0.475	0.600	0.625	0.862	0.926
tDQ	0.284	0.334	0.328	0.391	0.407	0.488	0.545	0.653	0.810	0.959
tDQN	0.379	0.372	0.413	0.411	0.479	0.487	0.609	0.637	0.872	0.937

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.317	0.328	0.373	0.357	0.413	0.422	0.444	0.450	0.463	0.467
tQ	0.318	0.380	0.384	0.445	0.437	0.497	0.482	0.540	0.517	0.572
tQN	0.403	0.400	0.469	0.466	0.521	0.519	0.564	0.564	0.597	0.598
tDQ	0.328	0.391	0.395	0.466	0.460	0.547	0.511	0.617	0.544	0.672
tDQN	0.413	0.411	0.487	0.478	0.567	0.543	0.635	0.594	0.689	0.627

Timing Numbers for SLNLB2:**Error Checks**

Pulse Width	EN (min)	0.298			
tH	EN -> D/SD/SC (min)	-0.156	tS	D/SD/SC -> EN (min)	0.324

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.276	0.279	0.295	0.302	0.329	0.342	0.395	0.418	0.524	0.568
tQ	0.272	0.322	0.303	0.360	0.352	0.419	0.435	0.519	0.577	0.686
tQN	0.412	0.400	0.430	0.420	0.462	0.459	0.526	0.535	0.654	0.685
tDQ	0.283	0.333	0.313	0.372	0.361	0.433	0.444	0.536	0.586	0.705
tDQN	0.424	0.409	0.442	0.430	0.475	0.469	0.539	0.545	0.667	0.695

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.295	0.302	0.351	0.357	0.391	0.396	0.422	0.424	0.441	0.440
tQ	0.303	0.360	0.369	0.425	0.423	0.477	0.468	0.520	0.502	0.552
tQN	0.430	0.420	0.496	0.486	0.547	0.540	0.590	0.585	0.622	0.620
tDQ	0.313	0.372	0.380	0.447	0.447	0.531	0.501	0.605	0.536	0.662
tDQN	0.442	0.430	0.517	0.498	0.601	0.567	0.674	0.622	0.731	0.660

Timing Numbers for SLNLB4:**Error Checks**

Pulse Width	EN (min)	0.444			
tH	EN -> D/SD/SC (min)	-0.210	tS	D/SD/SC -> EN (min)	0.472

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

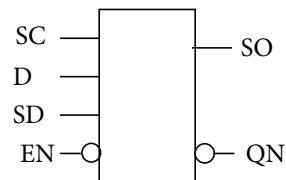
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.290	0.298	0.302	0.312	0.323	0.338	0.358	0.380	0.422	0.458
tQ	0.308	0.364	0.328	0.388	0.361	0.428	0.414	0.491	0.502	0.597
tQN	0.523	0.509	0.532	0.521	0.548	0.540	0.578	0.577	0.638	0.651
tDQ	0.316	0.379	0.336	0.403	0.369	0.444	0.421	0.508	0.509	0.617
tDQN	0.540	0.516	0.549	0.528	0.565	0.547	0.596	0.584	0.656	0.658

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.302	0.312	0.358	0.368	0.398	0.406	0.428	0.434	0.447	0.450
tQ	0.328	0.388	0.393	0.453	0.446	0.505	0.491	0.548	0.525	0.580
tQN	0.532	0.521	0.598	0.587	0.650	0.640	0.693	0.685	0.726	0.719
tDQ	0.336	0.403	0.404	0.479	0.476	0.567	0.535	0.646	0.574	0.707
tDQN	0.549	0.528	0.625	0.596	0.713	0.669	0.794	0.732	0.857	0.776

Muxed Scan Latches with Low Enable and QN only SLN1N1 , SLN1N2 and SLN1N4

The SLN1N1 (1x drive), SLN1N2 (2x drive) and SLN1N4 (4x drive) are active-low multiplexed latch flip-flop scan cells. The scan control input, SC, selects either the data input, D, or the scan data input, SD. These cells are level-sensitive in capture and edge-sensitive in scan shift. When EN is Low, the latch is transparent and data present at the D or SD inputs is transferred to the QN output. When EN is High, data on QN output is retained. Also data present at the D or SD inputs is clocked to the SO output on the rising edge of the clock, EN.



Function Table for SO output

INPUTS				OUTPUT
EN	SC	D	SD	SO
↑	L	L	X	L
↑	L	H	X	H
↑	H	X	L	L
↑	H	X	H	H
L	X	X	X	Q
H	X	X	X	Q

Function Table for QN output

INPUTS				OUTPUT
EN	SC	D	SD	QN
L	L	L	X	H
L	L	H	X	L
L	H	X	L	H
L	H	X	H	L
H	X	X	X	QN

Cell Description

Macro Name:	SLNLN1	SLNLN2	SLNLN4
Drive Capability:	1x	2x	4x
Gate Equivalents:	7.	7.	8.5
Leakage Power (pW):	238.7	274.7	379.3

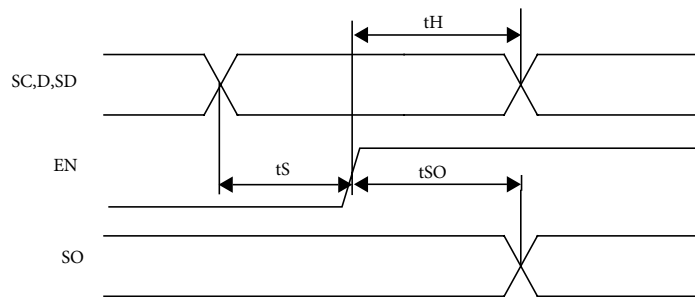
Pin Description

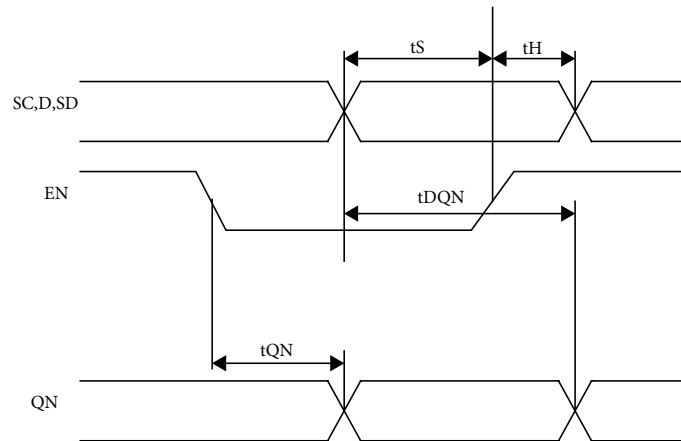
Name	Capacitance (pF)			Description
	SLNLN1	SLNLN2	SLNLN4	
D	0.004	0.004	0.004	Data Input
SD	0.004	0.004	0.004	Scan Data Input
SC	0.004	0.004	0.004	Scan Control
EN	0.004	0.004	0.004	Clock Input
Maximum capacitance				
QN	0.3	0.6	1.2	Inverted Output
SO	0.3	0.6	1.2	Scan Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	SLNLN1	SLNLN2	SLNLN4
D	0.023	0.024	0.027
SD	0.002	0.002	0.002
SC	0.044	0.044	0.044
EN	0.067	0.064	0.088
QN	0.075	0.089	0.121
SO	0.062	0.074	0.109

Waveforms for SO output



Waveforms for QN output

Timing Numbers for SLNLN1:**Error Checks**

Pulse Width	EN (min)	0.153			
tH	EN -> D/SD/SC (min)	-0.082	tS	D/SD/SC -> EN (min)	0.166

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.294	0.302	0.330	0.342	0.400	0.418	0.539	0.568	0.818	0.869
tQN	0.321	0.306	0.357	0.346	0.429	0.422	0.570	0.572	0.852	0.873
tDQN	0.308	0.310	0.345	0.350	0.417	0.426	0.558	0.576	0.839	0.877

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.330	0.342	0.388	0.370	0.430	0.440	0.462	0.470	0.482	0.487
tQN	0.357	0.346	0.426	0.414	0.480	0.471	0.526	0.518	0.560	0.553
tDQN	0.345	0.350	0.418	0.414	0.492	0.471	0.553	0.515	0.602	0.542

Timing Numbers for SLNLN2:**Error Checks**

Pulse Width	EN (min)	0.176			
tH	EN -> D/SD/SC (min)	-0.082	tS	D/SD/SC -> EN (min)	0.186

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.282	0.289	0.302	0.313	0.336	0.353	0.402	0.429	0.533	0.579
tQN	0.315	0.294	0.335	0.318	0.370	0.358	0.436	0.435	0.567	0.585
tDQN	0.301	0.302	0.322	0.327	0.357	0.367	0.423	0.443	0.554	0.593

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.302	0.313	0.361	0.370	0.402	0.410	0.434	0.441	0.454	0.458
tQN	0.335	0.318	0.402	0.385	0.456	0.440	0.500	0.486	0.533	0.520
tDQN	0.322	0.327	0.395	0.391	0.469	0.448	0.531	0.493	0.580	0.520

Timing Numbers for SLNLN4:**Error Checks**

Pulse Width	EN (min)	0.219			
tH	EN -> D/SD/SC (min)	-0.086	tS	D/SD/SC -> EN (min)	0.227

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

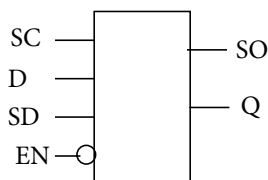
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.315	0.327	0.328	0.342	0.350	0.368	0.387	0.410	0.455	0.488
tQN	0.345	0.331	0.358	0.346	0.380	0.373	0.417	0.415	0.485	0.493
tDQN	0.327	0.329	0.340	0.344	0.363	0.370	0.400	0.412	0.468	0.491

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.328	0.342	0.389	0.402	0.433	0.444	0.467	0.476	0.488	0.495
tQN	0.358	0.346	0.428	0.416	0.485	0.474	0.532	0.523	0.567	0.560
tDQN	0.340	0.344	0.414	0.408	0.488	0.466	0.552	0.512	0.602	0.540

Muxed Scan Latches with low Enable and Q only SLNLQ1 , SLNLQ2 and SLNLQ4

The SLNLQ1 (1x drive), SLNLQ2 (2x drive) and SLNLQ4 (4x drive) are active-low multiplexed latch flip-flop scan cells. The scan control input, SC, selects either the data input, D, or the scan data input, SD. These cells are level-sensitive in capture and edge-sensitive in scan shift. When EN is Low, the latch is transparent and data present at the D or SD inputs is transferred to the Q output. When EN is High, data on Q output is retained. Also data present at the D or SD inputs is clocked to the SO output on the rising edge of the clock, EN.



Function Table for SO output

INPUTS				OUTPUT
EN	SC	D	SD	SO
↑	L	L	X	L
↑	L	H	X	H
↑	H	X	L	L
↑	H	X	H	H
L	X	X	X	Q
H	X	X	X	Q

Function Table for Q output

INPUTS				OUTPUT
EN	SC	D	SD	Q
L	L	L	X	L
L	L	H	X	H
L	H	X	L	L
L	H	X	H	H
H	X	X	X	Q

Cell Description

Macro Name:	SLNLQ1	SLNLQ2	SLNLQ4
Drive Capability:	1x	2x	4x
Gate Equivalents:	6.75	7.	8.
Leakage Power (pW):	261.4	327.1	485.1

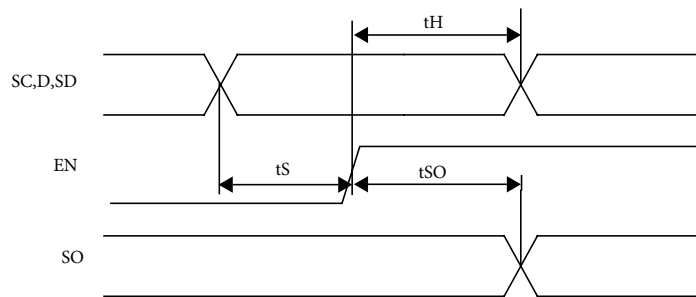
Pin Description

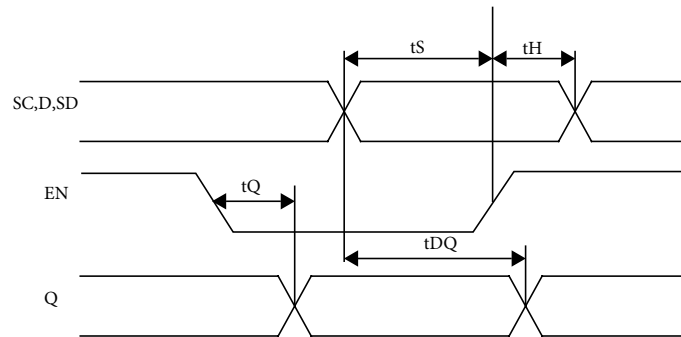
Name	Capacitance (pF)			Description
	SLNLQ1	SLNLQ2	SLNLQ4	
D	0.004	0.004	0.004	Data Input
SD	0.004	0.004	0.004	Scan Data Input
SC	0.004	0.004	0.004	Scan Control
EN	0.004	0.004	0.004	Clock Input
Maximum capacitance				
Q	0.3	0.6	1.2	Output
SO	0.3	0.6	1.2	Scan Output

**Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns**

Name	Pin Power (pW/Hz)		
	SLNLQ1	SLNLQ2	SLNLQ4
D	0.023	0.025	0.031
SD	0.002	0.002	0.002
SC	0.045	0.045	0.045
EN	0.066	0.065	0.06
Q	0.075	0.094	0.16
SO	0.062	0.076	0.109

Waveforms for SO output



Waveforms for Q output

Timing Numbers for SLNLQ1:**Error Checks**

Pulse Width	EN (min)	0.174			
tH	EN -> D/SD/SC (min)	-0.113	tS	D/SD/SC -> EN (min)	0.191

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.293	0.298	0.327	0.337	0.393	0.413	0.525	0.564	0.788	0.864
tQ	0.267	0.314	0.311	0.367	0.388	0.458	0.528	0.617	0.801	0.920
tDQ	0.269	0.314	0.313	0.369	0.390	0.463	0.530	0.625	0.802	0.930

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.327	0.337	0.385	0.371	0.426	0.434	0.457	0.464	0.477	0.481
tQ	0.311	0.367	0.378	0.434	0.433	0.487	0.479	0.531	0.514	0.564
tDQ	0.313	0.369	0.378	0.443	0.440	0.522	0.488	0.590	0.519	0.643

Timing Numbers for SLNLQ2:**Error Checks**

Pulse Width	EN (min)	0.217			
tH	EN -> D/SD/SC (min)	-0.137	tS	D/SD/SC -> EN (min)	0.236

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.287	0.290	0.307	0.314	0.342	0.354	0.408	0.432	0.540	0.585
tQ	0.268	0.312	0.299	0.349	0.347	0.405	0.431	0.500	0.579	0.662
tDQ	0.269	0.313	0.300	0.351	0.348	0.409	0.431	0.506	0.579	0.672

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.307	0.314	0.366	0.371	0.407	0.412	0.439	0.442	0.459	0.459
tQ	0.299	0.349	0.367	0.416	0.423	0.470	0.469	0.515	0.504	0.548
tDQ	0.300	0.351	0.366	0.425	0.431	0.507	0.482	0.579	0.515	0.634

Timing Numbers for SLNLQ4:**Error Checks**

Pulse Width	EN (min)	0.318			
tH	EN -> D/SD/SC (min)	-0.187	tS	D/SD/SC -> EN (min)	0.336

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.306	0.310	0.319	0.325	0.341	0.351	0.377	0.393	0.444	0.472
tQ	0.314	0.377	0.333	0.400	0.367	0.440	0.420	0.502	0.508	0.605
tDQ	0.309	0.380	0.328	0.404	0.361	0.444	0.414	0.507	0.502	0.612

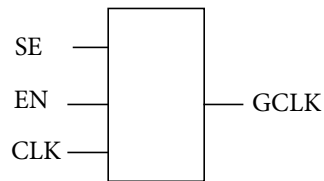
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tSO	0.319	0.325	0.379	0.384	0.422	0.426	0.455	0.458	0.476	0.476
tQ	0.333	0.400	0.401	0.468	0.457	0.523	0.504	0.568	0.540	0.602
tDQ	0.328	0.404	0.396	0.479	0.466	0.566	0.524	0.646	0.562	0.707

CLOCK-GATING CELLS

Latch Based Clock Gating Cells with Scan Enable GCLFSN1 , GCLFSN2, GCLFSN4, GCLFSN7 and GCLFSNA

The GCLFSN1 (1x drive), GCLFSN2 (2x drive), GCLFSN4 (4x drive), GCLFSN7 (7x drive) and GCLFSNA (10x drive) cells are latch-based gating logic. Logic appropriate for negative-edge-triggered registers. Test-control logic located after the latch.



Function Table

INPUTS			OUTPUT
SE	EN	CLK	GCLK
H	X	L	L
H	X	H	H
L	L	X	H
L	H	L	L
L	H	H	H

Cell Description

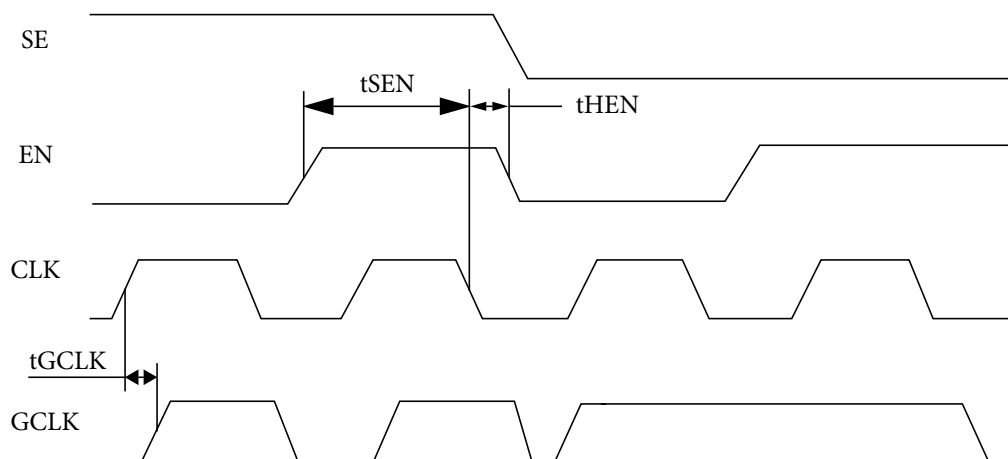
Macro Name:	GCLFSN1	GCLFSN2	GCLFSN4	GCLFSN7	GCLFSNA
Drive Capability:	1x	2x	4x	7x	10x
Gate Equivalents:	4.75	4.75	5.25	6.	7.25
Leakage Power (pW):	219.1	245.4	300.1	387.8	479.9

Pin Description

Name	Capacitance (pF)					Description
	GCLFSN1	GCLFSN2	GCLFSN4	GCLFSN7	GCLFSNA	
SE	0.005	0.004	0.004	0.004	0.004	Test pin
EN	0.003	0.003	0.003	0.003	0.003	Enable
CLK	0.004	0.005	0.005	0.005	0.004	Clock Input
Maximum capacitance						
GCLK	0.3	0.6	1.2	2.1	3	Clock Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)				
	GCLFSN1	GCLFSN2	GCLFSN4	GCLFSN7	GCLFSNA
SE	0.028	0.009	0.009	0.009	0.012
EN	0.05	0.047	0.047	0.047	0.05
CLK	0.05	0.049	0.049	0.049	0.057
GCLK	0.029	0.056	0.087	0.156	0.222

Waveforms

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Timing Numbers for GCLFSN1:**Error Checks**

Pulse Width	CLK (min)	0.134			
tHEN	CLK -> EN (min)	-0.047	tSEN	EN -> CLK (min)	0.096

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.169	0.150	0.205	0.187	0.271	0.258	0.403	0.397	0.663	0.675

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.205	0.187	0.275	0.253	0.337	0.303	0.386	0.342	0.422	0.368

Timing Numbers for GCLFSN2:**Error Checks**

Pulse Width	CLK (min)	0.132			
tHEN	CLK -> EN (min)	-0.067	tSEN	EN -> CLK (min)	0.092

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.245	0.229	0.265	0.252	0.300	0.291	0.366	0.364	0.498	0.504

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.265	0.252	0.322	0.306	0.364	0.346	0.397	0.376	0.421	0.397

Timing Numbers for GCLFSN4:**Error Checks**

Pulse Width	CLK (min)	0.13			
tHEN	CLK -> EN (min)	-0.068	tSEN	EN -> CLK (min)	0.092

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.275	0.248	0.289	0.263	0.314	0.289	0.355	0.330	0.433	0.404

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.289	0.263	0.345	0.316	0.388	0.354	0.423	0.384	0.448	0.403

Timing Numbers for GCLFSN7:**Error Checks**

Pulse Width	CLK (min)	0.13			
tHEN	CLK -> EN (min)	-0.067	tSEN	EN -> CLK (min)	0.092

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.329	0.298	0.339	0.310	0.356	0.328	0.386	0.359	0.435	0.412

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.339	0.310	0.397	0.364	0.439	0.402	0.474	0.432	0.499	0.451

Timing Numbers for GCLFSNA:**Error Checks**

Pulse Width	CLK (min)	0.134			
tHEN	CLK -> EN (min)	-0.064	tSEN	EN -> CLK (min)	0.092

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.304	0.288	0.311	0.295	0.324	0.310	0.345	0.332	0.378	0.369

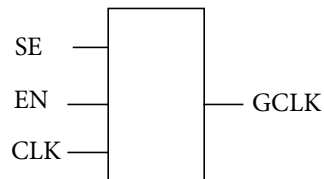
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.311	0.295	0.368	0.350	0.410	0.390	0.445	0.422	0.469	0.443

Latch Based Clock Gating Cells with Scan Enable

GCLRSN1 , GCLRSN2, GCLRSN4, GCLRSN7 and GCLRSNA

The GCLRSN1 (1x drive), GCLRSN2 (2x drive), GCLRSN4 (4x drive), GCLRSN7 (7x drive) and GCLRSNA (10x drive) cells are latch-based gating logic. Logic appropriate for positive-edge-triggered registers. Test-control logic located after the latch.



Function Table

INPUTS			OUTPUT
SE	EN	CLK	GCLK
H	X	L	L
H	X	H	H
L	L	X	L
L	H	L	L
L	H	H	H

Cell Description

Macro Name:	GCLRSN1	GCLRSN2	GCLRSN4	GCLRSN7	GCLRSNA
Drive Capability:	1x	2x	4x	7x	10x
Gate Equivalents:	4.75	5.	5.75	6.25	6.75
Leakage Power (pW):	144.2	248.2	316.1	388.9	480.6

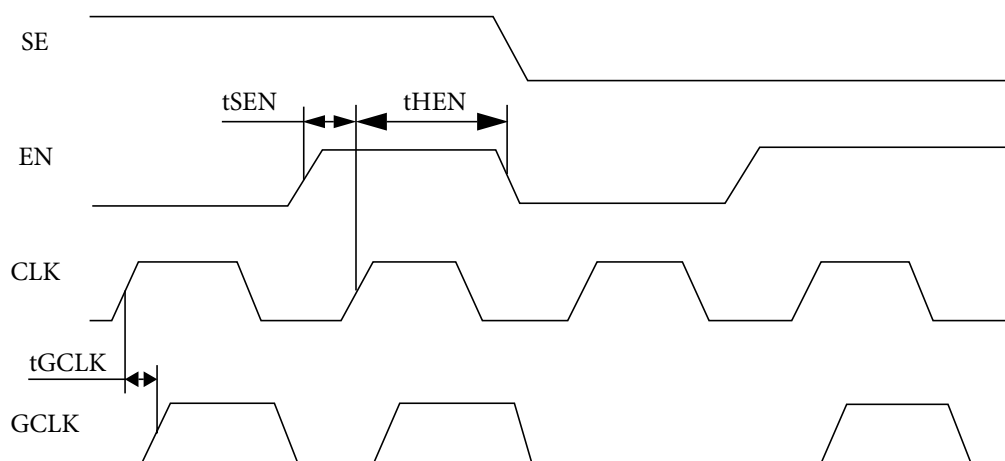
Pin Description

Name	Capacitance (pF)					Description
	GCLRSN1	GCLRSN2	GCLRSN4	GCLRSN7	GCLRSNA	
SE	0.004	0.004	0.004	0.004	0.004	Test pin
EN	0.003	0.003	0.003	0.003	0.003	Enable
CLK	0.004	0.005	0.005	0.005	0.005	Clock Input
Maximum capacitance						
GCLK	0.3	0.6	1.2	2.1	3	Clock Output

Pin Powers for Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)				
	GCLRSN1	GCLRSN2	GCLRSN4	GCLRSN7	GCLRSNA
SE	0.01	0.025	0.025	0.025	0.025
EN	0.052	0.05	0.05	0.051	0.051
CLK	0.053	0.05	0.051	0.051	0.052
GCLK	0.034	0.061	0.097	0.145	0.205

Waveforms



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Timing Numbers for GCLRSN1:**Error Checks**

Pulse Width	CLK (min)	0.111			
tHEN	CLK -> EN (min)	-0.032	tSEN	EN -> CLK (min)	0.070

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.165	0.160	0.202	0.198	0.271	0.270	0.407	0.408	0.679	0.682

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.202	0.198	0.272	0.267	0.333	0.321	0.381	0.364	0.417	0.392

Timing Numbers for GCLRSN2:**Error Checks**

Pulse Width	CLK (min)	0.127			
tHEN	CLK -> EN (min)	-0.072	tSEN	EN -> CLK (min)	0.115

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.228	0.223	0.247	0.242	0.281	0.278	0.347	0.346	0.474	0.479

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.247	0.242	0.305	0.298	0.347	0.338	0.382	0.369	0.406	0.390

Timing Numbers for GCLRSN4:**Error Checks**

Pulse Width	CLK (min)	0.13			
tHEN	CLK -> EN (min)	-0.077	tSEN	EN -> CLK (min)	0.119

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.244	0.239	0.256	0.251	0.278	0.273	0.314	0.309	0.379	0.374

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.256	0.251	0.310	0.308	0.349	0.350	0.379	0.384	0.399	0.408

Timing Numbers for GCLRSN7:**Error Checks**

Pulse Width	CLK (min)	0.13			
tHEN	CLK -> EN (min)	-0.077	tSEN	EN -> CLK (min)	0.119

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.266	0.259	0.276	0.268	0.292	0.284	0.319	0.310	0.367	0.354

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.276	0.268	0.329	0.324	0.368	0.366	0.398	0.400	0.419	0.424

Timing Numbers for GCLRSNA:**Error Checks**

Pulse Width	CLK (min)	0.13			
tHEN	CLK -> EN (min)	-0.076	tSEN	EN -> CLK (min)	0.119

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.283	0.291	0.290	0.299	0.304	0.314	0.326	0.337	0.363	0.375

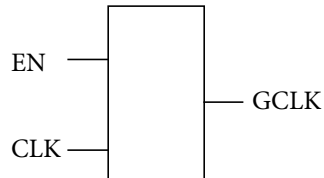
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.290	0.299	0.343	0.356	0.382	0.398	0.413	0.432	0.433	0.456

Latch-Free Clock Gating Cells

GCNFNN1 , GCNFNN2, GCNFNN4, GCNFNN7 and GCNFNNA

The GCNFNN1 (1x drive), GCNFNN2 (2x drive), GCNFNN4 (4x drive), GCNFNN7 (7x drive) and GCNFNNA (10x drive) cells are latch and flip-flop free gating logic. Logic appropriate for negative-edge-triggered registers.



Function Table

INPUTS		OUTPUT
EN	CLK	GCLK
L	X	L
H	L	L
H	H	H

Cell Description

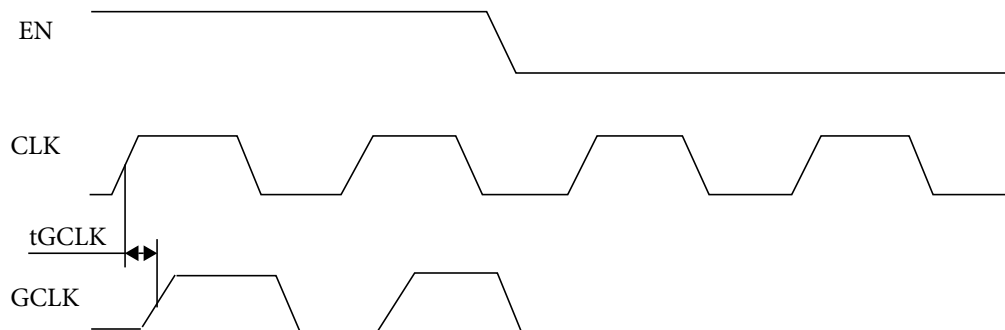
Macro Name:	GCNFNN1	GCNFNN2	GCNFNN4	GCNFNN7	GCNFNNA
Drive Capability:	1x	2x	4x	7x	10x
Gate Equivalents:	1.25	1.5	2.25	2.75	3.5
Leakage Power (pW):	43.6	81.5	146.2	242.9	322.2

Pin Description

Name	Capacitance (pF)					Description
	GCNFNN1	GCNFNN2	GCNFNN4	GCNFNN7	GCNFNNA	
EN	0.004	0.004	0.005	0.005	0.006	Enable
CLK	0.004	0.004	0.005	0.005	0.005	Clock Input
Maximum capacitance						
GCLK	0.3	0.6	1.2	2.1	3	Clock Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)				
	GCNFNN1	GCNFNN2	GCNFNN4	GCNFNN7	GCNFNNA
EN	0	0	0	0	0
CLK	0.001	0.001	0.002	0.002	0.002
GCLK	0.028	0.043	0.082	0.132	0.206

Waveforms

Timing Numbers for GCNFNN1:**Error Checks**

Pulse Width-Low	CLK (min)	0.319
Pulse Width-High	CLK(min)	0.414

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.131	0.121	0.166	0.157	0.232	0.227	0.363	0.365	0.624	0.642

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.166	0.157	0.218	0.214	0.256	0.257	0.288	0.292	0.310	0.317

Timing Numbers for GCNFNN2:**Error Checks**

Pulse Width-Low	CLK (min)	0.377
Pulse Width-High	CLK(min)	0.477

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.134	0.125	0.155	0.148	0.189	0.188	0.253	0.266	0.377	0.421

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.155	0.148	0.215	0.216	0.261	0.268	0.299	0.311	0.325	0.340

Timing Numbers for GCNFFNN4:**Error Checks**

Pulse Width-Low	CLK (min)	0.421
Pulse Width-High	CLK(min)	0.555

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.143	0.128	0.156	0.140	0.178	0.162	0.214	0.201	0.278	0.273

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.156	0.140	0.221	0.213	0.275	0.272	0.317	0.318	0.347	0.350

Timing Numbers for GCNFNN7:**Error Checks**

Pulse Width-Low	CLK (min)	0.488
Pulse Width-High	CLK(min)	0.683

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.165	0.140	0.175	0.149	0.191	0.165	0.219	0.192	0.265	0.238

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.175	0.149	0.244	0.228	0.306	0.295	0.355	0.348	0.388	0.384

Timing Numbers for GCNFNNA:**Error Checks**

Pulse Width-Low	CLK (min)	0.716
Pulse Width-High	CLK(min)	0.824

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.199	0.187	0.206	0.194	0.221	0.209	0.244	0.231	0.281	0.268

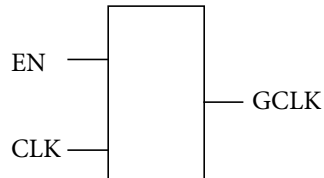
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.206	0.194	0.277	0.278	0.347	0.364	0.401	0.432	0.437	0.481

Latch-Free Clock Gating Cells

GCNRNN1 , GCNRNN2, GCNRNN4, GCNRNN7 and GCNRNNA

The GCNRNN1 (1x drive), GCNRNN2 (2x drive), GCNRNN4 (4x drive), GCNRNN7 (7x drive) and GCNRNNA (10x drive) cells are latch and flip-flop free gating logic. Logic appropriate for positive-edge-triggered registers.



Function Table

INPUTS		OUTPUT
EN	CLK	GCLK
L	X	H
H	L	L
H	H	H

Cell Description

Macro Name:	GCNRNN1	GCNRNN2	GCNRNN4	GCNRNN7	GCNRNNA
Drive Capability:	1x	2x	4x	7x	10x
Gate Equivalents:	1.5	2.	2.75	3.25	4.
Leakage Power (pW):	78.1	120.1	186.2	242.5	369.5

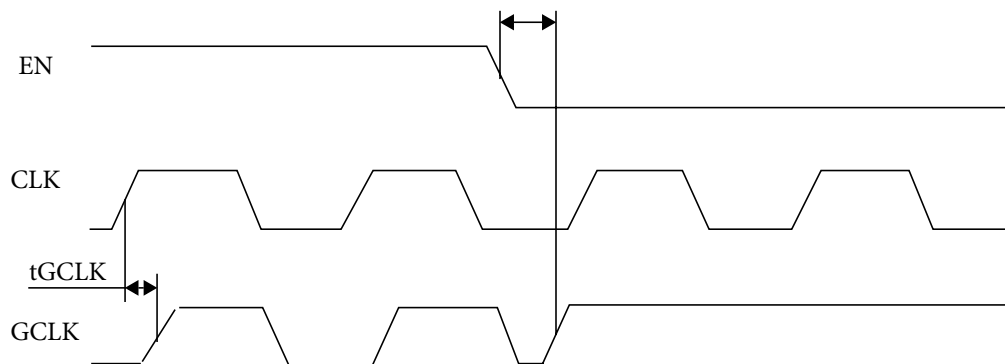
Pin Description

Name	Capacitance (pF)					Description
	GCNRNN1	GCNRNN2	GCNRNN4	GCNRNN7	GCNRNNA	
EN	0.006	0.003	0.004	0.004	0.004	Enable
CLK	0.004	0.005	0.005	0.005	0.005	Clock Input
Maximum capacitance						
GCLK	0.3	0.6	1.2	2.1	3	Clock Output

Pin Powers for Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)				
	GCNRNN1	GCNRNN2	GCNRNN4	GCNRNN7	GCNRNNA
EN	0	0.016	0.017	0.017	0.018
CLK	0.017	0.002	0.002	0.002	0.002
GCLK	0.017	0.053	0.086	0.137	0.262

Waveforms



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Timing Numbers for GCNRNN1:**Error Checks**

Pulse Width	CLK (min)	0.29
Pulse Width-High	CLK(min)	0.406

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.130	0.111	0.168	0.143	0.240	0.206	0.385	0.332	0.674	0.582

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.168	0.143	0.222	0.190	0.264	0.223	0.298	0.249	0.324	0.266

Timing Numbers for GCNRNN2:**Error Checks**

Pulse Width	CLK (min)	0.431
Pulse Width-High	CLK(min)	0.492

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.137	0.131	0.155	0.151	0.185	0.185	0.241	0.248	0.350	0.367

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.155	0.151	0.224	0.209	0.285	0.252	0.335	0.285	0.372	0.306

Timing Numbers for GCNRNN4:**Error Checks**

Pulse Width	CLK (min)	0.552
Pulse Width-High	CLK(min)	0.681

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.172	0.154	0.186	0.168	0.208	0.193	0.244	0.233	0.309	0.304

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.186	0.168	0.259	0.235	0.334	0.290	0.394	0.331	0.439	0.357

Timing Numbers for GCNRNN7:**Error Checks**

Pulse Width	CLK (min)	0.731
Pulse Width-High	CLK(min)	0.845

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.206	0.189	0.218	0.201	0.236	0.219	0.266	0.249	0.317	0.302

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.218	0.201	0.295	0.273	0.379	0.341	0.447	0.393	0.497	0.427

Timing Numbers for GCNRNNA:**Error Checks**

Pulse Width	CLK (min)	1.014
Pulse Width-High	CLK(min)	1.163

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.274	0.251	0.283	0.260	0.298	0.275	0.322	0.300	0.360	0.339

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tGCLK	0.283	0.260	0.361	0.334	0.456	0.417	0.540	0.485	0.598	0.529

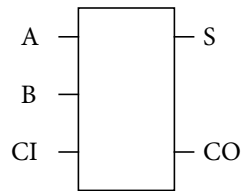
ADDERS

1-Bit Full Adders

AD01D0, AD01D1, AD01D2 and AD01D4

The AD01D0 (0.5x drive), AD01D1 (1x drive), AD01D2 (2x drive) and AD01D4 (4x drive) cells are a 1-bit full adders with buffered sum (S) and carry (CO) outputs.

Function Table



INPUTS			OUTPUTS	
CI	A	B	CO	S
L	L	L	L	L
L	L	H	L	H
L	H	L	L	H
L	H	H	H	L
H	L	L	L	H
H	L	H	H	L
H	H	L	H	L
H	H	H	H	H

Cell Description

Macro Name:	AD01D0	AD01D1	AD01D2	AD01D4
Drive Capability:	0.5x	1x	2x	4x
Gate Equivalents:	4.75	4.75	5.	6.75
Leakage Power (pW):	99.1	115.2	146.3	247.2

Pin Description

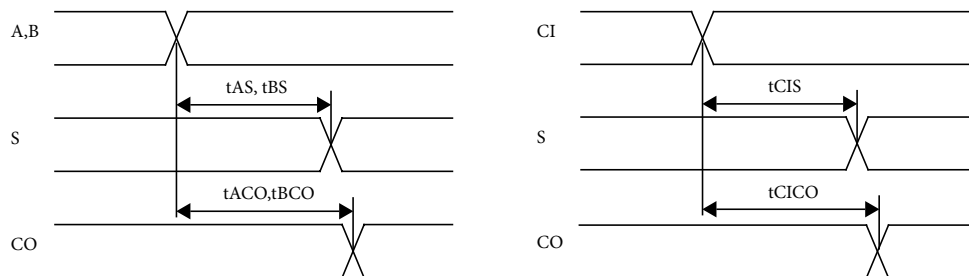
Name	Capacitance (pF)				Description
	AD01D0	AD01D1	AD01D2	AD01D4	
A	0.017	0.017	0.017	0.015	Data Input
B	0.017	0.017	0.017	0.015	Data Input
CI	0.013	0.013	0.013	0.011	Carry Input
Maximum capacitance					
S	0.15	0.3	0.6	1.2	Sum Output
CO	0.15	0.3	0.6	1.2	Carry Output

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Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)			
	AD01D0	AD01D1	AD01D2	AD01D4
S	0.034	0.042	0.056	0.112
CO	0.034	0.042	0.055	0.108

Waveforms



Timing Numbers for AD01D0:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.187	0.310	0.277	0.369	0.456	0.469	0.814	0.655	1.529	1.022
tBS	0.192	0.320	0.283	0.379	0.462	0.479	0.821	0.665	1.538	1.033
tCIS	0.185	0.285	0.275	0.344	0.454	0.444	0.812	0.630	1.527	0.998
tACO	0.211	0.313	0.303	0.376	0.482	0.478	0.841	0.664	1.558	1.031
tBCO	0.205	0.314	0.296	0.377	0.476	0.479	0.835	0.665	1.552	1.032
tCICO	0.197	0.275	0.289	0.336	0.468	0.437	0.827	0.623	1.544	0.990

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.277	0.369	0.305	0.422	0.308	0.496	0.302	0.571	0.283	0.641
tBS	0.283	0.379	0.312	0.419	0.319	0.471	0.316	0.529	0.302	0.588
tCIS	0.275	0.344	0.293	0.423	0.287	0.521	0.272	0.610	0.245	0.690
tACO	0.303	0.376	0.351	0.457	0.377	0.564	0.390	0.663	0.390	0.747
tBCO	0.296	0.377	0.329	0.434	0.343	0.511	0.347	0.587	0.338	0.657
tCICO	0.289	0.336	0.333	0.418	0.350	0.517	0.356	0.605	0.348	0.680

Timing Numbers for AD01D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.144	0.302	0.190	0.342	0.282	0.408	0.463	0.520	0.825	0.730
tBS	0.148	0.311	0.195	0.351	0.288	0.417	0.470	0.530	0.832	0.739
tCIS	0.142	0.277	0.189	0.317	0.280	0.383	0.462	0.495	0.824	0.704
tACO	0.166	0.301	0.211	0.341	0.300	0.409	0.474	0.519	0.821	0.718
tBCO	0.160	0.301	0.205	0.341	0.294	0.409	0.468	0.519	0.815	0.718
tCICO	0.152	0.261	0.197	0.301	0.286	0.367	0.460	0.477	0.807	0.675

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.190	0.342	0.221	0.393	0.227	0.468	0.222	0.543	0.204	0.613
tBS	0.195	0.351	0.226	0.391	0.236	0.443	0.235	0.500	0.222	0.559
tCIS	0.189	0.317	0.208	0.395	0.205	0.495	0.192	0.587	0.166	0.668
tACO	0.211	0.341	0.263	0.422	0.291	0.530	0.307	0.630	0.308	0.714
tBCO	0.205	0.341	0.240	0.398	0.256	0.475	0.262	0.551	0.253	0.621
tCICO	0.197	0.301	0.244	0.382	0.265	0.483	0.274	0.572	0.266	0.648

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Timing Numbers for AD01D2:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.134	0.319	0.160	0.349	0.207	0.393	0.300	0.465	0.484	0.583
tBS	0.138	0.329	0.164	0.359	0.212	0.403	0.305	0.475	0.489	0.593
tCIS	0.130	0.295	0.156	0.325	0.204	0.369	0.297	0.441	0.480	0.559
tACO	0.156	0.305	0.184	0.336	0.233	0.382	0.329	0.458	0.518	0.580
tBCO	0.149	0.304	0.178	0.335	0.227	0.381	0.323	0.457	0.512	0.579
tCICO	0.141	0.265	0.169	0.295	0.218	0.339	0.313	0.415	0.502	0.536

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.160	0.349	0.195	0.400	0.208	0.475	0.209	0.551	0.195	0.622
tBS	0.164	0.359	0.199	0.398	0.215	0.449	0.220	0.505	0.210	0.564
tCIS	0.156	0.325	0.181	0.400	0.185	0.504	0.178	0.601	0.157	0.683
tACO	0.184	0.336	0.241	0.415	0.275	0.524	0.296	0.627	0.301	0.712
tBCO	0.178	0.335	0.215	0.391	0.237	0.468	0.246	0.544	0.240	0.613
tCICO	0.169	0.295	0.222	0.376	0.250	0.478	0.263	0.570	0.260	0.646

Timing Numbers for AD01D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.230	0.433	0.246	0.454	0.273	0.491	0.315	0.549	0.387	0.649
tBS	0.245	0.446	0.261	0.467	0.289	0.505	0.331	0.563	0.405	0.663
tCIS	0.216	0.407	0.232	0.428	0.259	0.465	0.301	0.523	0.375	0.623
tACO	0.243	0.295	0.259	0.314	0.287	0.347	0.330	0.399	0.405	0.489
tBCO	0.232	0.303	0.248	0.322	0.276	0.355	0.319	0.407	0.394	0.497
tCICO	0.221	0.272	0.237	0.291	0.265	0.323	0.307	0.375	0.382	0.463

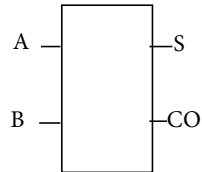
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.246	0.454	0.293	0.504	0.341	0.571	0.377	0.639	0.397	0.700
tBS	0.261	0.467	0.310	0.506	0.361	0.548	0.402	0.591	0.428	0.633
tCIS	0.232	0.428	0.276	0.497	0.316	0.593	0.347	0.688	0.361	0.765
tACO	0.259	0.314	0.328	0.389	0.404	0.484	0.464	0.568	0.505	0.633
tBCO	0.248	0.322	0.295	0.372	0.347	0.434	0.389	0.491	0.418	0.539
tCICO	0.237	0.291	0.307	0.368	0.378	0.459	0.433	0.539	0.466	0.598

1-Bit Half Adders

AH01D0, AH01D1, AH01D2 and AH01D4

The AH01D0 (0.5x drive), AH01D1 (1x drive), AH01D2 (2x drive) and AH01D4 (4x drive) cells are a 1-bit half adders with buffered sum (S) and carry (CO) outputs.



Function Table

INPUTS		OUTPUTS	
A	B	CO	S
L	L	L	L
L	H	L	H
H	L	L	H
H	H	H	L

Cell Description

Macro Name:	AH01D0	AH01D1	AH01D2	AH01D4
Drive Capability:	0.5x	1x	2x	4x
Gate Equivalents:	3.	2.75	3.5	4.
Leakage Power (pW):	60.5	75.8	107.3	219.4

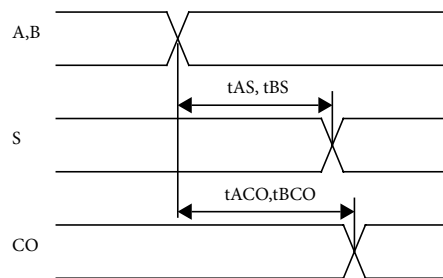
Pin Description

Name	Capacitance (pF)				Description
	AH01D0	AH01D1	AH01D2	AH01D4	
A	0.006	0.006	0.006	0.007	Data Input
B	0.006	0.006	0.006	0.007	Data Input
Maximum capacitance					
S	0.15	0.3	0.6	1.2	Sum Output
CO	0.15	0.3	0.6	1.2	Carry Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)			
	AH01D0	AH01D1	AH01D2	AH01D4
S	0.021	0.028	0.041	0.091
CO	0.019	0.027	0.04	0.086

Waveforms



Timing Numbers for AH01D0:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.209	0.218	0.314	0.276	0.523	0.382	0.941	0.589	1.775	1.002
tBS	0.206	0.233	0.312	0.290	0.521	0.396	0.939	0.603	1.773	1.016
tACO	0.193	0.202	0.299	0.261	0.507	0.367	0.925	0.573	1.759	0.985
tBCO	0.194	0.223	0.299	0.282	0.508	0.389	0.926	0.596	1.760	1.009

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.314	0.276	0.347	0.355	0.360	0.437	0.364	0.509	0.358	0.571
tBS	0.312	0.290	0.335	0.345	0.342	0.412	0.341	0.477	0.330	0.536
tACO	0.299	0.261	0.327	0.353	0.330	0.451	0.323	0.537	0.305	0.611
tBCO	0.299	0.282	0.320	0.378	0.319	0.483	0.308	0.577	0.285	0.656

Timing Numbers for AH01D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.157	0.205	0.206	0.241	0.302	0.304	0.493	0.417	0.874	0.637
tBS	0.152	0.221	0.201	0.257	0.297	0.319	0.488	0.433	0.870	0.652
tACO	0.141	0.195	0.191	0.232	0.289	0.296	0.481	0.409	0.864	0.629
tBCO	0.141	0.213	0.191	0.251	0.289	0.315	0.481	0.429	0.864	0.649

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.206	0.241	0.241	0.322	0.258	0.408	0.266	0.482	0.261	0.546
tBS	0.201	0.257	0.227	0.312	0.237	0.380	0.238	0.445	0.228	0.504
tACO	0.191	0.232	0.226	0.325	0.234	0.427	0.232	0.516	0.216	0.592
tBCO	0.191	0.251	0.216	0.346	0.218	0.456	0.211	0.552	0.189	0.633

Timing Numbers for AH01D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.153	0.222	0.182	0.248	0.234	0.288	0.332	0.355	0.528	0.472
tBS	0.144	0.236	0.172	0.263	0.223	0.303	0.322	0.369	0.518	0.486
tACO	0.132	0.210	0.160	0.237	0.211	0.277	0.310	0.345	0.506	0.462
tBCO	0.133	0.229	0.161	0.257	0.213	0.297	0.312	0.366	0.508	0.484

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.182	0.248	0.224	0.330	0.249	0.423	0.264	0.504	0.264	0.570
tBS	0.172	0.263	0.204	0.318	0.221	0.388	0.227	0.455	0.221	0.516
tACO	0.160	0.237	0.206	0.330	0.225	0.440	0.230	0.535	0.219	0.614
tBCO	0.161	0.257	0.194	0.353	0.204	0.467	0.203	0.569	0.186	0.653

Timing Numbers for AH01D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

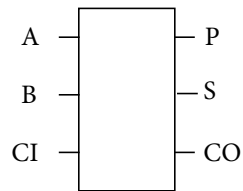
Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.205	0.321	0.221	0.340	0.249	0.373	0.291	0.426	0.366	0.516
tBS	0.222	0.338	0.238	0.357	0.266	0.390	0.308	0.443	0.383	0.533
tACO	0.178	0.151	0.192	0.165	0.218	0.191	0.258	0.231	0.327	0.308
tBCO	0.178	0.155	0.193	0.169	0.218	0.195	0.258	0.236	0.328	0.313

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.221	0.340	0.281	0.420	0.337	0.520	0.379	0.613	0.402	0.684
tBS	0.238	0.357	0.300	0.411	0.362	0.477	0.410	0.541	0.440	0.593
tACO	0.192	0.165	0.260	0.244	0.325	0.312	0.374	0.365	0.410	0.403
tBCO	0.193	0.169	0.239	0.250	0.284	0.323	0.321	0.380	0.347	0.423

1-Bit Full Adders with Propagate ADP1D0, ADP1D1, ADP1D2 and ADP1D4

The ADP1D0 (0.5x drive), ADP1D1 (1x drive), ADP1D2 (2x drive) and ADP1D4 (4x drive) cells are a 1-bit full adders with buffered sum (S), carry (CO), and propagate (P) outputs.



Function Table

INPUTS			OUTPUTS		
CI	A	B	CO	S	P
L	L	L	L	L	L
L	L	H	L	H	H
L	H	L	L	H	H
L	H	H	H	L	L
H	L	L	L	H	L
H	L	H	H	L	H
H	H	L	H	L	H
H	H	H	H	H	L

Cell Description

Macro Name:	ADP1D0	ADP1D1	ADP1D2	ADP1D4
Drive Capability:	0.5x	1x	2x	4x
Gate Equivalents:	6.75	6.75	7.	9.
Leakage Power (pW):	179.2	192.6	209.6	461.3

Pin Description

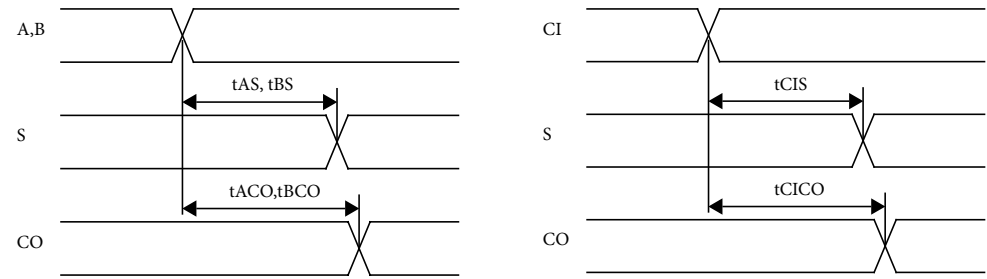
Name	Capacitance (pF)				Description
	ADP1D0	ADP1D1	ADP1D2	ADP1D4	
A	0.01	0.01	0.01	0.011	Data Input
B	0.011	0.011	0.011	0.011	Data Input
CI	0.013	0.013	0.013	0.011	Carry Input
Maximum capacitance					
S	0.15	0.3	0.6	1.2	Sum Output
CO	0.15	0.3	0.6	1.2	Carry Output
P	0.15	0.3	0.6	1.2	Propagate Output

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Pin Powers for
Standard Load = 0.032 pF Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)			
	ADP1D0	ADP1D1	ADP1D2	ADP1D4
S	0.049	0.054	0.056	0.135
CO	0.039	0.043	0.047	0.119
P	0.054	0.059	0.062	0.148

Waveforms



Timing numbers for ADP1D0:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.411	0.375	0.456	0.406	0.545	0.459	0.721	0.559	1.072	0.757
tBS	0.426	0.395	0.471	0.426	0.560	0.479	0.736	0.579	1.087	0.777
tCIS	0.160	0.177	0.206	0.209	0.294	0.263	0.470	0.364	0.820	0.563
tACO	0.449	0.380	0.581	0.436	0.844	0.547	1.369	0.768	2.421	1.210
tBCO	0.455	0.400	0.587	0.457	0.851	0.567	1.376	0.789	2.425	1.232
tCICO	0.227	0.215	0.359	0.276	0.622	0.389	1.147	0.612	2.196	1.055
tAP	0.184	0.180	0.233	0.210	0.324	0.255	0.506	0.329	0.869	0.473
tBP	0.200	0.201	0.249	0.231	0.339	0.277	0.521	0.351	0.884	0.495

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.456	0.406	0.491	0.483	0.518	0.563	0.538	0.634	0.543	0.692
tBS	0.471	0.426	0.508	0.474	0.542	0.529	0.567	0.580	0.581	0.624
tCIS	0.206	0.209	0.241	0.256	0.266	0.307	0.283	0.355	0.289	0.397
tACO	0.581	0.436	0.615	0.515	0.642	0.599	0.659	0.673	0.664	0.734
tBCO	0.587	0.457	0.625	0.507	0.659	0.564	0.684	0.618	0.697	0.664
tCICO	0.359	0.276	0.382	0.356	0.387	0.440	0.384	0.513	0.372	0.578
tAP	0.233	0.210	0.268	0.287	0.296	0.366	0.314	0.434	0.319	0.492
tBP	0.249	0.231	0.286	0.279	0.319	0.332	0.344	0.382	0.357	0.425

Timing numbers for ADP1D1:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.410	0.375	0.454	0.404	0.543	0.458	0.720	0.558	1.070	0.755
tBS	0.425	0.395	0.470	0.424	0.559	0.478	0.735	0.578	1.085	0.775
tCIS	0.161	0.178	0.205	0.209	0.294	0.263	0.471	0.364	0.821	0.560
tACO	0.385	0.354	0.442	0.376	0.555	0.413	0.781	0.480	1.230	0.609
tBCO	0.393	0.375	0.449	0.397	0.563	0.434	0.788	0.501	1.237	0.630
tCICO	0.162	0.185	0.219	0.211	0.332	0.254	0.558	0.325	1.006	0.456
tAP	0.184	0.180	0.233	0.208	0.324	0.255	0.506	0.329	0.870	0.465
tBP	0.200	0.202	0.248	0.230	0.339	0.277	0.521	0.351	0.886	0.487

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.454	0.404	0.489	0.482	0.517	0.562	0.535	0.632	0.541	0.691
tBS	0.470	0.424	0.506	0.473	0.540	0.528	0.565	0.578	0.579	0.623
tCIS	0.205	0.209	0.241	0.255	0.266	0.307	0.283	0.355	0.289	0.396
tACO	0.442	0.376	0.476	0.454	0.503	0.539	0.520	0.613	0.526	0.674
tBCO	0.449	0.397	0.487	0.446	0.521	0.504	0.546	0.558	0.559	0.604
tCICO	0.219	0.211	0.247	0.293	0.257	0.382	0.259	0.460	0.248	0.528
tAP	0.233	0.208	0.268	0.286	0.295	0.365	0.313	0.434	0.318	0.491
tBP	0.248	0.230	0.286	0.278	0.319	0.332	0.343	0.381	0.356	0.425

Timing numbers for ADP1D2:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.411	0.376	0.457	0.406	0.545	0.459	0.721	0.560	1.071	0.756
tBS	0.425	0.395	0.471	0.426	0.559	0.479	0.736	0.579	1.086	0.776
tCIS	0.160	0.177	0.206	0.209	0.294	0.263	0.471	0.363	0.821	0.560
tACO	0.368	0.373	0.401	0.392	0.462	0.423	0.583	0.476	0.823	0.571
tBCO	0.376	0.393	0.408	0.412	0.469	0.443	0.590	0.496	0.831	0.591
tCICO	0.141	0.205	0.175	0.230	0.235	0.266	0.356	0.324	0.597	0.424
tAP	0.185	0.180	0.234	0.210	0.324	0.255	0.506	0.331	0.871	0.466
tBP	0.200	0.201	0.250	0.232	0.340	0.276	0.521	0.352	0.886	0.488

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.457	0.406	0.490	0.484	0.518	0.564	0.538	0.634	0.543	0.693
tBS	0.471	0.426	0.508	0.475	0.542	0.529	0.567	0.580	0.580	0.625
tCIS	0.206	0.209	0.242	0.256	0.266	0.308	0.284	0.357	0.290	0.398
tACO	0.401	0.392	0.435	0.470	0.462	0.555	0.480	0.629	0.485	0.691
tBCO	0.408	0.412	0.446	0.461	0.480	0.520	0.506	0.574	0.519	0.620
tCICO	0.175	0.230	0.206	0.313	0.221	0.408	0.225	0.491	0.215	0.562
tAP	0.234	0.210	0.269	0.287	0.297	0.367	0.316	0.436	0.321	0.494
tBP	0.250	0.232	0.287	0.280	0.321	0.334	0.346	0.383	0.358	0.427

Timing numbers for ADP1D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.522	0.749	0.533	0.761	0.552	0.784	0.590	0.828	0.657	0.908
tBS	0.539	0.777	0.550	0.788	0.570	0.811	0.607	0.855	0.674	0.935
tCIS	0.209	0.272	0.224	0.291	0.251	0.322	0.291	0.372	0.362	0.458
tACO	0.527	0.714	0.538	0.725	0.560	0.746	0.600	0.787	0.670	0.864
tBCO	0.541	0.742	0.552	0.753	0.572	0.773	0.612	0.814	0.682	0.891
tCICO	0.214	0.284	0.230	0.303	0.258	0.336	0.301	0.388	0.375	0.476
tAP	0.243	0.415	0.260	0.438	0.290	0.477	0.336	0.537	0.413	0.635
tBP	0.260	0.442	0.277	0.465	0.307	0.505	0.354	0.565	0.431	0.663

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

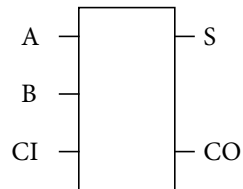
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.533	0.761	0.578	0.841	0.629	0.947	0.672	1.054	0.695	1.143
tBS	0.550	0.788	0.596	0.842	0.649	0.914	0.694	0.987	0.721	1.052
tCIS	0.224	0.291	0.269	0.342	0.316	0.403	0.353	0.458	0.377	0.504
tACO	0.538	0.725	0.583	0.805	0.633	0.912	0.675	1.020	0.697	1.111
tBCO	0.552	0.753	0.598	0.806	0.651	0.878	0.695	0.952	0.722	1.019
tCICO	0.230	0.303	0.276	0.382	0.324	0.477	0.362	0.563	0.385	0.629
tAP	0.260	0.438	0.305	0.518	0.353	0.625	0.390	0.732	0.408	0.818
tBP	0.277	0.465	0.323	0.519	0.373	0.591	0.413	0.663	0.436	0.726

SUBTRACTORS

1-Bit Full Subtractors

SU01D0, SU01D1, SU01D2 and SU01D4

The SU01D0 (0.5x drive), SU01D1(1x drive), SU01D2(2x drive) and SU01D4 (4x drive) cells are a 1-bit full subtractor with buffered sum (S) and carry (CO) outputs.



Function Table

INPUTS			OUTPUTS	
CI	A	B	CO	S
L	L	L	L	H
L	L	H	L	L
L	H	L	H	L
L	H	H	L	H
H	L	L	H	L
H	L	H	L	H
H	H	L	H	H
H	H	H	H	L

Cell Description

Macro Name:	SU01D0	SU01D1	SU01D2	SU01D4
Drive Capability:	0.5x	1x	2x	4x
Gate Equivalents:	5.	5.	5.5	6.5
Leakage Power (pW):	100.5	118.1	153.2	234.4

Pin Description

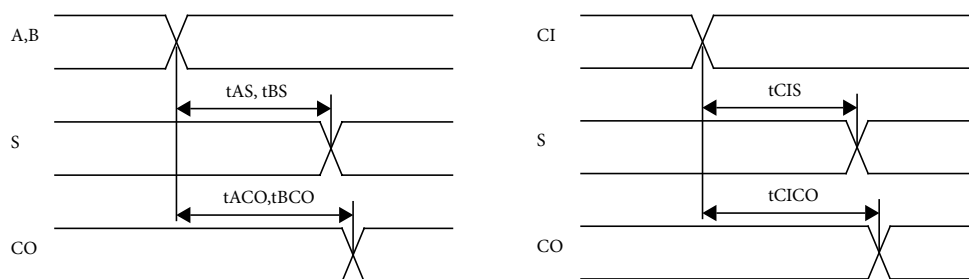
Name	Capacitance (pF)				Description
	SU01D0	SU01D1	SU01D2	SU01D4	
A	0.016	0.016	0.016	0.015	Data Input
B	0.004	0.004	0.004	0.003	Data Input
CI	0.011	0.011	0.011	0.011	Carry Input
Maximum capacitance					
S	0.15	0.3	0.6	1.2	Sum Output
CO	0.15	0.3	0.6	1.2	Carry Output

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Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)			
	SU01D0	SU01D1	SU01D2	SU01D4
S	0.041	0.049	0.066	0.113
CO	0.038	0.046	0.063	0.109

Waveforms



Timing numbers for SU01D0:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.405	0.250	0.510	0.306	0.719	0.404	1.137	0.588	1.971	0.953
tBS	0.499	0.410	0.604	0.471	0.813	0.574	1.231	0.761	2.065	1.129
tCIS	0.392	0.236	0.497	0.288	0.706	0.381	1.124	0.564	1.958	0.927
tACO	0.216	0.279	0.323	0.338	0.533	0.438	0.951	0.623	1.785	0.990
tBCO	0.349	0.371	0.456	0.432	0.665	0.534	1.083	0.720	1.917	1.087
tCICO	0.208	0.275	0.315	0.336	0.524	0.438	0.942	0.624	1.776	0.991

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.510	0.306	0.562	0.348	0.624	0.364	0.686	0.367	0.745	0.358
tBS	0.604	0.471	0.659	0.523	0.689	0.551	0.706	0.565	0.754	0.565
tCIS	0.497	0.288	0.579	0.319	0.672	0.326	0.755	0.322	0.828	0.305
tACO	0.323	0.338	0.367	0.420	0.387	0.522	0.394	0.612	0.389	0.690
tBCO	0.456	0.432	0.542	0.487	0.630	0.517	0.705	0.535	0.765	0.538
tCICO	0.315	0.336	0.355	0.419	0.367	0.520	0.368	0.610	0.355	0.687

Timing numbers for SU01D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.373	0.242	0.416	0.277	0.502	0.336	0.676	0.437	1.024	0.626
tBS	0.468	0.403	0.510	0.443	0.597	0.509	0.770	0.616	1.118	0.810
tCIS	0.360	0.232	0.403	0.263	0.489	0.316	0.663	0.412	1.011	0.599
tACO	0.157	0.266	0.204	0.304	0.297	0.368	0.480	0.475	0.842	0.674
tBCO	0.290	0.358	0.337	0.398	0.430	0.464	0.613	0.574	0.975	0.774
tCICO	0.150	0.262	0.197	0.302	0.289	0.368	0.472	0.478	0.834	0.678

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.416	0.277	0.468	0.322	0.533	0.342	0.596	0.349	0.657	0.342
tBS	0.510	0.443	0.565	0.496	0.595	0.523	0.612	0.538	0.634	0.538
tCIS	0.403	0.263	0.486	0.298	0.583	0.310	0.669	0.309	0.745	0.295
tACO	0.204	0.304	0.252	0.385	0.275	0.488	0.285	0.580	0.281	0.659
tBCO	0.337	0.398	0.425	0.452	0.513	0.482	0.587	0.500	0.648	0.503
tCICO	0.197	0.302	0.240	0.384	0.256	0.486	0.259	0.578	0.247	0.656

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Timing numbers for SU01D2:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.435	0.273	0.458	0.297	0.499	0.334	0.584	0.396	0.756	0.499
tBS	0.519	0.427	0.542	0.457	0.582	0.501	0.667	0.572	0.839	0.685
tCIS	0.423	0.262	0.446	0.283	0.487	0.316	0.572	0.372	0.744	0.469
tACO	0.144	0.274	0.170	0.301	0.216	0.341	0.305	0.406	0.481	0.512
tBCO	0.278	0.357	0.304	0.386	0.349	0.427	0.438	0.495	0.615	0.604
tCICO	0.136	0.273	0.162	0.302	0.207	0.343	0.296	0.411	0.473	0.521

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.458	0.297	0.510	0.349	0.579	0.377	0.647	0.391	0.710	0.389
tBS	0.542	0.457	0.589	0.502	0.611	0.521	0.621	0.528	0.625	0.521
tCIS	0.446	0.283	0.527	0.327	0.631	0.348	0.728	0.355	0.808	0.346
tACO	0.170	0.301	0.224	0.381	0.253	0.486	0.269	0.583	0.268	0.663
tBCO	0.304	0.386	0.392	0.433	0.483	0.455	0.562	0.465	0.627	0.460
tCICO	0.162	0.302	0.211	0.383	0.234	0.487	0.244	0.584	0.236	0.664

Timing numbers for SU01D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.564	0.519	0.578	0.537	0.601	0.567	0.638	0.616	0.706	0.702
tBS	0.648	0.524	0.662	0.545	0.684	0.583	0.720	0.641	0.787	0.741
tCIS	0.553	0.501	0.567	0.517	0.590	0.546	0.626	0.591	0.694	0.672
tACO	0.242	0.344	0.258	0.365	0.287	0.401	0.331	0.457	0.407	0.551
tBCO	0.339	0.449	0.355	0.470	0.383	0.505	0.427	0.561	0.503	0.656
tCICO	0.225	0.284	0.241	0.303	0.269	0.336	0.313	0.389	0.389	0.479

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

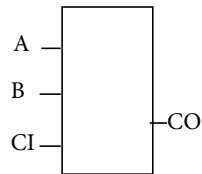
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tAS	0.578	0.537	0.628	0.605	0.686	0.683	0.741	0.751	0.787	0.799
tBS	0.662	0.545	0.728	0.611	0.779	0.660	0.818	0.695	0.842	0.718
tCIS	0.567	0.517	0.642	0.585	0.734	0.659	0.821	0.721	0.886	0.762
tACO	0.258	0.365	0.328	0.442	0.403	0.543	0.462	0.640	0.502	0.715
tBCO	0.355	0.470	0.433	0.536	0.502	0.587	0.557	0.626	0.599	0.651
tCICO	0.241	0.303	0.310	0.379	0.381	0.472	0.435	0.553	0.469	0.614

CARRY GENERATORS

1-Bit Carry Generators

CG01D0, CG01D1, CG01D2 and CG01D4

The CG01D0(0.5 x drive), CG01D1 (1x drive), CG01D2 (2x drive) and CG01D4 (4x drive) cells are a 1-bit carry generators with carry (CO) output.



Function Table

INPUTS			OUTPUTS
A	B	CI	CO
L	L	L	L
L	H	L	L
H	L	L	L
H	H	L	H
L	L	H	L
L	H	H	H
H	L	H	H
H	H	H	H

Cell Description

Macro Name:	CG01D0	CG01D1	CG01D2	CG01D4
Drive Capability:	0.5x	1x	2x	4x
Gate Equivalents:	2.25	2.25	2.5	3.
Leakage Power (pW)	40.0	49.2	76.5	161.7

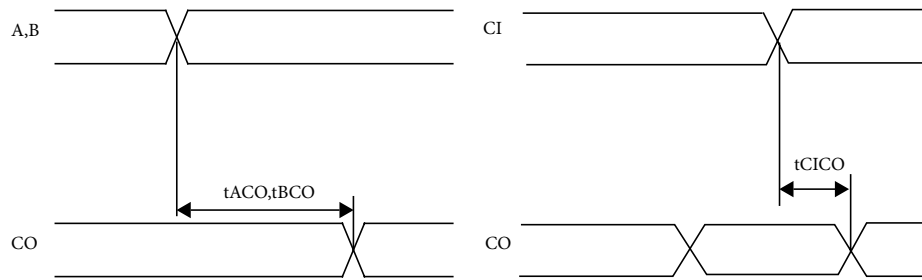
Pin Description

Name	Capacitance (pF)				Description
	CG01D0	CG01D1	CG01D2	CG01D4	
A	0.006	0.006	0.006	0.009	Data Input
B	0.006	0.006	0.006	0.008	Data Input
CI	0.003	0.003	0.003	0.004	Carry Input
Maximum capacitanc					
CO	0.15	0.3	0.6	1.2	Carry Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)			
	CG01D0	CG01D1	CG01D2	CG01D4
A	0.001	0.001	0.001	0.002
B	-0	-0	-0	-0
CI	0.002	0.002	0.002	0.003
CO	0.02	0.027	0.043	0.091

Waveforms



Timing numbers for CG01D0:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tACO	0.225	0.203	0.326	0.256	0.528	0.354	0.931	0.546	1.735	0.930
tBCO	0.217	0.213	0.319	0.266	0.521	0.364	0.923	0.556	1.727	0.940
tCICO	0.211	0.186	0.312	0.239	0.514	0.338	0.916	0.529	1.720	0.912

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Fanout	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tACO	0.326	0.256	0.379	0.332	0.413	0.410	0.435	0.477	0.446	0.536
tBCO	0.319	0.266	0.354	0.317	0.377	0.374	0.390	0.429	0.394	0.477
tCICO	0.312	0.239	0.360	0.314	0.386	0.386	0.402	0.449	0.406	0.503

Timing numbers for CG01D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tACO	0.183	0.203	0.232	0.238	0.328	0.298	0.518	0.406	0.896	0.618
tBCO	0.176	0.214	0.225	0.249	0.320	0.308	0.510	0.417	0.888	0.629
tCICO	0.169	0.185	0.218	0.220	0.313	0.280	0.503	0.389	0.880	0.600

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transitio	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tACO	0.232	0.238	0.289	0.316	0.328	0.400	0.355	0.473	0.369	0.535
tBCO	0.225	0.249	0.262	0.301	0.289	0.361	0.306	0.417	0.311	0.467
tCICO	0.218	0.220	0.271	0.298	0.303	0.377	0.324	0.446	0.330	0.503

Timing numbers for CG01D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tACO	0.154	0.238	0.181	0.263	0.230	0.301	0.323	0.365	0.508	0.472
tBCO	0.162	0.245	0.190	0.270	0.238	0.308	0.332	0.372	0.517	0.480
tCICO	0.154	0.218	0.182	0.243	0.230	0.281	0.324	0.345	0.509	0.453

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tACO	0.181	0.263	0.238	0.342	0.274	0.438	0.296	0.521	0.302	0.590
tBCO	0.190	0.270	0.236	0.328	0.270	0.394	0.291	0.457	0.298	0.516
tCICO	0.182	0.243	0.235	0.325	0.268	0.421	0.287	0.503	0.289	0.572

Timing numbers for CG01D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tACO	0.223	0.229	0.238	0.246	0.264	0.275	0.304	0.321	0.375	0.402
tBCO	0.209	0.235	0.224	0.252	0.250	0.281	0.290	0.328	0.361	0.408
tCICO	0.201	0.213	0.215	0.230	0.241	0.259	0.281	0.305	0.352	0.386

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

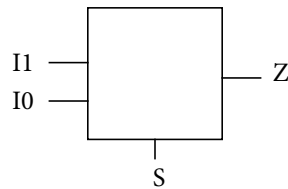
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tACO	0.238	0.246	0.305	0.321	0.377	0.405	0.434	0.474	0.474	0.528
tBCO	0.224	0.252	0.270	0.302	0.321	0.357	0.362	0.404	0.391	0.443
tCICO	0.215	0.230	0.283	0.306	0.351	0.386	0.402	0.452	0.436	0.502

MULTIPLEXERS

2-to-1 Multiplexers

MX02D0, MX02D1, MX02D2 and MX02D4

The MX02D0 (0.5x drive), MX02D1 (1x drive), MX02D2 (2x drive) and MX02D4 (4x drive) cells are 2-to-1 multiplexers. The state of the select input determines which data is present at the output.



Function Table

INPUTS			OUTPUT
S	I0	I1	Z
L	L	X	L
L	H	X	H
H	X	L	L
H	X	H	H

Cell Description

Macro Name:	MX02D0	MX02D1	MX02D2	MX02D4
Drive Capability:	0.5x	1x	2x	4x
Gate Equivalents:	2.25	2.	2.25	2.5
Leakage Power (pW):	44.8	51.8	78.0	98.3

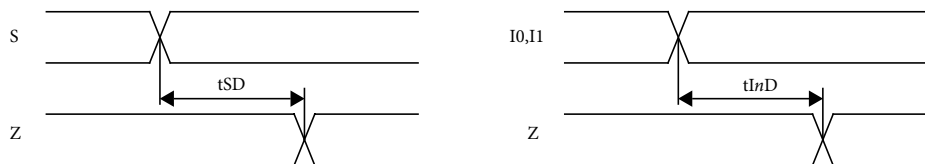
Pin Description

Name	Capacitance (pF)				Description
	MX02D0	MX02D1	MX02D2	MX02D4	
S	0.004	0.006	0.006	0.006	Select Input
I0	0.003	0.004	0.004	0.004	Data Input
I1	0.003	0.004	0.004	0.004	Data Input
Maximum capacitance					
Z	0.15	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)			
	MX02D0	MX02D1	MX02D2	MX02D4
S	0.009	0.013	0.013	0.013
I0	0	0	0	0
I1	0	0	0	0
Z	0.023	0.031	0.046	0.064

Waveforms



Timing numbers for MX02D0:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.175	0.268	0.258	0.312	0.423	0.382	0.753	0.506	1.412	0.750
tI1D	0.172	0.248	0.255	0.290	0.420	0.358	0.750	0.481	1.409	0.721
tSD	0.249	0.258	0.332	0.300	0.497	0.367	0.827	0.490	1.486	0.733

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.258	0.312	0.275	0.370	0.270	0.452	0.255	0.531	0.230	0.605
tI1D	0.255	0.290	0.275	0.347	0.273	0.426	0.261	0.503	0.238	0.574
tSD	0.332	0.300	0.415	0.314	0.493	0.309	0.563	0.299	0.625	0.279

Timing numbers for MX02D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.153	0.192	0.209	0.223	0.318	0.278	0.537	0.377	0.972	0.568
tI1D	0.147	0.196	0.203	0.228	0.312	0.283	0.531	0.382	0.966	0.573
tSD	0.234	0.214	0.290	0.246	0.399	0.301	0.617	0.400	1.053	0.591

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.209	0.223	0.234	0.275	0.244	0.337	0.246	0.397	0.238	0.451
tI1D	0.203	0.228	0.230	0.281	0.238	0.346	0.238	0.409	0.226	0.466
tSD	0.290	0.246	0.374	0.268	0.454	0.270	0.525	0.264	0.586	0.249

Timing numbers for MX02D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.134	0.199	0.162	0.221	0.214	0.255	0.315	0.312	0.516	0.408
tI1D	0.132	0.204	0.159	0.226	0.211	0.260	0.312	0.317	0.513	0.413
tSD	0.216	0.221	0.244	0.243	0.296	0.277	0.396	0.334	0.598	0.430

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.162	0.221	0.191	0.273	0.206	0.337	0.211	0.398	0.205	0.454
tI1D	0.159	0.226	0.193	0.279	0.210	0.344	0.216	0.406	0.210	0.463
tSD	0.244	0.243	0.330	0.266	0.411	0.269	0.483	0.263	0.546	0.247

Timing numbers for MX02D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.140	0.222	0.160	0.240	0.196	0.270	0.265	0.314	0.401	0.387
tI1D	0.139	0.227	0.159	0.245	0.196	0.275	0.265	0.319	0.401	0.393
tSD	0.223	0.245	0.243	0.264	0.279	0.293	0.348	0.337	0.484	0.411

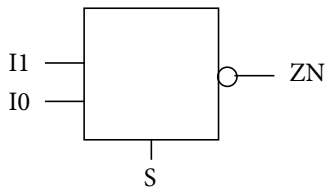
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.160	0.240	0.194	0.293	0.213	0.358	0.223	0.421	0.220	0.479
tI1D	0.159	0.245	0.197	0.299	0.219	0.366	0.230	0.430	0.227	0.489
tSD	0.243	0.264	0.329	0.287	0.411	0.291	0.484	0.286	0.547	0.271

2-to-1 Multiplexers with Inverted Output

MI02D0, MI02D1, MI02D2 and MI02D4

The MI02D0 (0.5x drive), MI02D1 (1x drive), MI02D2 (2x drive) and MI02D4 (4x drive) cells are 2-to-1 multiplexers with an inverted output. The state of the select inputs determines which data is present at the output.



Function Table

INPUTS			OUTPUT
S	I0	I1	ZN
L	L	X	H
L	H	X	L
H	X	L	H
H	X	H	L

Cell Description

Macro Name:	MI02D0	MI02D1	MI02D2	MI02D4
Drive Capability:	0.5x	1x	2x	4x
Gate Equivalents:	2.5	2.5	2.75	3.5
Leakage Power (pW)	53.2	66.9	77.9	102.1

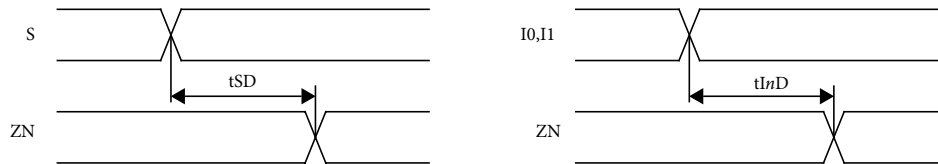
Pin Description

Name	Capacitance (pF)				Description
	MI02D0	MI02D1	MI02D2	MI02D4	
S	0.005	0.007	0.007	0.007	Select Input
I0	0.003	0.003	0.003	0.003	Data Input
I1	0.004	0.004	0.004	0.004	Data Input
Maximum capacitance					
ZN	0.15	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)			
	MI02D0	MI02D1	MI02D2	MI02D4
S	0.011	0.014	0.014	0.014
I0	0	0	0	0
I1	0	0	0	0
ZN	0.03	0.039	0.051	0.093

Waveforms



Timing numbers for MI02D0:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.360	0.188	0.464	0.248	0.670	0.368	1.082	0.608	1.907	1.087
tI1D	0.323	0.188	0.427	0.248	0.633	0.368	1.045	0.608	1.870	1.087
tSD	0.337	0.274	0.441	0.335	0.647	0.455	1.059	0.694	1.884	1.172

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.464	0.248	0.522	0.262	0.600	0.254	0.675	0.236	0.746	0.208
tI1D	0.427	0.248	0.483	0.269	0.552	0.267	0.621	0.257	0.683	0.236
tSD	0.441	0.335	0.462	0.420	0.463	0.501	0.456	0.573	0.440	0.636

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Timing numbers for MI02D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.322	0.167	0.368	0.192	0.459	0.238	0.639	1.521	1.000	0.508
tI1D	0.279	0.163	0.324	0.188	0.415	0.234	0.596	0.325	0.957	0.504
tSD	0.273	0.224	0.319	0.249	0.410	0.295	0.590	0.386	0.951	0.565

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.368	0.192	0.426	0.208	0.504	0.200	0.580	0.184	0.652	0.156
tI1D	0.324	0.188	0.382	0.206	0.454	0.203	0.524	0.190	0.589	0.167
tSD	0.319	0.249	0.322	0.323	0.310	0.392	0.292	0.456	0.268	0.513

Timing numbers for MI02D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.329	0.181	0.355	0.200	0.403	0.228	0.499	0.279	0.690	0.373
tI1D	0.283	0.177	0.309	0.196	0.358	0.224	0.453	0.275	0.644	0.369
tSD	0.277	0.238	0.303	0.257	0.352	0.285	0.447	0.336	0.638	0.430

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.355	0.200	0.413	0.216	0.492	0.209	0.570	0.193	0.642	0.166
tI1D	0.309	0.196	0.367	0.214	0.440	0.211	0.511	0.199	0.577	0.176
tSD	0.303	0.257	0.307	0.331	0.294	0.400	0.276	0.464	0.252	0.521

Timing numbers for MI02D4:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.370	0.230	0.385	0.243	0.412	0.265	0.459	0.298	0.548	0.353
tI1D	0.321	0.226	0.336	0.240	0.364	0.262	0.410	0.295	0.499	0.349
tSD	0.316	0.287	0.331	0.300	0.358	0.322	0.404	0.355	0.493	0.410

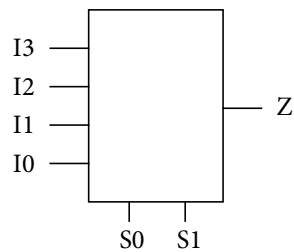
**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.385	0.243	0.443	0.260	0.524	0.253	0.602	0.238	0.676	0.211
tI1D	0.336	0.240	0.394	0.259	0.469	0.255	0.541	0.243	0.608	0.221
tSD	0.331	0.300	0.334	0.375	0.321	0.444	0.303	0.508	0.279	0.566

4-to-1 Multiplexers

MX04D0, MX04D1, MX04D2 and MX04D4

The MX04D0 (0.5x drive), MX04D1 (1x drive), MX04D2 (2x drive) and MX04D4 (4x drive) cells are 4-to-1 multiplexers. The state of the select inputs determines which data is present at the output.



Function Table

INPUTS						OUTPUT
S0	S1	I0	I1	I2	I3	Z
L	L	L	X	X	X	L
L	L	H	X	X	X	H
H	L	X	L	X	X	L
H	L	X	H	X	X	H
L	H	X	X	L	X	L
L	H	X	X	H	X	H
H	H	X	X	X	L	L
H	H	X	X	X	H	H

Cell Description

Macro Name:	MX04D0	MX04D1	MX04D2	MX04D4
Drive Capability:	0.5x	1x	2x	4x
Gate Equivalents:	4.75	4.75	5.25	6.25
Leakage Power (pW):	76.1	86.5	112.6	158.5

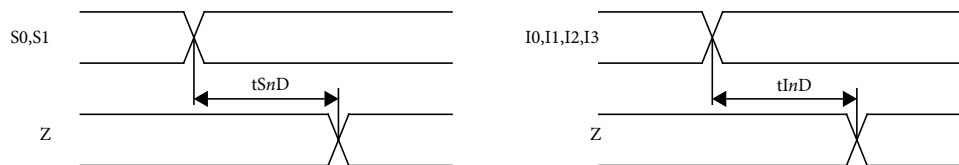
Pin Description

Name	Capacitance (pF)				Description
	MX04D0	MX04D1	MX04D2	MX04D4	
S0	0.009	0.009	0.009	0.009	Select Input
S1	0.011	0.011	0.011	0.011	Select Input
I0	0.004	0.004	0.004	0.004	Data Input
I1	0.004	0.004	0.004	0.004	Data Input
I2	0.004	0.004	0.004	0.004	Data Input
I3	0.004	0.004	0.004	0.004	Data Input
Maximum capacitance					
Z	0.15	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)			
	MX04D0	MX04D1	MX04D2	MX04D4
S0	0.025	0.025	0.026	0.025
S1	0.018	0.018	0.018	0.018
I0	0.001	0.001	0.001	0.001
I1	0	0	0	0
I2	0	0	0	0
I3	0	0	0	0
Z	0.047	0.056	0.08	0.111

Waveforms



Timing numbers for MX04D0:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.289	0.528	0.402	0.612	0.623	0.736	1.064	0.939	1.944	1.333
tI1D	0.294	0.523	0.407	0.606	0.628	0.730	1.069	0.934	1.949	1.330
tI2D	0.291	0.536	0.405	0.620	0.626	0.745	1.067	0.950	1.947	1.346
tI3D	0.294	0.531	0.407	0.613	0.628	0.737	1.069	0.941	1.949	1.335
tS0D	0.354	0.513	0.468	0.593	0.688	0.717	1.129	0.920	2.009	1.314
tS1D	0.391	0.574	0.504	0.657	0.725	0.781	1.166	0.985	2.049	1.381

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**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.402	0.612	0.423	0.650	0.436	0.702	0.440	0.764	0.429	0.832
tI1D	0.407	0.606	0.431	0.648	0.446	0.702	0.453	0.764	0.444	0.832
tI2D	0.405	0.620	0.428	0.661	0.441	0.717	0.446	0.781	0.435	0.850
tI3D	0.407	0.613	0.432	0.652	0.447	0.706	0.454	0.768	0.445	0.835
tS0D	0.468	0.593	0.549	0.622	0.621	0.637	0.683	0.645	0.735	0.643
tS1D	0.504	0.657	0.594	0.696	0.682	0.714	0.758	0.722	0.820	0.717

Timing numbers for MX04D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.224	0.518	0.276	0.575	0.375	0.665	0.565	0.800	0.943	1.023
tI1D	0.231	0.514	0.283	0.570	0.381	0.660	0.571	0.795	0.949	1.019
tI2D	0.227	0.526	0.279	0.584	0.378	0.675	0.568	0.810	0.946	1.035
tI3D	0.231	0.521	0.283	0.578	0.381	0.668	0.571	0.803	0.949	1.026
tS0D	0.290	0.502	0.342	0.558	0.440	0.646	0.630	0.781	1.008	1.004
tS1D	0.327	0.564	0.379	0.620	0.477	0.710	0.667	0.845	1.045	1.069

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.276	0.575	0.299	0.614	0.312	0.666	0.316	0.727	0.306	0.794
tI1D	0.283	0.570	0.308	0.612	0.324	0.667	0.330	0.728	0.321	0.796
tI2D	0.279	0.584	0.303	0.626	0.317	0.681	0.322	0.744	0.312	0.813
tI3D	0.283	0.578	0.308	0.617	0.325	0.670	0.331	0.732	0.322	0.798
tS0D	0.342	0.558	0.423	0.589	0.496	0.605	0.558	0.615	0.610	0.613
tS1D	0.379	0.620	0.469	0.661	0.557	0.678	0.634	0.688	0.696	0.682

Timing numbers for MX04D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.213	0.542	0.244	0.585	0.297	0.646	0.394	0.741	0.581	0.882
tI1D	0.213	0.546	0.244	0.589	0.296	0.650	0.393	0.745	0.580	0.887
tI2D	0.211	0.555	0.242	0.598	0.294	0.659	0.391	0.755	0.578	0.897
tI3D	0.213	0.554	0.244	0.597	0.296	0.658	0.393	0.753	0.580	0.895
tS0D	0.277	0.531	0.308	0.572	0.361	0.631	0.458	0.724	0.645	0.865
tS1D	0.316	0.590	0.347	0.632	0.399	0.692	0.497	0.787	0.683	0.929

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.244	0.585	0.268	0.623	0.284	0.674	0.290	0.735	0.282	0.802
tI1D	0.244	0.589	0.270	0.631	0.287	0.685	0.294	0.747	0.286	0.815
tI2D	0.242	0.598	0.266	0.639	0.281	0.693	0.286	0.757	0.276	0.827
tI3D	0.244	0.597	0.270	0.635	0.288	0.689	0.295	0.751	0.287	0.819
tS0D	0.308	0.572	0.389	0.605	0.462	0.623	0.524	0.639	0.576	0.637
tS1D	0.347	0.632	0.437	0.675	0.526	0.692	0.602	0.702	0.665	0.697

Timing numbers for MX04D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.312	0.573	0.329	0.585	0.361	0.606	0.422	0.639	0.545	0.702
tI1D	0.318	0.568	0.334	0.580	0.366	0.601	0.428	0.634	0.550	0.697
tI2D	0.315	0.581	0.331	0.593	0.363	0.614	0.424	0.647	0.547	0.709
tI3D	0.318	0.575	0.335	0.588	0.367	0.608	0.428	0.642	0.550	0.704
tS0D	0.378	0.559	0.394	0.571	0.426	0.592	0.488	0.625	0.610	0.688
tS1D	0.415	0.619	0.431	0.631	0.463	0.652	0.525	0.685	0.647	0.748

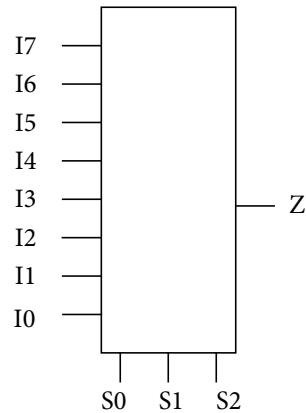
Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.329	0.585	0.351	0.623	0.362	0.676	0.364	0.736	0.352	0.802
tI1D	0.334	0.580	0.359	0.622	0.373	0.676	0.377	0.736	0.367	0.802
tI2D	0.331	0.593	0.355	0.634	0.367	0.689	0.369	0.751	0.357	0.819
tI3D	0.335	0.588	0.360	0.627	0.374	0.680	0.378	0.740	0.368	0.805
tS0D	0.394	0.571	0.475	0.600	0.548	0.617	0.609	0.629	0.661	0.627
tS1D	0.431	0.631	0.521	0.671	0.610	0.689	0.686	0.699	0.749	0.694

8-to-1 Multiplexers

MX08D1, MX08D2 and MX08D4

The MX08D1 (1x drive), MX08D2 (2x drive) and MX08D4 (4x drive) cells are 8-to-1 multiplexers. The state of the select inputs determines which data is present at the output.



Function Table

INPUTS											OUTPUT
S0	S1	S2	I0	I1	I2	I3	I4	I5	I6	I7	Z
L	L	L	L	X	X	X	X	X	X	X	L
L	L	L	H	X	X	X	X	X	X	X	H
H	L	L	X	L	X	X	X	X	X	X	L
H	L	L	X	H	X	X	X	X	X	X	H
L	H	L	X	X	L	X	X	X	X	X	L
L	H	L	X	X	H	X	X	X	X	X	H
H	H	L	X	X	X	L	X	X	X	X	L
H	H	L	X	X	X	H	X	X	X	X	H
L	L	H	X	X	X	X	L	X	X	X	L
L	L	H	X	X	X	X	H	X	X	X	H
H	L	H	X	X	X	X	X	L	X	X	L
H	L	H	X	X	X	X	X	H	X	X	H
L	H	H	X	X	X	X	X	X	L	X	L
L	H	H	X	X	X	X	X	X	H	X	H
H	H	H	X	X	X	X	X	X	X	L	L
H	H	H	X	X	X	X	X	X	X	H	H

Cell Description

Macro Name:	MX08D1	MX08D2	MX08D4
Drive Capability:	1x	2x	4x
Gate Equivalents:	9.	9.25	9.5
Leakage Power (pW):	193.1	223.0	278.2

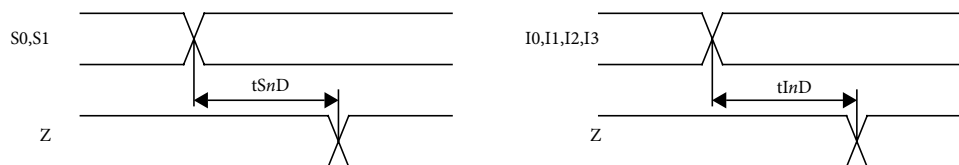
Pin Description

Name	Capacitance (pF)			Description
	MX08D1	MX08D2	MX08D4	
S0	0.004	0.004	0.004	Select Input
S1	0.005	0.005	0.005	Select Input
S2	0.004	0.004	0.004	Select Input
I0	0.003	0.003	0.003	Data Input
I1	0.003	0.003	0.003	Data Input
I2	0.004	0.004	0.004	Data Input
I3	0.003	0.003	0.003	Data Input
I4	0.004	0.004	0.004	Data Input
I5	0.003	0.003	0.003	Data Input
I6	0.003	0.003	0.003	Data Input
I7	0.003	0.003	0.003	Data Input
Maximum capacitance				
Z	0.3	0.6	1.2	Output

Pin Powers for
Standard Load = 0.032 pF, Standard input transition = 0.1 ns

Name	Pin Power (pW/Hz)		
	MX08D1	MX08D2	MX08D4
S0	0.069	0.069	0.069
S1	0.008	0.008	0.009
S2	0.006	0.006	0.006
I0	0.014	0.014	0.014
I1	0	0	0
I2	0.021	0.021	0.021
I3	0.001	0.001	0.001
I4	0.018	0.018	0.018
I5	0.001	0.001	0.001
I6	0.017	0.017	0.017
I7	0	0	0
Z	0.075	0.113	0.252

Waveforms



Timing numbers for MX08D1:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.516	0.412	0.576	0.458	0.671	0.538	0.828	0.681	1.130	0.958
tI1D	0.515	0.410	0.575	0.456	0.669	0.536	0.827	0.679	1.129	0.956
tI2D	0.475	0.386	0.531	0.432	0.623	0.512	0.779	0.655	1.081	0.931
tI3D	0.474	0.386	0.530	0.432	0.622	0.512	0.779	0.655	1.081	0.932
tI4D	0.479	0.391	0.535	0.437	0.627	0.517	0.784	0.660	1.086	0.937
tI5D	0.477	0.390	0.533	0.435	0.625	0.515	0.782	0.658	1.084	0.935
tI6D	0.531	0.428	0.591	0.474	0.687	0.554	0.845	0.697	1.147	0.974
tI7D	0.529	0.425	0.589	0.471	0.684	0.551	0.842	0.694	1.144	0.971
tS0D	0.626	0.504	0.686	0.550	0.780	0.630	0.937	0.773	1.239	1.050
tS1D	0.458	0.343	0.518	0.389	0.612	0.469	0.769	0.612	1.071	0.888
tS2D	0.419	0.329	0.478	0.375	0.573	0.455	0.730	0.598	1.032	0.875

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.576	0.458	0.643	0.533	0.735	0.611	0.836	0.677	0.938	0.730
tI1D	0.575	0.456	0.642	0.532	0.733	0.610	0.834	0.676	0.937	0.729
tI2D	0.531	0.432	0.600	0.499	0.687	0.560	0.776	0.613	0.860	0.654
tI3D	0.530	0.432	0.600	0.499	0.687	0.561	0.776	0.613	0.860	0.654
tI4D	0.535	0.437	0.604	0.505	0.691	0.567	0.781	0.619	0.865	0.660
tI5D	0.533	0.435	0.603	0.504	0.689	0.566	0.779	0.619	0.863	0.661
tI6D	0.591	0.474	0.660	0.550	0.752	0.630	0.852	0.698	0.954	0.753
tI7D	0.589	0.471	0.658	0.548	0.749	0.628	0.849	0.696	0.951	0.751
tS0D	0.686	0.550	0.767	0.610	0.842	0.650	0.906	0.678	0.956	0.693
tS1D	0.518	0.389	0.574	0.497	0.622	0.653	0.666	0.810	0.703	0.948
tS2D	0.478	0.375	0.527	0.470	0.570	0.623	0.610	0.779	0.645	0.918

Timing numbers for MX08D2:

**Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF**

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.578	0.458	0.629	0.491	0.697	0.543	0.799	0.631	0.962	0.784
tI1D	0.576	0.457	0.627	0.489	0.695	0.541	0.797	0.630	0.960	0.783
tI2D	0.523	0.429	0.570	0.462	0.633	0.514	0.732	0.602	0.893	0.755
tI3D	0.523	0.429	0.569	0.462	0.632	0.514	0.731	0.602	0.892	0.755
tI4D	0.527	0.434	0.574	0.467	0.637	0.519	0.737	0.607	0.898	0.760
tI5D	0.525	0.432	0.572	0.465	0.635	0.517	0.735	0.605	0.896	0.758
tI6D	0.592	0.475	0.643	0.508	0.711	0.560	0.814	0.648	0.977	0.801
tI7D	0.590	0.472	0.641	0.505	0.709	0.557	0.812	0.645	0.975	0.798
tS0D	0.687	0.550	0.738	0.583	0.806	0.635	0.908	0.724	1.071	0.877
tS1D	0.520	0.384	0.571	0.417	0.639	0.469	0.741	0.557	0.904	0.710
tS2D	0.482	0.370	0.533	0.403	0.601	0.455	0.703	0.543	0.866	0.696

**Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns**

Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.629	0.491	0.695	0.568	0.783	0.650	0.881	0.719	0.985	0.776
tI1D	0.627	0.489	0.693	0.566	0.781	0.649	0.879	0.718	0.983	0.775
tI2D	0.570	0.462	0.638	0.530	0.724	0.595	0.813	0.650	0.899	0.693
tI3D	0.569	0.462	0.637	0.531	0.723	0.595	0.812	0.650	0.898	0.693
tI4D	0.574	0.467	0.642	0.536	0.727	0.601	0.817	0.655	0.903	0.699
tI5D	0.572	0.465	0.640	0.535	0.725	0.600	0.814	0.655	0.901	0.699
tI6D	0.643	0.508	0.711	0.585	0.799	0.669	0.896	0.741	0.999	0.799
tI7D	0.641	0.505	0.709	0.583	0.797	0.667	0.893	0.739	0.996	0.797
tS0D	0.738	0.583	0.819	0.644	0.895	0.684	0.958	0.712	1.008	0.727
tS1D	0.571	0.417	0.628	0.519	0.675	0.673	0.718	0.833	0.755	0.978
tS2D	0.533	0.403	0.581	0.491	0.624	0.638	0.662	0.796	0.698	0.943

Timing numbers for MX08D4:

Propagation Delays (ns) for sample loads:
Standard input transition = 0.1 ns, load unit = 0.004 pF

Fanout load	4		8		16		32		64	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.758	0.584	0.792	0.605	0.849	0.643	0.930	0.701	1.044	0.798
tI1D	0.757	0.583	0.791	0.604	0.848	0.641	0.928	0.699	1.043	0.797
tI2D	0.674	0.549	0.705	0.570	0.757	0.607	0.830	0.665	0.938	0.763
tI3D	0.673	0.549	0.704	0.570	0.757	0.608	0.829	0.665	0.938	0.763
tI4D	0.709	0.554	0.709	0.575	0.762	0.612	0.834	0.670	0.943	0.767
tI5D	0.676	0.552	0.707	0.573	0.760	0.611	0.832	0.668	0.941	0.766
tI6D	0.772	0.601	0.806	0.622	0.863	0.660	0.944	0.717	1.059	0.815
tI7D	0.770	0.598	0.804	0.619	0.861	0.657	0.942	0.714	1.057	0.812
tS0D	0.867	0.677	0.901	0.698	0.958	0.735	1.038	0.793	1.153	0.890
tS1D	0.703	0.502	0.737	0.523	0.794	0.560	0.874	0.618	0.988	0.715
tS2D	0.665	0.487	0.699	0.508	0.756	0.546	0.836	0.603	0.951	0.701

Propagation Delays (ns) for sample input transitions:
Standard load = 0.032 pF, input unit = 0.1 ns

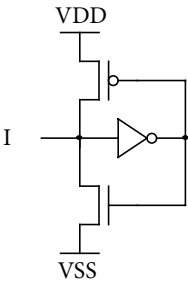
Input transition	1		5		10		15		20	
Edge	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL	RISE	FALL
tI0D	0.792	0.605	0.857	0.684	0.938	0.771	1.028	0.847	1.134	0.908
tI1D	0.791	0.604	0.855	0.682	0.937	0.770	1.027	0.846	1.132	0.907
tI2D	0.705	0.570	0.772	0.640	0.853	0.709	0.938	0.767	1.029	0.814
tI3D	0.704	0.570	0.772	0.640	0.853	0.709	0.938	0.767	1.028	0.814
tI4D	0.709	0.575	0.776	0.645	0.857	0.715	0.942	0.773	1.033	0.820
tI5D	0.707	0.573	0.774	0.644	0.855	0.714	0.940	0.772	1.031	0.819
tI6D	0.806	0.622	0.872	0.700	0.954	0.789	1.044	0.867	1.148	0.930
tI7D	0.804	0.619	0.870	0.698	0.952	0.787	1.041	0.865	1.146	0.928
tS0D	0.901	0.698	0.981	0.758	1.057	0.799	1.120	0.826	1.169	0.841
tS1D	0.737	0.523	0.794	0.618	0.841	0.762	0.883	0.918	0.919	1.073
tS2D	0.699	0.508	0.746	0.586	0.788	0.721	0.826	0.872	0.861	1.027

MISCELLANEOUS FUNCTIONS

Three-State Bus Holder

BH01D1

The BH01D1 cell is a high-impedance bus holder used to prevent an internal 3-state line going into an unknown state that can consume power. This cell is only needed when all 3-state drivers on a line can be disabled at the same time. The cell holds the last value that was driven onto the line.



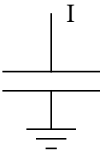
Cell Description

Gate Equivalents:	2.
Leakage Power (pW):	8.8
Capacitance (pF) of pin I:	0.004
Pin Power (pW/Hz) of pin I for Standard Input Transition = 0.1 ns	0.001

Unit Capacitor Load

CLOAD1

The CLOAD1 cell is a capacitive load equivalent of one flip-flop clock input. It is applied to clock driver outputs to balance capacitive loads to minimize clock skew.



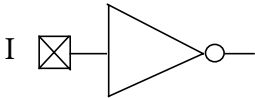
Cell Description

Gate Equivalents:	0.75
Leakage Power (pW):	8.8
Capacitance (pF) of pin I:	0.004
Pin Power (pW/Hz) of pin I for Standard Input Transition = 0.1 ns	0.001

Diode Cell

ADIODE

The ADIODE cell has its input I connected to a P and N type diode. This cell can be added in a routed netlist on a net to avoid antenna rule errors.



Cell Description

Gate Equivalents:	0.5
Leakage Power (pW):	8.8
Capacitance (pF) of pin I:	0.004
Pin Power (pW/Hz) of pin I for Standard Input Transition = 0.1 ns	0.001

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